From Complex CAPM to a Financial Gauge–Dirac System - Charge, Spin, Margin Gates, and Recursive Ledger Closure in Constraint-Bearing Finance

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Abstract

Modern finance already contains several mathematically mature layers.

CAPM relates systematic market exposure to required return. Discounted-cash-flow valuation converts expected cash flows into present value. Margin systems convert asset value, liabilities, collateral haircuts, and maintenance rules into admissible or inadmissible account states. Clearing, settlement, risk, treasury, accounting, legal, and regulatory systems then represent the same financial position through different operational frames.

These layers are usually studied separately.

This article asks whether they can be organized into one disciplined architecture of **identity-bearing financial transformation**.

The starting point is Complex CAPM:

$A_t^2 = R_t^2 + Q_t^2.$ (0.1)

$R_t = A_t \cos \theta_t.$ (0.2)

$Q_t = A_t \sin \theta_t.$ (0.3)

$Z_t = R_t + iQ_t = A_t \exp(i\theta_t).$ (0.4)

Here A_t is the baseline value amplitude, R_t is CAPM-admitted value, Q_t is the conjugate pressure coordinate implied by the declared valuation filter, and θ_t is the valuation phase.

This complex completion does not alter CAPM's scalar valuation. It preserves an orthogonal coordinate that scalar valuation normally compresses. The central local relation is:

$\partial R/\partial \theta = -Q.$ (0.5)

Thus Q is the first-order phase exposure of admitted value. It is not automatically realized loss, volatility, beta, margin shortfall, ledger residual, or financial charge.

Complex CAPM alone, however, remains a valuation geometry. It does not explain how a leveraged financial subject behaves when valuation movement encounters contractual constraints.

To make the problem operational, the article introduces a calibration case:

> A leveraged financial account holds a CAPM-valued risky asset against a funding liability under a collateral agreement containing an enforceable margin-call

mechanism.

This subject carries several stable relational orientations:

- an asset-claim orientation;
- a funding-obligation orientation;
- a contingent collateral obligation.

These orientations are candidates for **financial charge** only if they possess declared carriers, fields, signs, coupling laws, transport rules, interaction vertices, balance rules, and residual registers. Otherwise they remain sensitivities or exposures.

The margin mechanism supplies an authoritative gate. When the collateral buffer becomes negative, a dormant obligation becomes operational. Yet issuance of the margin call does not complete the financial event. The subject must post collateral, deleverage, undergo liquidation, or enter default and recovery. These consequences must then be reconciled across collateral, funding, risk, accounting, legal, and regulatory ledgers.

The account therefore possesses a candidate two-component identity:

$$\Psi_S = \begin{bmatrix} Z_{market} \\ Z_{ledger} \end{bmatrix}. \qquad (0.6)$$

The first component represents outward market and balance-sheet action. The second represents collateral admission, settlement, recognition, reconciliation, and future-conditioning trace.

This construction adapts the action–ledger spinor proposed in the generalized macro-Dirac framework:

$$\Psi_B = [\psi_{action}, \psi_{ledger}]^{T}. \qquad (0.7)$$

That source interprets macro spin not as literal physical rotation, but as the fact that one outward action cycle does not restore accountable identity. A second return-to-ledger cycle is required.

The article then develops governed transport among financial frames. A market value, collateral value, accounting amount, risk exposure, and regulatory exposure may differ while referring to the same underlying position. A valid transport system must therefore preserve a declared identity kernel while allowing frame-local representations to change.

The proposed continuous kernel is:

$$[i\Gamma^{0}D_{\tau} + ic_{P}\Gamma^{1}\mathfrak{D}_{G} - M_S]\Psi_S = \mathscr{R}_S. \qquad (0.8)$$

Here:

- Ψ_S is the charged market–ledger financial identity;
- D_{τ} is the field-coupled financial derivative;
- $\mathfrak{D}_G$ is the governed cross-frame transport operator;
- Γ^{0} and Γ^{1} distinguish and couple the two closure components;

- c_P is the maximum coherent rate of market-to-ledger propagation under protocol P;
- M_S is the identity-preserving mass operator;
- $\mathscr{R}$_S is unresolved valuation, transport, gate, or ledger residual.

The equation is only the continuous kernel. Margin finance is a hybrid system. At a binding constraint, a discrete gate acts:

$$\Psi_S(\tau_k^+) = G_{margin}[\Psi_S(\tau_k^-), L_k] + \eta_k. \qquad (0.9)$$

The ledger then updates:

$$L_{k+1} = \text{Update}(L_k, \text{Trace}_k, \text{ChargeFlow}_k, \text{Residual}_k). \qquad (0.10)$$

The resulting architecture is therefore a **Financial Gauge–Dirac–Gate–Ledger system**, not merely one continuous equation.

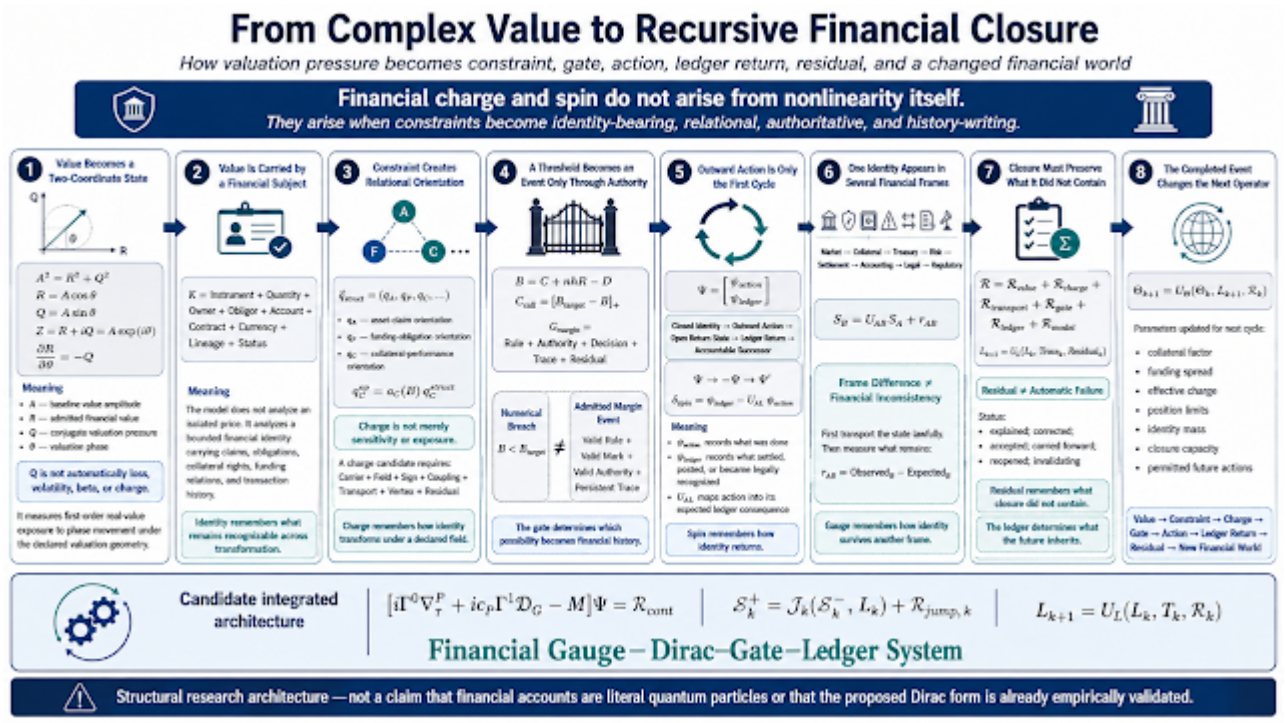
Its wider thesis is:

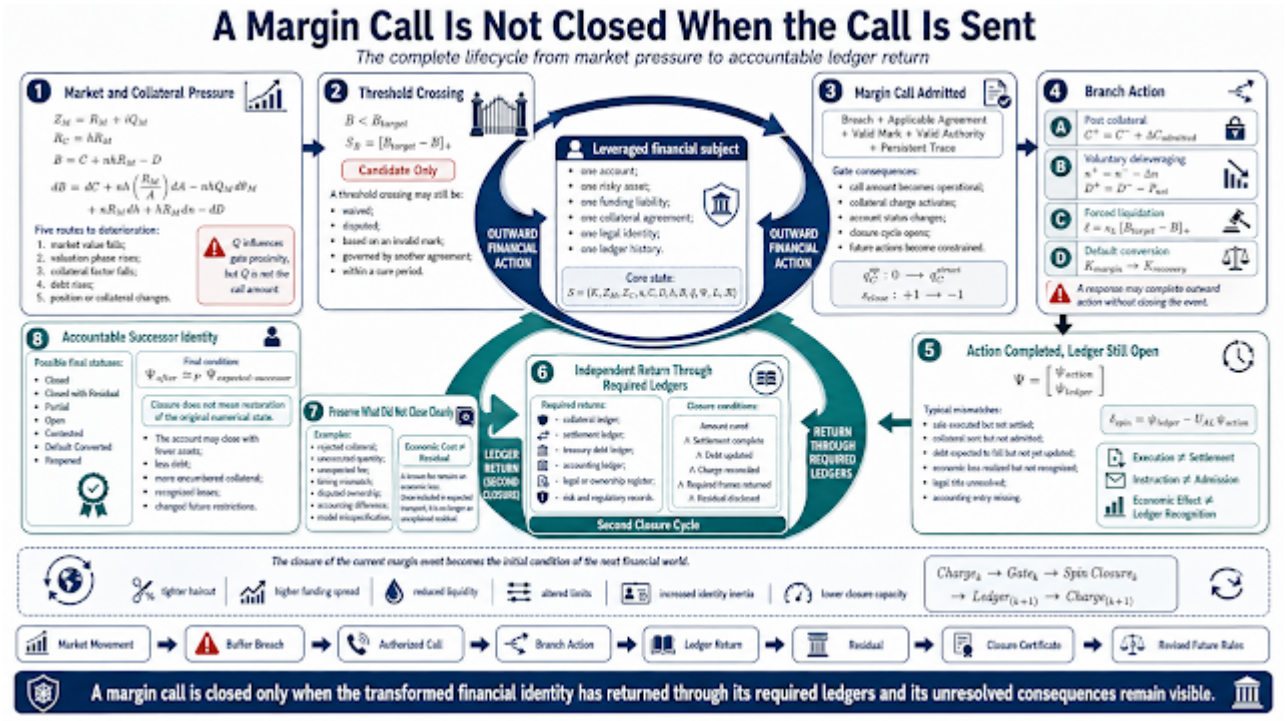
> Financial charge and spin do not arise merely because finance is nonlinear. They arise when constraints become identity-bearing, relational, authoritative, cross-frame, and history-writing.

Nonlinearity frequently follows through leverage, thresholds, positive-part functions, state-dependent collateral, forced liquidation, market impact, and recursive ledger feedback. But nonlinearity is neither necessary nor sufficient for charge or spin.

The proposed system is a formal research architecture, not a validated universal financial law. Its advanced terminology must be removed whenever simpler real-variable, state-space, hybrid-automaton, or reconciliation models perform equally well.







Source Grounding and Status of the Present Construction

This article combines three previously developed structures.

Complex Finance Geometry

The first structure is the complex completion:

$$Z = R + iQ. \qquad (0.11)$$

The Finance Geometry framework interprets R as value admitted by a declared mature financial filter and Q as the orthogonal pressure coordinate implied by that same filter. It expressly states that Q is not the ordinary scalar haircut A − R and that its practical legitimacy depends on incremental diagnostic or predictive value beyond scalar finance.

Transformation Memory in the Financial Standard Model

The second structure distinguishes:

- charge as remembered coupling orientation;
- spin as remembered return-to-self closure;
- mass as the cost of remaining the same identity;
- gates as admission mechanisms;
- ledgers as inherited historical constraint.

The source proposes the recurrence:

$$(I_p, q_p, \Psi_p) \to G_p \to (T_p, R_p) \to L_{p+1} \to (I_{p+1}, q_{p+1}, \Psi_{p+1}). \qquad (0.12)$$

It also imposes strict admission tests. A candidate charge must possess a transformation law, coupling law, transport law, vertex rule, and residual register. A candidate spinor must possess bounded identity, outward action, consequence, independent ledger return, double closure, residual, and cross-frame transport.

The Generalized Dirac Archetype

The third structure proposes:

$$(i\Gamma^a \nabla^p_a - M_B)\Psi_B = \mathscr{R}_P. \qquad (0.13)$$

The generalized Dirac paper interprets this as first-order propagation of accountable identity across frames. It does not claim that macro systems are physical particles. Its spinor is an action–ledger identity; its mass is identity inertia; its frame covariance is A-B Fixedness; and its right-hand side is unclosed residual.

Status of the New Material

The following are **new proposals developed in the present article**:

1. the leveraged margin account as the calibration subject;
2. the market–collateral complex spinor;
3. the separation of structural charge from CAPM factor loading and phase sensitivity;
4. the margin gate as contingent-charge activation;
5. the cross-frame transport graph;
6. loop residual as candidate financial curvature;
7. the Financial Gauge–Dirac continuous kernel;
8. the hybrid gate and ledger equations;

9. the general promotion ladder from identity-bearing constraint to charge, spin, gauge, and Dirac-like propagation.

These constructions are not stated as established results in the source articles. They are proposed syntheses that must be compared with simpler null models.

Part I — Why Smooth Valuation Is Not Enough

0. Reader's Guide

0.1 The question

The article asks:

> What additional structure becomes necessary when a CAPM-valued asset is held by a leveraged financial subject whose liabilities, collateral rules, margin gates, and ledgers affect its future state?

Ordinary valuation asks:

What is the asset worth?

The present framework adds:

- Who carries the claim?
- Who carries the corresponding obligation?
- Which field acts on that identity?
- What constraint defines admissibility?
- Which authority turns a threshold into an event?
- What action follows?
- Which independent ledgers must close?
- What remains unresolved?
- How does that closure alter future exposure?

These are not merely extra pricing variables. They concern the continued identity of the financial subject.

0.2 The calibration subject

The minimum subject is:

S := one leveraged account holding one risky asset against one funding liability under one collateral and margin agreement. (0.14)

The initial model deliberately excludes:

- multiple risky assets;
- portfolio cross-margining;
- options;
- multiple currencies;
- multiple collateral classes;
- stochastic recovery;
- regulatory capital;
- endogenous counterparties;
- network contagion.

These will be added only after the minimum architecture is clear.

0.3 The article's central distinctions

The framework depends on several distinctions that must not be collapsed.

Value is not identity

Two frames may assign different values to the same claim.

Q is not residual

Q belongs inside the declared complex valuation closure:

$$Q = \sqrt{(A^2 - R^2)}. \qquad (0.15)$$

Residual belongs outside the closure:

$$\varepsilon_P = \text{ObservedConsequence} - \text{ModelledOrRecognizedConsequence}. \qquad (0.16)$$

Therefore:

$$Q_P \neq \varepsilon_P. \qquad (0.17)$$

The source framework treats this distinction as necessary for falsifiability.

Charge is not beta

Beta is a statistical factor loading. It may help determine coupling strength, but it is not automatically a stable claim–obligation orientation.

Spin is not price rotation

Financial spin concerns the completion of a charged identity through action and independent ledger return.

A threshold is not automatically a gate

A threshold becomes a gate only when it has:

- authority;
- an admission or rejection rule;
- consequential state change;
- trace;
- residual handling.

Frame comparison is not automatically gauge transport

Gauge terminology requires:

- a declared source frame;
- a declared target frame;
- an invariant identity kernel;
- a connection or transport rule;
- a covariance condition;
- a loop test.

Two variables are not automatically a spinor

A spinor claim requires irreducible coupled components and a failure mode that a scalar representation does not capture.

0.4 Epistemic labels

Every major equation in this article belongs to one of four classes.

Exact identity under declared definitions

Example:

$A^2 = R^2 + Q^2.$ (0.18)

Definition

Example:

$B := \text{EligibleCollateral} - \text{FundingLiability}.$ (0.19)

Proposed mathematical construction

Example:

$\Psi_S := [Z_{market}, Z_{collateral}]^T.$ (0.20)

Empirical hypothesis

Example:

A rising market–ledger spinor split predicts unresolved margin closure. (0.21)

The article will not treat all four classes as equally established.

0.5 The claim ceiling

The present article may responsibly claim that:

- Complex CAPM supplies a rigorous local valuation geometry;
- a margin account provides a clean identity-bearing constraint system;
- contractual rights and obligations provide plausible charge candidates;
- margin calls provide authoritative gate events;
- margin resolution contains an independent ledger-return cycle;
- market, collateral, risk, and accounting frames require governed transport;
- a first-order multicomponent equation is a coherent formal candidate;
- the complete system generates testable hypotheses and reduction rules.

It may not presently claim that:

- financial subjects are physical fermions;
- CAPM pressure is physical gauge charge;
- margin closure is literally quantum spin;
- the proposed Γ algebra is uniquely determined;
- the model improves forecasting;
- the model is a universal financial law;
- the system guarantees profit or risk control.

The Financial Standard Model source states the same methodological ceiling: the current result is a testable architecture, not a validated universal model.

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1. The Limits of Linear and Scalar Finance

1.1 The linear expected-return kernel

The standard CAPM relation is:

$r_{CAPM} = r_f + \beta ERP.$ (1.1)

where:

- r_CAPM = required return for the risky claim;
- r_f = declared baseline or risk-free rate;
- β = systematic market-risk loading;
- ERP = equity risk premium.

This relation is linear in β and ERP under the declared model.

The familiar simplicity of equation (1.1) can create the impression that the whole CAPM valuation system is linear.

It is not.

1.2 Discounted valuation is nonlinear in required return

Consider one future cash flow CF_t payable at horizon t.

Its CAPM-discounted value is:

$R_t = CF_t/(1 + r_CAPM)^t.$ (1.2)

Substituting CAPM:

$R_t = CF_t/(1 + r_f + \beta ERP)^t.$ (1.3)

Even though required return is linear in β and ERP, present value is nonlinear in the combined rate.

The derivative with respect to required return is:

$\partial R_t/\partial r_CAPM = -tR_t/(1 + r_CAPM).$ (1.4)

The second derivative is:

$\partial^2 R_t/\partial r_CAPM^2 = t(t + 1)R_t/(1 + r_CAPM)^2.$ (1.5)

Thus ordinary discounted valuation already contains convexity.

The present article therefore does not begin from the claim:

> Finance is linear until constraints make it nonlinear.

The more accurate claim is:

> Finance often begins with a locally smooth valuation kernel. Constraints, gates, forced actions, and recursive ledgers then transform that smooth kernel into a hybrid identity-bearing system.

1.3 A scalar valuation does not describe the entire subject

Suppose a risky asset is valued at R.

A scalar model may represent the account as:

Equity = AssetValue − Debt. (1.6)

This is correct as far as it goes.

But it does not directly encode:

- the risk pressure implied by the valuation filter;
- the signed claim orientation;
- the funding obligation;
- the contingent collateral obligation;
- the distance to the margin boundary;
- the authority of the broker or clearing entity;
- the difference between call issuance and call resolution;
- the translation between market and collateral values;
- the reconciliation between collateral and accounting ledgers;
- the residual left after liquidation or default.

These structures cannot all be recovered from one scalar net-value number.

1.4 A nonlinear formula does not automatically create charge

Consider a nonlinear utility function:

$U(W) = W - aW^2$. (1.7)

The function contains curvature, but it does not by itself identify:

- a stable carrier;
- a claim–obligation polarity;
- a field-specific transformation;
- a charge-transfer vertex;
- a conservation or balance rule;
- a cross-frame transport law.

Therefore:

Nonlinear Function $\nRightarrow$ Financial Charge. (1.8)

A candidate charge requires considerably more structure.

1.5 A constraint does not automatically create spin

Consider a simple bound:

$x \leq x_{max}.$ (1.9)

This creates a feasible region.

But it does not necessarily create:

- an authoritative event;
- an outward action;
- an enforceable obligation;
- a distinct ledger-return process;
- residual if return fails.

Therefore:

$\text{Constraint} \not\Rightarrow \text{Financial Spin}.$ (1.10)

Spin-like closure becomes plausible only when a constraint is attached to a bounded identity and when crossing it activates consequential action that must return through an independent ledger process.

1.6 A locally linear process may still require spin-like closure

A trade can be represented through locally simple operations:

$\text{Order} \rightarrow \text{Execution} \rightarrow \text{Confirmation} \rightarrow \text{Settlement} \rightarrow \text{Accounting}.$ (1.11)

Each step may be approximately linear or rule-based.

Yet execution does not complete the transaction's accountable identity.

Until settlement and ledger recognition occur, the trade leaves:

- cash obligations;
- delivery obligations;
- counterparty exposure;
- reconciliation requirements;
- accounting consequences;
- possible residual.

Thus:

Local Linearity ⇏ Single-Cycle Closure. (1.12)

Nonlinearity is not necessary for spin-like structure.

1.7 The stronger transition

The relevant promotion sequence is:

Smooth Value
→ Bounded Financial Identity
→ Relational Right or Obligation
→ Constraint
→ Conditional Obligation
→ Authoritative Gate
→ Outward Action
→ Independent Ledger Return
→ Revised Identity. (1.13)

Charge becomes plausible at the stage where stable relational orientation appears.

Spin becomes plausible at the stage where action and return become independently necessary.

Gauge structure becomes plausible when the same identity must survive translation among multiple financial frames.

Dirac-like structure becomes plausible only when the identity is irreducibly multicomponent, first-order propagation is useful, a mass-like term binds the components, and the frame transformations obey explicit constraints.

1.8 The margin account as the calibration atom

A leveraged margin account contains all the required stages in a compact form.

Before the margin boundary is crossed, the subject carries:

- a risky asset claim;
- a funding obligation;
- a dormant collateral obligation;
- a CAPM valuation phase;
- leverage;
- a collateral haircut;
- a margin buffer.

At the boundary:

- the margin authority issues a call;

- the dormant collateral obligation becomes active;
- withdrawals may be restricted;
- collateral must be posted;
- the position may be reduced;
- forced liquidation may begin.

After the call:

- the collateral ledger must update;
- funding balances may change;
- asset quantity may change;
- realized loss may be recognized;
- accounting and risk systems must reconcile;
- residual shortfall may remain.

The account is therefore neither merely a priced asset nor merely a workflow.

It is an identity-bearing constrained financial subject.

Part II — Complex CAPM as the Valuation Kernel

2. The Complex CAPM Plane

2.1 Baseline valuation amplitude

Consider one positive future cash flow CF_t payable at horizon t.

Define the baseline value amplitude:

$A_t = CF_t/(1 + r_base)^t.$ (2.1)

Here r_base is the declared baseline discount rate.

The word amplitude does not imply a physical quantum amplitude. It means that A_t is the pre-CAPM value magnitude used to construct the declared two-dimensional geometry.

2.2 CAPM-admitted value

Define the CAPM required return:

$r_t = r_base + \beta ERP.$ (2.2)

The CAPM-admitted value is:

$R_t = CF_t/(1 + r_t)^t.$ (2.3)

When:

$r_t \geq r_base, \qquad (2.4)$

and:

$CF_t > 0, \qquad (2.5)$

then:

$0 < R_t \leq A_t. \qquad (2.6)$

The ratio is:

$R_t/A_t = [(1 + r_base)/(1 + r_t)]^t. \qquad (2.7)$

2.3 Valuation phase

Define:

$\cos \theta_t = R_t/A_t. \qquad (2.8)$

Therefore:

$\theta_t = \arccos(R_t/A_t). \qquad (2.9)$

Substituting equations (2.1)–(2.3):

$\theta_t = \arccos\{[(1 + r_base)/(1 + r_base + \beta ERP)]^t\}. \qquad (2.10)$

Under the positive-value convention:

$0 \leq \theta_t \leq \pi/2. \qquad (2.11)$

The angle θ_t records the deviation between baseline amplitude and CAPM-admitted value.

It should initially be called **valuation phase** or **filter phase**.

It is not yet:

- gauge phase;
- closure phase;
- spin phase;
- secondary time.

Those stronger interpretations require additional evidence.

2.4 Conjugate pressure coordinate

Define:

$Q_t = \sqrt{(A_t^2 - R_t^2)}. \qquad (2.12)$

Equivalently:

$R_t = A_t \cos \theta_t.$ (2.13)

$Q_t = A_t \sin \theta_t.$ (2.14)

The completed state is:

$Z_t = R_t + iQ_t.$ (2.15)

or:

$Z_t = A_t \exp(i\theta_t).$ (2.16)

Its norm is:

$|Z_t|^2 = R_t^2 + Q_t^2 = A_t^2.$ (2.17)

A_t, R_t, and Q_t all possess monetary units.

This dimensional compatibility is one reason the CAPM construction is stronger than arbitrarily combining price with a standardized indicator.

2.5 Q is not the scalar valuation haircut

The ordinary real-axis haircut is:

$H_t = A_t - R_t.$ (2.18)

The conjugate pressure coordinate is:

$Q_t = \sqrt{(A_t^2 - R_t^2)}.$ (2.19)

Therefore:

$Q_t \neq H_t.$ (2.20)

Indeed:

$Q_t^2 = (A_t - R_t)(A_t + R_t).$ (2.21)

So:

$Q_t^2 = H_t(A_t + R_t).$ (2.22)

For small θ:

$R \approx A(1 - \theta^2/2).$ (2.23)

$Q \approx A\theta.$ (2.24)

Hence:

$H = A - R \approx A\theta^2/2.$ (2.25)

and:

$H \approx Q^2/(2A).$ (2.26)

The scalar haircut is locally second-order in phase, while Q is locally first-order.

This is why Q may be numerically much larger than A − R without representing a larger monetary loss. The Finance Geometry source makes this distinction explicit.

2.6 The conjugate risk theorem

Theorem 2.1 — CAPM Phase Delta

Along a fixed-amplitude valuation orbit:

$$\partial R/\partial \theta = -Q. \qquad (2.27)$$

Proof

From:

$$R = A \cos \theta, \qquad (2.28)$$

holding A fixed:

$$\partial R/\partial \theta = -A \sin \theta. \qquad (2.29)$$

Since:

$$Q = A \sin \theta, \qquad (2.30)$$

it follows that:

$$\partial R/\partial \theta = -Q. \qquad (2.31)$$

■

Define the signed phase delta:

$$\Delta_\theta := \partial R/\partial \theta. \qquad (2.32)$$

Then:

$$\Delta_\theta = -Q. \qquad (2.33)$$

For a small phase movement:

$$dR = -Q d\theta. \qquad (2.34)$$

When amplitude also changes:

$$dR = (R/A)dA - Q d\theta. \qquad (2.35)$$

Equation (2.35) separates:

- radial value-amplitude movement;
- angular valuation-filter movement.

This separation will later allow the article to distinguish ordinary cash-flow change from CAPM phase movement and collateral-filter change.

2.7 Q is exposure, not consequence

Equation (2.34) does not say that Q is a realized loss.

It says that Q is the first-order coefficient relating phase movement to admitted-value movement.

The correct sequence is:

Q Exposure
→ Actual Phase Movement
→ Economic Value Change
→ Margin or Recognition Gate
→ Ledgered Consequence
→ Residual. (2.36)

Without actual movement:

Q alone does not create P&L. (2.37)

Without a gate:

Economic movement does not automatically become recognized history. (2.38)

Without ledger return:

Recognition does not necessarily complete accountable closure. (2.39)

These distinctions are essential for the later Gauge–Dirac construction.

2.8 CAPM phase sensitivity

Differentiate equation (2.8):

$-\sin\theta \cdot d\theta = d(R/A).$ (2.40)

Holding A fixed:

$d\theta = -dR/Q.$ (2.41)

From the CAPM value sensitivity:

$\partial R/\partial r = -tR/(1 + r),$ (2.42)

we obtain:

$\partial\theta/\partial r = tR/[(1 + r)Q].$ (2.43)

Since:

$r = r_base + \beta ERP,$ (2.44)

the phase sensitivity to ERP is:

$\partial\theta/\partial ERP = tR\beta/[(1 + r)Q].$ (2.45)

The phase sensitivity to beta is:

$$\partial\theta/\partial\beta = tRERP/[(1 + r)Q]. \qquad (2.46)$$

These are local sensitivities.

They are not automatically structural charges.

A structural charge should remain identifiable across transformations and vertices. Equations (2.45) and (2.46) vary with:

- R;
- Q;
- horizon;
- required return;
- beta;
- ERP.

They are state-dependent coupling responses.

This distinction will become central in Part III.

2.9 The role of Complex CAPM in the larger theory

Complex CAPM supplies:

1. a monetary amplitude A;
2. an admitted value R;
3. a conjugate coordinate Q;
4. a valuation phase θ;
5. exact local sensitivity identities;
6. a clean distinction between exposure and movement.

It does not yet supply:

- a financial subject;
- a funding liability;
- leverage;
- collateral eligibility;
- an authoritative gate;
- action–ledger double closure;
- a charge-transfer vertex;

- a frame graph;
- a gauge connection;
- a Dirac operator.

Complex CAPM is therefore the **valuation kernel**, not the complete Financial Gauge–Dirac system.

The next part will construct the bounded financial subject that carries this valuation state.

3. The Leveraged Financial Subject

3.1 Why the primitive object cannot be only an asset

An asset is a contractual or economic object.

A financial subject is the bounded entity that carries the asset, finances it, becomes constrained by it, and must account for the resulting consequences.

Consider one risky security.

The security may possess:

- expected cash flows;
- market value;
- beta;
- liquidity characteristics;
- collateral eligibility;
- legal rights.

But the security alone does not specify:

- who owns it;
- whether it is held long or short;
- how many units are held;
- whether the position is leveraged;
- which liability finances it;
- which margin agreement applies;
- which collateral has been posted;
- which ledger recognizes the position;
- which party can demand additional collateral;
- whether a margin call remains unresolved.

The Financial Gauge–Dirac construction therefore begins not from an isolated asset, but from a bounded financial subject.

Define:

S := one leveraged account carrying one risky claim, one funding liability, and one margin agreement. (3.1)

This subject is the minimum object capable of supporting all of the following in one model:

- CAPM valuation phase;
- signed claim orientation;
- funding obligation;
- contingent collateral obligation;
- margin-gate activation;
- action–ledger separation;
- cross-frame transport;
- recursive balance-sheet change.

3.2 The identity kernel

A useful financial model must distinguish the subject's identity from its current numerical valuation.

Let the identity kernel be:

$K_S := (x,n,o,a,c,T,e,\kappa,\nu)$. (3.2)

where:

- x = instrument or contract identity;
- n = signed quantity;
- o = legal owner or obligated entity;
- a = account, fund, portfolio, or balance-sheet boundary;
- c = denomination and settlement currency;
- T = maturity or contractual horizon;
- e = originating execution or transaction lineage;
- κ = settlement, collateral, netting, and legal terms;
- ν = current lifecycle status.

The lifecycle status may be drawn from:

$\nu \in$

{ordered,executed,confirmed,cleared,settled,recognized,called,liquidating,defaulted,closed}. (3.3)

The identity kernel answers:

> Which financial object is being carried, by whom, under which contractual and ledger conditions?

It does not yet answer:

> What is that object currently worth?

That second question belongs to the valuation state.

3.3 Identity is not numerical sameness

A position may preserve its identity while its value changes sharply.

For example:

- one bond remains the same legal bond after a market-price decline;
- one share remains the same share after a change in beta;
- one collateral lot remains legally identifiable after a haircut increase;
- one loan remains the same contractual claim after impairment recognition;
- one trade remains the same originating transaction while moving from execution to settlement.

Therefore:

Identity Preservation ≠ Numerical State Preservation. (3.4)

Let T_AB denote transport from frame A to frame B.

A valid transport may satisfy:

K_B[T_AB(S_A)] ≃ K_A(S_A). (3.5)

while:

Value_B[T_AB(S_A)] ≠ Value_A(S_A). (3.6)

The symbol ≃ means identity-equivalent under the declared protocol, not numerically equal in every coordinate.

This distinction is essential for gauge-like finance.

The same financial subject may legitimately possess:

- one market value;
- another collateral value;
- another accounting carrying amount;

- another regulatory exposure;
- another liquidation value.

The values differ, but the bounded identity remains transportable.

3.4 The declared protocol

Before charge, spin, transport, or gate status can be assigned, the financial protocol must be declared.

Define:

$P_S := (B,\Delta,h,u,\Phi,G,T_r,R_r,V).$ (3.7)

where:

- B = system boundary;
- Δ = observation and aggregation rule;
- h = horizon or state window;
- u = admissible intervention family;
- Φ = feature map;
- G = gate rule;
- T_r = trace rule;
- R_r = residual rule;
- V = transport or invariance rule.

For the minimum margin-account model:

$B :=$ one legal margin account. (3.8)

$\Delta :=$ declared marking, valuation, and collateral rules. (3.9)

$h :=$ the selected valuation and margin horizon. (3.10)

$u := \{\text{hold,post collateral,delever,liquidate,default}\}.$ (3.11)

$\Phi := \{R,Q,\theta,n,D,C,h_c,B_m,L\}.$ (3.12)

Here h_c denotes the collateral-admission factor and B_m denotes the margin buffer, avoiding collision with the general horizon h.

The source architecture insists that no financial charge is well-defined without a declared field, no invariance is well-defined without a declared frame, and no event is well-defined without a declared gate.

3.5 The complete subject state

The complete financial state is not exhausted by the Complex CAPM number Z.

Define:

$\mathcal{S}_t := (K_S, Z_{M,t}, D_t, C_t, h_{c,t}, B_t, L_t, \mathscr{R}_t).$ (3.13)

where:

- K_S = identity kernel;
- $Z_{M,t}$ = market-side Complex CAPM state;
- D_t = funding liability;
- C_t = independently posted collateral;
- $h_{c,t}$ = collateral-admission factor;
- B_t = margin buffer;
- L_t = ledger state;
- $\mathscr{R}_t$ = unresolved residual.

The market-side complex state is:

$Z_{M,t} = n_t(R_t + iQ_t).$ (3.14)

The admitted market value of the risky position is:

$V_t = n_t R_t.$ (3.15)

The corresponding market-pressure coordinate is:

$P_{Q,t} = n_t Q_t.$ (3.16)

The notation $P_{Q,t}$ is used here to prevent confusion between the CAPM coordinate Q and financial charge q.

3.6 Net equity

Let D_t denote the account's funding liability.

Let C_t denote cash or other independently posted collateral that is admitted at full value in the minimum model.

The account's real net equity is:

$E_t = V_t + C_t - D_t.$ (3.17)

Substituting V_t:

$E_t = n_t R_t + C_t - D_t.$ (3.18)

This is an ordinary balance-sheet identity.

It is not yet the margin constraint.

The margin mechanism may admit only part of the risky asset's market value as collateral.

3.7 Collateral-admitted value

Let:

$0 \le h_{c,t} \le 1.$ (3.19)

where $h_{c,t}$ is the fraction of market value admitted for collateral purposes.

The admitted risky-asset collateral value is:

$V_{C,t} = h_{c,t} V_t.$ (3.20)

Therefore:

$V_{C,t} = n_{th_{c,t}} R_t.$ (3.21)

Total eligible collateral is:

$K_t = C_t + V_{C,t}.$ (3.22)

Hence:

$K_t = C_t + n_{th_{c,t}} R_t.$ (3.23)

The margin buffer is defined as:

$B_t := K_t - D_t.$ (3.24)

Therefore:

$B_t = C_t + n_{th_{c,t}} R_t - D_t.$ (3.25)

The account is inside its declared collateral constraint when:

$B_t \ge 0.$ (3.26)

It is outside the constraint when:

$B_t < 0.$ (3.27)

The minimum shortfall is:

$C_{call,t} = [-B_t]_+.$ (3.28)

where:

$[x]_+ := \max(x,0).$ (3.29)

Equations (3.24)–(3.29) define the minimum margin geometry.

They do not yet describe the authority, timing, cure procedure, or liquidation rights that turn the numerical breach into a financial event.

3.8 Four kinds of financial property

The subject’s attributes should be divided into four classes.

Identity invariants

These should survive ordinary transport:

- instrument identity;
- signed contractual units;
- execution lineage;
- legal owner;
- obligated party;
- maturity;
- payoff rights;
- settlement terms.

Write:

$$I_{inv} := \{x,n,o,e,T,\kappa_{contract}\}. \qquad (3.30)$$

Frame-covariant quantities

These may change under declared transport:

- market value;
- collateral value;
- accounting carrying amount;
- regulatory exposure;
- risk-weighted value;
- reporting-currency amount;
- R/Q decomposition.

Write:

$$X_{cov} := \{R,Q,\theta,V,V_C,BookValue,RiskExposure\}. \qquad (3.31)$$

Gate-mutable properties

These change only through a valid event or vertex:

- ownership after settlement;
- open versus closed status;
- uncalled versus called margin status;

- option versus delivered underlying;
- performing debt versus defaulted recovery claim;
- pledged versus seized collateral.

Write:

$$X_{gate} := \{\nu, \text{OwnershipStatus}, \text{ClaimClass}, \text{CollateralStatus}\}. \quad (3.32)$$

Residual properties

These remain unresolved after valuation, transport, or gate processing:

- settlement exception;
- collateral disagreement;
- stale mark;
- legal dispute;
- unrecognized P&L;
- accounting-risk mismatch;
- model uncertainty;
- unpaid shortfall.

Write:

$$\mathscr{R} := \{\mathscr{R}_{\text{settle}}, \mathscr{R}_{\text{collateral}}, \mathscr{R}_{\text{mark}}, \mathscr{R}_{\text{legal}}, \mathscr{R}_{\text{ledger}}, \mathscr{R}_{\text{model}}\}. \quad (3.33)$$

The complete subject can therefore be described schematically as:

$$\text{Financial Subject} := \text{Identity Kernel} + \text{Covariant State} + \text{Gate Status} + \text{Residual}. \quad (3.34)$$

Equation (3.34) is a conceptual assembly, not an arithmetic equality.

3.9 The subject's two clocks

The margin account possesses at least two forms of time.

Market time

Market time records:

- price updates;
- valuation changes;
- CAPM phase movement;
- changes in beta or ERP;
- changes in volatility and liquidity.

Denote market time by:

$t_M.$ (3.35)

Ledger time

Ledger time advances when consequential states are written:

- trade executed;
- margin call issued;
- collateral received;
- liquidation completed;
- ownership settled;
- loss recognized;
- default recorded.

Denote ordered ledger ticks by:

$\tau_L.$ (3.36)

In general:

$\Delta t_M \neq \Delta \tau_L.$ (3.37)

Many market updates may occur without a ledger event.

Conversely, one legal or accounting gate may reclassify a large amount of prior market history in one ledger tick.

The generalized Dirac framework treats this distinction as central: signal speed is not equivalent to stable trace formation, and action may outrun the ledger that must make the event accountable.

3.10 Why the margin account is a clean calibration subject

The margin account is unusually useful because its major structures are observable.

Required structure	Margin-account realization
Bounded identity	Legal account and position lot
Complex valuation	$Z_M = n(R + iQ)$
Asset orientation	Long or short risky claim
Funding orientation	Borrower or lender
Conditional obligation	Collateral duty
Constraint	$B \geq 0$
Authoritative gate	Margin agreement
Outward event	Margin call
Independent return	Collateral, deleveraging, liquidation, or default

Required structure	Margin-account realization
Multiple frames	Market, broker, treasury, risk, accounting, legal
Ledger	Position, collateral, cash, P&L, and status records
Residual	Shortfall, reconciliation failure, or legal dispute

The subject is therefore complex enough to support the proposed theory, but narrow enough to remain testable.

Part III — From Constraint to Charge

4. Structural Financial Charge

4.1 Three quantities that must not share one symbol

The developing framework contains three different objects that may superficially resemble charge.

They must be separated.

Pressure tilt

Define:

$$\rho_Q := Q/R. \qquad (4.1)$$

Since:

$$R = A \cos \theta, \qquad (4.2)$$

and:

$$Q = A \sin \theta, \qquad (4.3)$$

we obtain:

$$\rho_Q = \tan \theta. \qquad (4.4)$$

The pressure tilt measures conjugate pressure relative to admitted value.

It is a state ratio.

It is not structural charge.

CAPM phase sensitivity

Define:

$$\kappa_{ERP} := \partial\theta/\partial ERP. \qquad (4.5)$$

From the Complex CAPM geometry:

$$\kappa_{ERP} = tR\beta/[(1 + r)Q]. \qquad (4.6)$$

Similarly:

$\kappa_\beta := \partial\theta/\partial\beta.$ (4.7)

and:

$\kappa_\beta = tRERP/[(1 + r)Q].$ (4.8)

These quantities measure local phase response.

They are sensitivities.

They vary with the current state and protocol.

They are not automatically identity-bearing charges.

Structural charge

Structural charge records a stable relational orientation carried by the financial subject.

Denote it by:

$q.$ (4.9)

A candidate structural charge must answer:

- What identity carries it?
- Under which field does it transform?
- What is its sign convention?
- What are its units?
- What couples to it?
- How is it transported?
- Which gate transfers or converts it?
- What balance rule applies?
- What residual remains?

The main article's charge-admission rule requires transformation, coupling, transport, vertex, and residual structure. A candidate failing these tests should remain a sensitivity, exposure, or descriptive coordinate.

4.2 The elementary claim–obligation pair

A financial claim is relational.

One party's right generally corresponds to another party's obligation.

Let the elementary claim charge be normalized as:

$q_{claim} = +1.$ (4.10)

Let the corresponding obligation charge be:

$q_{obligation} = -1.$ (4.11)

Then:

$q_{claim} + q_{obligation} = 0.$ (4.12)

Equation (4.12) does not say that the parties have zero economic value.

It says that the elementary relational identity contains a matched right and obligation.

Examples include:

- lender and borrower;
- bondholder and issuer;
- option holder and option writer;
- receivable owner and payable obligor;
- security receiver and delivery obligor;
- collateral taker and collateral provider.

The sign convention is relational, not moral.

Positive does not mean beneficial.

Negative does not mean harmful.

4.3 Quantity-scaled structural charge

Let n denote signed units of the claim.

Define:

$q_K(S) := nq_0(x).$ (4.13)

where:

- $q_0(x)$ = unit charge class of instrument x;
- n = signed quantity held by the subject.

For a normalized long claim:

$q_0(x) = +1.$ (4.14)

For a corresponding short or delivery obligation:

$q_0(x) = -1.$ (4.15)

The charge is additive across separately identified lots:

$q_{K,\mathrm{total}} = \Sigma_j q_K(S_j).$ (4.16)

This additivity is one reason signed claim units are a stronger charge candidate than beta or phase sensitivity.

4.4 The leveraged account's charge vector

The minimum margin subject carries at least three different orientations.

Define:

$\vec{q}_S := (q_A, q_F, q_C).$ (4.17)

where:

- q_A = risky-asset claim charge;
- q_F = funding charge;
- q_C = contingent collateral charge.

For a leveraged long account:

$q_A > 0.$ (4.18)

$q_F < 0.$ (4.19)

$q_C < 0$ when operationally activated. (4.20)

The coordinates belong to different fields.

They should not be added as though:

$q_A + q_F = 0$ (4.21)

would imply that the account is financially neutral.

A subject may be neutral under one field and highly exposed under another.

4.5 Field-indexed charge

Let the financial fields be:

$\mathbb{F} := \{\mathbb{F}_A, \mathbb{F}_F, \mathbb{F}_C\}.$ (4.22)

where:

- $\mathbb{F}_A$ = risky-asset or market field;
- $\mathbb{F}_F$ = funding field;
- $\mathbb{F}_C$ = collateral-admissibility field.

The corresponding charges are:

$q_A \leftrightarrow \mathbb{F}_A.$ (4.23)

$q_F \leftrightarrow \mathbb{F}_F.$ (4.24)

$q_C \leftrightarrow \mathbb{F}_C.$ (4.25)

A state is neutral under field r only if its net charge in that field is zero:

$q_{net,r} = \Sigma_j q_{j,r} = 0.$ (4.26)

Therefore:

$q_{net,A} = 0 \nRightarrow q_{net,F} = 0.$ (4.27)

and:

$q_{net,F} = 0 \nRightarrow q_{net,C} = 0.$ (4.28)

This prevents false cancellation across economically different obligations.

4.6 Dormant and operational charge

The collateral obligation exists in the margin contract before a call is issued.

However, it is not yet payable.

Define the activation indicator:

$$a_C(B) := \begin{cases} 0 \text{ when } B \geq 0; \\ 1 \text{ when } B < 0. \end{cases}$$ (4.29)

The operational collateral charge is:

$q_C^{op} := a_C(B) q_C.$ (4.30)

When:

$B \geq 0,$ (4.31)

the contingent charge is dormant.

When:

$B < 0,$ (4.32)

the margin gate activates it.

Thus:

Dormant Contractual Charge → Gate → Operational Obligation. (4.33)

This is one of the cleanest ways in which a constraint turns a latent relational property into an enforceable financial event.

4.7 Structural charge versus coupling strength

Structural charge tells us the subject's orientation.

It does not by itself tell us how strongly the subject responds.

Let:

g_A = asset-field coupling strength. (4.34)

g_F = funding-field coupling strength. (4.35)

g_C = collateral-field coupling strength. (4.36)

For the market sector, beta may contribute to g_A:

$g_A \sim \beta$. (4.37)

But beta is not the whole coupling.

The effective market response may also depend on:

- leverage;
- horizon;
- liquidity;
- concentration;
- collateral;
- regime.

Define leverage:

$\lambda := V/E$. (4.38)

A schematic effective market coupling is:

$g_A^{eff} := \beta\lambda\chi_A$. (4.39)

where χ_A collects additional protocol-dependent amplification.

The observed field response is then schematically:

$\text{Response}_A \approx q_A g_A^{eff} \Delta\mathbb{F}_A$. (4.40)

Equation (4.40) is a modelling proposal, not a standard CAPM identity.

Its purpose is to separate:

- orientation q_A;
- strength g_A^{eff};
- field movement $\Delta\mathbb{F}_A$.

4.8 Structural charge versus effective charge

The subject's structural orientation may remain stable while its effective response changes.

Define:

$$q_{eff,A} := \Lambda_A q_A. \qquad (4.41)$$

where:

$$\Lambda_A := f(\beta,\lambda,h_c,\text{Liquidity},\text{Collateral},\text{Ledger},\text{Regime}). \qquad (4.42)$$

As equity falls:

$$E \downarrow. \qquad (4.43)$$

If risky-asset value remains positive:

$$\lambda = V/E \uparrow. \qquad (4.44)$$

Then:

$$|q_{eff,A}| \uparrow. \qquad (4.45)$$

The account remains structurally long the asset, but its balance sheet becomes more sensitive to further adverse movement.

This is the recursive distinction emphasized in the main framework: charge governs coupling orientation, while ledgered consequences can alter the next period's effective coupling.

4.9 CAPM phase sensitivity is part of the response chain

The exact CAPM phase sensitivity is:

$$\kappa_{ERP} = \partial\theta/\partial ERP. \qquad (4.46)$$

A small ERP movement produces:

$$d\theta = \kappa_{ERP}\, dERP. \qquad (4.47)$$

The resulting admitted-value movement is:

$$dR = -Q\kappa_{ERP}\, dERP, \qquad (4.48)$$

when $dA = 0$.

For n units:

$$dV = -nQ\kappa_{ERP}\, dERP. \qquad (4.49)$$

Therefore, the account's market response chain is:

Structural Orientation
→ Factor Coupling
→ Phase Sensitivity

→ Admitted-Value Movement. (4.50)

A schematic decomposition is:

$dV \propto q_A \times \beta \times \lambda \times \kappa_{ERP} \times dERP.$ (4.51)

Equation (4.51) should not replace the exact valuation derivative.

It is a typed decomposition showing that structural charge and state-dependent sensitivity occupy different roles.

4.10 Charge does not imply conservation of value

Suppose a bond claim loses half of its market value.

Its structural claim charge may remain:

$q_{claim} = +1.$ (4.52)

The issuer's corresponding obligation charge may remain:

$q_{obligation} = -1.$ (4.53)

Yet:

$R_{new} < R_{old}.$ (4.54)

Therefore:

Charge Balance ≠ Value Conservation. (4.55)

Likewise:

Charge Persistence ≠ No Economic Loss. (4.56)

Charge tracks relational identity.

Valuation tracks current economic admission.

The two interact but must not be identified.

4.11 Charge does not imply quantization in every sector

The elementary claim count may be discrete:

$n \in \mathbb{Z}.$ (4.57)

But many financial quantities are not naturally quantized:

- beta;
- duration;
- leverage;
- haircut;

- phase sensitivity;
- liquidity amplification.

Therefore the article should distinguish:

Discrete Claim Charge. (4.58)

Continuous Effective Coupling. (4.59)

A Financial Gauge–Dirac system does not require every exposure to come in indivisible units.

The stronger statement is only:

> Some claim identities have discrete contractual units, while their effective field responses remain continuous and state-dependent.

4.12 Charge-admission rule for the margin account

The account's proposed charge q_r should be admitted only if:

AdmitCharge(q_r)
:= Carrier_r ∧ Field_r ∧ Sign_r ∧ Coupling_r ∧ Transport_r ∧ Vertex_r ∧ Residual_r. (4.60)

In particular:

- q_A must survive transport from market to collateral and ledger frames;
- q_F must remain linked to the funding obligation;
- q_C must correspond to a real contractual collateral duty;
- each charge must possess defined gate behavior;
- unreconciled charge must be recorded as residual.

Otherwise the terminology should be reduced:

q_A → directional exposure. (4.61)

q_F → funding sensitivity. (4.62)

q_C → contingent collateral obligation. (4.63)

The weaker terminology remains scientifically acceptable.

---

5. Charge Balance and Financial Vertices

5.1 Why a vertex is needed

A financial charge becomes operationally meaningful when it constrains permitted transformations.

A vertex is a declared event at which:

- identities enter;
- identities leave;
- rights or obligations transfer;
- charges activate, convert, or extinguish;
- a trace is written;
- residual is preserved.

Denote a financial vertex by:

$\mathcal{V}_k$. (5.1)

The generic charge-balance rule is:

$\Sigma q_{in} + q_{gate} = \Sigma q_{out} + r_q$. (5.2)

where:

- q_{in} = charges entering the vertex;
- q_{gate} = charge contributed or absorbed by the conversion mechanism;
- q_{out} = charges leaving the vertex;
- r_q = unresolved charge residual.

Equation (5.2) is a proposed balance schema.

It is not asserted as a universal physical conservation law.

5.2 Issuance vertex

At issuance, a new financial claim and a corresponding obligation appear together.

The vertex is:

Issuer + Investor + Contract → Claim + Obligation + CashTransfer. (5.3)

The elementary claim balance is:

$0 \rightarrow (+q_{claim}) + (-q_{obligation})$. (5.4)

Therefore:

$q_{before} = q_{after}$. (5.5)

if the paired right and obligation are both included within the declared boundary.

If the boundary contains only the investor, the account appears to acquire positive claim charge.

If the boundary contains only the issuer, the issuer appears to acquire negative obligation charge.

Charge neutrality is therefore boundary-dependent.

5.3 Leveraged acquisition vertex

The margin subject acquires the risky claim using borrowed funds.

The simplified vertex is:

InvestorEquity + Funding → RiskyAssetPosition. (5.6)

The subject acquires:

q_A > 0. (5.7)

and:

q_F < 0. (5.8)

The resulting charge signature is:

$\vec{q}_S = (q_A, q_F, 0)$. (5.9)

The collateral charge is present contractually but remains dormant:

q_C^op = 0. (5.10)

The leverage relation is:

V = E + D − C. (5.11)

In the minimum case C = 0:

V = E + D. (5.12)

The subject has not created a net universal charge.

It has entered two distinct relational fields:

- an asset-claim field;
- a funding-obligation field.

5.4 Secondary-market transfer vertex

Suppose seller S transfers one unit of the claim to buyer B.

Before settlement:

q_S = +1. (5.13)

q_B = 0. (5.14)

After completed settlement:

q_S′ = 0. (5.15)

q_B′ = +1. (5.16)

Therefore:

$$q_S + q_B = q_S' + q_B'. \qquad (5.17)$$

The claim charge has moved.

It has not been duplicated.

However, between execution and settlement, the system also contains temporary charges:

- buyer cash obligation;
- seller delivery obligation;
- clearing exposure;
- replacement-cost exposure.

Thus execution creates a temporary charge configuration that remains open until settlement.

This is already an action–ledger double closure.

5.5 Margin-call activation vertex

The margin buffer is:

$$B = C + nh_cR - D. \qquad (5.18)$$

The call condition is:

$$B < 0. \qquad (5.19)$$

At the gate:

$$\text{DormantCollateralDuty} \rightarrow \text{ActiveCollateralObligation}. \qquad (5.20)$$

The activation rule is:

$$q_C^{op,-} = 0. \qquad (5.21)$$

$$q_C^{op,+} = q_C. \qquad (5.22)$$

The gate therefore changes operational charge status:

$$\Delta q_C^{op} = q_C. \qquad (5.23)$$

But the underlying contractual obligation was already encoded in K_S.

The gate does not create the agreement retrospectively.

It activates one of its conditional branches.

This distinction is important:

$$\text{Contractual Existence} \neq \text{Operational Activation}. \qquad (5.24)$$

5.6 Collateral-posting vertex

Suppose the subject meets the call by posting cash ΔC.

Then:

$C^+ = C^- + \Delta C.$ (5.25)

The margin buffer becomes:

$B^+ = B^- + \Delta C.$ (5.26)

The call is cured if:

$B^+ \geq 0.$ (5.27)

The operational collateral obligation is discharged:

$q_C^{op} \to 0.$ (5.28)

But posting collateral changes the subject's future state.

It may reduce:

- free liquidity;
- ability to meet other obligations;
- optionality;
- funding capacity.

Thus:

Collateral Charge Discharged
$\to$ Liquidity State Altered. (5.29)

The event closes one obligation while changing future effective coupling.

5.7 Deleveraging vertex

Suppose the subject sells Δn units.

Then:

$n^+ = n^- - \Delta n.$ (5.30)

Let p_exec denote execution price.

The sale proceeds are:

$CashProceeds = p_exec\Delta n.$ (5.31)

If proceeds repay debt:

$D^+ = D^- - p_exec\Delta n.$ (5.32)

The asset charge changes:

$q_A^+ = q_A^- - \Delta n q_0(x).$ (5.33)

The funding charge changes as debt is reduced:

$|q_F^+| < |q_F^-|.$ (5.34)

This is a charge-transforming vertex.

It also changes leverage:

$\lambda^+ = V^+/E^+.$ (5.35)

The post-gate subject therefore carries a new effective response profile.

5.8 Forced-liquidation vertex

If the subject does not cure the call, the broker may liquidate under contractual authority.

Let:

$\ell :=$ liquidation quantity. (5.36)

Then:

$n^+ = n^- - \ell.$ (5.37)

If liquidation occurs below the pre-gate mark:

$p_exec < R^-.$ (5.38)

the realized execution loss is:

$Loss_exec = \ell(R^- - p_exec).$ (5.39)

The loss belongs to the economic and ledger sectors.

It is not itself charge residual.

Charge residual asks whether rights and obligations reconciled.

Economic residual asks whether losses, costs, and exposures were fully recognized.

Therefore:

$r_q \neq \mathscr{R}_P\&L.$ (5.40)

A forced liquidation may have:

$r_q = 0,$ (5.41)

while:

$\mathscr{R}_P\&L > 0.$ (5.42)

For example, all contractual units may reconcile correctly while the subject suffers a large realized loss.

5.9 Default vertex

If the account remains deficient after liquidation:

$B^{+} < 0.$ (5.43)

the system may enter default.

The original funding and collateral identities may transform into:

- deficiency claim;
- recovery claim;
- collateral-enforcement claim;
- litigation claim;
- guarantor claim;
- extinguished amount.

Schematically:

$q_F + q_C \rightarrow q_{recovery} + q_{legal} + q_{extinguished} + r_q.$ (5.44)

The vertex must specify:

- which obligations survive;
- which are accelerated;
- which are netted;
- which are legally extinguished;
- which become contingent litigation claims;
- which remain disputed.

Default therefore demonstrates why charge cannot be reduced to price sensitivity.

It is an identity-conversion problem.

5.10 Charge residual

Define charge residual at vertex k:

$r_{q,k} := \Sigma q_{in,k} + q_{gate,k} - \Sigma q_{out,k}.$ (5.45)

Possible causes of nonzero r_q include:

- missing position;
- duplicate booking;

- failed settlement;
- unmatched collateral;
- disputed ownership;
- legal novation failure;
- incorrect netting;
- incomplete exercise allocation;
- unrecorded extinguishment.

A nonzero charge residual does not necessarily imply financial loss.

It implies that the identity transformation has not fully reconciled.

5.11 Charge ledger

The ledger should preserve:

$$L_{q,k} := (\text{Carrier},\text{Field},\text{Sign},\text{Units},\text{Vertex},\text{ChargeIn},\text{ChargeOut},r_q,\text{Authority},\text{Timestamp}). \quad (5.46)$$

This record makes charge claims auditable.

It prevents retrospective relabelling such as:

- redefining a shortfall as a different obligation;
- changing the carrier after the event;
- silently altering the boundary;
- discarding an unmatched claim;
- treating a missing position as mere noise.

The charge ledger will later become one component of full spin closure.

5.12 Provisional result of Part III

The margin subject now possesses a candidate charge architecture:

$$\vec{q}_S = (q_A,q_F,q_C). \quad (5.47)$$

These charges:

- belong to declared fields;
- possess relational signs;
- can activate or transform at gates;
- can be transported across frames;

- can be tested at vertices;
- can leave measurable residual.

The architecture is still provisional.

Its empirical usefulness must be compared against ordinary contractual labels and exposure measures.

But it now contains more structure than simply renaming beta, Q, or leverage as charge.

The next part will construct the collateral complex plane and show how CAPM phase movement and haircut movement jointly drive the charged subject toward the margin gate.

Part IV — Margin as an Authoritative Financial Gate

6. The Collateral Plane

6.1 Why the market value is not the collateral value

The leveraged subject holds a risky claim whose market-side complex state is:

$$Z_M = n(R_M + iQ_M). \qquad (6.1)$$

where:

- n = signed asset units;
- R_M = CAPM-admitted market value per unit;
- Q_M = retained market-risk pressure per unit.

For the minimum long-position model:

$$n > 0. \qquad (6.2)$$

The market frame asks:

> What value is presently admitted under the declared valuation protocol?

The collateral frame asks a different question:

> How much of that market value may support a funding obligation under the declared margin agreement?

These are related but not identical questions.

A broker, clearing house, or secured lender may admit less than the current market value because of:

- liquidation risk;
- price volatility;
- bid–ask cost;
- concentration;

- wrong-way risk;
- settlement status;
- legal eligibility;
- currency mismatch;
- operational delay.

Define the collateral-admission factor:

$$0 \le h \le 1. \qquad (6.3)$$

Then collateral-admitted value per unit is:

$$R_C = hR_M. \qquad (6.4)$$

For n units:

$$V_C = nR_C = nhR_M. \qquad (6.5)$$

The quantity h is commonly described as one minus the haircut in the minimum model.

If the haircut rate is k_h:

$$h = 1 - k_h. \qquad (6.6)$$

Thus:

$$R_C = (1 - k_h)R_M. \qquad (6.7)$$

The market and collateral frames are therefore connected by a declared institutional filter.

Finance Geometry already treats collateral and liquidity haircuts as mature financial filters capable of defining an admitted value and retained-pressure coordinate. The present use of that principle as a second filter applied after CAPM is a new construction.

6.2 Sequential filtering

The valuation sequence is:

$$A \rightarrow \text{CAPM Filter} \rightarrow R_M \rightarrow \text{Collateral Filter} \rightarrow R_C. \qquad (6.8)$$

Here:

- A = baseline value amplitude;
- R_M = value admitted by the CAPM filter;
- R_C = value admitted by the collateral protocol.

The two filter weights are:

$$w_M = R_M/A. \qquad (6.9)$$

$$w_{C|M} = R_C/R_M = h. \qquad (6.10)$$

The combined filter weight is:

$w_C = R_C/A.$ (6.11)

Therefore:

$w_C = hw_M.$ (6.12)

Since:

$w_M = \cos \theta_M,$ (6.13)

the combined collateral weight is:

$w_C = h \cos \theta_M.$ (6.14)

Define the collateral angle:

$\cos \theta_C = R_C/A.$ (6.15)

Then:

$\cos \theta_C = h \cos \theta_M.$ (6.16)

Therefore:

$\theta_C = \arccos(h \cos \theta_M).$ (6.17)

Equation (6.17) is exact under the declared sequential-filter construction.

It shows that collateral phase can change for two distinct reasons:

1. the underlying market valuation phase θ_M changes;
2. the collateral-admission factor h changes.

6.3 The collateral pressure coordinate

Using the same baseline amplitude A, define:

$Q_C = \sqrt{(A^2 - R_C^2)}.$ (6.18)

Substituting $R_C = hR_M$:

$Q_C = \sqrt{(A^2 - h^2R_M^2)}.$ (6.19)

The collateral complex state is:

$Z_C = n(R_C + iQ_C).$ (6.20)

or:

$Z_C = nA \exp(i\theta_C).$ (6.21)

The collateral geometry is:

$A^2 = R_C^2 + Q_C^2.$ (6.22)

$$R_C = A \cos \theta_C. \qquad (6.23)$$

$$Q_C = A \sin \theta_C. \qquad (6.24)$$

The article should be explicit about the status of this construction.

The market-side Q_M is derived from the CAPM filter.

The collateral-side Q_C is derived from the sequential CAPM-plus-collateral filter.

It is not asserted that brokers presently calculate Q_C, or that Q_C is a standard margin variable. It is a proposed diagnostic coordinate implied by the declared filter geometry.

6.4 How collateral pressure extends market pressure

The market geometry gives:

$$Q_M^2 = A^2 - R_M^2. \qquad (6.25)$$

The collateral geometry gives:

$$Q_C^2 = A^2 - h^2 R_M^2. \qquad (6.26)$$

Subtracting:

$$Q_C^2 - Q_M^2 = (1 - h^2) R_M^2. \qquad (6.27)$$

Therefore:

$$Q_C^2 = Q_M^2 + (1 - h^2) R_M^2. \qquad (6.28)$$

Since:

$$0 \le h \le 1, \qquad (6.29)$$

it follows that:

$$Q_C \ge Q_M. \qquad (6.30)$$

under the nonnegative principal-root convention.

Equation (6.28) gives the collateral pressure coordinate two components:

1. pressure already implied by CAPM;
2. additional pressure introduced by collateral restriction.

Define the additional squared collateral pressure:

$$Q_h^2 := (1 - h^2) R_M^2. \qquad (6.31)$$

Then:

$$Q_C^2 = Q_M^2 + Q_h^2. \qquad (6.32)$$

This resembles an orthogonal decomposition, but Q_h should not yet be interpreted as an independently measured physical component. It is a derived contribution under the sequential-filter geometry.

6.5 Collateral pressure is not the haircut amount

The scalar haircut amount per unit is:

$$H_C = R_M - R_C. \quad (6.33)$$

Since $R_C = hR_M$:

$$H_C = (1 - h)R_M. \quad (6.34)$$

But:

$$Q_C - Q_M \neq H_C. \quad (6.35)$$

and:

$$Q_h = R_M\sqrt{(1 - h^2)}. \quad (6.36)$$

Therefore:

$$Q_h \neq H_C. \quad (6.37)$$

The haircut amount is a real-axis reduction.

The collateral pressure coordinate is the conjugate coordinate implied by the combined filter.

As in the original Finance Geometry, the real-axis reduction and the orthogonal pressure coordinate should not be confused.

6.6 Collateral phase transmission

Differentiate:

$$\cos \theta_C = h \cos \theta_M. \quad (6.38)$$

Then:

$$-\sin \theta_C \, d\theta_C = \cos \theta_M \, dh - h \sin \theta_M \, d\theta_M. \quad (6.39)$$

Therefore:

$$d\theta_C = [h \sin \theta_M \, d\theta_M - \cos \theta_M \, dh]/\sin \theta_C. \quad (6.40)$$

Equation (6.40) separates two sources of collateral-phase movement.

Market-transmission term

$$d\theta_C|\text{market} = [h \sin \theta_M/\sin \theta_C]d\theta_M. \quad (6.41)$$

Haircut term

$$d\theta_C|\text{haircut} = -[\cos \theta_M/\sin \theta_C]dh. \quad (6.42)$$

Thus:

$$d\theta_C = d\theta_C|_{market} + d\theta_C|_{haircut}. \qquad (6.43)$$

If:

$$dh = 0, \qquad (6.44)$$

then collateral phase moves only because market phase moves.

If:

$$d\theta_M = 0, \qquad (6.45)$$

then:

$$d\theta_C = -[\cos \theta_M / \sin \theta_C]dh. \qquad (6.46)$$

When collateral eligibility deteriorates:

$$dh < 0, \qquad (6.47)$$

we obtain:

$$d\theta_C > 0. \qquad (6.48)$$

Thus a collateral phase can deteriorate even when the market valuation phase is unchanged.

6.7 Two independent stress routes

The margin account can therefore approach distress through two conceptually different routes.

Route A — Market deterioration

ERP rises, beta rises, expected cash flow falls, or market conditions otherwise reduce R_M:

$$\theta_M \uparrow \rightarrow R_M \downarrow \rightarrow R_C \downarrow \rightarrow \text{Margin Buffer} \downarrow. \qquad (6.49)$$

Route B — Collateral deterioration

Market value remains approximately unchanged, but the lender reduces h:

$$h \downarrow \rightarrow R_C \downarrow \rightarrow \theta_C \uparrow \rightarrow \text{Margin Buffer} \downarrow. \qquad (6.50)$$

The second route is especially important.

A subject may appear stable in the market frame while becoming increasingly fragile in the collateral frame.

This is a direct example of why one financial identity may require several local representations.

The source phase-world framework similarly emphasizes that quoted, funded, collateral,

accounting, regulatory, and liquidation values belong to different financial frames and become consequential through different gates.

6.8 Three distinct pressures

The minimum model now contains three different quantities.

Market pressure

$$Q_M = \sqrt{(A^2 - R_M^2)}. \qquad (6.51)$$

Collateral pressure

$$Q_C = \sqrt{(A^2 - R_C^2)}. \qquad (6.52)$$

Margin shortfall

$$C_{call} = [D - C - nR_C]_+. \qquad (6.53)$$

These answer different questions.

Q_M asks:

> What conjugate pressure is implied by the CAPM valuation filter?

Q_C asks:

> What conjugate pressure is implied after the collateral-admission filter?

C_call asks:

> How much admitted collateral is missing under the margin rule?

Therefore:

$$Q_M \neq Q_C. \qquad (6.54)$$

$$Q_C \neq C_{call}. \qquad (6.55)$$

$$Q_M \neq C_{call}. \qquad (6.56)$$

A high Q_C does not automatically mean that a margin call has occurred.

The account may still possess substantial collateral and equity.

A low current shortfall does not imply low Q_C.

The account may remain just above the gate while carrying large retained pressure.

6.9 The market–collateral mismatch

Define the real-axis valuation mismatch:

$$\Delta R_{MC} := R_M - R_C. \qquad (6.57)$$

Since:

$$R_C = hR_M, \qquad (6.58)$$

we obtain:

$$\Delta R_{MC} = (1 - h)R_M. \qquad (6.59)$$

Define the pressure mismatch:

$$\Delta Q_{MC} := Q_C - Q_M. \qquad (6.60)$$

Define the phase mismatch:

$$\Delta\theta_{MC} := \theta_C - \theta_M. \qquad (6.61)$$

Under:

$$0 \le h \le 1, \qquad (6.62)$$

we have:

$$\theta_C \ge \theta_M. \qquad (6.63)$$

Therefore:

$$\Delta\theta_{MC} \ge 0. \qquad (6.64)$$

The three mismatch measures describe different aspects of the same cross-frame relation:

- ΔR_{MC} = real value withheld by the collateral rule;
- ΔQ_{MC} = additional conjugate pressure implied by the collateral filter;
- $\Delta\theta_{MC}$ = angular separation between market and collateral admission.

These quantities will later contribute to the financial spinor split and gauge-transport residual.

6.10 A caution about the shared amplitude

The construction above uses the same amplitude A for both market and collateral planes.

This choice means:

> The collateral filter is interpreted as a second admission rule applied to the same pre-filter value amplitude.

This is useful because it gives:

$$\cos\theta_C = h\cos\theta_M. \qquad (6.65)$$

However, another protocol might define a separate collateral amplitude:

$$A_C \ne A. \qquad (6.66)$$

For example, a liquidation model may begin from stressed executable value rather than the original baseline-discounted cash flow.

In that case:

$$A_C^2 = R_C^2 + Q_C^2. \qquad (6.67)$$

and the simple relation in equation (6.16) may not hold.

Therefore the shared-amplitude model is a **minimum calibration convention**, not a universal law.

Every empirical implementation must disclose:

- the source amplitude;
- the market filter;
- the collateral filter;
- whether the amplitude is shared;
- the residual left outside both filters.

---

7. Margin Buffer and Gate Activation

7.1 The margin buffer

Total eligible collateral is:

$$K = C + nR_C. \qquad (7.1)$$

Using $R_C = hR_M$:

$$K = C + nhR_M. \qquad (7.2)$$

Let D denote the funding liability recognized by the margin protocol.

Define the margin buffer:

$$B := K - D. \qquad (7.3)$$

Therefore:

$$B = C + nhR_M - D. \qquad (7.4)$$

Equivalently, using the collateral angle:

$$B = C + nA\cos\theta_C - D. \qquad (7.5)$$

The feasible collateral region is:

$$B \geq 0. \qquad (7.6)$$

The breach region is:

$$B < 0. \qquad (7.7)$$

Equation (7.4) is a balance-sheet constraint.

It does not yet describe the institutional response.

7.2 The constraint surface

The margin boundary is:

$$B = 0. \qquad (7.8)$$

Substituting the state variables:

$$C + nhR_M - D = 0. \qquad (7.9)$$

The boundary may be solved for several critical values.

Critical market value

$$R_M^* = (D - C)/(nh). \qquad (7.10)$$

Critical haircut factor

$$h^* = (D - C)/(nR_M). \qquad (7.11)$$

Critical debt level

$$D^* = C + nhR_M. \qquad (7.12)$$

Critical collateral amount

$$C^* = D - nhR_M. \qquad (7.13)$$

These equations show that the same margin gate can be reached through:

- lower asset value;
- lower collateral eligibility;
- higher debt;
- withdrawal of posted collateral;
- increased position size.

7.3 Minimum call amount

Under the simplest cure rule, the account must restore the buffer to zero.

Define:

$$C_{call} = [-B]_+. \qquad (7.14)$$

Thus:

$$C_{call} = [D - C - nhR_M]_+. \qquad (7.15)$$

More generally, the contract may require restoration to a positive target buffer B_{target}:

$$B_{target} \geq 0. \qquad (7.16)$$

Then:

$$C_{call} = [B_{target} - B]_+. \qquad (7.17)$$

The zero-target model is recovered when:

$$B_{target} = 0. \qquad (7.18)$$

Actual margin agreements may use initial margin, maintenance margin, variation margin, stress add-ons, concentration add-ons, or discretionary requirements. The present equation is a minimum model, not a description of every real margin methodology.

7.4 Threshold versus gate

A numerical condition such as:

$$B < 0 \qquad (7.19)$$

is a threshold.

It becomes a financial gate only if a recognized authority can convert the breach into consequential action.

Define a margin gate:

$$G_{margin} := (Rule, Authority, Decision, Trace, Residual). \qquad (7.20)$$

The gate must specify:

- the measured buffer;
- the threshold or target;
- the authorized caller;
- the cure deadline;
- eligible collateral;
- permitted response branches;
- liquidation rights;
- ledger-writing rules;
- dispute handling;
- residual treatment.

A minimum gate decision function is:

$$d_k = G_{margin}(B_k, L_k, P_k). \qquad (7.21)$$

where:

- B_k = current buffer;
- L_k = prior ledger;
- P_k = governing margin protocol;
- d_k = gate decision.

A possible decision set is:

$$d_k \in \{\text{NoAction},\text{Call},\text{Restrict},\text{Deleverage},\text{Liquidate},\text{Default}\}. \quad (7.22)$$

The source Periodic Grammar repeatedly distinguishes a mere crossing from an admitted event and requires a gate to possess functional authority, trace, and residual governance.

7.5 Margin-gate trace

When a call is issued, the trace should preserve at least:

$$T_{\text{call},k} := (\text{Account},\text{Authority},\text{Time},B_k,B_{\text{target}},C_{\text{call}},\text{Deadline},\text{Eligibility},\text{Response},\text{Status}). \quad (7.23)$$

This record establishes:

- what was measured;
- which rule applied;
- who possessed authority;
- what obligation was created;
- when it had to be satisfied;
- what action occurred;
- whether closure was completed.

Without this trace, later interpretation may not distinguish:

- a valid call;
- an operational error;
- a discretionary request;
- a disputed valuation;
- an unauthorized liquidation;
- a retrospective reconstruction.

7.6 Activation of contingent collateral charge

Before the gate, the collateral duty exists contractually but is dormant.

Define:

a_C(B) = 0 when B ≥ 0. (7.24)

Define:

a_C(B) = 1 when B < 0. (7.25)

Then:

q_C^op = a_C(B)q_C. (7.26)

The activation vertex is:

G_margin : q_C^dormant → q_C^operational. (7.27)

This does not mean that the numerical charge q_C must be binary in every implementation.

The binary variable records operational activation.

The size of the required performance is measured separately by:

C_call. (7.28)

Thus:

Charge Status ≠ Obligation Amount. (7.29)

The charge describes the kind and orientation of obligation.

The call amount describes its current required magnitude.

7.7 The natural dual variable created by a constraint

A binding constraint naturally creates a shadow value.

Define the constraint function:

g_B := D − C − nhR_M. (7.30)

The admissibility rule is:

g_B ≤ 0. (7.31)

Introduce a nonnegative margin multiplier:

ζ_B ≥ 0. (7.32)

The constrained objective may be written schematically as:

$\mathscr{L}$_margin = $\mathscr{L}$0 + ζ_B(D − C − nhR_M). (7.33)

The complementarity conditions are:

B ≥ 0. (7.34)

ζ_B ≥ 0. (7.35)

$\zeta_{BB} = 0.$ (7.36)

The multiplier ζ_B measures the marginal value of relaxing the margin constraint.

This is an important conceptual correction:

> A constraint naturally produces a dual or shadow-pressure variable. It does not automatically produce structural charge.

Therefore:

$\zeta_B \neq q_C.$ (7.37)

The contingent collateral charge comes from the contractual right–obligation structure.

The multiplier ζ_B comes from the binding constraint.

The two interact at the gate but should not be identified.

7.8 Gate residual

Suppose the required call amount is C_{call}.

Let C_{admit} denote the amount actually delivered and accepted by the margin authority.

Define the immediate call residual:

$\varepsilon_{call} = C_{call} - C_{admit}.$ (7.38)

If:

$\varepsilon_{call} \leq 0,$ (7.39)

the minimum call amount has been met, subject to the quality and timing rules of the protocol.

If:

$\varepsilon_{call} > 0,$ (7.40)

an unresolved shortfall remains.

However, a complete residual vector is richer:

$\mathscr{R}_{gate} := (\varepsilon_{call}, \varepsilon_{time}, \varepsilon_{eligibility}, \varepsilon_{mark}, \varepsilon_{legal}, \varepsilon_{ledger}).$ (7.41)

where:

- ε_{call} = amount shortfall;
- ε_{time} = timing failure;
- $\varepsilon_{eligibility}$ = ineligible collateral;
- ε_{mark} = valuation disagreement;
- ε_{legal} = disputed authority or enforceability;
- ε_{ledger} = recording or reconciliation mismatch.

The gate may be consequential even when residual remains.

Therefore:

Commitment ≠ Exhaustion. (7.42)

The Periodic Grammar source makes the same distinction: a gate may write an event into the ledger while leaving substantial unresolved consequence.

7.9 Four response branches

Once the gate activates, the subject must follow one or more permitted branches.

Branch A — Post collateral

$C^{+} = C^{-} + \Delta C.$ (7.43)

Branch B — Deleverage voluntarily

$n^{+} = n^{-} - \Delta n.$ (7.44)

$D^{+} = D^{-} - p_exec\Delta n.$ (7.45)

Branch C — Forced liquidation

$n^{+} = n^{-} - \ell.$ (7.46)

$D^{+} = D^{-} - p_liq\ell.$ (7.47)

Branch D — Default or recovery

Unresolved deficiency becomes a new legal and financial identity:

Deficiency → RecoveryClaim + LegalTrace + Residual. (7.48)

The gate is therefore not one scalar reset.

It selects a branch in the subject's identity evolution.

7.10 Why gate activation is a nonlinear event

The positive-part function:

$C_call = [B_target - B]_{+}$ (7.49)

is piecewise linear.

But the complete gate is nonlinear for deeper reasons:

- the response depends on prior ledger state;
- collateral eligibility is state-dependent;
- liquidation changes n and D;

- execution price may depend on liquidation size;
- losses change equity and leverage;
- leverage changes future sensitivity;
- haircuts may rise during stress;
- unresolved calls may change legal status.

Therefore:

Gate Output ≠ Function of Current Price Alone. (7.50)

A more complete expression is:

d_k = G_margin(B_k,K_S,L_k,MarketDepth_k,Authority_k,Residual_k). (7.51)

This state dependence is one reason the margin account is a useful calibration example for recursive financial closure.

---

8. How Q Drives the Subject Toward the Gate

8.1 The full margin-buffer differential

Recall:

B = C + nhR_M − D. (8.1)

Differentiate:

dB = dC + nh dR_M + nR_M dh + hR_M dn − dD. (8.2)

The equation contains five immediate channels:

1. posted-collateral movement dC;
2. market-value movement dR_M;
3. haircut movement dh;
4. position movement dn;
5. debt movement dD.

The market component is only one part of margin dynamics.

8.2 Substitute the Complex CAPM differential

The market value satisfies:

dR_M = (R_M/A)dA − Q_M dθ_M. (8.3)

Substitute equation (8.3) into equation (8.2):

$$dB = dC + nh[(R_M/A)dA - Q_Md\theta_M] + nR_Mdh + hR_Mdn - dD. \quad (8.4)$$

Therefore:

$$dB = dC + nh(R_M/A)dA - nhQ_Md\theta_M + nR_Mdh + hR_Mdn - dD. \quad (8.5)$$

Equation (8.5) is the principal local decomposition of margin-buffer movement.

8.3 Six channels of buffer movement

Define:

$$dB_{cash} := dC. \quad (8.6)$$

$$dB_{amplitude} := nh(R_M/A)dA. \quad (8.7)$$

$$dB_{phase} := -nhQ_Md\theta_M. \quad (8.8)$$

$$dB_{haircut} := nR_Mdh. \quad (8.9)$$

$$dB_{quantity} := hR_Mdn. \quad (8.10)$$

$$dB_{debt} := -dD. \quad (8.11)$$

Then:

$$dB = dB_{cash} + dB_{amplitude} + dB_{phase} + dB_{haircut} + dB_{quantity} + dB_{debt}. \quad (8.12)$$

This decomposition prevents all margin stress from being attributed to price.

An account may deteriorate because:

- market phase worsens;
- the base cash-flow amplitude falls;
- haircut becomes more severe;
- debt increases;
- collateral is withdrawn;
- the position is enlarged.

8.4 Phase-to-margin exposure

Hold A, h, n, C, and D locally fixed.

Then:

$$dB = -nhQ_Md\theta_M. \quad (8.13)$$

Therefore:

$$\partial B/\partial\theta_M = -nhQ_M. \quad (8.14)$$

Define the market-phase margin delta:

$$\Delta_{B,\theta M} := \partial B / \partial \theta_M. \qquad (8.15)$$

Then:

$$\Delta_{B,\theta M} = -nhQ_M. \qquad (8.16)$$

This is one of the central results of the calibration model.

It says:

> The CAPM pressure coordinate Q_M is the first-order coefficient through which a market-phase movement changes the account's collateral buffer.

Q_M is not the margin call.

Q_M is not the shortfall.

Q_M is the phase exposure that can move the subject toward or away from the gate.

8.5 Position-level phase exposure

The position-level market pressure is:

$$P_Q = nQ_M. \qquad (8.17)$$

Therefore:

$$\partial B / \partial \theta_M = -hP_Q. \qquad (8.18)$$

The collateral factor scales the amount of market phase exposure admitted into the margin buffer.

When:

$$h = 1, \qquad (8.19)$$

the full market position contributes to collateral.

When:

$$h = 0, \qquad (8.20)$$

the risky asset contributes nothing to eligible collateral.

Then:

$$\partial B / \partial \theta_M = 0. \qquad (8.21)$$

This does not mean the market position has no risk.

It means that market-value changes no longer alter the declared collateral buffer because the asset has already become fully ineligible.

The economic loss and the collateral effect are different objects.

8.6 Collateral-phase formulation

Using:

$$B = C + nA\cos\theta_C - D, \qquad (8.22)$$

differentiate:

$$dB = dC + n\cos\theta_C\, dA - nA\sin\theta_C\, d\theta_C + A\cos\theta_C\, dn - dD. \qquad (8.23)$$

Since:

$$R_C = A\cos\theta_C, \qquad (8.24)$$

and:

$$Q_C = A\sin\theta_C, \qquad (8.25)$$

we obtain:

$$dB = dC + n(R_C/A)dA - nQ_C d\theta_C + R_C dn - dD. \qquad (8.26)$$

Holding A, n, C, and D fixed:

$$dB = -nQ_C d\theta_C. \qquad (8.27)$$

Therefore:

$$\partial B/\partial\theta_C = -nQ_C. \qquad (8.28)$$

This mirrors the original Complex CAPM identity:

$$\partial R_M/\partial\theta_M = -Q_M. \qquad (8.29)$$

At the collateral level:

$$\partial B/\partial\theta_C = -nQ_C. \qquad (8.30)$$

The two equations form a nested chain:

Market Admitted Value Sensitivity: $\partial R_M/\partial\theta_M = -Q_M.$ (8.31)

Margin Buffer Sensitivity: $\partial B/\partial\theta_C = -nQ_C.$ (8.32)

This nested phase geometry is a new proposal of the present article.

8.7 Chain transmission from market phase to margin buffer

From equation (6.40):

$$\partial\theta_C/\partial\theta_M = h\sin\theta_M/\sin\theta_C. \qquad (8.33)$$

Using:

$$\partial B/\partial\theta_C = -nQ_C, \qquad (8.34)$$

the chain rule gives:

$$\partial B/\partial \theta_M = (\partial B/\partial \theta_C)(\partial \theta_C/\partial \theta_M). \qquad (8.35)$$

Therefore:

$$\partial B/\partial \theta_M = -nQ_C[h \sin \theta_M/\sin \theta_C]. \qquad (8.36)$$

Since:

$$Q_C = A \sin \theta_C, \qquad (8.37)$$

and:

$$Q_M = A \sin \theta_M, \qquad (8.38)$$

equation (8.36) reduces to:

$$\partial B/\partial \theta_M = -nhQ_M. \qquad (8.39)$$

This confirms the direct derivation.

The transmission chain is:

$$\theta_M \rightarrow \theta_C \rightarrow B. \qquad (8.40)$$

The market phase acts through the collateral filter before becoming margin-buffer movement.

8.8 Haircut-to-margin exposure

From:

$$B = C + nhR_M - D, \qquad (8.41)$$

holding other variables fixed:

$$\partial B/\partial h = nR_M. \qquad (8.42)$$

Therefore:

$$dB_{haircut} = nR_M dh. \qquad (8.43)$$

If:

$$dh < 0, \qquad (8.44)$$

then:

$$dB_{haircut} < 0. \qquad (8.45)$$

The haircut sensitivity is proportional to current admitted market value, not Q_M.

This gives two different local pressure channels:

Market-phase channel: $-nhQ_M d\theta_M$. (8.46)

Haircut channel: nR_Mdh. (8.47)

The first depends on retained CAPM pressure.

The second depends on currently admitted market value.

A highly valued asset may create a large margin shock when its haircut changes, even if its CAPM phase is temporarily stable.

8.9 Debt-to-margin exposure

Holding other variables fixed:

$\partial B/\partial D = -1.$ (8.48)

Thus:

dB_debt = −dD. (8.49)

Funding accumulation moves the subject directly toward the margin boundary.

If interest capitalizes into the debt balance:

dD = r_FD dt. (8.50)

then:

dB_funding = −r_FD dt. (8.51)

This creates a slow margin erosion channel even when:

dR_M = 0. (8.52)

and:

dh = 0. (8.53)

The subject may therefore approach the gate through funding time rather than market shock.

8.10 Quantity-to-margin exposure

Holding other variables fixed:

$\partial B/\partial n = hR_M.$ (8.54)

Therefore:

dB_quantity = hR_Mdn. (8.55)

For a long position:

dn > 0 (8.56)

increases eligible collateral.

But acquiring the additional position normally also requires cash or debt.

If the purchase is debt-financed:

$$dD \approx p_{buy}\, dn. \qquad (8.57)$$

Then:

$$dB \approx hR_M dn - p_{buy} dn. \qquad (8.58)$$

If:

$$p_{buy} > hR_M, \qquad (8.59)$$

the leveraged acquisition reduces the buffer:

$$dB < 0. \qquad (8.60)$$

Thus more collateralizable assets can still make the account less safe when their admitted collateral value is below their funding cost.

8.11 Distance to the gate

Define the normalized margin distance:

$$b := B/D, \qquad D > 0. \qquad (8.61)$$

Then:

$$b > 0 \qquad (8.62)$$

indicates positive buffer relative to debt.

The gate occurs when:

$$b < 0. \qquad (8.63)$$

A first-order phase movement required to reach the gate may be approximated by:

$$\Delta\theta_M^* \approx B/(nhQ_M). \qquad (8.64)$$

under the assumptions:

- Q_M locally constant;
- A fixed;
- h fixed;
- n fixed;
- C fixed;
- D fixed;
- adverse phase movement positive.

Equation (8.64) defines a local **phase distance to margin**.

It is only a first-order estimate.

Near the gate, Q_M and other state variables may change rapidly, so the linear approximation may fail.

8.12 Margin hazard as a phase-conditioned quantity

A conventional model may estimate:

Pr(Call within horizon H | B,Volatility,Leverage). (8.65)

The complex extension proposes testing:

Pr(Call within H | B,Q_M,θ_M,Δθ_MC,ΔQ_MC,Controls). (8.66)

The incremental hypothesis is:

Information(Call | B,Q_M,θ_M,Controls)

Information(Call | B,Controls). (8.67)

If Q_M and θ_M add no out-of-sample information, the complex extension should be removed from the margin model.

This follows the source discipline that complex notation earns its place only through explanatory, diagnostic, comparative, or predictive gain.

8.13 The pre-gate warning hierarchy

The model now distinguishes several levels.

Level 1 — High retained market pressure

Q_M is large.

This indicates high phase exposure but not an imminent call.

Level 2 — High collateral pressure

Q_C is large.

This indicates stronger restriction under the collateral frame.

Level 3 — Large market–collateral mismatch

Δθ_MC or ΔQ_MC is large.

This indicates increasing disagreement between market and collateral admission.

Level 4 — Small positive buffer

$0 < B \ll D$. (8.68)

The subject is near the gate.

Level 5 — Breach

$B < 0.$ (8.69)

The threshold has been crossed.

Level 6 — Admitted margin event

G_margin confirms the call under valid authority.

Level 7 — Open closure cycle

The call remains unresolved.

Level 8 — Closed or defaulted ledger state

Collateral, liquidation, or recovery has been recorded and reconciled.

This hierarchy prevents pressure, warning, breach, event, and closure from being treated as interchangeable.

8.14 Reflexive amplification near the boundary

Suppose losses reduce net equity:

$E = nR_M + C - D.$ (8.70)

Define leverage:

$\lambda = nR_M/E.$ (8.71)

A decline in R_M reduces both the numerator and denominator, but the denominator may fall proportionally faster.

In many leveraged states:

$dR_M < 0$
$\rightarrow E \downarrow$
$\rightarrow \lambda \uparrow.$ (8.72)

Higher leverage increases the sensitivity of equity to further market movement.

If the lender also reduces h during stress:

$dR_M < 0$
$\rightarrow h \downarrow$
$\rightarrow$ B falls through two channels. (8.73)

The combined buffer movement is:

$dB_{stress} \approx -nhQ_M d\theta_M + nR_M dh.$ (8.74)

When:

$d\theta_M > 0,$ (8.75)

and:

$dh < 0,$ (8.76)

both terms are negative.

This creates a coupled market–collateral spiral.

8.15 The gate does not create the original pressure

A margin call may appear to cause the crisis because the call forces liquidation.

But the gate should not be confused with the pressure accumulated before it.

The sequence is:

Market and Funding Conditions
→ Q_M and Q_C
→ Buffer Erosion
→ Threshold Breach
→ Margin Gate
→ Forced Action
→ Ledger Consequence. (8.77)

The gate activates and channels pressure.

It does not necessarily create the pressure originally.

However, after activation, forced action can feed back into the market and create new pressure.

Thus:

Pre-Gate Pressure ≠ Post-Gate Backreaction. (8.78)

Both belong to the complete recursive model.

8.16 Transition to financial spin

The margin call is now issued.

An enforceable obligation exists.

But the subject has not yet returned to accountable closure.

The account may still contain:

- unpaid collateral;
- unsettled sales;
- unresolved ownership;
- unrecognized losses;
- debt still outstanding;

- disputed marks;
- open legal rights;
- incomplete accounting entries.

Therefore:

Margin Call Issued ≠ Margin Event Closed. (8.79)

This is the transition from charge and gate to spin.

Charge identifies the orientations and obligations carried into the event.

The gate activates consequential action.

Spin will describe whether the charged subject can return through collateral, settlement, accounting, and legal ledgers as a coherent post-event identity.

Part V — Financial Spin as Action–Ledger Closure

9. The Financial Spinor

9.1 Why the margin account cannot be represented by one scalar

Immediately before a margin event, a conventional account model may be summarized by:

$X^{-} = (n,R_M,D,C,h,B).$ (9.1)

The margin rule evaluates:

$B = C + nhR_M - D.$ (9.2)

When:

$B < 0,$ (9.3)

the account enters breach.

A scalar or ordinary vector model can record the breach correctly.

However, after the call is issued, two different facts coexist:

1. an outward institutional action has occurred;
2. the resulting financial obligations have not necessarily returned through the required ledgers.

The broker may have issued a valid call while:

- no collateral has yet arrived;
- the asset sale remains unsettled;
- debt has not been repaid;

- the collateral ledger remains stale;
- accounting has not recognized the loss;
- legal title remains disputed;
- residual shortfall remains open.

Therefore:

Call Issuance ≠ Margin Closure. (9.4)

The source Financial Standard Model expresses the same general principle as:

Action Completion ≠ Identity Closure. (9.5)

It proposes that a purpose-bearing financial identity contains an outward action component and an inward ledger component, and that one outward cycle leaves an unresolved return-to-ledger obligation.

9.2 Closure space and internal financial state

A careful construction should distinguish two mathematical spaces.

Internal financial state space

Let:

$\mathscr{H}$_fin := the declared space of valuation, balance-sheet, charge, gate, and ledger variables. (9.6)

A minimum internal state may contain:

x_fin = (Z_M,Z_C,n,D,C,h,B,$\vec{q}$,v,L,$\mathscr{R}$). (9.7)

Closure-component space

Let:

$\mathbb{C}^2$_closure := span{e_action,e_ledger}. (9.8)

The basis vectors identify:

e_action = outward action component. (9.9)

e_ledger = inward recognition and return component. (9.10)

The complete state belongs to:

Ψ_S ∈ $\mathbb{C}^2$_closure ⊗ $\mathscr{H}$_fin. (9.11)

This construction is important.

The two-dimensional spinor index records the action–ledger distinction.

The internal state space carries the actual financial variables.

The Γ operators introduced later act primarily on the closure-component index rather than pretending that price, collateral, debt, and legal status are all ordinary spatial coordinates.

9.3 Common embedding of action and ledger states

The action and ledger records may contain different raw variables.

To compare them, define two protocol-dependent feature maps:

$$\Phi_A : \mathrm{RawActionState} \to \mathscr{H}_{\mathrm{fin}}. \qquad (9.12)$$

$$\Phi_L : \mathrm{RawLedgerState} \to \mathscr{H}_{\mathrm{fin}}. \qquad (9.13)$$

Then define:

$$\psi_A := \Phi_A(S). \qquad (9.14)$$

$$\psi_L := \Phi_L(S). \qquad (9.15)$$

The financial spinor is:

$$\Psi_S := \begin{bmatrix} \psi_A \\ \psi_L \end{bmatrix}. \qquad (9.16)$$

The maps Φ_A and Φ_L must place economically corresponding objects into comparable coordinates.

For example:

Action coordinate	Ledger coordinate
Executed asset units	Settled asset units
Market value	Collateral or accounting value
Call amount	Accepted collateral amount
Sale instruction	Settled sale quantity
Expected debt repayment	Recorded debt repayment
Economic loss	Recognized P&L
Contractual obligation	Ledgered payable or claim
Market ownership state	Legal ownership state

Without such a common embedding, the difference between ψ_A and ψ_L would be undefined.

9.4 The minimum market–collateral representation

For the narrow calibration model, define:

$$\psi_A = Z_M. \qquad (9.17)$$

$$\psi_L = Z_C. \qquad (9.18)$$

Then:

$$
\Psi_{MC} :=
\begin{bmatrix}
Z_M \\
Z_C
\end{bmatrix}. \qquad (9.19)
$$

This is the simplest market–collateral doublet.

It records:

- the market/CAPM representation;
- the collateral-admitted representation.

However, equation (9.19) is not yet sufficient for the strongest spin claim.

Two different valuations do not automatically constitute a spinor.

The stronger state must also include:

- gate status;
- obligations created by the event;
- ledger completion;
- charge reconciliation;
- residual.

Therefore the practical spinor should be written more generally as:

$$
\Psi_S :=
\begin{bmatrix}
\Phi_A(Z_M,B,\vec{q},\nu_A) \\
\Phi_L(Z_C,C,D,L,\mathrm{Trace},\mathscr{R},\nu_L)
\end{bmatrix}. \qquad (9.20)
$$

Equation (9.20) is a proposed construction.

9.5 The action component

The action component records what has happened on the outward economic and institutional surface.

A minimum action state is:

$$
\psi_A := \Phi_A(Z_M,n,D,B,\vec{q},d_{gate}). \qquad (9.21)
$$

It may contain:

- current Complex CAPM state;
- asset quantity;
- current debt;

- margin buffer;
- active structural charges;
- gate decision;
- liquidation or collateral instruction.

Before breach:

d_gate = NoAction. (9.22)

After valid call issuance:

d_gate = Call. (9.23)

After liquidation instruction:

d_gate = Liquidate. (9.24)

The action component may therefore update rapidly when the authorized gate acts.

9.6 The ledger component

The ledger component records whether the outward action has become an accountable financial fact.

Define:

ψ_L := Φ_L(Z_C,C,D,L,Trace,$\mathscr{R}$,ν_L). (9.25)

It may contain:

- collateral actually received;
- asset units legally settled;
- debt actually reduced;
- realized P&L recognized;
- ownership updated;
- collateral title perfected;
- risk exposure recorded;
- accounting entry posted;
- residual classified;
- audit trace available.

The ledger component is not merely passive memory.

It determines:

- whether the call remains open;

- whether further liquidation is authorized;
- whether collateral can be reused;
- whether loss enters capital;
- whether the account may continue trading;
- whether default or recovery procedures begin.

Thus the ledger changes the subject's future action space.

9.7 Action-to-ledger transport

Define the expected action-to-ledger transport:

$$T_{A\to L} : \psi_A \to \hat{\psi}_L. \qquad (9.26)$$

Here:

$$\hat{\psi}_L := T_{A\to L}\psi_A \qquad (9.27)$$

is the ledger state expected if the outward action closes correctly.

Examples include:

- executed quantity → settled quantity;
- call amount → accepted collateral;
- liquidation instruction → completed sale;
- sale proceeds → debt repayment;
- realized loss → accounting entry;
- collateral receipt → treasury and risk update.

The actual ledger state is ψ_L.

Define the spinor residual:

$$\delta_{spin} := \psi_L - T_{A\to L}\psi_A. \qquad (9.28)$$

A weighted spinor split is:

$$\Delta_{spin} := \|\psi_L - T_{A\to L}\psi_A\|_W. \qquad (9.29)$$

where W is a declared positive weighting operator.

A large Δ_{spin} means that outward action has outrun inward consequence integration.

This specializes the source proposal:

$$\Delta_{spinor} = \|\psi_{action} - \psi_{ledger}\|. \qquad (9.30)$$

The source interprets a large split as identity-drift risk and gives examples such as trading outrunning risk reconciliation and market movement outrunning settlement capacity.

9.8 Why raw subtraction is insufficient

The expression:

$\psi_L - \psi_A$ (9.31)

is generally invalid because market and ledger states use different conventions.

For example:

- market value and collateral value are not expected to be equal;
- trade date and settlement date differ;
- accounting value may follow a different recognition rule;
- currency translation may be required;
- collateral haircuts are deliberately asymmetric.

Therefore the correct residual is not:

$\mathrm{RawDifference} := \psi_L - \psi_A.$ (9.32)

It is:

$\mathrm{TransportResidual} := \psi_L - T_{A\to L}\psi_A.$ (9.33)

Only the difference remaining after lawful transport should count as closure failure.

This is where financial spin begins to require gauge structure.

The action–ledger split cannot be measured correctly without a declared frame map.

9.9 Pre-gate synchronized state

Before a margin breach, assume the account is reconciled within tolerance:

$B^- \geq 0.$ (9.34)

$\nu_A^- = \mathrm{ordinary}.$ (9.35)

$\nu_L^- = \mathrm{reconciled}.$ (9.36)

and:

$\Delta_{\mathrm{spin}}^- \leq \varepsilon_{\mathrm{spin}}.$ (9.37)

The spinor is not required to have identical components.

Instead:

$\psi_L^- \approx T_{A\to L}\psi_A^-.$ (9.38)

The account is then operationally closed under the declared protocol.

9.10 First-cycle transition

Suppose a shock produces:

$B^{-} < 0.$ (9.39)

The margin authority issues a call:

$\psi_A^{call} = G_{call}\psi_A^{-} + \eta_{call}.$ (9.40)

where:

- G_{call} = authorized call operator;
- η_{call} = call-specific update, notice, timing, or residual.

The ledger does not instantaneously become the resolved state:

$\psi_L^{call} \neq T_{A\to L}\psi_A^{call}.$ (9.41)

Therefore:

$\Delta_{spin}^{call} > \varepsilon_{spin}.$ (9.42)

The account has entered an open-return state.

The outward event is complete enough to be consequential, but the identity has not yet returned to ledger closure.

9.11 The open-return sign

The source macro-Dirac notation writes:

$\Psi \to -\Psi \to \Psi.$ (9.43)

It explicitly states that the minus sign is not literal physical quantum phase. It denotes the same outward identity carrying an unresolved return-to-ledger obligation.

For the financial model, introduce a closure sign:

$s_C \in \{+1,-1\}.$ (9.44)

Define:

$s_C = +1$ when the subject is ledger-closed. (9.45)

Define:

$s_C = -1$ when outward action has created unresolved return obligations. (9.46)

The call transition is:

$s_C^{-} = +1 \to s_C^{call} = -1.$ (9.47)

Successful return gives:

$s_C^{call} = -1 \to s_C^{+} = +1.$ (9.48)

This sign is a closure-status variable.

It is not a market-position sign and should not be confused with:

- long versus short;
- positive versus negative charge;
- profit versus loss;
- bullish versus bearish direction.

9.12 Operational spin versus physical spin

The minimum defensible claim is:

Financial Spin_op
:= independent action–ledger double closure of a bounded financial identity. (9.49)

This requires:

AdmitSpin_op
:= Identity
∧ Action
∧ Consequence
∧ IndependentLedgerReturn
∧ Transport
∧ Residual. (9.50)

This matches the source's terminology-admission rule, which requires action, independent ledger return, double closure, and identity continuity before the word spin is used.

The present article does not initially claim:

Financial Spin_op = Physical Spin-½. (9.51)

The correspondence is currently structural and operational.

9.13 Conditional double-cover extension

A stronger mathematical extension may define a closure rotation:

$S_C(\varphi) := \exp[-i\varphi(\hat{n}\cdot\sigma)/2]$. (9.52)

where:

- φ = closure angle;
- $\hat{n}$ = declared closure axis;
- σ = Pauli-matrix vector acting on action–ledger component space.

Then:

$S_C(2\pi) = -I_2$. (9.53)

$S_C(4\pi) = I_2$. (9.54)

Under this representation:

$\varphi = 0$ corresponds to closed pre-action identity. (9.55)

$\varphi = 2\pi$ corresponds to outwardly complete but return-obligated identity. (9.56)

$\varphi = 4\pi$ corresponds to completed action–ledger return. (9.57)

This is a proposed topological compression.

It should be retained only if it adds measurable structure beyond the operational sign s_C and an ordinary two-stage workflow model.

The main source classifies financial spin-½ as a macro closure analogy rather than a physical identity.

9.14 Closure phase is not CAPM phase

The CAPM valuation phase is:

$\theta_M = \arccos(R_M/A)$. (9.58)

The closure phase is:

φ_C = progress through action–ledger return. (9.59)

Therefore:

$\theta_M \neq \varphi_C$. (9.60)

A market phase movement may trigger a call.

But the subsequent closure phase depends on:

- notice;
- collateral delivery;
- settlement;
- debt repayment;
- liquidation;
- accounting;
- legal resolution.

The same θ_M may coexist with several closure states:

$\varphi_C = 0$, (9.61)

$\varphi_C = 2\pi$, (9.62)

or:

$$\varphi_C = 4\pi. \qquad (9.63)$$

The article must keep these phases separate.

9.15 Spinor norm and financial conservation

A physical Dirac spinor possesses a probability-current structure.

A financial spinor does not automatically preserve an analogous norm.

Collateral posting, liquidation, fees, taxes, defaults, and write-offs may change the financial magnitude.

Therefore:

$$\|\Psi_{S^+}\| \neq \|\Psi_{S^-}\| \quad \text{in general.} \qquad (9.64)$$

Financial closure does not require numerical norm preservation.

It requires:

- identity continuity or valid conversion;
- lawful charge transfer;
- accountable ledger return;
- explicit residual.

The relevant invariant is closer to the claim–obligation kernel than to the Euclidean magnitude of the state.

9.16 The accountable financial current

The generalized Dirac article proposes an accountable current of the form:

$$J_B^a = \Psi_B^\dagger \Gamma^a \Psi_B. \qquad (9.65)$$

In the financial specialization, define provisionally:

$$J_S^a := \Psi_S^\dagger W \Gamma^a \Psi_S. \qquad (9.66)$$

where W converts financial components into declared comparable units or normalized scores.

Possible interpretations include:

- rate of economically valid action;
- rate of ledgered recognition;
- rate of collateral transport;
- rate of identity-preserving closure.

Equation (9.66) is a new candidate construction.

No claim is made yet that J_S^a obeys a universal conservation equation.

A testable closure relation might be:

$\partial_aJ_S^a = -\Lambda_S + Source_S + Residual_S.$ (9.67)

where:

- Λ_S = leakage through fees, default, write-off, or identity conversion;
- Source_S = new capital, collateral, or contractual issuance;
- Residual_S = unexplained mismatch.

The exact form remains to be derived empirically.

9.17 The closure defect

A complete closure metric should combine more than the spinor split.

Define the identity-kernel defect:

$\delta_K := d_K[K(S^+),K_expected(S^-)].$ (9.68)

Define the charge residual:

$r_q := \Sigma q_in + q_gate - \Sigma q_out.$ (9.69)

Define the gate residual:

$\mathscr{R}_gate :=$ unresolved gate consequence. (9.70)

Define the transport residual:

$\mathscr{R}_T := \psi_L - T_A \to L\psi_A.$ (9.71)

Define the loop residual:

$\mathscr{R}_loop :=$ failure after transport through the required financial-frame loop. (9.72)

A combined closure defect is:

$$\Delta_close^2 := w_s\Delta_spin^2$$

- $w_K\delta_K^2$
- $w_q|r_q|^2$
- $w_g\|\mathscr{R}_gate\|^2$
- $w_T\|\mathscr{R}_T\|^2$
- $w_loop\|\mathscr{R}_loop\|^2.$ (9.73)

The weights must be declared by protocol.

They should not be selected retrospectively to force closure.

9.18 Operational closure criterion

Define:

SpinClosed_P(S) = 1 (9.74)

only when all of the following hold:

GateStatus ∈ {Resolved,ValidlyDefaulted,ValidlyConverted}. (9.75)

IdentityStatus = Continuous or ValidlyTransformed. (9.76)

ChargeStatus = ReconciledWithinTolerance. (9.77)

LedgerStatus = PostedAndAuditable. (9.78)

Δ_close ≤ ε_P. (9.79)

Otherwise:

SpinClosed_P(S) = 0. (9.80)

This criterion allows a defaulted account to reach valid closure.

Closure does not mean that the outcome was favorable.

It means that the event became an accountable post-event financial identity.

9.19 Partial closure

An account may satisfy some but not all closure conditions.

Define a closure vector:

$\vec{c}$:=
(c_amount,c_time,c_title,c_risk,c_accounting,c_legal,c_charge,c_residual). (9.81)

Each coordinate may lie in:

0 ≤ c_j ≤ 1. (9.82)

Examples include:

- collateral amount received but received late;
- sale executed but not settled;
- debt reduced but P&L not posted;
- accounting completed but legal dispute remains;
- call cured economically but charge ledger mismatched.

A scalar completion score may be defined:

C_total := Σ_jw_jc_j. (9.83)

But a scalar should not replace the vector when different failure types have different consequences.

The spinor representation earns its place only if it preserves distinctions that a single completion percentage loses.

9.20 Spin falsifier

The spin terminology should be removed if:

- ψ_A and ψ_L cannot be independently observed;
- T_A→L cannot be defined;
- action creates no meaningful unresolved obligation;
- one scalar workflow status performs equally well;
- closure residual has no predictive or governance value;
- the same bounded identity does not persist across the two stages.

Then use:

Two-Stage Margin Workflow. (9.84)

or:

Call-and-Resolution Process. (9.85)

The source framework explicitly requires reduction to weaker terms when double closure and identity continuity are not demonstrated.

---

10. The Two Margin Cycles

10.1 Cycle zero: reconciled pre-event identity

Before the shock, let:

$$\Psi_0 := \begin{bmatrix} \psi_{A,0} \\ \psi_{L,0} \end{bmatrix}. \quad (10.1)$$

Assume:

$$B_0 \geq 0. \quad (10.2)$$

$$q_{C^{op},0} = 0. \quad (10.3)$$

$$s_{C,0} = +1. \quad (10.4)$$

$$\Delta_{close,0} \leq \varepsilon_P. \quad (10.5)$$

The account is financially open as a trading subject but closed with respect to the latest completed margin cycle.

10.2 The initiating field movement

A market or collateral movement changes the buffer.

Using the previous result:

$$dB = dC + nh(R_M/A)dA - nhQ_Md\theta_M + nR_Mdh + hR_Mdn - dD. \qquad (10.6)$$

A breach occurs when:

$$B_1 < 0. \qquad (10.7)$$

The breach may be caused by:

- market-phase deterioration;
- cash-flow amplitude deterioration;
- haircut reduction;
- debt growth;
- collateral withdrawal;
- position expansion.

The breach is a measured state.

It is not yet an admitted event.

10.3 Cycle one: call issuance

The authorized margin gate evaluates:

$$d_1 = G_{margin}(B_1,K_S,L_0,P_S). \qquad (10.8)$$

If the gate admits the event:

$$d_1 = Call. \qquad (10.9)$$

The required amount is:

$$C_{call,1} = [B_{target} - B_1]_+. \qquad (10.10)$$

The contingent collateral charge activates:

$$q_C^{op,1} = q_C. \qquad (10.11)$$

The action state becomes:

$$\psi_{A,1} = G_{call}\psi_{A,0} + \eta_{call}. \qquad (10.12)$$

The ledger state initially remains incomplete:

ψ_L,1 ≠ T_A→Lψ_A,1. (10.13)

Therefore:

s_C,1 = −1. (10.14)

and normally:

Δ_close,1 > ε_P. (10.15)

This completes the first cycle:

Reconciled Account
→ Breach
→ Authorized Margin Call
→ Open Return Obligation. (10.16)

10.4 Why the first cycle is incomplete

After call issuance, the financial subject may carry:

- active collateral obligation;
- withdrawal restriction;
- pending liquidation authority;
- counterparty credit exposure;
- unresolved funding requirement;
- open timing risk;
- disputed mark;
- unsettled ownership consequences.

The call trace proves that the event occurred.

It does not prove that the event closed.

Thus:

Trace Written ≠ Residual Eliminated. (10.17)

and:

Gate Commitment ≠ Ledger Exhaustion. (10.18)

10.5 Cycle-two branch A: collateral posting

Suppose the subject posts eligible collateral ΔC.

The action is:

C_instruction = ΔC. (10.19)

The ledger admits:

$C_{admit} = EligibilityRule(\Delta C).$ (10.20)

The buffer becomes:

$B_2 = B_1 + C_{admit}.$ (10.21)

The amount condition is:

$B_2 \geq B_{target}.$ (10.22)

But complete closure additionally requires:

- collateral delivered by deadline;
- title or control perfected;
- treasury record updated;
- risk system updated;
- funding system updated;
- collateral not double-pledged;
- legal eligibility confirmed.

The expected ledger state is:

$\psi\hat{}_{L,2} = T_{A \to L}^{collateral} \psi_{A,2}.$ (10.23)

The branch residual is:

$\mathscr{R}_{collateral} = \psi_{L,2} - \psi\hat{}_{L,2}.$ (10.24)

If all tests pass:

$q_C^{op,2} \to 0.$ (10.25)

$s_{C,2} \to +1.$ (10.26)

10.6 Collateral posting can close one charge and amplify another

Posting cash cures the margin obligation but reduces free liquidity.

Let free cash be F.

Then:

$F_2 = F_1 - C_{admit}.$ (10.27)

The collateral charge may close:

$q_C^{op,2} = 0.$ (10.28)

while effective funding or liquidity charge rises:

$$|q_{eff,F,2}| > |q_{eff,F,1}|. \qquad (10.29)$$

Therefore:

Closure of Current Obligation
$\nRightarrow$ Reduction of All Future Exposure. (10.30)

This is the first clear recursive charge–spin relation.

Spin closure changes the balance sheet that determines the next effective charge.

10.7 Cycle-two branch B: voluntary deleveraging

Suppose the subject sells Δn units voluntarily.

Then:

$$n_2 = n_1 - \Delta n. \qquad (10.31)$$

Gross sale proceeds are:

$$P_{sale} = p_{exec}\Delta n. \qquad (10.32)$$

After transaction costs κ_{tx}:

$$P_{net} = p_{exec}\Delta n - \kappa_{tx}. \qquad (10.33)$$

If proceeds reduce debt:

$$D_2 = D_1 - P_{net}. \qquad (10.34)$$

The post-sale buffer is:

$$B_2 = C_1 + n_2 h_2 R_{M,2} - D_2. \qquad (10.35)$$

The asset charge changes:

$$q_{A,2} = q_{A,1} - \Delta n q_0(x). \qquad (10.36)$$

The funding charge magnitude may fall:

$$|q_{F,2}| < |q_{F,1}|. \qquad (10.37)$$

Closure also requires settlement:

$$\text{ExecutedSale} \neq \text{SettledSale}. \qquad (10.38)$$

Until settlement:

- proceeds may not be available;
- debt may not be legally reduced;
- delivery obligation remains;
- replacement-cost exposure remains.

Thus the deleveraging branch itself contains a trade-settlement sub-spinor.

10.8 Nested spin structure

The margin cycle can contain smaller closure cycles.

For example:

Margin Call
→ Asset Sale Instruction
→ Trade Execution
→ Settlement
→ Debt Repayment
→ Accounting Recognition
→ Margin Resolution. (10.39)

Define the outer spinor:

Ψ_{margin}. (10.40)

Define the nested trade spinor:

Ψ_{trade}. (10.41)

Then:

Closure(Ψ_{margin}) requires Closure(Ψ_{trade}). (10.42)

This produces a recursive closure hierarchy.

A subject may appear closed at the broker-notice level while remaining open at the settlement or accounting level.

10.9 Cycle-two branch C: forced liquidation

If the subject does not cure the call, the broker may liquidate under contractual authority.

Let liquidation quantity be:

$\ell = \kappa_L[B_{target} - B]_+$. (10.43)

The position update is:

$dn/d\tau = -\ell$. (10.44)

Debt reduction is:

$dD/d\tau = -p_{exec}\ell + Fees_L$. (10.45)

The market impact law may be written:

$dR_M/d\tau = dR_{CAPM}/d\tau - \kappa_{impact}\ell$. (10.46)

The buffer then obeys:

dB/dτ
= dC/dτ

- nh dR_M/dτ
- nR_M dh/dτ
- hR_M dn/dτ
  − dD/dτ. (10.47)

Substituting liquidation effects creates recursive feedback.

10.10 The liquidation spiral

A possible sequence is:

B ↓
→ ℓ ↑
→ R_M ↓
→ V_C ↓
→ B ↓ further. (10.48)

If market stress also reduces the collateral factor:

B ↓
→ h ↓
→ V_C ↓
→ ℓ ↑. (10.49)

The combined dynamic is nonlinear because:

- liquidation activates only after the gate;
- liquidation size depends on the shortfall;
- execution price depends on liquidation size;
- haircuts may depend on stress;
- updated values feed back into the buffer.

This is not merely nonlinear pricing.

It is nonlinear identity-bearing closure.

10.11 Forced liquidation does not guarantee closure

A broker may complete the sale but still fail to close the account.

For example:

SaleProceeds < Debt + Fees + RequiredBuffer. (10.50)

Then:

$B_2 < 0$. (10.51)

The subject still carries:

- deficiency;
- recovery claim;
- legal obligation;
- residual counterparty exposure.

Therefore:

Liquidation Completed ≠ Margin Closure. (10.52)

The liquidation action may be complete while the return-to-ledger cycle remains open.

10.12 Cycle-two branch D: default and recovery

If the account cannot restore admissibility:

v_2 = Defaulted. (10.53)

The original margin identity transforms into a recovery identity.

Let the conversion operator be:

G_default : K_margin → K_recovery. (10.54)

The charge balance becomes:

q_A + q_F + q_C
→ q_remaining-asset + q_deficiency + q_recovery + q_extinguished + r_q. (10.55)

A valid default closure requires:

- deficiency quantified;
- collateral ownership resolved;
- recovery claim established;
- legal authority recorded;
- accounting loss recognized;
- residual disputes preserved.

Default closure is therefore a valid form of identity return:

Identity Continuity
→ Valid Identity Conversion. (10.56)

The post-event identity need not be the same economic instrument.

It must be the lawfully traceable successor.

10.13 Branch operator

Let the resolution branch be:

$b_k \in \{\text{Collateral},\text{Deleverage},\text{Liquidation},\text{Default}\}.$ (10.57)

Define the branch operator:

$G_{\text{resolve}}^{(b_k)}.$ (10.58)

The post-branch action state is:

$\psi_{A,2} = G_{\text{resolve}}^{(b_k)}\psi_{A,1} + \eta_b.$ (10.59)

The expected ledger return is:

$\hat{\psi}_{L,2} = T_{A\to L}^{(b_k)}\psi_{A,2}.$ (10.60)

The actual ledger state is:

$\psi_{L,2}.$ (10.61)

The branch residual is:

$\delta_b = \psi_{L,2} - \hat{\psi}_{L,2}.$ (10.62)

Branch-specific transport is necessary because collateral posting, asset sale, and default do not produce the same ledger consequences.

10.14 The complete closure operator

Define the complete margin closure operator:

$$\mathcal{C}_{\text{margin}} := T_{\text{return}} \circ G_{\text{resolve}}^{(b)} \circ G_{\text{call}}.$$ (10.63)

Applied to the pre-event state:

$\Psi_2 = \mathcal{C}_{\text{margin}}\Psi_0 + \mathscr{R}_{\text{margin}}.$ (10.64)

The identity requirement is:

$K_{\text{expected}}(\Psi_2) \simeq K_{\text{after-valid-branch}}(\Psi_0).$ (10.65)

The charge requirement is:

$\Sigma q_{\text{in}} + q_{\text{gate}} = \Sigma q_{\text{out}} + r_q.$ (10.66)

The ledger requirement is:

$\Delta_{\text{spin},2} \le \varepsilon_{\text{spin}}.$ (10.67)

The total requirement is:

$\Delta_{\text{close},2} \le \varepsilon_P.$ (10.68)

Only then is the margin cycle closed.

10.15 Closure time

Let τ_call be the ledger tick at which the call is admitted.

Let τ_close be the tick at which the event becomes closed.

Define closure duration:

T_close := τ_close − τ_call. (10.69)

In calendar time:

T_close^clock := t_close − t_call. (10.70)

The two are different.

A long calendar interval may contain few ledger updates.

A short crisis interval may contain many.

Closure-time analysis may therefore use:

- clock duration;
- event count;
- settlement stages;
- phase depth;
- residual decay.

10.16 Closure velocity

Define normalized closure progress:

0 ≤ C(τ) ≤ 1. (10.71)

Then closure velocity is:

v_C := dC/dτ. (10.72)

A high v_C means the system rapidly integrates the event into accountable ledgers.

A low v_C means obligations remain open.

But rapid closure is not always good.

A forced liquidation may close quickly while producing severe loss.

Therefore:

Closure Speed ≠ Closure Quality. (10.73)

10.17 Closure quality

Define closure quality:

Q_close
:=
f(IdentityContinuity,ChargeReconciliation,ResidualMagnitude,LegalValidity,Auditability).
(10.74)

To avoid confusion with the CAPM coordinate Q, use:

$\mathcal{Q}$_close := closure quality. (10.75)

A rapid but destructive liquidation may have:

T_close small, (10.76)

but:

$\mathcal{Q}$_close moderate or low. (10.77)

A carefully reconciled collateral cure may have:

T_close larger, (10.78)

but:

$\mathcal{Q}$_close high. (10.79)

10.18 False closure

A system may report the call as closed even when important residual remains.

Define reported closure:

C_reported = 1. (10.80)

Define actual protocol closure:

SpinClosed_P = 0. (10.81)

This produces false closure:

FalseClosure := C_reported − SpinClosed_P = 1. (10.82)

Examples include:

- collateral received but legally ineligible;
- sale executed but unsettled;
- debt marked down but not contractually discharged;
- accounting entry posted without collateral reconciliation;
- call removed manually while shortfall remains;

- default status avoided by reclassification.

False closure is an important empirical target.

10.19 Residual after valid closure

Even valid closure may leave residual.

Suppose the call is cured, but the account suffers:

- lost liquidity;
- realized loss;
- reduced position;
- higher future funding spread;
- damaged credit standing.

Then:

SpinClosed_P = 1, (10.83)

while:

$\mathscr{R}$_post ≠ 0. (10.84)

This is not contradiction.

Closure means the event has entered accountable history.

Residual means the achieved closure did not eliminate every consequence.

The main framework treats residual as what the declared closure failed to contain and insists that commitment and exhaustion remain distinct.

10.20 Recursive update after closure

After the second cycle, the subject has a new state:

$\mathcal{S}_3$ = (K3,Z_M,3,D3,C3,h3,B3,L3,$\mathscr{R}_3$). (10.85)

Its structural charge may be unchanged or validly transformed.

Its effective charge is updated:

$\vec{q}$_eff,3
= U_q($\vec{q}$_struct,3,n3,D3,C3,h3,L3,$\mathscr{R}_3$,Field3). (10.86)

For example:

- asset sale reduces q_eff,A;
- debt repayment reduces q_eff,F;

- cash posting may increase liquidity sensitivity;
- default may replace margin charge with recovery charge;
- residual may increase future haircuts.

The source Financial Standard Model gives the same recursive pattern: a leveraged loss enters the risk ledger, collateral capacity declines, forced selling becomes more likely, and the original directional orientation becomes effectively amplified.

10.21 The specialized master recurrence

The complete margin recurrence is:

$$
\begin{aligned}
&(K_k, \vec{q}_k, \Psi_k, L_k) \\
&\to \text{MarketAndCollateralMovement} \\
&\to \text{G_margin} \\
&\to \text{BranchAction} \\
&\to (\text{Trace}_k, \text{Residual}_k) \\
&\to L_{k+1} \\
&\to (K_{k+1}, \vec{q}_{k+1}, \Psi_{k+1}).
\end{aligned}
\tag{10.87}
$$

In compact form:

$$
\begin{aligned}
&\text{Charge} \\
&\to \text{Field Response} \\
&\to \text{Buffer Erosion} \\
&\to \text{Gate} \\
&\to \text{Open Spin State} \\
&\to \text{Ledger Return} \\
&\to \text{Updated Charge}.
\end{aligned}
\tag{10.88}
$$

This is the central recursive closure law of the margin calibration model.

10.22 Result of Part V

Financial spin is now defined operationally.

It is not:

- market direction;
- momentum;
- clockwise or counterclockwise CAPM rotation;
- merely having two state variables.

It is:

> the closure structure through which a charged financial identity moves from consequential outward action to independent ledger return, preserving or validly converting its identity while reconciling charge and retaining residual.

The margin account passes the provisional admission test because:

- it has a bounded identity;
- a market or collateral event produces enforceable action;
- the first cycle creates new obligations;
- a separate return process is required;
- closure can fail measurably;
- the event must survive transport across several financial frames.

The next part develops that frame transport explicitly and asks when the stronger word **gauge** is justified.

Part VI — Gauge Transport Across Financial Frames

11. The Financial Frame Graph

11.1 Why one position appears as several different objects

The leveraged margin subject does not exist inside only one informational frame.

The same underlying position may appear as:

- a market-valued asset to the trading desk;
- eligible collateral to the broker;
- a funded exposure to treasury;
- a risk-weighted position to risk management;
- a carrying amount to accounting;
- a contractual claim to legal;
- a regulated exposure to a supervisory authority.

These are not necessarily errors or competing descriptions.

Each frame applies a different admissibility rule.

Let the principal frames be:

$$\mathbb{F} := \{F_M, F_C, F_R, F_T, F_A, F_L, F_G\}. \qquad (11.1)$$

where:

- F_M = market frame;
- F_C = collateral frame;
- F_R = risk frame;

- F_T = treasury and funding frame;
- F_A = accounting frame;
- F_L = legal frame;
- F_G = regulatory frame.

The Financial Standard Model source already requires governed transport because the same transaction, obligation, exposure, or ownership state may appear differently in trading, treasury, risk, accounting, legal, tax, and regulatory frames. It also distinguishes gauge transport from double-entry accounting: double entry protects balance and trace, whereas gauge-like transport concerns lawful redescription of the same economic relation.

11.2 Frame-local financial states

Let the complete financial subject be S.

Its representation in frame f is:

$S_f := \Pi_f(S).$ (11.2)

where Π_f is the projection and admission protocol of frame f.

Examples include:

S_M = market representation. (11.3)

S_C = collateral representation. (11.4)

S_R = risk representation. (11.5)

S_A = accounting representation. (11.6)

The representations need not be numerically equal:

$S_M \neq S_C \neq S_R \neq S_A.$ (11.7)

Yet they may still refer to the same bounded financial subject:

$K(S_M) \simeq K(S_C) \simeq K(S_R) \simeq K(S_A).$ (11.8)

Here K denotes the identity kernel.

Equation (11.8) is the foundational gauge requirement.

The coordinates may vary.

The identity must remain recognizable.

11.3 Frame-specific questions

Each frame answers a different operational question.

Market frame

What value is currently admitted by the market or declared valuation model?

Its main coordinates include:

Z_M = n(R_M + iQ_M). (11.9)

Collateral frame

What value is admissible as security for the funding obligation?

Its principal state includes:

Z_C = n(R_C + iQ_C). (11.10)

Risk frame

What loss or exposure distribution is recognized under the declared risk model?

Write:

X_R = RiskMap(S_M,S_C,ScenarioSet). (11.11)

Treasury frame

What funding, liquidity, cash, and encumbrance consequences follow?

Write:

X_T = TreasuryMap(D,C,Maturity,Liquidity). (11.12)

Accounting frame

What amount, classification, P&L, impairment, or collateral entry is recognized?

Write:

X_A = AccountingMap(S,Standard,RecognitionDate). (11.13)

Legal frame

Who owns the asset, who owes performance, and which rights are enforceable?

Write:

X_L = LegalMap(Contract,Title,SecurityInterest,DefaultStatus). (11.14)

Regulatory frame

What exposure, capital, liquidity, reporting, or concentration treatment applies?

Write:

X_G = RegulatoryMap(S,EntityBoundary,RuleSet). (11.15)

These maps may disagree numerically without referring to different underlying instruments.

But if they disagree about:

- the instrument;
- quantity;
- owner;
- obligor;
- settlement status;
- collateral title;
- event identity,

then cross-frame objectivity has failed.

11.4 The frame graph

Represent the financial system as a directed graph:

$\mathcal{G}_F := (\mathbb{F}, \mathscr{E})$. (11.16)

where:

- $\mathbb{F}$ is the set of financial frames;
- $\mathscr{E}$ is the set of permitted transport edges.

A minimum margin-account graph may contain:

F_M → F_C. (11.17)

F_M → F_R. (11.18)

F_C → F_T. (11.19)

F_C → F_A. (11.20)

F_L → F_C. (11.21)

F_A → F_G. (11.22)

The legal-to-collateral edge matters because an asset may possess market value while being legally ineligible as collateral.

The market-to-risk edge matters because the same position may generate a risk exposure different from its current price.

The collateral-to-treasury edge matters because posted collateral may become encumbered and unavailable for other obligations.

The graph records which transformations are institutionally meaningful.

Not every frame can transport directly into every other frame.

11.5 Edge transport

For an admissible edge A → B, define:

$$T_{AB} : S_A \to \hat{S}_B. \qquad (11.23)$$

where:

$$\hat{S}_B := T_{AB}(S_A) \qquad (11.24)$$

is the target-frame state expected from the source-frame state.

The actual observed target state is:

$$S_B. \qquad (11.25)$$

The edge residual is:

$$r_{AB} := S_B - T_{AB}(S_A). \qquad (11.26)$$

Equation (11.26) is meaningful only after:

- source and target variables are embedded into comparable coordinates;
- units are aligned;
- timing conventions are declared;
- expected costs and adjustments are included;
- the relevant identity kernel is matched.

Otherwise, the subtraction compares different objects.

11.6 Components of a financial connection

A transport map requires a connection.

Denote the connection on edge A → B by:

$$\mathcal{A}_{AB}. \qquad (11.27)$$

The connection may contain:

$$\mathcal{A}_{AB} := (FX_{AB}, Discount_{AB}, Haircut_{AB}, Netting_{AB}, Recognition_{AB}, Legal_{AB}, Timing_{AB}). \qquad (11.28)$$

Possible components include:

- foreign-exchange translation;
- discount curve;
- collateral haircut;
- liquidity adjustment;
- netting rule;

- transfer-pricing rule;
- legal-equivalence rule;
- accounting classification;
- settlement-date mapping;
- regulatory conversion.

The source framework lists FX rates, discount curves, benchmark mappings, consolidation rules, hedge ratios, and legal-equivalence maps as candidate financial connection objects. Without such a rule, a claim that two frames contain the “same exposure” is underdefined.

11.7 The minimum market-to-collateral connection

For the calibration subject:

$$T_{MC} : Z_M \to Z_C. \qquad (11.29)$$

The minimum real-axis rule is:

$$R_C = hR_M. \qquad (11.30)$$

Under the shared-amplitude convention:

$$Q_C = \sqrt{(A^2 - h^2 R_M^2)}. \qquad (11.31)$$

Therefore:

$$T_{MC}(R_M,Q_M;h,A) = (hR_M,\sqrt{(A^2 - h^2 R_M^2)}). \qquad (11.32)$$

In complex notation:

$$T_{MC}(Z_M;h,A) = h\mathrm{Re}(Z_M) + i\sqrt{[A^2 - h^2 \mathrm{Re}(Z_M)^2]}. \qquad (11.33)$$

This transport is nonlinear.

It is not ordinary multiplication of the full complex number by h:

$$Z_C \neq hZ_M \quad \text{in general.} \qquad (11.34)$$

Indeed:

$$hZ_M = hR_M + ihQ_M, \qquad (11.35)$$

while:

$$Z_C = hR_M + iQ_C. \qquad (11.36)$$

and generally:

$$Q_C \neq hQ_M. \qquad (11.37)$$

The collateral filter does not simply shrink the market complex vector.

It recomputes the conjugate coordinate under the target protocol.

11.8 Market-to-collateral transport residual

Let the expected collateral state be:

$\hat{Z}_C := T_{MC}(Z_M;h,A).$ (11.38)

Let the observed collateral state be:

$Z_C^{obs}.$ (11.39)

Define:

$r_{MC} := Z_C^{obs} - \hat{Z}_C.$ (11.40)

The real and imaginary components are:

$r_{MC}^R := R_C^{obs} - hR_M.$ (11.41)

$r_{MC}^Q := Q_C^{obs} - \sqrt{(A^2 - h^2R_M^2)}.$ (11.42)

Possible causes include:

- stale market mark;
- incorrect haircut;
- settlement delay;
- ineligible collateral;
- concentration adjustment omitted from h;
- currency mismatch;
- disputed asset identity;
- inconsistent source amplitude;
- operational booking failure.

The residual should not be forced to zero by redefining h after the event.

Such retrospective calibration would destroy falsifiability.

11.9 Time alignment

A source frame and target frame may update at different times.

Let:

t_M = market observation time. (11.43)

t_C = collateral recognition time. (11.44)

Then:

$t_C \geq t_M$ in many processes. (11.45)

A valid transport should therefore include a timing operator:

$T_{MC}^{(\Delta t)}$. (11.46)

where:

$\Delta t_{MC} := t_C - t_M$. (11.47)

The expected collateral state is:

$$\hat{Z}_C(t_C) = T_{MC}[Z_M(t_M), MarketPath(t_M, t_C), h(t_C), \Delta t_{MC}]. \quad (11.48)$$

Without timing alignment, a fast market movement may appear as a collateral-system error even when the target system is updating according to its declared cycle.

Conversely, excessive delay may itself be a genuine transport failure.

11.10 Transport status

Each edge should be assigned one of the following statuses:

- survives;
- covariantly survives;
- partially survives;
- local only;
- fails;
- indeterminate;
- non-comparable.

A claim **survives** when both identity and expected target relation remain within tolerance.

A claim **covariantly survives** when numerical coordinates change according to the declared connection.

A claim is **local only** when it remains valid in the source frame but has no justified target-frame extension.

A claim **fails** when the observed target state contradicts the expected transport.

A claim is **non-comparable** when no lawful common coordinate exists.

A failed transport does not always invalidate the local claim.

It may instead limit its scope.

11.11 Transport and charge

Structural charge should survive ordinary representational transport.

For charge coordinate r:

q_r,B[T_AB(S_A)] = q_r,A(S_A). (11.49)

unless the edge contains a valid charge-conversion vertex.

For example:

- market-to-collateral transport should not reverse a long claim into a short claim;
- accounting recognition should not silently change asset ownership;
- regulatory netting should not erase the underlying contractual obligations;
- currency translation should not change who owes payment.

Therefore:

Ordinary Frame Transport → Charge Preserved. (11.50)

A charge change requires:

Declared Conversion Gate → Charge Transformed. (11.51)

Examples include:

- exercise;
- settlement;
- novation;
- default;
- debt-to-equity conversion;
- collateral seizure.

11.12 Transport and the financial spinor

The action component and ledger component may be located in different frames.

Write:

Ψ_S :=
[
ψ_A in F_A
ψ_L in F_L
]. (11.52)

Their direct difference is not frame-valid.

Define the expected ledger component:

$\psi\hat{}_L := T_AL\psi_A.$ (11.53)

Then the covariant spinor split is:

$\delta_spin\hat{}cov := \psi_L - T_AL\psi_A.$ (11.54)

The corresponding norm is:

$\Delta_spin\hat{}cov := \|\psi_L - T_AL\psi_A\|_W.$ (11.55)

This replaces the naive expression:

$\|\psi_L - \psi_A\|.$ (11.56)

Financial spin closure therefore depends on gauge transport.

Without a frame map:

No Valid Spinor Split. (11.57)

Without an independent ledger surface:

No Spinor Claim. (11.58)

Without an invariant identity kernel:

No Gauge Claim. (11.59)

This dependency order follows directly from the source research architecture.

---

12. A-B Fixedness and Financial Gauge Covariance

12.1 A-B Fixedness

The generalized Dirac source defines A-B Fixedness as the condition under which two frames can continue identifying the same event, obligation, trace, or object after lawful translation.

A general expression is:

$$ABFix_P(e) \Leftrightarrow T_AB(e_A) \approx e_B \wedge Inv_A(e_A) = Inv_B(e_B) \wedge Rec_AB(e). \quad (12.1)$$

Here:

- T_AB = frame-transport map;
- Inv_A and Inv_B = invariant relations;
- Rec_AB = accessible or reconstructable cross-frame record.

The source stresses that macro A-B Fixedness additionally requires compatible observation, accessible trace, invariant preservation, and residual honesty. A financial position is not A-B fixed when trading, accounting, risk, treasury, collateral, and regulatory frames cannot preserve the same exposure identity.

12.2 Financial A-B Fixedness

For financial subject S:

ABFix_P(S;A,B)
⇔ d_B[T_AB(S_A),S_B] ≤ ε_AB
∧ Inv_B[T_AB(K_A)] = Inv_A(K_A)
∧ Rec_AB(S) = 1
∧ Residual_AB disclosed. (12.2)

The terms mean:

Transport agreement

The transported source state is sufficiently close to the observed target state.

Invariant preservation

The same claim, obligation, owner, quantity, or settlement identity remains recognizable.

Record accessibility

Both frames can inspect or reconstruct the relevant trace.

Residual honesty

Unresolved differences remain explicitly recorded.

A-B Fixedness does not require perfect numerical equality.

It requires controlled disagreement about the same object.

12.3 The invariant kernel

For the margin subject, define:

Inv_K(S)
:=
(InstrumentID,SignedUnits,Owner,Obligor,PayoffRights,Maturity,Currency,ExecutionLineage).
(12.3)

A stronger collateral-sensitive invariant may include:

Inv_C(S)
:= (Inv_K,CollateralAgreement,SecurityInterest,EligibilityClass,SettlementStatus). (12.4)

A transport is identity-preserving when:

Inv_B[T_AB(S_A)] = Inv_A(S_A). (12.5)

A valid conversion vertex may change part of the invariant, but it must preserve lineage:

Lineage(K_before → K_after) = auditable. (12.6)

For example, option exercise changes the instrument identity.

It does not erase the originating option contract and exercise trace.

12.4 Frame-local value is not the invariant

The following are generally not gauge invariants:

R_M. (12.7)

R_C. (12.8)

BookValue. (12.9)

RiskExposure. (12.10)

RegulatoryExposure. (12.11)

They are frame-local or frame-covariant quantities.

The stronger invariant is the relation they represent.

Examples include:

- the same contractual payoff rights;
- the same legal ownership;
- the same originating trade;
- the same funding obligation;
- the same collateral pledge;
- the same netted exposure set.

Thus:

Value_A ≠ Value_B may be valid. (12.12)

But:

Identity_A ≠ Identity_B without a conversion gate is a failure. (12.13)

12.5 General local frame transformations

Let a local redescription in frame A be:

S_A′ = G_AS_A. (12.14)

Let the corresponding redescription in frame B be:

S_B′ = G_BS_B. (12.15)

The transport operator must transform as:

$$T_{AB'} = G_B T_{ABG} A^{-1}. \qquad (12.16)$$

Then:

$$T_{AB'} S_{A'} = G_B(T_{ABS} A). \qquad (12.17)$$

Equation (12.17) is the general covariance condition.

It says that performing the transport before or after local redescription yields equivalent target-frame content.

The diagram commutes:

$$\begin{array}{ccc} S_A & \xrightarrow{T_{AB}} & S_B \\ \downarrow & & \downarrow \\ G_A & & G_B \\ \downarrow & & \downarrow \\ S_{A'} & \xrightarrow{T_{AB'}} & S_{B'}. \end{array} \qquad (12.18)$$

This is the minimum mathematical content required for a gauge-like claim.

12.6 Financial examples of local redescription

Possible G_A transformations include:

- changing reporting currency;
- changing units from shares to lots;
- decomposing a hedge into cash and derivative legs;
- moving from gross to net presentation;
- reallocating the position between desks;
- consolidating legal entities;
- changing accounting presentation while preserving transaction substance.

A lawful local redescription should not alter the underlying economic relation.

For example, currency translation changes numbers:

$$\text{Value}_{GBP} \neq \text{Value}_{USD}. \qquad (12.19)$$

But, after applying the declared FX connection:

$$FX_{USD \to GBP}(\text{Value}_{USD}) = \text{Value}_{GBP}. \qquad (12.20)$$

The identity and exposure survive covariantly.

12.7 General representation of financial charge

Let the subject carry charge vector:

$\vec{q} = (q_A, q_F, q_C, \ldots)$. (12.21)

A frame transformation acts through a charge representation:

$\rho_q(G_f)$. (12.22)

Then:

$\Psi_f' = \rho_q(G_f)\Psi_f$. (12.23)

The connection transforms as:

$U_{AB}' = \rho_q(G_B)U_{AB}\rho_q(G_A)^{-1}$. (12.24)

The covariant edge difference is:

$D_{AB}\Psi := \Psi_B - U_{AB}\Psi_A$. (12.25)

Under local transformations:

$D_{AB}'\Psi' = \rho_q(G_B)D_{AB}\Psi$. (12.26)

Therefore the residual transforms as a target-frame object rather than changing arbitrarily.

12.8 Restricted Abelian approximation

A simple candidate sector uses a phase representation.

Let:

$G_f(\chi_f) := \exp(iq\chi_f)$. (12.27)

Then:

$\Psi_f' = \exp(iq\chi_f)\Psi_f$. (12.28)

Define the edge transporter:

$U_{AB} := \exp(iq\mathcal{A}_{AB})T_{AB}^0$. (12.29)

where:

- T_{AB}^0 = ordinary declared financial transport;
- $\mathcal{A}_{AB}$ = connection phase or orientation adjustment;
- q = charge carried by the identity.

Covariance requires:

$U_{AB}' = \exp(iq\chi_B)U_{AB}\exp(-iq\chi_A)$. (12.30)

For an Abelian sector:

$\mathcal{A}_{AB}' = \mathcal{A}_{AB} + \chi_B - \chi_A$. (12.31)

Then:

$D_{AB}\Psi' = \exp(iq\chi_B)D_{AB}\Psi.$ (12.32)

Equations (12.27)–(12.32) form a candidate U(1)-like financial transport sector.

They are proposed mathematical constructions.

They are not yet empirical laws of financial markets.

12.9 Gauge phase is not CAPM phase

The local frame phase χ_f is not automatically the CAPM valuation phase θ_M.

CAPM phase records valuation orientation:

$\theta_M = \arccos(R_M/A).$ (12.33)

Gauge phase records local representation:

χ_f = orientation convention of frame f. (12.34)

Therefore:

$\chi_f \neq \theta_M.$ (12.35)

A CAPM phase movement changes the financial state.

A pure gauge redescription should leave properly constructed economic observables invariant.

This distinction is essential.

Otherwise, a real market-risk movement would be confused with a change of notation.

12.10 Gauge-invariant relative phase

Suppose frame A records phase θ_A and frame B records phase θ_B.

Define the covariant relative phase:

$\Theta_{AB} := \theta_B - \theta_A - q\mathcal{A}_{AB}.$ (12.36)

Under:

$\theta_A' = \theta_A + q\chi_A,$ (12.37)

$\theta_B' = \theta_B + q\chi_B,$ (12.38)

and:

$\mathcal{A}_{AB}' = \mathcal{A}_{AB} + \chi_B - \chi_A,$ (12.39)

we obtain:

$\Theta_{AB}' = \Theta_{AB}.$ (12.40)

Thus Θ_{AB} is gauge-invariant under the restricted construction.

Financially, Θ_AB may measure disagreement remaining after the expected frame relation has been removed.

For the market–collateral edge:

$$\Theta_{MC} = \theta_C - \theta_M - q\mathcal{A}_{MC}. \qquad (12.41)$$

If the connection fully explains the frame difference:

$$\Theta_{MC} \approx 0. \qquad (12.42)$$

If:

$$|\Theta_{MC}| \gg 0, \qquad (12.43)$$

then the frames disagree beyond the declared connection.

Possible causes include:

- model misspecification;
- stale records;
- hidden haircut adjustment;
- disputed eligibility;
- wrong identity match;
- unrecorded authority decision.

12.11 The gauge-covariant spinor split

Let action state ψ_A be in frame A and ledger state ψ_L be in frame L.

Define:

U_{AL} := charge-sensitive action-to-ledger transporter. (12.44)

Then:

$$\delta_{spin}^{G} := \psi_L - U_{AL}\psi_A. \qquad (12.45)$$

Under local transformations:

$$\psi_A' = G_A\psi_A. \qquad (12.46)$$

$$\psi_L' = G_L\psi_L. \qquad (12.47)$$

$$U_{AL}' = G_L U_{AL} G_A^{-1}. \qquad (12.48)$$

Therefore:

$$\delta_{spin}^{\{G'\}} = G_L\delta_{spin}^{G}. \qquad (12.49)$$

A gauge-invariant defect norm requires a compatible target metric W_L:

$$\Delta_{spin,G}^2 := (\delta_{spin}^{G})^\dagger W_L \delta_{spin}^{G}. \qquad (12.50)$$

with:

$W_{L'} = (G_{L^{-1}})^{\dagger} W_{LG} L^{-1}$. (12.51)

Then:

$\Delta_{spin,G'^2} = \Delta_{spin,G^2}$. (12.52)

This construction gives financial spin closure a frame-independent diagnostic.

12.12 Charge conversion and non-preserving edges

Ordinary transport preserves charge.

A conversion vertex may not.

Let charge operator in the source frame be:

$\hat{Q}_A$. (12.53)

Let charge operator in the target frame be:

$\hat{Q}_B$. (12.54)

Charge-preserving transport satisfies:

$\hat{Q}_B U_{AB} = U_{AB} \hat{Q}_A$. (12.55)

Equivalently:

$[\hat{Q}, U_{AB}] = 0$ (12.56)

when the same operator acts in both frames.

A charge-changing vertex satisfies:

$\hat{Q}_B U_{AB} - U_{AB} \hat{Q}_A = J_{AB}$. (12.57)

where J_{AB} is the declared charge-conversion current.

Examples include:

- option exercise;
- collateral seizure;
- default conversion;
- novation;
- debt-to-equity exchange.

The conversion must be reconciled:

$q_{in} + q_{gate} = q_{out} + r_q$. (12.58)

Without an explicit vertex, a charge-changing edge is an identity error.

12.13 A-B Fixedness score

An empirical A-B Fixedness score may combine:

- identity agreement;
- quantity agreement;
- timing agreement;
- charge agreement;
- trace accessibility;
- residual agreement.

Define:

Score_AB
:= w_KS_K

- w_qS_q
- w_tS_t
- w_RS_R
- w_recS_rec
  − λ_AB‖r_AB‖. (12.59)

where each component lies between zero and one.

A possible interpretation is:

Score_AB = 0 no shared financial object. (12.60)

Score_AB ≈ 1 shared identity, lawful transport, accessible trace, and residual agreement. (12.61)

The exact score is an empirical engineering proposal.

It should not be mistaken for a universal invariant.

12.14 Failure modes of financial A-B Fixedness

Identity mismatch

The two frames refer to different instruments or lots.

Quantity mismatch

Trading records 1,000 units while collateral records 900.

Timing mismatch

One frame includes a settlement that another has not yet admitted.

Charge mismatch

One frame records ownership while another records only a delivery obligation.

Authority mismatch

The broker regards a margin call as valid while legal regards it as unauthorized.

Trace failure

The relevant trade, call, or collateral record is inaccessible.

Residual denial

One frame reports closure while another preserves an unresolved exception.

A-B Fixedness fails whenever the system cannot establish that the frames are disagreeing about the same bounded identity.

12.15 Objectivity in the financial system

Financial objectivity does not require one privileged numerical value.

It requires that frame-local values remain connected through lawful transport.

Thus:

FinancialObjectivity_P(S)
:= Invariance of S across admissible financial frames under P. (12.62)

A market value may differ from a collateral value.

A collateral value may differ from an accounting carrying amount.

The system remains objective when:

- the identity is shared;
- the connection is declared;
- the trace is accessible;
- the residual is honest.

The generalized Dirac source describes this as the higher-order role of A-B Fixedness: stable internal trace may exist while cross-frame recognition still fails.

---

13. Financial Holonomy, Curvature, and Loop Residual

13.1 Why pairwise transport is not enough

A position may reconcile pairwise between:

- market and collateral;
- collateral and accounting;
- accounting and market.

Yet the full closed loop may still fail.

Consider:

F_M → F_C → F_A → F_M. (13.1)

The loop tests whether repeated lawful translations return the subject to an equivalent representation in the starting frame.

This is stronger than testing one edge.

13.2 Loop transporter

Define:

U_MC : F_M → F_C. (13.2)

U_CA : F_C → F_A. (13.3)

U_AM : F_A → F_M. (13.4)

The loop transporter is:

H_MCA := U_AMU_CAU_MC. (13.5)

Applied to the initial market state:

S_M^return = H_MCAS_M^start. (13.6)

If transport is path-independent and fully reconciled:

S_M^return ≃ S_M^start. (13.7)

If not:

S_M^return ≠ S_M^start. (13.8)

The operator H_MCA is the financial loop holonomy.

13.3 Naive loop residual

A simple loop residual is:

$\mathscr{R}$_MCA^naive := H_MCAS_M − S_M. (13.9)

This follows the candidate curvature concept in the Financial Standard Model source, where transport around A → B → C → A produces residual when the state does not return to its starting representation. The source identifies transaction costs, legal asymmetry, timing

mismatch, funding differences, inconsistent valuation, arbitrage restrictions, and settlement friction as possible causes.

However, equation (13.9) is too strict for many real financial systems.

A lawful loop may intentionally change value because of:

- transaction fees;
- accrued funding;
- tax;
- bid–ask spread;
- elapsed time;
- realized cash flow;
- authorized write-off.

These should not automatically be classified as unexplained curvature.

13.4 Expected loop transport

Define the expected loop operator:

H_MCA^P. (13.10)

This operator includes all protocol-authorized changes:

- known fees;
- declared timing;
- expected accrual;
- valid tax treatment;
- authorized haircut;
- lawful accounting recognition;
- permitted identity conversion.

The expected return state is:

Ŝ_M^return := H_MCA^PS_M^start. (13.11)

The observed return state is:

S_M^return,obs. (13.12)

Define the governed loop residual:

$\mathscr{R}$_loop := S_M^return,obs – Ŝ_M^return. (13.13)

Equation (13.13) is the stronger financial curvature candidate.

It measures mismatch after known path effects have been removed.

13.5 Normalized loop curvature

Define:

$\kappa_{loop} := \|\mathscr{R}_{loop}\|_W / [\|\hat{S}_M^{return}\|_W + \varepsilon]$. (13.14)

where ε prevents division by zero.

Interpretation:

$\kappa_{loop} \approx 0$ transport loop closes within tolerance. (13.15)

$\kappa_{loop} > 0$ unexplained path dependence remains. (13.16)

The term curvature should remain provisional.

It earns its place only if κ_{loop} provides information beyond ordinary reconciliation-error measures.

13.6 Abelian loop phase

Under the restricted Abelian model:

$U_{AB} = \exp(iq\mathcal{A}_{AB})T_{AB}^0$. (13.17)

The total loop connection is:

$\Phi_{MCA} := \mathcal{A}_{MC} + \mathcal{A}_{CA} + \mathcal{A}_{AM}$. (13.18)

The loop phase factor is:

$W_{MCA} := \exp(iq\Phi_{MCA})$. (13.19)

Under local frame transformations, endpoint phases cancel around the closed loop:

$\Phi_{MCA}' = \Phi_{MCA}$. (13.20)

Therefore:

$W_{MCA}' = W_{MCA}$. (13.21)

This is a candidate financial Wilson-loop-like quantity.

It measures net orientation accumulated around the frame loop.

The terminology is purely structural unless a stable empirical phase connection can be measured.

13.7 Zero-curvature condition

A flat or path-independent financial connection satisfies:

$$H_{loop}^{P} \simeq I \qquad (13.22)$$

after all authorized adjustments.

In the Abelian phase sector:

$$\Phi_{loop} \equiv 0 \bmod 2\pi. \qquad (13.23)$$

Financially, this means:

- the same identity returns;
- all expected conversions reconcile;
- no unexplained frame disagreement remains;
- residual is below tolerance.

A nonzero loop phase or residual may indicate:

- inconsistent valuation;
- missing position;
- hidden funding cost;
- incorrect netting;
- settlement failure;
- legal-title mismatch;
- accounting recognition error;
- unauthorized collateral use.

13.8 Curvature and ordinary economic cost

Curvature must not be confused with ordinary cost.

Suppose a loop includes a lawful transaction fee F.

Then:

$$\text{Value}_{return} = \text{Value}_{start} - F. \qquad (13.24)$$

If F is included in H_{loop}^{P}:

$$\mathscr{R}_{loop} = 0. \qquad (13.25)$$

even though:

$$\text{Value}_{return} \neq \text{Value}_{start}. \qquad (13.26)$$

Thus:

$$\text{Economic Change} \neq \text{Gauge Failure}. \qquad (13.27)$$

Gauge failure exists when the observed return differs from the declared, lawful transformed return.

13.9 Curvature and margin closure

For the margin account, consider:

Market
→ Margin Call
→ Collateral or Liquidation
→ Accounting and Risk
→ Market/Reconciled Position. (13.28)

The loop closes only when:

- post-event asset units reconcile;
- debt reconciles;
- collateral reconciles;
- ownership and settlement reconcile;
- realized P&L reconciles;
- active charges reconcile;
- residual is declared.

Therefore:

Spin Closure requires Loop Closure. (13.29)

More precisely:

SpinClosed_P(S) = 1
⇒ κ_required-loops ≤ ε_loop. (13.30)

The converse need not hold.

One frame loop may reconcile while another required legal or regulatory loop remains open.

13.10 Multiple loops

A financial subject may participate in several loops.

Market–collateral–market loop

F_M → F_C → F_M. (13.31)

Trade–settlement–accounting loop

F_M → F_Ledger → F_A → F_M. (13.32)

Collateral–treasury–risk loop

$F_C \to F_T \to F_R \to F_C.$ (13.33)

Accounting–regulatory–legal loop

$F_A \to F_G \to F_L \to F_A.$ (13.34)

Define the required loop set:

$\mathscr{L}_P(S) := \{\ell_1,\ell_2,\dots,\ell_m\}.$ (13.35)

The aggregate loop defect is:

$\Delta_{loop}^2 := \Sigma_{\ell} w_\ell \kappa_\ell^2.$ (13.36)

A high-impact account may require closure across more loops than a simple retail position.

13.11 Local flatness and global curvature

Every pairwise edge may appear acceptable:

$\|r_{AB}\| \le \varepsilon_{AB}.$ (13.37)

Yet the accumulated loop may be material:

$\kappa_{loop} > \varepsilon_{loop}.$ (13.38)

Small errors can compound.

Alternatively, each edge may use a locally valid but mutually inconsistent convention.

Examples include:

- one frame uses trade-date recognition;
- another uses settlement-date recognition;
- another uses cash-date recognition.

Each may be locally coherent.

The closed loop may still fail.

Therefore:

Pairwise Consistency $\not\Rightarrow$ Global Loop Consistency. (13.39)

This is one reason gauge geometry may add diagnostic value beyond isolated reconciliations.

13.12 Path dependence

Suppose two routes connect market frame M to accounting frame A.

Route 1:

$M \to C \to A.$ (13.40)

Route 2:

$M \to R \to A.$ (13.41)

The expected accounting states are:

$\hat{S}_A^{(1)} = U_{CAU_MCS_M}.$ (13.42)

$\hat{S}_A^{(2)} = U_{RAU_MRS_M}.$ (13.43)

Path independence requires:

$\hat{S}_A^{(1)} \simeq \hat{S}_A^{(2)}.$ (13.44)

Define path residual:

$r_{path} := \hat{S}_A^{(1)} - \hat{S}_A^{(2)}.$ (13.45)

A nonzero r_{path} may be legitimate if the two routes represent different recognized processes.

But if both claim to produce the same accounting object under the same protocol, r_{path} identifies connection inconsistency.

13.13 Curvature as institutional disagreement

The loop residual can be decomposed:

$\mathscr{R}_{loop}$
$= \mathscr{R}_{identity}$

- $\mathscr{R}_{quantity}$
- $\mathscr{R}_{value}$
- $\mathscr{R}_{charge}$
- $\mathscr{R}_{timing}$
- $\mathscr{R}_{authority}$
- $\mathscr{R}_{trace}.$ (13.46)

where:

- $\mathscr{R}_{identity}$ = different instruments or lots;
- $\mathscr{R}_{quantity}$ = unmatched units;
- $\mathscr{R}_{value}$ = unexplained valuation mismatch;
- $\mathscr{R}_{charge}$ = unreconciled rights or obligations;
- $\mathscr{R}_{timing}$ = incompatible event dates;
- $\mathscr{R}_{authority}$ = disagreement about valid gate status;

- $\mathscr{R}$_trace = missing or inaccessible records.

This decomposition is more informative than one scalar reconciliation difference.

13.14 Curvature as an early-warning signal

A possible empirical hypothesis is:

κ_loop,t ↑
⇒ Pr[MarginFailure or ReconciliationBreak within H] ↑. (13.47)

A stronger hypothesis is:

κ_loop adds predictive information beyond B, leverage, and ordinary exception counts. (13.48)

This must be tested against simpler alternatives.

Possible null models include:

- total reconciliation-error count;
- stale-data indicator;
- settlement-failure rate;
- ordinary operational-risk score;
- multivariate state-space residual.

If κ_loop adds no incremental value, use the weaker term:

Cross-Ledger Reconciliation Residual. (13.49)

13.15 Curvature and residual honesty

A system may reduce reported curvature by deleting difficult records or forcing one frame to adopt another's numbers.

That is not genuine closure.

Let:

κ_reported = reported loop defect. (13.50)

Let:

κ_audited = loop defect using preserved source traces. (13.51)

Residual denial occurs when:

κ_reported ≪ κ_audited. (13.52)

Examples include:

- manual position overrides;
- backdated journal entries;
- suppressed valuation disputes;
- unrecorded collateral substitutions;
- off-ledger obligations;
- model changes applied retrospectively.

The generalized Dirac source treats residual honesty as a condition of A-B Fixedness. Without it, apparent agreement may be produced by concealment rather than lawful transport.

13.16 Gauge-closure certificate

A completed transport loop should produce a certificate:

GaugeClosureCertificate
:=
(SubjectID,LoopID,Frames,Protocol,Connections,Invariant,ExpectedReturn,ObservedReturn,Resi
(13.53)

The certificate should state:

- which subject was transported;
- which frames were traversed;
- which transport rules applied;
- which invariant was preserved;
- which costs or transformations were expected;
- what residual remained;
- who authorized closure;
- when the certificate was issued.

This is an engineering proposal.

Its purpose is auditability, not metaphysical proof.

13.17 Gauge falsifier

The gauge terminology should be removed when:

- no stable identity kernel exists;
- source and target frames cannot be declared;
- no lawful transport map can be specified;

- local transformations possess no covariance rule;
- path or loop residual adds no information beyond ordinary reconciliation;
- frame differences are arbitrary rather than governed;
- residual can be eliminated only through retrospective redefinition.

Then use:

Multi-Frame Financial Reconciliation. (13.54)

or:

Cross-Ledger Transport Audit. (13.55)

The source framework's methodological rule is that gauge terminology creates obligations: frames, a connection, an invariant, and transport tests must all be demonstrated.

13.18 Result of Part VI

The financial subject now has:

- a bounded identity kernel;
- frame-local state representations;
- declared transport edges;
- connection objects;
- charge-sensitive transformation rules;
- A-B Fixedness conditions;
- covariant spinor residual;
- closed-loop holonomy;
- candidate curvature;
- residual-preserving closure tests.

The relationship among the major concepts can now be stated:

Charge remembers how the identity transforms under a declared field. (13.56)

Spin remembers whether outward action returns through an independent ledger cycle. (13.57)

Gauge transport determines whether the same charged identity remains recognizable across financial frames. (13.58)

Curvature records the mismatch remaining after transport around a declared closed loop. (13.59)

The next task is to determine whether these structures can be assembled into a genuine first-order **Financial Gauge–Dirac kernel**, rather than remaining a collection of separate

analogies.

Part VII — The Financial Gauge–Dirac Kernel

14. Why a Dirac-Like Form Is Considered

14.1 The source-derived structural archetype

The generalized macro-Dirac framework proposes:

$(i\Gamma^{a}\nabla^{p}_a - M_B)\Psi_B = \mathscr{R}_P.$ (14.1)

Its central claim is not that institutions, markets, legal systems, or accounting ledgers are physical particles.

Its narrower structural claim is:

Dirac-Like Structure
= Identity-Bearing State

- Coupled Internal Components
- First-Order Propagation
- Frame Covariance
- Mass-Like Identity Constraint
- Residual. (14.2)

The source interprets Ψ_B as an action–ledger spinor, Γ^{a} as frame-fixedness operators, ∇^{p}_a as a protocol-covariant derivative, M_B as identity-preserving Purpose Belt mass, and $\mathscr{R}_P$ as residual produced by failed transport, excessive transformation speed, frame mismatch, or unclosed audit.

The financial construction must now determine whether these objects can be defined specifically enough to justify a corresponding equation.

14.2 The financial specialization

For the leveraged margin subject, define:

$$\Psi_S := \begin{bmatrix} \psi_A \\ \psi_L \end{bmatrix}.$$ (14.3)

where:

- ψ_A = outward market, funding, collateral, and gate-action state;
- ψ_L = inward settlement, recognition, reconciliation, and residual-integration state.

The subject also carries:

- identity kernel K_S;
- structural charge vector $\vec{q}$_S;
- margin buffer B;
- ledger L;
- frame graph $\mathcal{G}$_F;
- transport operators U_AB;
- residual $\mathscr{R}$_S.

The candidate equation must describe:

> How does a charged financial identity evolve through market time, cross financial frames, encounter a margin constraint, and remain recognizably itself while its action and ledger components attempt to close?

14.3 Why an ordinary scalar differential equation may be insufficient

A scalar state x(τ) may describe:

- market value;
- equity;
- leverage;
- buffer;
- call amount.

But after a margin call, two independently consequential states coexist:

x_action ≠ x_ledger. (14.4)

For example:

- liquidation instructed but not settled;
- collateral transferred but not legally perfected;
- debt economically reduced but not posted;
- loss incurred but not recognized;
- call marked resolved by one system but open in another.

A scalar model can add status variables to represent these differences.

Therefore, scalar inadequacy is not assumed.

The spinor model earns its place only if the coupled action–ledger representation provides measurable gain over:

- a scalar hybrid automaton;
- a conventional multivariate state-space model;
- an ordinary workflow model;
- a reconciliation dashboard.

The Financial Standard Model source explicitly classifies the present spinor framework as a testable architecture rather than an established forecasting law.

14.4 The minimum Dirac admission conditions

Define the Financial Dirac admission rule:

AdmitDirac_fin
:= Identity
∧ IrreducibleDoublet
∧ FirstOrderPropagation
∧ FrameCovariance
∧ MassConstraint
∧ Residual
∧ IncrementalGain. (14.5)

The terms mean:

Identity

The same bounded financial subject can be followed through the transformation.

Irreducible doublet

Action and ledger components are independently measurable and cannot be compressed without material loss.

First-order propagation

The next local state depends upon the current state and current field or transport inputs.

Frame covariance

The same financial identity remains recognizable under declared frame transformations.

Mass constraint

A measurable operator resists arbitrary identity change and binds the two components.

Residual

Closure failure remains visible on the right-hand side.

Incremental gain

The representation improves explanation, diagnosis, transport, or prediction relative to simpler

alternatives.

14.5 Weak and strong Dirac claims

The article should distinguish two levels.

Weak Financial Dirac Claim

A first-order, two-component, frame-covariant equation governs action–ledger identity propagation:

$(i\not{\mathcal{D}}_{fin} - M_S)\Psi_S = \mathscr{R}_S.$ (14.6)

The slash notation $\not{\mathcal{D}}_{fin}$ denotes contraction of the financial derivatives with Γ operators.

Strong Financial Dirac Claim

In addition to equation (14.6):

1. Γ operators satisfy a meaningful Clifford-like algebra;
2. the squared first-order operator produces an interpretable second-order financial relation;
3. charge connections generate measurable curvature;
4. mass eigenmodes correspond to distinguishable closure behaviour;
5. the structure outperforms simpler coupled models.

The present article constructs the weak model and derives the requirements for the strong model.

It does not assume that the strong model is already empirically valid.

14.6 Closure coordinates

A physical Dirac equation is defined over spacetime.

The financial equation requires a different domain.

Let:

τ = ordered financial closure time. (14.7)

The coordinate τ may advance through:

- market observation ticks;
- margin decisions;
- collateral events;
- settlement events;
- accounting events;

- legal-status events.

Let:

s = action–ledger closure depth. (14.8)

The minimum convention is:

s = 0 for the outward action surface. (14.9)

s = 1 for the inward ledger-return surface. (14.10)

The margin account therefore occupies a minimum closure space:

X_fin := (τ,s). (14.11)

The coordinate s is not physical distance.

It indexes progression through the declared closure architecture.

14.7 Continuous and discrete closure space

Real financial systems contain both continuous and discrete structure.

Continuous approximation

When updates occur frequently:

ψ(τ,s) may be treated as a continuously evolving field. (14.12)

Discrete frame graph

When closure occurs through identifiable institutional systems:

s is better represented by graph nodes and edges. (14.13)

The minimum graph contains:

Action Node A ↔ Ledger Node L. (14.14)

A richer graph contains:

Market → Collateral → Treasury → Risk → Accounting → Legal. (14.15)

The Financial Gauge–Dirac equation should therefore admit both:

- a continuous closure-coordinate form;
- a discrete graph-Dirac form.

The margin calibration will use the two-node graph as its minimum implementation.

14.8 Financial propagation speed

The generalized Dirac source proposes c_P as the maximum rate at which a purpose-bearing

system can convert change into stable, reusable trace. Information may arrive faster, but coherent commitment and ledger integration remain bounded.

Define the financial closure capacity:

c_P := maximum coherent action-to-ledger propagation rate under protocol P. (14.16)

Possible bottlenecks include:

- collateral verification;
- settlement;
- treasury funding;
- risk ingestion;
- accounting recognition;
- legal validation;
- regulatory reporting.

A provisional bottleneck rule is:

$c_P = \min(c_{collateral}, c_{settlement}, c_{treasury}, c_{risk}, c_{accounting}, c_{legal})$. (14.17)

Equation (14.17) is a modelling hypothesis.

It states that the slowest mandatory closure channel limits coherent end-to-end financial propagation.

14.9 The financial closure cone

Let Δs measure required closure depth traversed during ledger interval $\Delta\tau$.

The coherent-closure condition is:

$|\Delta s| \le c_P \Delta\tau$. (14.18)

Equivalently, define closure velocity:

$v_{close} := |\Delta s|/\Delta\tau$. (14.19)

Then:

$v_{close} \le c_P$ implies admissible closure propagation. (14.20)

$v_{close} > c_P$ implies closure overload. (14.21)

In a margin account, overload may appear when:

- market value changes faster than collateral systems update;
- liquidation volume exceeds settlement capacity;
- margin calls exceed operational processing capacity;

- accounting recognition lags realized losses;
- legal validation lags collateral seizure;
- risk records lag trading activity.

The generalized source gives the finance-specific failure condition:

Trading Velocity > Risk-Ledger Velocity
→ Hidden Exposure. (14.22)

This is precisely the type of imbalance the present model attempts to formalize.

14.10 Financial cone residual

Define action velocity:

$v_A := \|\Delta\psi_A\|/\Delta\tau.$ (14.23)

Define ledger integration velocity:

$v_L := \|\Delta\psi_L\|/\Delta\tau.$ (14.24)

A simple velocity mismatch is:

$\delta v := v_A - v_L.$ (14.25)

Define cone residual:

$\mathscr{R}_{cone} := [v_A - c_P]_+.$ (14.26)

A broader spinor-flow residual is:

$\mathscr{R}_{flow} := [v_A - v_L]_+.$ (14.27)

When $\mathscr{R}_{flow}$ remains positive:

- unsettled actions accumulate;
- collateral exceptions accumulate;
- risk capture lags;
- reconciliation queues grow;
- hidden leverage may increase.

The cone condition is not a statement about physical relativity.

It is an operational capacity constraint.

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15. The Financial Γ Algebra

15.1 Why Γ operators are needed

The Γ operators should perform four tasks:

1. distinguish the action and ledger components;
2. couple them under propagation;
3. assign opposite closure orientation where appropriate;
4. prevent the equation from reducing to two unrelated scalar equations.

If Γ operators merely decorate the notation, the word Dirac should be removed.

15.2 Closure basis

Use the closure basis:

e_A :=
[
1
0
]. (15.1)

e_L :=
[
0
1
]. (15.2)

Then:

Ψ_S = ψ_Ae_A + ψ_Le_L. (15.3)

The projectors are:

P_A := (I2 + σ_z)/2. (15.4)

P_L := (I2 − σ_z)/2. (15.5)

Therefore:

P_AΨ_S =
[
ψ_A
0
]. (15.6)

P_LΨ_S =
[
0
ψ_L
]. (15.7)

15.3 Pauli operators in closure space

Define:

σ_x :=
[
0 1
1 0
]. (15.8)

σ_y :=
[
0 −i
i 0
]. (15.9)

σ_z :=
[
1 0
0 −1
]. (15.10)

Their provisional financial interpretations are:

σ_z — closure orientation

σ_z distinguishes outward action from inward ledger return.

σ_x — component exchange

σ_x couples action to ledger and ledger to action.

σ_y — phase-sensitive exchange

σ_y couples the components with relative complex orientation.

These interpretations are constructions of the present article.

They are not source-established empirical facts.

15.4 Minimum 1+1 Γ representation

Choose:

$\Gamma^0 := \sigma_z$. (15.11)

Choose:

$\Gamma^1 := i\sigma_y$. (15.12)

Explicitly:

$\Gamma^0 =$
[
1 0
0 −1
]. (15.13)

$\Gamma^1 =$

$$
\begin{bmatrix}
0 & 1 \\
-1 & 0
\end{bmatrix}. \qquad (15.14)
$$

Then:

$$(\Gamma^0)^2 = I_2. \qquad (15.15)$$

$$(\Gamma^1)^2 = -I_2. \qquad (15.16)$$

$$\Gamma^0\Gamma^1 + \Gamma^1\Gamma^0 = 0. \qquad (15.17)$$

Therefore:

$$\{\Gamma^a,\Gamma^b\} = 2\eta^{ab}I_2. \qquad (15.18)$$

with:

$$\eta^{ab} = \mathrm{diag}(1,-1). \qquad (15.19)$$

This is an exact algebraic statement for the chosen representation.

Its financial meaning remains a hypothesis.

15.5 Interpretation of the two directions

The positive-sign direction Γ^0 acts on ordered closure time τ.

The negative-sign direction Γ^1 acts on action–ledger transport depth s.

The sign difference does not mean that financial closure space is physical Minkowski spacetime.

It means that the model distinguishes:

- propagation through event order;
- exchange across closure surfaces.

The algebra forces the two directions to be structurally different.

15.6 The action–ledger exchange operator

The matrix:

$$\alpha_{fin} := \Gamma^0\Gamma^1. \qquad (15.20)$$

Using equations (15.13) and (15.14):

$$
\alpha_{fin} =
\begin{bmatrix}
0 & 1 \\
1 & 0
\end{bmatrix}. \qquad (15.21)
$$

Therefore:

$$\alpha_{fin} = \sigma_x. \qquad (15.22)$$

This is useful.

The financial velocity or transport operator becomes the component-exchange operator.

It maps:

$$\text{Action} \rightarrow \text{Ledger}. \qquad (15.23)$$

$$\text{Ledger} \rightarrow \text{Action}. \qquad (15.24)$$

This matches the recursive architecture:

- action must enter the ledger;
- the updated ledger constrains the next action.

15.7 The financial β operator

Define:

$$\beta_{fin} := \Gamma^0. \qquad (15.25)$$

Then:

$$\beta_{fin} = \sigma_z. \qquad (15.26)$$

This symbol β_{fin} must not be confused with CAPM beta.

To avoid collision, the article will normally use Γ^0 rather than β_{fin}.

The operator assigns opposite closure orientation to:

- outward action;
- inward ledger return.

15.8 The closure generator

Define the closure-rotation generator:

$$\Sigma_C := (i/2)\Gamma^0\Gamma^1. \qquad (15.27)$$

Since $\Gamma^0\Gamma^1 = \sigma_x$:

$$\Sigma_C = (i/2)\sigma_x. \qquad (15.28)$$

Alternatively, a unitary closure rotation may use:

$$S_C(\varphi) = \exp(-i\varphi\sigma_x/2). \qquad (15.29)$$

Then:

$S_C(2\pi) = -I_2.$ (15.30)

$S_C(4\pi) = I_2.$ (15.31)

This exact double-cover property belongs to the chosen mathematical representation.

Its interpretation as financial closure remains conditional.

Operational double closure is source-grounded; exact SU(2)-type closure is a stronger proposal.

15.9 Why the Γ algebra may matter

If the action and ledger components were governed by two independent scalar equations, then:

ψ_A and ψ_L could evolve without structural coupling. (15.32)

The Γ algebra instead enforces:

- opposite closure orientation;
- component exchange;
- first-order propagation;
- a specific second-order structure after squaring.

This gives the model falsifiable mathematical consequences.

The strong Dirac claim should survive only if those consequences prove useful.

15.10 Two-node graph difference

Define the expected action-to-ledger transporter:

$U_{AL} : \psi_A \to \psi\hat{}_L.$ (15.33)

Define the expected ledger-to-action transporter:

$U_{LA} : \psi_L \to \psi\hat{}_A.$ (15.34)

The covariant edge defects are:

$\delta_L := \psi_L - U_{AL}\psi_A.$ (15.35)

$\delta_A := \psi_A - U_{LA}\psi_L.$ (15.36)

Introduce a declared closure length ℓ_P.

Define the graph difference operator:

$$\mathfrak{D}_{G\Psi_S} := (1/\ell_P) \begin{bmatrix} \delta_A \\ \delta_L \end{bmatrix}$$

]. (15.37)

The quantity ℓ_P is not physical distance.

It normalizes the institutional separation between action and ledger surfaces.

15.11 Perfect graph closure

If both transports close exactly:

ψ_L = U_ALψ_A. (15.38)

ψ_A = U_LAψ_L. (15.39)

then:

𝔇_GΨ_S = 0. (15.40)

This is the flat two-node closure state.

During an open margin cycle:

𝔇_GΨ_S ≠ 0. (15.41)

The graph derivative therefore measures unresolved action–ledger displacement.

15.12 General frame-graph Dirac operator

For a larger financial frame graph, let f index nodes and e index directed edges.

Let B_G be the oriented incidence matrix of the graph.

Let U_e be the transport operator attached to edge e.

A covariant graph derivative may be written schematically:

𝔇_GΨ := B_G^UΨ. (15.42)

where B_G^U is the incidence matrix modified by edge transport.

For an edge e : A → B:

(B_G^UΨ)_e := Ψ_B − U_ABΨ_A. (15.43)

A graph-Dirac operator may then combine node and edge states:

𝒟_G :=
[
0 (B_G^U)†
B_G^U 0
]. (15.44)

Equation (15.44) is a new graph-theoretic extension.

The minimum two-component model is recovered when the graph contains only one action node

and one ledger node.

15.13 Γ admission test

The Γ terminology should be retained only if:

AdmitGamma
:= ComponentDistinction
∧ AnticommutationRole
∧ FirstOrderConstraint
∧ InterpretableSquare
∧ EmpiricalGain. (15.45)

If these conditions fail, use:

Coupling Matrix. (15.46)

or:

Action–Ledger Transition Operator. (15.47)

---

16. The Financial Mass Operator

16.1 Mass is not monetary size

Financial mass is not:

- market capitalization;
- asset value;
- debt balance;
- position quantity;
- accounting total.

The source Financial Standard Model defines mass as the cost of preserving identity while changing. Its closing formulation is:

Mass remembers the cost of remaining itself.

For the margin subject:

> Financial mass measures the resistance, cost, and institutional work required to change the account while preserving or validly converting its accountable identity.

16.2 Scalar identity mass

A simple scalar mass is:

$m_I \geq 0.$ (16.1)

A provisional operational definition is:

m_I
:= ResourceCost of Identity-Preserving Change
÷ Magnitude of Admissible State Change. (16.2)

Possible resource costs include:

- market impact;
- transaction fees;
- settlement effort;
- legal work;
- tax cost;
- funding friction;
- collateral substitution cost;
- accounting and audit work;
- regulatory approval.

The units depend on the declared normalization.

Mass cannot be compared across protocols unless the units and feature maps are aligned.

16.3 Why a matrix mass is needed

The action and ledger components may have different inertia.

For example:

- market action may change quickly;
- accounting recognition may change slowly;
- collateral transfer may be operationally fast;
- legal title perfection may be slow.

The mass term should therefore be allowed to act differently on each component and to bind them together.

Define:

M_S := m_I I_2 + m_Cσ_x + m_Δσ_z. (16.3)

Explicitly:

M_S =
[
m_I + m_Δ m_C
m_C m_I – m_Δ

]. (16.4)

where:

- m_I = common identity inertia;
- m_C = action–ledger binding mass;
- m_Δ = component asymmetry.

16.4 Interpretation of m_I

The term m_I acts equally on both components.

It represents the general resistance of the financial subject to identity-preserving change.

A highly standardized, liquid, legally simple position may have low m_I.

A concentrated, illiquid, legally complex, tax-sensitive position may have high m_I.

16.5 Interpretation of m_C

The term m_Cσ_x couples action and ledger.

A higher m_C means that a change in one component creates stronger required response in the other.

Examples include:

- real-time margining;
- automatic collateral calls;
- straight-through settlement;
- tightly integrated risk and accounting systems;
- immediate capital consequences.

A very low m_C allows:

Market Action ≫ Ledger Response. (16.5)

This may produce hidden exposure.

A very high m_C may produce excessive institutional rigidity:

Small Market Movement
→ Large Immediate Ledger Intervention. (16.6)

The healthy range is protocol-dependent.

16.6 Interpretation of m_Δ

The term m_Δσ_z records asymmetry between the two closure surfaces.

If:

m_Δ > 0, (16.7)

the action component carries greater effective inertia.

If:

m_Δ < 0, (16.8)

the ledger component carries greater effective inertia.

In many institutions:

|m_ledger| > |m_action|. (16.9)

Trades can be executed faster than legal, accounting, and regulatory recognition can be completed.

This asymmetry can be represented through m_Δ and c_P jointly.

16.7 State-dependent mass

Margin constraints create state-dependent coupling.

Therefore:

M_S = M_S(B,L,h,ν,$\mathscr{R}$). (16.10)

A candidate bounded coupling function is:

m_C(B)
= m_C,0 + m_C,1/[1 + exp(B/b0)]. (16.11)

where:

- m_C,0 = ordinary action–ledger coupling;
- m_C,1 = additional coupling near the margin boundary;
- b0 = buffer scale.

When:

B ≫ b0, (16.12)

the additional coupling is small.

When:

B ≈ 0, (16.13)

the coupling strengthens.

When:

$B < 0$, (16.14)

the system enters the high-coupling margin regime.

Equation (16.11) is an illustrative hypothesis, not a source-derived law.

16.8 Residual-dependent mass

Prior closure failures may increase future rigidity.

Define:

$m_{I,k+1} = m_{I,k} + \alpha_R \|\mathscr{R}_k\| - \alpha_C \mathrm{ClosureQuality}_k$. (16.15)

Possible interpretations include:

- repeated failures cause stricter haircuts;
- unresolved disputes cause legal friction;
- settlement failures cause operational restrictions;
- poor credit history increases funding constraints.

Thus the ledger can increase the future mass of the subject.

16.9 Mass eigenvalues

The eigenvalues of equation (16.4) are:

$m_\pm = m_I \pm \sqrt{(m_C^2 + m_\Delta^2)}$. (16.16)

The corresponding eigenvectors represent two combined closure modes.

Lower-mass mode

The lower-mass mode changes more easily.

It may correspond to a synchronized action–ledger mode.

Higher-mass mode

The higher-mass mode resists transformation.

It may correspond to an opposed or mismatched action–ledger mode.

This interpretation remains provisional.

The empirical question is whether the eigenmodes correspond to distinguishable closure behaviour.

16.10 Stability condition

To avoid a negative effective mass eigenvalue in the minimum real-mass model, require:

$m_I \geq \sqrt{(m_C^2 + m_\Delta^2)}.$ (16.17)

This is a modelling stability condition.

If equation (16.17) fails, the meaning of the negative mode must be declared rather than ignored.

It may indicate:

- unstable identity;
- mis-specified mass matrix;
- active amplification;
- missing damping;
- inappropriate normalization.

16.11 Mass and liquidity

Liquidity affects mass but is not identical to it.

A liquid asset may be easy to sell:

m_{market} low. (16.18)

Yet its legal or accounting closure may remain difficult:

m_{ledger} high. (16.19)

Conversely, an illiquid asset may have:

m_{market} high, (16.20)

while legal transfer is straightforward.

Therefore:

Financial Mass = Multi-Channel Identity Inertia. (16.21)

It cannot be reduced to bid–ask spread alone.

16.12 Mass and leverage

Leverage does not automatically increase identity mass.

It increases the sensitivity of the subject to changes and reduces its freedom near constraints.

A useful distinction is:

Leverage = amplification of response. (16.22)

Mass = resistance to identity-preserving change. (16.23)

A highly leveraged liquid position may respond strongly but be easy to liquidate.

A low-leverage illiquid position may respond less strongly but be costly to transform.

16.13 Mass and the margin boundary

Near the margin boundary:

- permitted actions narrow;
- timing becomes critical;
- collateral substitution becomes restricted;
- liquidation may become compulsory;
- cross-ledger consequences accelerate.

The account may therefore exhibit both:

Higher effective coupling

Market movement rapidly enters the ledger.

Lower discretionary freedom

The subject loses the ability to choose its own transformation path.

The mass operator should distinguish coupling strength from freedom of action.

A separate control or constraint operator may therefore be required in the full model.

16.14 Dissipation is not mass

Financial processes are non-unitary.

They contain:

- fees;
- loss;
- default;
- taxes;
- slippage;
- write-offs;
- irreversible legal change.

These should not all be hidden inside M_S.

Define a dissipation operator:

$\Lambda_S \geq 0$. (16.24)

A more complete propagation equation may contain:

$-i\Lambda_S\Psi_S$. (16.25)

Alternatively, dissipation may be recorded in $\mathscr{R}_S$.

The minimum article keeps M_S as identity inertia and places unmodelled loss and irreversibility in the residual.

A later extension may explicitly separate:

Mass Operator M_S. (16.26)

Dissipation Operator Λ_S. (16.27)

16.15 Empirical calibration of mass

Possible mass proxies include:

- time required to reduce the position by a fixed percentage;
- expected market-impact cost;
- settlement duration;
- collateral-substitution delay;
- legal-transfer cost;
- accounting restatement cost;
- number of mandatory approval gates;
- closure residual after standardized intervention.

A candidate effective mass score is:

m_{eff}
$:= w_1 ImpactCost$

- $w_2 SettlementDelay$
- $w_3 LegalFriction$
- $w_4 AccountingFriction$
- $w_5 CollateralRigidity$. (16.28)

The weights must be fixed before testing.

If mass cannot be measured more reliably than a generic friction score, the safer term is:

Identity Inertia Index. (16.29)

---

17. The Gauge-Covariant Financial Derivative

17.1 Charge operators

Let the subject carry the structural charge vector:

$\vec{q}_S = (q_A, q_F, q_C).$ (17.1)

For each field r, define a charge operator:

$\hat{Q}_r \Psi_S = q_r \Psi_S.$ (17.2)

In the minimum action–ledger model, both components carry the same underlying structural charge:

$\hat{Q}_r = q_r I_2.$ (17.3)

This expresses the fact that the action and ledger surfaces represent the same bounded financial subject.

17.2 Equal charge across closure surfaces

For ordinary frame transport:

$q_{r,A} = q_{r,L}.$ (17.4)

Therefore:

$[\hat{Q}_r, M_S] = 0$ (17.5)

for charge-preserving mass coupling.

This permits direct action–ledger binding.

If the two components carried different charges without an explicit conversion field, the off-diagonal mass term would violate charge consistency.

17.3 Charge conversion at a gate

When a valid gate transforms the identity:

$q_{r,out} \neq q_{r,in}.$ (17.6)

Then a conversion operator G_k must satisfy:

$\hat{Q}_{out} G_k - G_k \hat{Q}_{in} = J_{q,k}.$ (17.7)

where $J_{q,k}$ is the declared charge transferred, activated, extinguished, or reassigned by the gate.

The balance rule remains:

$\Sigma q_{in} + q_{gate} = \Sigma q_{out} + r_q.$ (17.8)

Margin-call activation is a simple example:

$q_C^{dormant} \to q_C^{operational}.$ (17.9)

17.4 Continuous local gauge transformation

Let the local financial frame transformation be:

$U(\chi) := \exp[i\Sigma_r \chi_r(\tau) \hat{Q}_r].$ (17.10)

The spinor transforms as:

$\Psi_S' = U(\chi)\Psi_S.$ (17.11)

Introduce financial connections:

$\mathcal{A}_{r,\tau}.$ (17.12)

Define coupling strengths:

$g_r.$ (17.13)

The gauge-covariant time derivative is:

$$D_\tau^G := \partial_\tau + i\Sigma_r g_r \mathcal{A}_{r,\tau} \hat{Q}_r.$$ (17.14)

For covariance:

$D_\tau^{G'}\Psi_S' = U(\chi) D_\tau^G \Psi_S.$ (17.15)

The connection transformation is:

$\mathcal{A}_{r,\tau}' = \mathcal{A}_{r,\tau} - (1/g_r)\partial_\tau \chi_r.$ (17.16)

Equations (17.10)–(17.16) are exact within the proposed Abelian gauge construction.

Their empirical interpretation is not yet established.

17.5 The relevant financial connections

For the minimum margin account:

$$D_\tau^G = \partial_\tau$$

- $ig_A \mathcal{A}_{A,\tau} \hat{Q}_A$
- $ig_F \mathcal{A}_{F,\tau} \hat{Q}_F$
- $ig_C \mathcal{A}_{C,\tau} \hat{Q}_C.$ (17.17)

Possible interpretations are:

Asset connection $\mathcal{A}_{A,\tau}$

- market-risk environment;
- benchmark mapping;
- valuation-frame orientation;
- price-to-collateral transport convention.

Funding connection $\mathcal{A}_{F,\tau}$

- funding curve;
- refinancing convention;
- maturity transformation;
- currency funding basis.

Collateral connection $\mathcal{A}_{C,\tau}$

- haircut protocol;
- eligibility rule;
- concentration adjustment;
- collateral substitution map.

The connections are protocol maps, not generic “market forces.”

17.6 CAPM phase remains dynamical

The CAPM valuation phase satisfies:

$$Z_M = nA\exp(i\theta_M). \qquad (17.18)$$

Therefore:

$$\partial_\tau Z_M = [\dot{A}/A + i\Omega_M + \dot{n}/n]Z_M. \qquad (17.19)$$

where:

$$\Omega_M := d\theta_M/d\tau. \qquad (17.20)$$

The CAPM phase velocity Ω_M is a dynamical state change.

It is not a pure gauge transformation.

The gauge connection $\mathcal{A}_{A,\tau}$ describes the frame rule under which the state is represented or transported.

Therefore:

$$\Omega_M \neq g_{Aq_A}\mathcal{A}_{A,\tau} \quad \text{in general.} \qquad (17.21)$$

A calibrated model may relate them locally, but they should not be identified by definition.

17.7 Full local derivative of the market component

Applying equation (17.14):

$$D_\tau^{GZ_M} = [\dot{A}/A + \dot{n}/n + i\Omega_M + i\Sigma_{rg_{rq_r}}\mathcal{A}_{r,\tau}]Z_M. \quad (17.22)$$

The terms have different meanings:

- $\dot{A}/A$ = amplitude growth or contraction;
- $\dot{n}/n$ = quantity change;
- Ω_M = real market-state phase motion;
- $g_{rq_r}\mathcal{A}_{r,\tau}$ = frame-covariant connection contribution.

This decomposition prevents a true market loss from being treated as a mere frame change.

17.8 Discrete gauge derivative

For an edge $A \to B$:

$$D_{AB}^{G\Psi} := \Psi_B - U_{AB}\Psi_A. \quad (17.23)$$

The edge transporter is:

$$U_{AB} := \rho_q(G_{AB})T_{AB}^0. \quad (17.24)$$

Under local transformations:

$$U_{AB'} = G_B U_{AB} G_A^{-1}. \quad (17.25)$$

Therefore:

$$D_{AB}^{\{G'\}\Psi'} = G_B D_{AB}^{G\Psi}. \quad (17.26)$$

The two-node graph operator $\mathfrak{D}_G$ in equation (15.37) is assembled from these discrete covariant differences.

17.9 Gauge curvature

In a continuous notation, define:

$$F_{ab} := (1/i)[D_a^G, D_b^G]. \quad (17.27)$$

For an Abelian field:

$$F_{ab} = \partial_a\mathcal{A}_b - \partial_b\mathcal{A}_a. \quad (17.28)$$

For a multi-charge system, include the corresponding coupling and charge operators.

In the frame graph, the curvature analogue is loop holonomy:

$$H_{loop} := \prod_{(A \to B \in loop)} U_{AB}. \qquad (17.29)$$

Flat transport satisfies:

$$H_{loop} \simeq I. \qquad (17.30)$$

Non-flat transport leaves:

$$\mathscr{R}_{loop} = \text{ObservedReturn} - \text{ExpectedReturn}. \qquad (17.31)$$

The continuous commutator and discrete loop residual are two representations of the same proposed idea:

> Transport order matters when the financial connection has curvature.

17.10 Curvature acting upon charge

A charged identity responds to curvature through:

$$Q\hat{}_{r} F_{ab}{}^{(r)} \Psi_S. \qquad (17.32)$$

Financially, this means that frame inconsistency matters only insofar as it acts upon a subject carrying the relevant relational orientation.

Examples include:

- a borrower exposed to inconsistent funding and collateral rules;
- a long asset holder exposed to market–accounting mismatch;
- a collateral provider exposed to legal–treasury disagreement;
- a derivative writer exposed to exercise–settlement inconsistency.

A frame loop with no relevant carrier may be operationally irrelevant to that subject.

17.11 Protocol-covariant derivative

Gauge covariance is only one part of financial admissibility.

The derivative must also respect:

- event timing;
- gate status;
- ledger accessibility;
- identity lineage;
- residual preservation.

Define the fuller protocol-covariant derivative:

$$\nabla_{\tau}{}^{P} := D_{\tau}{}^{G} + \Omega\hat{}_{gate} + \Omega\hat{}_{trace} + \Omega\hat{}_{ledger}. \qquad (17.33)$$

where:

- $\Omega\hat{}_{gate}$ = gate-status update structure;
- $\Omega\hat{}_{trace}$ = trace-preservation structure;
- $\Omega\hat{}_{ledger}$ = inherited-ledger constraint.

Equation (17.33) is schematic.

The central Financial Gauge–Dirac equation will use $\nabla_\tau\hat{}P$ when referring to the complete institutional process and $D_\tau\hat{}G$ when isolating gauge transport.

17.12 The financial cone in covariant form

Define the covariant closure displacement:

$$\Delta_P\Psi := \Psi(\tau + \Delta\tau) - \mathrm{Transport}_P[\Psi(\tau)]. \qquad (17.34)$$

The coherent propagation condition is:

$$\|\Delta_P\Psi\| \le c_P\Delta\tau. \qquad (17.35)$$

If:

$$\|\Delta_P\Psi\| > c_P\Delta\tau, \qquad (17.36)$$

define cone violation:

$$\mathscr{R}_{cone} := \|\Delta_P\Psi\| - c_P\Delta\tau. \qquad (17.37)$$

The excess becomes part of the right-hand-side residual.

---

18. The Central Financial Gauge–Dirac Equation

18.1 Compact form

Define the financial slash operator:

$$\not{D}_{fin} := \Gamma^0\nabla_\tau\hat{}P + c_P\Gamma^1\mathfrak{D}_G. \qquad (18.1)$$

The proposed continuous kernel is:

$$(i\not{D}_{fin} - M_S)\Psi_S = \mathscr{R}_S. \qquad (18.2)$$

Expanded:

$$[i\Gamma^0\nabla_\tau\hat{}P + ic_P\Gamma^1\mathfrak{D}_G - M_S]\Psi_S = \mathscr{R}_S. \qquad (18.3)$$

Equation (18.3) is the central Financial Gauge–Dirac equation.

It is a proposed construction.

It is not an established financial law.

18.2 Definition of the objects

The state is:

$$\Psi_S = \begin{bmatrix} \psi_A \\ \psi_L \end{bmatrix}. \qquad (18.4)$$

The Γ operators are:

$$\Gamma^0 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \qquad (18.5)$$

$$\Gamma^1 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}. \qquad (18.6)$$

The mass operator is:

$$M_S = \begin{bmatrix} m_I + m_\Delta & m_C \\ m_C & m_I - m_\Delta \end{bmatrix}. \qquad (18.7)$$

The graph derivative is:

$$\mathfrak{D}_G\Psi_S = (1/\ell_P) \begin{bmatrix} \psi_A - U_{LA}\psi_L \\ \psi_L - U_{AL}\psi_A \end{bmatrix}. \qquad (18.8)$$

The residual is:

$$\mathscr{R}_S = \begin{bmatrix} \mathscr{R}_A \\ \mathscr{R}_L \end{bmatrix}. \qquad (18.9)$$

18.3 Component form

Define:

$$\delta_A := (\psi_A - U_{LA}\psi_L)/\ell_P. \qquad (18.10)$$

Define:

$$\delta_L := (\psi_L - U_{AL}\psi_A)/\ell_P. \qquad (18.11)$$

Then equation (18.3) becomes:

$$i\nabla_\tau{}^P\psi_A + ic_P\delta_L - (m_I + m_\Delta)\psi_A - m_C\psi_L = \mathscr{R}_A. \qquad (18.12)$$

The ledger component is:

$$-i\nabla_\tau{}^P\psi_L - ic_P\delta_A - m_C\psi_A - (m_I - m_\Delta)\psi_L = \mathscr{R}_L. \qquad (18.13)$$

These two equations show what the compact notation means.

Action equation

The outward state changes through:

- protocol-covariant evolution;
- unresolved ledger mismatch;
- identity inertia;
- action–ledger binding;
- action residual.

Ledger equation

The inward state changes through:

- protocol-covariant ledger evolution;
- unresolved action mismatch;
- reciprocal binding;
- ledger inertia;
- ledger residual.

18.4 Interpretation of the action equation

Equation (18.12) may be read:

Action Change

- Ledger-Return Pressure
  – Identity Inertia
  – Closure Binding
  = Unresolved Action Consequence. (18.14)

In the margin example, $\mathscr{R}_A$ may include:

- unmodelled price impact;

- incomplete liquidation instruction;
- unexpected funding withdrawal;
- unauthorized action;
- omitted market response;
- model error.

18.5 Interpretation of the ledger equation

Equation (18.13) may be read:

Ledger Change

- Unintegrated Action Pressure
  – Identity Inertia
  – Reciprocal Binding
  = Unresolved Ledger Consequence. (18.15)

$\mathscr{R}$_L may include:

- unsettled quantity;
- missing collateral entry;
- unrecognized loss;
- legal-title dispute;
- reconciliation break;
- reporting delay;
- residual shortfall.

The generalized macro-Dirac source treats the residual as central rather than defective: macro systems normally live through residual management rather than exact zero-residual propagation.

18.6 Residual decomposition

Define:

$\mathscr{R}$_S
:= $\mathscr{R}$_value

- $\mathscr{R}$_charge
- $\mathscr{R}$_transport
- $\mathscr{R}$_gate
- $\mathscr{R}$_ledger

- $\mathscr{R}$_cone
- $\mathscr{R}$_model. (18.16)

where:

Valuation residual

Observed financial change not explained by the declared value dynamics.

Charge residual

Unreconciled rights or obligations.

Transport residual

Failure of lawful frame translation.

Gate residual

Difference between required and admitted gate consequence.

Ledger residual

Failure of expected recognition or settlement.

Cone residual

Change exceeding coherent closure capacity.

Model residual

Error caused by misspecification or omitted variables.

These components should be separately recorded where data permit.

18.7 The free-closure limit

Suppose:

- no margin constraint binds;
- frame transport is perfect;
- mass is constant;
- residual is zero.

Then:

$\mathscr{R}$_S = 0. (18.17)

and:

$(i\not{D}_{fin} - M_S)\Psi_S = 0.$ (18.18)

This is the ideal free-closure equation.

Real financial systems will rarely satisfy equation (18.18) exactly.

Its role is to define the zero-residual benchmark.

18.8 The synchronized limit

If:

$\psi_L = U_{AL}\psi_A,$ (18.19)

and:

$\psi_A = U_{LA}\psi_L,$ (18.20)

then:

$\mathfrak{D}_G\Psi_S = 0.$ (18.21)

The equation reduces to:

$(i\Gamma^0\nabla_\tau{}^P - M_S)\Psi_S = \mathscr{R}_S.$ (18.22)

This describes synchronized action and ledger evolution without cross-surface displacement.

18.9 The scalar reduction

If the two components remain functionally dependent:

$\psi_L = T\psi_A \quad$ for all admissible states, (18.23)

and the residual contains no independent ledger information, define one scalar state:

$x_S := \psi_A.$ (18.24)

Then the spinor model reduces to:

$dx_S/d\tau = f(x_S,u_S) + \varepsilon_S.$ (18.25)

The word spin should then be removed.

18.10 The no-gauge reduction

If no lawful frame connection is required:

$U_{AB} = I.$ (18.26)

Then:

$D_\tau{}^G = \partial_\tau.$ (18.27)

The model reduces to an ordinary two-component first-order system.

The word gauge should then be removed.

18.11 The no-mass reduction

If identity-preserving friction is negligible:

$$M_S \to 0. \qquad (18.28)$$

Then:

$$i\mathcal{D}_{fin}\Psi_S = \mathscr{R}_S. \qquad (18.29)$$

This describes freely propagating action–ledger change.

If this limit fits the data, the mass concept adds no value.

18.12 Squaring the first-order operator

The strong Dirac claim requires that the first-order equation imply an interpretable second-order relation.

Let:

$$\mathcal{D} := \mathcal{D}_{fin}. \qquad (18.30)$$

Starting from:

$$(i\mathcal{D} - M_S)\Psi_S = \mathscr{R}_S, \qquad (18.31)$$

left-multiply by:

$$(i\mathcal{D} + M_S). \qquad (18.32)$$

Then:

$$[\mathcal{D}^2 + M_S^2 - i[\mathcal{D},M_S]]\Psi_S$$
$$= -(i\mathcal{D} + M_S)\mathscr{R}_S. \qquad (18.33)$$

Equation (18.33) is exact algebraically under the declared operator ordering.

18.13 Expansion of the squared derivative

Using the Γ algebra:

$$\mathcal{D}^2$$
$$= (\nabla_\tau{}^P)^2$$
$$- c_P^2 \mathfrak{D}_G^2$$

- $(c_P/2)[\Gamma^0,\Gamma^1][\nabla_\tau{}^P,\mathfrak{D}_G]. \qquad (18.34)$

Define financial field curvature:

$\mathfrak{F}_\tau G := [\nabla_\tau \hat{P}, \mathfrak{D}_G]. \qquad (18.35)$

Then:

$$\mathcal{D}^2 = (\nabla_\tau \hat{P})^2 - c_P^2 \mathfrak{D}_G^2$$

- $(c_P/2)[\Gamma^0,\Gamma^1]\mathfrak{F}_\tau G. \qquad (18.36)$

Substituting into equation (18.33):

$$[(\nabla_\tau \hat{P})^2 - c_P^2 \mathfrak{D}_G^2$$

- M_S^2
- $(c_P/2)[\Gamma^0,\Gamma^1]\mathfrak{F}_\tau G - i[\mathcal{D}, M_S]]\Psi_S = -(i\mathcal{D} + M_S)\mathscr{R}_S. \qquad (18.37)$

Equation (18.37) is the proposed second-order financial closure relation.

18.14 Meaning of the second-order terms

$(\nabla_\tau \hat{P})^2 \Psi_S$

Closure acceleration through ordered financial time.

$-c_P^2 \mathfrak{D}_G^2 \Psi_S$

Restoring or dispersive pressure across action–ledger space.

$M_S^2 \Psi_S$

Squared identity inertia.

$[\Gamma^0,\Gamma^1]\mathfrak{F}_\tau G \Psi_S$

Interaction between closure orientation and financial curvature.

$-i[\mathcal{D}, M_S]\Psi_S$

Effect of changing identity mass.

Right-hand side

Propagation of residual into future closure dynamics.

This is where the strong Dirac model becomes empirically distinguishable from arbitrary two-component notation.

18.15 Constant-mass flat-connection limit

If:

$M_S = \text{constant}, \qquad (18.38)$

$\mathfrak{F}_{\tau G} = 0, \qquad (18.39)$

$\mathscr{R}_S = 0, \qquad (18.40)$

then equation (18.37) reduces to:

$[(\nabla_{\tau^P})^2 - c_P^2 \mathfrak{D}_G^2 + M_S^2]\Psi_S = 0. \qquad (18.41)$

This resembles a financial Klein–Gordon-type closure relation.

Its financial interpretation would be:

Closure Acceleration
– Cross-Surface Propagation

- Identity Inertia
  = 0. (18.42)

The resemblance is structural only.

18.16 Why the squared equation matters

If equation (18.41) produces no interpretable or testable financial consequence, the Clifford structure has not earned its place.

Possible empirical implications include:

- characteristic closure times;
- oscillatory action–ledger mismatch;
- overdamped or underdamped closure modes;
- mass-dependent propagation delay;
- curvature-sensitive residual growth;
- identifiable synchronized and opposed eigenmodes.

These must be tested against ordinary second-order state-space systems.

18.17 Accountable financial current

Define the financial adjoint:

$\Psi_S := \Psi_S^{\dagger} W \Gamma^0. \qquad (18.43)$

where W is a declared positive metric or unit-normalization operator.

Define the accountable current:

$J^a_S := \Psi_S \Gamma^a \Psi_S. \qquad (18.44)$

The temporal component is:

$J^0_S = \Psi_S^\dagger W \Psi_S.$ (18.45)

The closure-flow component is:

$J^1_S = \Psi_S^\dagger W \Gamma^0 \Gamma^1 \Psi_S.$ (18.46)

Since $\Gamma^0\Gamma^1 = \sigma_x$:

$J^1_S = \Psi_S^\dagger W \sigma_x \Psi_S.$ (18.47)

This term measures coupled action–ledger flow under the declared normalization.

The generalized source similarly proposes an accountable current and interprets healthy flow as co-propagation of action and ledger; high action with low ledger creates drift, while high ledger with low action creates paralysis.

18.18 Financial continuity equation

For a closed, constant-mass, Hermitian model, one may seek:

$\nabla_a J^a_S = 0.$ (18.48)

Real financial systems contain sources, sinks, and residual.

Therefore the more appropriate balance is:

$$\nabla_a J^a_S = \mathcal{S}_{capital} - \mathcal{D}_{loss} - \mathcal{D}_{fee} - \mathcal{D}_{default} + \mathcal{J}_{conversion} + \mathscr{R}_{current}.$$ (18.49)

where:

- $\mathcal{S}_{capital}$ = new capital or collateral source;
- $\mathcal{D}_{loss}$ = realized economic loss;
- $\mathcal{D}_{fee}$ = transaction and funding cost;
- $\mathcal{D}_{default}$ = extinguished or impaired value;
- $\mathcal{J}_{conversion}$ = valid identity conversion;
- $\mathscr{R}_{current}$ = unexplained flow mismatch.

Equation (18.49) is a proposed financial current-balance law.

18.19 Current imbalance diagnostics

Define action current:

$J_A := \psi_A^\dagger W_A \psi_A.$ (18.50)

Define ledger current:

$J_L := \psi_L^\dagger W_L \psi_L.$ (18.51)

Define current imbalance:

$\Delta J := J_A - J_L.$ (18.52)

Possible interpretations are:

$\Delta J \gg 0$
$\rightarrow$ action outruns ledger. (18.53)

$\Delta J \ll 0$
$\rightarrow$ ledger rigidity exceeds productive action. (18.54)

$\Delta J \approx 0$
$\rightarrow$ synchronized flow, subject to residual tests. (18.55)

A zero current imbalance does not guarantee closure.

Both currents may be equally wrong or equally stale.

Therefore:

$\Delta J \approx 0 \nRightarrow \mathrm{SpinClosed}_P = 1.$ (18.56)

18.20 The complete continuous kernel

The most explicit continuous system developed so far is:

$$[i\Gamma^0(\partial_\tau + i\Sigma_{rg} r\mathcal{A}_{r,\tau}\hat{Q}_r + \hat{\Omega}_{\mathrm{gate}} + \hat{\Omega}_{\mathrm{trace}} + \hat{\Omega}_{\mathrm{ledger}})$$

- $$ic_P\Gamma^1\mathfrak{D}_G - M_S(B,L)]\Psi_S = \mathscr{R}_S.$$ (18.57)

For the minimum margin account:

$r \in \{\mathrm{Asset},\mathrm{Funding},\mathrm{Collateral}\}.$ (18.58)

The equation contains:

- market and financial time evolution;
- charge-sensitive frame connection;
- gate status;
- trace preservation;
- inherited ledger constraints;

- action–ledger transport;
- identity mass;
- explicit residual.

Equation (18.57) is the fullest continuous Financial Gauge–Dirac kernel of the present article.

18.21 Why equation (18.57) is still incomplete

Margin finance contains discontinuous events.

At:

$$B \le 0, \qquad (18.59)$$

the system may jump through:

- call issuance;
- collateral posting;
- voluntary deleveraging;
- forced liquidation;
- default conversion.

A continuous equation cannot by itself determine all such branch changes.

Therefore, equation (18.57) must be embedded inside a hybrid system.

The next Part will add:

- the margin guard;
- the branch operator;
- charge-conversion rules;
- ledger update;
- forced-liquidation feedback;
- recursive effective-charge update.

18.22 Scientific status of Part VII

The following are source-grounded structural ideas:

- action–ledger spinor;
- first-order identity propagation;
- frame-fixedness operators;

- identity-preserving mass;
- finite coherent closure rate;
- residual as a central diagnostic;
- accountable current.

The following are new constructions of this article:

- the market–collateral financial spinor;
- the explicit Γ^0 and Γ^1 representation;
- the two-node covariant graph derivative;
- the financial charge operators;
- the matrix mass M_S;
- the margin-state-dependent mass function;
- the squared financial closure equation;
- the financial current-balance equation.

These proposals must be reduced or removed if simpler models perform equally well.

18.23 Result of Part VII

The Financial Gauge–Dirac equation can now be stated in its compact form:

$(i\not{D}_{fin} - M_S)\Psi_S = \mathscr{R}_S.$ (18.60)

Its financial meaning is:

> A charged action–ledger identity propagates through ordered financial time and governed frame transport, constrained by identity inertia and finite closure capacity, while every failure of valuation, transport, gate, ledger, charge, or model closure remains visible as residual.

The equation becomes genuinely Dirac-like only when:

- the doublet is irreducible;
- Γ operators constrain the dynamics;
- the first-order structure matters;
- the squared equation yields interpretable modes;
- charge transport is covariant;
- mass is measurable;
- residual improves diagnosis;
- the full system defeats simpler benchmarks.

The next Part turns this continuous kernel into the complete **Financial Gauge–Dirac–Gate–Ledger system**.

Part VIII — Gate Dynamics and Recursive Closure

19. The Hybrid Margin Jump

19.1 Why the continuous equation is not enough

The continuous Financial Gauge–Dirac kernel is:

$$(i\not{D}_{fin} - M_S)\Psi_S = \mathscr{R}_S. \qquad (19.1)$$

This equation describes propagation while the financial subject remains inside one declared dynamical regime.

It can represent:

- market-phase movement;
- collateral-phase movement;
- funding accrual;
- action–ledger transport;
- frame mismatch;
- state-dependent identity inertia;
- residual accumulation.

A margin account, however, does not evolve entirely through smooth propagation.

When the account reaches a contractual boundary, its permitted actions change discontinuously.

The subject may move from:

- ordinary trading to restricted trading;
- dormant collateral obligation to active call;
- voluntary control to broker-directed liquidation;
- performing account to defaulted recovery process.

These are changes of status, authority, and admissible transition—not merely large continuous movements.

The complete system must therefore distinguish:

Flow

Continuous or quasi-continuous evolution inside one regime.

Guard

A declared condition that determines whether the current regime remains admissible.

Gate

An authorized decision that promotes the guard condition into a consequential event.

Jump

A discontinuous state transition produced by the admitted event.

Ledger return

The process that makes the jump accountable and updates the next admissible state.

The Financial Standard Model source treats this distinction as fundamental: a gate decides what becomes history, trace records what was admitted, residual preserves what closure failed to contain, and the ledger determines what the next state inherits.

19.2 The hybrid state

Define the full pre-gate state:

$$\mathcal{S}^{-}_{k} := (K^{-}_{k}, \Psi^{-}_{k}, X^{-}_{k}, \vec{q}^{-}_{k}, L^{-}_{k}, \mathscr{R}^{-}_{k}). \qquad (19.2)$$

where:

- K^{-}_{k} = identity kernel;
- Ψ^{-}_{k} = action–ledger spinor;
- X^{-}_{k} = balance-sheet and margin variables;
- $\vec{q}^{-}_{k}$ = structural and effective charges;
- L^{-}_{k} = inherited ledger;
- $\mathscr{R}^{-}_{k}$ = accumulated residual.

The ordinary financial variables are:

$$X_k := (n_k, R_k, Q_k, D_k, C_k, h_k, B_k). \qquad (19.3)$$

The margin buffer is:

$$B_k = C_k + n_k h_k R_k - D_k. \qquad (19.4)$$

The gate guard is:

$$g_margin(\mathcal{S}_k) := B_target,_k - B_k. \qquad (19.5)$$

The safe regime satisfies:

$$g_margin(\mathcal{S}_k) \le 0. \qquad (19.6)$$

A candidate breach satisfies:

$g_margin(\mathcal{S}_k) > 0.$ (19.7)

This condition identifies a threshold crossing.

It does not yet prove that an authoritative margin event has occurred.

19.3 The gate-admission function

Define the margin-admission function:

$a_k := Admit_margin(\mathcal{S}^{-}_k, P_k, A_k).$ (19.8)

where:

- P_k = governing margin protocol;
- A_k = authorized decision-maker or automated authority;
- a_k = admission result.

Let:

$a_k \in \{Reject, Observe, Call, Restrict, Liquidate, Default\}.$ (19.9)

A valid call requires:

$$\begin{aligned} Call_k \\ &:= Breach_k \\ &\wedge ValidAuthority_k \\ &\wedge ApplicableRule_k \\ &\wedge ValidMark_k \\ &\wedge TraceCreated_k. \end{aligned}$$ (19.10)

Thus:

$B_k < B_target,_k \nRightarrow ValidCall_k.$ (19.11)

A numerical shortfall may be:

- based on a stale mark;
- calculated under the wrong agreement;
- disputed;
- waived;
- subject to a cure convention;
- not yet admitted by the authorized system.

The Periodic Grammar explicitly distinguishes a numerical threshold from a committed event: promotion requires a gate, authority, trace, residual, and ledger consequence.

19.4 The margin jump operator

Once admitted, the gate acts through a jump operator:

$\mathcal{S}^{+}_{k} = \mathcal{J}_margin^{(a_k)}(\mathcal{S}^{-}_{k}, L^{-}_{k}) + \eta_k.$ (19.12)

The jump may change:

- gate status;
- operational charge;
- permissible actions;
- collateral availability;
- position quantity;
- debt balance;
- legal status;
- ledger obligations.

The term η_k records jump residual or exogenous intervention.

The spinor jump is:

$\Psi^{+}_{k} = G_margin^{(a_k)}\Psi^{-}_{k} + \eta_\Psi,_k.$ (19.13)

The balance-sheet jump is:

$X^{+}_{k} = J_X^{(a_k)}(X^{-}_{k}) + \eta_X,_k.$ (19.14)

The charge jump is:

$\vec{q}^{+}_{k} = J_q^{(a_k)}(\vec{q}^{-}_{k}) + \vec{r}_q,_k.$ (19.15)

The identity update is:

$K^{+}_{k} = J_K^{(a_k)}(K^{-}_{k}) + r_K,_k.$ (19.16)

Not every branch changes every component.

A call may change operational charge and status while leaving asset quantity unchanged.

A liquidation changes quantity, debt, value, and ledger state.

A default may change the identity class itself.

19.5 Pre-jump and post-jump notation

For a gate occurring at ledger time τ_k:

$\Psi(\tau_k^{-})$:= state immediately before admission. (19.17)

$\Psi(\tau_k^{+})$:= state immediately after the jump. (19.18)

The jump magnitude is:

$\Delta\Psi_k\text{^jump} := \Psi(\tau_k^+) - \Psi(\tau_k^-).$ (19.19)

For an instantaneous idealization:

$d\Psi/d\tau$ contains an impulse at τ_k. (19.20)

A distributional representation is:

$(i\mathcal{D}_\text{fin} - M_S)\Psi$
$= \mathscr{R}_\text{cont} + \Sigma_k \delta(\tau - \tau_k) J_k \Psi.$ (19.21)

Here:

- $\mathscr{R}_\text{cont}$ = continuous residual;
- $\delta(\cdot)$ = Dirac delta distribution;
- J_k = gate-jump generator.

Equation (19.21) uses a mathematical impulse representation.

It does not imply that the financial event is a physical particle interaction.

19.6 Flow–jump decomposition

Between gates:

$\tau \in (\tau_k, \tau_{k+1}).$ (19.22)

The subject obeys:

$(i\mathcal{D}_\text{fin} - M_S)\Psi = \mathscr{R}_\text{cont}.$ (19.23)

At the gate:

$\Psi(\tau_k^+) = G_k \Psi(\tau_k^-) + \eta_k.$ (19.24)

After the gate:

$L_{k+1} = U_L(L_k, \text{Trace}_k, \text{Residual}_k).$ (19.25)

The next continuous regime begins from:

$\Psi_{k+1}\text{^initial} = \text{Init}(\Psi(\tau_k^+), L_{k+1}).$ (19.26)

This gives the hybrid architecture:

Continuous Flow
→ Guard
→ Gate
→ Jump
→ Ledger Update
→ New Continuous Flow. (19.27)

19.7 The call branch

If the authority issues a margin call:

$a_k = \text{Call}.$ (19.28)

The call amount is:

$C_call,_k = [B_target,_k - B_k]_+.$ (19.29)

The collateral obligation activates:

$q_C\hat{}op,^+ = q_C.$ (19.30)

The call status becomes:

$\nu_call,^+ = \text{Open}.$ (19.31)

The closure sign changes:

$s_C : +1 \to -1.$ (19.32)

The outward action component updates:

$\psi_A^+ = G_call\psi_A^-.$ (19.33)

The ledger component may initially remain:

$\psi_L^+ \approx \psi_L^-.$ (19.34)

Therefore the spinor split increases:

$\Delta_spin,G^+ > \Delta_spin,G^-.$ (19.35)

This is expected.

The call deliberately creates an open obligation.

A high post-call spinor split is not automatically a system error.

Failure arises when the split persists beyond the permitted closure interval or becomes hidden.

19.8 Cure deadline and temporal gate

A margin call usually includes a cure deadline.

Let:

$t_due,_k = \text{call deadline}.$ (19.36)

Define time remaining:

$\Delta t_due := t_due,_k - t.$ (19.37)

The temporal guard is:

$g_time := -\Delta t_due.$ (19.38)

Before deadline:

$g_time < 0.$ (19.39)

At or after deadline:

$g_time \geq 0.$ (19.40)

The account may therefore face two linked guards:

Amount guard

$g_amount := B_target - B.$ (19.41)

Time guard

$g_time := t - t_due.$ (19.42)

Forced action may require:

$g_amount > 0 \wedge g_time \geq 0.$ (19.43)

A contractual protocol may permit immediate liquidation, in which case the time guard differs.

The rule must be declared rather than inferred retrospectively.

19.9 Collateral-posting jump

Suppose accepted collateral ΔC_admit is posted.

Then:

$C^{+} = C^{-} + \Delta C_admit.$ (19.44)

The buffer becomes:

$B^{+} = B^{-} + \Delta C_admit.$ (19.45)

The amount residual is:

$\varepsilon_amount = C_call - \Delta C_admit.$ (19.46)

The gate may close if:

$B^{+} \geq B_target.$ (19.47)

But ledger closure additionally requires:

- collateral receipt confirmed;
- legal control established;
- eligibility validated;
- treasury record updated;
- collateral ledger updated;
- duplicate pledge excluded.

Define collateral closure indicator:

c_collateral
:= 1[Amount ∧ Eligibility ∧ Control ∧ Ledger ∧ Trace]. (19.48)

The call status becomes resolved only when:

c_collateral = 1. (19.49)

Thus:

Cash Sent ⇏ Collateral Admitted. (19.50)

and:

Collateral Admitted ⇏ Entire Account Closed. (19.51)

19.10 Voluntary-deleveraging jump

Suppose the subject sells Δn units.

The position update is:

$n^+ = n^- - \Delta n.$ (19.52)

Gross proceeds are:

$P_{gross} = p_{exec}\Delta n.$ (19.53)

Net proceeds are:

$P_{net} = p_{exec}\Delta n - Fees - Tax - Slippage.$ (19.54)

If net proceeds repay debt:

$D^+ = D^- - P_{net}.$ (19.55)

The asset charge changes:

$q_{A^+} = q_{A^-} - \Delta n q_0(x).$ (19.56)

The funding charge changes according to the debt reduction:

$q_{F^+} = ChargeMap_F(D^+).$ (19.57)

The new buffer is:

$B^+ = C^+ + n^+ h^+ R^+ - D^+.$ (19.58)

A sale instruction does not immediately imply equation (19.55).

Before settlement, the system may carry:

- delivery obligation;
- receivable;

- counterparty exposure;
- unsettled cash;
- replacement-cost risk.

Therefore the deleveraging jump opens a nested settlement cycle.

19.11 Forced-liquidation jump

When the protocol authorizes liquidation:

a_k = Liquidate. (19.59)

The broker chooses liquidation quantity:

ℓ_k = LiquidationRule($\mathcal{S}^-_k$,P_k,MarketDepth_k). (19.60)

The quantity jump is:

$n^+ = n^- - \ell_k$. (19.61)

The debt update is:

$D^+ = D^- - P_net,liq.$ (19.62)

The realized execution residual is:

ε_exec
$:= \ell_k R_pre - P_net,liq.$ (19.63)

The subject's control authority changes:

$\text{Authority_subject} \downarrow$. (19.64)

$\text{Authority_broker} \uparrow$. (19.65)

This authority transfer is part of the identity change.

It is not represented by price alone.

19.12 Default jump

Default occurs when the account cannot satisfy the permitted cure process.

Define the default guard:

$g_default$
$:= \text{ShortfallRemaining}$
$\wedge \text{CureExpired}$
$\wedge \text{RecoveryAuthorityActivated}.$ (19.66)

When admitted:

a_k = Default. (19.67)

The identity kernel transforms:

K_margin → K_recovery. (19.68)

The post-default charge set may include:

q_deficiency. (19.69)

q_collateral-enforcement. (19.70)

q_guarantor. (19.71)

q_litigation. (19.72)

q_extinguished. (19.73)

The charge balance is:

q_margin,in + q_gate
= q_recovery,out + q_extinguished + r_q. (19.74)

A valid default jump must preserve transaction lineage even when the original account identity terminates.

19.13 Jump commutation with charge

For an ordinary charge-preserving jump G_k:

$Q\hat{}_outG_k = G_kQ\hat{}_in.$ (19.75)

For a charge-converting gate:

$Q\hat{}_outG_k - G_kQ\hat{}_in = J_q,_k.$ (19.76)

The conversion current $J_q,_k$ must be traceable to:

- contract;
- settlement;
- exercise;
- liquidation;
- default;
- novation;
- legal order.

If no conversion authority exists:

$J_q,_k \neq 0$ indicates charge failure. (19.77)

19.14 Jump covariance

A valid gate should produce equivalent outcomes under lawful changes of financial representation.

Let:

$\Psi' = U_frame\Psi.$ (19.78)

The gate is gauge-covariant when:

$G_k' = U_frame,out G_k U_frame,in^{-1}.$ (19.79)

Then:

$G_k'\Psi'^{-} = U_frame,out(G_k\Psi^{-}).$ (19.80)

Financially, this means that changing:

- reporting currency;
- account presentation;
- unit scale;
- permissible aggregation;

should not change whether the same legally defined margin event occurred, once the protocol is translated correctly.

19.15 Jump residual

Define the expected post-jump state:

$\mathcal{S}^{+}_k := \mathcal{J}_margin\hat{}(a_k)(\mathcal{S}^{-}_k, L^{-}_k).$ (19.81)

The observed post-jump state is:

$\mathcal{S}^{+}_k,obs.$ (19.82)

The jump residual is:

$\mathcal{R}_jump,_k := \mathcal{S}^{+}_k,obs - \mathcal{S}^{+}_k.$ (19.83)

Possible sources include:

- execution slippage;
- rejected collateral;
- incorrect liquidation quantity;
- delayed settlement;
- unrecorded fee;
- disputed authority;
- stale debt balance;

- missing accounting entry;
- omitted legal claim.

The jump residual should not be merged automatically with continuous model error.

Its source is a specific admitted transition.

19.16 The ledger-update operator

After the jump, define:

$$L_{k+1} = U_L(L_k,T_k,\mathscr{R}_k,\text{Authority}_k,\text{RevisionRule}_k). \qquad (19.84)$$

The ledger update should record:

- pre-gate state;
- gate condition;
- authority;
- branch selected;
- charge flows;
- economic consequence;
- post-gate identity;
- unresolved residual;
- closure status.

A minimum ledger tuple is:

$$L_{k+1} := (\text{EventID},\text{SubjectID},\text{Gate},\text{Authority},\text{StateBefore},\text{StateAfter},\text{ChargeFlow},\text{Trace},\text{Residual},\text{Status}). \qquad (19.85)$$

The ledger is not merely an archive.

It constrains the next state.

19.17 The ledger as the next-period boundary condition

Let Θ_k denote the parameters governing future evolution:

$$\Theta_k := (\beta_k,h_k,g_k,M_k,c_P,_k,\text{Limits}_k,\text{Permissions}_k). \qquad (19.86)$$

The post-event ledger updates them:

$$\Theta_{k+1} = U_\Theta(\Theta_k,L_{k+1},\mathscr{R}_k). \qquad (19.87)$$

Examples include:

- a margin failure reduces collateral eligibility;
- a settlement failure raises operational mass;
- a large loss reduces risk limits;
- a default ends trading permission;
- a successful cure may still reduce liquidity capacity;
- a disputed call may increase legal friction.

Thus:

Ledger Update → New Dynamical Law. (19.88)

The next period does not evolve under the same effective environment as the previous one.

19.18 The hybrid Financial Gauge–Dirac system

The complete system can now be written:

Continuous regime

$$(i\mathcal{D}\!\!\!/_{\mathrm{fin}}[\Theta_k] - M_S[\Theta_k])\Psi = \mathscr{R}_{\mathrm{cont}}, \quad \tau \notin \{\tau_k\}. \tag{19.89}$$

Guard

$$g_j(\mathcal{S}) \geq 0 \quad \text{for some gate } j. \tag{19.90}$$

Gate admission

$$a_k = \mathrm{Admit}_j(\mathcal{S}^-_k, P_k, A_k). \tag{19.91}$$

Jump

$$\mathcal{S}^+_k = \mathcal{J}_{j^\wedge(a_k)}(\mathcal{S}^-_k, L_k) + \mathscr{R}_{\mathrm{jump},k}. \tag{19.92}$$

Ledger update

$$L_{k+1} = U_L(L_k, T_k, \mathscr{R}_k). \tag{19.93}$$

Parameter recursion

$$\Theta_{k+1} = U_\Theta(\Theta_k, L_{k+1}, \mathscr{R}_k). \tag{19.94}$$

Equations (19.89)–(19.94) form the full hybrid **Financial Gauge–Dirac–Gate–Ledger system**.

---

20. Forced-Liquidation Nonlinearity

20.1 Why forced liquidation is the decisive nonlinear branch

The basic CAPM return relation is linear in β and ERP.

The discounted value relation is smooth but nonlinear.

The margin gate adds a piecewise constraint.

Forced liquidation adds a stronger form of nonlinearity because the account's required response affects the market variable that generated the response.

The sequence becomes:

Market Decline
→ Buffer Breach
→ Liquidation
→ Price Impact
→ Further Buffer Decline. (20.1)

This is endogenous feedback.

The financial subject no longer responds passively to an external field.

Its gate-triggered action changes the field.

20.2 Liquidation demand

Let the margin deficiency be:

$S_B := [B_{target} - B]_+$. (20.2)

A minimum liquidation rule is:

$\ell = \kappa_{LS} B$. (20.3)

where κ_L converts monetary shortfall into units to be sold.

A more realistic rule is:

$\ell = \min\{n, \Phi_L(S_B, p_{depth}, h, Volatility, Deadline)\}$. (20.4)

The rule may depend on:

- shortfall;
- current price;
- market depth;
- collateral factor;
- volatility;
- cure deadline;

- permitted liquidation set.

The liquidation quantity is therefore state-dependent.

20.3 Price-impact law

Let the pre-liquidation CAPM value evolution be:

dR_CAPM/dτ. (20.5)

Let liquidation cause impact:

I(ℓ,Depth,Volatility). (20.6)

Then:

dR_M/dτ
= dR_CAPM/dτ − I(ℓ,Depth,Volatility). (20.7)

A simple linear-impact approximation is:

I = κ_Iℓ. (20.8)

A concave impact model may be:

I = κ_I√ℓ. (20.9)

A nonlinear depth-adjusted model may be:

I = κ_Iℓ^α/Depth^γ. (20.10)

where:

0 < α ≤ 1. (20.11)

γ > 0. (20.12)

The article does not select one universal impact function.

The mechanism should be estimated from domain data.

20.4 Position dynamics

Under liquidation:

dn/dτ = −ℓ. (20.13)

The risky-asset value is:

V = nR_M. (20.14)

Therefore:

dV/dτ = n dR_M/dτ + R_M dn/dτ. (20.15)

Substituting equations (20.7) and (20.13):

$$dV/d\tau = n[dR_{CAPM}/d\tau - I(\ell)] - R_M\ell. \qquad (20.16)$$

The first liquidation effect is price impact on remaining units.

The second is removal of sold units.

20.5 Debt dynamics

Let p_{exec} be net executable price per unit after fees.

Then:

$$dD/d\tau = -p_{exec}\ell + dD_{accrual}/d\tau + dD_{fee}/d\tau. \qquad (20.17)$$

If funding accrues at rate r_F:

$$dD_{accrual}/d\tau = r_F D. \qquad (20.18)$$

If liquidation fees are proportional to quantity:

$$dD_{fee}/d\tau = \kappa_{fee}\ell. \qquad (20.19)$$

Thus:

$$dD/d\tau = -p_{exec}\ell + r_F D + \kappa_{fee}\ell. \qquad (20.20)$$

Liquidation reduces debt only to the extent that net proceeds exceed costs and contemporaneous accrual.

20.6 Haircut dynamics

Collateral eligibility may deteriorate during stress.

Let:

$$dh/d\tau = -H(\text{Volatility},\text{Liquidity},\text{Concentration},\text{DefaultRisk}). \qquad (20.21)$$

A simple specification is:

$$dh/d\tau = -\kappa_{h,\sigma}\sigma - \kappa_{h,\text{LLiqStress}} - \kappa_{h,\text{CC concentration}}. \qquad (20.22)$$

The exact function is protocol-dependent.

The important point is:

Market stress can reduce both R_M and h. (20.23)

This creates a double contraction of admitted collateral.

20.7 Buffer dynamics under liquidation

Recall:

$$B = C + nhR_M - D. \qquad (20.24)$$

Differentiate:

$$\frac{dB}{d\tau} = \frac{dC}{d\tau}$$

- $nh\,\frac{dR_M}{d\tau}$
- $nR_M\,\frac{dh}{d\tau}$
- $hR_M\,\frac{dn}{d\tau} - \frac{dD}{d\tau}. \qquad (20.25)$

Substitute equations (20.7), (20.13), and (20.20):

$$\frac{dB}{d\tau} = \frac{dC}{d\tau}$$

- $nh\left[\frac{dR_{CAPM}}{d\tau} - I(\ell)\right]$
- $nR_M\,\frac{dh}{d\tau} - hR_M\ell$
- $p_{exec}\ell - r_{FD} - \kappa_{fee}\ell. \qquad (20.26)$

Collect liquidation terms:

$$\frac{dB}{d\tau} = \frac{dC}{d\tau}$$

- $nh\,\frac{dR_{CAPM}}{d\tau}$
- $nR_M\,\frac{dh}{d\tau} - r_{FD}$
- $\ell[p_{exec} - hR_M - \kappa_{fee}] - nhI(\ell). \qquad (20.27)$

Equation (20.27) is the central nonlinear buffer equation.

20.8 The liquidation-benefit condition

Ignoring price impact momentarily, liquidation improves the margin buffer if:

$$p_{exec} - hR_M - \kappa_{fee} > 0. \qquad (20.28)$$

Thus:

$$p_{exec} > hR_M + \kappa_{fee}. \qquad (20.29)$$

This condition is intuitive.

Selling one unit removes collateral value hR_M but generates net debt repayment approximately p_exec − κ_fee.

If net proceeds exceed the collateral value removed, the buffer improves.

If:

p_exec < hR_M + κ_fee, (20.30)

liquidation may reduce the buffer before considering other effects.

20.9 Impact-adjusted cure condition

Including price impact, liquidation improves the buffer only if:

ℓ[p_exec − hR_M − κ_fee]

nhI(ℓ)
− dC/dτ
− nh dR_CAPM/dτ
− nR_M dh/dτ

- r_FD. (20.31)

During severe stress:

dR_CAPM/dτ < 0. (20.32)

dh/dτ < 0. (20.33)

I(ℓ) > 0. (20.34)

The right-hand side can become large.

The amount of liquidation needed to cure the account may therefore rise as liquidation proceeds.

20.10 Self-defeating liquidation

Define the marginal buffer effect:

MBE(ℓ) := ∂(dB/dτ)/∂ℓ. (20.35)

From equation (20.27):

MBE(ℓ)
= p_exec − hR_M − κ_fee − nhI′(ℓ) + Terms_from_p_exec(ℓ). (20.36)

Liquidation becomes locally self-defeating when:

MBE(ℓ) < 0. (20.37)

This may occur when:

- market depth is low;
- the position is concentrated;
- executable price falls quickly;
- remaining collateral value is heavily impacted;
- liquidation fees are high;
- haircuts worsen simultaneously.

Then:

More Liquidation
→ Less Margin Improvement. (20.38)

20.11 The fire-sale fixed point

A cure quantity ℓ^* satisfies:

$$B_{after}(\ell^*) = B_{target}. \quad (20.39)$$

If no solution exists for:

$$0 \le \ell^* \le n, \quad (20.40)$$

then the account cannot cure through liquidation alone.

Define the cure function:

$$F_{cure}(\ell) := C + [n - \ell]h(\ell)R_M(\ell) - [D - P_{net}(\ell)] - B_{target}. \quad (20.41)$$

A cure exists when:

$$\exists \ell \in [0,n] \text{ such that } F_{cure}(\ell) \ge 0. \quad (20.42)$$

No liquidation cure exists when:

$$F_{cure}(\ell) < 0 \quad \text{for all } \ell \in [0,n]. \quad (20.43)$$

This gives a precise transition from liquidation branch to default branch.

20.12 Local stability of the margin boundary

Let the deficiency be:

$$x := B_{target} - B. \quad (20.44)$$

Then:

$$x > 0 \quad \text{indicates breach.} \quad (20.45)$$

Suppose liquidation responds:

$$\ell = \kappa_L[x]_+. \qquad (20.46)$$

The deficiency dynamics are:

$$dx/d\tau = -dB/d\tau. \qquad (20.47)$$

A locally stable cure requires:

$$\partial(dx/d\tau)/\partial x < 0 \quad \text{near } x = 0^+. \qquad (20.48)$$

If:

$$\partial(dx/d\tau)/\partial x > 0, \qquad (20.49)$$

small breaches grow rather than shrink.

The margin boundary then becomes dynamically unstable.

This is a more meaningful crisis condition than merely observing:

$$B < 0. \qquad (20.50)$$

20.13 Constraint-induced phase acceleration

The market complex state is:

$$Z_M = nA\exp(i\theta_M). \qquad (20.51)$$

Liquidation may increase phase velocity through impact:

$$d\theta_M/d\tau = \Omega_{exogenous} + \Omega_{impact}(\ell). \qquad (20.52)$$

If:

$$\Omega_{impact}'(\ell) > 0, \qquad (20.53)$$

then forced sale accelerates movement around the valuation plane.

The margin buffer responds:

$$dB/d\tau \approx -nhQ_M[d\theta_M/d\tau] + \text{OtherChannels}. \qquad (20.54)$$

Substituting equation (20.52):

$$dB/d\tau \approx -nhQ_M\Omega_{exogenous} - nhQ_M\Omega_{impact}(\ell)$$

- $\text{OtherChannels}. \qquad (20.55)$

The gate-triggered action therefore feeds back into the same phase channel that originally moved the subject toward the gate.

20.14 Effective-charge amplification

Recall the schematic effective market charge:

$$q_{eff,A} = q_A \beta \lambda \chi_A. \qquad (20.56)$$

where:

$$\lambda = nR_M/E. \qquad (20.57)$$

During adverse movement:

$$R_M \downarrow. \qquad (20.58)$$

$$E \downarrow. \qquad (20.59)$$

In many leveraged states:

$$\lambda \uparrow. \qquad (20.60)$$

Therefore:

$$|q_{eff,A}| \uparrow. \qquad (20.61)$$

The subject's structural orientation remains long:

$$q_A > 0. \qquad (20.62)$$

But its effective response becomes more extreme.

This produces the feedback:

Loss
→ Lower Equity
→ Higher Effective Coupling
→ Faster Buffer Erosion
→ Larger Forced Action. (20.63)

The main Financial Standard Model uses this same leveraged-position example to illustrate how ledgered losses can increase future effective charge while structural orientation remains stable.

20.15 Collateral-charge amplification

The contingent collateral obligation is activated when:

$$B < B_{target}. \qquad (20.64)$$

Its required amount is:

$$C_{call} = [B_{target} - B]_+. \qquad (20.65)$$

As the buffer deteriorates:

$$\partial C_{call}/\partial B = -1 \quad \text{in the breach region.} \qquad (20.66)$$

If price impact further reduces B:

$$C_{call} \uparrow. \qquad (20.67)$$

Thus the active collateral obligation grows with the action taken to resolve it.

This is a charge-amplification loop:

Operational Collateral Charge
→ Liquidation
→ Lower Admitted Collateral
→ Larger Operational Obligation. (20.68)

20.16 Cross-account market impact

A single subject may be small.

Many similarly constrained subjects can produce collective liquidation.

Let i index accounts.

Aggregate liquidation is:

$L_{total} = \Sigma_i \ell_i$. (20.69)

Market impact becomes:

$I_{total} = I(L_{total}, Depth)$. (20.70)

Each account's price changes according to:

$$\frac{dR_i}{d\tau} = \frac{dR_{i,exogenous}}{d\tau} - \beta_{i,I} I_{total}. \quad (20.71)$$

The collective loop is:

Price Decline
→ Many Margin Breaches
→ Aggregate Liquidation
→ Further Price Decline. (20.72)

This is the transition from one financial subject to a collective market mode.

The present article uses one account as the calibration atom, but the architecture naturally extends to a network.

20.17 Network charge and spin

For accounts $i = 1,\ldots,N$:

$\vec{q}_{total} = \Sigma_i \vec{q}_i$. (20.73)

The aggregate action–ledger state is:

$$\Psi_{network} = \begin{bmatrix} \Psi_1 \\ \Psi_2 \\ \vdots \\ \Psi_N \end{bmatrix}. \quad (20.74)$$

The network transport operator contains:

- common market field;
- shared clearing infrastructure;
- common collateral rules;
- funding links;
- counterparty connections.

A failure in one account may create ledger obligations in others.

The collective closure problem is therefore not the scalar sum of individual closures.

20.18 Margin contagion as recursive curvature

Suppose one account liquidates and depresses price.

Another account then breaches.

The event path is:

Account 1 Gate
→ Market Impact
→ Account 2 Gate
→ Further Market Impact. (20.75)

The system's own closure path changes the field through which later subjects are transported.

This can be described as recursive curvature:

Curvature_k
→ Gate Action_k
→ Connection_{k+1}
→ $\text{Greater Curvature}_{k+1}$. (20.76)

The generalized macro-Dirac source describes governance failure similarly: cone violation creates residual, residual weakens cross-frame fixedness, spinor split grows, and the resulting panic reduces future integration capacity.

20.19 Ledger latency and hidden leverage

Suppose the market state updates at τ_M, while the risk ledger updates at τ_R.

Define latency:

$\Delta\tau_{RM} := \tau_R - \tau_M.$ (20.77)

During latency, the recorded leverage may be:

$\lambda_{recorded} = V_{old}/E_{old}.$ (20.78)

The true current leverage may be:

$$\lambda_{current} = V_{new}/E_{new}. \qquad (20.79)$$

Hidden leverage is:

$$\Delta\lambda_{hidden} := \lambda_{current} - \lambda_{recorded}. \qquad (20.80)$$

When:

$$\Delta\lambda_{hidden} > 0, \qquad (20.81)$$

the ledger understates effective coupling.

The generalized Dirac source gives the banking failure pattern directly:

Trading Velocity > Risk-Ledger Velocity
→ Hidden Exposure.

20.20 Residual growth equation

Let total residual magnitude be:

$$R_{\mathscr{R}} := \|\mathscr{R}_S\|. \qquad (20.82)$$

Define residual growth:

$$g_{\mathscr{R}} := dR_{\mathscr{R}}/d\tau. \qquad (20.83)$$

Let integration capacity be:

$$c_{\mathscr{R}} := \text{maximum residual-processing rate.} \qquad (20.84)$$

A residual crisis begins when:

$$g_{\mathscr{R}} > c_{\mathscr{R}}. \qquad (20.85)$$

Then unresolved mismatches accumulate faster than the system can classify and close them.

Possible outcomes include:

- failed margin processing;
- settlement backlog;
- accounting restatement;
- legal dispute;
- uncontrolled liquidation;
- default.

The generalized source treats residual denial as especially dangerous because false closure converts short-term concealment into long-term curvature.

20.21 Nonlinearity taxonomy

The margin system contains several distinct nonlinearities.

Valuation nonlinearity

$$R = CF/(1 + r)^t. \qquad (20.86)$$

Complex-completion nonlinearity

$$Q = \sqrt{(A^2 - R^2)}. \qquad (20.87)$$

Constraint nonlinearity

$$C_call = [B_target - B]_+. \qquad (20.88)$$

Leverage nonlinearity

$$\lambda = V/E. \qquad (20.89)$$

State-dependent connection

$$h = h(Volatility,Liquidity,Concentration). \qquad (20.90)$$

Gate discontinuity

$$\Psi^+ = G_margin\Psi^-. \qquad (20.91)$$

Impact nonlinearity

$$I = I(\ell,Depth). \qquad (20.92)$$

Recursive ledger nonlinearity

$$\Theta_{k+1} = U_\Theta(\Theta_k,L_{k+1},\mathscr{R}_k). \qquad (20.93)$$

These nonlinearities arise at different logical levels.

They should not be compressed into one generic nonlinear term.

20.22 The correct generalization

The margin example supports the following proposition:

> Identity-bearing constraints generate charge and spin not because the equations are nonlinear, but because the constraints encode relational obligations and require authoritative action followed by independent ledger return.

Nonlinearity then emerges through:

- activation;
- branching;
- feedback;

- path dependence;
- state-dependent coupling;
- recursive inheritance.

Thus:

Identity-Bearing Constraint
→ Conditional Charge
→ Authoritative Gate
→ Spin Closure
→ Recursive Nonlinearity. (20.94)

Not:

Nonlinearity
→ Charge and Spin Automatically. (20.95)

---

21. Recursive Charge–Spin Closure

21.1 The closure does not end the dynamics

A margin event may close operationally.

Yet the subject that emerges is not identical to the pre-event subject in every numerical coordinate.

After closure, it may hold:

- fewer assets;
- less cash;
- less debt;
- different collateral;
- recognized loss;
- altered legal status;
- reduced risk limit;
- higher funding cost;
- new residual.

Therefore:

Closure ≠ Restoration of the Old Numerical State. (21.1)

Closure means:

> The system has produced an accountable successor identity whose relation to the prior identity is traceable under the declared protocol.

21.2 Post-closure identity

Let the pre-event identity be:

K_k. (21.2)

Let the valid conversion map be:

$C_K,_k$. (21.3)

Then:

$K_{k+1} = C_K,_k(K_k,Branch_k,Trace_k)$. (21.4)

Possible cases include:

Identity preserved

The account remains active after posting collateral.

Identity modified

The account remains active with fewer asset units and less debt.

Identity converted

The account becomes a recovery or default identity.

Identity extinguished

The contract is closed and no successor claim remains inside the boundary.

Each case requires lineage.

21.3 Structural-charge update

Let structural charge be:

$\vec{q}_struct,_k$. (21.5)

The gate and closure branch produce:

$\vec{q}_struct,_{k+1}$
$= C_q(\vec{q}_struct,_k,Branch_k,K_{k+1}) + \vec{r}_q,_k$. (21.6)

For collateral posting:

q_A may remain unchanged. (21.7)

q_F may remain unchanged. (21.8)

q_C^{op} returns to dormant status. (21.9)

For deleveraging:

$|q_A|$ decreases. (21.10)

$|q_F|$ may decrease. (21.11)

For default:

q_F and q_C may convert into recovery and legal charges. (21.12)

Structural charge changes only through declared vertices.

21.4 Effective-charge update

Effective charge depends on the post-event state.

Define:

$$\vec{q}_{\mathrm{eff},k} := \Lambda(\beta_k,\lambda_k,h_k,\mathrm{Liquidity}_k,\mathrm{Funding}_k,L_k,\mathscr{R}_k)\vec{q}_{\mathrm{struct},k}. \quad (21.13)$$

After closure:

$$\vec{q}_{\mathrm{eff},k+1} := \Lambda(\beta_{k+1},\lambda_{k+1},h_{k+1},\mathrm{Liquidity}_{k+1},\mathrm{Funding}_{k+1},L_{k+1},\mathscr{R}_{k+1})\vec{q}_{\mathrm{struct},k+1}. \quad (21.14)$$

Even when:

$$\vec{q}_{\mathrm{struct},k+1} = \vec{q}_{\mathrm{struct},k}, \quad (21.15)$$

it may be true that:

$$\vec{q}_{\mathrm{eff},k+1} \neq \vec{q}_{\mathrm{eff},k}. \quad (21.16)$$

This is the principal recursive charge law.

21.5 Spin closure updates charge

The direction of causality is now two-way.

Charge to spin

Charge determines how the subject responds to:

- market field;
- funding field;
- collateral field.

This response may move the subject to the gate.

Spin to charge

The resolution branch changes:

- asset units;
- debt;
- cash;
- collateral;
- ledger status;
- residual.

These changes determine future effective charge.

Therefore:

$$\text{Charge}_k \rightarrow \text{Gate}_k \rightarrow \text{SpinClosure}_k \rightarrow \text{Charge}_{k+1}. \quad (21.17)$$

Charge and spin remain conceptually distinct.

They are dynamically coupled.

21.6 The recursive spinor state

The post-closure spinor is:

$$\Psi_{k+1} = \begin{bmatrix} \psi_A,_{k+1} \\ \psi_L,_{k+1} \end{bmatrix}. \quad (21.18)$$

Closure requires:

$$\psi_L,_{k+1} \approx U_A \rightarrow L,_{k+1} \psi_A,_{k+1}. \quad (21.19)$$

But the transport operator may itself change:

$$U_A \rightarrow L,_{k+1} \neq U_A \rightarrow L,_k. \quad (21.20)$$

For example:

- haircut rules tighten;
- accounting classification changes;
- settlement permissions narrow;
- legal status changes;
- risk treatment becomes more conservative.

Thus the closure process updates not only the state but also the geometry of future closure.

21.7 Recursive mass update

Identity inertia changes after the event.

Define:

$$M_{k+1} = U_M(M_k, L_{k+1}, \mathscr{R}_k, \text{ClosureQuality}_k). \quad (21.21)$$

Examples include:

Mass increase

- higher collateral requirements;
- trading restrictions;
- additional approvals;
- legal dispute;
- operational remediation;
- greater funding friction.

Mass decrease

- standardized process;
- improved automation;
- better collateral pre-positioning;
- resolved legal uncertainty;
- increased liquidity.

A possible local update is:

$$M_{k+1} = M_k$$

- $\alpha_\mathscr{R}\|\mathscr{R}_k\|$
- $\alpha_\text{FFailure}_k - \alpha_Q\mathcal{Q}_\text{close},_k. \quad (21.22)$

This is a modelling proposal.

21.8 Recursive closure capacity

The coherent closure rate may also change:

$$c_P,_{k+1} = U_c(c_P,_k, L_{k+1}, \text{Backlog}_k, \text{Resources}_k, \text{Residual}_k). \quad (21.23)$$

After a crisis:

- operational backlog may reduce c_P;
- emergency staffing may increase c_P;
- legal restrictions may reduce c_P;
- automation may increase c_P;
- system outages may reduce c_P.

Thus:

Closure Event
→ Changed Future Closure Capacity. (21.24)

This creates an institutional memory.

21.9 Recursive margin protocol

The margin parameters may update:

$h_{k+1} = U_h(h_k, Volatility_k, Loss_k, Residual_k, Policy_k)$. (21.25)

$B_{target,k+1}$
$= U_B(B_{target,k}, Risk_k, History_k, Authority_k)$. (21.26)

Funding spread may update:

$s_{F,k+1}$
$= U_F(s_{F,k}, CreditHistory_k, Collateral_k, Residual_k)$. (21.27)

The gate therefore changes the future gate.

This is recursive market closure.

21.10 Trace becomes next-period load

The completed event writes:

$Trace_k$. (21.28)

That trace enters the next ledger:

$L_{k+1} = L_k \oplus Trace_k \oplus Residual_k$. (21.29)

The next state is conditioned by:

$Load_{k+1} := f(L_{k+1})$. (21.30)

Examples of next-period load include:

- lower available liquidity;
- higher margin requirement;

- tighter risk limit;
- reduced market confidence;
- larger legal reserve;
- higher expected funding cost.

The Periodic Grammar states this recursion explicitly:

Committed trace at period p becomes part of the operative load at period p + 1.

21.11 Master recurrence

The complete margin-account recurrence is:

$$
\begin{aligned}
&(K_k, \vec{q}_k, \Psi_k, \Theta_k, L_k) \\
&\to \mathrm{FieldMovement}_k \\
&\to \mathrm{ConstraintEncounter}_k \\
&\to G_k \\
&\to (\mathrm{Trace}_k, \mathrm{Residual}_k) \\
&\to L_{k+1} \\
&\to (K_{k+1}, \vec{q}_{k+1}, \Psi_{k+1}, \Theta_{k+1}).
\end{aligned}
\tag{21.31}
$$

In compact form:

$$
\begin{aligned}
&(I_k, q_k, \Psi_k) \\
&\to G_k \\
&\to (T_k, R_k) \\
&\to L_{k+1} \\
&\to (I_{k+1}, q_{k+1}, \Psi_{k+1}).
\end{aligned}
\tag{21.32}
$$

Equation (21.32) is the specialized application of the main Financial Standard Model recurrence.

21.12 Recursive equation family

The continuous operator in period k is:

$$
\begin{aligned}
&\mathcal{L}_k \\
&:= i\mathcal{D}_\mathrm{fin}[\Theta_k] - M[\Theta_k].
\end{aligned}
\tag{21.33}
$$

Then:

$$
\mathcal{L}_k \Psi_k = \mathscr{R}_k. \tag{21.34}
$$

After the gate and ledger update:

$$
\Theta_{k+1} = U_\Theta(\Theta_k, L_{k+1}, \mathscr{R}_k). \tag{21.35}
$$

Therefore:

$$
\mathcal{L}_{k+1} \neq \mathcal{L}_k \quad \text{in general.} \tag{21.36}
$$

The subject does not merely move inside a fixed equation.

Its closure history changes the coefficients and sometimes the admissible state space of the next equation.

This is one of the deepest differences between ordinary smooth finance and recursive constraint-bearing finance.

21.13 Path dependence

Suppose two accounts arrive at the same current balance sheet:

$$X_{A,t} = X_{B,t}. \qquad (21.37)$$

But their ledgers differ:

$$L_{A,t} \neq L_{B,t}. \qquad (21.38)$$

Then they may have different:

- haircuts;
- funding spreads;
- risk limits;
- collateral permissions;
- legal status;
- closure capacity;
- effective mass.

Therefore:

$$\text{Same Current Variables} \not\Rightarrow \text{Same Future Dynamics}. \qquad (21.39)$$

A Markov model using only X_t may be insufficient.

The augmented state must include ledger memory:

$$\text{State}_t := (X_t, L_t, \mathscr{R}_t). \qquad (21.40)$$

21.14 Recursive residual

Residual not integrated at period k becomes inherited load:

$$\mathscr{R}_k \to L_{k+1} \to \Theta_{k+1}. \qquad (21.41)$$

Define residual carry-forward:

$$\mathscr{R}_{k+1}^{\text{inherited}} = \rho_{\mathscr{R}} \mathscr{R}_k - \text{Integration}_k. \qquad (21.42)$$

where:

$0 \le \rho_{\mathscr{R}} \le 1.$ (21.43)

If integration is insufficient:

$\|\mathscr{R}_{k+1}\| > \|\mathscr{R}_k\|.$ (21.44)

Residual debt accumulates.

The system may report repeated local closure while becoming globally less stable.

21.15 False recursive closure

A dangerous case is:

$\mathrm{ReportedClosure}_k = 1.$ (21.45)

$\mathrm{ActualResidual}_k > \varepsilon_P.$ (21.46)

The next period then uses incorrect initial conditions:

$\hat{\Theta}_{k+1} \neq \Theta_{\mathrm{true},k+1}.$ (21.47)

This may cause:

- understated leverage;
- excessive collateral reuse;
- incorrect risk limit;
- hidden legal exposure;
- false capital adequacy;
- repeated margin failure.

The generalized macro-Dirac source calls this residual denial and warns that it creates accumulated curvature and eventual crisis, restatement, litigation, or intervention.

21.16 Closure quality and future effective charge

Let closure quality be:

$\mathcal{Q}_{\mathrm{close},k} \in [0,1].$ (21.48)

A possible effective-charge update is:

$$q_{\mathrm{eff},k+1} = q_{\mathrm{eff},k}$$

- $\alpha_L \Delta \mathrm{Leverage}_k$
- $\alpha_H \Delta \mathrm{Haircut}_k$
- $\alpha_R \|\mathscr{R}_k\| - \alpha_Q \mathcal{Q}_{\mathrm{close},k}.$ (21.49)

The equation is illustrative.

Its qualitative meaning is:

- higher leverage increases future sensitivity;
- worse haircut increases constraint pressure;
- residual increases uncertainty and friction;
- high-quality closure reduces future amplification.

21.17 Spin-closure quality and mass

A low-quality closure may increase identity mass:

$$Q_\text{close} \downarrow \rightarrow M_{k+1} \uparrow. \qquad (21.50)$$

The account becomes harder to transform because:

- more approvals are required;
- collateral rules tighten;
- legal review increases;
- risk limits shrink;
- market confidence falls.

But excessive mass can create paralysis.

Therefore the objective is not:

$$\text{Maximize } M. \qquad (21.51)$$

It is:

$$\text{Choose sufficient mass to preserve identity without preventing adaptive closure.} \qquad (21.52)$$

21.18 Stable recursive closure

A stable recursive system should satisfy:

1. gate events reduce rather than amplify expected future residual;
2. ledger updates improve rather than degrade frame agreement;
3. closure does not create unbounded effective charge;
4. mass remains within an adaptive range;
5. closure capacity remains above required action velocity.

One possible stability condition is:

$\rho(U_\Theta) < 1$ near the operating state. (21.53)

where ρ denotes the spectral radius of the local parameter-update Jacobian.

If:

$\rho(U_\Theta) > 1,$ (21.54)

small closure disturbances may amplify across periods.

This is a proposed dynamical-systems criterion.

21.19 Recursive crisis condition

A margin crisis may be characterized by the joint condition:

$q_{eff} \uparrow.$ (21.55)

$c_P \downarrow.$ (21.56)

$M_{rigidity} \uparrow.$ (21.57)

$\mathscr{R} \uparrow.$ (21.58)

$\kappa_{loop} \uparrow.$ (21.59)

Then:

Field Response Accelerates
while
Closure Capacity Deteriorates. (21.60)

The subject becomes both more sensitive and less able to integrate the consequences.

This is a stronger crisis definition than price decline alone.

21.20 Recursive repair

Repair must act on several layers.

Reduce effective charge

- reduce leverage;
- diversify;
- hedge;
- reduce concentration;
- improve collateral quality.

Increase closure capacity

- pre-position collateral;

- improve settlement infrastructure;
- integrate risk and accounting systems;
- increase operational resources.

Reduce curvature

- align marks;
- clarify legal identity;
- improve transport maps;
- reconcile frame differences.

Integrate residual

- disclose exceptions;
- classify disputes;
- preserve uncertainty;
- revise the protocol.

Adjust mass

- strengthen weak identity controls;
- relax unnecessary rigidity;
- use reversible intervention where possible.

A one-dimensional repair—such as demanding more collateral—may solve one branch while worsening another.

21.21 The full closure law

The margin calibration now yields:

Declared Financial Subject
→ Structural Charge
→ Effective Coupling
→ CAPM and Collateral Phase Movement
→ Constraint Encounter
→ Authoritative Margin Gate
→ Action–Ledger Spin Split
→ Gauge Transport
→ Branch Resolution
→ Trace + Residual
→ Ledger Update
→ New Mass, Connection, Capacity, and Effective Charge. (21.61)

In compact form:

Charge
→ Motion
→ Constraint
→ Gate
→ Spin
→ Gauge Return
→ Ledger
→ Updated Charge. (21.62)

21.22 Scientific interpretation

The model does not say that every margin process requires quantum terminology.

It says that the margin system supplies a clean test bed in which the proposed terms can be given operational content:

- **charge** identifies stable asset, funding, and collateral orientations;
- **spin** distinguishes outward call or liquidation action from independent ledger return;
- **gauge transport** maps the same subject across market, collateral, risk, accounting, and legal frames;
- **mass** measures identity-preserving transformation inertia;
- **gate** converts a contractual condition into consequential history;
- **residual** preserves what remains unresolved;
- **ledger recursion** changes the next period's dynamics.

The terms survive only if they add structure beyond ordinary financial language.

The main source itself imposes that ceiling: financial charge has not yet been validated as a gauge algebra, action–ledger closure is not established as physical spin-½, and advanced terms must earn value through clarity, classification, transport, gate discipline, and empirical testing.

21.23 Result of Part VIII

The Financial Gauge–Dirac equation is now embedded inside its complete operating system:

Continuous kernel

$(i\not{D}_{fin} - M_S)\Psi_S = \mathscr{R}_{cont}$. (21.63)

Gate guard

$g_j(\mathcal{S}) \geq 0$. (21.64)

Gate admission

$a_k = Admit_j(\mathcal{S}^-_k, P_k, A_k)$. (21.65)

Jump

$\mathcal{S}^{+}_{k} = \mathcal{J}_j\hat{}(a_k)(\mathcal{S}^{-}_{k}, L_k) + \mathscr{R}_\text{jump},_k.$ (21.66)

Ledger update

$L_{k+1} = U_L(L_k, \text{Trace}_k, \text{Residual}_k).$ (21.67)

Recursive operator update

$\mathcal{L}_{k+1} = U_\mathcal{L}(\mathcal{L}_k, L_{k+1}, \mathscr{R}_k).$ (21.68)

This hybrid equation family is the article's central engineering proposal.

The next Part will move from the margin-account calibration case to the broader theory of **constraint-bearing finance** and determine when charge, spin, gauge structure, and Dirac-like propagation should—or should not—generalize beyond margin systems.

Part IX — General Theory of Constraint-Bearing Finance

22. Why Nonlinearity Is Neither Necessary nor Sufficient

22.1 The tempting but incorrect generalization

The margin-account example contains many nonlinear elements:

- nonlinear discounted valuation;
- complex completion;
- leverage ratios;
- collateral haircuts;
- positive-part functions;
- margin thresholds;
- state-dependent liquidation;
- market impact;
- recursive ledger updates.

It is therefore tempting to conclude:

Nonlinear Finance → Charge + Spin. (22.1)

This conclusion is too strong.

Nonlinearity is neither sufficient nor necessary for the appearance of financial charge or financial spin.

The stronger source architecture does not derive charge and spin from mathematical curvature alone. It defines charge through transformation and coupling orientation, and spin through action–ledger double closure.

22.2 A nonlinear model without charge

Consider quadratic utility:

$$U(W) = W - aW^2. \qquad (22.2)$$

The model is nonlinear.

Yet equation (22.2) does not by itself identify:

- a bounded contractual carrier;
- a claim–obligation polarity;
- a field-specific transformation law;
- a charge-transfer vertex;
- a cross-frame transport rule;
- a ledger-return cycle.

Therefore:

$$\text{Nonlinearity} \not\Rightarrow \text{Structural Charge}. \qquad (22.3)$$

A nonlinear payoff has curvature.

It does not automatically possess charge.

22.3 A nonlinear model without spin

Consider a nonlinear transaction-cost function:

$$\text{Cost}(x) = \kappa_1|x| + \kappa_2 x^2. \qquad (22.4)$$

The function may describe increasing market impact.

But unless the trade creates a consequential obligation that must return through an independently measurable ledger surface, the model does not require a spinor.

Therefore:

$$\text{Nonlinear Cost} \not\Rightarrow \text{Action–Ledger Spin}. \qquad (22.5)$$

The function may improve execution modelling while remaining scalar.

22.4 Charge-like structure in a locally linear system

Suppose a lender provides $100 to a borrower under a simple one-period contract.

The relation may be locally represented by linear accounting:

Asset_lender = +$100 receivable. (22.6)

Liability_borrower = −$100 payable. (22.7)

The system possesses a stable claim–obligation orientation even if all local equations are linear.

Thus:

Local Linearity ⇏ Absence of Charge-Like Structure. (22.8)

The relational identity exists because one party has a right and another has an obligation.

It does not depend on nonlinear pricing.

22.5 Spin-like closure in a locally linear process

Consider a standard trade:

Order → Execution → Settlement → Accounting. (22.9)

Each step may be represented by linear transfer and bookkeeping rules.

Yet execution does not complete accountable identity.

The trade creates:

- delivery obligation;
- cash obligation;
- counterparty exposure;
- settlement trace;
- accounting consequence.

Therefore:

Locally Linear Process → Possible Double Closure. (22.10)

Spin-like structure can arise without strong mathematical nonlinearity.

22.6 Constraint is also insufficient

Consider:

$x \leq 100$. (22.11)

This is a constraint.

But equation (22.11) does not specify:

- who imposed the limit;
- what identity is constrained;
- what happens at breach;
- whether an obligation activates;
- whether an authoritative gate exists;
- whether a trace is written;
- whether a ledger return is required.

Therefore:

Constraint Alone ⇏ Charge. (22.12)

Constraint Alone ⇏ Spin. (22.13)

A passive numerical bound may be only a modelling restriction.

22.7 The stronger causal sequence

The margin example suggests the following order:

Bounded Identity
→ Relational Right or Obligation
→ Constraint on Admissible State
→ Conditional Obligation
→ Authoritative Gate
→ Consequential Action
→ Independent Ledger Return
→ Recursive State Update. (22.14)

Charge becomes plausible at the relational-right stage.

Spin becomes plausible at the independent-return stage.

Gauge becomes plausible when several frames must preserve the same identity.

Dirac-like structure becomes plausible only when the multicomponent first-order representation adds genuine value.

22.8 The primary generator is not nonlinearity

The deeper generator is:

Identity-Bearing Constraint Architecture. (22.15)

Such an architecture contains:

- a carrier;
- a boundary;

- rights and obligations;
- admissibility conditions;
- authority;
- transition rules;
- trace;
- residual;
- future inheritance.

Nonlinearity commonly appears because these structures introduce:

- thresholds;
- branching;
- complementarity;
- activation;
- path dependence;
- state-dependent coupling;
- market feedback;
- recursive parameters.

Thus:

Identity-Bearing Constraint
→ Charge and Closure Structure
→ Nonlinear Dynamics. (22.16)

The reverse implication does not generally hold.

22.9 Constraint duality is not charge

A binding inequality often generates a Lagrange multiplier.

Let:

$g(X) \leq 0.$ (22.17)

Introduce:

$\zeta \geq 0.$ (22.18)

with complementarity:

$\zeta g(X) = 0.$ (22.19)

The multiplier ζ is a shadow value or marginal pressure.

It measures the local value of relaxing the constraint.

It does not automatically identify:

- a contractual carrier;
- a relational sign;
- a transferable right;
- a charge-conversion vertex.

Therefore:

Shadow Price $\zeta \neq$ Structural Charge q. (22.20)

A constraint naturally generates a dual variable.

A relational contract may generate charge.

The two may interact at a gate but must not be conflated.

22.10 Constraint activation is not yet spin

Suppose a risk limit becomes binding:

$\text{Risk}(X) > \text{Limit}$. (22.21)

A model may respond by setting:

$\text{Position}_{new} = \text{Position}_{max}$. (22.22)

This is a constrained update.

But it becomes spin-like only if:

1. an authorized action occurs;
2. that action creates independently measurable obligations;
3. those obligations require return through settlement, accounting, legal, or risk ledgers;
4. failure to return leaves measurable residual.

Thus:

Binding Constraint + Automatic Reset $\nRightarrow$ Spin Necessarily. (22.23)

Binding Constraint + Consequential Action + Independent Ledger Return $\rightarrow$ Candidate Spin. (22.24)

22.11 The role of authority

Authority separates a numerical condition from a consequential event.

A covenant ratio may fall below its threshold.

But several possibilities remain:

- no one has yet measured it;
- the lender waives it;
- the borrower disputes the calculation;
- the breach triggers renegotiation;
- the breach accelerates repayment;
- the breach changes collateral rights.

The same numerical condition can produce different financial worlds because the governing authority and protocol differ.

Therefore:

Threshold + Authority + Trace = Candidate Gate. (22.25)

The source Periodic Grammar makes precisely this distinction: a signal or relation is not an event until an admission gate creates consequential trace and changed future admissibility.

22.12 The role of history

A pure nonlinear function may depend only on the current state:

$Y_t = f(X_t)$. (22.26)

A constraint-bearing financial system may depend on inherited ledger history:

$Y_{t+1} = f(X_{t+1}, L_t, \mathscr{R}_t)$. (22.27)

Two subjects with the same current balance sheet may face different:

- haircuts;
- funding costs;
- permissions;
- collateral rights;
- risk limits;
- legal classifications.

Therefore:

X_A,t = X_B,t (22.28)

does not imply:

FutureLaw_A = FutureLaw_B. (22.29)

when:

$L_{A,t} \neq L_{B,t}$. (22.30)

Ledger history creates transformation memory.

22.13 Proposed Constraint-Generation Principle

The article proposes:

> **Constraint-Generation Principle**
> A financial constraint generates candidate charge when it is attached to a bounded relational identity and distinguishes stable rights, obligations, or coupling orientations. It generates candidate spin when breach creates a consequential outward action whose obligations require independently observable return through one or more ledgers.

In compact form:

Constraint + Identity + Relational Orientation → Candidate Charge. (22.31)

Constraint + Authority + Action + Independent Return → Candidate Spin. (22.32)

Multiple Frames + Invariant + Transport → Candidate Gauge. (22.33)

Irreducible Doublet + First-Order Covariant Propagation + Mass → Candidate Dirac Structure. (22.34)

These are promotion conditions, not universal equivalences.

22.14 The correct generalization

The margin example therefore supports the following conclusion:

> Extra nonlinear elements do not automatically introduce charge and spin. Identity-bearing constraints, enforceable obligations, authoritative gates, and independent ledger return create the structural conditions under which charge and spin become natural mathematical candidates.

The most general sequence is:

Linear or Smooth Financial Kernel
→ Constraint-Bearing Financial Subject
→ Relational Charge
→ Gate Activation
→ Action–Ledger Spin
→ Gauge Transport
→ Recursive Nonlinearity. (22.35)

---

23. The Constraint-to-Dirac Promotion Ladder

23.1 Why a promotion ladder is needed

Advanced terminology should not be assigned all at once.

A model may legitimately possess:

- a constraint but no charge;
- charge but no spin;
- spin but no gauge;
- gauge transport but no useful Dirac algebra.

The framework therefore requires a graded promotion ladder.

Each higher level adds conditions that can fail independently.

This follows the source's minimal-terminology rule: use the weaker term unless the stronger structure has been demonstrated.

23.2 Level 0 — Smooth financial state

At the base level, the subject has ordinary state variables:

$X_t \in \mathcal{X}$. (23.1)

Its evolution is:

$dX/dt = f(X,u,t) + \varepsilon$. (23.2)

Examples include:

- price;
- value;
- debt;
- cash;
- leverage;
- duration;
- volatility.

No charge, spin, gauge, or Dirac language is required.

23.3 Level 1 — Constraint

Introduce an admissibility condition:

$g_j(X) \le 0$. (23.3)

The feasible state space is:

$\mathcal{X}_{adm} := \{X \in \mathcal{X} \mid g_j(X) \le 0 \text{ for all } j\}$. (23.4)

Examples include:

- margin requirement;
- loan-to-value limit;
- capital ratio;
- liquidity coverage threshold;
- covenant;
- concentration cap.

At this level, the safe term is:

Constrained Financial System. (23.5)

23.4 Level 2 — Identity-bearing relation

Define a bounded financial identity:

I := (Carrier,Owner,Obligor,Rights,Terms,Lineage). (23.6)

The constraint is identity-bearing when breach or compliance affects:

- who owes;
- who may claim;
- which asset is pledged;
- which contract survives;
- which party gains authority;
- which successor identity appears.

Then:

g_j = g_j(I,X,L). (23.7)

The safe term is:

Identity-Bearing Constraint. (23.8)

23.5 Level 3 — Candidate charge

A candidate charge q_r requires:

Carrier_r. (23.9)

Field_r. (23.10)

Orientation_r. (23.11)

CouplingLaw_r. (23.12)

TransportLaw_r. (23.13)

VertexRule_r. (23.14)

ResidualRule_r. (23.15)

Define:

AdmitCharge(q_r)
:= Carrier_r
∧ Field_r
∧ Orientation_r
∧ CouplingLaw_r
∧ TransportLaw_r
∧ VertexRule_r
∧ ResidualRule_r. (23.16)

If equation (23.16) fails, use:

Exposure. (23.17)

Sensitivity. (23.18)

Contractual Position. (23.19)

The source Financial Standard Model likewise treats charge as remembered transformation or coupling orientation and warns that a candidate lacking transport and vertex structure should not be promoted.

23.6 Level 4 — Authoritative gate

A constraint becomes a gate only when the system specifies:

- rule;
- authority;
- admissible evidence;
- decision;
- consequence;
- trace;
- residual.

Define:

G_j := (g_j,A_j,D_j,T_j,R_j). (23.20)

The gate-admission condition is:

AdmitGate(G_j)
:= ConstraintMeasured
∧ AuthorityValid
∧ DecisionCommitted

∧ StateChanged
∧ TraceWritten. (23.21)

If equation (23.21) fails, use:

Threshold. (23.22)

Warning. (23.23)

Candidate Event. (23.24)

23.7 Level 5 — Two-stage closure

Suppose gate action creates a new state:

$X^{+} = G_j(X^{-})$. (23.25)

The event becomes two-stage when the outward action creates obligations that remain open:

ActionCompleted = 1. (23.26)

LedgerReturned = 0. (23.27)

The safe term is:

Two-Stage Financial Closure. (23.28)

Examples include:

- margin call followed by collateral settlement;
- trade execution followed by settlement;
- exercise followed by delivery;
- default declaration followed by recovery allocation;
- impairment decision followed by accounting and capital recognition.

23.8 Level 6 — Candidate spin

Define:

Ψ :=
[
ψ_action
ψ_ledger
]. (23.29)

Candidate spin is admitted when:

AdmitSpin
:= BoundedIdentity
∧ IndependentComponents
∧ ConsequentialAction

∧ IncompleteFirstCycle
∧ IndependentLedgerReturn
∧ Residual
∧ IncrementalGain. (23.30)

The operational closure is:

Ψ_closed
→ Ψ_open-return
→ Ψ_closed′. (23.31)

The compact sign notation is:

Ψ → −Ψ → Ψ′. (23.32)

The final state Ψ′ need not equal the original numerical state.

It must be identity-continuous or validly converted.

The generalized macro-Dirac source defines precisely this action–ledger double cycle and treats the spin-½ interpretation as structural rather than physical.

23.9 Level 7 — Multi-frame identity

Suppose the same subject appears in frames:

$\mathbb{F} = \{F_1, F_2, \ldots, F_m\}$. (23.33)

Frame-local representations are:

$S_f = \Pi_f(S)$. (23.34)

A common invariant kernel exists:

$Inv_f(S_f) = Inv(S)$. (23.35)

The safe term is:

Multi-Frame Financial Identity. (23.36)

This does not yet establish gauge structure.

23.10 Level 8 — Governed transport

For each admissible edge A → B, define:

$T_{AB} : F_A \to F_B$. (23.37)

Expected target state is:

$\hat{S}_B = T_{AB}(S_A)$. (23.38)

Residual is:

$r_{AB} = S_B - \hat{S}_B$. (23.39)

The safe term is:

Governed Frame Transport. (23.40)

If no lawful transport exists, use:

Cross-System Comparison. (23.41)

23.11 Level 9 — Candidate gauge structure

Gauge terminology is admitted when:

AdmitGauge
:= DeclaredFrames
∧ LocalRepresentations
∧ InvariantKernel
∧ Connection
∧ CovarianceLaw
∧ LoopTest
∧ ResidualHonesty. (23.42)

The covariance condition is:

$T_{AB'} = G_B T_{AB} G_A^{-1}$. (23.43)

The loop test is:

$H_{loop} = \prod U_{AB}$. (23.44)

The governed loop residual is:

$\mathscr{R}_{loop} = \text{ObservedReturn} - \text{ExpectedReturn}$. (23.45)

If these conditions fail, use:

Multi-Ledger Reconciliation. (23.46)

The main article explicitly presents gauge structure as lawful redescription across financial frames and requires a frame map, connection, invariant, and loop test.

23.12 Level 10 — Identity mass

A multicomponent financial identity may resist transformation.

Define identity mass:

$M_I \approx$ Cost of Identity-Preserving Change / Admissible Identity Change. (23.47)

A matrix form is:

$$M = \begin{bmatrix} m_A & m_C \\ m_C & m_L \end{bmatrix}.$$ (23.48)

Mass is admitted when:

- a measurable transformation cost exists;
- the cost is identity-specific;
- it affects propagation or closure;
- it is not merely another name for size or volatility.

The safe fallback term is:

Identity Inertia. (23.49)

23.13 Level 11 — First-order coupled propagation

Let the multicomponent identity evolve through:

$$i\Gamma^a D_a \Psi - M\Psi = \mathscr{R}. \quad (23.50)$$

The weak Dirac claim requires:

- first-order dynamics;
- coupled components;
- meaningful Γ operators;
- covariant derivative;
- mass term;
- residual.

The safe term is:

First-Order Coupled Identity Model. (23.51)

23.14 Level 12 — Candidate Financial Dirac structure

The stronger term is admitted when:

$$\begin{aligned}&\text{AdmitDirac}\\ &:= \text{AdmitSpin}\\ &\wedge \text{AdmitGauge}\\ &\wedge \text{AdmitMass}\\ &\wedge \text{MeaningfulGammaAlgebra}\\ &\wedge \text{InterpretableSquaredOperator}\\ &\wedge \text{EmpiricalGain}.\end{aligned} \quad (23.52)$$

The central equation is:

$$(i\not{D}_{fin} - M)\Psi = \mathscr{R}. \quad (23.53)$$

The squared operator must yield additional testable structure:

$[(D)^2 - c_P^2\mathfrak{D}_G^2 + M^2 + \text{CurvatureTerms}]\Psi = \text{ResidualDynamics}.$ (23.54)

If the Γ algebra adds no value, reduce the model to:

Gauge-Coupled Hybrid State-Space System. (23.55)

23.15 Level 13 — Recursive closure

The most advanced level allows the ledger to change the next operator:

$L_{k+1} = \text{Update}(L_k, \text{Trace}_k, \text{Residual}_k).$ (23.56)

$\Theta_{k+1} = U_\Theta(\Theta_k, L_{k+1}, \mathscr{R}_k).$ (23.57)

Therefore:

$\mathcal{L}_{k+1} \neq \mathcal{L}_k.$ (23.58)

The subject's history changes:

- its charge amplification;
- mass;
- connections;
- constraints;
- closure capacity;
- admissible actions.

This is:

Recursive Constraint-Bearing Finance. (23.59)

23.16 The full promotion ladder

The complete ladder is:

Smooth State
→ Constraint
→ Identity-Bearing Relation
→ Candidate Charge
→ Authoritative Gate
→ Two-Stage Closure
→ Candidate Spin
→ Multi-Frame Identity
→ Governed Transport
→ Candidate Gauge
→ Identity Mass
→ First-Order Coupled Propagation
→ Candidate Dirac Structure
→ Recursive Closure. (23.60)

Every promotion requires new evidence.

No higher level is implied automatically by the lower levels.

23.17 The corresponding reduction ladder

The reverse ladder is:

Recursive Gauge–Dirac System
→ Hybrid Gauge-Coupled Model. (23.61)

Hybrid Gauge-Coupled Model
→ Multi-Frame Reconciliation Model. (23.62)

Multi-Frame Reconciliation Model
→ Two-Stage Workflow Model. (23.63)

Two-Stage Workflow Model
→ Constrained State-Space Model. (23.64)

Constrained State-Space Model
→ Ordinary Financial Model. (23.65)

The source Periodic Grammar treats such reduction as a scientific virtue rather than failure: a mature theory should become smaller when its stronger layers are unsupported.

23.18 Promotion is protocol-relative

Let P denote the declared protocol.

Then:

AdmitCharge_P(q) may hold. (23.66)

while:

AdmitCharge_P′(q) may fail. (23.67)

For example, an instrument may carry a clear claim charge in a legal protocol but only a statistical exposure in a short-window trading model.

Likewise:

AdmitSpin_P(Ψ) may hold for settlement operations. (23.68)

while:

AdmitSpin_P′(Ψ) may fail for daily price analysis. (23.69)

Therefore:

Strong Terminology Is Protocol-Bound. (23.70)

23.19 Promotion is evidence-filtered

A model may be conceptually eligible for a stronger term but not empirically validated.

Define:

Eligible(Term) := structural conditions satisfied. (23.71)

Define:

Validated(Term) := empirical tests passed. (23.72)

Then:

Validated(Term) ⇒ Eligible(Term). (23.73)

But:

Eligible(Term) ⇏ Validated(Term). (23.74)

The present Financial Gauge–Dirac system is structurally eligible as a research model.

It is not yet empirically validated.

---

24. General Constraint-Bearing Financial Subjects

24.1 Generic financial subject

Define a general financial subject:

$S := (K,X,\vec{q},\Psi,G,L,\mathscr{R},P).$ (24.1)

where:

- K = identity kernel;
- X = ordinary financial state;
- $\vec{q}$ = structural and effective charge coordinates;
- Ψ = action–ledger state;
- G = gates and vertices;
- L = ledger;
- $\mathscr{R}$ = residual;
- P = protocol.

The subject is constraint-bearing when:

$g_j(K,X,L,P) \le 0 \quad \text{for } j = 1,\dots,m.$ (24.2)

A breach candidate is:

$g_j(K,X,L,P) > 0.$ (24.3)

An admitted event requires:

$G_j = \mathrm{Admit}[g_j,\mathrm{Authority}_j,\mathrm{Evidence}_j]. \qquad (24.4)$

24.2 Generic hybrid equation

Between gates:

$(i\mathcal{D}_{\mathrm{fin}}[\Theta] - M[\Theta])\Psi = \mathscr{R}_{\mathrm{cont}}. \qquad (24.5)$

At gate k:

$\Psi^{+}_{k} = G_k\Psi^{-}_{k} + \eta_k. \qquad (24.6)$

Charge reconciles through:

$\Sigma q_{\mathrm{in}} + q_{\mathrm{gate}} = \Sigma q_{\mathrm{out}} + r_q. \qquad (24.7)$

The ledger updates:

$L_{k+1} = U_L(L_k,\mathrm{Trace}_k,\mathrm{Residual}_k). \qquad (24.8)$

The next operator is:

$\Theta_{k+1} = U_\Theta(\Theta_k,L_{k+1},\mathscr{R}_k). \qquad (24.9)$

Equations (24.5)–(24.9) define the general Financial Gauge–Dirac–Gate–Ledger family.

24.3 Secured loan

Identity

Borrower–lender loan contract secured by collateral.

Charge

- borrower funding obligation;
- lender claim;
- collateral-provider obligation;
- security-interest right.

Constraint

Loan-to-value ratio:

$\mathrm{LTV} = D/V_C. \qquad (24.10)$

The admissibility condition is:

$\mathrm{LTV} \le \mathrm{LTV}_{\max}. \qquad (24.11)$

Gate

Breach may trigger:

- additional collateral;
- mandatory repayment;
- repricing;
- acceleration;
- enforcement.

Spin

Notice or acceleration is the outward action.

Payment, collateral transfer, restructuring, or legal enforcement is the ledger return.

Gauge frames

- lending;
- collateral valuation;
- legal title;
- accounting;
- regulatory capital.

The secured loan therefore fits the same architecture as the margin account but usually evolves more slowly.

24.4 Repurchase agreement

Identity

Cash lender and securities provider connected by sale-and-repurchase obligations.

Charge

- cash claim;
- securities-return obligation;
- collateral charge;
- funding charge.

Constraint

Repo margin or haircut:

$$\text{CashAdvanced} \le h_{\text{repo}} \times \text{CollateralValue}. \qquad (24.12)$$

Gate

Variation-margin call, substitution request, or default.

Spin

The first leg alone does not close the repo identity.

The repurchase leg and collateral return form the required second closure.

Gauge frames

- market;
- collateral;
- treasury;
- settlement;
- legal ownership;
- accounting.

Repo may provide an even cleaner financial double-cycle than an ordinary loan because its identity explicitly spans two legs.

24.5 Collateralized derivative

Identity

Derivative contract plus collateral-support agreement.

Charge

- payoff orientation;
- variation-margin obligation;
- initial-margin obligation;
- funding charge;
- close-out right.

Constraint

Exposure after collateral:

$$E_{\text{net}} = \text{MarkToMarket} - \text{Collateral}. \qquad (24.13)$$

A margin rule may require:

Collateral ≥ f(MarkToMarket,Volatility,NettingSet). (24.14)

Gate

Margin call, dispute, close-out, or default.

Spin

Revaluation creates an outward collateral obligation.

Payment, settlement, dispute resolution, and accounting recognition complete the return.

Gauge frames

- pricing;
- collateral;
- treasury;
- legal netting;
- accounting;
- regulatory exposure.

This system contains richer charge conversion because derivative payoff orientation may change with delta, gamma, and exercise status.

24.6 Covenant-bearing bond or loan

Identity

Debt claim subject to contractual covenants.

Charge

- lender claim;
- issuer obligation;
- contingent acceleration right;
- waiver or amendment authority.

Constraint

A covenant ratio:

Debt/EBITDA ≤ κ_max. (24.15)

or:

InterestCoverage ≥ κ_min. (24.16)

Gate

Breach notice, waiver, amendment, acceleration, or restructuring.

Spin

The numerical breach does not close the event.

The lender decision and subsequent legal, funding, and accounting consequences form the double cycle.

Important lesson

The same numerical breach may produce different outcomes depending on authority.

This makes covenant systems especially useful for distinguishing threshold from gate.

24.7 Insurance reserve system

Identity

Insurance contract, claim, reserve obligation, and policyholder right.

Charge

- insurer payment obligation;
- policyholder claim;
- reinsurance recovery right;
- reserve recognition obligation.

Constraint

Capital or reserve adequacy:

$$\text{AvailableCapital} - \text{RequiredCapital} \geq 0. \qquad (24.17)$$

Gate

Claim admission, reserve revision, impairment, settlement, or regulatory intervention.

Spin

Claim event and claim settlement are distinct.

Reserve recognition, payment, reinsurance recovery, and audit trace form the ledger return.

Gauge frames

- actuarial;

- claims;
- accounting;
- legal;
- regulatory;
- reinsurance.

The phase-world source already identifies insurance claims and accounting recognition as natural domains for R/Q, gate, and settlement-phase analysis.

24.8 Bank capital constraint

Identity

Regulated banking entity with assets, liabilities, capital, and risk-weighted exposures.

Charge

- credit claim;
- deposit obligation;
- funding orientation;
- regulatory capital obligation;
- loss-absorption priority.

Constraint

Capital ratio:

$$\text{CapitalRatio} = \text{EligibleCapital}/\text{RiskWeightedAssets}. \qquad (24.18)$$

The admissibility condition is:

$$\text{CapitalRatio} \geq \text{MinimumRatio}. \qquad (24.19)$$

Gate

- management action;
- dividend restriction;
- capital raise;
- asset reduction;
- supervisory intervention;
- resolution.

Spin

A market or credit loss is not closed when the trade desk recognizes it.

It must propagate through:

- risk;
- provisioning;
- accounting;
- capital;
- treasury;
- regulatory reporting.

The generalized macro-Dirac paper specifically identifies bank risk as an example in which trading velocity can exceed risk-ledger velocity and create hidden exposure.

24.9 Liquidity coverage constraint

Identity

Financial institution whose funding and liquid assets must remain viable under stress.

Constraint

Liquidity ratio:

$$LCR = HighQualityLiquidAssets/NetCashOutflows. \qquad (24.20)$$

The admissibility rule is:

$$LCR \geq LCR_min. \qquad (24.21)$$

Charge

- liquidity-demand orientation;
- funding obligation;
- collateral encumbrance;
- withdrawal exposure.

Gate

- liquidity buffer draw;
- collateral mobilization;
- asset sale;

- central-bank facility;
- regulatory intervention.

Spin

Liquidity action must return through treasury, collateral, accounting, and regulatory ledgers.

The same asset may possess:

- market value;
- collateral value;
- regulatory liquidity value;
- liquidation value.

This makes gauge transport central.

24.10 Option exercise

Identity

Option contract with holder, writer, underlying, strike, maturity, and settlement terms.

Charge

- exercise right;
- delivery obligation;
- cash-settlement obligation;
- underlying market orientation.

Constraint or gate condition

Exercise rule:

$$\text{ExerciseValue} > \text{ExerciseThreshold}. \qquad (24.22)$$

But the gate also requires:

- holder instruction or automatic exercise;
- timing;
- contract validity;
- settlement authority.

Vertex conversion

Option identity transforms:

q_option + q_gate
→ q_underlying + q_cash + q_delivery + r_q. (24.23)

Spin

Exercise decision is the outward event.

Delivery, cash transfer, title update, and accounting form the return.

This example requires more complex charge-conversion algebra than the margin account.

24.11 Default and recovery

Identity

Debt contract before default and recovery claims after default.

Charge

- payment obligation;
- creditor claim;
- collateral right;
- guarantee;
- recovery priority.

Constraint

Payment or covenant failure.

Gate

Formal default recognition, acceleration, insolvency filing, or restructuring agreement.

Vertex conversion

q_debt
→ q_recovery + q_collateral + q_restructuring + q_extinguished + r_q. (24.24)

Spin

Default declaration is not closure.

Legal process, claim verification, collateral enforcement, recovery allocation, and accounting recognition complete the return.

Default may therefore provide the richest financial example of valid identity conversion rather than simple identity preservation.

24.12 Accounting impairment

Identity

Recognized asset or claim under an accounting entity and standard.

Charge

- ownership claim;
- recognition obligation;
- audit responsibility;
- capital consequence.

Constraint

Impairment criterion:

$$\text{RecoverableAmount} < \text{CarryingAmount}. \tag{24.25}$$

Gate

Authorized impairment decision.

Spin

Economic deterioration may occur before accounting recognition.

The first cycle is:

$$\text{Economic Loss} \rightarrow \text{Impairment Candidate}. \tag{24.26}$$

The second cycle is:

$$\text{Authorized Recognition} \rightarrow \text{Posted Entry} \rightarrow \text{Audit Trace} \rightarrow \text{Capital Consequence}. \tag{24.27}$$

Gauge frames

- market;
- valuation model;
- accounting;
- audit;
- regulatory capital.

Accounting provides a particularly clear ledger surface, but its values should not be mistaken for market invariants.

24.13 Regulatory concentration limit

Identity

Portfolio or institution subject to exposure limits.

Constraint

Concentration ratio:

$$\text{Exposure}_i/\text{EligibleCapital} \le \text{Limit}_i. \qquad (24.28)$$

Gate

Limit breach, supervisory notification, trading restriction, or forced reduction.

Charge

- exposure orientation;
- reporting obligation;
- capital burden;
- remediation duty.

Spin

Breach detection is not closure.

The institution must reduce exposure, raise capital, obtain exemption, or enter enforcement.

The resulting positions and capital consequences must be reconciled.

24.14 General mapping table

Financial subject	Candidate charge	Constraint	Gate	Ledger return
Margin account	asset, funding, collateral	buffer	margin call	collateral, liquidation, default
Secured loan	claim, debt, security	LTV	cure or acceleration	repayment, enforcement
Repo	cash, security return	repo margin	variation call	repurchase and collateral return
Derivative	payoff, collateral	exposure rule	margin or close-out	settlement and accounting
Covenant loan	lender claim, acceleration	covenant ratio	waiver or acceleration	amendment, repayment, restructuring
Insurance	claim and reserve	reserve/ capital	claim or intervention	settlement and audit
Bank	credit, funding, capital	capital ratio	management/ supervisory action	capital and regulatory ledger

Financial subject	Candidate charge	Constraint	Gate	Ledger return
Liquidity regime	funding and liquidity	LCR	buffer action	treasury/regulatory closure
Option	exercise and delivery	exercise condition	exercise	delivery or cash settlement
Default	debt and recovery	payment failure	default recognition	legal recovery and allocation

This table is a proposed classification.

It does not establish that every entry requires the full Gauge–Dirac architecture.

24.15 Generic questions for any candidate subject

Before applying the framework, ask:

1. What is the bounded identity?
2. Who carries the rights and obligations?
3. Which fields act on the identity?
4. Which candidate charges possess stable orientation?
5. Which quantities are merely sensitivities?
6. What constraint defines admissibility?
7. What authority converts breach into an event?
8. Which outward action follows?
9. What independent ledger return is required?
10. Which frames must preserve the identity?
11. What transport maps connect them?
12. What residual remains after expected transport?
13. Does a multicomponent state outperform a scalar workflow?
14. Does the Γ algebra produce testable consequences?
15. How does closure change future constraints and couplings?

24.16 Constraint families

The general theory should distinguish several constraint types.

Balance constraint

Assets − Liabilities − Equity = 0. (24.29)

Inequality constraint

$$g(X) \le 0. \qquad (24.30)$$

Complementarity constraint

$$\zeta \ge 0, \quad g(X) \le 0, \quad \zeta g(X) = 0. \qquad (24.31)$$

Conditional contractual constraint

$$\text{Condition} \rightarrow \text{Obligation}. \qquad (24.32)$$

Authority-bearing constraint

$$\text{Condition} + \text{ValidAuthority} \rightarrow \text{Gate}. \qquad (24.33)$$

Recursive constraint

$$g_{k+1} = F(g_k, L_{k+1}, \mathscr{R}_k). \qquad (24.34)$$

The last three are especially important for charge and spin.

24.17 Constraint families and terminology

A balance constraint may require no charge beyond ordinary accounting labels.

An inequality may create only a shadow price.

A conditional contract may create a dormant obligation.

An authority-bearing condition may create a gate.

A recursive constraint may change future effective charge and mass.

Therefore:

$$\text{More Complex Constraint Type} \rightarrow \text{Greater Possible Need for Transformation Memory}. \qquad (24.35)$$

But every promotion remains empirical.

24.18 The generic Constraint–Charge–Spin theorem candidate

The article proposes the following formal research conjecture.

Conjecture 24.1 — Constraint–Charge–Spin Emergence

Let a financial system contain:

1. a bounded identity I;

2. a relational right–obligation structure q;
3. an admissibility constraint g(I,X,L) ≤ 0;
4. an authority A capable of admitting breach;
5. an outward transition G;
6. an independently measurable return map T_A→L;
7. persistent trace and residual.

Then the minimum faithful state representation will generally require:

- a charge-bearing identity classification;
- a two-stage action–ledger closure representation;
- a recursive update conditioned by the post-event ledger.

Symbolically:

$$\{I,q,g,A,G,T_{A\to L},L,\mathscr{R}\} \Rightarrow \text{Candidate}\{\text{Charge},\text{Spin},\text{RecursiveClosure}\}. \qquad (24.36)$$

The implication is a research conjecture, not a proven theorem.

The word “generally” allows systems in which simpler compression remains sufficient.

24.19 Gauge extension conjecture

Conjecture 24.2 — Multi-Frame Gauge Necessity

If the same bounded identity must be recognized in several frames and frame-local quantities differ, then any faithful cross-frame model must specify:

- an invariant kernel;
- transport maps;
- connection rules;
- loop residuals.

Symbolically:

$$\text{SameIdentity} + \text{MultipleFrames} + \text{DifferentRepresentations} \Rightarrow \text{Need}\{\text{Invariant},\text{Transport},\text{Residual}\}. \qquad (24.37)$$

The word gauge is justified only if the transport additionally satisfies covariance and loop structure.

24.20 Dirac extension conjecture

Conjecture 24.3 — First-Order Identity Propagation

If:

- action and ledger are irreducible components;
- propagation is locally first-order;
- closure must remain frame-covariant;
- identity inertia is measurable;
- residual is dynamically consequential,

then a Dirac-like operator may provide a compact faithful representation:

$(i\Gamma^{a}D_a - M)\Psi = \mathscr{R}.$ (24.38)

The conjecture fails if an ordinary hybrid state-space model compresses the same information equally well.

24.21 The domain of the generalization

The strongest defensible generalization is therefore not:

All Finance Is Gauge–Dirac. (24.39)

It is:

> Financial systems become natural candidates for charge, spin, gauge transport, and Dirac-like representation when they contain bounded identities whose rights and obligations are activated by authoritative constraints, propagated across independent ledgers, and inherited recursively.

This defines a domain of application.

It does not claim universality.

24.22 Relation to the Financial Standard Model

The main Financial Standard Model source proposes a generative kernel containing:

- identity;
- charge;
- spin;
- mass;
- mediators;
- bindings;
- gates;
- ledgers;

- frame rules;
- residual.

The present article adds a more specific engineering interpretation:

> Identity-bearing financial constraints are one principal mechanism through which those abstract kernel objects become operationally visible.

A margin agreement reveals:

- charge through claim, funding, and collateral orientation;
- spin through call and ledger return;
- mass through transformation friction;
- gauge through cross-frame transport;
- gate through enforceable margin authority;
- residual through unresolved shortfall and reconciliation mismatch.

The margin account is therefore not the whole Financial Standard Model.

It is a compact **calibration atom** for the deeper kernel.

24.23 Transition to scientific testing

The generalized theory is now structurally complete.

The remaining questions are empirical:

- Does Q improve margin diagnostics beyond real variables?
- Does structural charge improve interaction accounting beyond ordinary labels?
- Does the action–ledger spinor improve closure diagnosis?
- Does gauge curvature improve reconciliation or crisis prediction?
- Does the Γ algebra add information beyond a state-space model?
- Can identity mass be measured consistently?
- Does the recursive model predict future effective coupling?

The next Part will establish the reduction ladder, falsifiers, benchmark models, data requirements, and empirical programme needed to answer those questions.

Part X — Reduction, Falsification, and Empirical Programme

25. The Reduction Ladder

25.1 A theory must know how to become smaller

The Financial Gauge–Dirac construction contains several layers:

- Complex CAPM;
- structural charge;
- action–ledger spin;
- governed frame transport;
- financial curvature;
- identity mass;
- Γ operators;
- margin gates;
- recursive ledger closure.

These layers should not be accepted or rejected as one indivisible package.

A dataset may support:

- Q but not privileged complex notation;
- two-stage closure but not spinor algebra;
- lawful transport but not gauge covariance;
- identity friction but not a mass operator;
- a hybrid state-space model but not a Dirac representation.

The governing rule is:

Retain the Least Complex Model that Preserves the Demonstrated Gain. (25.1)

The main Financial Standard Model gives the same discipline:

Model Complexity rises only when Residual Reduction exceeds Complexity Cost. (25.2)

Its appendix also requires charge, spinor, gauge, and complex representations to reduce when their admission conditions fail.

25.2 Model utility

Let model j have:

- predictive value P_j;
- diagnostic value D_j;
- gate-calibration value G_j;

- transport value T_j;
- revision value V_j;
- governance value H_j;
- complexity cost C_j.

Define model utility:

U_j
:= w_PP_j

- w_DD_j
- w_GG_j
- w_TT_j
- w_VV_j
- w_HH_j
  − λ_CC_j. (25.3)

The weights must be declared before model comparison.

The advanced model is retained when:

U_advanced > U_simple. (25.4)

A prediction-neutral model may still be retained if it produces a clearly demonstrated governance benefit, such as:

- better charge reconciliation;
- lower false-closure rate;
- improved auditability;
- earlier detection of ledger mismatch;
- more reliable cross-frame transport.

But such benefits must be measured rather than asserted.

25.3 The nested benchmark family

Define a nested family of models.

Model M0 — Conventional scalar margin model

State:

X0 := (R,n,D,C,h,B). (25.5)

Margin buffer:

$$B = C + nhR - D. \qquad (25.6)$$

Call rule:

$$C_{call} = [B_{target} - B]_+. \qquad (25.7)$$

This is the indispensable baseline.

Model M_1 — Real two-coordinate model

Add Q as an ordinary second real variable:

$$X_1 := (R,Q,n,D,C,h,B). \qquad (25.8)$$

No special complex privilege is assumed.

The model tests whether Q adds information.

Model M_2 — Complex CAPM model

Use:

$$Z = R + iQ = A \exp(i\theta). \qquad (25.9)$$

Test whether phase-based representation improves:

- estimation;
- episode alignment;
- gate prediction;
- diagnostic compression.

Model M_3 — Charge-typed model

Add:

$$\vec{q}_{struct} = (q_A,q_F,q_C). \qquad (25.10)$$

and:

$$\vec{q}_{eff} = \Lambda(X,L,Regime)\vec{q}_{struct}. \qquad (25.11)$$

Test whether typed relational orientation adds value beyond ordinary exposure labels.

Model M_4 — Action–ledger doublet

Add:

$$\Psi = \begin{bmatrix} \psi_A \end{bmatrix}$$

ψ_L
]. (25.12)

Test whether an independently measured action–ledger split improves closure diagnosis.

Model M5 — Multi-frame transport model

Add:

T_AB : S_A → Ŝ_B. (25.13)

and:

r_AB = S_B − T_ABS_A. (25.14)

Test whether governed frame transport improves upon ordinary reconciliation.

Model M6 — Gauge-covariant transport model

Add:

T_AB′ = G_BT_ABG_A⁻¹. (25.15)

and loop holonomy:

H_loop = ∏_(A→B∈loop)U_AB. (25.16)

Test covariance, loop residual, and frame-independent diagnostics.

Model M7 — Financial Gauge–Dirac system

Use:

(i𝒟_fin − M_S)Ψ_S = ℛ_S. (25.17)

together with:

$\mathcal{S}^{+}_{k} = \mathcal{J}_k(\mathcal{S}^{-}_{k}, L_k) + \mathcal{R}_\text{jump},_k$. (25.18)

and:

$\Theta_{k+1} = U_\Theta(\Theta_k, L_{k+1}, \mathcal{R}_k)$. (25.19)

This is the fullest proposed model.

It survives only if its additional structure performs identifiable work.

25.4 Complex reduction

The complex representation begins from:

Z = R + iQ. (25.20)

Its first reduction is:

$Z \rightarrow (R,Q).$ (25.21)

This retains both coordinates while removing complex privilege.

Use the real pair when:

- Q remains useful;
- phase adds no gain;
- complex multiplication has no operational meaning;
- normalization is unstable;
- episode alignment is not improved.

The next reduction is:

$(R,Q) \rightarrow R.$ (25.22)

Use the scalar when Q adds no reliable:

- predictive value;
- diagnostic value;
- intervention value;
- transport value.

The phase source explicitly requires reduction from a full phase–gate–ledger world through complex state, real pair, and finally real scalar as stronger layers fail.

25.5 Charge reduction

The strong term is:

Financial Charge. (25.23)

Its first reduction is:

Charge → Coupling Orientation. (25.24)

Use this when:

- sign and carrier are clear;
- a field response exists;
- conservation or vertex structure is incomplete.

The next reduction is:

Coupling Orientation → Exposure or Sensitivity. (25.25)

Use this when:

- the carrier is unstable;
- transport is undefined;
- no vertex rule exists;
- the coordinate is merely statistical.

The charge-admission rule in the source requires transformation, coupling, transport, vertex, and residual structure; otherwise the proposed charge should remain a sensitivity, exposure, or descriptive coordinate.

25.6 Spin reduction

The strong term is:

Financial Spinor. (25.26)

Its first reduction is:

Spinor → Action–Ledger Doublet. (25.27)

This retains two independently measured components but removes the stronger spin claim.

The next reduction is:

Action–Ledger Doublet → Two-Stage Workflow. (25.28)

Use this when:

- action and ledger stages exist;
- no meaningful transformation algebra is demonstrated.

The final reduction is:

Two-Stage Workflow → Scalar Status Machine. (25.29)

Use this when:

- ledger return is not independent;
- action produces no meaningful residual;
- cross-frame identity is irrelevant;
- the split cannot be measured.

This is the reduction rule stated by the source spinor appendix.

25.7 Gauge reduction

The strong term is:

Financial Gauge Structure. (25.30)

Its first reduction is:

Gauge → Governed Frame Transport. (25.31)

Use this when:

- source and target frames are defined;
- a transport map exists;
- a full covariance group has not been demonstrated.

The next reduction is:

Governed Transport → Multi-Ledger Reconciliation. (25.32)

Use this when:

- the same position is compared across systems;
- no invariant kernel or lawful transformation rule is established.

The final reduction is:

Multi-Ledger Reconciliation → Ordinary Data Matching. (25.33)

Use this when the task is only:

- matching identifiers;
- reconciling units;
- comparing balances.

The source rule is direct:

No Frame Map + No Invariant ⇒ No Gauge Claim. (25.34)

25.8 Curvature reduction

The strong term is:

Financial Curvature. (25.35)

Its safer form is:

Loop Transport Residual. (25.36)

Use “curvature” only when:

- a connection is defined;
- transport order matters;
- a closed loop exists;
- expected path effects are removed;

- the remaining loop defect is stable and meaningful.

If the result is only a mismatch among systems, use:

Closed-Loop Reconciliation Error. (25.37)

25.9 Mass reduction

The strong term is:

Financial Mass. (25.38)

Its first reduction is:

Mass → Identity Inertia. (25.39)

Use this when transformation resistance is observable but no operator structure is justified.

The next reduction is:

Identity Inertia → Friction Index. (25.40)

Use this when the quantity is only a weighted combination of:

- liquidity cost;
- legal delay;
- transaction cost;
- settlement delay;
- operational burden.

If the proposed mass is merely another name for size, leverage, volatility, or illiquidity, remove it.

25.10 Γ and Dirac reduction

The strongest equation is:

$(i\Gamma^{a}D_a - M)\Psi = \mathscr{R}$. (25.41)

Its first reduction is:

Dirac System → First-Order Coupled Identity Model. (25.42)

Use this when:

- two components are useful;
- first-order coupling is useful;
- the Γ algebra adds no distinct result.

The next reduction is:

First-Order Coupled Identity Model
→ Hybrid State-Space Model. (25.43)

Use this when:

- ordinary transition matrices;
- switching regimes;
- jump operators;
- latent states

capture the process equally well.

The final reduction is:

Hybrid State-Space Model → Conventional Margin Model. (25.44)

No conceptual prestige is lost by this reduction.

A simpler model is scientifically superior when it provides equal utility at lower complexity.

25.11 Recursive reduction

The recursive model uses:

$\Theta_{k+1} = U_\Theta(\Theta_k, L_{k+1}, \mathscr{R}_k)$. (25.45)

If ledger history adds no out-of-sample information:

$\Theta_{k+1} = \Theta_k$. (25.46)

The model reduces to fixed-parameter dynamics.

If only a small number of discrete states matter:

$\Theta_k \in \{\text{Normal}, \text{Stressed}, \text{Default}\}$. (25.47)

use a regime-switching model.

If a finite lag vector is sufficient:

$\Theta_{k+1} = f(X_k, X_{k-1}, \ldots, X_{k-p})$. (25.48)

use an ordinary autoregressive or state-space representation.

Ledger ontology is justified only when it captures relevant history not already contained in conventional state variables.

25.12 Local rather than global survival

A strong term may survive in one part of the system and fail in another.

For example:

- collateral obligations may support charge;
- beta may remain only a factor loading;
- settlement may support spin;
- daily price updates may not;
- market-to-collateral translation may support governed transport;
- the whole market may not support one gauge algebra.

Therefore:

Claim_global → Claim_local,P when transport fails. (25.49)

A local surviving claim is preferable to a forced universal claim.

25.13 Partial model survival

Suppose empirical testing finds:

- Q useful;
- phase not useful;
- action–ledger split useful;
- Γ algebra not useful;
- loop residual useful;
- gauge covariance not established.

The surviving model is then:

Real R–Q Pair

- Action–Ledger Doublet
- Governed Loop Reconciliation
- Hybrid Margin Gates. (25.50)

This is already a meaningful architecture.

The failure of stronger layers should not erase the useful lower structure.

25.14 Preserve failed traces

When an advanced branch fails:

Model_v1 → ReducedModel_v2. (25.51)

The research ledger should preserve:

- original hypothesis;
- original model;
- original parameter declarations;
- failed tests;
- residual;
- reason for reduction;
- reduced model.

The source explicitly requires reopening and revision to preserve prior trace rather than rewriting history as if the earlier closure never occurred.

25.15 Reduction certificate

A formal reduction record may be:

ReductionCertificate
:=
(ModelBefore,FailedCondition,Evidence,Residual,ModelAfter,RetainedClaims,RemovedClaims). (25.52)

This makes reduction auditable.

It also prevents a failed strong claim from surviving informally after its formal terminology has been removed.

25.16 The full reduction ladder

The full ladder is:

Financial Gauge–Dirac–Gate–Ledger System
→ Hybrid Gauge-Coupled System
→ Governed Multi-Frame Transport Model
→ Action–Ledger Doublet
→ Two-Stage Hybrid Workflow
→ Complex or Real Two-Coordinate Margin Model
→ Conventional Scalar Margin Model. (25.53)

The correct stopping point is:

Highest Level with Reproducible Incremental Utility. (25.54)

---

26. Falsification Tests

26.1 Falsification must be term-specific

The complete framework should not be tested by one question such as:

“Does the Financial Gauge–Dirac equation work?”

That question is too broad.

Each promoted term creates its own burden:

- Q burden;
- phase burden;
- charge burden;
- spin burden;
- gauge burden;
- curvature burden;
- mass burden;
- Γ burden;
- recursion burden.

A failure at one layer should remove that layer without automatically destroying all lower layers.

26.2 Complex-CAPM falsifiers

The complex interpretation fails when any of the following holds.

Artificial closure

The relation:

$A^2 = R^2 + Q^2$ (26.1)

exists only because the researcher imposed a convenient normalization with no operational meaning.

Q as error bucket

Q is defined as:

Q := Everything not explained by R. (26.2)

This is invalid because unexplained remainder belongs in residual.

No independent proxy

Q cannot be estimated independently of the outcome being predicted.

Scaling instability

Small changes in A or normalization cause large, arbitrary changes in:

- Q;

- θ;
- episode order;
- gate classification.

No incremental value

Performance(R,Q) ≤ Performance(R,Controls). (26.3)

No gain over real pair

Performance_complex ≤ Performance_real-pair. (26.4)

Then retain the ordered pair rather than the complex representation.

These conditions follow the phase source's explicit falsification and reduction rules.

26.3 Phase falsifiers

The phase interpretation fails when:

No coherent phase order

θ does not evolve reproducibly within episodes.

Excessive branching

Phase order changes arbitrarily under minor protocol changes.

No gate concentration

Margin calls or closure events do not occur more reliably in declared phase regions than under ordinary state variables.

No episode alignment

Let D_t be between-episode distance after calendar-time alignment.

Let D_θ be distance after phase alignment.

Phase compression fails when:

D_θ ≥ D_t. (26.5)

No derivative simplification

The phase equation:

dZ/dθ (26.6)

is no simpler, more stable, or more useful than:

dZ/dt. (26.7)

Then θ remains a descriptive coordinate rather than an internal clock.

26.4 Q-to-margin falsifier

The calibration model proposes:

$\partial B/\partial \theta_M \approx -nhQ_M.$ (26.8)

This relation is an exact local identity under its declared simplifying conditions.

Its practical relevance fails if:

- amplitude movement dominates phase movement;
- actual margin rules do not use the assumed value;
- Q is too unstable to estimate;
- Q adds no information about buffer movement;
- haircut and debt channels explain all relevant variation.

A direct empirical test is:

ΔB_t
$= \alpha$

- $\beta_Q(-n_{th}_tQ_t\Delta\theta_t)$
- $\beta_A[n_{th}_t(R_t/A_t)\Delta A_t]$
- $\beta_h(n_tR_t\Delta h_t)$
- $\beta_D(-\Delta D_t)$
- $\varepsilon_t.$ (26.9)

The phase-margin channel is unsupported when:

β_Q is unstable, incorrectly signed, or adds no out-of-sample gain. (26.10)

26.5 Charge falsifiers

A proposed financial charge fails if:

Carrier instability

The identity carrying q cannot be defined consistently.

Field ambiguity

No declared field determines what q couples to.

Sign instability

The sign convention changes to fit the outcome.

No transport rule

q cannot be followed across market, collateral, accounting, or legal frames.

No vertex rule

There is no explicit rule for:

- transfer;
- activation;
- conversion;
- extinguishment.

No balance meaning

The expression:

$$\Sigma q_{in} + q_{gate} = \Sigma q_{out} + r_q \qquad (26.11)$$

has no observable interpretation.

No distinction from exposure

q is fully explained by an ordinary position sign or sensitivity.

Then use:

Exposure, Position, Obligation, or Sensitivity. (26.12)

26.6 Charge-conservation falsifier

Suppose charge is claimed to be preserved across ordinary transport.

Then test:

$$r_{q,k} := \Sigma q_{in,k} + q_{gate,k} - \Sigma q_{out,k}. \qquad (26.13)$$

The conservation-like claim fails when:

$$E[r_q \mid \text{OrdinaryTransport}] \neq 0 \qquad (26.14)$$

after accounting for:

- known conversion gates;
- timing;
- unit changes;
- netting;

- settlement.

Persistent unexplained charge residual suggests:

- incorrect carrier definition;
- missing vertices;
- misclassified conversion;
- non-conserved candidate coordinate.

26.7 Spin falsifiers

The financial spin claim fails when:

No independent components

ψ_A and ψ_L cannot be observed separately.

No lawful return map

T_A→L cannot be defined.

No first-cycle incompleteness

Outward action already completes all relevant consequences.

No residual from delayed return

Incomplete ledger return creates no measurable failure mode.

Scalar sufficiency

A scalar completion status predicts and diagnoses equally well.

No identity continuity

The action and ledger records do not refer to the same bounded subject.

Then reduce:

Spinor → Two-Stage Workflow or Scalar State Machine. (26.15)

26.8 Spinor-split falsifier

The proposed split is:

Δ_spin,G
:= ‖ψ_L − U_ALψ_A‖_W. (26.16)

Its diagnostic claim fails if:

- $\Delta_{spin,G}$ does not rise during genuinely open events;
- it remains high after verified closure;
- it does not distinguish resolved from unresolved calls;
- it is dominated by routine timing differences already known to the protocol;
- a simple age-of-call variable performs equally well.

A primary test is:

$$\Pr(\text{Unresolved at } H \mid \Delta_{spin,G}, \text{Controls}) \quad \Pr(\text{Unresolved at } H \mid \text{Controls}). \qquad (26.17)$$

If equation (26.17) fails out of sample, the spinor split adds no demonstrated value.

26.9 Double-cover falsifier

The exact representation:

$$S_C(2\pi) = -I_2. \qquad (26.18)$$

$$S_C(4\pi) = I_2. \qquad (26.19)$$

is mathematically true for the chosen SU(2)-like construction.

Its financial interpretation fails if:

- no measurable intermediate sign reversal exists;
- ordinary two-stage status represents the process fully;
- no observable distinguishes 2π from 4π closure;
- closure paths do not compose according to the proposed group law.

Then retain operational double closure without the exact spin-½ analogy.

26.10 Gauge falsifiers

Gauge terminology fails if:

No invariant kernel

The subject cannot be recognized across frames.

No lawful connection

T_{AB} is chosen retrospectively for each observation.

No covariance

Equivalent local redescriptions produce different substantive event classifications.

No loop test

There is no meaningful closed transport path.

Residual erasure

Mismatches are eliminated only by forcing one frame to copy another.

No gain over reconciliation

Gauge quantities add no information beyond ordinary data-quality measures.

Then use:

Governed Transport or Multi-Frame Reconciliation. (26.20)

26.11 Covariance falsifier

For local transformations G_A and G_B, the model requires:

$T_{AB'} = G_B T_{AB} G_A^{-1}$. (26.21)

The covariance claim fails when:

$T_{AB'} G_A S_A \neq G_B T_{AB} S_A$ (26.22)

beyond declared tolerance.

Financial examples include:

- event classification changes merely because reporting currency changes;
- charge changes because units are rescaled;
- closure status changes after a presentation-only reclassification;
- the same legal position is mapped differently after desk reassignment.

Such failures indicate that the model confuses representation with substance.

26.12 Curvature falsifiers

The loop residual is:

$\mathscr{R}_{loop} = \text{ObservedReturn} - \text{ExpectedReturn}$. (26.23)

Curvature terminology fails if:

- no expected loop operator can be declared;
- lawful fees and timing explain the entire defect;

- path order has no effect;
- loop residual is unstable across minor protocol changes;
- ordinary reconciliation counts perform equally well.

Then use:

Loop Reconciliation Residual. (26.24)

26.13 Mass falsifiers

The mass interpretation fails if:

No measurable inertia

Transformation cost or delay cannot be estimated.

No identity specificity

The proposed mass is only generic illiquidity.

No dynamic role

Mass does not affect closure time, residual, or response.

Parameter instability

The mass estimate changes arbitrarily with normalization.

Complete redundancy

Mass is fully explained by:

- market value;
- leverage;
- volatility;
- bid–ask spread;
- transaction cost.

Then use:

Friction or Operational Burden. (26.25)

26.14 Mass-mode falsifier

The proposed mass eigenvalues are:

$$m_{\pm} = m_I \pm \sqrt{(m_C^2 + m_\Delta^2)}. \quad (26.26)$$

The eigenmode interpretation fails if:

- estimated modes are not stable;
- modes have no distinguishable closure behaviour;
- closure times do not vary with $m_{\pm}$;
- ordinary principal components perform equally well;
- parameter estimates are non-identifiable.

Then the matrix mass should be replaced by an ordinary coupling matrix or friction score.

26.15 Γ-algebra falsifiers

The Γ representation fails as a substantive financial structure if:

- anticommutation is imposed but does no empirical work;
- another arbitrary matrix basis performs equally well;
- the component signs have no financial interpretation;
- the squared equation provides no testable consequence;
- Γ parameters cannot be identified;
- predictive or diagnostic results depend on arbitrary basis selection.

Then:

Γ Operators → Generic Coupling Matrices. (26.27)

26.16 Dirac falsifiers

The Financial Dirac claim fails if:

No irreducible spinor

The two components can be collapsed without loss.

No first-order advantage

A conventional transition model performs equally well.

No useful squared relation

The second-order equation adds no interpretable modes.

No measurable mass

M cannot be estimated.

No gauge covariance

D_a is merely an ordinary derivative.

No incremental utility

U_Dirac ≤ U_HybridBaseline. (26.28)

Then use:

Hybrid Constraint-Bearing Financial System. (26.29)

26.17 Closure-capacity falsifier

The model proposes a finite coherent closure rate:

c_P = sup{v | Residual remains bounded and ABFix remains stable}. (26.30)

This concept fails if:

- no threshold behaviour exists;
- residual grows independently of action velocity;
- A-B Fixedness does not deteriorate at high throughput;
- c_P cannot be estimated reproducibly;
- an ordinary queueing model explains the process fully.

Then use:

Operational Processing Capacity. (26.31)

The generalized Dirac source treats the estimation of c_P, spinor split, A-B Fixedness, mass, and residual as preliminary empirical protocols rather than established measurements.

26.18 Recursive-closure falsifiers

The recursive claim is:

$\Theta_{k+1} = U_\Theta(\Theta_k, L_{k+1}, \mathscr{R}_k)$. (26.32)

It fails if:

No ledger dependence

Future parameters do not depend on prior trace after current state variables are controlled.

No residual inheritance

Prior unresolved residual does not affect later closure.

No path dependence

Subjects with identical current states but different histories behave identically.

Fixed-model sufficiency

A stationary model performs equally well.

Then use:

Fixed-Parameter or Finite-Lag Dynamics. (26.33)

26.19 Constraint-generation falsifier

The general theory proposes:

Identity-Bearing Constraint
→ Candidate Charge and Spin. (26.34)

This generalization weakens if:

- relational obligation adds no useful classification;
- action–ledger separation adds no diagnostic value;
- constraint-bearing and ordinary nonlinear systems behave equivalently;
- no systematic distinction appears across financial domains.

The claim should then be narrowed to:

Some contractually governed financial processes exhibit useful charge-like and double-closure structure. (26.35)

26.20 Full-system falsifier

The complete model is retained only if:

$U_{M7} > \max(U_{M0}, U_{M1}, \ldots, U_{M6})$. (26.36)

or if M_7 produces a distinct, measured governance benefit unavailable to the simpler models.

The complete model fails when:

Advanced Performance − Complexity Penalty
≤ Simple Performance − Simple Complexity Penalty. (26.37)

This benchmark rule is stated explicitly in the source Financial Standard Model.

26.21 Failed prediction versus failed ontology

A prediction may fail because:

- estimated parameters were poor;
- the event was rare;
- the sample changed;
- the outcome was noisy.

An ontology fails more deeply when:

- the carrier cannot be defined;
- the frames cannot be distinguished;
- the gate cannot be identified;
- the residual cannot be typed;
- the alleged components are not independently measurable.

Therefore:

Predictive Failure ≠ Automatic Ontological Failure. (26.38)

and:

Ontological Failure ⇒ Strong Terminology Must Be Removed. (26.39)

26.22 Prospective invalidation

Every empirical study should declare in advance:

- which result would remove Q;
- which result would remove phase;
- which result would remove charge;
- which result would remove spin;
- which result would remove gauge;
- which result would remove Dirac;
- which simpler model would remain.

This converts falsification from a rhetorical promise into an operational gate.

26.23 Falsification ledger

Define:

L_falsification
:= (Claim,Protocol,Test,Threshold,Outcome,Residual,Decision,ReducedClaim). (26.40)

A failed claim should remain in the ledger.

The model should not be silently redefined after the outcome.

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27. Empirical Calibration Programme

27.1 The empirical objective

The empirical programme does not begin by asking whether finance is secretly quantum.

It asks narrower questions:

1. Can the proposed variables be measured?
2. Are the components identifiable?
3. Do they improve diagnosis or prediction?
4. Do transport and closure rules survive protocol changes?
5. Does the advanced model outperform simpler benchmarks?
6. Which layers must be reduced?

The Financial Standard Model's empirical contract requires classification reliability, incremental information, gate separation, residual recall, and comparison with complexity-penalized simpler models.

27.2 Unit of analysis

Several observational units are possible.

Account-time panel

One row per account per observation interval.

Useful for:

- buffer dynamics;
- leverage;
- phase exposure;
- call hazard.

Margin-event panel

One row per margin call.

Useful for:

- call resolution;
- closure time;

- branch selection;
- residual.

Transport-edge panel

One row per subject-frame-edge observation.

Useful for:

- market-to-collateral mapping;
- risk-to-accounting transport;
- edge residual.

Closed-loop panel

One row per completed transport loop.

Useful for:

- holonomy;
- loop curvature;
- false closure.

Subject-history panel

One sequence per bounded financial identity.

Useful for:

- recursive charge;
- mass evolution;
- path dependence.

A complete study may use all five units.

27.3 Minimum data schema

The calibration dataset should include the following groups.

Identity data

- account identifier;
- instrument identifier;
- contract identifier;
- legal entity;

- counterparty;
- signed quantity;
- currency;
- maturity;
- execution lineage.

Market data

- price or declared market mark;
- cash-flow assumptions;
- baseline rate;
- beta;
- ERP;
- volatility;
- liquidity;
- bid–ask spread;
- market depth.

Collateral data

- eligible asset quantity;
- haircut or admission factor;
- concentration adjustment;
- currency adjustment;
- collateral received;
- collateral rejected;
- encumbrance status.

Funding data

- debt balance;
- funding rate;
- maturity;
- repayment;
- refinancing;

- funding withdrawal.

Margin data

- target buffer;
- maintenance rule;
- call amount;
- call time;
- deadline;
- authority;
- waiver;
- liquidation authority.

Action data

- collateral instruction;
- sale instruction;
- executed quantity;
- execution price;
- forced-liquidation quantity;
- default declaration.

Ledger data

- settled quantity;
- collateral posting;
- debt repayment;
- realized P&L;
- accounting entry;
- risk update;
- legal status;
- regulatory status.

Residual data

- settlement break;
- collateral dispute;

- mark dispute;
- missing position;
- timing exception;
- legal dispute;
- unrecognized loss;
- residual shortfall.

27.4 Protocol preregistration

Before outcome evaluation, declare:

$$P := (B,\Delta,h,u). \qquad (27.1)$$

For the margin study:

- B = account and legal boundary;
- Δ = valuation and aggregation rule;
- h = prediction and closure horizon;
- u = admissible interventions.

Also declare:

- the CAPM specification;
- the baseline value amplitude;
- the collateral rule;
- the gate rule;
- the identity kernel;
- the charge signs;
- the closure conditions;
- the frame graph;
- the falsifiers;
- the benchmark models.

No element should be selected after observing whether the call succeeded or failed.

27.5 Evidence-time consistency

Every feature used to predict an event at time t must have been available at or before t.

Define the information set:

$\mathcal{I}_t$:= information available by prediction time t. (27.2)

A valid predictor satisfies:

$X_{predictor,t} \in \mathcal{I}_t$. (27.3)

Future leakage occurs when:

$X_{predictor,t}$ uses information from $t' > t$. (27.4)

Examples of leakage include:

- using final liquidation price to estimate pre-call Q;
- using later collateral acceptance to classify initial eligibility;
- using eventual default status to redefine the original gate;
- using revised accounting data without preserving the original version.

27.6 Calculation of Complex CAPM variables

For each instrument and horizon:

$r_t = r_{base,t} + \beta_t ERP_t$. (27.5)

$A_t = CF_t/(1 + r_{base,t})^T$. (27.6)

$R_t = CF_t/(1 + r_t)^T$. (27.7)

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (27.8)

$\theta_t = \arccos(R_t/A_t)$. (27.9)

For n_t units:

$Z_{M,t} = n_t(R_t + iQ_t)$. (27.10)

The study must record:

- beta estimation window;
- ERP source;
- cash-flow forecast source;
- horizon;
- parameter vintage;
- revisions.

Complex CAPM should not replace the actual contractual margin mark.

It provides the proposed risk-phase environment.

The real margin engine remains the source of truth for:

- margin amount;
- haircut;
- eligibility;
- call status.

27.7 Collateral-plane variables

Given h_t:

$$R_{C,t} = h_t R_t. \qquad (27.11)$$

$$Q_{C,t} = \sqrt{(A_t^2 - R_{C,t}^2)}. \qquad (27.12)$$

$$\theta_{C,t} = \arccos(R_{C,t}/A_t). \qquad (27.13)$$

The market–collateral phase difference is:

$$\Delta\theta_{MC,t} = \theta_{C,t} - \theta_{M,t}. \qquad (27.14)$$

The pressure difference is:

$$\Delta Q_{MC,t} = Q_{C,t} - Q_{M,t}. \qquad (27.15)$$

The actual margin buffer is:

$$B_t = C_t + n_t h_t R_{mark,t} - D_t. \qquad (27.16)$$

Here $R_{mark,t}$ should use the mark defined by the actual margin protocol.

The proposed CAPM buffer may be calculated separately:

$$B_{CAPM,t} = C_t + n_t h_t R_t - D_t. \qquad (27.17)$$

The difference is:

$$\mathscr{R}_{mark,t} = B_t - B_{CAPM,t}. \qquad (27.18)$$

This residual should remain visible.

27.8 Charge coding

Structural charge should be coded from legal and transactional data, not inferred from later price movement.

For each subject:

$q_A :=$ signed contractual asset units. (27.19)

$q_F :=$ signed funding obligation class. (27.20)

$q_C :=$ signed contingent collateral-obligation class. (27.21)

The operational collateral charge is:

q_C^op,t = 1[CallActive_t]q_C. (27.22)

A screening operator may be estimated:

$\vec{q}$_eff,t = S_t$\vec{q}$_struct,t. (27.23)

Candidate elements of S_t include:

- leverage;
- hedge ratio;
- collateral;
- netting;
- insurance;
- liquidity;
- concentration.

The source describes such screening as a reduction in effective charge rather than annihilation and treats effective charge as recursively updated through trace and residual.

27.9 Spinor construction

Define a common feature space.

The action vector may be:

ψ_A,t
:= Normalize(
MarketValue_t,
Position_t,
CallAmount_t,
SaleInstruction_t,
ExpectedDebtRepayment_t,
EconomicLoss_t,
GateStatus_t
). (27.24)

The ledger vector may be:

ψ_L,t
:= Normalize(
CollateralAdmitted_t,
SettledPosition_t,
CallLedger_t,
SettledSale_t,
RecordedDebtRepayment_t,
RecognizedP&L_t,
LedgerStatus_t
). (27.25)

Define the expected action-to-ledger map:

$\psi\hat{}_{L,t} = U_{AL,t}\psi_{A,t}.$ (27.26)

The covariant split is:

$\Delta_{spin,t} = \|\psi_{L,t} - \psi\hat{}_{L,t}\|_W.$ (27.27)

The feature mapping, normalization, and weights must be frozen before outcome testing.

27.10 Closure label

Define verified closure:

$Y_{close,t} = 1$ (27.28)

when all required conditions hold:

- amount reconciled;
- deadline satisfied or validly amended;
- collateral legally controlled;
- sale settled;
- debt updated;
- P&L recognized;
- charge reconciled;
- required frame loops closed;
- residual within tolerance.

Otherwise:

$Y_{close,t} = 0.$ (27.29)

A separate valid-default closure label should be used:

$Y_{valid\text{-}default,t} = 1$ (27.30)

when the account has lawfully converted into a reconciled recovery identity.

Closure should not be equated with cure alone.

27.11 Closure time

For call k:

$T_{close,k} = t_{close,k} - t_{call,k}.$ (27.31)

Possible censoring occurs when the event remains unresolved at the end of observation.

A survival model may estimate:

Pr(T_close > t | X). (27.32)

The primary spinor hypothesis is:

∂Hazard(Unresolved)/∂Δ_spin > 0. (27.33)

Equivalently, larger spinor split should predict slower or less successful closure, after controls.

27.12 Edge transport residual

For each frame edge A → B:

r_AB,t = S_B,t − T_AB,tS_A,t. (27.34)

The edge residual should be decomposed into:

r_AB
= r_identity

- r_quantity
- r_value
- r_timing
- r_charge
- r_authority
- r_trace. (27.35)

Each component requires its own unit or normalized score.

The decomposition prevents a small monetary difference from concealing a large legal-identity failure.

27.13 Loop curvature estimate

For loop ℓ:

H_ℓ,t = ∏_(A→B∈ℓ)U_AB,t. (27.36)

Expected return state:

Ŝ_start,t^return = H_ℓ,t^PS_start,t. (27.37)

Observed return state:

S_start,t^return,obs. (27.38)

Loop residual:

$\mathscr{R}$_ℓ,t = S_start,t^return,obs − Ŝ_start,t^return. (27.39)

Normalized loop defect:

$$\kappa_{\ell,t} = \|\mathscr{R}_{\ell,t}\|_W / [\|\hat{S}_{start,t}^{return}\|_W + \varepsilon]. \quad (27.40)$$

The empirical question is not whether κ is ever nonzero.

It is whether κ:

- is stable;
- is interpretable;
- predicts failure;
- adds value beyond ordinary reconciliation metrics.

27.14 Closure-capacity estimate

The generalized operational definition is:

$$c_P = \sup\{v_{close} \mid \mathscr{R} \text{ remains bounded and ABFix remains stable}\}. \quad (27.41)$$

For each time window, estimate:

$$v_{action,t} := \text{ActionDisplacement}_t / \Delta\tau. \quad (27.42)$$

$$v_{ledger,t} := \text{LedgerIntegration}_t / \Delta\tau. \quad (27.43)$$

Define backlog growth:

$$g_{backlog,t} := \text{NewOpenObligations}_t - \text{ClosedObligations}_t. \quad (27.44)$$

A candidate empirical c_P is the throughput level beyond which:

- backlog accelerates;
- residual grows;
- cross-frame agreement falls;
- closure time increases sharply.

Change-point or threshold models may estimate this boundary.

27.15 Identity-mass estimate

A practical mass proxy may combine:

$$m_{eff,t} := w_1 \text{ImpactCost}_t$$

- w_2SettlementDelay_t
- w_3CollateralSubstitutionDelay_t

- w_4LegalFriction_t
- w_5AccountingFriction_t
- w_6ApprovalCount_t. (27.45)

A more structural estimate treats M as a parameter in:

$(i\mathcal{D}_{fin} - M)\Psi = \mathscr{R}$. (27.46)

Possible estimation approaches include:

- state-space maximum likelihood;
- Bayesian filtering;
- penalized least squares;
- system identification;
- neural differential equations with strong structural constraints.

The operational proxy should be tested before attempting a fully latent mass operator.

27.16 Estimating the mass matrix

Let:

$$
M = \begin{bmatrix} m_A & m_C \\ m_C & m_L \end{bmatrix}. \quad (27.47)
$$

A discrete first-order approximation is:

$$i\Gamma^0(\Psi_{t+1} - \Psi_t)/\Delta\tau$$

- $$ic_P\Gamma^1\mathfrak{D}_G\Psi_t - M\Psi_t = \mathscr{R}_t. \quad (27.48)$$

Given observed Ψ, transport, and residual proxies, estimate:

$$\theta_M := (m_A, m_C, m_L, c_P). \quad (27.49)$$

Identification requires variation in:

- action intensity;
- ledger delay;
- transport mismatch;
- closure outcomes.

If m_C cannot be distinguished from transport or residual terms, the matrix-mass interpretation

is not identified.

27.17 Primary hypotheses

H1 — Q incremental value

Q improves prediction of future buffer erosion beyond R and conventional controls:

$$I(\Delta B_{future};Q \mid R,B,Leverage,Volatility,h) > 0. \qquad (27.50)$$

H2 — Phase-to-buffer transmission

The phase channel has the expected sign:

$$\beta_Q < 0 \qquad (27.51)$$

in equation:

$$\Delta B = \alpha + \beta_Q(nhQ\Delta\theta) + Controls + \varepsilon. \qquad (27.52)$$

H3 — Collateral-phase value

θ_C and $\Delta\theta_{MC}$ improve call prediction beyond market phase alone:

$$Performance(\theta_M,\theta_C,\Delta\theta_{MC},Controls)$$
$$> Performance(\theta_M,Controls). \qquad (27.53)$$

H4 — Charge typing

Charge-typed exposure predicts interaction outcomes better than unsigned notional:

$$Performance(\vec{q}_{struct},\vec{q}_{eff},Controls)$$
$$> Performance(Notional,Controls). \qquad (27.54)$$

H5 — Spinor split

Δ_{spin} predicts unresolved closure:

$$Pr(Unresolved_H \mid \Delta_{spin},Controls)$$
increases with Δ_{spin}. (27.55)

H6 — Loop curvature

κ_{loop} predicts:

- reconciliation break;
- call dispute;

- settlement delay;
- false closure;
- default escalation.

Formally:

$$I(\text{Failure}_H;\kappa_{\text{loop}} \mid \text{OrdinaryExceptions},\text{Controls}) > 0. \qquad (27.56)$$

H7 — Finite closure capacity

Residual accelerates after action velocity exceeds c_P:

$$E[g_{\mathscr{R}} \mid v_{\text{action}} > c_P] \neq E[g_{\mathscr{R}} \mid v_{\text{action}} \le c_P]. \qquad (27.57)$$

H8 — Identity mass

Higher m_{eff} predicts longer closure time after controlling for event size:

$$\partial E[T_{\text{close}}]/\partial m_{\text{eff}} > 0. \qquad (27.58)$$

H9 — Recursive ledger effect

Prior residual changes future effective coupling:

$$I(q_{\text{eff},t+1};\mathscr{R}_t \mid X_{t+1}) > 0. \qquad (27.59)$$

H10 — Full-model utility

The Financial Gauge–Dirac system provides greater complexity-adjusted utility:

$$U_{M7} > \max(U_{M0},\ldots,U_{M6}). \qquad (27.60)$$

27.18 Benchmark models

The advanced model should be compared with mature alternatives.

Scalar margin rule

Uses current buffer and contract terms.

Logistic call model

Estimates probability of call or failure.

Survival model

Estimates time to call resolution or default.

State-space model

Estimates latent stress and observation error.

Hidden Markov model

Estimates normal, stressed, and default regimes.

Hybrid automaton

Represents continuous dynamics plus discrete margin gates.

Queueing model

Represents action arrival and ledger-processing capacity.

Graph reconciliation model

Represents cross-system mismatches.

Machine-learning benchmark

Uses flexible nonlinear features without physics-derived terminology.

The advanced system succeeds only when it adds value beyond these comparators.

The draft Periodic Grammar specifically identifies state-space models, hidden Markov regimes, change-point detection, event studies, survival analysis, microstructure, and institutional accounting or legal-event models as relevant comparison frameworks.

27.19 Stage 0 — Simulation identifiability

Before testing real data, construct simulated systems with known ground truth.

Simulate:

- one asset;
- CAPM phase process;
- haircut process;
- debt accrual;
- margin gate;
- collateral or liquidation branches;
- ledger delay;
- transport errors;
- recursive parameter updates.

The first question is:

Can the estimation method recover parameters known to exist? (27.61)

Simulation can test:

- charge recovery;
- spinor split;
- c_P threshold;
- mass parameters;
- loop curvature;
- gate identification.

Simulation success does not establish real-market validity.

The source empirical programme makes the same distinction:

Simulation Identifiability
→ Real-Data Applicability
→ Cross-Domain Transport.

27.20 Stage 1 — Retrospective event study

Collect historical margin events.

For each event:

1. reconstruct the information available before call;
2. calculate conventional variables;
3. calculate Q and phase variables;
4. code structural charges;
5. reconstruct action and ledger states;
6. calculate transport and loop residuals;
7. record closure branch and outcome;
8. compare nested models.

Important outcomes include:

- call occurrence;
- call amount;
- cure method;
- closure time;

- liquidation severity;
- default;
- residual;
- false closure.

27.21 Stage 2 — Prospective validation

Freeze:

- protocol;
- variables;
- parameters;
- gates;
- thresholds;
- benchmark set.

Then evaluate future events without retrospective adjustment.

Prospective validation is essential because many advanced constructs can otherwise be fitted to known closure histories.

27.22 Stage 3 — Natural experiments

Useful external changes include:

- margin-rule changes;
- haircut revisions;
- collateral-eligibility changes;
- settlement-cycle changes;
- disclosure rules;
- accounting-standard changes;
- regulatory capital changes.

A natural experiment may identify:

GateRuleChange
→ ActionSetChange
→ LedgerBackreaction. (27.62)

Such designs can help separate:

- market shock;

- rule shock;
- authority shock;
- processing-capacity shock.

The source draft identifies regulatory changes, trading-halt rules, margin changes, and accounting transitions as especially useful natural experiments because the gate or authority changes externally.

27.23 Stage 4 — Cross-institution transport

Estimate the model across:

- different brokers;
- different clearing systems;
- different asset classes;
- different legal entities;
- different regulatory regimes.

A claim survives transport when:

Performance_P′(T_P→P′Model_P) remains above the declared threshold. (27.63)

Failure should localize the claim:

Claim_global → Claim_local,P. (27.64)

27.24 Predictive metrics

For binary outcomes:

- area under ROC curve;
- precision–recall area;
- Brier score;
- log loss;
- calibration slope;
- calibration intercept.

For closure time:

- concordance index;
- integrated Brier score;
- time-dependent calibration;

- survival-curve error.

For continuous buffer movement:

- mean absolute error;
- root mean square error;
- tail error;
- directional accuracy;
- conditional quantile loss.

Predictive metrics should always be reported out of sample.

27.25 Diagnostic metrics

Prediction is not the only target.

Define residual recall:

ResidualRecall
:= FutureFailureModes anticipated in $\mathscr{R}_k$
÷ Total FutureFailureModes. (27.65)

Define false-closure rate:

FalseClosureRate
:= ReportedClosed but ProtocolOpen
÷ Total ReportedClosed. (27.66)

Define charge-reconciliation accuracy:

ChargeAccuracy
:= 1 − |r_q|/[TotalChargeFlow + ε]. (27.67)

Define transport accuracy:

TransportAccuracy_AB
:= 1 − ‖r_AB‖/[‖S_B‖ + ε]. (27.68)

Define A-B Fixedness stability:

ABFixStability
:= Proportion of observations preserving invariant and trace within tolerance. (27.69)

27.26 Gate metrics

A gate model should distinguish:

- candidate breach;
- admitted call;

- resolved call;
- defaulted closure.

Define call-admission precision:

Precision_gate
:= ValidAdmittedCalls/AllPredictedCalls. (27.70)

Define call-admission recall:

Recall_gate
:= ValidAdmittedCalls/AllActualAdmittedCalls. (27.71)

Define gate-separation gain:

G_gate
:= Pr(PersistentConsequence | Breach,Gate)
− Pr(PersistentConsequence | Breach,NoGate). (27.72)

The source empirical contract expects a genuine gate to separate persistent events from ungated candidates.

27.27 Complexity cost

The complexity penalty should include:

C_model
:= α_1ParameterCount

- α_2DataRequirement
- α_3CalibrationInstability
- α_4InterpretationCost
- α_5OperationalCost
- α_6GovernanceRisk. (27.73)

A model that improves fit slightly while requiring fragile, opaque, or unavailable data may have lower total utility.

27.28 Robustness tests

The study should vary:

- CAPM estimation window;
- ERP estimate;
- risk-free baseline;
- cash-flow forecast;

- collateral amplitude convention;
- haircut rule;
- call threshold;
- closure tolerance;
- frame graph;
- residual weights;
- normalization;
- event horizon.

A strong result should survive reasonable protocol changes.

A claim that survives only one arbitrary normalization should be narrowed.

27.29 Negative controls

Use variables that should not predict closure after proper alignment.

Examples include:

- future ledger data incorrectly shifted backward;
- random charge signs;
- shuffled frame mappings;
- unrelated instrument phase;
- arbitrary loop order with no institutional meaning.

The advanced features should outperform these negative controls.

Otherwise the result may be a generic complexity artifact.

27.30 Placebo gates

Construct placebo thresholds that lack contractual authority.

Compare:

Actual Margin Gate. (27.74)

Placebo Numerical Threshold. (27.75)

If both predict the same consequence, the claimed authority effect may be weak.

The gate hypothesis expects:

Pr(ConsequentialClosure | ActualGate)

Pr(ConsequentialClosure | PlaceboCrossing). (27.76)

27.31 Causal versus predictive interpretation

A variable may predict a call without causing it.

For example:

Q may predict buffer erosion because it summarizes valuation geometry.

It does not necessarily cause the call.

Likewise:

κ_loop may predict failure because it measures institutional disagreement.

It may not be the original cause of the disagreement.

Therefore every result should be labelled as:

- descriptive;
- predictive;
- causal;
- normative;
- engineering.

Prediction should not be silently promoted into mechanism.

27.32 Mixed-method evidence

Some variables cannot be recovered from prices alone.

Qualitative evidence may be needed for:

- legal authority;
- waiver;
- dispute;
- collateral eligibility;
- gate reasoning;
- identity conversion;
- closure classification.

A mixed record may contain:

Quantitative State

- Structured Gate Evidence
- Residual Audit. (27.77)

Qualitative evidence must still be:

- sourced;
- timestamped;
- typed;
- challengeable.

The source empirical programme specifically permits structured qualitative evidence for legal authority, institutional decisions, ambiguous branches, and model revisions.

27.33 Missing data as residual

Missing data should not automatically be imputed away.

Define:

$\mathscr{R}$_missing := unavailable state required by the declared model. (27.78)

Examples include:

- missing collateral timestamp;
- unavailable legal status;
- unknown execution price;
- missing ledger entry;
- unclear authority.

Imputation may be used for estimation, but the original missingness should remain in the residual register.

27.34 Model-version ledger

For each model version v, preserve:

L_model,v
:=
(Protocol_v,Variables_v,Parameters_v,TrainingWindow_v,Tests_v,Failures_v,Revision_v). (27.79)

When the model changes:

Model_v → Model_v+1. (27.80)

The prior version should remain accessible.

This prevents retrospective rewriting of:

- Q definition;
- charge convention;
- gate threshold;
- closure criteria;
- frame transport.

27.35 Research decision rule

After testing, choose the retained model:

$M^* = \arg\max_j U_j.$ (27.81)

subject to:

- calibration validity;
- leakage control;
- residual disclosure;
- falsifier compliance;
- interpretability threshold.

The model may be retained for one purpose and rejected for another.

For example:

- M_7 may improve governance;
- M_2 may be best for prediction;
- M_0 may remain best for operational simplicity.

There is no requirement that one model dominate every objective.

27.36 Possible empirical outcomes

Outcome A — Full support

Q, charge, spinor split, gauge residual, mass, and Γ structure all add reproducible value.

This would support continued development of the Financial Gauge–Dirac system.

Outcome B — Partial support

Q and action–ledger closure add value, but gauge and Dirac layers fail.

Retain:

Complex Margin Geometry + Two-Stage Closure Model. (27.82)

Outcome C — Governance-only support

Advanced variables do not improve prediction but improve:

- reconciliation;
- audit;
- residual recall;
- false-closure detection.

Retain the framework as a governance architecture.

Outcome D — Conventional-model dominance

Ordinary margin variables perform equally well.

Reduce to the conventional model.

Outcome E — Protocol instability

Results depend excessively on arbitrary definitions.

Revise the measurement architecture before further prediction claims.

27.37 Success through disciplined failure

The empirical programme succeeds even when several advanced branches fail, provided that it produces:

- clear reduction;
- preserved trace;
- typed residual;
- better definitions;
- stronger benchmarks;
- narrower claims.

The source architecture treats such reduction as scientific discipline rather than defeat.

27.38 Minimum empirical contract

A study claiming support for the Financial Gauge–Dirac architecture should commit to:

1. declared protocol before outcome evaluation;
2. evidence-time consistency;

3. preserved identity lineage;
4. separate measurement of action and ledger;
5. explicit gate authority;
6. residual preservation;
7. fixed charge conventions;
8. declared frame maps;
9. simpler benchmarks;
10. prospective falsifiers;
11. out-of-sample testing;
12. explicit reduction after failure.

In compressed form:

No Hidden Boundary.
No Future Leakage.
No Ungated Event.
No Residual Erasure.
No Silent Revision.
No Advanced Model Without a Simpler Benchmark. (27.83)

27.39 Result of Part X

The Financial Gauge–Dirac proposal is now scientifically bounded.

Its strongest version is not accepted merely because the equations are coherent.

It must pass a sequence of independent tests:

Complex State
→ Incremental Q Test
→ Phase Test
→ Charge Test
→ Spin Test
→ Transport Test
→ Gauge Test
→ Mass Test
→ Γ Test
→ Recursive Test
→ Full Utility Test. (27.84)

Failure at one layer produces reduction rather than conceptual collapse.

The central scientific rule is:

Claim Strength
≤ Weakest Mandatory Demonstrated Support. (27.85)

The next Part will interpret the completed architecture more directly:

- what financial charge means;
- what financial spin means;
- what financial gauge transport means;
- what “Dirac” can responsibly mean;
- where the physics comparison ends;
- what the final general thesis of constraint-bearing finance should be.

Part XI — Interpretation, Limits, and Final Thesis

28. What Financial Charge Means

28.1 Charge is transformation memory

The word **charge** should not initially be understood as a mysterious substance stored inside a financial instrument.

Its minimum meaning is:

> Financial charge is a compact description of how a bounded financial identity is oriented and how it responds under a declared field, transport rule, or interaction vertex.

Let x be a bounded financial identity.

Let g be an admissible transformation.

Let $\rho_x(g)$ be the representation carried by x.

Then:

$$x \to \rho_x(g)x. \qquad (28.1)$$

Two objects may appear numerically similar at one moment while responding differently under the same transformation:

$$x_1 \to \rho_1(g)x_1. \qquad (28.2)$$

$$x_2 \to \rho_2(g)x_2. \qquad (28.3)$$

If:

$$\rho_1(g) \neq \rho_2(g), \qquad (28.4)$$

then x_1 and x_2 carry different transformation identities.

Charge is the compact memory of that difference.

The main Financial Standard Model expresses the principle directly:

Identity remembers what remains recognizable.

Charge remembers how that identity rotates.

28.2 “Rotation” does not require literal spatial rotation

The word rotation can refer to a change in orientation under a declared internal or institutional transformation.

For a one-dimensional phase representation:

$Z' = \exp(iq\alpha)Z.$ (28.5)

where:

- α = applied frame or field transformation;
- q = transformation weight;
- $q\alpha$ = phase accumulated by the identity.

Then:

$\theta' = \theta + q\alpha.$ (28.6)

The idealized charge relation is:

$q = \Delta\theta_{identity}/\Delta\alpha_{field}.$ (28.7)

Equation (28.7) defines a structural criterion.

It does not establish that every signed financial exposure is a gauge charge.

The candidate must still satisfy:

- carrier stability;
- declared field;
- sign convention;
- coupling law;
- transport law;
- vertex rule;
- residual register.

28.3 Charge is not general financial risk

The term risk may include:

- volatility;
- default probability;
- drawdown;
- liquidity loss;

- model uncertainty;
- legal uncertainty;
- operational failure;
- tail loss.

Charge is narrower.

It asks:

> What relational orientation does this identity carry under a particular field or transformation?

Therefore:

Financial Risk ≠ Financial Charge. (28.8)

A long asset claim may carry positive asset charge while possessing low or high risk.

A borrower may carry negative funding charge while possessing excellent credit quality.

An option writer may carry a contractual delivery obligation even when the probability of exercise is low.

Charge identifies orientation.

Risk describes uncertainty and consequence.

28.4 Charge is not Q

The Complex CAPM pressure coordinate is:

$$Q = \sqrt{(A^2 - R^2)}. \quad (28.9)$$

It describes the conjugate coordinate implied by a declared valuation filter.

Structural charge is:

q = stable relational transformation orientation. (28.10)

Therefore:

Q ≠ q. (28.11)

Q may affect the magnitude of the account's response to phase movement.

But Q does not determine whether the subject is:

- asset holder or asset deliverer;
- lender or borrower;
- collateral taker or collateral provider;
- option holder or option writer.

Those distinctions belong to charge-like relational identity.

28.5 Charge is not beta

CAPM beta is:

β = systematic return sensitivity to the market factor. (28.12)

Beta contributes to coupling strength.

But:

$\beta \neq q_A$. (28.13)

Two long asset holders may carry the same structural asset orientation:

$q_{A,1} = q_{A,2} > 0$. (28.14)

while possessing different betas:

$\beta_1 \neq \beta_2$. (28.15)

Similarly, the same instrument may retain its contractual claim orientation while its estimated beta changes through time.

Therefore:

Structural Orientation = q. (28.16)

Statistical Factor Loading = β. (28.17)

Effective Response may depend on both:

$\text{Response}_{\text{eff}} \sim g\beta q$. (28.18)

The source framework makes this separation explicit: charge determines representation or direction, while coupling strength determines how intensely the field acts.

28.6 Charge is not leverage

Leverage is:

$\lambda = \text{GrossExposure}/\text{Equity}$. (28.19)

It amplifies the financial consequence of movement.

But:

$\lambda \neq q$. (28.20)

A lightly leveraged and highly leveraged long account may have the same asset-charge sign.

Their effective sensitivities differ.

A schematic relation is:

$q_{eff} = \Lambda q_{struct}$. (28.21)

where Λ may contain:

- leverage;
- beta;
- liquidity;
- concentration;
- collateral;
- funding;
- regime.

Thus:

Structural Charge remains relational. (28.22)

Effective Charge is state-dependent. (28.23)

28.7 Charge is not load

Load measures carried magnitude or inherited memory.

Charge measures coupling-relevant orientation.

A long and short position may carry similar absolute notional load:

$|Load_{long}| \approx |Load_{short}|$. (28.24)

But their market orientations are opposite:

$q_{long} = -q_{short}$. (28.25)

Likewise, a small option position may carry low notional load while possessing high convexity or volatility sensitivity.

Therefore:

Load ≠ Charge. (28.26)

The distinction prevents the Load column of the Periodic Grammar from being silently renamed as charge.

28.8 Charge is not economic value

A contractual claim may preserve its orientation while losing most of its market value.

For example:

$q_{claim,t_1} = q_{claim,t_2}$. (28.27)

while:

$R_{t_2} \ll R_{t_1}$. (28.28)

Therefore:

Charge Preservation ≠ Value Preservation. (28.29)

Likewise, extinguishing or transferring a claim can change charge while creating no immediate profit or loss.

Charge and value belong to different layers.

28.9 Charge is field-indexed

A financial subject may carry several charges:

$\vec{q} = (q_A, q_F, q_C, q_L, q_R, \ldots)$. (28.30)

Possible coordinates include:

- q_A = asset-claim orientation;
- q_F = funding orientation;
- q_C = collateral orientation;
- q_L = liquidity-demand orientation;
- q_R = regulatory or reporting obligation.

Neutrality in one field does not imply global neutrality.

For example:

$q_{net,A} = 0$ (28.31)

does not imply:

$q_{net,F} = 0$. (28.32)

A market-neutral portfolio may remain highly exposed to funding, collateral, liquidity, or legal fields.

Thus:

Neutrality Is Field-Relative. (28.33)

28.10 Charge is boundary-relative

Consider a loan.

Inside a boundary containing both lender and borrower:

$q_{claim} + q_{obligation} = 0$. (28.34)

Inside a boundary containing only the lender:

$q_{net} = +q_{claim}.$ (28.35)

Inside a boundary containing only the borrower:

$q_{net} = -q_{obligation}.$ (28.36)

Charge accounting therefore requires an explicit boundary.

A statement such as:

“The system is charge-neutral” (28.37)

is incomplete unless it declares:

- which system;
- which field;
- which counterparties;
- which claims;
- which lifecycle stage.

28.11 Charge is protocol-relative but not arbitrary

A protocol determines:

- the system boundary;
- the field;
- the sign convention;
- the transport rule;
- the relevant vertices.

Therefore:

q_P may differ from $q_{P'}$. (28.38)

But this does not make charge arbitrary.

Within one declared protocol, the rules must remain stable.

The researcher may not change:

- carrier;
- sign;
- field;
- boundary;

- vertex interpretation

after observing the outcome.

Protocol relativity is disciplined contextuality, not freedom to relabel.

28.12 Charge must survive ordinary transport

For an ordinary frame transport A → B:

q_B[T_AB(S_A)] = q_A(S_A). (28.39)

unless the edge includes a valid conversion vertex.

A market-to-collateral translation may change:

- admitted value;
- pressure coordinate;
- haircut;
- reporting amount.

It should not silently change:

- who owns the asset;
- who owes funding;
- who must provide collateral.

If charge changes without a gate:

ChargeTransportFailure = 1. (28.40)

28.13 Charge changes at declared vertices

A charge-changing event must be explicit.

Examples include:

- issuance;
- transfer;
- settlement;
- exercise;
- novation;
- collateral seizure;
- default;

- redemption;
- extinguishment.

The generic balance rule is:

$$\Sigma q_{in} + q_{gate} = \Sigma q_{out} + r_q. \qquad (28.41)$$

A nonzero r_q may identify:

- missing claim;
- duplicated obligation;
- disputed ownership;
- unrecorded extinguishment;
- incomplete novation;
- failed settlement.

The main article's charge audit requires a candidate charge to identify its carrier, field, sign, units, mediator, transport, conversion gate, conservation status, quantization status, and residual. A candidate unable to do so should remain a sensitivity or descriptive coordinate.

28.14 Minimum responsible definition

The strongest currently defensible definition is:

> **Financial charge is a protocol-bound, field-indexed relational orientation carried by a bounded financial identity and constrained by declared transformation, transport, and vertex rules.**

In compact form:

$$q_{fin} := \text{Carrier}$$

- RelationalOrientation
- FieldCoupling
- TransportRule
- VertexRule
- ResidualRegister. (28.42)

This is an operational research definition.

It is not a claim that financial charge is physically identical to electric charge.

---

29. What Financial Spin Means

29.1 Spin is closure memory

Charge records how an identity responds.

Spin records how it returns.

The main source describes:

Financial Closure
= Outward Action

- Independent Ledger Return. (29.1)

Its most important practical distinction is:

Action Completed ≠ Identity Closed. (29.2)

A trade is not closed at execution.

An option is not closed at exercise instruction.

A margin call is not closed when the call is issued.

A default is not closed when the trigger is declared.

An accounting event is not closed when the economic loss occurs.

The return cycle matters.

29.2 Spin is not market rotation

A price may move around the Complex CAPM plane:

$Z = Ae^{i\theta}$. (29.3)

That rotation is a valuation-state movement.

Financial spin is a different concept.

It concerns:

Potential
→ Outward Action
→ Unresolved Consequence
→ Ledger Return
→ Accountable Identity. (29.4)

Therefore:

CAPM Phase Rotation ≠ Financial Spin. (29.5)

The two may interact.

A CAPM phase movement can trigger a margin gate.

But the subsequent settlement and ledger return constitute the spin-like closure cycle.

29.3 Spin is not bullish versus bearish direction

Bullish and bearish directions concern market orientation.

They belong more naturally to:

- charge;
- sign;
- directional exposure;
- motion.

Financial spin concerns whether the subject has completed the required return cycle.

A bullish position may be ledger-closed.

A bearish position may be ledger-closed.

Either may become spin-open after an unsettled transaction or margin event.

Therefore:

Market Direction ≠ Closure State. (29.6)

29.4 Spin is not merely two variables

A financial object should not receive a spinor merely because it has:

$x = (x_1, x_2)$. (29.7)

The two components must represent an irreducible closure distinction.

The source admission rule requires:

SpinorEligible_P(x)
:= Identity
∧ Action
∧ Consequence
∧ IndependentLedger
∧ DoubleClosure
∧ Residual
∧ FrameTransport. (29.8)

A price and volume pair does not automatically satisfy this rule.

An executed trade and its settlement ledger may.

A margin call and its collateral-resolution ledger may.

A default declaration and recovery process may.

The source explicitly warns that a spinor should not be assigned merely because an object has two measured variables.

29.5 The minimum financial spinor

The minimum representation is:

$$\Psi_S = \begin{bmatrix} \psi_{action} \\ \psi_{ledger} \end{bmatrix}. \qquad (29.9)$$

where:

- ψ_{action} = outward intervention or status-changing act;
- ψ_{ledger} = trace, reconciliation, audit, residual integration, and future-conditioning consequence.

The action component changes the world.

The ledger component determines whether the changed world can be inherited as an accountable state.

29.6 The double cycle

The generic cycle is:

$$\Psi_S \rightarrow -\Psi_S \rightarrow \Psi_S'. \qquad (29.10)$$

Here:

- Ψ_S = pre-action accountable identity;
- $-\Psi_S$ = action-complete but ledger-incomplete identity;
- Ψ_S' = post-return accountable successor identity.

The final state need not be numerically identical:

$$\Psi_S' \neq \Psi_S. \qquad (29.11)$$

But it may remain identity-equivalent under protocol P:

$$\Psi_S' \simeq_P \Psi_S. \qquad (29.12)$$

The purpose of closure is not to reverse history.

It is to preserve continuity through history.

This exact interpretation appears in the source Financial Standard Model.

29.7 The minus sign

The intermediate minus sign does not mean:

- negative monetary value;

- short market exposure;
- bearish direction;
- physical antiparticle state.

It means:

> The identity has performed an outward act but carries an unresolved obligation to return through the relevant ledgers.

Define:

$s_{close} = +1$ for accountable closure. (29.13)

$s_{close} = -1$ for action-complete but ledger-open state. (29.14)

Then:

$+1 \to -1 \to +1.$ (29.15)

This is an operational closure sign.

29.8 Spin closure may transform identity

Suppose a margin account posts collateral successfully.

The same account survives.

Suppose it deleverages.

The same account survives with fewer units.

Suppose it defaults.

The margin identity may convert into a recovery identity.

Thus closure may take three forms:

Preservation

$K_{after} \simeq K_{before}.$ (29.16)

Modification

$K_{after} = \mathrm{Modified}(K_{before}).$ (29.17)

Valid conversion

$K_{after} = \mathrm{Conversion}(K_{before}, \mathrm{Gate}, \mathrm{Trace}).$ (29.18)

Spin closure requires lineage, not numerical sameness.

29.9 Spin residual

Define the expected ledger return:

$\psi\hat{}_ledger = ReturnMap(\psi_action).$ (29.19)

The spin residual is:

$\mathscr{R}_spin = Diff(\psi_ledger,\psi\hat{}_ledger).$ (29.20)

Examples include:

- unsettled trade;
- missing collateral;
- unrecognized liability;
- unpaid shortfall;
- unrecorded loss;
- unreconciled accounting difference;
- failed legal perfection.

A large residual indicates that outward action has not yet become an accountable successor identity.

The source defines the same object as the difference between the ledger surface and the return map of the action surface.

29.10 Spin closure does not require a favorable outcome

An account may close through:

- full collateral cure;
- loss-making liquidation;
- valid default;
- recovery allocation;
- contract extinguishment.

Therefore:

$SpinClosed = 1$ (29.21)

does not imply:

$Profit > 0.$ (29.22)

or:

$Residual = 0.$ (29.23)

Closure means that the event has become traceable, reconciled, and recursively inheritable.

It does not mean that no adverse consequence remains.

29.11 Spin closure and residual coexist

The mature closure relation is:

Closure
= Admitted Trace

- Preserved Nonclosure. (29.24)

A validly resolved margin event may still leave:

- realized loss;
- lower liquidity;
- higher funding spread;
- reduced risk limits;
- reputational damage;
- future collateral restrictions.

The event is closed because these consequences have become part of the ledger.

It is not closed because they have disappeared.

The Periodic Grammar explicitly states that commitment does not exhaust the world and treats residual as a future-bearing object.

29.12 Spin creates recursive inheritance

Once ledger return is completed:

$L_{k+1} = L_k \oplus \text{Trace}_k \oplus \text{Residual}_k$. (29.25)

The new ledger affects:

- future leverage;
- future haircut;
- future permissions;
- future funding cost;
- future effective charge;
- future mass;
- future closure capacity.

Therefore:

$$\text{Spin Closure}_k \rightarrow \text{Conditions of Motion}_{k+1}. \qquad (29.26)$$

This is why financial spin is more than a workflow metaphor.

It connects action, memory, and future admissibility.

29.13 Operational spin versus exact spin-½

The operational claim is:

$$\text{Financial Spin_op} := \text{Action–Ledger Double Closure}. \qquad (29.27)$$

A stronger representation uses:

$$S_C(\varphi) = \exp(-i\varphi\sigma/2). \qquad (29.28)$$

with:

$$S_C(2\pi) = -I. \qquad (29.29)$$

$$S_C(4\pi) = I. \qquad (29.30)$$

This exact double-cover property is mathematical.

Its financial interpretation remains hypothetical.

The stronger claim survives only if:

- 2π and 4π states correspond to measurably distinct closure conditions;
- the composition law adds value;
- ordinary two-stage models are insufficient.

Otherwise retain only operational double closure.

29.14 Minimum responsible definition

The strongest currently defensible definition is:

> **Financial spin is the protocol-bound closure class of a bounded financial identity whose outward action creates consequential obligations that require an independent return-to-ledger cycle before accountable successor identity is restored.**

In compact form:

$$\text{Spin_fin} := \text{Identity}$$

- OutwardAction

- UnresolvedConsequence
- IndependentLedgerReturn
- Residual
- AccountableSuccessor. (29.31)

---

30. What Financial Gauge Means

30.1 Gauge begins with multiple legitimate descriptions

The same financial identity may possess:

- market value;
- collateral value;
- liquidation value;
- accounting value;
- risk exposure;
- legal status;
- regulatory exposure.

These values need not be equal.

The question is not:

> Which frame is universally correct?

The question is:

> Can the same bounded identity be transported lawfully among the frames while preserving the appropriate invariant?

This is the starting point of financial gauge structure.

30.2 Gauge is not ordinary relabelling

Changing a column name is not gauge transport.

Moving a number from one report to another is not necessarily gauge transport.

A valid gauge-like system requires:

- declared source frame;
- declared target frame;

- identity kernel;
- transport map;
- connection;
- covariance law;
- residual;
- loop test.

Without these:

Gauge → Multi-Frame Comparison. (30.1)

30.3 The identity invariant

Let K(S) be the financial identity kernel.

A valid transport A → B satisfies:

K_B[T_AB(S_A)] ≃ K_A(S_A). (30.2)

The invariant may include:

- instrument identity;
- signed quantity;
- owner;
- obligor;
- maturity;
- payoff right;
- execution lineage;
- collateral agreement;
- settlement status.

Value is generally covariant, not invariant.

Thus:

R_A ≠ R_B may be valid. (30.3)

But:

K_A ≠ K_B without conversion is failure. (30.4)

30.4 The connection

The transport rule is:

$$T_{AB} : S_A \to \hat{S}_B. \qquad (30.5)$$

Its connection may include:

$$\mathcal{A}_{AB} = (\mathrm{FX},\mathrm{Discount},\mathrm{Haircut},\mathrm{Timing},\mathrm{Netting},\mathrm{Recognition},\mathrm{LegalRule}). \qquad (30.6)$$

The expected target state is:

$$\hat{S}_B = T_{AB}(S_A;\mathcal{A}_{AB}). \qquad (30.7)$$

The observed target state is:

$$S_B^{\mathrm{obs}}. \qquad (30.8)$$

The transport residual is:

$$r_{AB} = S_B^{\mathrm{obs}} - \hat{S}_B. \qquad (30.9)$$

A residual is meaningful only after known frame differences have been included in the connection.

30.5 Gauge covariance

Let local frame redescriptions be:

$$S_A' = G_A S_A. \qquad (30.10)$$

$$S_B' = G_B S_B. \qquad (30.11)$$

The transport must transform as:

$$T_{AB}' = G_B T_{AB} G_A^{-1}. \qquad (30.12)$$

Then:

$$T_{AB}' S_A' = G_B(T_{AB} S_A). \qquad (30.13)$$

This means that lawful changes of representation do not change the substantive transported relation.

Examples include:

- reporting currency change;
- unit rescaling;
- gross-to-net presentation;
- desk reassignment;
- consolidation convention.

If event status changes merely because a presentation convention changes, covariance has failed.

30.6 Gauge is not CAPM phase

CAPM phase is:

$\theta_M = \arccos(R_M/A).$ (30.14)

It is a valuation-state coordinate.

Gauge phase χ_A is a local representation parameter.

Therefore:

$\theta_M \neq \chi_A.$ (30.15)

A change in θ_M may represent a real economic change.

A pure gauge transformation should not change gauge-invariant financial observables.

Confusing these two phases would convert real risk into a notation artifact.

30.7 Gauge transport is not double-entry accounting

Double-entry accounting enforces balanced entries within an accounting framework.

Gauge transport concerns:

- the same identity across several frames;
- differences in local representation;
- lawful mappings among them;
- invariants preserved by the mappings.

Double entry may be one ledger mechanism inside a gauge-transport architecture.

But:

Double Entry ≠ Full Gauge Transport. (30.16)

A transaction may balance inside accounting while remaining inconsistent with:

- market position;
- legal ownership;
- collateral ledger;
- risk system;
- regulatory report.

30.8 A-B Fixedness

A-B Fixedness means that two frames can still identify the same event or obligation after

translation.

A financial version is:

ABFix_P(S;A,B)
⇔ d_B[T_AB(S_A),S_B] ≤ ε_AB
∧ Inv_B[T_AB(K_A)] = Inv_A(K_A)
∧ Rec_AB(S)
∧ Residual_AB disclosed. (30.17)

It requires:

- frame map;
- compatible observation;
- accessible record;
- invariant relation;
- residual honesty.

The generalized Dirac source uses exactly these requirements for cross-frame identity.

30.9 Loop holonomy

Consider a closed frame loop:

A → B → C → A. (30.18)

The loop transporter is:

H_ABC = U_ACU_CBU_BA. (30.19)

The expected return is:

Ŝ_A^return = H_ABC^PS_A^start. (30.20)

The governed loop residual is:

$\mathscr{R}$_loop = S_A^return,obs − Ŝ_A^return. (30.21)

A flat loop satisfies:

$\mathscr{R}$_loop ≈ 0. (30.22)

A nonzero loop residual may indicate:

- timing mismatch;
- stale marks;
- missing trade;
- hidden funding cost;
- inconsistent netting;

- title disagreement;
- accounting error;
- unrecorded conversion.

30.10 Curvature is not every economic difference

Suppose a lawful loop incurs a $100 transaction fee.

Then:

Value_return = Value_start − $100. (30.23)

If the expected loop operator includes the fee:

$\mathscr{R}$_loop = 0. (30.24)

Therefore:

Economic Cost ≠ Gauge Curvature. (30.25)

Curvature refers to unexplained or path-dependent mismatch after authorized transformations have been included.

30.11 Gauge structure and residual honesty

A system may create apparent agreement by:

- overwriting one frame;
- deleting exceptions;
- backdating entries;
- forcing target values;
- suppressing disputes.

This is not gauge closure.

A genuine gauge-like architecture preserves:

- source state;
- transport rule;
- target state;
- difference;
- unresolved residual.

Therefore:

Apparent Agreement without Trace

≠ A-B Fixedness. (30.26)

The generalized framework treats hidden residual—not residual itself—as failure.

30.12 Minimum responsible definition

The strongest currently defensible definition is:

> **Financial gauge structure is the protocol-governed transport of a bounded financial identity across multiple legitimate frames, preserving a declared invariant kernel while allowing frame-local values to transform covariantly and recording any remaining loop residual.**

In compact form:

Gauge_fin
:= Frames

- IdentityInvariant
- Connection
- CovariantTransport
- LoopTest
- ResidualHonesty. (30.27)

---

31. What “Dirac” Can Responsibly Mean in Finance

31.1 The structural archetype

The physical Dirac equation describes relativistic spin-½ particles.

The present financial article does not claim that leveraged accounts are physical fermions.

The generalized source proposes a broader structural archetype:

> A system carrying identity must propagate across changing frames without dissolving, while coupled internal components and a mass-like term constrain that propagation.

Its macro equation is:

$(i\Gamma^{a}\nabla^{p}_a - M_B)\Psi_B = \mathscr{R}_P.$ (31.1)

The source explicitly states that this is not an equation of particles but an equation of accountable identity.

31.2 The financial specialization

The proposed financial equation is:

$$(i\mathcal{D}\!\!\!/_{fin} - M_S)\Psi_S = \mathscr{R}_S. \qquad (31.2)$$

where:

$$\Psi_S = \begin{bmatrix} \psi_{action} \\ \psi_{ledger} \end{bmatrix}. \qquad (31.3)$$

and:

$$\mathcal{D}\!\!\!/_{fin} = \Gamma^0 \nabla_\tau \hat{P} + c_P \Gamma^1 \mathfrak{D}_G. \qquad (31.4)$$

The complete system also includes gates:

$$\mathcal{S}^+_k = \mathcal{J}_k(\mathcal{S}^-_k, L_k) + \mathscr{R}_{jump,k}. \qquad (31.5)$$

and recursive ledger update:

$$L_{k+1} = U_L(L_k, Trace_k, Residual_k). \qquad (31.6)$$

Thus the Financial Gauge–Dirac equation is the continuous kernel of a larger hybrid system.

31.3 Why first-order propagation matters

A Dirac-like model is first-order.

The local next state depends on:

- current financial identity;
- current field;
- current connection;
- current mass;
- current residual.

Schematically:

$$\Psi(\tau + \Delta\tau) = \Psi(\tau) + \Delta\tau F[\Psi(\tau), \Theta(\tau)] + O(\Delta\tau^2). \qquad (31.7)$$

First-order propagation is appropriate when the financial subject evolves:

- event by event;
- margin call by margin call;
- settlement tick by settlement tick;
- ledger update by ledger update.

But first-order structure alone does not justify the word Dirac.

Many ordinary state-space models are first-order.

31.4 Why the doublet matters

The Dirac-like claim becomes stronger when:

- ψ_action and ψ_ledger are independently measurable;
- neither can be discarded;
- their coupling generates observable closure modes;
- a scalar model loses material information.

If:

ψ_ledger = Tψ_action for every admissible state, (31.8)

and no independent residual exists, then:

Ψ_S → x_S. (31.9)

The spinor and Dirac claims should be removed.

31.5 Why Γ operators matter

The Γ operators should:

- distinguish action and ledger orientation;
- mix the components;
- constrain first-order propagation;
- produce an interpretable squared relation.

A candidate representation is:

$\Gamma^0 = \sigma_z$. (31.10)

$\Gamma^1 = i\sigma_y$. (31.11)

with:

$\{\Gamma^a,\Gamma^b\} = 2\eta^{ab}I$. (31.12)

This algebra is mathematically exact for the selected matrices.

Its financial meaning is not automatically exact.

The representation is justified only if the algebra produces stable, measurable consequences.

31.6 Why mass matters

The mass operator is:

M_S =
[
m_I + m_Δ m_C
m_C m_I – m_Δ
]. (31.13)

Its intended roles are:

- m_I = common identity inertia;
- m_C = action–ledger binding;
- m_Δ = component asymmetry.

The source framework defines mass as the cost of changing while remaining recognizable and distinguishes institutional identity mass from market-depth resistance.

A financial mass claim is responsible only when:

- transformation cost is measurable;
- it is identity-specific;
- it affects propagation;
- it is not merely renamed illiquidity or leverage.

31.7 Why the covariant derivative matters

The derivative is not merely:

∂_τΨ. (31.14)

It includes governed frame transport:

D_τ^G
= ∂_τ + iΣ_rg_r𝒜_r,τQ̂_r. (31.15)

The connection determines how the same charged identity is compared across frames.

Without a lawful connection:

Gauge-Covariant Derivative
→ Ordinary Time Derivative. (31.16)

Then the gauge claim disappears.

31.8 Why residual belongs on the right-hand side

The physical free Dirac equation is often written with zero on the right-hand side.

A financial system normally contains:

- incomplete information;

- disputes;
- stale data;
- timing mismatch;
- model error;
- settlement failure;
- unrecognized consequence.

Therefore:

$\mathscr{R}_S \neq 0$ in general. (31.17)

Residual is not an embarrassing defect to hide.

It is part of the financial object.

The generalized source defines residual as the unclosed difference after projection, action, transport, gate, and ledger and states that residual is not automatically failure; hidden residual is failure.

31.9 Why the squared equation matters

Starting from:

$(i\not{D}_{fin} - M_S)\Psi_S = \mathscr{R}_S,$ (31.18)

the second-order form is:

$[\not{D}_{fin}^2 + M_S^2 - i[\not{D}_{fin}, M_S]]\Psi_S$
$= -(i\not{D}_{fin} + M_S)\mathscr{R}_S.$ (31.19)

Expanding:

$[(\nabla_\tau \hat{} P)^2$
$- c_P^2 \mathfrak{D}_G^2$

- M_S^2
- CurvatureCoupling
- MassVariation]Ψ_S
  = ResidualDynamics. (31.20)

The strong Dirac claim requires these terms to correspond to observable phenomena such as:

- closure acceleration;
- action–ledger oscillation;
- synchronized modes;
- opposed modes;
- curvature-sensitive residual;

- mass-dependent closure time.

If equation (31.20) adds no measurable structure, Γ should be reduced to an ordinary coupling matrix.

31.10 Why the model is hybrid rather than purely Dirac

Margin systems contain discontinuous gates.

The central continuous equation applies between events.

At events:

$\Psi^{+} = G_k \Psi^{-} + \eta_k.$ (31.21)

Therefore the full architecture is:

Continuous Gauge–Dirac Propagation

- Discrete Gate Jumps
- Ledger Recursion. (31.22)

A pure differential equation is insufficient because:

- margin calls activate obligations;
- liquidation changes authority;
- default changes identity class;
- settlement changes legal status.

The hybrid nature is not a correction added after the theory.

It is part of the financial ontology.

31.11 Dirac does not mean unitary

Financial systems contain irreversible processes:

- transaction cost;
- slippage;
- default;
- taxation;
- legal extinguishment;
- write-off;
- loss.

Therefore:

$$\|\Psi(\tau_2)\| \neq \|\Psi(\tau_1)\| \quad \text{in general.} \qquad (31.23)$$

The model is not required to preserve a quantum probability norm.

Its relevant continuity concerns:

- identity;
- charge balance;
- trace;
- residual;
- lawful conversion.

A non-Hermitian or dissipative extension may eventually be required.

31.12 Dirac does not mean universal Lorentz symmetry

The chosen Γ algebra uses a 1+1 signature:

$$\eta = \mathrm{diag}(1,-1). \qquad (31.24)$$

This creates a mathematically useful distinction between:

- ordered closure time;
- action–ledger transport direction.

It does not establish physical spacetime.

The financial protocol may possess:

- asymmetric frames;
- irreversible gates;
- nonuniform capacities;
- state-dependent connections.

Thus any Lorentz-like language must remain restricted and operational.

31.13 Dirac does not imply antimatter

A negative closure state:

$$-\Psi \qquad (31.25)$$

does not denote an antiparticle.

It denotes unresolved return obligation.

Similarly, a short position is not the antiparticle of a long position.

Financial oppositions may include:

- claim versus obligation;
- holder versus writer;
- lender versus borrower;
- asset versus liability.

These relational dualities should not be promoted automatically into physical antiparticle correspondence.

31.14 Dirac does not imply quantum mechanics

The complex CAPM state:

$$Z = R + iQ \qquad (31.26)$$

and the action–ledger spinor:

$$\Psi = \begin{bmatrix} \psi_A \\ \psi_L \end{bmatrix} \qquad (31.27)$$

do not by themselves imply:

- quantum superposition;
- wavefunction collapse;
- Planck-scale quantization;
- physical uncertainty relations;
- quantum entanglement;
- microscopic fermionic statistics.

The architecture borrows mathematical organization.

It does not inherit every physical interpretation attached to similar symbols.

31.15 Weak versus strong financial Dirac claims

Weak claim

A charged, multicomponent financial identity can be represented through a first-order, frame-covariant, mass-constrained, residual-bearing equation.

Intermediate claim

The representation improves:

- closure diagnosis;
- transport auditing;
- margin-event modelling;
- residual classification.

Strong claim

The Γ algebra, mass modes, curvature terms, and squared equation produce reproducible financial laws unavailable to simpler models.

The current article establishes the weak claim as a coherent construction.

It proposes tests for the intermediate and strong claims.

It does not claim that they have already passed.

31.16 Minimum responsible definition

The strongest currently defensible definition is:

> **A Financial Gauge–Dirac system is a hybrid first-order model of a charged, multicomponent financial identity whose action and ledger surfaces propagate through governed financial frames under identity-preserving inertia, authoritative gates, and explicit residual.**

In compact form:

Financial Gauge–Dirac
:= ChargedIdentity

- ActionLedgerDoublet
- CovariantTransport
- IdentityMass
- FirstOrderPropagation
- GateJumps
- Residual
- LedgerRecursion. (31.28)

---

32. Final Thesis

32.1 The question revisited

The article began with a general question:

> Do nonlinear elements or constraints added to ordinary finance naturally introduce

charge and spin?

The answer is now more precise.

No

Nonlinearity alone does not create charge.

A nonlinear function may contain no relational identity.

Constraint alone does not create spin.

A numerical bound may create no authoritative action or ledger return.

Yes, under stronger conditions

Charge becomes natural when a constraint is attached to:

- a bounded identity;
- stable rights and obligations;
- field-indexed orientation;
- transformation and transport rules;
- conversion vertices.

Spin becomes natural when breach creates:

- an authoritative event;
- consequential outward action;
- unresolved obligations;
- independent return through ledgers;
- residual if the return fails.

Therefore:

Nonlinearity ⇏ Charge or Spin. (32.1)

Constraint ⇏ Charge or Spin. (32.2)

Identity-Bearing Relational Constraint → Candidate Charge. (32.3)

Authoritative Action + Independent Ledger Return → Candidate Spin. (32.4)

32.2 The causal order

The proposed causal sequence is:

Bounded Identity
→ Relational Orientation

→ Constraint
→ Conditional Obligation
→ Authoritative Gate
→ Outward Action
→ Ledger Return
→ Residual
→ Recursive Future State. (32.5)

Nonlinearity usually appears later through:

- threshold activation;
- complementarity;
- leverage;
- branching;
- market impact;
- path dependence;
- recursive parameter change.

Therefore:

Identity-Bearing Constraint
→ Transformation Memory
→ Nonlinear Recursive Dynamics. (32.6)

This is stronger and more defensible than:

Nonlinearity → Charge and Spin. (32.7)

32.3 The four memories

The Financial Gauge–Dirac system can be summarized through four forms of memory.

Identity memory

Identity remembers what remains recognizable.

Charge memory

Charge remembers how the identity transforms and couples.

Spin memory

Spin remembers how the identity returns through action and ledger.

Mass memory

Mass remembers the cost of remaining the same identity while changing.

The source Financial Standard Model develops precisely this sequence.

32.4 The role of the gate

The gate is the boundary between possibility and admitted history.

Before the gate:

- pressure may exist;
- warning may exist;
- candidate breach may exist;
- obligations may remain dormant.

At the gate:

- authority acts;
- a decision is committed;
- a trace is written;
- a branch becomes consequential.

After the gate:

- obligations must return through ledgers;
- residual must be preserved;
- the future state changes.

Thus:

Measured Condition ≠ Event. (32.8)

Threshold Crossing ≠ Admitted Gate. (32.9)

Gate Admission ≠ Complete Closure. (32.10)

The Periodic Grammar defines a gate as more than a threshold and insists that gate validity and later outcome remain separate.

32.5 The role of residual

Residual is:

$\mathscr{R}$_P = unresolved consequence under protocol P. (32.11)

It may include:

- model error;
- missing evidence;
- legal dispute;

- frame conflict;
- timing mismatch;
- unrecognized loss;
- incomplete settlement;
- residual shortfall.

Residual is not automatically failure.

The failure is:

Residual Denied or Hidden. (32.12)

A mature financial system writes:

Trace + Residual. (32.13)

It does not force every event into false completeness.

32.6 The role of the ledger

The ledger transforms past action into future constraint.

The recurrence is:

$L_{k+1} = L_k \oplus \text{Trace}_k \oplus \text{Residual}_k.$ (32.14)

The next period's parameters are:

$\Theta_{k+1} = U_\Theta(\Theta_k, L_{k+1}, \mathscr{R}_k).$ (32.15)

Therefore:

Past Closure
→ Future Admissibility. (32.16)

The market does not merely move through time.

It inherits its own committed history.

32.7 The margin account as calibration atom

The leveraged margin account was selected because it contains, in one narrow system:

- a CAPM-valued asset;
- a funding liability;
- a collateral agreement;
- structural asset, funding, and collateral orientations;
- a measurable constraint;

- an authoritative gate;
- several closure branches;
- independent ledgers;
- forced-action feedback;
- recursive state change.

Its minimum state is:

$$\mathcal{S}_t = (K,Z_M,Z_C,n,D,C,h,B,\vec{q},\Psi,L,\mathscr{R}). \qquad (32.17)$$

Its market state is:

$$Z_M = n(R_M + iQ_M). \qquad (32.18)$$

Its collateral state is:

$$Z_C = n(R_C + iQ_C). \qquad (32.19)$$

Its buffer is:

$$B = C + nhR_M - D. \qquad (32.20)$$

Its phase-to-buffer sensitivity is:

$$\partial B/\partial \theta_M = -nhQ_M. \qquad (32.21)$$

Its action–ledger state is:

$$\Psi = \begin{bmatrix} \psi_action \\ \psi_ledger \end{bmatrix}. \qquad (32.22)$$

Its continuous kernel is:

$$(i\not{D}_fin - M_S)\Psi = \mathscr{R}_cont. \qquad (32.23)$$

Its gate jump is:

$$\mathcal{S}^{+}_{k} = \mathcal{J}_margin(\mathcal{S}^{-}_{k},L_k) + \mathscr{R}_jump,_k. \qquad (32.24)$$

Its ledger recursion is:

$$L_{k+1} = U_L(L_k,Trace_k,Residual_k). \qquad (32.25)$$

This is the article's complete calibration system.

32.8 The full Financial Gauge–Dirac–Gate–Ledger law

The system can be compressed into six equations.

Valuation geometry

$$Z_M = R_M + iQ_M = A\exp(i\theta_M). \qquad (32.26)$$

Charge-covariant propagation

$$D_\tau^G = \partial_\tau + i\Sigma_r g_r \mathcal{A}_{r,\tau} \hat{Q}_r. \qquad (32.27)$$

Financial Gauge–Dirac kernel

$$[i\Gamma^0 \nabla_\tau^P + ic_P \Gamma^1 \mathfrak{D}_G - M_S]\Psi_S = \mathscr{R}_{cont}. \qquad (32.28)$$

Gate condition

$$g_j(\mathcal{S}) \geq 0. \qquad (32.29)$$

Gate jump

$$\mathcal{S}^+_k = \mathcal{J}_j(\mathcal{S}^-_k, L_k) + \mathscr{R}_{jump,k}. \qquad (32.30)$$

Recursive ledger update

$$(K_{k+1}, \vec{q}_{k+1}, \Psi_{k+1}, \Theta_{k+1}, L_{k+1}) = U_{close}(K_k, \vec{q}_k, \Psi_k, \Theta_k, L_k, Trace_k, Residual_k). \qquad (32.31)$$

Equations (32.26)–(32.31) constitute the proposed **Financial Gauge–Dirac–Gate–Ledger family**.

32.9 The master recurrence

The conceptual process is:

Charge
→ Field Coupling
→ Complex Phase Movement
→ Constraint Encounter
→ Gate Activation
→ Spinor Split
→ Branch Action
→ Gauge Transport
→ Ledger Return
→ Residual
→ Updated Charge and Mass. (32.32)

In one line:

$$(K_k, \vec{q}_k, \Psi_k, \Theta_k, L_k) \rightarrow G_k \rightarrow (Trace_k, Residual_k) \rightarrow (K_{k+1}, \vec{q}_{k+1}, \Psi_{k+1}, \Theta_{k+1}, L_{k+1}). \qquad (32.33)$$

32.10 The general constraint-bearing law

The wider theory can be expressed as:

Constraint-Bearing Finance
:= SmoothFinancialKernel

- BoundedIdentity
- RightsAndObligations
- AuthoritativeGates
- IndependentLedgers
- FrameTransport
- Residual
- RecursiveInheritance. (32.34)

Where these structures are absent, the full Gauge–Dirac system should not be used.

Where only some are present, the model should stop at the corresponding lower level.

32.11 The strongest coherent theoretical claim

The strongest coherent theoretical claim of this article is:

> **A class of financial systems can be reconstructed as protocol-bound identity-propagation systems in which contractual claims and obligations define charge-like transformation orientations, authoritative constraints activate gates, outward actions create ledger-return obligations, cross-frame transport preserves bounded identity, residual records nonclosure, and completed closure recursively changes the parameters governing the subject's future behaviour.**

This is a structural claim.

It is stronger than metaphor.

It is weaker than a validated universal law.

32.12 The strongest mathematical claim

The strongest mathematical claim is:

> **When action and ledger are irreducible components, frame transport is lawfully covariant, identity-preserving inertia is measurable, and local change is first-order, the system admits a candidate hybrid Gauge–Dirac representation.**

Formally:

$(i\mathcal{D}_{fin} - M_S)\Psi_S = \mathscr{R}_S$ between gates. (32.35)

$\Psi_S^{+} = G_k \Psi_S^{-} + \eta_k$ at gates. (32.36)

$L_{k+1} = \mathrm{Update}(L_k, \mathrm{Trace}_k, \mathrm{Residual}_k)$ after gates. (32.37)

This claim remains a proposed model family until tested.

32.13 The strongest empirical claim currently permitted

The strongest current empirical claim is not that the model works.

It is:

> The margin-account system provides measurable variables and falsifiable comparisons through which the charge, spin, gauge, mass, and Dirac layers can be tested separately against simpler financial models.

The empirical programme can test:

- Q incremental value;
- phase-to-buffer transmission;
- charge reconciliation;
- spinor split;
- loop residual;
- closure capacity;
- mass modes;
- recursive ledger effects;
- full complexity-adjusted utility.

No stronger empirical conclusion is presently supported by the conceptual construction alone.

32.14 The strongest engineering claim

The architecture may be valuable even before full predictive validation if it improves:

- separation of warning from event;
- distinction between action and closure;
- charge reconciliation;
- cross-frame identity mapping;
- residual preservation;
- false-closure detection;
- revision without trace erasure.

Its engineering principle is:

Do not declare a financial event closed merely because its outward action has completed.

Instead require:

Closure
= ValidAction

- LawfulTransport
- LedgerReturn
- ChargeReconciliation
- ResidualDisclosure. (32.38)

32.15 The lexical discipline

The article's terminology should obey:

Strength of Terminology
≤ Strength of Demonstrated Structure. (32.39)

Therefore:

No Stable Carrier → No Charge. (32.40)

No Independent Return → No Spin. (32.41)

No Frame Map and Invariant → No Gauge. (32.42)

No Measurable Inertia → No Mass. (32.43)

No Meaningful Γ Algebra → No Dirac Claim. (32.44)

No Incremental Utility → Reduce the Model. (32.45)

The source's final lexical rule is to use the strongest term only after the object has passed the gate giving that term meaning.

32.16 The final scientific rule

Whenever the theory appears more impressive than the evidence:

Reduce the Word before Enlarging the Theory. (32.46)

Whenever a frame fails:

Localize the Claim. (32.47)

Whenever new evidence invalidates closure:

Reopen the Gate with Prior Trace Preserved. (32.48)

Whenever residual remains:

Record It rather than Reclassify It as Success. (32.49)

Whenever the advanced representation adds no value:

Return to the Simpler Model. (32.50)

These are not merely editorial cautions.

They are part of the proposed Financial Standard Model's governance architecture.

Conclusion

Classical finance is highly effective when the primary question is:

> What value should be assigned under a declared model?

But many important financial processes ask additional questions:

- Who carries the claim?
- Who carries the obligation?
- What conditional responsibility is dormant?
- Which authority activates it?
- What event changes the subject's status?
- Which ledgers must recognize the consequence?
- Does the same identity survive translation across frames?
- What remains unresolved?
- How does the event alter future admissibility?

The leveraged margin account reveals these questions in their simplest integrated form.

Complex CAPM supplies the valuation geometry.

Contractual positions supply relational orientation.

Margin rules supply identity-bearing constraints.

Broker authority supplies the gate.

Collateral posting, deleveraging, liquidation, and default supply branch actions.

Settlement, treasury, risk, accounting, legal, and regulatory systems supply the return ledgers.

Cross-frame consistency supplies the gauge problem.

Transformation friction supplies the mass problem.

Unresolved consequence supplies residual.

Ledger inheritance supplies recursion.

The result is not a claim that finance secretly consists of physical particles.

It is a proposal that mature finance must sometimes model more than value:

> It must model the accountable propagation of identity through constrained transformation.

The deepest summary is:

Identity remembers what remains recognizable. (32.51)

Charge remembers how identity transforms. (32.52)

Spin remembers how identity returns. (32.53)

Mass remembers the cost of remaining itself. (32.54)

Gauge remembers how identity survives another frame. (32.55)

The gate determines what becomes history. (32.56)

The ledger determines what the future inherits. (32.57)

Residual remembers what closure did not contain. (32.58)

And the final law of constraint-bearing finance is:

Smooth Value
→ Identity-Bearing Constraint
→ Charge
→ Gate
→ Spin Closure
→ Gauge Return
→ Ledger
→ New Financial World. (32.59)

Appendix A — Compact Formula Compendium and Symbol Ledger

A.1 Purpose of this appendix

This appendix gathers the article's principal equations into one operational sequence.

Each formula is labelled as one of four types:

- **Identity** — exact under the declared definitions;
- **Definition** — establishes notation;
- **Construction** — proposed mathematical architecture;
- **Hypothesis** — requires empirical testing.

The sequence follows the system's causal order:

Complex Valuation
→ Collateral Admission

→ Margin Constraint
→ Structural Charge
→ Gate Activation
→ Action–Ledger Spin
→ Gauge Transport
→ Dirac Propagation
→ Ledger Recursion.

A.2 Protocol and Financial Subject

A.2.1 Declared protocol

Define the financial protocol:

P := (B_sys,Δ,h_T,u,Φ,G,T_r,R_r,V). (A.1)

where:

- B_sys = system boundary;
- Δ = observation and aggregation rule;
- h_T = declared horizon;
- u = admissible interventions;
- Φ = feature map;
- G = gate system;
- T_r = trace rule;
- R_r = residual rule;
- V = frame-transport rule.

Type: Definition.

A.2.2 Bounded financial subject

Define the subject:

S := one bounded financial identity carrying claims, obligations, constraints, and ledger history. (A.2)

For the minimum margin calibration:

S_margin := one leveraged account holding one risky asset against one funding liability under one margin agreement. (A.3)

Type: Definition.

A.2.3 Identity kernel

Define:

$K_S := (x,n,o,a,c,T,e,\kappa,\nu).$ (A.4)

where:

- x = instrument;
- n = signed quantity;
- o = owner or obligor;
- a = account boundary;
- c = currency;
- T = maturity or horizon;
- e = transaction lineage;
- κ = contractual terms;
- ν = lifecycle status.

Type: Construction.

A.2.4 Complete financial state

Define:

$\mathcal{S}_t := (K_t,Z_{M,t},Z_{C,t},D_t,C_t,h_t,B_t,\vec{q}_t,\Psi_t,L_t,\mathscr{R}_t).$ (A.5)

Type: Construction.

A.3 Complex CAPM Valuation Kernel

A.3.1 CAPM required return

$r_{CAPM} = r_{base} + \beta ERP.$ (A.6)

Type: Standard financial relation.

A.3.2 Baseline value amplitude

$A_t = CF_t/(1 + r_{base})^t.$ (A.7)

Type: Definition.

A.3.3 CAPM-admitted value

$R_t = CF_t/(1 + r_{CAPM})^t.$ (A.8)

Type: Definition.

A.3.4 Admission ratio

$c_t := R_t/A_t.$ (A.9)

Therefore:

$c_t = [(1 + r_{base})/(1 + r_{CAPM})]^t.$ (A.10)

Type: Exact identity.

A.3.5 Valuation phase

$\theta_t := \arccos(R_t/A_t).$ (A.11)

Therefore:

$R_t = A_t \cos \theta_t.$ (A.12)

Type: Definition and exact identity.

A.3.6 Conjugate pressure coordinate

$Q_t := \sqrt{(A_t^2 - R_t^2)}.$ (A.13)

Therefore:

$Q_t = A_t \sin \theta_t.$ (A.14)

Type: Definition and exact identity.

A.3.7 Complex valuation state

$Z_t := R_t + iQ_t.$ (A.15)

Equivalently:

$Z_t = A_t \exp(i\theta_t).$ (A.16)

and:

$|Z_t|^2 = R_t^2 + Q_t^2 = A_t^2.$ (A.17)

Type: Exact construction under equations (A.11)–(A.15).

A.3.8 Scalar haircut

Define:

$H_t := A_t - R_t.$ (A.18)

Then:

$Q_t^2 = H_t(A_t + R_t).$ (A.19)

Therefore:

$Q_t \neq H_t.$ (A.20)

Type: Exact identity.

A.3.9 Phase delta

Holding A fixed:

$\partial R/\partial \theta = -Q.$ (A.21)

For infinitesimal movement:

$dR = -Qd\theta.$ (A.22)

When A also changes:

$dR = (R/A)dA - Qd\theta.$ (A.23)

Type: Exact differential identity.

A.3.10 CAPM phase sensitivities

Let:

$r = r_{base} + \beta ERP.$ (A.24)

Then:

$\partial R/\partial r = -tR/(1 + r).$ (A.25)

Therefore:

$\partial \theta/\partial r = tR/[(1 + r)Q].$ (A.26)

The ERP phase sensitivity is:

$\partial \theta/\partial ERP = tR\beta/[(1 + r)Q].$ (A.27)

The beta phase sensitivity is:

$\partial \theta/\partial \beta = tRERP/[(1 + r)Q].$ (A.28)

Type: Exact local identities under the declared CAPM construction.

A.3.11 Interpretation limits

Q is:

- a conjugate pressure coordinate;
- a first-order phase exposure;
- measured in the same monetary units as A and R.

Q is not automatically:

- realized loss;
- volatility;
- beta;
- structural charge;
- margin shortfall;
- model residual.

A.4 Collateral Plane

A.4.1 Collateral-admission factor

Define:

$$0 \le h \le 1. \qquad \text{(A.29)}$$

The collateral-admitted value is:

$$R_C = hR_M. \qquad \text{(A.30)}$$

Type: Definition under the minimum collateral protocol.

A.4.2 Collateral phase

Using the shared amplitude A:

$$\cos \theta_C = R_C/A. \qquad \text{(A.31)}$$

Therefore:

$$\cos \theta_C = h \cos \theta_M. \qquad \text{(A.32)}$$

and:

$$\theta_C = \arccos(h \cos \theta_M). \qquad \text{(A.33)}$$

Type: Exact identity under the shared-amplitude convention.

A.4.3 Collateral pressure coordinate

$$Q_C := \sqrt{(A^2 - R_C^2)}. \qquad (A.34)$$

Therefore:

$$Z_C := R_C + iQ_C = Ae^{(i\theta_C)}. \qquad (A.35)$$

Type: Proposed sequential-filter construction.

A.4.4 Relation between market and collateral pressure

$$Q_M^2 = A^2 - R_M^2. \qquad (A.36)$$

$$Q_C^2 = A^2 - h^2 R_M^2. \qquad (A.37)$$

Therefore:

$$Q_C^2 = Q_M^2 + (1 - h^2)R_M^2. \qquad (A.38)$$

Define:

$$Q_{h^2} := (1 - h^2)R_M^2. \qquad (A.39)$$

Then:

$$Q_C^2 = Q_M^2 + Q_{h^2}. \qquad (A.40)$$

Type: Exact identity under the construction.

A.4.5 Collateral-phase differential

Differentiate:

$$\cos\theta_C = h\cos\theta_M. \qquad (A.41)$$

Then:

$$d\theta_C = [h\sin\theta_M\, d\theta_M - \cos\theta_M\, dh]/\sin\theta_C. \qquad (A.42)$$

Type: Exact differential identity.

A.5 Margin Geometry

A.5.1 Position market value

For n units:

$$V_M = nR_M. \qquad (A.43)$$

Type: Definition.

A.5.2 Position collateral value

$V_C = nhR_M.$ (A.44)

Type: Definition.

A.5.3 Eligible collateral

Let C be independently posted collateral.

Then:

$K = C + nhR_M.$ (A.45)

Type: Definition.

A.5.4 Margin buffer

Let D be the funding liability.

Define:

$B := C + nhR_M - D.$ (A.46)

The admissible region is:

$B \geq B_{target}.$ (A.47)

A breach candidate exists when:

$B < B_{target}.$ (A.48)

Type: Definition.

A.5.5 Call amount

$C_{call} := [B_{target} - B]_+.$ (A.49)

where:

$[x]_+ := \max(x,0).$ (A.50)

Type: Definition under the minimum cure rule.

A.5.6 Critical market value

At the margin boundary:

$B = B_{target}.$ (A.51)

Therefore:

$R_M^* = [D + B_{target} - C]/(nh).$ (A.52)

Type: Exact identity.

A.5.7 Critical collateral factor

$h^* = [D + B_{target} - C]/(nR_M).$ (A.53)

Type: Exact identity.

A.5.8 Buffer differential

Differentiate equation (A.46):

$dB = dC + nhdR_M + nR_Mdh + hR_Mdn - dD.$ (A.54)

Substitute equation (A.23):

$dB = dC + nh(R_M/A)dA - nhQ_Md\theta_M + nR_Mdh + hR_Mdn - dD.$ (A.55)

Type: Exact differential identity.

A.5.9 Phase-to-buffer sensitivity

Holding A, C, h, n, and D fixed:

$\partial B/\partial \theta_M = -nhQ_M.$ (A.56)

Using collateral phase:

$\partial B/\partial \theta_C = -nQ_C.$ (A.57)

Type: Exact local identities.

A.5.10 Local phase distance to the gate

For a small adverse phase movement:

$\Delta\theta_M^* \approx [B - B_{target}]/(nhQ_M).$ (A.58)

Type: First-order approximation.

A.6 Financial Charge

A.6.1 Structural charge vector

Define:

$\vec{q}_{struct} := (q_A,q_F,q_C).$ (A.59)

where:

- q_A = asset-claim orientation;
- q_F = funding orientation;
- q_C = contingent collateral orientation.

Type: Proposed classification.

A.6.2 Elementary claim–obligation pairing

Normalize:

$$q_{claim} = +1. \quad (A.60)$$

$$q_{obligation} = -1. \quad (A.61)$$

Therefore:

$$q_{claim} + q_{obligation} = 0. \quad (A.62)$$

when both sides lie within the declared boundary.

Type: Sign convention.

A.6.3 Quantity-scaled charge

$$q_K = nq_0(x). \quad (A.63)$$

Type: Proposed charge definition.

A.6.4 Operational collateral-charge activation

Define:

$$a_C(B) := \begin{cases} 0, & B \geq B_{target}; \\ 1, & B < B_{target}. \end{cases} \quad (A.64)$$

Then:

$$q_C^{op} = a_C(B) q_C. \quad (A.65)$$

Type: Proposed gate-activation rule.

A.6.5 Effective charge

Define schematically:

$$\vec{q}_{eff} = \Lambda(X,L,Regime)\vec{q}_{struct}. \quad (A.66)$$

A simplified market component may be:

$q_{eff,A} = q_A \beta \lambda \chi_A.$ (A.67)

where:

$\lambda = V_M/E.$ (A.68)

Type: Empirical model construction.

A.6.6 Charge balance at a vertex

$\Sigma q_{in} + q_{gate} = \Sigma q_{out} + r_q.$ (A.69)

where r_q is unresolved charge residual.

Type: Proposed accounting law.

A.6.7 Charge-admission rule

AdmitCharge(q_r)
:= Carrier_r
∧ Field_r
∧ Sign_r
∧ Coupling_r
∧ Transport_r
∧ Vertex_r
∧ Residual_r. (A.70)

Type: Methodological rule.

A.7 Financial Spin

A.7.1 Action–ledger spinor

Define:

$\Psi_S := [\psi_A, \psi_L]^T.$ (A.71)

where:

- ψ_A = outward action state;
- ψ_L = ledger-return state.

Type: Proposed construction.

A.7.2 Expected ledger return

$\hat{\psi}_L := U_{AL}\psi_A.$ (A.72)

Type: Definition.

A.7.3 Covariant spinor residual

$\delta_{spin} := \psi_L - U_{AL}\psi_A.$ (A.73)

The weighted split is:

$\Delta_{spin}^2 := \delta_{spin}^{\dagger} W \delta_{spin}.$ (A.74)

Type: Proposed diagnostic.

A.7.4 Operational double closure

The closure cycle is:

$\Psi_{closed} \rightarrow \Psi_{open\text{-}return} \rightarrow \Psi_{closed'}.$ (A.75)

Compactly:

$\Psi \rightarrow -\Psi \rightarrow \Psi'.$ (A.76)

The minus sign denotes unresolved return obligation.

Type: Operational interpretation.

A.7.5 Closure sign

$s_{close} = +1$ for ledger-closed identity. (A.77)

$s_{close} = -1$ for action-complete but ledger-open identity. (A.78)

Type: Definition.

A.7.6 Spin-admission rule

AdmitSpin
:= BoundedIdentity
∧ IndependentComponents
∧ ConsequentialAction
∧ IncompleteFirstCycle
∧ IndependentLedgerReturn
∧ Residual
∧ IncrementalGain. (A.79)

Type: Methodological rule.

A.8 Gauge Transport

A.8.1 Frame-local representation

Let:

$S_f := \Pi_f(S).$ (A.80)

Type: Definition.

A.8.2 Edge transport

For frame $A \to B$:

$\hat{S}_B := U_{ABS_A}.$ (A.81)

The observed target state is S_B.

The edge residual is:

$r_{AB} := S_B - U_{ABS_A}.$ (A.82)

Type: Definition.

A.8.3 Identity preservation

Ordinary transport requires:

$K_B(U_{ABS_A}) \simeq K_A(S_A).$ (A.83)

Type: Gauge-eligibility condition.

A.8.4 Local frame transformations

$S_{A'} = G_{AS_A}.$ (A.84)

$S_{B'} = G_{BS_B}.$ (A.85)

The transporter transforms as:

$U_{AB'} = G_B U_{ABG_A^{-1}}.$ (A.86)

Therefore:

$U_{AB'S_{A'}} = G_B(U_{ABS_A}).$ (A.87)

Type: Exact covariance requirement.

A.8.5 Discrete covariant derivative

$D_{AB\Psi} := \Psi_B - U_{AB\Psi_A}.$ (A.88)

Under lawful local transformations:

$D_{AB}'\Psi' = G_{BD}D_{AB}\Psi$. (A.89)

Type: Proposed gauge construction.

A.8.6 Loop holonomy

For loop ℓ:

$H_\ell := \prod_{(A\to B\in\ell)} U_{AB}$. (A.90)

The expected return is:

$\hat{S}_{start}^{return} = H_\ell^{PS_{start}}$. (A.91)

The governed loop residual is:

$\mathscr{R}_{loop} := S_{start}^{return,obs} - \hat{S}_{start}^{return}$. (A.92)

Type: Proposed curvature diagnostic.

A.8.7 Normalized loop defect

$\kappa_{loop} := \|\mathscr{R}_{loop}\|_W / [\|\hat{S}_{start}^{return}\|_W + \varepsilon]$. (A.93)

Type: Empirical construction.

A.8.8 Gauge-admission rule

AdmitGauge
:= DeclaredFrames
∧ InvariantKernel
∧ Connection
∧ CovarianceLaw
∧ LoopTest
∧ ResidualHonesty. (A.94)

Type: Methodological rule.

A.9 Γ Algebra and Identity Mass

A.9.1 Closure matrices

Use:

$\Gamma^0 := [[1,0],[0,-1]]$. (A.95)

$\Gamma^1 := [[0,1],[-1,0]]$. (A.96)

Then:

$(\Gamma^0)^2 = I_2$. (A.97)

$(\Gamma^1)^2 = -I_2$. (A.98)

$\Gamma^0\Gamma^1 + \Gamma^1\Gamma^0 = 0$. (A.99)

Therefore:

$\{\Gamma^a,\Gamma^b\} = 2\eta^{ab}I_2$. (A.100)

with:

$\eta = \mathrm{diag}(1,-1)$. (A.101)

Type: Exact algebra for the selected matrices.

A.9.2 Mass operator

Define:

$M_S := m_I I_2 + m_C \sigma_x + m_\Delta \sigma_z$. (A.102)

Explicitly:

$M_S = [[m_I + m_\Delta, m_C],[m_C, m_I - m_\Delta]]$. (A.103)

Type: Proposed construction.

A.9.3 Mass eigenvalues

$m_\pm = m_I \pm \sqrt{(m_C^2 + m_\Delta^2)}$. (A.104)

Type: Exact matrix identity.

A.9.4 Stability restriction

For nonnegative minimum eigenvalue:

$m_I \geq \sqrt{(m_C^2 + m_\Delta^2)}$. (A.105)

Type: Proposed model restriction.

A.9.5 State-dependent coupling mass

An illustrative boundary-sensitive coupling is:

$m_C(B) = m_{C,0} + m_{C,1}/[1 + \exp(B/b_0)]$. (A.106)

Type: Empirical hypothesis.

A.10 Gauge-Covariant Financial Derivative

A.10.1 Structural charge operators

For field r:

$\hat{Q}_r\Psi = q_r\Psi.$ (A.107)

Type: Proposed representation.

A.10.2 Local gauge transformation

$U(\chi) := \exp[i\Sigma_r \chi_r \hat{Q}_r].$ (A.108)

Then:

$\Psi' = U(\chi)\Psi.$ (A.109)

Type: Proposed restricted Abelian construction.

A.10.3 Covariant time derivative

$D_\tau^G := \partial_\tau + i\Sigma_r g_r \mathcal{A}_{r,\tau} \hat{Q}_r.$ (A.110)

Covariance requires:

$D_\tau^{G'}\Psi' = U(\chi) D_\tau^G \Psi.$ (A.111)

Type: Proposed gauge construction.

A.10.4 Protocol-covariant derivative

Define schematically:

$\nabla_\tau^P := D_\tau^G + \hat{\Omega}_{gate} + \hat{\Omega}_{trace} + \hat{\Omega}_{ledger}.$ (A.112)

Type: Proposed institutional extension.

A.11 Financial Gauge–Dirac Kernel

A.11.1 Closure graph derivative

For the minimum action–ledger graph:

$\mathfrak{D}_G\Psi := (1/\ell_P)[\psi_A - U_{LA}\psi_L, \psi_L - U_{AL}\psi_A]^T.$ (A.113)

Type: Proposed construction.

A.11.2 Slash operator

$\mathcal{D}$_fin := Γ⁰∇_τ^P + c_PΓ¹𝔇_G. (A.114)

Type: Definition.

A.11.3 Central continuous equation

(i$\mathcal{D}$_fin − M_S)Ψ_S = $\mathscr{R}$_S. (A.115)

Expanded:

[iΓ⁰∇_τ^P + ic_PΓ¹𝔇_G − M_S]Ψ_S = $\mathscr{R}$_S. (A.116)

Type: Central proposed equation.

A.11.4 Component equations

Let:

δ_A := (ψ_A − U_LAψ_L)/ℓ_P. (A.117)

δ_L := (ψ_L − U_ALψ_A)/ℓ_P. (A.118)

Then:

i∇_τ^Pψ_A + ic_Pδ_L − (m_I + m_Δ)ψ_A − m_Cψ_L = $\mathscr{R}$_A. (A.119)

−i∇_τ^Pψ_L − ic_Pδ_A − m_Cψ_A − (m_I − m_Δ)ψ_L = $\mathscr{R}$_L. (A.120)

Type: Exact expansion of the proposed equation.

A.11.5 Residual decomposition

$\mathscr{R}$_S
:= $\mathscr{R}$_value

- $\mathscr{R}$_charge
- $\mathscr{R}$_transport
- $\mathscr{R}$_gate
- $\mathscr{R}$_ledger
- $\mathscr{R}$_cone
- $\mathscr{R}$_model. (A.121)

Type: Proposed classification.

A.11.6 Squared equation

Left-multiplying equation (A.115) by i$\mathcal{D}$_fin + M_S gives:

$[\mathcal{D}_{fin}^2 + M_S^2 - i[\mathcal{D}_{fin}, M_S]]\Psi_S = -(i\mathcal{D}_{fin} + M_S)\mathcal{R}_S.$ (A.122)

Type: Exact operator identity.

A.12 Hybrid Gate System

A.12.1 Guard

For gate j:

$g_j(\mathcal{S}) \geq 0.$ (A.123)

Type: Definition.

A.12.2 Gate admission

$a_k = \mathrm{Admit}_j(\mathcal{S}_k^-, P_k, \mathrm{Authority}_k).$ (A.124)

Type: Proposed decision rule.

A.12.3 Jump

$\mathcal{S}_k^+ = \mathcal{J}_j^{(a_k)}(\mathcal{S}_k^-, L_k) + \mathcal{R}_{\mathrm{jump},k}.$ (A.125)

Type: Proposed hybrid-system equation.

A.12.4 Ledger update

$L_{k+1} = U_L(L_k, \mathrm{Trace}_k, \mathrm{Residual}_k).$ (A.126)

Type: Proposed recursion.

A.12.5 Parameter update

$\Theta_{k+1} = U_\Theta(\Theta_k, L_{k+1}, \mathcal{R}_k).$ (A.127)

Type: Proposed recursion.

A.12.6 Complete hybrid family

Between gates:

$(i\mathcal{D}_{fin}[\Theta_k] - M_S[\Theta_k])\Psi = \mathcal{R}_{\mathrm{cont}}.$ (A.128)

At gates:

$\mathcal{S}_k^+ = \mathcal{J}_k(\mathcal{S}_k^-, L_k) + \mathcal{R}_{\mathrm{jump},k}.$ (A.129)

After gates:

$L_{k+1} = U_L(L_k,\mathrm{Trace}_k,\mathrm{Residual}_k).$ (A.130)

For the next period:

$\mathcal{L}_{k+1} = U_{\mathcal{L}}(\mathcal{L}_k,L_{k+1},\mathcal{R}_k).$ (A.131)

Type: Full proposed Financial Gauge–Dirac–Gate–Ledger system.

A.13 Forced-Liquidation Dynamics

A.13.1 Deficiency

$S_B := [B_{\mathrm{target}} - B]_+.$ (A.132)

A.13.2 Liquidation quantity

$\ell = \kappa_{LS} B.$ (A.133)

Type: Minimum policy hypothesis.

A.13.3 Position change

$dn/d\tau = -\ell.$ (A.134)

A.13.4 Price impact

$dR_M/d\tau = dR_{CAPM}/d\tau - I(\ell,\mathrm{Depth},\mathrm{Volatility}).$ (A.135)

A.13.5 Debt change

$dD/d\tau = -p_{exec}\ell + r_{FD} + \kappa_{fee}\ell.$ (A.136)

A.13.6 Buffer dynamics

$dB/d\tau$
$= dC/d\tau$

- $nh dR_{CAPM}/d\tau$
- $nR_M dh/d\tau$
  $- r_{FD}$
- $\ell[p_{exec} - hR_M - \kappa_{fee}]$
  $- nhI(\ell).$ (A.137)

Type: Exact decomposition under the declared dynamic assumptions.

A.13.7 Marginal liquidation benefit

$$MBE(\ell) := \partial(dB/d\tau)/\partial\ell. \qquad (A.138)$$

Liquidation becomes locally self-defeating when:

$$MBE(\ell) < 0. \qquad (A.139)$$

Type: Proposed stability diagnostic.

A.14 Recursive Closure

A.14.1 Structural-charge update

$$\vec{q}_{struct,k+1} = C_q(\vec{q}_{struct,k}, Branch_k, K_{k+1}) + \vec{r}_{q,k}. \qquad (A.140)$$

A.14.2 Effective-charge update

$$\vec{q}_{eff,k+1} = \Lambda(X_{k+1}, L_{k+1}, \mathscr{R}_{k+1})\vec{q}_{struct,k+1}. \qquad (A.141)$$

A.14.3 Mass update

$$M_{k+1} = U_M(M_k, L_{k+1}, \mathscr{R}_k, \mathcal{Q}_{close,k}). \qquad (A.142)$$

A.14.4 Closure-capacity update

$$c_{P,k+1} = U_c(c_{P,k}, L_{k+1}, Backlog_k, Residual_k). \qquad (A.143)$$

A.14.5 Connection update

$$U_{AB,k+1} = U_{connection}(U_{AB,k}, L_{k+1}, RuleChange_k, \mathscr{R}_k). \qquad (A.144)$$

A.14.6 Master recurrence

$$\begin{aligned}&(K_k, \vec{q}_k, \Psi_k, \Theta_k, L_k)\\ &\to G_k\\ &\to (Trace_k, Residual_k)\\ &\to (K_{k+1}, \vec{q}_{k+1}, \Psi_{k+1}, \Theta_{k+1}, L_{k+1}).\end{aligned} \qquad (A.145)$$

Type: Central recursive construction.

A.15 Closure Conditions

A.15.1 Spinor closure

$\Delta_{spin,G} \le \varepsilon_{spin}$. (A.146)

A.15.2 Charge closure

$|r_q| \le \varepsilon_q$. (A.147)

A.15.3 Gate closure

$\|\mathscr{R}_{gate}\| \le \varepsilon_{gate}$. (A.148)

A.15.4 Transport closure

$\|r_{AB}\| \le \varepsilon_{AB}$. (A.149)

A.15.5 Loop closure

$\kappa_{loop} \le \varepsilon_{loop}$. (A.150)

A.15.6 Identity closure

$d_K(K_{after},K_{expected}) \le \varepsilon_K$. (A.151)

A.15.7 Complete closure

$SpinClosed_P(S) = 1$ (A.152)

only when:

GateResolved
∧ IdentityPreservedOrValidlyConverted
∧ ChargeReconciled
∧ RequiredLedgersPosted
∧ RequiredLoopsClosed
∧ ResidualDisclosed. (A.153)

Type: Proposed operational criterion.

A.16 Reduction Rules

A.16.1 Complex reduction

Z = R + iQ
→ (R,Q)
→ R. (A.154)

A.16.2 Charge reduction

Charge
→ Coupling Orientation
→ Exposure or Sensitivity. (A.155)

A.16.3 Spin reduction

Spinor
→ Action–Ledger Doublet
→ Two-Stage Workflow
→ Scalar Status. (A.156)

A.16.4 Gauge reduction

Gauge Structure
→ Governed Frame Transport
→ Multi-Ledger Reconciliation
→ Data Matching. (A.157)

A.16.5 Mass reduction

Mass
→ Identity Inertia
→ Friction Index. (A.158)

A.16.6 Dirac reduction

Financial Gauge–Dirac System
→ First-Order Coupled Identity Model
→ Hybrid State-Space Model
→ Conventional Financial Model. (A.159)

A.17 Master Symbol Table

Symbol	Meaning	Status
A	Baseline value amplitude	Defined valuation quantity
R_M	CAPM-admitted market value	Defined valuation quantity
Q_M	CAPM conjugate pressure coordinate	Derived coordinate

Symbol	Meaning	Status
θ_M	CAPM valuation phase	Derived coordinate
Z_M	Market complex state $R_M + iQ_M$	Complex construction
h	Collateral-admission factor	Protocol parameter
R_C	Collateral-admitted value	Defined target-frame value
Q_C	Collateral pressure coordinate	Derived construction
θ_C	Collateral phase	Derived construction
n	Signed position quantity	Contractual state
C	Independently posted collateral	Balance-sheet state
D	Funding liability	Balance-sheet state
B	Margin buffer	Constraint variable
C_{call}	Required call amount	Gate variable
q_A	Asset structural charge	Proposed relational classification
q_F	Funding structural charge	Proposed relational classification
q_C	Collateral structural charge	Proposed relational classification
q_{eff}	State-dependent effective charge	Empirical construction
ψ_A	Outward action component	Spinor component
ψ_L	Ledger-return component	Spinor component
Ψ	Financial action–ledger spinor	Proposed construction
U_{AB}	Frame transporter	Gauge candidate
r_{AB}	Edge transport residual	Diagnostic
H_{loop}	Loop holonomy	Gauge candidate
κ_{loop}	Normalized loop defect	Diagnostic
Γ^0, Γ^1	Closure-space operators	Proposed Dirac structure
M_S	Financial identity-mass operator	Proposed construction
c_P	Coherent closure capacity	Empirical hypothesis
$\mathscr{R}$	Residual vector	Explicit nonclosure
G_k	Gate or jump operator	Hybrid-system component
L_k	Ledger state	Recursive memory
Θ_k	Period-specific model parameters	Recursive environment

Appendix B — Worked Numerical Calibration of a Leveraged Margin Account

B.1 Purpose

This example demonstrates the complete minimum sequence:

Complex CAPM
→ Collateral Plane
→ Margin Buffer
→ Adverse Phase Movement
→ Haircut Deterioration
→ Margin Call
→ Collateral or Deleveraging Branch
→ Ledger Closure.

The numerical values are illustrative.

They are not market forecasts, margin recommendations, or investment advice.

B.2 Initial Contract and Valuation Inputs

Assume one future cash flow per unit:

$CF_1 = \$120.00.$ (B.1)

The horizon is:

$T = 1$ year. (B.2)

The baseline rate is:

$r_{base} = 4.00\%.$ (B.3)

The asset beta is:

$\beta = 1.20.$ (B.4)

The equity risk premium is:

$ERP_0 = 6.00\%.$ (B.5)

Therefore:

$r_{CAPM,0} = 4.00\% + 1.20 \times 6.00\%.$ (B.6)

Hence:

$r_{CAPM,0} = 11.20\%.$ (B.7)

B.3 Initial Complex CAPM State

B.3.1 Baseline amplitude

$A = \$120.00/1.04.$ (B.8)

Therefore:

$A = \$115.38.$ (B.9)

B.3.2 CAPM-admitted value

$R_{M,0} = \$120.00/1.112.$ (B.10)

Therefore:

$R_{M,0} = \$107.91.$ (B.11)

B.3.3 Pressure coordinate

$Q_{M,0} = \sqrt{(\$115.38^2 - \$107.91^2)}$. (B.12)

Therefore:

$Q_{M,0} = \$40.84$. (B.13)

B.3.4 Market phase

$\theta_{M,0} = \arccos(\$107.91/\$115.38)$. (B.14)

Therefore:

$\theta_{M,0} = 0.3618$ radians. (B.15)

or:

$\theta_{M,0} = 20.73^\circ$. (B.16)

B.3.5 Complex market state

$Z_{M,0} = \$107.91 + i\$40.84$. (B.17)

The magnitude remains:

$|Z_{M,0}| = \$115.38$. (B.18)

B.4 Initial Leveraged Account

Assume:

$n_0 = 100$ units. (B.19)

Posted cash collateral is:

$C_0 = \$300.00$. (B.20)

Funding debt is:

$D_0 = \$8{,}200.00$. (B.21)

The collateral-admission factor is:

$h_0 = 0.75$. (B.22)

The target buffer is:

$B_{target} = \$0.00$. (B.23)

B.4.1 Market value

$V_{M,0} = 100 \times \$107.91$. (B.24)

Therefore:

$V_{M,0} = \$10{,}791.37.$ (B.25)

B.4.2 Collateral-admitted value

$R_{C,0} = 0.75 \times \$107.91.$ (B.26)

Therefore:

$R_{C,0} = \$80.94$ per unit. (B.27)

For 100 units:

$V_{C,0} = \$8{,}093.53.$ (B.28)

B.4.3 Eligible collateral

$K_0 = \$300.00 + \$8{,}093.53.$ (B.29)

Therefore:

$K_0 = \$8{,}393.53.$ (B.30)

B.4.4 Initial margin buffer

$B_0 = \$8{,}393.53 - \$8{,}200.00.$ (B.31)

Therefore:

$B_0 = \$193.53.$ (B.32)

The account is inside the margin boundary:

$B_0 > 0.$ (B.33)

No call is active:

$C_{call,0} = \$0.00.$ (B.34)

The contingent collateral charge remains dormant:

$q_C^{op,0} = 0.$ (B.35)

B.5 Initial Collateral Complex State

B.5.1 Collateral pressure

$Q_{C,0} = \sqrt{(\$115.38^2 - \$80.94^2)}.$ (B.36)

Therefore:

$Q_{C,0} = \$82.24.$ (B.37)

B.5.2 Collateral phase

$\theta_{C,0} = \arccos(\$80.94/\$115.38).$ (B.38)

Therefore:

$\theta_{C,0} = 0.7934$ radians. (B.39)

or:

$\theta_{C,0} = 45.46°.$ (B.40)

B.5.3 Market–collateral phase separation

$\Delta\theta_{MC,0} = 45.46° - 20.73°.$ (B.41)

Therefore:

$\Delta\theta_{MC,0} = 24.73°.$ (B.42)

The account is solvent under the minimum margin rule, but the collateral frame is already substantially more restrictive than the market frame.

B.6 Adverse Market and Collateral Shock

Assume the ERP rises:

$ERP_1 = 8.00\%.$ (B.43)

Assume beta remains:

$\beta = 1.20.$ (B.44)

Then:

$r_{CAPM,1} = 4.00\% + 1.20 \times 8.00\%.$ (B.45)

Therefore:

$r_{CAPM,1} = 13.60\%.$ (B.46)

At the same time, the collateral-admission factor falls:

$h_1 = 0.70.$ (B.47)

Assume initially:

$n_1 = 100.$ (B.48)

$C_1 = \$300.00.$ (B.49)

$D_1 = \$8{,}200.00.$ (B.50)

B.7 Post-Shock Complex CAPM State

B.7.1 New market value

R_M,1 = $120.00/1.136. (B.51)

Therefore:

R_M,1 = $105.63. (B.52)

The real-axis value decline is:

ΔR_M = $105.63 − $107.91. (B.53)

Therefore:

ΔR_M = −$2.28 per unit. (B.54)

For 100 units:

ΔV_M = −$227.99. (B.55)

B.7.2 New pressure coordinate

Q_M,1 = √($115.38² − $105.63²). (B.56)

Therefore:

Q_M,1 = $46.42. (B.57)

The pressure coordinate rises:

ΔQ_M = $46.42 − $40.84. (B.58)

Therefore:

ΔQ_M = $5.58. (B.59)

B.7.3 New market phase

θ_M,1 = arccos($105.63/$115.38). (B.60)

Therefore:

θ_M,1 = 0.4141 radians. (B.61)

or:

θ_M,1 = 23.72°. (B.62)

The phase movement is:

Δθ_M = 23.72° − 20.73°. (B.63)

Therefore:

$\Delta\theta_M = 2.99^\circ$. (B.64)

In radians:

$\Delta\theta_M = 0.0522$. (B.65)

B.8 Post-Shock Collateral State

B.8.1 Collateral-admitted value

$R_{C,1} = 0.70 \times \$105.63$. (B.66)

Therefore:

$R_{C,1} = \$73.94$ per unit. (B.67)

For 100 units:

$V_{C,1} = \$7{,}394.37$. (B.68)

B.8.2 Collateral pressure coordinate

$Q_{C,1} = \sqrt{(\$115.38^2 - \$73.94^2)}$. (B.69)

Therefore:

$Q_{C,1} = \$88.58$. (B.70)

B.8.3 Collateral phase

$\theta_{C,1} = \arccos(\$73.94/\$115.38)$. (B.71)

Therefore:

$\theta_{C,1} = 0.8752$ radians. (B.72)

or:

$\theta_{C,1} = 50.15^\circ$. (B.73)

B.8.4 Phase separation after shock

$\Delta\theta_{MC,1} = 50.15^\circ - 23.72^\circ$. (B.74)

Therefore:

$\Delta\theta_{MC,1} = 26.43^\circ$. (B.75)

The market–collateral phase mismatch has widened.

B.9 Margin Breach

B.9.1 Post-shock eligible collateral

$K_1 = \$300.00 + \$7{,}394.37.$ (B.76)

Therefore:

$K_1 = \$7{,}694.37.$ (B.77)

B.9.2 Post-shock buffer

$B_1 = \$7{,}694.37 - \$8{,}200.00.$ (B.78)

Therefore:

$B_1 = -\$505.63.$ (B.79)

The account has crossed the threshold:

$B_1 < 0.$ (B.80)

B.9.3 Required call amount

$C_{call,1} = [-B_1]_+.$ (B.81)

Therefore:

$C_{call,1} = \$505.63.$ (B.82)

If the authorized margin system validates the mark and rule, a margin event is admitted.

The contingent collateral charge becomes operational:

$q_C^{op,1} = q_C.$ (B.83)

The closure sign becomes:

$s_{close} : +1 \rightarrow -1.$ (B.84)

B.10 Decomposition of the Buffer Loss

The total buffer change is:

$\Delta B = B_1 - B_0.$ (B.85)

Therefore:

$\Delta B = -\$505.63 - \$193.53.$ (B.86)

Hence:

$\Delta B = -\$699.16.$ (B.87)

This loss arose through two channels:

1. market-value decline;
2. collateral-factor decline.

B.10.1 Market channel at the original haircut

The market-only effect is approximately:

$\Delta B_{market} = nh_0 \Delta R_M.$ (B.88)

Therefore:

$\Delta B_{market} = 100 \times 0.75 \times (-\$2.28).$ (B.89)

Hence:

$\Delta B_{market} \approx -\$171.00.$ (B.90)

B.10.2 Haircut channel at the new market value

The haircut effect is approximately:

$\Delta B_{haircut} = nR_{M,1} \Delta h.$ (B.91)

Here:

$\Delta h = 0.70 - 0.75 = -0.05.$ (B.92)

Therefore:

$\Delta B_{haircut} = 100 \times \$105.63 \times (-0.05).$ (B.93)

Hence:

$\Delta B_{haircut} \approx -\$528.17.$ (B.94)

B.10.3 Total

$\Delta B_{market} + \Delta B_{haircut} \approx -\$699.17.$ (B.95)

This closely matches the exact buffer movement of $-\$699.16$, subject to rounding.

The example shows that collateral deterioration dominates the immediate breach even though the initiating market phase also worsens.

B.11 Phase-to-Buffer Approximation

Using the initial phase exposure:

$\partial B/\partial \theta_M \approx -n h_0 Q_{M,0}$. (B.96)

Therefore:

$\partial B/\partial \theta_M \approx -100 \times 0.75 \times \$40.84$. (B.97)

Hence:

$\partial B/\partial \theta_M \approx -\$3{,}063.32$ per radian. (B.98)

For:

$\Delta \theta_M = 0.0522$ radians, (B.99)

the first-order phase effect is:

$\Delta B_{phase} \approx -\$3{,}063.32 \times 0.0522$. (B.100)

Therefore:

$\Delta B_{phase} \approx -\$159.93$. (B.101)

The result is close to the exact market-value contribution of approximately −$171.00.

The difference arises because Q changes during the phase movement and the calculation is only first-order.

B.12 Branch A — Cure by Posting Collateral

Suppose the account posts exactly:

$\Delta C = \$505.63$. (B.102)

Then:

$C_2 = \$300.00 + \$505.63$. (B.103)

Therefore:

$C_2 = \$805.63$. (B.104)

The new buffer is:

$B_2 = \$805.63 + \$7{,}394.37 - \$8{,}200.00$. (B.105)

Therefore:

$B_2 = \$0.00$. (B.106)

The amount condition is satisfied.

But operational closure still requires:

- funds received;
- collateral eligible;
- control perfected;

- treasury updated;
- risk ledger updated;
- accounting ledger updated;
- trace preserved.

Until these conditions hold:

CallAmountCured = 1 (B.107)

does not necessarily imply:

SpinClosed_P = 1. (B.108)

B.12.1 Liquidity consequence

Assume the account had free liquidity before the call of:

F_1 = \$700.00. (B.109)

After posting:

F_2 = \$700.00 − \$505.63. (B.110)

Therefore:

F_2 = \$194.37. (B.111)

The collateral obligation closes, but the subject's future liquidity sensitivity increases.

Thus:

Current Margin Closure
→ Lower Future Liquidity Capacity. (B.112)

B.13 Branch B — Cure by Deleveraging

Assume the subject sells units at:

p_{exec} = \$104.00 per unit. (B.113)

Transaction cost is:

κ_{fee} = \$0.50 per unit. (B.114)

Net proceeds per unit are:

p_{net} = \$104.00 − \$0.50. (B.115)

Therefore:

p_{net} = \$103.50. (B.116)

The collateral value removed per sold unit is:

$h_1 R_{M,1} = 0.70 \times \$105.63.$ (B.117)

Therefore:

$h_1 R_{M,1} = \$73.94.$ (B.118)

The approximate buffer improvement per unit sold is:

$\Delta B_{\text{per-unit}} = p_{\text{net}} - h_1 R_{M,1}.$ (B.119)

Therefore:

$\Delta B_{\text{per-unit}} = \$103.50 - \$73.94.$ (B.120)

Hence:

$\Delta B_{\text{per-unit}} = \$29.56.$ (B.121)

B.13.1 Minimum units required

The approximate units required are:

$\ell^* = \$505.63/\$29.56.$ (B.122)

Therefore:

$\ell^* \approx 17.11 \text{ units}.$ (B.123)

If fractional units are not permitted, the account must sell at least:

$\ell = 18 \text{ units}.$ (B.124)

B.13.2 Post-sale quantity

$n_2 = 100 - 18.$ (B.125)

Therefore:

$n_2 = 82 \text{ units}.$ (B.126)

B.13.3 Net proceeds

$P_{\text{net}} = 18 \times \$103.50.$ (B.127)

Therefore:

$P_{\text{net}} = \$1{,}863.00.$ (B.128)

B.13.4 New debt balance

$D_2 = \$8{,}200.00 - \$1{,}863.00.$ (B.129)

Therefore:

$D_2 = \$6{,}337.00.$ (B.130)

B.13.5 Remaining collateral value

$V_{C,2} = 82 \times 0.70 \times \$105.63.$ (B.131)

Therefore:

$V_{C,2} \approx \$6{,}063.38.$ (B.132)

B.13.6 New buffer

$B_2 = \$300.00 + \$6{,}063.38 - \$6{,}337.00.$ (B.133)

Therefore:

$B_2 \approx \$26.38.$ (B.134)

The numerical margin condition is restored:

$B_2 > 0.$ (B.135)

But the sale must still settle before full spin closure.

B.14 Nested Trade-Settlement Spin

Immediately after execution:

$\text{ExecutedQuantity} = 18.$ (B.136)

SettledQuantity may still be:

$\text{SettledQuantity} = 0.$ (B.137)

Debt may not yet have legally fallen by $1,863.00.

The outward action component records:

$\psi_A = \text{SaleExecuted} + \text{ExpectedDebtReduction}.$ (B.138)

The ledger component may still record:

$\psi_L = \text{SaleUnsettled} + \text{DebtUnchanged}.$ (B.139)

The spinor residual includes:

$\delta_{\text{quantity}} = 18 - 0.$ (B.140)

$\delta_{\text{cash}} = \$1{,}863.00 - \$0.00.$ (B.141)

$\delta_{\text{debt}} = \text{ExpectedDebtReduction} - \text{RecordedDebtReduction}.$ (B.142)

Only after settlement and debt posting does:

$\psi_L \approx U_{AL}\psi_A.$ (B.143)

Then the nested trade spin closes.

The outer margin cycle may close only afterward.

B.15 Branch C — Impact-Adjusted Forced Liquidation

Suppose forced liquidation lowers executable price to:

$p_{exec} = \$101.00.$ (B.144)

Assume fees remain:

$\kappa_{fee} = \$0.50.$ (B.145)

Then:

$p_{net} = \$100.50.$ (B.146)

If the collateral factor remains 0.70, the direct buffer improvement per unit is:

$\Delta B_{direct} = \$100.50 - \$73.94.$ (B.147)

Therefore:

$\Delta B_{direct} = \$26.56.$ (B.148)

Ignoring further market impact, the required sale is:

$\ell^* \approx \$505.63/\$26.56.$ (B.149)

Therefore:

$\ell^* \approx 19.04$ units. (B.150)

The worse execution price requires more liquidation.

B.15.1 Remaining-position impact

Suppose the forced sale also lowers the market value of every remaining unit by:

$\text{Impact}_{\text{per-unit}} = \$1.50.$ (B.151)

If 19 units are sold, approximately 81 units remain.

The collateral-value loss on the remaining position is:

$$\text{ImpactLoss}_{\text{remaining}} = 81 \times 0.70 \times \$1.50.$$ (B.152)

Therefore:

$\text{ImpactLoss}_{\text{remaining}} = \$85.05.$ (B.153)

The original sale quantity may no longer cure the account.

Additional liquidation becomes necessary.

This is the beginning of the feedback loop:

Liquidation
→ Price Impact
→ Lower Remaining Collateral
→ Larger Required Liquidation. (B.154)

B.16 Charge Evolution Through the Example

B.16.1 Initial charge signature

Before the call:

$\vec{q}_{struct,0} = (q_A, q_F, q_C)$. (B.155)

For the leveraged long account:

$q_A > 0$. (B.156)

$q_F < 0$. (B.157)

$q_C < 0$. (B.158)

But operationally:

$q_C^{op,0} = 0$. (B.159)

B.16.2 At call admission

After the gate:

$q_C^{op,1} = q_C$. (B.160)

The account now carries an active collateral-performance obligation.

The call amount is:

$505.63. (B.161)

The amount and charge are related but not identical.

The amount measures current required performance.

The charge identifies the obligation's orientation and class.

B.16.3 After cash cure

If the account posts collateral:

q_A remains unchanged. (B.162)

q_F remains unchanged. (B.163)

q_C^op returns to dormant status. (B.164)

But effective liquidity charge may rise because free cash has fallen.

B.16.4 After deleveraging

If 18 units are sold:

q_A,2 = 82q0(x). (B.165)

rather than:

q_A,1 = 100q0(x). (B.166)

Debt falls from:

$8,200.00 to $6,337.00. (B.167)

Therefore the funding-charge magnitude also declines under the declared charge map.

B.17 Spin Evolution Through the Example

B.17.1 Pre-shock

The account is reconciled:

s_close,0 = +1. (B.168)

and:

Δ_spin,0 ≤ ε_spin. (B.169)

B.17.2 After call

The gate creates an open obligation:

s_close,1 = −1. (B.170)

The action state contains the call.

The ledger state does not yet contain completed cure.

Therefore:

Δ_spin,1 > ε_spin. (B.171)

B.17.3 After branch action but before ledger return

Suppose collateral is sent or a sale is executed.

Then:

ActionCompleted = 1. (B.172)

LedgerReturned = 0. (B.173)

The account remains spin-open.

B.17.4 After verified settlement

Once:

- collateral is admitted;
- sale is settled;
- debt is updated;
- P&L is recognized;
- charge balances;
- residual is below tolerance,

then:

$s_{close,2} = +1.$ (B.174)

and:

SpinClosed_P = 1. (B.175)

The post-closure account differs from the original account.

Closure restores accountable identity, not the original numerical state.

B.18 Gauge Transport Through the Example

B.18.1 Market frame

The market frame records:

$R_{M,1} = \$105.63.$ (B.176)

B.18.2 Collateral frame

The collateral frame expects:

$R_{C,1} = h_1 R_{M,1}.$ (B.177)

Therefore:

R_C,1 = $73.94. (B.178)

B.18.3 Observed collateral mark

Suppose the collateral system actually records:

R_C,1^obs = $72.50. (B.179)

Then the real-axis edge residual is:

r_MC^R = $72.50 − $73.94. (B.180)

Therefore:

r_MC^R = −$1.44 per unit. (B.181)

For 100 units:

PositionResidual_MC = −$144.00. (B.182)

Possible explanations include:

- additional concentration haircut;
- stale or different market mark;
- currency adjustment;
- operational error;
- omitted eligibility rule.

The residual should not be erased by silently redefining h.

B.19 Loop Closure

Consider the loop:

Market
→ Collateral
→ Treasury
→ Accounting
→ Market-Reconciled Position. (B.183)

Suppose expected post-sale debt is:

D_expected = $6,337.00. (B.184)

Accounting records:

D_accounting = $6,350.00. (B.185)

The debt-loop residual is:

$\mathscr{R}$_D = $6,350.00 − $6,337.00. (B.186)

Therefore:

$\mathscr{R}$_D = $13.00. (B.187)

Suppose the difference is an unrecorded fee.

Once the fee is identified and entered into the expected transport map:

D_expected,revised = $6,350.00. (B.188)

Then:

$\mathscr{R}$_D = $0.00. (B.189)

The $13.00 economic cost remains.

But it is no longer unexplained gauge residual.

This illustrates:

Economic Loss ≠ Transport Failure. (B.190)

B.20 Post-Closure Recursive State

Assume the account cures through sale.

The new state includes:

$n_2 = 82.$ (B.191)

$D_2 = \$6{,}337.00.$ (B.192)

$B_2 = \$26.38.$ (B.193)

Suppose the broker tightens the collateral factor for the next period:

$h_3 = 0.65.$ (B.194)

Suppose the funding spread rises because of the margin event.

The next-period operator differs:

$\mathcal{L}_3 \neq \mathcal{L}_0.$ (B.195)

The account has closed the immediate event, but its future financial world has changed.

The completed call becomes part of the next load:

$\text{Trace}_2 \rightarrow L_3 \rightarrow \Theta_3.$ (B.196)

B.21 Summary of the Worked Example

The example shows six distinct layers.

1. Complex pressure

Q_M rises from:

$40.84 to $46.42. (B.197)

This identifies stronger CAPM phase exposure.

---

2. Collateral restriction

h falls from:

0.75 to 0.70. (B.198)

This creates additional collateral pressure independently of the market move.

---

3. Margin breach

The buffer moves from:

+$193.53 to −$505.63. (B.199)

The required call is:

$505.63. (B.200)

---

4. Charge activation

The dormant collateral obligation becomes operational.

---

5. Spin opening

The call is issued, but the event remains open until collateral, sale, debt, accounting, and legal consequences return through the required ledgers.

---

6. Recursive closure

After cure, the subject carries:

- fewer assets or less liquidity;
- a different funding state;
- a changed collateral protocol;
- new ledger history;
- altered future effective charge.

The example therefore confirms the conceptual sequence:

Pressure

→ Constraint
→ Gate
→ Action
→ Ledger Return
→ New Financial State. (B.201)

Appendix C — Financial Charge Vertices and Transaction Grammar

C.1 Purpose

This appendix converts the article's charge concept into a transaction-level grammar.

The objective is not merely to label positions as positive or negative. It is to specify:

- which identity carries the charge;
- which field the charge belongs to;
- how the charge enters a transaction;
- whether it is transported, activated, converted, or extinguished;
- which authority admits the transformation;
- what trace must be written;
- what residual remains if the transaction fails to reconcile.

The source Financial Standard Model requires a candidate charge to possess a transformation law, coupling law, transport law, vertex rule, and residual register. Otherwise, the candidate should remain a sensitivity, exposure, or descriptive coordinate.

The constructions below develop that requirement specifically for the leveraged financial system.

C.2 The Financial Vertex

C.2.1 General vertex

Define a financial vertex:

$\mathcal{V}_k := (I_in, \vec{q}_in, G_k, I_out, \vec{q}_out, T_k, \mathcal{R}_k)$. (C.1)

where:

- I_in = identities entering the event;
- $\vec{q}$_in = charge vector entering the event;
- G_k = authorized gate or conversion operator;
- I_out = identities leaving the event;
- $\vec{q}$_out = post-event charge vector;

- T_k = persistent event trace;
- $\mathscr{R}_k$ = residual.

The vertex is not merely a trade timestamp.

It is the full identity transformation admitted by the declared protocol.

C.2.2 Vertex balance

The minimum proposed charge balance is:

$\Sigma\vec{q}_in + \vec{q}_gate = \Sigma\vec{q}_out + \vec{r}_q.$ (C.2)

A more complete open-boundary form is:

$\Sigma\vec{q}_in + \vec{q}_external$
$= \Sigma\vec{q}_out + \vec{q}_cost + \vec{q}_loss + \vec{r}_q.$ (C.3)

The terms must be typed.

For example:

- $\vec{q}_external$ may represent newly supplied funding or capital;
- $\vec{q}_cost$ may represent a separately identified fee obligation;
- $\vec{q}_loss$ may represent a valid write-off or extinguishment;
- $\vec{r}_q$ represents unexplained identity or obligation mismatch.

Equation (C.3) is a proposed accounting grammar, not a universal physical conservation law. The source article likewise treats financial conservation as boundary-sensitive and allows external flow, cost, loss, and residual.

C.2.3 Identity continuity

A charge-balanced vertex should also preserve or validly transform identity.

For an ordinary transfer:

$K_out \simeq T_K(K_in).$ (C.4)

For a conversion:

$K_out = C_K(K_in, G_k, T_k).$ (C.5)

For an extinguishment:

$K_out = \emptyset,$ (C.6)

but the trace must preserve:

$Lineage(K_in \rightarrow Extinguished).$ (C.7)

Charge balance without identity lineage is insufficient.

C.2.4 Vertex trace

Define the minimum vertex trace:

$$T_k := (\text{VertexID},\text{SubjectID},\text{InputIdentity},\text{OutputIdentity},\text{Authority},\text{Time},\text{ChargeIn},\text{ChargeOut},\text{Residual}) \tag{C.8}$$

A stronger trace may also contain:

- contract;
- instruction;
- mark;
- quantity;
- cash amount;
- settlement status;
- legal status;
- accounting status;
- dispute status.

C.2.5 Vertex admission rule

Define:

$$\begin{aligned}\text{AdmitVertex}(\mathcal{V}_k) &:= \text{IdentityMatched} \\ &\wedge \text{AuthorityValid} \\ &\wedge \text{InputChargesTyped} \\ &\wedge \text{OutputChargesTyped} \\ &\wedge \text{ConversionRuleDeclared} \\ &\wedge \text{TraceWritten} \\ &\wedge \text{ResidualPreserved}. \end{aligned} \tag{C.9}$$

If equation (C.9) fails, the event should be classified as:

- candidate transaction;
- unmatched event;
- operational exception;
- unresolved identity transformation.

C.3 Charge Sectors

C.3.1 Multi-sector charge vector

Let the charge vector be:

$\vec{q} := (q_A, q_F, q_C, q_K, q_L, q_O, q_R).$ (C.10)

where:

- q_A = asset-claim orientation;
- q_F = funding orientation;
- q_C = collateral orientation;
- q_K = cash-payment orientation;
- q_L = liquidity orientation;
- q_O = optionality or exercise orientation;
- q_R = reporting or recognition obligation.

These sectors are illustrative.

A real implementation should retain only sectors with:

- observable carriers;
- declared transport;
- defined vertices;
- measurable residual.

C.3.2 No cross-sector cancellation by default

Suppose a leveraged long account has:

$q_A > 0.$ (C.11)

$q_F < 0.$ (C.12)

It does not follow that:

$q_A + q_F = 0.$ (C.13)

because the charges belong to different fields.

The correct representation is:

$\vec{q} = (q_A, q_F, 0, \ldots).$ (C.14)

Neutrality is evaluated component by component:

$q_{net,r} = \Sigma_i q_{i,r}$. (C.15)

Therefore:

$q_{net,A} = 0 \nRightarrow q_{net,F} = 0$. (C.16)

C.3.3 Structural and operational charges

Define structural charge:

$\vec{q}_{struct}$:= rights and obligations encoded in the identity. (C.17)

Define operational charge:

$\vec{q}_{op}$:= currently activated rights and obligations. (C.18)

For contingent charge r:

$q_r^{op} = a_r(X,L,G) q_r^{struct}$. (C.19)

where:

$a_r \in [0,1]$. (C.20)

The binary minimum is:

$a_r \in \{0,1\}$. (C.21)

The distinction allows a collateral obligation to exist contractually before it becomes payable.

C.4 Ordinary Transport Vertex

C.4.1 Transport without identity conversion

An ordinary frame transport changes representation but not underlying charge.

Define:

$\mathcal{V}_{transport} : (K_A, \vec{q}_A) \to (K_B, \vec{q}_B)$. (C.22)

The identity condition is:

$K_B \simeq T_{AB} K_A$. (C.23)

The charge condition is:

$\vec{q}_B = \vec{q}_A$. (C.24)

The representation may change:

$X_B = T_{AB}(X_A; \mathcal{A}_{AB})$. (C.25)

but the relational orientation remains.

C.4.2 Transport residual

Define:

$\vec{r}_{q,AB} := \vec{q}_B - \vec{q}_A.$ (C.26)

For a non-converting edge:

$\vec{r}_{q,AB} = 0$ (C.27)

within declared tolerance.

A nonzero value may indicate:

- ownership mismatch;
- sign error;
- duplicate position;
- missing obligation;
- incorrect netting;
- unauthorized conversion.

C.5 Issuance Vertex

C.5.1 Claim creation

At issuance, a claim and matching obligation are created.

The minimum vertex is:

CashProvider + Issuer
→ FinancialClaim + FinancialObligation + CashTransfer. (C.28)

Normalize:

$q_{claim} = +1.$ (C.29)

$q_{obligation} = -1.$ (C.30)

Then:

$0 + q_{gate}$
$= q_{claim} + q_{obligation} + r_q.$ (C.31)

Under a closed claim–obligation boundary:

$q_{gate} = 0.$ (C.32)

Therefore:

$q_{claim} + q_{obligation} = 0.$ (C.33)

C.5.2 Cash sector

Issuance also creates a cash transfer.

Before settlement:

Investor carries a cash-payment obligation. (C.34)

Issuer carries a cash-receipt claim. (C.35)

After settlement:

q_cash-payment → 0. (C.36)

q_cash-receipt → 0. (C.37)

and the cash balances are updated.

The claim charge survives.

The temporary settlement charges close.

C.5.3 Issuance residual

Possible issuance residuals include:

- claim created without cash received;
- cash received without claim allocation;
- incorrect quantity;
- wrong legal issuer;
- unrecorded fee;
- settlement failure.

Define:

r_q,issue
:= q_in + q_gate − q_out. (C.38)

A nonzero issuance residual means the initial financial identity was not formed cleanly.

C.6 Secondary-Market Transfer Vertex

C.6.1 Execution state

Suppose seller S agrees to transfer n units to buyer B.

Before execution:

q_A,S = +nq0. (C.39)

$q_{A,B} = 0.$ (C.40)

After execution but before settlement, legal ownership may remain with the seller while contractual delivery obligations arise.

Temporary charges include:

$q_{delivery,S} < 0.$ (C.41)

$q_{receive,B} > 0.$ (C.42)

$q_{cash\text{-}pay,B} < 0.$ (C.43)

$q_{cash\text{-}receive,S} > 0.$ (C.44)

C.6.2 Settlement state

After valid delivery-versus-payment settlement:

$q_{A,S'} = 0.$ (C.45)

$q_{A,B'} = +nq_0.$ (C.46)

Temporary delivery and cash charges close:

$q_{delivery,S'} = 0.$ (C.47)

$q_{receive,B'} = 0.$ (C.48)

$q_{cash\text{-}pay,B'} = 0.$ (C.49)

$q_{cash\text{-}receive,S'} = 0.$ (C.50)

The permanent asset charge has moved from seller to buyer.

C.6.3 Transfer balance

Across seller and buyer:

$$q_{A,S} + q_{A,B} = q_{A,S'} + q_{A,B'} + r_q. \quad \text{(C.51)}$$

For successful settlement:

$r_q = 0.$ (C.52)

The trade demonstrates:

Execution Creates Temporary Charge Configuration. (C.53)

Settlement Closes Temporary Charges and Transfers Persistent Charge. (C.54)

C.6.4 Failed settlement

If execution occurs but settlement fails:

$q_{delivery,S} \neq 0.$ (C.55)

$q_{receive,B} \neq 0.$ (C.56)

$q_{cash\text{-}pay,B}$ may remain open. (C.57)

$q_{cash\text{-}receive,S}$ may remain open. (C.58)

The asset charge may remain legally unresolved.

The failed settlement is therefore not merely a delayed price record.

It is a persistent open-charge state.

C.7 Borrowing Vertex

C.7.1 Funding creation

At borrowing:

LenderCash + BorrowerPromise
→ BorrowerCash + LenderClaim + BorrowerDebt. (C.59)

Define:

$q_{F,lender} = +D.$ (C.60)

$q_{F,borrower} = -D.$ (C.61)

Within the combined lender–borrower boundary:

$q_{F,lender} + q_{F,borrower} = 0.$ (C.62)

C.7.2 Funding accrual

Interest accrual increases the monetary amount of the obligation.

Let debt evolve as:

$D_{t+1} = D_t(1 + r_F \Delta t).$ (C.63)

The structural orientation remains:

$\text{sign}(q_F)$ unchanged. (C.64)

The effective amount changes:

$|q_F^{amount,t+1}| > |q_F^{amount,t}|.$ (C.65)

This distinguishes:

- structural charge sign;

- monetary obligation magnitude;
- effective coupling.

C.7.3 Repayment vertex

At repayment amount P:

$D^+ = D^- - P.$ (C.66)

The funding-charge magnitude becomes:

$|q_F^+| = ChargeMap_F(D^+).$ (C.67)

When:

$D^+ = 0,$ (C.68)

the funding charge is extinguished:

$q_F^+ = 0.$ (C.69)

subject to fees, interest, and legal discharge being fully reconciled.

C.8 Collateral-Pledge Vertex

C.8.1 Creation of a security relation

At collateral pledge:

UnencumberedAsset
→ PledgedAsset + SecurityInterest. (C.70)

The provider retains some ownership rights but creates an enforceable collateral relation.

Define:

$q_C,provider < 0.$ (C.71)

$q_C,taker > 0.$ (C.72)

The signs represent:

- provider's obligation to maintain or surrender collateral under conditions;
- taker's conditional enforcement right.

C.8.2 Pledge balance

Within the provider–taker boundary:

$q_C,provider + q_C,taker = 0.$ (C.73)

The market asset charge may remain with the provider:

$$q_{A,provider} > 0. \quad (C.74)$$

Therefore:

Asset Ownership Charge
and
Collateral Encumbrance Charge (C.75)

coexist.

They must not be collapsed.

C.8.3 Collateral release

When the secured obligation is discharged:

$$SecurityInterest \rightarrow Released. \quad (C.76)$$

Then:

$$q_{C,provider} \rightarrow 0. \quad (C.77)$$

$$q_{C,taker} \rightarrow 0. \quad (C.78)$$

The release trace must establish:

- debt discharged;
- collateral identified;
- security interest removed;
- title restrictions updated.

C.9 Margin-Call Activation Vertex

C.9.1 Dormant obligation

Before breach:

$$q_C^{struct} \neq 0. \quad (C.79)$$

but:

$$q_C^{op} = 0. \quad (C.80)$$

The margin agreement already contains the collateral obligation.

It is not yet payable.

C.9.2 Gate condition

The buffer is:

$$B = C + nhR - D. \qquad (C.81)$$

The call guard is:

$$g_margin = B_target - B. \qquad (C.82)$$

A candidate breach exists when:

$$g_margin > 0. \qquad (C.83)$$

A valid margin-call vertex requires:

$$\begin{aligned} &\text{Breach} \\ &\wedge \text{ValidAuthority} \\ &\wedge \text{ApplicableAgreement} \\ &\wedge \text{ValidMark} \\ &\wedge \text{Trace}. \end{aligned} \qquad (C.84)$$

C.9.3 Activation

At call admission:

$$q_C^{op,-} = 0. \qquad (C.85)$$

$$q_C^{op,+} = q_C^{struct}. \qquad (C.86)$$

The operational-charge change is:

$$\Delta q_C^{op} = q_C^{struct}. \qquad (C.87)$$

The required monetary amount is:

$$C_call = [B_target - B]_{+}. \qquad (C.88)$$

Therefore:

$$\text{Charge Activation} \neq \text{Call Amount}. \qquad (C.89)$$

The former identifies obligation class and orientation.

The latter identifies current performance magnitude.

C.9.4 Activation balance

The gate contributes the conditional activation:

$$\begin{aligned} &q_C^{dormant} + q_gate \\ &\rightarrow q_C^{operational} + r_q. \end{aligned} \qquad (C.90)$$

The gate does not create a previously nonexistent contract.

It converts a dormant branch of the contract into an active branch.

C.10 Collateral-Posting Vertex

C.10.1 Posting action

Suppose collateral amount ΔC is delivered.

Then:

$C^{+} = C^{-} + \Delta C_admitted.$ (C.91)

The buffer becomes:

$B^{+} = B^{-} + \Delta C_admitted.$ (C.92)

C.10.2 Charge discharge

If the admitted amount satisfies the call:

$B^{+} \geq B_target,$ (C.93)

then:

$q_C\text{^}op \to 0.$ (C.94)

subject to:

- eligibility;
- timing;
- control;
- legal perfection;
- ledger recognition.

Cash transmission alone does not discharge the collateral charge if the asset is rejected or delivered to the wrong account.

C.10.3 New liquidity orientation

Posting collateral may create or amplify liquidity pressure.

Let free liquidity be F.

Then:

$F^{+} = F^{-} - \Delta C_admitted.$ (C.95)

A liquidity charge or effective liquidity coupling may rise:

$|q_L,eff^{+}| > |q_L,eff^{-}|.$ (C.96)

Thus one charge may close while another effective exposure intensifies.

C.11 Voluntary-Deleveraging Vertex

C.11.1 Sale and debt reduction

Suppose Δn units are sold at net price p_{net}.

Then:

$$n^+ = n^- - \Delta n. \qquad (C.97)$$

$$D^+ = D^- - p_{net}\Delta n. \qquad (C.98)$$

The asset-charge change is:

$$\Delta q_A = -\Delta n q_0(x). \qquad (C.99)$$

The funding-charge change is:

$$\Delta q_F = \text{ChargeMap}_F(D^+) - \text{ChargeMap}_F(D^-). \qquad (C.100)$$

C.11.2 Temporary settlement charges

Before settlement:

$$q_{delivery} < 0. \qquad (C.101)$$

$$q_{cash\text{-}receivable} > 0. \qquad (C.102)$$

Debt reduction may remain expected rather than ledgered.

Only after settlement:

$$q_{delivery} \to 0. \qquad (C.103)$$

$$q_{cash\text{-}receivable} \to 0. \qquad (C.104)$$

$$D_{ledger} \to D^+. \qquad (C.105)$$

The deleveraging vertex therefore contains a nested transfer vertex.

C.11.3 Deleveraging residual

Define:

$$r_{q,delev} := q_{in} - q_{out} - q_{closed} + q_{created}. \qquad (C.106)$$

Possible residuals include:

- executed but unsettled units;
- incorrect debt reduction;

- unmatched cash;
- residual delivery obligation;
- unrecorded transaction costs.

C.12 Forced-Liquidation Vertex

C.12.1 Authority change

At forced liquidation, control shifts.

Before the event:

Authority_subject = Active. (C.107)

After valid broker intervention:

Authority_broker = Active. (C.108)

This authority change should appear in the identity and trace.

C.12.2 Position conversion

For liquidation quantity ℓ:

$n^+ = n^- - \ell$. (C.109)

Debt changes:

$D^+ = D^- - P_{net,liq}$. (C.110)

The asset-charge reduction is:

$\Delta q_A = -\ell q_0(x)$. (C.111)

The funding-charge reduction depends on realized proceeds:

$\Delta q_F = ChargeMap_F(D^+) - ChargeMap_F(D^-)$. (C.112)

C.12.3 Economic loss versus charge residual

Execution loss is:

$Loss_{exec} = \ell(R_{mark} - p_{exec}) + Fees$. (C.113)

Charge residual is:

r_q = unexplained rights–obligations mismatch. (C.114)

Therefore:

$Loss_{exec} \neq r_q$. (C.115)

The liquidation may be charge-reconciled while economically destructive.

Conversely, it may have small economic loss but leave an unresolved ownership or settlement mismatch.

C.12.4 Incomplete liquidation

If liquidation does not cure the buffer:

$B^{+} < B_target,$ (C.116)

then the operational collateral charge remains active:

$q_C^{op,+} \neq 0.$ (C.117)

The vertex closes the sale but does not close the margin cycle.

C.13 Option-Exercise Vertex

C.13.1 Pre-exercise identity

Before exercise, the holder carries:

$q_O,holder > 0.$ (C.118)

The writer carries:

$q_O,writer < 0.$ (C.119)

The option charge belongs to the exercise field.

It is not identical to the underlying asset charge.

C.13.2 Exercise gate

Exercise requires:

ExerciseRightValid
∧ TimeValid
∧ InstructionValid
∧ ContractInForce. (C.120)

At exercise:

Option Identity
→ Underlying Delivery or Cash Settlement Identity. (C.121)

C.13.3 Physical settlement

For physical delivery:

q_O + q_gate
→ q_A + q_cash + q_delivery + r_q. (C.122)

The option charge is converted into:

- underlying-asset charge;
- cash-payment charge;
- delivery obligation.

The vertex is charge-converting rather than merely charge-transporting.

C.13.4 Cash settlement

For cash settlement:

q_O + q_gate
→ q_cash-receivable + q_cash-payable + r_q. (C.123)

After payment:

q_cash-receivable → 0. (C.124)

q_cash-payable → 0. (C.125)

The option identity is extinguished.

C.14 Novation Vertex

C.14.1 Counterparty replacement

Novation replaces one obligor or claimant with another.

Before novation:

Claimant A ↔ Obligor B. (C.126)

After novation:

Claimant A ↔ Obligor C. (C.127)

The economic terms may remain similar.

The identity relation changes.

C.14.2 Novation balance

The original obligation is extinguished:

q_B → 0. (C.128)

A new obligation is created:

$q_{C,new} \neq 0$. (C.129)

The balance is:

$$q_B + q_{gate} = q_{C,new} + q_{extinguished} + r_q. \quad (C.130)$$

Valid authority and consent are essential.

Without them, the transformation is not a lawful novation.

C.14.3 Novation residual

Possible residuals include:

- old obligation not legally released;
- new obligor not validly bound;
- accounting system still referencing old counterparty;
- collateral agreement not transferred;
- regulatory exposure double-counted.

Novation is therefore a particularly strong test of cross-frame charge identity.

C.15 Netting Vertex

C.15.1 Gross obligations

Suppose two counterparties owe:

A owes B amount X. (C.131)

B owes A amount Y. (C.132)

Gross charge representation is:

$$\vec{q}_{gross} = (-X, +X, +Y, -Y). \quad (C.133)$$

C.15.2 Net obligation

Under a valid netting agreement:

$$NetPayable_A = [X - Y]_+. \quad (C.134)$$

$$NetPayable_B = [Y - X]_+. \quad (C.135)$$

The gross identities are transformed into a net settlement identity.

C.15.3 Netting is not annihilation

Gross contractual charges may remain relevant for:

- legal enforceability;
- close-out;
- collateral;
- regulatory exposure;
- default scenarios.

Therefore:

NetPresentation ≠ Erasure of Gross Identity. (C.136)

A valid netting vertex must state:

- which obligations are legally nettable;
- under which event;
- in which jurisdiction;
- for which frame.

C.15.4 Netting residual

A residual may arise when:

- accounting nets but legal does not;
- collateral nets but regulatory reporting remains gross;
- settlement nets only same-day obligations;
- one leg is disputed.

Netting therefore illustrates field-indexed neutrality.

C.16 Default Vertex

C.16.1 Default admission

A default candidate may arise from:

- missed payment;
- covenant breach;
- failure to meet margin;

- insolvency event;
- acceleration;
- contractual credit event.

A valid default vertex requires:

Trigger
∧ ApplicableContract
∧ ValidAuthority
∧ Evidence
∧ Trace. (C.137)

C.16.2 Identity conversion

The performing debt identity transforms:

K_performing
→ K_defaulted
→ {K_recovery,K_collateral,K_legal,K_extinguished}. (C.138)

The charge conversion is:

q_debt + q_collateral + q_guarantee + q_gate
→ q_recovery + q_enforcement + q_litigation + q_extinguished + r_q. (C.139)

C.16.3 Recovery hierarchy

Let recovery classes be indexed by j.

Then:

q_recovery,total = Σ_jq_recovery,j. (C.140)

The distribution must obey:

PriorityRule(K_contract,K_law). (C.141)

A failure to allocate according to the declared priority produces:

r_q,priority ≠ 0. (C.142)

C.16.4 Default closure

Default declaration does not close the identity.

Closure requires:

- claims verified;
- collateral allocated;

- recovery rights recorded;
- extinguished amounts identified;
- accounting losses recognized;
- residual disputes preserved.

Thus:

Default Event ≠ Recovery Closure. (C.143)

C.17 Extinguishment Vertex

C.17.1 Valid extinguishment

A charge may be extinguished through:

- full payment;
- expiration;
- cancellation;
- release;
- write-off;
- legal discharge;
- merger;
- settlement agreement.

The vertex is:

q_in + q_gate
→ q_extinguished + r_q. (C.144)

For complete valid extinguishment:

r_q = 0. (C.145)

C.17.2 Accounting write-off versus legal extinguishment

An accounting write-off may remove an asset from the accounting frame.

It does not necessarily eliminate the legal claim.

Therefore:

AccountingWriteOff ⇏ LegalExtinguishment. (C.146)

Possible states include:

$q_accounting = 0.$ (C.147)

$q_legal > 0.$ (C.148)

This is not contradiction if the frame mapping is declared.

It becomes a gauge failure only when the frames incorrectly claim the same charge status.

C.18 Vertex Composition

C.18.1 Sequential vertices

Suppose:

$\mathcal{V}_1 : K_0 \to K_1.$ (C.149)

$\mathcal{V}_2 : K_1 \to K_2.$ (C.150)

The composite vertex is:

$\mathcal{V}_2 \circ \mathcal{V}_1 : K_0 \to K_2.$ (C.151)

Charge transport gives:

$$\vec{q}_2 = J_q,2[J_q,1(\vec{q}_0)] + \text{accumulated residual.} \quad \text{(C.152)}$$

C.18.2 Residual composition

Let:

$\vec{r}_q,1 = \vec{q}_{1,obs} - \vec{q}_{1,expected}.$ (C.153)

$\vec{r}_q,2 = \vec{q}_{2,obs} - \vec{q}_{2,expected}.$ (C.154)

Then total residual is not always a simple sum.

A transported composition is:

$$\vec{r}_q,total = \vec{r}_q,2 + T_12(\vec{r}_q,1). \quad \text{(C.155)}$$

The earlier residual must be translated into the later frame before aggregation.

C.18.3 Noncommuting vertices

In general:

$\mathcal{V}_A\mathcal{V}_B \neq \mathcal{V}_B\mathcal{V}_A.$ (C.156)

For example:

CollateralPosting followed by Liquidation (C.157)

may produce a different result from:

Liquidation followed by CollateralPosting. (C.158)

Likewise:

Netting before Default (C.159)

may differ from:

Default before Netting. (C.160)

The order matters because each vertex changes:

- identity;
- available assets;
- authority;
- future admissible branches.

This is one source of path dependence and candidate financial curvature.

C.19 Vertex Graph

C.19.1 Identity graph

Let identities be graph nodes:

$\mathcal{N}$:= {K1,K2,...,K_N}. (C.161)

Let admitted transformations be directed edges:

$\mathscr{E}$:= {$\mathcal{V}$1,$\mathcal{V}$2,...,$\mathcal{V}$_M}. (C.162)

The financial identity graph is:

$\mathcal{G}$_q := ($\mathcal{N}$,$\mathscr{E}$). (C.163)

C.19.2 Incidence matrix

Let B_q be the oriented incidence matrix.

For vertex edge e connecting source s(e) to target t(e):

(B_q)_{ve} =
{
−1, v = s(e);
+1, v = t(e);
0, otherwise.

} (C.164)

Let f_q be the vector of charge flows along edges.

Then the node imbalance is:

$\vec{r}_{node} = B_q f_q + \vec{q}_{external}.$ (C.165)

For a reconciled internal node:

$\vec{r}_{node} = 0.$ (C.166)

A nonzero node residual identifies:

- missing incoming charge;
- missing outgoing charge;
- duplicate transformation;
- unmatched identity.

C.19.3 Multi-sector incidence

For R charge sectors, define:

$f_q \in \mathbb{R}^{M \times R}.$ (C.167)

Then:

$R_{node} = B_q f_q + Q_{external}.$ (C.168)

Each column of R_node corresponds to one field:

- asset;
- funding;
- collateral;
- cash;
- liquidity;
- optionality;
- reporting.

This prevents a residual in one sector from being concealed by an opposite residual in another.

C.20 Discrete Charge-Continuity Equation

C.20.1 Local continuity

Let q_v,k be charge stored at node v at ledger period k.

Let f_in,v,k and f_out,v,k be incoming and outgoing charge flows.

Then:

$$q_{v,k+1} - q_{v,k} = f_{in,v,k} - f_{out,v,k} + s_{v,k} - r_{v,k}. \quad (C.169)$$

where:

- s_v,k = valid source, issuance, or conversion;
- r_v,k = unexplained residual or leakage.

Equation (C.169) is the discrete financial charge-continuity candidate.

C.20.2 Vector form

Let q_k be the vector of node charges.

Then:

$$q_{k+1} - q_k = -B_{qf} f_k + s_k - r_k. \quad (C.170)$$

For a closed charge-preserving system:

$$s_k = 0. \quad (C.171)$$

$$r_k = 0. \quad (C.172)$$

Therefore:

$$q_{k+1} - q_k = -B_{qf} f_k. \quad (C.173)$$

Total internal charge is preserved when the graph boundary is closed.

C.20.3 Open-boundary form

For an open system:

$$\Sigma_v q_{v,k+1} - \Sigma_v q_{v,k} = q_{external,in} - q_{external,out} - r_{total}. \quad (C.174)$$

A claim of neutrality is meaningful only after declaring:

- graph boundary;
- charge sector;
- period;
- external flows;
- valid conversions.

C.21 Charge and Spin at a Vertex

C.21.1 Charge describes the relational transformation

At a margin-call vertex:

q_C^dormant → q_C^operational. (C.175)

At settlement:

q_A,seller → q_A,buyer. (C.176)

At exercise:

q_option → q_underlying + q_cash. (C.177)

Charge identifies what relational orientation changes.

C.21.2 Spin describes completion

The same vertex may create an open action–ledger cycle.

At execution:

Action = trade matched. (C.178)

Ledger return = settlement and ownership update. (C.179)

At margin call:

Action = call issued. (C.180)

Ledger return = collateral, deleveraging, liquidation, or default closure. (C.181)

Therefore:

Charge Transformation ≠ Spin Closure. (C.182)

A charge may be transferred while its ledger return remains incomplete.

C.21.3 Vertex-spinor representation

For vertex k:

$\Psi_k :=$
[
ψ_action,k
ψ_ledger,k
]. (C.183)

The expected ledger return is:

ψ^_ledger,k = U_kψ_action,k. (C.184)

The vertex closure residual is:

$\mathscr{R}_{spin,k} = \psi_{ledger,k} - \psi\hat{}_{ledger,k}.$ (C.185)

A charge-balanced but spin-open event may satisfy:

$r_{q,k} \approx 0,$ (C.186)

while:

$\mathscr{R}_{spin,k} \neq 0.$ (C.187)

For example, the obligations may be correctly identified but not yet settled.

C.22 Charge and Gauge Transport

C.22.1 Charge-preserving connection

For ordinary frame edge $A \to B$:

$Q\hat{}_B U_{AB} = U_{AB} Q\hat{}_A.$ (C.188)

This means that transport respects the charge representation.

C.22.2 Charge-converting connection

For a conversion edge:

$Q\hat{}_B U_{AB} - U_{AB} Q\hat{}_A = J_{q,AB}.$ (C.189)

The conversion current $J_{q,AB}$ must correspond to a declared vertex.

Without such a vertex:

$J_{q,AB} \neq 0$ (C.190)

is a transport defect.

C.22.3 Loop charge residual

For a closed frame loop ℓ:

$H_\ell := \prod U_{AB}.$ (C.191)

The charge-loop residual is:

$r_{q,loop} := Q\hat{}_{start} H_\ell \Psi - H_\ell Q\hat{}_{start} \Psi.$ (C.192)

For charge-preserving flat transport:

$r_{q,loop} = 0.$ (C.193)

A nonzero value may indicate that different frames disagree about:

- owner;
- obligor;
- collateral provider;
- exercise status;
- netting status;
- extinguishment.

C.23 Charge Screening

C.23.1 Structural versus effective charge

The source model proposes:

$\vec{q}_{eff} = S_P \vec{q}_{struct}.$ (C.194)

where S_P is a screening operator. Possible screening mechanisms include hedging, collateral, netting, insurance, diversification, and legal guarantees. Screening reduces observed effective exposure but does not necessarily annihilate the structural relation.

C.23.2 Diagonal screening

A simple screening operator is:

$S_P = \mathrm{diag}(s_A, s_F, s_C, \ldots).$ (C.195)

where:

$0 \le s_r \le 1.$ (C.196)

Then:

$q_{eff,r} = s_r q_{struct,r}.$ (C.197)

Examples include:

- hedge reduces effective market charge;
- collateral reduces effective credit or funding charge;
- insurance reduces effective loss exposure;
- legal guarantee reduces recovery uncertainty.

C.23.3 Cross-sector screening

A more general operator is:

$S_P =$

[
s_AA s_AF s_AC
s_FA s_FF s_FC
s_CA s_CF s_CC
]. (C.198)

Off-diagonal terms allow one field to alter another.

For example:

- collateral may reduce funding exposure;
- leverage may amplify market exposure;
- liquidity stress may increase collateral sensitivity.

This matrix should be estimated rather than assumed.

C.23.4 Screening residual

Define:

r_screen := q_eff,obs − S_Pq_struct. (C.199)

A large screening residual may indicate:

- hidden leverage;
- hedge failure;
- wrong-way risk;
- legal ineligibility;
- basis risk;
- concentration;
- nonlinear feedback.

Screening is therefore another empirical layer, not a guaranteed cancellation.

C.24 Charge Renormalization Through Closure

C.24.1 Periodic update

Let $\vec{q}_k$ be the charge estimate in closure period k.

The source proposes a closure-renormalization map in which trace and residual alter effective charge across periods.

The present specialization is:

$\vec{q}_{eff,k+1}$

$= \mathscr{R}_q,k\to k_{+1}(\vec{q}_eff,k,Trace_k,Residual_k,X_k_{+1}).$ (C.200)

C.24.2 Example: margin failure

A margin failure may produce:

- lower collateral factor;
- higher funding spread;
- tighter position limit;
- greater liquidation probability.

Then:

$|q_eff,F,k_{+1}| > |q_eff,F,k|.$ (C.201)

$|q_eff,C,k_{+1}| > |q_eff,C,k|.$ (C.202)

even when the structural signs remain unchanged.

C.24.3 Stable charge class

A temporary orientation should not automatically become a permanent charge label.

A candidate stable charge may arise through repeated transport:

a_0
$\to$ $response_1$
$\to$ $trace_1$
$\to$ $response_2$
$\to$ $trace_2$
$\to$ ...
$\to$ stable class q. (C.203)

A possible definition is:

q := stable equivalence class of recursively transported coupling orientation. (C.204)

This remains a generative hypothesis rather than an established theorem.

C.25 Charge Residual Register

C.25.1 Residual vector

Define:

$\mathscr{R}_q$
:= (r_carrier,r_field,r_sign,r_units,r_transport,r_vertex,r_balance,r_timing,r_authority). (C.205)

where:

- r_carrier = uncertain identity carrier;
- r_field = unclear field;
- r_sign = disputed orientation;
- r_units = dimensional inconsistency;
- r_transport = cross-frame mismatch;
- r_vertex = unexplained conversion;
- r_balance = charge-flow imbalance;
- r_timing = lifecycle mismatch;
- r_authority = invalid or disputed gate authority.

C.25.2 Residual severity

Define a weighted residual score:

$R_q^2 = \mathscr{R}_q^T W_q \mathscr{R}_q.$ (C.206)

The weights should reflect consequence.

For example:

- an immaterial rounding mismatch may receive low weight;
- disputed collateral title may receive high weight;
- uncertain owner or obligor may receive critical weight.

C.25.3 Residual status

Each residual should have status:

$$\text{Status}(r_q) \in \{\text{Open},\text{Investigating},\text{Explained},\text{Corrected},\text{Accepted},\text{Escalated}\}. \quad \text{(C.207)}$$

An explained economic difference may remain in the trace but cease to be unexplained residual.

An accepted residual remains unresolved but consciously retained.

C.26 Charge Audit Template

For each proposed charge q_r, record:

Audit field	Required statement
Carrier	Which bounded identity carries q_r?

Audit field	Required statement
Field	Under which declared field does it transform?
Orientation	What do positive and negative signs mean?
Units	Is q_r discrete, monetary, normalized, or dimensionless?
Coupling	What acts differently because q_r exists?
Screening	Which hedges, collateral, or netting reduce q_eff?
Transport	How does q_r move across frames?
Vertex	Which event activates, transfers, converts, or extinguishes it?
Authority	Who may admit that vertex?
Balance	What is the proposed conservation or reconciliation rule?
Quantization	Are units discrete, continuous, or mixed?
Trace	What permanent record proves the transformation?
Residual	What mismatch remains?
Falsifier	Which evidence would demote q_r to exposure?

This operationalizes the source charge-audit questions.

C.27 Minimum Charge Ledger

C.27.1 Ledger record

Define:

$$
\begin{aligned}
&L_{q,k} \\
&:= (\\
&\text{EventID}, \\
&\text{CarrierID}, \\
&\text{Field}, \\
&\text{ChargeClass}, \\
&\text{ChargeIn}, \\
&\text{GateContribution}, \\
&\text{ChargeOut}, \\
&\text{Residual}, \\
&\text{Authority}, \\
&\text{Timestamp}, \\
&\text{Status} \\
&). \qquad (C.208)
\end{aligned}
$$

C.27.2 Ledger update

The charge ledger evolves as:

$$L_{q,k+1} = L_{q,k} \oplus T_{charge,k} \oplus \mathscr{R}_{q,k}. \qquad (C.209)$$

The next effective charge is:

$$\vec{q}_{eff,k+1} = U_q(\vec{q}_{struct,k+1}, L_{q,k+1}, MarketState_{k+1}). \qquad (C.210)$$

C.27.3 No silent correction

If a charge error is discovered:

$L_{q,k} \to L_{q,k+1}$. (C.211)

The corrected record should preserve:

- original entry;
- correction;
- authority;
- reason;
- impact;
- residual.

The original trace should not be overwritten as though it never existed.

C.28 Reduction Rules

C.28.1 Charge to obligation

If the candidate has:

- clear carrier;
- clear contractual duty;
- no meaningful field transformation;
- no transport algebra;

use:

Contractual Obligation. (C.212)

C.28.2 Charge to signed exposure

If the candidate has:

- stable sign;
- measurable response;
- no transfer or conversion vertex;

use:

Signed Exposure. (C.213)

C.28.3 Charge to sensitivity

If the candidate is only:

$$\partial Y/\partial X, \qquad (C.214)$$

and varies with the current state without stable relational identity, use:

$$\text{Sensitivity}. \qquad (C.215)$$

Examples include:

- beta;
- duration;
- delta;
- phase sensitivity.

They may contribute to coupling strength without constituting structural charge.

C.28.4 Charge to descriptive coordinate

If no carrier, field, or vertex exists, use:

$$\text{Descriptive Coordinate}. \qquad (C.216)$$

This reduction preserves useful measurement while removing unsupported ontology.

C.29 Summary Vertex Table

Vertex	Charge before	Gate	Charge after	Principal residual
Issuance	no claim	valid issue	claim + obligation	cash or allocation mismatch
Trade execution	seller claim	execution	temporary delivery/ cash charges	unmatched trade
Settlement	temporary obligations	settlement	claim transferred	failed delivery or cash
Borrowing	no debt relation	loan contract	lender claim + borrower debt	funding mismatch
Collateral pledge	unencumbered asset	pledge	provider duty + taker right	invalid security interest
Margin call	dormant collateral duty	valid call	active collateral obligation	disputed mark or authority
Collateral posting	active call	collateral admission	obligation discharged	rejected or late collateral
Deleveraging	asset + debt	sale	reduced asset and debt charges	unsettled sale
Forced liquidation	active call	broker authority	reduced position, possible deficiency	execution and settlement mismatch

Vertex	Charge before	Gate	Charge after	Principal residual
Exercise	option charge	exercise	underlying or cash charges	invalid exercise or delivery failure
Novation	original counterparty	valid consent	replacement counterparty	double obligation
Netting	gross obligations	valid netting	net settlement charge	cross-frame netting disagreement
Default	performing debt	default recognition	recovery/legal charges	disputed claims
Extinguishment	active claim	payment/ release	zero or successor identity	unrecorded surviving right

---

C.30 Final Charge Grammar

The complete charge process is:

Identity
→ Structural Orientation
→ Field Coupling
→ Screening
→ Gate
→ Vertex
→ Transport or Conversion
→ Charge Balance
→ Trace
→ Residual
→ Updated Effective Charge. (C.217)

In compact form:

$$(K_k,\vec{q}_{struct,k},\vec{q}_{eff,k}) \to \mathcal{V}_k \to (K_{k+1},\vec{q}_{struct,k+1},\vec{q}_{eff,k+1},T_k,\mathscr{R}_{q,k}). \quad (C.218)$$

The principal interpretive rule is:

> Charge identifies what relational orientation is carried through a financial transformation. The vertex identifies how that orientation is transferred, activated, converted, or extinguished. The ledger records whether the transformation actually closed.

The next appendix will construct the corresponding **action–ledger matrices and closure operators**, showing how charge vertices are embedded inside the financial spinor and how distinct closure modes arise from the matrix dynamics.

Appendix D — Action–Ledger Matrices and Financial Closure Operators

D.1 Purpose

Appendix C specified how financial charge is transported, activated, converted, and reconciled at transaction vertices.

This appendix develops the corresponding matrix architecture for **financial spin**.

Its central problem is:

> How should a financial subject be represented when outward action and ledger return are distinct, coupled, and independently capable of failure?

The source framework defines the governing doublet as:

$\Psi_B = [\psi_action, \psi_ledger]^T$. (D.1)

It also imposes a strict admission condition: the object must possess bounded identity, consequential outward action, an independent ledger-return process, identifiable residual, and cross-frame relevance. A pair of arbitrary variables is not sufficient.

The source's general closure law is:

Potential
→ Outward Action
→ Unresolved Consequence
→ Ledger Return
→ Accountable Identity. (D.2)

Its compressed double-cycle notation is:

$\Psi \rightarrow -\Psi \rightarrow \Psi'$. (D.3)

The final state Ψ' need not be numerically identical to Ψ. It must remain recognizably continuous with Ψ, or be validly converted into a traceable successor identity.

The present appendix converts that verbal structure into:

- closure-space basis vectors;
- action and ledger projectors;
- exchange and orientation matrices;
- branch-specific closure operators;
- transport-adjusted residuals;
- eigenmodes;
- discrete propagation laws;
- stability conditions;
- reduction rules.

These matrices are proposed constructions of the present article unless otherwise stated.

D.2 The Two-Component Closure Space

D.2.1 Closure basis

Define the action basis vector:

e_A :=
[
1
0
]. (D.4)

Define the ledger basis vector:

e_L :=
[
0
1
]. (D.5)

The two-dimensional closure space is:

$\mathbb{C}^2$_cl := span{e_A,e_L}. (D.6)

A financial closure state is:

Ψ :=
[
ψ_A
ψ_L
]. (D.7)

Equivalently:

Ψ = ψ_Ae_A + ψ_Le_L. (D.8)

The components ψ_A and ψ_L may themselves be:

- scalars;
- normalized feature vectors;
- complex valuation states;
- tensor-product states;
- frame-indexed ledger objects.

Therefore:

Ψ ∈ $\mathbb{C}^2$_cl ⊗ $\mathscr{H}$_fin. (D.9)

where $\mathscr{H}$_fin is the internal financial state space.

D.2.2 Action component

The action component may contain:

ψ_A
:= Φ_A(
MarketState,
GateDecision,
PositionInstruction,

CollateralInstruction,
LiquidationInstruction,
ExpectedCashFlow,
ExpectedDebtChange
). (D.10)

It records what the subject or authority has done outwardly.

D.2.3 Ledger component

The ledger component may contain:

ψ_L
:= Φ_L(
SettledPosition,
AdmittedCollateral,
RecordedCash,
RecordedDebt,
RecognizedP&L,
LegalStatus,
RiskStatus,
Residual
). (D.11)

It records what has become institutionally inherited.

D.2.4 Why the components require a common feature space

Raw action and ledger data may use different units and conventions.

For example:

- executed units versus settled units;
- market value versus collateral value;
- expected debt reduction versus booked debt reduction;
- economic loss versus accounting loss;
- contractual ownership versus legal title.

Therefore the feature maps Φ_A and Φ_L must embed both components into a declared comparable space.

Without a common embedding:

ψ_A – ψ_L (D.12)

is generally meaningless.

The correct comparison must include the expected transport:

ψ_L – U_ALψ_A. (D.13)

D.3 Projection Operators

D.3.1 Action projector

Define:

P_A :=
[
1 0
0 0
]. (D.14)

Then:

P_AΨ =
[
ψ_A
0
]. (D.15)

D.3.2 Ledger projector

Define:

P_L :=
[
0 0
0 1
]. (D.16)

Then:

P_LΨ =
[
0
ψ_L
]. (D.17)

D.3.3 Projector identities

The projectors satisfy:

$P_A^2 = P_A$. (D.18)

$P_L^2 = P_L$. (D.19)

$P_A P_L = 0$. (D.20)

$P_A + P_L = I_2$. (D.21)

These are exact matrix identities.

They formalize the statement that action and ledger are distinct but jointly exhaustive components of the minimum closure state.

D.3.4 Component extraction

The action state is:

$\psi_A = e_A^\dagger \Psi.$ (D.22)

The ledger state is:

$\psi_L = e_L^\dagger \Psi.$ (D.23)

A scalar closure score may be calculated from both components, but it should not replace them before their independent contribution has been tested.

D.4 Basic Closure Matrices

D.4.1 Identity matrix

The closure identity is:

$$I_2 := \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$ (D.24)

It preserves both components:

$I_2\Psi = \Psi.$ (D.25)

D.4.2 Orientation matrix

Define:

$$\sigma_z := \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$ (D.26)

Then:

$$\sigma_z\Psi = \begin{bmatrix} \psi_A \\ -\psi_L \end{bmatrix}.$$ (D.27)

The matrix σ_z distinguishes outward action from inward ledger return.

It does not imply that the ledger component is economically negative.

The sign records opposite closure orientation.

D.4.3 Exchange matrix

Define:

σ_x :=
[
0 1
1 0
]. (D.28)

Then:

σ_xΨ =
[
ψ_L
ψ_A
]. (D.29)

The matrix exchanges action and ledger components.

Financially, it represents the reciprocal principle:

Action must enter the ledger. (D.30)

Ledger must constrain the next action. (D.31)

D.4.4 Phase-sensitive exchange matrix

Define:

σ_y :=
[
0 −i
i 0
]. (D.32)

Then:

σ_yΨ =
[
−iψ_L
iψ_A
]. (D.33)

This matrix permits relative phase between action and ledger components.

Its use is optional.

It is justified only when complex relative orientation provides measurable value beyond a real

two-component coupling matrix.

D.4.5 Pauli identities

The matrices satisfy:

$\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = I_2.$ (D.34)

Their commutators are:

$[\sigma_i,\sigma_j] = 2i\varepsilon_{ijk}\sigma_k.$ (D.35)

Their anticommutators are:

$\{\sigma_i,\sigma_j\} = 2\delta_{ij}I_2.$ (D.36)

These are exact algebraic properties.

Their financial interpretation remains a proposed operator correspondence.

D.5 Action and Ledger Completion Operators

D.5.1 Outward-action operator

Define the outward-action operator:

$$\mathcal{A} := \begin{bmatrix} A_A & 0 \\ 0 & I_L \end{bmatrix}. \quad \text{(D.37)}$$

where:

- A_A acts on the action component;
- I_L leaves the ledger component unchanged initially.

Then:

$$\mathcal{A}\Psi^- = \begin{bmatrix} A_A\psi_A^- \\ \psi_L^- \end{bmatrix}. \quad \text{(D.38)}$$

This describes a margin call, sale instruction, exercise declaration, or default notice that changes outward state before ledger return has completed.

D.5.2 Ledger-return operator

Define:

$\mathscr{L}$:=
[
I_A 0
0 L_L
]. (D.39)

Then:

$\mathscr{L}\Psi$ =
[
ψ_A
L_Lψ_L
]. (D.40)

However, a genuine ledger return should usually depend on the action component.

The more complete operator is therefore:

$\mathscr{L}$_A→L :=
[
I_A 0
U_AL L_0
]. (D.41)

Applied to Ψ:

$\mathscr{L}$_A→LΨ =
[
ψ_A
U_ALψ_A + L_0ψ_L
]. (D.42)

Here:

- U_AL transports the action into expected ledger form;
- L_0 preserves or updates prior ledger content.

D.5.3 Recursive ledger-to-action operator

The updated ledger may constrain the next action.

Define:

$\mathcal{A}$_L→A :=
[
A_0 U_LA
0 I_L
]. (D.43)

Then:

$\mathcal{A}$_L→AΨ =
[
A_0ψ_A + U_LAψ_L
ψ_L

]. (D.44)

The two directions form a recursive pair:

Action → Ledger. (D.45)

Ledger → Next Action. (D.46)

This is the matrix form of recursive market closure.

D.6 One-Cycle and Two-Cycle Closure

D.6.1 First cycle

Let the first-cycle operator be:

$C_1 := \mathcal{A}$. (D.47)

Then:

$\Psi_1 = C_1\Psi_0$. (D.48)

The first cycle changes the outward component while leaving ledger return incomplete.

Normally:

$\Psi_1 \neq \Psi_0$. (D.49)

and:

$\text{SpinClosed}(\Psi_1) = 0$. (D.50)

D.6.2 Second cycle

Define the return operator:

$C_R := \mathscr{L}_A\to L$. (D.51)

Then the two-cycle operator is:

$C_2 := C_R\, C_1$. (D.52)

The completed state is:

$\Psi_2 = C_2\Psi_0$. (D.53)

Closure requires:

$\Psi_2 \simeq_P\ \Psi_\text{expected}$. (D.54)

The symbol $\simeq_P$ denotes identity equivalence under protocol P.

It does not require:

$\Psi_2 = \Psi_0$. (D.55)

D.6.3 Exact identity closure

The strongest special case is:

$C_2 = I_2.$ (D.56)

Then:

$\Psi_2 = \Psi_0.$ (D.57)

This exact return is rarely appropriate for real finance because:

- position quantity may change;
- debt may change;
- collateral may change;
- losses may be recognized;
- legal status may change.

The more realistic condition is:

$C_2\Psi_0 = V_{successor}\Psi_0 + \mathscr{R}_{close}.$ (D.58)

where:

- $V_{successor}$ is the valid identity-successor map;
- $\mathscr{R}_{close}$ is closure residual.

D.6.4 Valid transformed closure

Define:

$C_2\Psi_0 = \Psi_{successor} + \mathscr{R}_{close}.$ (D.59)

Closure is accepted when:

$d_K[K(\Psi_{successor}),K_{expected}] \le \varepsilon_K.$ (D.60)

and:

$\|\mathscr{R}_{close}\| \le \varepsilon_{close}.$ (D.61)

Thus:

Closure = Valid Return to Accountable Successor. (D.62)

not:

Closure = Erasure of Change. (D.63)

D.7 The Open-Return Operator

D.7.1 Closure-sign matrix

Define the closure-sign operator:

$S_{open} := -I_2.$ (D.64)

The notation:

$\Psi \to -\Psi$ (D.65)

marks the action-complete but ledger-open condition.

This is an abstract closure sign.

It does not imply that every financial coordinate changes sign.

D.7.2 Component-selective open state

A more operational matrix is:

$$S_{AL} := \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$ (D.66)

Then:

$$S_{AL}\Psi = \begin{bmatrix} \psi_A \\ -\psi_L \end{bmatrix}.$$ (D.67)

The action component remains outwardly committed.

The ledger component is marked as unresolved or oppositely oriented.

This is closer to the financial interpretation than multiplying every internal financial coordinate by −1.

D.7.3 Closure status operator

Define:

$$\hat{C} := \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$ (D.68)

The expectation-like closure orientation is:

$c_\Psi := \Psi^\dagger W \hat{C} \Psi / [\Psi^\dagger W \Psi]$. (D.69)

Possible interpretation:

$c_\Psi \approx +1$
→ action-dominant committed state. (D.70)

$c_\Psi \approx -1$
→ ledger-dominant return state. (D.71)

$c_\Psi \approx 0$
→ mixed action–ledger state. (D.72)

This is a proposed diagnostic.

It should not be interpreted as a probability without further normalization and empirical validation.

D.8 Transport-Normalized Closure Residual

D.8.1 Expected ledger image

Let:

$\psi\hat{}_L := U_{AL} \psi_A$. (D.73)

The ledger residual is:

$\delta_L := \psi_L - \psi\hat{}_L$. (D.74)

D.8.2 Expected action image

Let:

$\psi\hat{}_A := U_{LA} \psi_L$. (D.75)

The reciprocal action residual is:

$\delta_A := \psi_A - \psi\hat{}_A$. (D.76)

D.8.3 Residual spinor

Define:

$$\Delta\Psi_G := \begin{bmatrix} \delta_A \\ \delta_L \end{bmatrix}.$$ (D.77)

Equivalently:

$$\Delta\Psi_G = \begin{bmatrix} \psi_A - U_{LA}\psi_L \\ \psi_L - U_{AL}\psi_A \end{bmatrix}. \qquad (D.78)$$

This is the two-sided gauge-adjusted closure defect.

D.8.4 Weighted defect

Let W be a positive semidefinite metric.

Define:

$$\Delta_{close}^2 := \Delta\Psi_G^\dagger W \Delta\Psi_G. \qquad (D.79)$$

The closure condition is:

$$\Delta_{close} \le \varepsilon_P. \qquad (D.80)$$

The weights in W may reflect:

- monetary materiality;
- legal severity;
- timing significance;
- systemic importance;
- identity criticality.

A missing legal title may receive more weight than a small rounding difference.

D.8.5 Why raw action–ledger difference may mislead

Suppose market value is \$100 and collateral value is \$70 because the declared haircut is 30%.

Then:

$$\psi_L - \psi_A = -\$30. \qquad (D.81)$$

But if the expected transport is:

$$U_{AL}(\$100) = \$70, \qquad (D.82)$$

the lawful residual is:

$$\delta_L = \$70 - \$70 = \$0. \qquad (D.83)$$

Therefore:

$$\text{Frame Difference} \neq \text{Closure Failure}. \qquad (D.84)$$

Closure failure is the difference remaining after lawful transport.

D.9 Symmetric and Antisymmetric Closure Modes

D.9.1 Symmetric mode

Define:

$$e_+ := (1/\sqrt{2}) \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \quad \text{(D.85)}$$

This represents action and ledger moving together.

The corresponding amplitude is:

$$\psi_+ := (\psi_A + \psi_L)/\sqrt{2}. \quad \text{(D.86)}$$

D.9.2 Antisymmetric mode

Define:

$$e_- := (1/\sqrt{2}) \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \quad \text{(D.87)}$$

This represents action and ledger moving in opposition.

The corresponding amplitude is:

$$\psi_- := (\psi_A - \psi_L)/\sqrt{2}. \quad \text{(D.88)}$$

D.9.3 Mode transformation

Define the Hadamard-like matrix:

$$H_{cl} := (1/\sqrt{2}) \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}. \quad \text{(D.89)}$$

Then:

$$\Psi_{mode} = H_{cl}\Psi. \quad \text{(D.90)}$$

Therefore:

Ψ_mode =
[
ψ_+
ψ_-
]. (D.91)

D.9.4 Interpretation

The symmetric mode may represent:

- synchronized execution and settlement;
- action and ledger co-propagation;
- rapid accountable closure;
- low hidden exposure.

The antisymmetric mode may represent:

- action outrunning ledger;
- ledger resisting action;
- open settlement;
- unresolved margin call;
- frame fracture.

The source framework identifies the same practical polarity:

High action with low ledger → drift risk. (D.92)

Low action with high ledger → paralysis risk. (D.93)

Healthy operation requires coupled action and ledger flow.

D.10 Closure Coupling Matrix

D.10.1 General real coupling matrix

Define:

K_cl :=
[
k_AA k_AL
k_LA k_LL
]. (D.94)

A discrete closure update is:

$\Psi_{k+1} = K_{cl}\Psi_k + u_k + \mathscr{R}_k$. (D.95)

where:

- u_k = external gate or field input;
- $\mathscr{R}_k$ = unresolved residual.

D.10.2 Symmetric coupling

For reciprocal coupling:

k_AL = k_LA = k_C. (D.96)

Then:

K_cl =
[
k_A k_C
k_C k_L
]. (D.97)

D.10.3 Eigenvalues

The eigenvalues are:

λ_± = (k_A + k_L)/2 ± √[((k_A − k_L)/2)² + k_C²]. (D.98)

These determine the two principal closure modes.

D.10.4 Interpretation of λ_+

The λ_+ mode may correspond to:

- co-moving action and ledger;
- persistent synchronized activity;
- stable high-throughput closure.

If:

$|\lambda_+| < 1$, (D.99)

the mode decays in the absence of new input.

If:

$|\lambda_+| \approx 1$, (D.100)

the mode persists.

If:

$|\lambda_+| > 1$, (D.101)

the mode amplifies.

D.10.5 Interpretation of λ_-

The λ_- mode often corresponds to mismatch.

If:

$$|\lambda_-| < 1, \qquad (D.102)$$

action–ledger mismatch decays.

If:

$$|\lambda_-| \approx 1, \qquad (D.103)$$

mismatch persists.

If:

$$|\lambda_-| > 1, \qquad (D.104)$$

mismatch amplifies.

An unstable λ_- mode may indicate:

- growing settlement backlog;
- compounding collateral exceptions;
- hidden exposure;
- false closure;
- residual debt.

D.10.6 Stability condition

For the linear discrete model, local stability requires:

$$\rho(K_{cl}) < 1. \qquad (D.105)$$

where ρ denotes spectral radius.

This means:

$$\max\{|\lambda_+|, |\lambda_-|\} < 1. \qquad (D.106)$$

The condition is local and model-dependent.

It does not guarantee resilience under gate jumps or nonlinear liquidation feedback.

D.11 Action–Ledger Mass Matrix

D.11.1 Matrix definition

Define:

M :=
[
m_A m_C
m_C m_L
]. (D.107)

where:

- m_A = action-side inertia;
- m_L = ledger-side inertia;
- m_C = binding between the components.

D.11.2 Pauli decomposition

Let:

m_I := (m_A + m_L)/2. (D.108)

Let:

m_Δ := (m_A − m_L)/2. (D.109)

Then:

M = m_II₂ + m_Cσ_x + m_Δσ_z. (D.110)

This reproduces the mass operator used in the main article.

D.11.3 Eigenvalues

The eigenvalues are:

m_± = m_I ± √(m_C² + m_Δ²). (D.111)

The lower-mass mode is easier to transform.

The higher-mass mode resists change.

D.11.4 Financial interpretation

Possible sources of action-side mass include:

- market impact;
- concentration;
- mandate restrictions;
- execution limits.

Possible sources of ledger-side mass include:

- settlement delay;
- accounting procedure;
- legal validation;
- regulatory approval;
- audit requirements.

Possible sources of coupling mass include:

- automatic margining;
- straight-through processing;
- real-time risk updates;
- integrated collateral systems.

D.11.5 Mass is not always desirable

If ledger mass is too low:

Action
→ Immediate but weakly verified ledgering. (D.112)

This may create false closure.

If ledger mass is too high:

Action
→ Severe delay or paralysis. (D.113)

The goal is not maximum mass.

It is sufficient identity inertia with adaptive closure.

D.12 Continuous Closure Generator

D.12.1 First-order evolution

Define a continuous closure generator K(τ):

$d\Psi/d\tau = K(\tau)\Psi + u(\tau) + \mathscr{R}(\tau)$. (D.114)

A minimum real generator is:

$$K = \begin{bmatrix} -\alpha_A & \kappa_{LA} \\ \kappa_{AL} & -\alpha_L \end{bmatrix}.$$ (D.115)

where:

- α_A = action damping;
- α_L = ledger damping;
- κ_{AL} = action-to-ledger transfer;
- κ_{LA} = ledger-to-action feedback.

D.12.2 Action equation

$$d\psi_A/d\tau = -\alpha_A\psi_A + \kappa_{LA}\psi_L + u_A + \mathscr{R}_A. \qquad (D.116)$$

D.12.3 Ledger equation

$$d\psi_L/d\tau = \kappa_{AL}\psi_A - \alpha_L\psi_L + u_L + \mathscr{R}_L. \qquad (D.117)$$

D.12.4 Interpretation

The action component:

- decays or completes at rate α_A;
- receives feedback from the ledger through κ_{LA}.

The ledger component:

- receives action through κ_{AL};
- decays, settles, or integrates through α_L.

D.12.5 Mismatch dynamics

Define:

$$\delta := \psi_A - \psi_L. \qquad (D.118)$$

Then:

$$d\delta/d\tau$$
$$= -(\alpha_A + \kappa_{AL})\psi_A$$

- $(\alpha_L + \kappa_{LA})\psi_L$
- $u_A - u_L$
- $\mathscr{R}_A - \mathscr{R}_L$. (D.119)

In the symmetric case:

$$\alpha_A = \alpha_L = \alpha. \qquad (D.120)$$

$\kappa_{AL} = \kappa_{LA} = \kappa.$ (D.121)

Then:

$d\delta/d\tau = -(\alpha + \kappa)\delta + \Delta u + \Delta \mathscr{R}.$ (D.122)

Without continuing asymmetric input:

$\delta(\tau) = \delta(0)\exp[-(\alpha + \kappa)\tau].$ (D.123)

Thus stronger valid coupling κ reduces mismatch more quickly.

D.13 Delay and Ledger Latency

D.13.1 Delayed ledger return

Real ledgers do not update instantaneously.

A simple delay equation is:

$d\psi_L/d\tau = \kappa_{AL}\psi_A(\tau - \tau_d) - \alpha_L\psi_L(\tau) + \mathscr{R}_L.$ (D.124)

where τ_d is ledger latency.

D.13.2 Delay-induced instability

A large τ_d may generate:

- persistent mismatch;
- oscillation;
- overcorrection;
- duplicate action;
- false escalation.

A stylized characteristic equation is:

$s + \alpha_L - \kappa_{AL}\exp(-s\tau_d) = 0.$ (D.125)

The stability boundary depends on:

- κ_{AL};
- α_L;
- τ_d.

This is a control-theoretic hypothesis for empirical calibration.

D.13.3 Hidden exposure during latency

Let action exposure be:

$E_A(\tau)$. (D.126)

Let ledger-recognized exposure be:

$E_L(\tau)$. (D.127)

Define hidden exposure:

$E_{hidden}(\tau) := E_A(\tau) - E_L(\tau)$. (D.128)

During latency:

$E_{hidden} > 0$. (D.129)

The source describes the financial failure condition as trading velocity exceeding risk-ledger velocity, producing hidden exposure.

D.14 Closure-Velocity Matrix

D.14.1 Action and ledger velocities

Define:

$v_A := \|d\psi_A/d\tau\|$. (D.130)

Define:

$v_L := \|d\psi_L/d\tau\|$. (D.131)

D.14.2 Velocity vector

Define:

$$\vec{v} := \begin{bmatrix} v_A \\ v_L \end{bmatrix}.$$ (D.132)

D.14.3 Velocity imbalance

Define:

$\Delta v := v_A - v_L$. (D.133)

Interpretation:

$\Delta v > 0$
$\rightarrow$ action outruns ledger. (D.134)

$\Delta v < 0$
→ ledger activity exceeds outward action. (D.135)

$\Delta v \approx 0$
→ rates are aligned, subject to quality tests. (D.136)

D.14.4 Closure-capacity matrix

Let the component capacities be:

c_A = sustainable action-processing capacity. (D.137)

c_L = sustainable ledger-integration capacity. (D.138)

Define:

$C_P :=$
$\begin{bmatrix} c_A & 0 \\ 0 & c_L \end{bmatrix}$. (D.139)

The normalized velocity is:

$\vec{\kappa} :=$
$C_P^{-1}\vec{v}$. (D.140)

Thus:

$\kappa_A = v_A/c_A$. (D.141)

$\kappa_L = v_L/c_L$. (D.142)

D.14.5 Cone safety

A minimum cone-safety condition is:

$\max\{\kappa_A,\kappa_L\} \leq 1$. (D.143)

A cross-component condition is:

$v_A \leq c_A L v_L + c_0$. (D.144)

The source defines cone safety as action or semantic displacement relative to coherent integration capacity and treats values above one as action outrunning stable trace formation.

D.15 Branch-Specific Closure Matrices

D.15.1 General branch matrix

Let branch b belong to:

b ∈ {Collateral,Deleverage,Liquidation,Default}. (D.145)

Define:

G_b :=
[
G_AA^(b) G_AL^(b)
G_LA^(b) G_LL^(b)
]. (D.146)

Then:

$\Psi^{+} = G_b\Psi^{-} + \eta_b$. (D.147)

The branch matrices need not be unitary or invertible.

Financial gates are often irreversible.

D.15.2 Collateral-posting branch

A minimum collateral-posting matrix is:

G_coll :=
[
1 0
κ_coll 1 − α_coll
]. (D.148)

Then:

$\psi_A^{+} = \psi_A^{-}$. (D.149)

$\psi_L^{+} = \kappa_{coll}\psi_A^{-} + (1 - \alpha_{coll})\psi_L^{-}$. (D.150)

Interpretation:

- the outward call remains the reference action;
- the ledger absorbs admitted collateral;
- prior mismatch decays at rate α_coll.

D.15.3 Deleveraging branch

Define:

G_delev :=
[
r_n 0
κ_settle r_L
]. (D.151)

where:

- $0 \le r_n < 1$ reflects reduced position and action exposure;

- κ_settle transfers executed sale into ledger;
- 0 ≤ r_L ≤ 1 retains prior ledger state.

Then:

$\psi_A^{+} = r_n\psi_A^{-}$. (D.152)

$\psi_L^{+} = \kappa_settle\psi_A^{-} + r_L\psi_L^{-}$. (D.153)

D.15.4 Forced-liquidation branch

Define:

$$G_liq := \begin{bmatrix} r_liq & -\mu_back \\ \kappa_liq & r_ledger \end{bmatrix}.$$ (D.154)

The term μ_back permits ledger state to feed back negatively into action through:

- liquidation authority;
- collateral deficiency;
- risk restriction;
- execution impact.

The matrix is non-normal in general:

$G_liq G_liq^{\dagger} \neq G_liq^{\dagger} G_liq$. (D.155)

Non-normality can create transient amplification even when eigenvalues are locally stable.

This is a candidate mathematical representation of fire-sale escalation.

D.15.5 Default branch

Default changes identity class.

Define a conversion matrix:

$$G_def : \mathscr{H}_margin \to \mathscr{H}_recovery.$$ (D.156)

In a minimum two-component representation:

$$G_def := \begin{bmatrix} 0 & 0 \\ \kappa_rec & r_rec \end{bmatrix}.$$ (D.157)

The ordinary action component of the performing margin account terminates:

$$\psi_{A,\mathrm{margin}}^{+} = 0. \qquad (D.158)$$

A recovery-ledger component is created:

$$\psi_{L,\mathrm{recovery}}^{+} = \kappa_{\mathrm{rec}}\psi_{A,\mathrm{margin}}^{-} + r_{\mathrm{rec}}\psi_{L,\mathrm{margin}}^{-}. \qquad (D.159)$$

This is not return to the original state.

It is valid identity conversion.

D.16 Non-Normal Closure and Transient Amplification

D.16.1 Why eigenvalues are not sufficient

A matrix may have:

$$\rho(G) < 1, \qquad (D.160)$$

yet still produce temporary amplification:

$$\|G^k\| > 1 \quad \text{for some } k. \qquad (D.161)$$

This occurs when G is non-normal.

D.16.2 Financial interpretation

A margin system may be ultimately stabilizing while initially worsening:

- a liquidation cures debt eventually;
- the first sales depress market value;
- collateral value falls before settlement proceeds arrive;
- reported shortfall increases temporarily.

Thus:

$$\text{Long-Run Stability} \not\Rightarrow \text{No Short-Run Amplification.} \qquad (D.162)$$

D.16.3 Transient amplification factor

Define:

$$\mathcal{G}_{\max} := \sup_{k\geq 0}\|G^k\|. \qquad (D.163)$$

A large $\mathcal{G}_{\max}$ indicates substantial short-run amplification.

Possible empirical interpretation:

$\mathcal{G}_{\max}$ high
→ small margin breach may create a large temporary closure disturbance. (D.164)

This may help distinguish ordinary cure processes from fire-sale-prone processes.

D.17 Closure Hamiltonian Candidate

D.17.1 Cautious terminology

A matrix generating time evolution is sometimes called a Hamiltonian.

In finance, that word should be used cautiously because the system is:

- open;
- dissipative;
- non-unitary;
- gate-driven;
- residual-bearing.

The safer term is:

Closure Generator. (D.165)

D.17.2 Complex first-order generator

A proposed generator is:

$$i\, d\Psi/d\tau = H_{cl}\Psi + i\mathscr{R}. \quad (D.166)$$

where:

$$H_{cl} := h_0 I_2 + h_x \sigma_x + h_y \sigma_y + h_z \sigma_z. \quad (D.167)$$

The coefficients may represent:

- h_0 = common closure drift;
- h_x = direct action–ledger coupling;
- h_y = phase-sensitive exchange;
- h_z = action–ledger asymmetry.

D.17.3 Non-Hermitian extension

To represent irreversible loss and delay:

$$H_{eff} := H_{cl} - i\Lambda. \quad (D.168)$$

where:

Λ :=
[
λ_A λ_C
λ_C* λ_L
]. (D.169)

Then:

i dΨ/dτ = H_effΨ + i$\mathscr{R}$. (D.170)

The anti-Hermitian part produces:

- damping;
- leakage;
- fee loss;
- default loss;
- irreversible recognition.

This is an optional extension.

It should not be introduced unless the simpler real closure generator is inadequate.

D.18 Dirac-Compatible Matrix Form

D.18.1 Γ matrices

Use:

$\Gamma^0 = \sigma_z$. (D.171)

$\Gamma^1 = i\sigma_y$. (D.172)

Therefore:

$\Gamma^0\Gamma^1 = \sigma_x$. (D.173)

D.18.2 Financial Dirac equation

The minimum continuous equation is:

$[i\Gamma^0\nabla_\tau \hat{} P + ic_P\Gamma^1\mathfrak{D}_G - M]\Psi = \mathscr{R}$. (D.174)

D.18.3 Matrix expansion

Using the definitions above:

iσ_z∇_τ^PΨ

– c_Pσ_y𝔇_GΨ
– MΨ
= 𝓡. (D.175)

The σ_z term distinguishes action and ledger orientation.

The σ_y term transports and mixes the components.

The mass matrix binds the identity against arbitrary drift.

D.18.4 Component form

Let:

δ_A := ψ_A – U_LAψ_L. (D.176)

δ_L := ψ_L – U_ALψ_A. (D.177)

Then:

i∇_τ^Pψ_A + ic_Pδ_L/ℓ_P – m_Aψ_A – m_Cψ_L = 𝓡_A. (D.178)

–i∇_τ^Pψ_L – ic_Pδ_A/ℓ_P – m_Cψ_A – m_Lψ_L = 𝓡_L. (D.179)

These equations are the matrix-level operational form of the Financial Gauge–Dirac kernel.

D.19 Gauge Transformation of Closure Matrices

D.19.1 Local frame transformation

Let the action and ledger frames transform separately:

ψ_A′ = G_Aψ_A. (D.180)

ψ_L′ = G_Lψ_L. (D.181)

Define the block transformation:

G_cl :=
[
G_A 0
0 G_L
]. (D.182)

Then:

Ψ′ = G_clΨ. (D.183)

D.19.2 Transporter transformation

The action-to-ledger transporter transforms as:

U_AL′ = G_LU_ALG_A⁻¹. (D.184)

The ledger-to-action transporter transforms as:

$U_{LA'} = G_A U_{LA} G_L^{-1}$. (D.185)

D.19.3 Graph derivative covariance

The graph defect transforms as:

$\Delta\Psi_{G'} = G_{cl}\Delta\Psi_G$. (D.186)

A compatible metric W satisfies:

$W' = (G_{cl}^{-1})^\dagger W G_{cl}^{-1}$. (D.187)

Then:

$\Delta_{close}'^2 = \Delta_{close}^2$. (D.188)

Thus the closure defect is independent of lawful local representation.

D.19.4 Gate covariance

A branch operator transforms as:

$G_b' = G_{cl,out} G_b G_{cl,in}^{-1}$. (D.189)

Then:

$\Psi'^+ = G_b' \Psi'^-$ (D.190)

is equivalent to:

$\Psi'^+ = G_{cl,out}(G_b \Psi^-)$. (D.191)

A margin call should therefore not change substantive status merely because:

- currency changes;
- unit scale changes;
- presentation changes;
- a position moves between reporting systems.

D.20 Charge-Compatible Closure Matrices

D.20.1 Charge operator

Let:

$\hat{Q} =$
$[$
$q_A \quad 0$

0 q_L
]. (D.192)

For ordinary action–ledger representations of the same identity:

q_A = q_L = q. (D.193)

Therefore:

Q^ = qI2. (D.194)

D.20.2 Charge-preserving closure

A closure matrix G preserves charge when:

Q^_outG = GQ^_in. (D.195)

For equal charge on both components:

[Q^,G] = 0. (D.196)

D.20.3 Charge-converting gate

A conversion branch satisfies:

Q^_outG_b − G_bQ^_in = J_q,b. (D.197)

where J_q,b records:

- activation;
- transfer;
- conversion;
- extinguishment.

The charge residual is:

r_q,b = q_in + q_gate − q_out. (D.198)

D.20.4 Margin-call example

Before call:

Q^_in =
[
q_A 0
0 q_C^dormant
]. (D.199)

After call:

$$\hat{Q}_{out} = \begin{bmatrix} q_A & 0 \\ 0 & q_C^{operational} \end{bmatrix}. \quad (D.200)$$

The gate activation current is:

$$J_{q,call} = \hat{Q}_{out} G_{call} - G_{call} \hat{Q}_{in}. \quad (D.201)$$

This formalizes the distinction between:

- pre-existing contractual obligation;
- operationally active obligation.

D.21 Closure Quality Matrix

D.21.1 Multi-dimensional closure

Closure quality should not be reduced immediately to one number.

Define the quality vector:

$$\vec{c} := \begin{bmatrix} c_{amount} \\ c_{time} \\ c_{settlement} \\ c_{title} \\ c_{risk} \\ c_{accounting} \\ c_{legal} \\ c_{charge} \end{bmatrix}. \quad (D.202)$$

Each coordinate satisfies:

$$0 \le c_j \le 1. \quad (D.203)$$

D.21.2 Quality matrix

Define:

$$Q_{cl} := \mathrm{diag}(c_{amount}, c_{time}, c_{settlement}, c_{title}, c_{risk}, c_{accounting}, c_{legal}, c_{charge}). \quad (D.204)$$

A weighted scalar summary is:

$$\mathcal{Q}_{close} := \mathrm{Tr}(W_Q Q_{cl}) / \mathrm{Tr}(W_Q). \quad (D.205)$$

The vector or diagonal matrix should remain accessible even when a scalar summary is reported.

D.21.3 False closure

Define reported closure:

$C_{reported} \in \{0,1\}$. (D.206)

Define protocol closure:

$C_{verified} := 1[\Delta_{close} \le \varepsilon_P \wedge RequiredConditions]$. (D.207)

False closure is:

$F_{close} := C_{reported} - C_{verified}$. (D.208)

A dangerous case is:

$C_{reported} = 1$. (D.209)

$C_{verified} = 0$. (D.210)

The source identifies false return as declared closure while A-B Fixedness remains absent.

D.22 Residual Matrices

D.22.1 Residual covariance matrix

Let residual observations be:

$\mathscr{R}_1, \mathscr{R}_2, \ldots, \mathscr{R}_N$. (D.211)

Define:

$\Sigma_{\mathscr{R}} := E[(\mathscr{R} - \mu_{\mathscr{R}})(\mathscr{R} - \mu_{\mathscr{R}})^\dagger]$. (D.212)

This matrix records:

- residual scale;
- correlation across action and ledger;
- repeated failure modes;
- systemic residual structure.

D.22.2 Principal residual modes

Let:

$\Sigma_{\mathscr{R}} v_j = \lambda_j v_j$. (D.213)

Large-eigenvalue modes may correspond to:

- settlement–cash mismatch;

- collateral–legal mismatch;
- accounting–risk mismatch;
- repeated timing failure;
- authority disagreement.

This is an empirical diagnostic, not a claim of physical eigenstates.

D.22.3 Hidden residual

Let recorded residual be:

$\mathscr{R}$_recorded. (D.214)

Let audited residual be:

$\mathscr{R}$_audited. (D.215)

Define hidden residual:

$\mathscr{R}$_hidden := $\mathscr{R}$_audited − $\mathscr{R}$_recorded. (D.216)

The dangerous condition is:

‖$\mathscr{R}$_hidden‖ ≫ 0. (D.217)

The source repeatedly distinguishes residual from residual denial: residual itself is not automatically failure; hidden residual is the dangerous failure.

D.23 Closure Observables

D.23.1 Action intensity

Define:

J_A := ψ_A†W_Aψ_A. (D.218)

D.23.2 Ledger intensity

Define:

J_L := ψ_L†W_Lψ_L. (D.219)

D.23.3 Coupled current

Define:

J_AL := Ψ†Wσ_xΨ. (D.220)

For scalar components and W = I:

J_AL = ψ_A*ψ_L* + *ψ_L*ψ_A. (D.221)

Therefore:

J_AL = 2Re(ψ_A*ψ_L). (D.222)

A high positive value may indicate co-propagation.

A negative value may indicate opposition or phase conflict.

D.23.4 Closure polarization

Define:

P_cl := Ψ†Wσ_zΨ/[Ψ†WΨ]. (D.223)

Interpretation:

P_cl → +1
→ action-dominant state. (D.224)

P_cl → −1
→ ledger-dominant state. (D.225)

P_cl → 0
→ balanced component magnitudes. (D.226)

Balanced magnitude does not guarantee lawful transport or closure.

D.23.5 Closure coherence

Define:

C_AL := |ψ_A†W_ALψ_L|/[√(ψ_A†W_Aψ_A)√(ψ_L†W_Lψ_L) + ε]. (D.227)

Then:

0 ≤ C_AL ≤ 1. (D.228)

High C_AL means action and ledger are strongly aligned in the declared feature space.

Low C_AL means they are weakly aligned or orthogonal.

This metric must be interpreted together with magnitude and residual.

Two empty components can appear perfectly aligned but be economically irrelevant.

D.24 Margin-Account Closure Matrix

D.24.1 State vector

For the minimum margin example, define:

ψ_A :=
[
C_call
ℓ_instruction
ΔD_expected
Loss_economic
]. (D.229)

Define:

ψ_L :=
[
C_admitted
ℓ_settled
ΔD_recorded
Loss_recognized
]. (D.230)

The full spinor is:

Ψ_margin :=
[
ψ_A
ψ_L
]. (D.231)

Here each component is a four-dimensional vector, so:

Ψ_margin ∈ $\mathbb{C}^2$_cl ⊗ $\mathbb{R}^4$. (D.232)

D.24.2 Expected transport matrix

Define:

U_AL :=
[
u_CC o o o
o u_ℓℓ o o
o o u_DD o
o o o u_LL
]. (D.233)

The entries represent expected conversion from:

- call amount to admitted collateral;
- liquidation instruction to settled quantity;
- expected debt reduction to recorded debt reduction;
- economic loss to recognized loss.

D.24.3 Margin closure residual

The margin residual is:

$$\delta_{\text{margin}} := \psi_L - U_{AL}\psi_A. \qquad (D.234)$$

Explicitly:

$$\delta_{\text{margin}} = \begin{bmatrix} C_{\text{admitted}} - u_{CCC_call} \\ \ell_{\text{settled}} - u_{\ell\ell\ell_instruction} \\ \Delta D_{\text{recorded}} - u_{DD\Delta D_expected} \\ \text{Loss}_{\text{recognized}} - u_{LL\text{Loss}_economic} \end{bmatrix}. \qquad (D.235)$$

D.24.4 Weighted margin defect

Define:

$$\Delta_{\text{margin}}^2 := \delta_{\text{margin}}^{\top} W_{\text{margin}} \delta_{\text{margin}}. \qquad (D.236)$$

Possible weights include:

- high weight on unpaid collateral;
- high weight on ownership or settlement failure;
- moderate weight on timing delay;
- lower weight on immaterial rounding differences.

D.25 Closure Through Nested Subsystems

D.25.1 Outer and inner spinors

The outer margin spinor is:

$$\Psi_{\text{margin}}. \qquad (D.237)$$

The inner trade-settlement spinor is:

$$\Psi_{\text{trade}}. \qquad (D.238)$$

The inner collateral-transfer spinor is:

$$\Psi_{\text{collateral}}. \qquad (D.239)$$

The accounting-recognition spinor is:

$$\Psi_{\text{accounting}}. \qquad (D.240)$$

D.25.2 Composite closure space

The total state may be:

Ψ_total
:= Ψ_margin ⊗ Ψ_trade ⊗ Ψ_collateral ⊗ Ψ_accounting. (D.241)

This full tensor representation may become too large for practical use.

A reduced hierarchical representation is often preferable.

D.25.3 Hierarchical closure condition

Outer closure requires all mandatory inner closures:

Closed_margin
⇒ Closed_trade
∧ Closed_collateral
∧ Closed_accounting
∧ Closed_legal. (D.242)

The converse need not hold.

A settled trade may exist inside an unresolved margin cycle.

D.25.4 Closure aggregation matrix

Let the subsystem closure indicators be:

c_sub :=
[
c_trade
c_collateral
c_accounting
c_legal
]. (D.243)

Define the outer closure score:

C_outer = $w_{sub}^{T} c_{sub}$. (D.244)

For mandatory conjunctive closure, use:

C_outer = min{c_trade,c_collateral,c_accounting,c_legal}. (D.245)

This prevents a high average from concealing one critical open subsystem.

D.26 Repair Operators

D.26.1 Slow-action operator

When action outruns ledger:

v_A > c_P. (D.246)

Define:

R_slow :=
[
r_A 0
0 1
]. (D.247)

where:

0 < r_A < 1. (D.248)

Then:

ψ_A′ = r_Aψ_A. (D.249)

This may represent:

- reducing trading rate;
- splitting liquidation;
- staging implementation;
- delaying promotion;
- limiting transaction size.

The source recommends slowing the outward cycle when action exceeds ledger capacity.

D.26.2 Ledger-capacity operator

Define:

R_capacity :=
[
1 0
κ_upgrade r_L
]. (D.250)

This may represent:

- automation;
- improved reconciliation;
- added collateral-processing capacity;
- faster legal verification;
- stronger audit.

D.26.3 Residual-integration operator

Define:

R_res :=

[
1 0
κ_res 1 − α_res
]. (D.251)

It transfers unresolved consequence into the ledger while reducing hidden residual.

D.26.4 Reopen operator

If new evidence invalidates closure:

Closed_v1 → Reopened_v2. (D.252)

Define:

R_reopen :=
[
1 0
0 r_open
]. (D.253)

with:

r_open restoring an open ledger status while preserving prior trace.

The source requires reopening to preserve the original decision, original evidence, new evidence, residual, and revision rather than rewriting history.

D.27 Empirical Estimation

D.27.1 Discrete observed system

A practical observed system is:

$\Psi_{k+1} = K_k\Psi_k + B_ku_k + \mathscr{R}_k$. (D.254)

The matrices may depend on regime:

$K_k = K(Regime_k, Gate_k, Ledger_k)$. (D.255)

D.27.2 State-space observation equation

Observed data y_k may be:

$y_k = H_k\Psi_k + \varepsilon_k$. (D.256)

where:

- H_k = measurement matrix;
- ε_k = observation noise.

This permits action and ledger components to be partially latent.

D.27.3 Estimation approaches

Possible methods include:

- Kalman filtering;
- switching state-space models;
- hidden Markov models;
- Bayesian filtering;
- constrained least squares;
- system identification;
- survival models for closure time;
- graph neural models with fixed transport constraints.

The method should not determine the ontology retrospectively.

The component definitions and gate rules should be declared before fitting.

D.27.4 Identifiability requirement

The parameters are identifiable only if the data contain independent variation in:

- action intensity;
- ledger latency;
- gate branches;
- transport mappings;
- closure outcomes;
- residual.

If action and ledger always update simultaneously, the two-component model cannot be identified.

D.28 Matrix Falsifiers

D.28.1 Projection failure

The matrix model fails if:

P_A and P_L do not correspond to independently observable states. (D.257)

D.28.2 Transport failure

The model fails if:

U_AL cannot be specified prospectively. (D.258)

D.28.3 Coupling failure

The off-diagonal terms fail if:

k_AL = k_LA = 0 (D.259)

without loss of explanatory or predictive value.

Then action and ledger are separate bookkeeping channels rather than a coupled spinor.

D.28.4 Eigenmode failure

The mode interpretation fails if:

- eigenvectors are unstable;
- modes do not correspond to distinct closure behaviour;
- no relation exists between λ_- and unresolved closure;
- ordinary principal components perform equally well.

D.28.5 Γ failure

The Dirac-compatible matrices fail if:

- the anticommutation structure adds no restriction;
- arbitrary matrices perform equally well;
- the squared operator has no financial meaning;
- basis selection determines the result.

Then use:

Coupled Closure Matrix Model. (D.260)

D.28.6 Scalar sufficiency

If:

Performance_scalar ≥ Performance_matrix – ComplexityPenalty, (D.261)

reduce:

$\Psi \to x.$ (D.262)

The source explicitly requires scalar reduction when ledger return is not independent, action creates no meaningful residual, cross-frame identity is irrelevant, or spinor split cannot be measured.

D.29 Reduction Ladder

The matrix architecture reduces as follows:

Financial Dirac Spinor
→ Γ-Coupled Action–Ledger Matrix. (D.263)

Γ-Coupled Matrix
→ Ordinary Two-Component State Space. (D.264)

Two-Component State Space
→ Two-Stage Workflow. (D.265)

Two-Stage Workflow
→ Scalar Closure Status. (D.266)

The correct representation is the lowest level that preserves:

- independent consequences;
- measurable residual;
- cross-frame relevance;
- incremental utility.

D.30 Matrix Audit Template

For each proposed closure matrix, record:

Audit field	Required statement
State basis	What do e_A and e_L represent?
Component units	Are ψ_A and ψ_L comparable?
Feature maps	How are raw states embedded?
Projectors	How are action and ledger separated?
Transporter	What is U_AL?
Reciprocal map	What is U_LA?
Gate branch	Which G_b applies?
Charge rule	Does the matrix preserve or convert charge?
Mass	What resists identity change?
Capacity	What limits action-to-ledger propagation?
Residual	What remains after expected transport?
Closure tolerance	What ε_P is used?
Eigenmodes	What do λ_+ and λ_- mean operationally?
Falsifier	What result would reduce the representation?

D.31 Compact Matrix Summary

D.31.1 State

$\Psi = [\psi_A,\psi_L]^T.$ (D.267)

D.31.2 Projectors

$P_A = (I_2 + \sigma_z)/2.$ (D.268)

$P_L = (I_2 - \sigma_z)/2.$ (D.269)

D.31.3 Exchange

$\sigma_x\Psi = [\psi_L,\psi_A]^T.$ (D.270)

D.31.4 Gauge-adjusted mismatch

$\Delta\Psi_G = [\psi_A - U_{LA}\psi_L,\psi_L - U_{AL}\psi_A]^T.$ (D.271)

D.31.5 Closure defect

$\Delta_{close}^2 = \Delta\Psi_G^\dagger W \Delta\Psi_G.$ (D.272)

D.31.6 Discrete propagation

$\Psi_{k+1} = K_k\Psi_k + B_k u_k + \mathscr{R}_k.$ (D.273)

D.31.7 Gate jump

$\Psi^+_k = G_b\Psi^-_k + \eta_b.$ (D.274)

D.31.8 Continuous propagation

$d\Psi/d\tau = K(\tau)\Psi + u(\tau) + \mathscr{R}(\tau).$ (D.275)

D.31.9 Dirac-compatible kernel

$[i\Gamma^0\nabla_\tau\hat{P} + ic_P\Gamma^1\mathfrak{D}_G - M]\Psi = \mathscr{R}.$ (D.276)

D.31.10 Closure criterion

SpinClosed_P = 1 (D.277)

only if:

Δ_close ≤ ε_P
∧ ChargeReconciled
∧ GateResolved
∧ IdentityPreservedOrConverted
∧ ResidualDisclosed. (D.278)

D.32 Final Interpretation

The financial spinor is not simply a sophisticated notation for two columns of data.

Its two components represent two logically distinct facts:

ψ_A records that the financial world was changed. (D.279)

ψ_L records whether that changed world became an accountable inherited state. (D.280)

The off-diagonal matrices represent the two directions of institutional causality:

Action writes into the ledger. (D.281)

The ledger constrains the next action. (D.282)

The symmetric mode represents co-propagation.

The antisymmetric mode represents unresolved transformation.

The branch matrices represent collateral cure, deleveraging, liquidation, and default.

The mass matrix represents resistance to identity-preserving change.

The gauge-adjusted defect distinguishes lawful frame difference from genuine nonclosure.

The residual matrix preserves what the declared closure failed to integrate.

The resulting matrix grammar is:

Action
→ Consequence
→ Transport
→ Ledger
→ Residual
→ Revised Action. (D.283)

Its strongest compact form is:

Ψ_{k+1}
= U_close[
G_gateΨ_k,
L_k,
$\vec{q}_k$,

$\mathscr{R}_k$
]. (D.284)

The next appendix will develop the **financial frame graph, connection catalogue, and loop-closure matrices**, specifying in greater detail how market, collateral, treasury, risk, accounting, legal, and regulatory representations are transported and reconciled.

Appendix E — Financial Frame Graphs, Connection Catalogues, and Loop-Closure Matrices

E.1 Purpose

A financial identity rarely exists in only one operational frame.

The same position may be represented simultaneously as:

- a market-valued asset;
- eligible collateral;
- a funded balance;
- a risk exposure;
- an unsettled transaction;
- an accounting asset or liability;
- a legal claim;
- a regulatory exposure.

These representations need not be numerically identical.

They must, however, remain connected to the same bounded financial identity through declared transport rules.

The source Financial Standard Model therefore requires governed transport across trading, treasury, risk, accounting, legal, tax, collateral, and regulatory frames. It distinguishes this from double-entry bookkeeping: double entry preserves balance and trace inside accounting, while financial gauge transport preserves an economic relation across different representations.

This appendix constructs:

- the financial frame graph;
- frame-local state vectors;
- identity invariants;
- edge connections;
- nonlinear and linearized transport;
- block transport matrices;
- path composition;

- loop holonomy;
- loop residuals;
- timing and authority adjustments;
- cross-ledger reconciliation gates;
- empirical tests and reduction rules.

The source-derived requirement is that A-B Fixedness needs a frame map, compatible observation, accessible trace, invariant preservation, and residual honesty.

The particular matrices and connection catalogues below are proposed constructions of the present article.

E.2 The Financial Frame Registry

E.2.1 Frame set

Define the principal financial frames:

$$\mathbb{F} := \{F_M, F_C, F_T, F_R, F_S, F_A, F_L, F_G\}. \qquad (E.1)$$

where:

- F_M = market and trading frame;
- F_C = collateral frame;
- F_T = treasury and funding frame;
- F_R = risk frame;
- F_S = settlement and clearing frame;
- F_A = accounting frame;
- F_L = legal and contractual frame;
- F_G = regulatory frame.

Additional frames may include:

- tax;
- investor reporting;
- management reporting;
- rating agency;
- resolution authority;
- central counterparty;
- portfolio-construction frame.

A frame should be added only when it applies a distinct:

- observation rule;
- admissibility rule;
- valuation rule;
- authority;
- trace;
- consequence.

E.2.2 Frame definition

Define frame f as:

$F_f := (B_f, \Delta_f, h_f, u_f, Authority_f, Trace_f).$ (E.2)

where:

- B_f = frame boundary;
- Δ_f = observation and measurement rule;
- h_f = relevant horizon;
- u_f = permitted intervention set;
- $Authority_f$ = recognized decision-maker;
- $Trace_f$ = frame-local record.

A frame is not merely a database.

It is a protocol determining what counts as:

- an observable object;
- an admitted value;
- a recognized event;
- an actionable state;
- an authoritative trace.

E.2.3 Frame-local projection

Let S be the underlying bounded financial subject.

Its representation in frame f is:

$S_f := \Pi_f(S).$ (E.3)

The projection Π_f may:

- select variables;
- revalue quantities;
- aggregate positions;
- apply haircuts;
- apply netting;
- change recognition date;
- change legal classification;
- change reporting units.

Thus:

$S_f \neq S_g$ in general. (E.4)

But lawful representation requires:

$K_f(S_f) \simeq K_g(S_g)$. (E.5)

unless a declared identity-conversion gate lies between them.

E.3 The Frame Graph

E.3.1 Directed frame graph

Define:

$\mathcal{G}_F := (\mathbb{F}, \mathscr{E}_F)$. (E.6)

where $\mathscr{E}_F$ contains permitted directed transport edges.

A minimum margin-system graph may contain:

$F_M \to F_C$. (E.7)

$F_M \to F_R$. (E.8)

$F_L \to F_C$. (E.9)

$F_C \to F_T$. (E.10)

$F_M \to F_S$. (E.11)

$F_S \to F_A$. (E.12)

$F_A \to F_G$. (E.13)

$F_R \to F_G$. (E.14)

$F_T \to F_A$. (E.15)

The source framework explicitly recommends cross-ledger experiments using trading, settlement, risk, accounting, legal, and regulatory frames and requires shared invariants and accessible records.

E.3.2 Edge meaning

An edge $A \to B$ means:

> Frame B possesses a declared procedure for constructing or evaluating its representation from information admitted in frame A.

An edge does not imply:

- numerical equality;
- immediate recognition;
- invertibility;
- symmetry;
- absence of residual.

Therefore:

$$A \to B \not\Rightarrow B \to A. \qquad \text{(E.16)}$$

For example:

- legal eligibility may determine collateral admission;
- collateral admission does not fully determine legal ownership;
- market price may feed accounting valuation;
- accounting carrying value may not uniquely recover market state.

E.3.3 Edge catalogue

Define the edge registry:

$$\mathscr{E}_F := \{e_AB \mid \text{Source_A}, \text{Target_B}, \text{TransportRule}, \text{Authority}, \text{Timing}, \text{ResidualRule}\}. \qquad \text{(E.17)}$$

Every edge should record:

Field	Meaning
Source	Origin frame
Target	Destination frame
Carrier	Financial identity being transported
Inputs	Source variables used
Connection	Translation rule
Authority	Who validates the map
Timing	Effective and recognition dates

Field	Meaning
Expected output	Target state implied by the source
Residual	Observed minus expected target state
Falsifier	Condition under which the edge is rejected

E.4 Frame-Local State Vectors

E.4.1 Common embedded state

To compare frames, define a common embedded feature space:

$\mathcal{H}_F := \text{span}\{\text{Identity},\text{Quantity},\text{Value},\text{Charge},\text{Status},\text{Time},\text{Trace},\text{Residual}\}.$ (E.18)

A frame-local embedded state is:

$x_f := [k_f,n_f,v_f,q_f,\nu_f,t_f,\tau_f,r_f]^T.$ (E.19)

where:

- k_f = encoded identity kernel;
- n_f = signed quantity;
- v_f = frame-local value coordinates;
- q_f = charge coordinates;
- ν_f = lifecycle or gate status;
- t_f = economic observation time;
- τ_f = ledger-recognition time;
- r_f = frame-local residual.

Not every frame observes every coordinate.

Missing coordinates should be marked:

Missing or NotApplicable. (E.20)

They should not be silently set to zero.

E.4.2 Market-frame state

A minimum market vector is:

$x_M := [K,n,R_M,Q_M,\theta_M,p_{bid},p_{ask},t_M,\nu_M]^T.$ (E.21)

It answers:

- what is the instrument?
- how many units are held?

- what value is admitted?
- what conjugate pressure is implied?
- what phase is occupied?
- at what market time is the state observed?

E.4.3 Collateral-frame state

Define:

$x_C := [K,n,R_C,Q_C,h,Eligibility,Encumbrance,B,t_C,\nu_C]^T.$ (E.22)

It answers:

- is the asset eligible?
- how much value is admitted?
- what haircut or admission factor applies?
- is the asset already pledged?
- what margin buffer results?

E.4.4 Treasury-frame state

Define:

$x_T := [K,D,C_{cash},FundingRate,Maturity,Liquidity,Encumbrance,t_T,\nu_T]^T.$ (E.23)

It answers:

- how much funding is outstanding?
- what cash is available?
- what liquidity remains free?
- which assets are encumbered?
- what refinancing obligations exist?

E.4.5 Risk-frame state

Define:

$x_R := [K,Exposure,Delta,Gamma,Vega,LossDistribution,StressLoss,LimitUsage,t_R,\nu_R]^T.$ (E.24)

The exact coordinates depend on the instrument.

Risk coordinates are normally frame-local.

They are not automatically structural charge coordinates.

E.4.6 Settlement-frame state

Define:

$$x_S := [K,\text{TradeID},\text{ExecutedQuantity},\text{SettledQuantity},\text{CashDue},\text{CashSettled},\text{FailStatus},t_S,\nu_S]^{\mathsf{T}}. \tag{E.25}$$

It distinguishes:

$$\text{Execution} \neq \text{Settlement}. \tag{E.26}$$

The source framework gives the chain:

$$\text{Order} \to \text{Execution} \to \text{Confirmation} \to \text{Clearing} \to \text{Settlement} \to \text{Recognition}. \tag{E.27}$$

and warns that success at an early gate does not imply success at the final gate.

E.4.7 Accounting-frame state

Define:

$$x_A := [K,\text{CarryingAmount},\text{Classification},\text{RecognizedP\&L},\text{Impairment},\text{JournalStatus},\text{ReportingPeriod},t_ \tag{E.28}$$

Accounting recognition is controlled by:

- standards;
- control transfer;
- measurement reliability;
- reporting period;
- evidence;
- classification rules.

Economic occurrence may precede accounting recognition.

E.4.8 Legal-frame state

Define:

$$x_L := [K,\text{Owner},\text{Obligor},\text{Title},\text{SecurityInterest},\text{ContractStatus},\text{DefaultStatus},\text{Jurisdiction},t_L,\nu_L]^{\mathsf{T}}. \tag{E.29}$$

Legal state determines:

- enforceable ownership;
- contractual obligations;
- collateral rights;
- default rights;
- priority;
- jurisdiction.

A market position may possess value while failing legal eligibility for collateral.

E.4.9 Regulatory-frame state

Define:

x_G :=
[K,ExposureClass,RiskWeight,CapitalCharge,LiquidityTreatment,Concentration,ReportingStatus,t
(E.30)

The regulatory frame may aggregate or transform positions according to:

- entity perimeter;
- netting recognition;
- capital rules;
- liquidity rules;
- large-exposure rules;
- reporting standards.

E.5 The Invariant Kernel

E.5.1 Core identity invariant

Define:

I_core :=
(InstrumentID,ContractID,SignedUnits,Owner,Obligor,Currency,Maturity,ExecutionLineage).
(E.31)

This is the minimum candidate invariant across frames.

E.5.2 Extended invariant

For collateral-sensitive systems:

I_ext := (I_core,CollateralAgreement,SecurityInterest,EligibilityClass,SettlementStatus).

(E.32)

For derivatives:

I_deriv := (I_core,NettingSet,ExerciseTerms,SettlementMethod,UnderlyingID). (E.33)

For default and recovery:

I_rec := (OriginatingClaim,Seniority,CollateralRights,DefaultEvent,RecoveryLineage). (E.34)

E.5.3 Invariant-preservation test

For edge A → B:

Inv_B[T_AB(x_A)] = Inv_A(x_A). (E.35)

If a valid conversion occurs:

Inv_B[T_AB(x_A)] = Convert_AB[Inv_A(x_A),Gate_AB]. (E.36)

Without a conversion gate:

Inv_B ≠ Inv_A indicates transport failure. (E.37)

E.5.4 A-B Fixedness

Define:

ABFix_P(S;A,B) := M_AB × I_AB × R_AB × H_AB. (E.38)

where:

- M_AB = identity-match score;
- I_AB = invariant-preservation score;
- R_AB = record-accessibility score;
- H_AB = residual-honesty score.

Each component may lie in:

0 ≤ M_AB,I_AB,R_AB,H_AB ≤ 1. (E.39)

The multiplicative form ensures that a zero in one mandatory dimension collapses the total score.

A simpler source-derived operational form is:

ABFix(e) = IdentityMatch × InvariantPreservation × RecordAccessibility. (E.40)

The addition of residual honesty follows the broader A-B Fixedness requirements.

E.6 Connection Catalogue

E.6.1 General connection

For edge A → B, define:

$\mathcal{A}$_AB :=
($\mathcal{A}$_AB^value,$\mathcal{A}$_AB^units,$\mathcal{A}$_AB^time,$\mathcal{A}$_AB^legal,$\mathcal{A}$_AB^recognition,$\mathcal{A}$_AB^charge).
(E.41)

The transport is:

x^_B = T_AB(x_A;$\mathcal{A}$_AB). (E.42)

The observed target state is:

x_B^obs. (E.43)

The edge residual is:

r_AB := x_B^obs – x^_B. (E.44)

When direct subtraction is not meaningful, use a declared difference operator:

r_AB := Diff_B[x_B^obs,T_AB(x_A)]. (E.45)

E.6.2 Market-to-collateral connection

The minimum connection contains:

$\mathcal{A}$_MC := (h,Eligibility,Concentration,Liquidity,Currency,Timing). (E.46)

The real-value transport is:

R_C = hR_M. (E.47)

Under the shared-amplitude complex completion:

Q_C = √(A² – h²R_M²). (E.48)

Therefore:

T_MC(R_M,Q_M;A,h) = [hR_M,√(A² – h²R_M²)]ᵀ. (E.49)

This map is nonlinear.

It should not be replaced by:

Z_C = hZ_M (E.50)

unless an alternative protocol explicitly defines such scaling.

E.6.3 Legal-to-collateral connection

Define eligibility indicator:

e_LC ∈ {0,1}. (E.51)

A richer eligibility score may satisfy:

0 ≤ e_LC ≤ 1. (E.52)

Then:

R_C = e_LC hR_M. (E.53)

Legal ineligibility gives:

e_LC = 0. (E.54)

even when:

R_M > 0. (E.55)

This edge prevents market value from being mistaken for legally usable collateral.

E.6.4 Market-to-risk connection

Define:

x^_R = T_MR(x_M;ScenarioSet,Horizon,Model). (E.56)

A local linear approximation may be:

Δx_R ≈ J_MRΔx_M. (E.57)

where:

J_MR := ∂T_MR/∂x_M. (E.58)

Possible outputs include:

- delta exposure;
- stress loss;
- expected shortfall;
- limit usage;
- factor exposure.

A risk model is one connection, not the universal identity of the position.

E.6.5 Collateral-to-treasury connection

Posted collateral changes treasury availability.

Let:

C_posted = collateral admitted. (E.59)

Let:

F_free = free liquidity. (E.60)

A minimum relation is:

F_free,T = F_total − C_posted − OtherEncumbrance. (E.61)

The edge map is:

x̂_T = T_CT(x_C;FundingRules,ReuseRights,SettlementStatus). (E.62)

The legal right of reuse may materially alter the connection.

E.6.6 Settlement-to-accounting connection

Define:

x̂_A = T_SA(x_S;RecognitionRule,ReportingDate,AccountingStandard). (E.63)

Possible relations include:

SettledQuantity → RecognizedPosition. (E.64)

SettledCash → RecordedCash. (E.65)

RealizedExecutionLoss → RecognizedP&L. (E.66)

But accounting may sometimes recognize on trade date rather than settlement date.

Therefore the connection must declare the recognition convention.

E.6.7 Treasury-to-accounting connection

Funding balances transport as:

D_A = T_TA(D_T;Accrual,Fees,FX,Classification). (E.67)

A minimum relation is:

D_A = D_T + AccruedInterest + RecognizedFees + FXAdjustment. (E.68)

The residual is:

r_TA^D = D_A^obs − D_A^expected. (E.69)

E.6.8 Accounting-to-regulatory connection

Define:

x̂_G = T_AG(x_A;CapitalRules,ExposureClass,NettingRecognition,EntityPerimeter). (E.70)

Accounting carrying amount may be adjusted for:

- regulatory deductions;
- risk weights;
- eligible capital treatment;
- prudential valuation;
- exposure conversion factors.

Therefore:

RegulatoryExposure ≠ CarryingAmount in general. (E.71)

E.6.9 Risk-to-regulatory connection

Define:

x̂_G^risk = T_RG(x_R;RegulatoryModel,StressRules,CapitalFormula). (E.72)

Two routes may therefore produce regulatory representations:

F_A → F_G. (E.73)

F_R → F_G. (E.74)

Their agreement becomes a path-consistency test.

E.6.10 Legal-to-accounting connection

Legal state may determine:

- ownership;
- control;
- enforceability;
- recognition boundary;
- derecognition;
- consolidation.

Define:

x̂_A = T_LA(x_L;AccountingStandard,ControlTest,ContractSubstance). (E.75)

An accounting write-off need not extinguish legal charge.

Thus the map may produce:

CarryingAmount = 0. (E.76)

while:

LegalClaim > 0. (E.77)

This is lawful only when the connection and frame distinction are explicit.

E.7 Linearized Transport Matrices

E.7.1 Why linearization is needed

Many financial transport maps are nonlinear.

For local analysis around state x_A^0:

$$T_{AB}(x_A) \approx T_{AB}(x_A^0) + J_{AB}(x_A - x_A^0). \quad (E.78)$$

where:

$$J_{AB} := \partial T_{AB}/\partial x_A |_{\{x_A^0\}}. \quad (E.79)$$

The Jacobian J_{AB} is the local transport matrix.

It should not be mistaken for the full global connection.

E.7.2 Market-to-collateral Jacobian

For:

$$R_C = hR_M, \quad (E.80)$$

$$Q_C = \sqrt{(A^2 - h^2 R_M^2)}, \quad (E.81)$$

the derivatives are:

$$\partial R_C/\partial R_M = h. \quad (E.82)$$

$$\partial R_C/\partial h = R_M. \quad (E.83)$$

$$\partial Q_C/\partial R_M = -h^2 R_M/Q_C. \quad (E.84)$$

$$\partial Q_C/\partial h = -hR_M^2/Q_C. \quad (E.85)$$

Holding A fixed, the local Jacobian is:

$$J_{MC} = [[h, R_M], [-h^2 R_M/Q_C, -hR_M^2/Q_C]]. \quad (E.86)$$

The input vector is:

$$\delta x_{MC}^{in} := [\delta R_M, \delta h]^T. \quad (E.87)$$

The output change is:

$$[\delta R_C, \delta Q_C]^T \approx J_{MC} \delta x_{MC}^{in}. \quad (E.88)$$

E.7.3 Buffer-transport row

The margin buffer is:

$$B = C + nhR_M - D. \quad (E.89)$$

Its local sensitivity row is:

$$J_B = [nh, nR_M, hR_M, 1, -1]. \quad (E.90)$$

acting on:

$$\delta x_B := [\delta R_M, \delta h, \delta n, \delta C, \delta D]^T. \quad (E.91)$$

Therefore:

$$\delta B \approx J_B \delta x_B. \quad (E.92)$$

This row is exact for first-order changes and locally exact for the bilinear terms under differential interpretation.

E.7.4 Settlement-to-accounting matrix

For the simplified vector:

$$x_S := [q_{exec}, q_{settle}, cash_{settle}, loss_{exec}]^T, \quad (E.93)$$

define:

$$J_{SA} := [[0,1,0,0],[0,0,1,0],[0,0,0,\rho_L]]. \quad (E.94)$$

The accounting output is:

$$x_A := [position_{recognized}, cash_{recorded}, loss_{recognized}]^T. \quad (E.95)$$

Then:

$$\hat{x}_A = J_{SA} x_S. \quad (E.96)$$

The coefficient ρ_L represents the declared recognition ratio or timing convention.

A complete empirical model would also include classifications, currencies, and reporting dates.

E.8 Block Connection Matrix

E.8.1 Stacked frame state

Stack all frame states:

$$X_F := [x_M^T, x_C^T, x_T^T, x_R^T, x_S^T, x_A^T, x_L^T, x_G^T]^T. \quad (E.97)$$

E.8.2 Block transport matrix

Define block matrix $\mathbb{U}_F$ whose block U_{BA} maps frame A into frame B:

$$\mathbb{U}_F := [U_{BA}]_{B,A\in\mathbb{F}}. \qquad (E.98)$$

A simplified synchronous update is:

$$X_{F,k+1} = \mathbb{U}_{F,k}X_{F,k} + b_k + \mathscr{R}_{F,k}. \qquad (E.99)$$

where:

- b_k = external transactions, fields, and gate inputs;
- $\mathscr{R}_{F,k}$ = stacked frame residual.

E.8.3 Sparse structure

The block matrix should normally be sparse.

For example:

$$U_{CM} \neq 0. \qquad (E.100)$$

$$U_{RM} \neq 0. \qquad (E.101)$$

$$U_{TC} \neq 0. \qquad (E.102)$$

$$U_{AS} \neq 0. \qquad (E.103)$$

$$U_{GA} \neq 0. \qquad (E.104)$$

$U_{GL} \neq 0$ only if legal state directly informs regulation. (E.105)

An unsupported direct edge should remain zero rather than being filled for mathematical convenience.

E.8.4 Self-updates

Diagonal blocks U_{AA} represent frame-local persistence and processing.

For example:

U_{MM} = market-state evolution. (E.106)

U_{CC} = collateral-ledger persistence. (E.107)

U_{AA} = accounting roll-forward. (E.108)

U_{LL} = legal-status persistence. (E.109)

E.8.5 Residual propagation

Residual may propagate across frames.

Define residual-transition matrix $D_{\mathscr{R}}$:

$\mathscr{R}_{F,k+1} = D_{\mathscr{R}\mathscr{R}}{}_{F,k} + \varepsilon_{F,k+1}$. (E.110)

For stable residual integration:

$\rho(D_{\mathscr{R}}) < 1$. (E.111)

If:

$\rho(D_{\mathscr{R}}) > 1$, (E.112)

unresolved mismatch amplifies across the frame network.

E.9 Incidence and Connection Operators

E.9.1 Graph incidence matrix

Let B_F be the oriented incidence matrix of the frame graph.

For edge $e : A \to B$:

$(B_F)_{fe} = -1$ if $f = A$. (E.113)

$(B_F)_{fe} = +1$ if $f = B$. (E.114)

$(B_F)_{fe} = 0$ otherwise. (E.115)

E.9.2 Covariant incidence operator

Ordinary graph differences compare:

$x_B - x_A$. (E.116)

Financial frames require transport-adjusted differences:

$d_e^G := x_B - U_{AB}x_A$. (E.117)

Stacking all edge defects gives:

$D_{GX_F} := B_F^U X_F$. (E.118)

where B_F^U is the connection-modified incidence operator.

E.9.3 Graph Laplacian

Define the covariant graph Laplacian:

$L_G^U := (B_F^U)^\dagger W_E B_F^U$. (E.119)

where W_E weights edges by:

- materiality;
- reliability;

- authority;
- systemic importance.

The global reconciliation energy is:

E_recon := X_F†L_G^UX_F. (E.120)

Equivalently:

E_recon = Σ_e w_e‖x_B − U_ABx_A‖². (E.121)

A low E_recon indicates broad cross-frame agreement under the declared connections.

It does not prove that the underlying economic model is correct.

All frames could share the same incorrect source.

E.9.4 Anchored reconciliation

Choose one or more high-authority anchors:

𝔽_anchor ⊆ 𝔽. (E.122)

Examples include:

- legal ownership register;
- clearing record;
- audited ledger;
- contractual margin engine.

Define anchored reconciliation:

E_anchor := Σ_{A∈𝔽_anchor}Σ_Bw_AB‖x_B − U_ABx_A‖². (E.123)

Anchor choice is protocol-dependent.

No frame is universally privileged for every question.

E.10 Path Composition

E.10.1 Path operator

For path:

p : F0 → F1 → … → F_m, (E.124)

define:

U_p := U_{F_mF_{m−1}}⋯U_{F2F1}U_{F1F0}. (E.125)

The transported state is:

$\hat{x}_{F_m}^{(p)} = U_{px_{F_0}}$. (E.126)

E.10.2 Path residual

Given observed target state x_{F_m}:

$r_p := x_{F_m} - U_{px_{F_0}}$. (E.127)

This residual includes accumulated edge effects.

To identify the source, decompose:

$r_p = r_{e_m} + U_{e_m} r_{e_{m-1}} + \dots + U_{e_m} \cdots U_{e_2} r_{e_1}$. (E.128)

Earlier residuals must be transported into the final frame before aggregation.

E.10.3 Path dependence

Suppose two paths connect A to D:

$p_1 : A \to B \to D$. (E.129)

$p_2 : A \to C \to D$. (E.130)

Then:

$\hat{x}_D^{(1)} = U_{DBU_{BAx_A}}$. (E.131)

$\hat{x}_D^{(2)} = U_{DCU_{CAx_A}}$. (E.132)

Define path disagreement:

$r_{path} := \hat{x}_D^{(1)} - \hat{x}_D^{(2)}$. (E.133)

If both routes claim to construct the same target object:

$r_{path} \approx 0$ should hold. (E.134)

If the routes represent different legitimate recognition rules, their difference must be declared rather than classified automatically as error.

E.11 Loop-Closure Matrices

E.11.1 Closed loop

For loop:

$\ell : F_0 \to F_1 \to \dots \to F_m = F_0$, (E.135)

define the holonomy:

$H_\ell := U_{F_0 F_{m-1}} \cdots U_{F_2 F_1} U_{F_1 F_0}$. (E.136)

E.11.2 Naive loop defect

The naive return defect is:

$r_\ell^{naive} := H_\ell x_0 - x_0.$ (E.137)

This is too strict when the loop legitimately includes:

- elapsed time;
- fees;
- interest;
- tax;
- realized cash flow;
- authorized identity conversion.

E.11.3 Expected loop matrix

Define protocol-authorized loop operator:

$H_\ell^P := U_{cost}U_{time}U_{conversion}H_\ell.$ (E.138)

The expected return is:

$\hat{x}_0^{return} = H_\ell^P x_0^{start}.$ (E.139)

The observed return is:

$x_0^{return,obs}.$ (E.140)

The governed loop residual is:

$\mathscr{R}_\ell := x_0^{return,obs} - H_\ell^P x_0^{start}.$ (E.141)

E.11.4 Loop-closure matrix

Define:

$C_\ell := I - H_\ell^P.$ (E.142)

For a state transported only by the declared loop:

$\mathscr{R}_\ell = C_\ell x_0$ (E.143)

when observed return is identified with the initial representation basis.

A perfectly closing loop satisfies:

$C_\ell = 0.$ (E.144)

More generally:

$\|C_{\ell x_0}\| \le \varepsilon_\ell$. (E.145)

E.11.5 Normalized curvature candidate

Define:

$\kappa_\ell := \|\mathscr{R}_\ell\|_W / [\|H_\ell^P x_0\|_W + \varepsilon]$. (E.146)

The term curvature is justified only if:

- the connection is stable;
- the loop is meaningful;
- authorized path effects have been removed;
- transport order matters;
- κ_ℓ adds value beyond ordinary reconciliation error.

Otherwise use:

Normalized Loop Residual. (E.147)

E.11.6 Holonomy eigenvalues

Let:

$H_\ell v_j = \lambda_j v_j$. (E.148)

Interpretations may include:

$\lambda_j \approx 1$
→ mode returns consistently. (E.149)

$|\lambda_j| < 1$
→ mode loses magnitude through recognized or unrecognized leakage. (E.150)

$|\lambda_j| > 1$
→ mode amplifies through the loop. (E.151)

$\arg(\lambda_j) \neq 0$
→ mode accumulates phase or orientation shift. (E.152)

These interpretations require normalization and protocol analysis.

A fee-bearing loop may legitimately have:

$|\lambda_j| < 1$. (E.153)

That is not unexplained curvature if the fee is included in H_ℓ^P.

E.12 Principal Financial Loops

E.12.1 Market–collateral–market loop

The loop is:

F_M → F_C → F_M. (E.154)

The holonomy is:

H_MCM := U_MCU_CM. (E.155)

The connection tests whether the collateral representation can be translated back into the expected market representation.

Because collateral filtering loses information, U_MC may not be a true inverse.

Therefore:

U_MCU_CM ≠ I in general. (E.156)

The loop should compare only transportable invariants and declared reconstructed coordinates.

E.12.2 Market–risk–regulatory–market loop

Define:

H_MRGM := U_MGU_GRU_RM. (E.157)

This loop compares:

- market exposure;
- risk transformation;
- regulatory treatment;
- reconstructed economic exposure.

It may reveal:

- inconsistent netting;
- risk-weight mismatch;
- missing hedge;
- regulatory perimeter mismatch.

E.12.3 Trade–settlement–accounting–trading loop

Define:

H_MSAM := U_MAU_ASU_SM. (E.158)

This loop tests whether:

- executed trade;
- settled position;
- accounting recognition;
- reconstructed market inventory

refer to the same transaction and quantity.

Possible residuals include:

- unsettled units;
- cash mismatch;
- missing journal entry;
- duplicate trade;
- timing disagreement.

E.12.4 Legal–collateral–treasury–accounting–legal loop

Define:

H_LCTAL := U_LAU_ATU_TCU_CL. (E.159)

This loop tests whether:

- legal collateral rights;
- collateral admission;
- treasury encumbrance;
- accounting classification;
- legal reconstruction

remain coherent.

It is especially relevant for:

- rehypothecation;
- secured lending;
- repo;
- collateral substitution;
- bankruptcy remoteness.

E.12.5 Margin-event loop

A practical margin-event loop is:

Market → Collateral → Margin Gate → Settlement/Treasury → Accounting/Risk → Reconciled Market Position. (E.160)

Let the corresponding operator be:

H_margin := U_returnU_ARU_SAU_gateU_CM. (E.161)

The exact operator order depends on the institutional process.

Margin closure requires:

$\|\mathscr{R}_\text{margin-loop}\| \le \varepsilon_\text{margin-loop}$. (E.162)

E.13 Three-Frame Symbolic Example

E.13.1 Frames

Consider:

- market frame M;
- collateral frame C;
- accounting frame A.

Let the scalar market value be R_M.

E.13.2 Market-to-collateral transport

Define:

R_C = hR_M. (E.163)

Thus:

U_CM = h. (E.164)

E.13.3 Collateral-to-accounting transport

Suppose accounting recognizes collateral at factor a_C:

R_A = a_CR_C. (E.165)

Thus:

U_AC = a_C. (E.166)

E.13.4 Accounting-to-market reconstruction

Suppose the expected market reconstruction uses factor m_A:

R_M^return = m_AR_A. (E.167)

Thus:

U_MA = m_A. (E.168)

E.13.5 Loop holonomy

The loop operator is:

H_MCAM = m_Aa_Ch. (E.169)

The naive loop closes when:

m_Aa_Ch = 1. (E.170)

But this may be impossible if h < 1 because collateral filtering discards market-value content.

A valid expected loop must therefore include the intended reconstruction rule rather than assume invertibility.

E.13.6 Loop residual

For initial value R_M:

$\mathscr{R}$_MCA = R_M^return,obs − m_Aa_ChR_M. (E.171)

If:

R_M^return,obs = m_Aa_ChR_M, (E.172)

then:

$\mathscr{R}$_MCA = 0. (E.173)

even though:

R_M^return,obs ≠ R_M. (E.174)

This demonstrates:

Loop Closure ≠ Numerical Return to Original Value. (E.175)

It means return to the expected transformed state.

E.14 Timing and Asynchronous Frames

E.14.1 Multiple times

Each frame may carry:

- economic event time t_e;
- observation time t_o;
- recognition time t_r;
- settlement time t_s;
- reporting time t_p.

Thus:

t_e ≤ t_o ≤ t_r ≤ t_s ≤ t_p may hold. (E.176)

But the exact order depends on the event.

For a market trade:

t_execution < t_settlement. (E.177)

For an impairment:

t_economic-deterioration may precede t_accounting-recognition. (E.178)

For a legal default:

market pricing may precede legal recognition. (E.179)

The source default example explicitly notes that market, accounting, contractual, legal, rating, and regulatory frames may recognize the same underlying deterioration at different times.

E.14.2 Timing connection

Define delay:

Δt_AB := t_B − t_A. (E.180)

The transport becomes:

x^_B(t_B) = T_AB[x_A(t_A),Path(t_A,t_B),Δt_AB]. (E.181)

A stale source error is:

r_stale,AB := x_B(t_B) − T_AB[x_A(t_A),Δt_AB]. (E.182)

If Δt_AB is omitted, ordinary latency may be misclassified as transport failure.

E.14.3 Delay matrix

A discrete delay of d periods may be represented by:

x_B,k = U_ABx_A,k−d + r_AB,k. (E.183)

A distributed delay is:

$$x_{B,k} = \Sigma_{j=0}^{d} W_j U_{AB} x_{A,k-j} + r_{AB,k}. \qquad (E.184)$$

where:

$$\Sigma_{j=0}^{d} W_j = I. \qquad (E.185)$$

The weights encode staged recognition or settlement.

E.14.4 Timing residual

Define:

$$r_{time,AB} := t_B^{obs} - t_B^{expected}. \qquad (E.186)$$

Timing residual may exist even when quantities reconcile.

Examples include:

- collateral posted late;
- accounting entry delayed;
- regulatory filing late;
- legal perfection incomplete.

Timing should remain a distinct residual coordinate.

E.15 Authority and Recognition Vectors

E.15.1 Recognition vector

For event e, define:

$$\vec{g}_e := [g_M, g_C, g_T, g_R, g_S, g_A, g_L, g_G]^T. \qquad (E.187)$$

where:

$$g_f \in \{0,1\} \qquad (E.188)$$

indicates whether frame f has admitted the event.

A softer score may use:

$$0 \le g_f \le 1. \qquad (E.189)$$

for confidence, partial recognition, or staged processing.

E.15.2 Examples of recognition states

Market-leading state

$g_M = 1$, while $g_A = g_L = 0$. (E.190)

The market has priced an event before accounting or legal recognition.

Institution-leading state

$g_L = 1$ or $g_G = 1$, while g_M is incomplete. (E.191)

A legal or regulatory gate changes before market repricing is complete.

Fragmented state

Relevant g_f values disagree materially. (E.192)

Delayed state

Recognition is expected but has not yet been written. (E.193)

Contested state

Authorities disagree about whether the event occurred. (E.194)

The draft Periodic Grammar uses the same categories—aligned, market-leading, institution-leading, fragmented, delayed, and contested—for cross-ledger reconciliation states.

E.15.3 Authority vector

Define:

$\vec{a}_e := [a_M, a_C, a_T, a_R, a_S, a_A, a_L, a_G]^T$. (E.195)

where a_f records the authority weight of frame f for the declared question.

For margin calls:

a_C may dominate. (E.196)

For legal default:

a_L may dominate. (E.197)

For financial-statement recognition:

a_A may dominate. (E.198)

For capital treatment:

a_G may dominate. (E.199)

No universal authority vector exists.

E.15.4 Authority-weighted recognition

Define:

$R_auth(e) := \vec{a}_e^{T}\vec{g}_e.$ (E.200)

subject to:

$\Sigma_fa_f = 1.$ (E.201)

A high R_auth means the event has been recognized by the frames possessing declared authority.

But full closure may still require non-authoritative operational frames to complete their return.

E.15.5 Recognition disagreement

Define:

$D_rec(e) := \vec{g}_e^{T}L_F\vec{g}_e.$ (E.202)

where L_F is a graph Laplacian connecting frames expected to agree.

A high D_rec indicates fragmented recognition.

A low D_rec may indicate alignment—or shared omission.

Therefore recognition scores must be combined with trace and evidence.

E.16 Cross-Ledger Reconciliation Gate

E.16.1 Reconciliation gate

Define:

G_recon(e)
:= Admit[
IdentityMatch,
InvariantPreservation,
RecognitionAlignment,
ChargeBalance,
TraceAccessibility,
ResidualDisclosure
]. (E.203)

The output is:

$G_recon(e) \rightarrow (Decision, Strength, ResidualVector, Invalidation, Authority).$ (E.204)

The draft Periodic Grammar proposes this residual-bearing gate format as an improvement over simple yes/no event labels.

E.16.2 Gate outcomes

Let:

$$\text{Decision_recon} \in \{\text{Open},\text{Partial},\text{Closed},\text{Contested},\text{Failed}\}. \qquad \text{(E.205)}$$

Open

Required frame returns remain incomplete.

Partial

Some required frames reconcile, but material residual remains.

Closed

All mandatory transport and trace conditions pass within tolerance.

Contested

Authorities disagree about the event or identity.

Failed

Required transport cannot preserve the identity.

E.16.3 Closure strength

Define:

$$S_{recon} := \Sigma_{fw_fg_f} - \lambda_{DD_rec} - \lambda_{\mathscr{R}} \|\mathscr{R}_F\|. \qquad \text{(E.206)}$$

This is a proposed score.

It should not replace the underlying vector of frame states.

E.16.4 Mandatory-frame rule

Let:

$$\mathbb{F}_{req}(e) := \text{required frames for event } e. \qquad \text{(E.207)}$$

Then closure requires:

$$g_f = 1 \quad \text{for all } f \in \mathbb{F}_{req}(e). \qquad \text{(E.208)}$$

or, under tolerances:

$$g_f \geq \gamma_{f}^{\min} \quad \text{for all } f \in \mathbb{F}_{req}(e). \qquad \text{(E.209)}$$

A weighted average should not allow a critical missing legal or settlement gate to be hidden by strong agreement elsewhere.

E.17 Loop-Closure Certificate

E.17.1 Certificate structure

Define:

Cert_ℓ := (SubjectID,LoopID,Frames,Path,Protocol,Invariant,Connections,StartState,ExpectedReturn,Obse
(E.210)

E.17.2 Certificate conditions

A valid certificate requires:

IdentityMatched = 1. (E.211)

RequiredEdgesValidated = 1. (E.212)

InvariantPreserved = 1. (E.213)

TraceAccessible = 1. (E.214)

ResidualDisclosed = 1. (E.215)

AuthorityValid = 1. (E.216)

E.17.3 Certificate status

Status(Cert_ℓ) ∈ {Closed,ClosedWithResidual,Open,Contested,Invalid}. (E.217)

Closed

Residual lies below tolerance.

ClosedWithResidual

The event is accountable, but accepted consequence remains.

Open

Mandatory return is incomplete.

Contested

Frame or authority disagreement remains.

Invalid

Identity or connection cannot be established.

E.17.4 Certificate recursion

The certificate enters the ledger:

$L_{k+1} = L_k \oplus \mathrm{Cert}_{\ell,k}.$ (E.218)

It may change:

- future haircut;
- collateral eligibility;
- risk limits;
- accounting classification;
- legal permissions;
- regulatory treatment.

Thus:

Loop Closure → New Connection Conditions. (E.219)

E.18 Connection Curvature and Order Effects

E.18.1 Noncommuting transports

For frames A, B, and C:

$U_{CBU_{BA}} \neq U_{CAU_{AB}}$ in general. (E.220)

More simply, two transformations may fail to commute:

$U_X U_Y \neq U_Y U_X.$ (E.221)

Examples include:

- netting before default versus default before netting;
- collateral posting before liquidation versus liquidation before collateral posting;
- legal reclassification before accounting recognition versus the reverse;
- currency conversion before haircut versus haircut before currency conversion.

E.18.2 Commutator

Define:

$F_{XY} := [U_X, U_Y] = U_X U_Y - U_Y U_X.$ (E.222)

A nonzero commutator means order affects the resulting representation.

This is a candidate discrete curvature measure.

It does not automatically imply error.

The order effect may be:

- legally required;
- economically real;
- institutionally intended.

The protocol must state whether the operations are expected to commute.

E.18.3 Order residual

For initial state x:

r_order := U_XU_Yx – U_YU_Xx. (E.223)

If both sequences claim to implement the same process:

r_order should be small. (E.224)

If they represent different legal or economic sequences, r_order is a substantive path-dependence measure.

E.18.4 Margin example

Compare:

Path 1: Post Collateral → Revalue → Recalculate Margin. (E.225)

Path 2: Revalue → Liquidate → Post Remaining Collateral. (E.226)

The final states may differ because:

- liquidation changes quantity;
- impact changes price;
- collateral changes debt need;
- authority changes after breach.

Thus:

G_postG_liq ≠ G_liqG_post. (E.227)

This noncommutativity is one source of recursive margin dynamics.

E.19 Connection Repair

E.19.1 Local-state repair versus connection repair

Suppose frames disagree:

$$x_B^{obs} \neq U_{AB}x_A. \qquad (E.228)$$

There are at least three possible causes:

1. source state x_A is wrong;
2. target state x_B is wrong;
3. connection U_{AB} is wrong.

Correcting only the source or target values may fail when the connection itself is defective.

E.19.2 Connection residual decomposition

Define:

$$r_{AB} = r_{source} + r_{target} + r_{connection} + r_{time} + r_{authority}. \qquad (E.229)$$

where:

- r_{source} = incorrect source state;
- r_{target} = incorrect target record;
- $r_{connection}$ = wrong mapping rule;
- r_{time} = asynchronous recognition;
- $r_{authority}$ = disagreement about valid rule.

E.19.3 Repair operator

Define repair:

$$R_{AB} := (R_{source}, R_{target}, R_{connection}, R_{time}, R_{authority}). \qquad (E.230)$$

The repaired target estimate is:

$$\hat{x}_B^{repair} = T_{AB}^{repair}(x_A^{repair}; \mathcal{A}_{AB}^{repair}). \qquad (E.231)$$

The repair should preserve:

- original source state;
- original target state;
- original connection;
- reason for revision;
- authority;

- resulting residual.

E.19.4 Lowest sufficient repair

The intervention principle is:

Choose the Lowest Repair Level Capable of Closing the Relevant Residual. (E.232)

Possible levels are:

1. data correction;
2. timing correction;
3. connection recalibration;
4. gate revision;
5. identity reclassification;
6. legal or policy change.

The draft Periodic Grammar likewise treats some local–system disagreements as connection failures rather than local-variable failures.

E.20 Empirical Estimation of Frame Connections

E.20.1 Edge dataset

For each edge A → B, collect observations:

$\mathcal{D}_{AB} := \{(x_A,t,x_B,t',\text{Protocol}_t,\text{Trace}_t)\}.$ (E.233)

The dataset must preserve:

- source timestamp;
- target timestamp;
- source version;
- target version;
- applicable rule;
- later revisions.

E.20.2 Connection estimation

Estimate:

$x_B = T_{AB}(x_A;\theta_{AB}) + \varepsilon_{AB}.$ (E.234)

Possible models include:

- deterministic rules;
- linear regression;
- generalized additive models;
- state-space models;
- constrained neural networks;
- legal rule engines;
- accounting transformation engines.

The model class should respect known institutional constraints.

E.20.3 Prospective residual

After fitting on training data:

$$r_{AB,t}^{test} = x_{B,t}^{obs} - \hat{T}_{AB}(x_{A,t}). \qquad (E.235)$$

Connection quality should be evaluated out of sample.

E.20.4 Edge metrics

Possible metrics include:

- identity-match accuracy;
- quantity reconciliation;
- monetary error;
- timing error;
- charge-preservation rate;
- trace-accessibility rate;
- false-closure rate.

Define normalized edge accuracy:

$$Acc_{AB} := 1 - \|r_{AB}\| / [\|x_B\| + \varepsilon]. \qquad (E.236)$$

This scalar should be accompanied by residual components.

E.20.5 Path test

For two routes p_1 and p_2:

$r_{path} = U_{p_1x_A} - U_{p_2x_A}.$ (E.237)

The path-consistency score is:

$PC_{p_1,p_2} := 1 - \|r_{path}\| / [\|U_{p_1x_A}\| + \|U_{p_2x_A}\| + \varepsilon].$ (E.238)

E.20.6 Loop test

For loop ℓ:

$\kappa_\ell = \|x_{return,obs} - H_\ell^{Px_{start}}\| / [\|H_\ell^{Px_{start}}\| + \varepsilon].$ (E.239)

Test whether κ_ℓ predicts:

- settlement failure;
- margin escalation;
- accounting restatement;
- legal dispute;
- regulatory correction;
- default.

E.20.7 A-B Fixedness test

A practical score is:

$ABFix_{AB} = M_{AB} \times I_{AB} \times R_{AB} \times H_{AB}.$ (E.240)

The empirical hypothesis is:

Higher $ABFix_{AB}$
→ lower future reconciliation failure. (E.241)

A stronger hypothesis is:

ABFix adds value beyond ordinary matching and exception counts. (E.242)

E.21 Frame-Graph Falsifiers

E.21.1 Identity failure

The frame graph fails when no stable shared identity can be specified.

Then the systems may be observing different objects.

Use:

Related-System Comparison. (E.243)

not:

Transport of the Same Financial Identity. (E.244)

E.21.2 Connection failure

A connection claim fails when:

- T_AB is selected after observing x_B;
- rules vary without trace;
- timing is omitted;
- units are inconsistent;
- authority is unclear;
- the edge cannot be reproduced.

E.21.3 Covariance failure

The transport fails gauge-like covariance when presentation-only changes alter substantive results:

T_AB′G_Ax_A ≠ G_BT_ABx_A. (E.245)

Examples include:

- unit rescaling changes margin status;
- currency presentation changes charge sign;
- desk transfer changes legal ownership;
- reporting format changes event identity.

E.21.4 Loop failure

The loop claim fails when:

- no meaningful closed path exists;
- expected transformations are unknown;
- authorized fees explain the entire difference;
- the starting and ending objects are not comparable;
- loop residual adds no value beyond ordinary reconciliation.

Then use:

Cross-System Reconciliation. (E.246)

E.21.5 Graph-overreach failure

The graph becomes unnecessarily complex when:

- many nodes duplicate the same frame;
- unused edges are added;
- every institutional function is treated as a gauge frame;
- no distinct admission or authority rule exists.

Then merge equivalent nodes.

E.22 Reduction Ladder

The frame architecture reduces as follows:

Gauge-Covariant Frame Graph
→ Governed Financial Transport Graph. (E.247)

Governed Transport Graph
→ Multi-Ledger Reconciliation Network. (E.248)

Multi-Ledger Network
→ Pairwise Reconciliation Rules. (E.249)

Pairwise Rules
→ Ordinary Data Matching. (E.250)

The strongest level is retained only if:

- identity invariants survive;
- connections are prospectively defined;
- covariance matters;
- path and loop tests add information;
- residual remains interpretable.

E.23 Frame and Connection Audit Template

For each frame f, record:

Audit field	Required statement
Boundary	What system does the frame contain?
Observer	Who or what constructs the state?
Authority	Which decisions can the frame make?
Coordinates	Which variables are local to the frame?
Time	Which event and recognition times apply?

Audit field	Required statement
Trace	What record is written?
Invariant	What identity relation should survive?
Residual	What does the frame leave unresolved?

For each edge A → B, record:

Audit field	Required statement
Source	Which frame supplies the input?
Target	Which frame receives the transported state?
Carrier	Which bounded identity is being transported?
Map	What is T_AB?
Connection	Which haircut, FX, timing, legal, or recognition rules apply?
Charge	Which charge sectors must be preserved or converted?
Authority	Who validates the map?
Delay	What recognition lag is expected?
Residual	How is mismatch classified?
Falsifier	What evidence rejects the edge?

E.24 Compact Matrix Catalogue

E.24.1 Frame state

$X_F = [x_M^T, x_C^T, x_T^T, x_R^T, x_S^T, x_A^T, x_L^T, x_G^T]^T.$ (E.251)

E.24.2 Edge transport

$\hat{x}_B = U_{AB} x_A.$ (E.252)

E.24.3 Edge residual

$r_{AB} = x_B - U_{AB} x_A.$ (E.253)

E.24.4 Covariant graph defect

$D_{GX_F} = B_{F^\wedge U} X_F.$ (E.254)

E.24.5 Reconciliation energy

$E_{recon} = X_F^\dagger (B_{F^\wedge U})^\dagger W_{EB_{F^\wedge U}} X_F.$ (E.255)

E.24.6 Path operator

$$U_p = U_{F_mF_{m-1}}\cdots U_{F_1F_0}. \qquad (E.256)$$

E.24.7 Loop holonomy

$$H_\ell = \prod_{e\in\ell} U_e. \qquad (E.257)$$

E.24.8 Governed loop residual

$$\mathscr{R}_\ell = x_{return,obs} - H_\ell^P x_{start}. \qquad (E.258)$$

E.24.9 Loop-closure matrix

$$C_\ell = I - H_\ell^P. \qquad (E.259)$$

E.24.10 Normalized loop defect

$$\kappa_\ell = \|\mathscr{R}_\ell\| / [\|H_\ell^P x_{start}\| + \varepsilon]. \qquad (E.260)$$

E.24.11 Recognition vector

$$\vec{g}_e = [g_M, g_C, g_T, g_R, g_S, g_A, g_L, g_G]^T. \qquad (E.261)$$

E.24.12 Authority-weighted recognition

$$R_{auth}(e) = \vec{a}_e^T \vec{g}_e. \qquad (E.262)$$

E.24.13 A-B Fixedness

$$ABFix_{AB} = IdentityMatch \times InvariantPreservation \times RecordAccessibility \times ResidualHonesty. \qquad (E.263)$$

E.25 Final Interpretation

A financial frame is not merely a viewpoint.

It is an operational protocol determining:

- what object is recognized;
- how it is measured;
- which authority may act;

- what trace is written;
- what future state inherits the result.

The market frame asks:

What is currently valued? (E.264)

The collateral frame asks:

What value may support an obligation? (E.265)

The treasury frame asks:

What funding and liquidity remain usable? (E.266)

The risk frame asks:

What loss and limit structure is recognized? (E.267)

The settlement frame asks:

What has actually transferred? (E.268)

The accounting frame asks:

What has been recognized and classified? (E.269)

The legal frame asks:

Who owns, owes, controls, and may enforce? (E.270)

The regulatory frame asks:

What exposure and capital consequence is institutionally admitted? (E.271)

These frames need not speak identical numerical languages.

They must remain able to establish that they refer to the same bounded identity.

The complete frame grammar is:

Local Representation
→ Declared Connection
→ Covariant Transport
→ Target Recognition
→ Edge Residual
→ Path Composition
→ Loop Return
→ Closure Certificate
→ Ledger Update. (E.272)

Its central object is:

$\mathscr{R}_\ell$ = Observed Return − Protocol-Expected Return. (E.273)

Its central scientific discipline is:

> Do not call two frames inconsistent merely because their numbers differ. First

transport the identity lawfully. Then measure what remains.

Its central governance discipline is:

> Do not call a loop closed merely because one system reports completion. Require invariant preservation, accessible trace, authority alignment, charge reconciliation, and residual honesty.

The resulting frame graph connects the financial spinor to the wider institutional world:

Action occurs in one frame. (E.274)

Consequence appears in another. (E.275)

Transport determines whether they belong to the same identity. (E.276)

The loop determines whether the identity returns accountably. (E.277)

The ledger determines what the next loop inherits. (E.278)

The next appendix will construct a **simulation and estimation blueprint** for the complete margin-account system, including synthetic data generation, nested benchmark models, parameter recovery, gate-event experiments, and prospective falsification tests.

Appendix F — Simulation, Identification, and Estimation Blueprint

F.1 Purpose

The Financial Gauge–Dirac–Gate–Ledger system contains several objects that may be mathematically coherent yet empirically non-identifiable:

- the Complex CAPM pressure coordinate Q;
- structural and effective financial charge;
- the action–ledger spinor;
- frame connections;
- loop curvature;
- identity mass;
- coherent closure capacity c_P;
- recursive ledger backreaction.

Before applying the complete architecture to confidential broker, clearing, treasury, accounting, and legal data, the model should first be tested in a synthetic environment where the true generating process is known.

The purpose of simulation is not to prove that the theory describes real finance.

It is to answer four preliminary questions:

1. Can the proposed variables be recovered when they truly exist?

2. Can the model reject them when they do not exist?

3. Can the stronger architecture be distinguished from simpler alternatives?

4. Which observations are required for identification?

The source Financial Standard Model explicitly presents the wider architecture as a testable research programme rather than a validated physical or financial theory. It also requires advanced models to compete against simpler real-vector, state-space, regime-switching, and conventional statistical alternatives.

The simulation programme therefore follows:

Structural Construction
→ Synthetic Identifiability
→ Benchmark Competition
→ Falsification
→ Reduction
→ Real-Data Pilot. (F.1)

F.2 Epistemic Status of Simulation

F.2.1 What simulation can establish

A successful simulation can establish that:

- the equations are internally executable;
- parameter recovery is possible under declared conditions;
- the proposed diagnostics respond to known ground truth;
- the estimation method does not mechanically invent the desired structure;
- certain data frequencies and sample sizes are sufficient;
- model layers can be compared under controlled misspecification.

F.2.2 What simulation cannot establish

Simulation cannot establish that:

- real financial charge exists;
- real margin systems obey the proposed Γ algebra;
- Q improves actual forecasting;
- financial identity mass is universal;
- c_P is a natural constant;
- gauge curvature predicts real crises;
- the model guarantees risk-adjusted profit.

Therefore:

Simulation Recovery ≠ Real-World Validation. (F.2)

The role of simulation is:

Test the Measurement Architecture before Testing the Market Ontology. (F.3)

F.2.3 Source-derived discipline

The related Periodic Grammar research programme adopts a staged evidential order in which gate, residual, transport, complex eligibility, and internal phase are tested separately. It also requires chronological isolation, equal treatment of benchmarks, negative-result reporting, and removal of unsupported complexity.

The present simulation blueprint applies the same discipline to the margin-account system.

F.3 Simulation Levels

F.3.1 Level S_0 — Scalar margin system

The minimum generating process contains:

- market value R;
- quantity n;
- collateral factor h;
- posted collateral C;
- debt D;
- buffer B;
- margin call.

No Q, charge, spinor, gauge, mass, or recursion is generated.

This level tests whether the advanced estimation procedure falsely detects structures that are absent.

F.3.2 Level S_1 — Real two-coordinate valuation system

Generate:

(R,Q), (F.4)

as two ordinary real state variables.

Do not impose:

$Z = R + iQ$ (F.5)

as a privileged complex structure.

This level tests whether complex notation adds anything beyond a flexible real-pair model.

F.3.3 Level S2 — Genuine complex valuation system

Generate:

$Z = Ae^{i\theta}.$ (F.6)

with:

$R = A\cos\theta.$ (F.7)

$Q = A\sin\theta.$ (F.8)

The phase relation is part of the true data-generating process.

This level tests whether:

- θ can be recovered;
- Q improves state reconstruction;
- the complex model defeats the real-pair benchmark.

F.3.4 Level S3 — Charge-typed constraint system

Generate stable structural charges:

$\vec{q}_{struct} = (q_A, q_F, q_C).$ (F.9)

Let state-dependent screening produce:

$\vec{q}_{eff,k} = S_k \vec{q}_{struct,k}.$ (F.10)

This level tests whether the estimation system can distinguish:

- structural orientation;
- coupling magnitude;
- ordinary sensitivity;
- current monetary exposure.

F.3.5 Level S4 — Action–ledger doublet

Generate independently evolving action and ledger states:

$\Psi_k = [\psi_{A,k}, \psi_{L,k}]^T.$ (F.11)

Allow:

$\psi_{A,k} \neq \psi_{L,k}$. (F.12)

Include measurable ledger latency and residual.

This level tests whether a spinor-like representation is identifiable and whether a scalar workflow model is sufficient.

F.3.6 Level S5 — Multi-frame transport system

Generate several frames:

$\mathbb{F} = \{M,C,T,R,S,A,L,G\}$. (F.13)

Define true transport operators:

U_{AB}^*. (F.14)

Generate observed frame states with known residual:

$x_{B,k} = U_{AB}^* x_{A,k} - d_{AB} + r_{AB,k}$. (F.15)

This level tests recovery of:

- frame connections;
- edge residual;
- A-B Fixedness;
- loop closure.

F.3.7 Level S6 — Gauge-covariant system

Generate local frame redescriptions:

$x_{A'} = G_A x_A$. (F.16)

Transform connections according to:

$U_{AB'} = G_B U_{AB} G_A^{-1}$. (F.17)

This level tests whether the estimation procedure recognizes representation-invariant transport rather than treating every coordinate change as economic movement.

F.3.8 Level S7 — Full hybrid Gauge–Dirac system

Generate:

$(i\mathcal{D}_{fin} - M_k)\Psi_k = \mathscr{R}_k$ (F.18)

between gates, together with:

$\mathcal{S}_k^+ = \mathcal{J}_k(\mathcal{S}_k^-, L_k) + \mathscr{R}_{jump,k}$ (F.19)

at gates and:

$\Theta_{k+1} = U_\Theta(\Theta_k, L_{k+1}, \mathscr{R}_k)$ (F.20)

after ledger update.

This is the maximum simulation level.

The estimator should not require Level S_7 to explain data generated at lower levels.

F.4 Simulation Clock and Event Order

F.4.1 Calendar time

Let:

$t = 0,1,2,\ldots,T.$ (F.21)

Calendar time represents equally spaced market observation intervals.

Possible intervals include:

- seconds;
- minutes;
- days;
- margin cycles;
- settlement days.

F.4.2 Ledger time

Let:

$\tau = 0,1,2,\ldots,K.$ (F.22)

Ledger time advances when a recognized trace-writing event occurs.

Examples include:

- margin calculation;
- call issuance;
- collateral admission;
- execution;
- settlement;
- accounting posting;
- default recognition.

Calendar time and ledger time need not advance together:

$\Delta t \neq \Delta \tau$ in general. (F.23)

F.4.3 Event queue

Define the event queue:

$\mathcal{Q}$_event,k := ordered set of pending institutional events. (F.24)

Possible event types are:

e_k ∈
{MarketUpdate,HaircutUpdate,DebtAccrual,Call,CollateralPost,Sale,Settlement,Accounting,Defau
(F.25)

Each event contains:

e_k := (Type,Subject,Time,Authority,Inputs,Status,Trace,Residual). (F.26)

F.4.4 Event priority

A simple priority order is:

Market Update
→ Margin Calculation
→ Call Admission
→ Cure Instruction
→ Execution
→ Settlement
→ Ledger Recognition. (F.27)

The order may be randomized or changed in stress experiments.

Noncommuting branch order is one source of path dependence.

F.5 Latent Market Process

F.5.1 Baseline amplitude process

Let the baseline value amplitude evolve as:

$\ln A_{k+1} = \ln A_k + \mu_A \Delta t + \sigma_A \sqrt{\Delta t}\, \varepsilon_{A,k}$. (F.28)

where:

$\varepsilon_{A,k} \sim N(0,1)$. (F.29)

This process may represent changing expected cash flow or baseline discounting.

F.5.2 Market-phase process

Define:

$\theta_{k+1} = \theta_k + \mu_\theta \Delta t + \sigma_\theta \sqrt{\Delta t}\, \varepsilon_{\theta,k} + \Omega_{\text{impact},k} \Delta t.$ (F.30)

The terms are:

- μ_θ = average phase drift;
- σ_θ = exogenous phase volatility;
- $\Omega_{\text{impact},k}$ = endogenous liquidation-induced phase acceleration.

To preserve the first-quadrant minimum model:

$0 \le \theta_k \le \pi/2.$ (F.31)

Reflection, clipping, or a bounded transformed process may enforce this range.

F.5.3 Complex market state

Generate:

$R_k = A_k \cos \theta_k.$ (F.32)

$Q_k = A_k \sin \theta_k.$ (F.33)

$Z_k = R_k + iQ_k.$ (F.34)

Then:

$A_k^2 = R_k^2 + Q_k^2.$ (F.35)

This is the true complex model for Levels S_2 and above.

F.5.4 Scalar null model

For Level S_0, generate R directly:

$\ln R_{k+1} = \ln R_k + \mu_R \Delta t + \sigma_R \sqrt{\Delta t}\, \varepsilon_{R,k}.$ (F.36)

No latent A, Q, or θ exists.

If the estimation method still recovers a stable Q or phase advantage, it is overfitting the imposed geometry.

F.5.5 Real-pair null model

For Level S_1:

$R_{k+1} = f_R(R_k, Q_k) + \varepsilon_{R,k}.$ (F.37)

$Q_{k+1} = f_Q(R_k, Q_k) + \varepsilon_{Q,k}.$ (F.38)

Do not require:

$R_k^2 + Q_k^2 = A_k^2$. (F.39)

This tests whether the complex constraint helps only when it is genuinely present.

F.6 Collateral Process

F.6.1 Latent stress state

Define a latent collateral-stress variable:

s_k. (F.40)

Let:

$s_{k+1} = \rho_s s_k + \beta_{s\theta}\Delta\theta_k + \beta_{s\sigma}\sigma_k + \sigma_s\varepsilon_{s,k}$. (F.41)

where:

- $\Delta\theta_k$ = market-phase movement;
- σ_k = market volatility or stress;
- ρ_s = stress persistence.

F.6.2 Collateral factor

Map stress into the admissible interval through:

$h_k = h_{min} + (h_{max} - h_{min})/[1 + \exp(s_k)]$. (F.42)

where:

$0 \le h_{min} < h_{max} \le 1$. (F.43)

Higher stress produces lower h.

F.6.3 Independent haircut shock

To distinguish market stress from protocol shock, include:

$h_k^+ = \mathrm{Clip}[h_k + \xi_{h,k}, h_{min}, h_{max}]$. (F.44)

where $\xi_{h,k}$ may represent:

- rule change;
- concentration penalty;
- legal ineligibility;
- discretionary margin tightening.

F.6.4 Collateral plane

Generate:

$R_{C,k} = h_k R_k.$ (F.45)

$Q_{C,k} = \sqrt{(A_k^2 - R_{C,k}^2)}.$ (F.46)

$\theta_{C,k} = \arccos(R_{C,k}/A_k).$ (F.47)

The market–collateral phase difference is:

$\Delta\theta_{MC,k} = \theta_{C,k} - \theta_k.$ (F.48)

F.6.5 Legal eligibility

Let:

$e_{LC,k} \in \{0,1\}.$ (F.49)

Then:

$R_{C,k} = e_{LC,k} h_k R_k.$ (F.50)

Legal ineligibility may arrive through a separate gate:

$\Pr(e_{LC,k} = 0) = \text{Logistic}(\alpha_L + \beta_{LL} k + \beta_D \text{DefaultRisk}_k).$ (F.51)

This separates:

Economic Haircut
from
Legal Ineligibility. (F.52)

F.7 Funding and Balance-Sheet Process

F.7.1 Quantity

Before intervention:

$n_{k+1} = n_k + u_{n,k}.$ (F.53)

where $u_{n,k}$ may be:

- voluntary purchase;
- voluntary sale;
- forced liquidation.

F.7.2 Debt accrual

Let debt evolve as:

$D_{k+1}^{-} = D_k(1 + r_{F,k}\Delta t) + u_{D,k}.$ (F.54)

where:

- $r_{F,k}$ = funding rate;
- $u_{D,k}$ = new borrowing or repayment before gate action.

F.7.3 Posted collateral

Let:

$C_{k+1}^{-} = C_k + u_{C,k}.$ (F.55)

where $u_{C,k}$ may be zero before a margin call.

F.7.4 Margin buffer

The buffer is:

$B_k = C_k + n_k h_k R_k - D_k.$ (F.56)

Define target:

$B_{target,k} \geq 0.$ (F.57)

Deficiency is:

$S_{B,k} = [B_{target,k} - B_k]_+.$ (F.58)

F.8 Gate Generation

F.8.1 Candidate breach

Define:

$b_k := 1[B_k < B_{target,k}].$ (F.59)

This is the numerical crossing indicator.

F.8.2 Valid gate authority

Let:

$a_k^{auth} \in \{0,1\}.$ (F.60)

A valid margin call is generated by:

$g_k^{call} = b_k a_k^{auth} v_k^{mark} p_k^{rule}.$ (F.61)

where:

- v_k^{mark} = mark validity;
- p_k^{rule} = rule applicability.

This creates simulations in which:

$$\text{Breach} \neq \text{Call}. \qquad (F.62)$$

F.8.3 Gate error

Allow:

$$\Pr(\text{FalsePositiveCall}) = p_{FP}. \qquad (F.63)$$

$$\Pr(\text{MissedCall} \mid \text{Breach}) = p_{FN}. \qquad (F.64)$$

These parameters test whether the model can distinguish:

- economic breach;
- operational gate error;
- authority failure.

F.8.4 Call amount

When $g_k^{call} = 1$:

$$C_{call,k} = S_{B,k}. \qquad (F.65)$$

When $g_k^{call} = 0$:

$$C_{call,k} = 0. \qquad (F.66)$$

F.8.5 Branch selection

Let branch be:

$$b_k^{branch} \in \{\text{Collateral}, \text{Deleverage}, \text{Liquidation}, \text{Default}\}. \qquad (F.67)$$

A multinomial selection model may be:

$$\Pr(b_k^{branch} = j) = \exp(\eta_{j,k}) / \Sigma_m \exp(\eta_{m,k}). \qquad (F.68)$$

Possible predictors include:

- liquidity;
- shortfall;
- market depth;

- legal permissions;
- time remaining;
- collateral availability;
- prior ledger history.

F.9 Branch Dynamics

F.9.1 Collateral branch

Let intended collateral be:

$C_{instr,k} = \min(C_{available,k}, C_{call,k}).$ (F.69)

Let eligibility and operational acceptance be:

$\alpha_{C,k} \in [0,1].$ (F.70)

Then admitted collateral is:

$C_{admit,k} = \alpha_{C,k} C_{instr,k}.$ (F.71)

The updated cash collateral is:

$C_k^+ = C_k^- + C_{admit,k}.$ (F.72)

F.9.2 Deleveraging branch

Let intended sale quantity be:

$\ell_{instr,k} = \min[n_k, \kappa_{LS_B,k}].$ (F.73)

Execution quantity is:

$\ell_{exec,k} = \alpha_{exec,k} \ell_{instr,k}.$ (F.74)

with:

$0 \le \alpha_{exec,k} \le 1.$ (F.75)

The position becomes:

$n_k^+ = n_k^- - \ell_{exec,k}.$ (F.76)

F.9.3 Price impact

Let aggregate liquidation be:

$L_k = \Sigma_i \ell_{exec,i,k}.$ (F.77)

Define impact:

$I_k = \kappa_{IL} k^{\alpha} / Depth_k^{\gamma}$. (F.78)

Then phase impact is:

$\Omega_{impact,k} = \kappa_{\theta II} k$. (F.79)

or real-value impact may be applied directly:

$R_k^{+} = R_k^{-} - I_k$. (F.80)

Only one convention should be used in a given simulation design unless their interaction is explicitly studied.

F.9.4 Net proceeds

Let:

$p_{exec,k} = R_k^{-} - Slippage_k - ImpactPerUnit_k$. (F.81)

Net proceeds are:

$P_{net,k} = \ell_{exec,k}(p_{exec,k} - FeePerUnit_k)$. (F.82)

Debt becomes:

$D_k^{+} = \max(0, D_k^{-} - P_{net,k})$. (F.83)

F.9.5 Default branch

Default occurs when:

$S_{B,k} > 0$ (F.84)

and:

$NoPermittedCure_k = 1$. (F.85)

Then:

$\nu_k^{+} = Defaulted$. (F.86)

The identity transforms:

$K_{margin,k} \rightarrow K_{recovery,k}$. (F.87)

Generate recovery value:

$R_{rec,k} = RecoveryRate_k \times OutstandingDebt_k$. (F.88)

F.10 Action–Ledger Data Generation

F.10.1 Action vector

For each margin event, define:

ψ_A,k :=
[
C_call,k
C_instr,k
ℓ_instr,k
ℓ_exec,k
ΔD_expected,k
Loss_economic,k
GateStatus_k
]. (F.89)

F.10.2 Ledger vector

Define:

ψ_L,k :=
[
C_admit,k
C_recorded,k
ℓ_settled,k
ΔD_recorded,k
Loss_recognized,k
LegalStatus_k
LedgerStatus_k
]. (F.90)

F.10.3 Expected action-to-ledger transport

Let the true transporter be:

U_AL* = U_AL(Protocol,Branch,Delay,Eligibility). (F.91)

The expected ledger image is:

ψ^_L,k+d = U_AL*ψ_A,k. (F.92)

F.10.4 Ledger delay

For each component j, generate delay:

d_j ~ DelayDistribution_j. (F.93)

Possible distributions include:

- deterministic;
- geometric;
- Poisson;
- lognormal;

- branch-dependent mixture.

The observed ledger component is:

$$\psi_{L,j,k+d_j} = \hat{\psi}_{L,j,k} + \varepsilon_{L,j,k}. \qquad (F.94)$$

F.10.5 Spinor residual

Generate:

$$\delta_{spin,k} = \psi_{L,k} - U_{AL}\psi_{A,k}. \qquad (F.95)$$

Residual may contain:

- ordinary observation noise;
- operational failure;
- legal rejection;
- missing transaction;
- hidden fee;
- deliberate suppression.

F.10.6 Hidden residual

Let the true residual be:

$$\mathscr{R}_{true,k}. \qquad (F.96)$$

Let the recorded residual be:

$$\mathscr{R}_{recorded,k} = M_{record,k}\mathscr{R}_{true,k}. \qquad (F.97)$$

where $M_{record,k}$ is a residual-disclosure mask.

Hidden residual is:

$$\mathscr{R}_{hidden,k} = \mathscr{R}_{true,k} - \mathscr{R}_{recorded,k}. \qquad (F.98)$$

This permits experiments on false closure and residual denial.

The generalized Dirac source treats residual denial as especially dangerous because it allows apparent short-term closure while creating longer-term curvature and learning failure.

F.11 Charge Data Generation

F.11.1 Structural charge

For a leveraged long account:

$q_{A,k} = +n_k q_0$. (F.99)

$q_{F,k} = -\mathrm{ChargeMap}_F(D_k)$. (F.100)

$q_{C,k}^{\mathrm{struct}} = -q_{C,0}$. (F.101)

The signs are declared before simulation.

F.11.2 Operational charge

The collateral charge activates when the gate is admitted:

$q_{C,k}^{\mathrm{op}} = g_k^{\mathrm{call}} q_{C,k}^{\mathrm{struct}}$. (F.102)

F.11.3 Screening operator

Generate:

$\vec{q}_{\mathrm{eff},k} = S_k \vec{q}_{\mathrm{struct},k}$. (F.103)

A minimum diagonal screening matrix is:

$S_k = \mathrm{diag}(s_{A,k}, s_{F,k}, s_{C,k})$. (F.104)

Possible rules are:

$s_{A,k} = \beta_k \lambda_k \chi_{A,k}$. (F.105)

$s_{F,k} = f_F(\mathrm{Collateral}_k, \mathrm{Credit}_k, \mathrm{Liquidity}_k)$. (F.106)

$s_{C,k} = f_C(\mathrm{CallStatus}_k, \mathrm{Deadline}_k, \mathrm{Eligibility}_k)$. (F.107)

F.11.4 Charge-conversion vertices

At deleveraging:

$q_{A,k^+} = q_{A,k^-} - \ell_{\mathrm{exec},k} q_0$. (F.108)

At debt repayment:

$q_{F,k^+} = \mathrm{ChargeMap}_F(D_{k^+})$. (F.109)

At default:

$\vec{q}_{\mathrm{margin},k} \to \vec{q}_{\mathrm{recovery},k} + \vec{q}_{\mathrm{extinguished},k} + \vec{r}_{q,k}$. (F.110)

F.11.5 Charge error injection

Inject known errors:

- sign reversal;

- duplicate obligation;
- omitted claim;
- incorrect boundary;
- unauthorized extinguishment;
- unmatched novation.

This tests whether charge-residual diagnostics identify the correct failure type.

F.12 Frame-System Generation

F.12.1 True market frame

Generate:

x_M,k = Φ_M(K_k,R_k,Q_k,n_k,t_k). (F.111)

F.12.2 True collateral frame

Generate:

x_C,k = U_CM*x_M,k + ε_C,k. (F.112)

F.12.3 True treasury frame

Generate:

x_T,k = U_TC*x_C,k* + *U_TD*D_k + ε_T,k. (F.113)

F.12.4 True settlement frame

Generate:

x_S,k = U_SM*x_M,k−d_SM + ε_S,k. (F.114)

F.12.5 True accounting frame

Generate:

x_A,k = U_AS*x_S,k−d_AS* + *U_AT*x_T,k−d_AT + ε_A,k. (F.115)

F.12.6 True legal frame

Generate:

x_L,k = LegalState(K_k,Contract_k,Gate_k,Settlement_k) + ε_L,k. (F.116)

F.12.7 True regulatory frame

Generate:

x_G,k = U_GA*x_A,k−d_GA* + *U_GR*x_R,k−d_GR + ε_G,k. (F.117)

F.12.8 Connection misspecification

The estimator may be given:

U_AB^model ≠ U_AB*. (F.118)

Types of misspecification include:

- omitted haircut;
- wrong timing;
- incorrect netting;
- wrong FX rate;
- missing fee;
- wrong legal eligibility;
- wrong recognition date.

This tests whether the model places error correctly into:

- source state;
- target state;
- connection;
- residual.

F.13 Gauge-Covariance Experiment

F.13.1 Original representation

Generate true states:

x_A,x_B. (F.119)

with:

x_B = U_ABx_A. (F.120)

F.13.2 Local reparameterization

Apply invertible transformations:

$x_{A'} = G_A x_A.$ (F.121)

$x_{B'} = G_B x_B.$ (F.122)

Examples include:

- currency conversion;
- unit rescaling;
- gross/net presentation;
- normalization.

F.13.3 Correct transformed connection

Generate:

$U_{AB'} = G_B U_{AB} G_A^{-1}.$ (F.123)

Then:

$x_{B'} = U_{AB'} x_{A'}.$ (F.124)

F.13.4 Covariance test

Estimate the substantive event classification in both representations.

Covariance succeeds when:

$\text{Classification}(x_A, x_B, U_{AB}) = \text{Classification}(x_{A'}, x_{B'}, U_{AB'}).$ (F.125)

It fails when presentation changes alter:

- call status;
- charge sign;
- identity match;
- closure status;
- residual classification.

F.14 Loop-Curvature Generation

F.14.1 Flat loop

For frames A → B → C → A, choose:

U_ACU_CBU_BA = I. (F.126)

after authorized adjustments.

Then:

$\mathscr{R}$_loop = 0 (F.127)

apart from observation noise.

F.14.2 Lawful cost loop

Let a known cost operator be:

U_cost. (F.128)

Set:

H_loop^P = U_cost. (F.129)

The observed return is:

x_A^return = U_costx_A^start + ε. (F.130)

The economic value changes, but governed residual remains near zero.

This tests whether the estimator wrongly interprets lawful cost as curvature.

F.14.3 Curved loop

Inject an unexplained path effect:

H_loop^obs = U_hiddenH_loop^P. (F.131)

Then:

$\mathscr{R}$_loop = (U_hidden − I)H_loop^Px_A. (F.132)

Possible hidden effects include:

- unrecorded fee;
- stale quantity;
- legal-title mismatch;
- missing collateral adjustment;
- incorrect netting;
- delayed accounting entry.

F.14.4 Order-effect experiment

Generate noncommuting operators:

$$U_XU_Y \neq U_YU_X. \quad (F.133)$$

Compare:

$$Path_1 = U_XU_Yx. \quad (F.134)$$

$$Path_2 = U_YU_Xx. \quad (F.135)$$

The order residual is:

$$r_{order} = Path_1 - Path_2. \quad (F.136)$$

The estimator should identify order dependence when present and reject it when operations commute.

F.15 Identity-Mass Generation

F.15.1 Scalar mass

Generate identity inertia:

$$m_I > 0. \quad (F.137)$$

Let transformation velocity decline with mass:

$$v_{change,k} = Force_k/(m_I + \varepsilon). \quad (F.138)$$

F.15.2 Matrix mass

Generate:

$$M^* = [[m_A,m_C],[m_C,m_L]]. \quad (F.139)$$

The spinor dynamics are:

$$i\Gamma^0\Delta\Psi_k/\Delta\tau + ic_P\Gamma^1\mathfrak{D}_G\Psi_k - M^*\Psi_k = \mathscr{R}_k. \quad (F.140)$$

F.15.3 Mass regimes

Create three regimes.

Low mass

M_{low} permits rapid change but may allow identity drift.

Moderate mass

M_mid supports adaptive closure.

High mass

M_high produces slow response or paralysis.

The simulation should not assume that greater mass is always safer.

F.15.4 Mass proxy observations

Generate observable proxies:

ImpactCost_k. (F.141)

SettlementDelay_k. (F.142)

LegalFriction_k. (F.143)

AccountingDelay_k. (F.144)

ApprovalCount_k. (F.145)

Let:

$m_{proxy,k} = w_1 ImpactCost_k + w_2 SettlementDelay_k + w_3 LegalFriction_k + w_4 AccountingDelay_k + w_5 ApprovalCount_k.$ (F.146)

Test whether m_proxy recovers the latent M* modes.

F.16 Closure-Capacity Generation

F.16.1 Action arrival rate

Let open actions arrive at rate:

$\lambda_{A,k}.$ (F.147)

F.16.2 Ledger service rate

Let ledger capacity be:

$\mu_{L,k}.$ (F.148)

A simple queue evolves as:

$Backlog_{k+1} = [Backlog_k + Arrivals_k - Completions_k]_+.$ (F.149)

F.16.3 Coherent capacity

Define:

$$c_{P,k} := \mu_{L,k} \times \mathrm{QualityFactor}_k. \qquad (F.150)$$

where:

$$0 \le \mathrm{QualityFactor}_k \le 1. \qquad (F.151)$$

Processing more records with lower verification quality should not automatically count as greater coherent capacity.

F.16.4 Cone ratio

Define:

$$\kappa_{\mathrm{cone},k} = v_{A,k}/c_{P,k}. \qquad (F.152)$$

The source macro-Dirac framework treats the corresponding ratio as a preliminary operational diagnostic rather than a finalized universal metric.

F.16.5 Capacity threshold

Generate residual growth as:

$$g_{\mathscr{R},k} = \alpha_0 + \alpha_1[\kappa_{\mathrm{cone},k} - 1]_+ + \varepsilon_{\mathscr{R},k}. \qquad (F.153)$$

Then:

$$\kappa_{\mathrm{cone}} \le 1 \quad \rightarrow \text{residual remains controlled.} \qquad (F.154)$$

$$\kappa_{\mathrm{cone}} > 1 \quad \rightarrow \text{residual growth accelerates.} \qquad (F.155)$$

The estimator should recover the threshold without being told its exact location.

F.17 Recursive Ledger Generation

F.17.1 Ledger state

Define:

$$L_k := (\mathrm{TraceHistory}_k, \mathrm{ResidualHistory}_k, \mathrm{ClosureHistory}_k, \mathrm{DefaultHistory}_k). \qquad (F.156)$$

F.17.2 Parameter recursion

Generate:

$$h_{k+1} = h_k - \alpha_h \mathrm{Failure}_k - \beta_h \|\mathscr{R}_k\| + \varepsilon_{h,k}. \qquad (F.157)$$

Generate funding spread:

$$s_{F,k+1} = s_{F,k} + \alpha_F \mathrm{Failure}_k + \beta_F \|\mathscr{R}_k\| + \varepsilon_{F,k}. \qquad (F.158)$$

Generate mass:

$M_{k+1} = M_k + \alpha_M \|\mathcal{R}_k\| - \beta_M Q_{\text{close},k}.$ (F.159)

Generate capacity:

$c_{P,k+1} = c_{P,k} - \alpha_B \text{Backlog}_k + \beta_R \text{Resources}_k.$ (F.160)

F.17.3 No-recursion null

For the null experiment:

$\Theta_{k+1} = \Theta_k + \varepsilon_{\Theta,k}.$ (F.161)

independent of prior trace and residual.

If the recursive estimator detects strong ledger backreaction under this null, it is overfitting historical features.

F.17.4 Path-dependence pair

Generate two subjects with identical current states:

$X_{A,k} = X_{B,k}.$ (F.162)

But different ledgers:

$L_{A,k} \neq L_{B,k}.$ (F.163)

Under true recursion:

$\Pr(\text{FutureEvent}_A \mid X_k, L_{A,k}) \neq \Pr(\text{FutureEvent}_B \mid X_k, L_{B,k}).$ (F.164)

Under the null:

$\Pr(\text{FutureEvent}_A \mid X_k) = \Pr(\text{FutureEvent}_B \mid X_k).$ (F.165)

This is a direct test of ledger memory.

F.18 Complete Synthetic State Transition

At each event step k, the full state is:

$\mathcal{S}_k := (K_k, A_k, R_k, Q_k, \theta_k, h_k, n_k, C_k, D_k, B_k, \vec{q}_k, \Psi_k, X_{F,k}, L_k, \mathcal{R}_k, \Theta_k).$ (F.166)

The transition is:

$\mathcal{S}_k$
→ MarketUpdate
→ CollateralUpdate
→ BufferCalculation
→ GateAdmission
→ BranchAction
→ FrameTransport

→ LedgerReturn
→ ResidualUpdate
→ ParameterRecursion
→ $\mathcal{S}_k{}_{+1}$. (F.167)

This is the synthetic implementation of the article's master recurrence.

F.19 Simulation Algorithm

F.19.1 Initialization

For each account i:

1. generate identity K_i;
2. assign structural charges $\vec{q}_i$;
3. generate A_i,0 and θ_i,0;
4. compute R_i,0 and Q_i,0;
5. assign n_i,0, C_i,0, D_i,0, and h_i,0;
6. calculate B_i,0;
7. initialize frame states;
8. initialize ledger L_i,0;
9. initialize mass M_i,0 and capacity c_P,i,0.

F.19.2 Continuous update

For each calendar interval:

1. update A and θ;
2. compute R and Q;
3. update stress and h;
4. accrue debt;
5. update buffer;
6. update effective charge;
7. propagate action and ledger states;
8. generate frame-local observations.

F.19.3 Gate update

When:

B < B_target, (F.168)

evaluate:

- authority;
- mark validity;
- rule applicability;
- waiver;
- call status.

If admitted, write the call trace and activate q_C^op.

F.19.4 Branch update

Choose:

- collateral posting;
- deleveraging;
- forced liquidation;
- default.

Update:

- n;
- C;
- D;
- market impact;
- action vector;
- expected ledger return.

F.19.5 Ledger update

After branch-specific delays:

- settle quantity;
- admit collateral;
- update debt;
- recognize loss;

- update legal state;
- calculate spin residual;
- issue closure status.

F.19.6 Recursive update

Use trace and residual to update:

- h;
- funding spread;
- mass;
- capacity;
- limits;
- transport maps;
- effective charge.

F.20 Simulation Scenarios

F.20.1 Scenario A — Stable account

Parameters:

- low σ_θ;
- high h;
- low leverage;
- no gate;
- low residual;
- stable frame connections.

Expected result:

All advanced diagnostics remain quiet.

F.20.2 Scenario B — Pure market-phase shock

Parameters:

- θ rises;
- h fixed;

- D fixed;
- no legal shock.

Expected result:

Buffer erosion occurs primarily through:

$-nhQ\Delta\theta$. (F.169)

F.20.3 Scenario C — Pure haircut shock

Parameters:

- θ fixed;
- h falls.

Expected result:

Buffer erosion occurs through:

$nR\Delta h$. (F.170)

The model should not attribute this event to market phase.

F.20.4 Scenario D — Funding creep

Parameters:

- R fixed;
- h fixed;
- debt accrues.

Expected result:

B declines through:

$-\Delta D$. (F.171)

Q should provide little incremental information.

F.20.5 Scenario E — Valid call and rapid collateral cure

Parameters:

- breach;
- call admitted;
- collateral available;

- low ledger delay;
- low residual.

Expected result:

- spinor split rises briefly;
- closure time is short;
- q_C^op activates then returns to dormant;
- no persistent curvature.

F.20.6 Scenario F — Execution without settlement

Parameters:

- sale executed;
- settlement delayed;
- debt not yet reduced.

Expected result:

- action component updates;
- ledger component lags;
- Δ_spin increases;
- outer margin closure remains open.

F.20.7 Scenario G — Fire-sale feedback

Parameters:

- low market depth;
- high κ_I;
- many correlated accounts;
- falling h.

Expected result:

$B \downarrow \rightarrow \ell \uparrow \rightarrow R \downarrow \rightarrow B \downarrow$ further. (F.172)

The estimator should detect nonlinear branch feedback.

F.20.8 Scenario H — Legal ineligibility shock

Parameters:

$e_{LC} : 1 \to 0.$ (F.173)

Market value remains positive.

Expected result:

- collateral value collapses;
- legal-to-collateral edge changes;
- market frame remains stable;
- margin call arises from frame-specific gate.

F.20.9 Scenario I — False closure

Parameters:

- ledger reports Closed;
- unsettled quantity remains;
- residual mask suppresses the exception.

Expected result:

$C_{reported} = 1.$ (F.174)

$SpinClosed_P = 0.$ (F.175)

The model should identify hidden residual and loop failure.

F.20.10 Scenario J — Path-dependent recovery

Two accounts end with the same n, D, C, h, and B.

One has:

- prior failed settlement;
- residual legal dispute;
- tighter future haircut.

The other has clean history.

Expected result:

Same X
but
Different Future Dynamics. (F.176)

F.21 Nested Estimation Models

F.21.1 Model M0 — Scalar baseline

Use:

$X_k^{M0} = (R_k,n_k,h_k,C_k,D_k,B_k).$ (F.177)

Estimate:

$\Pr(Call_{k+1} = 1 \mid X_k^{M0}).$ (F.178)

F.21.2 Model M1 — Real-pair model

Add Q:

$X_k^{M1} = (X_k^{M0},Q_k).$ (F.179)

Permit nonlinear interactions.

This prevents an unfairly weak benchmark.

The related source programme specifically requires complex-state studies to compete with flexible real-vector alternatives.

F.21.3 Model M2 — Complex phase model

Use:

$Z_k = R_k + iQ_k.$ (F.180)

Add:

$\theta_k,\Delta\theta_k,\theta_{C,k},\Delta\theta_{MC,k}.$ (F.181)

Test whether phase provides gain beyond the real pair.

F.21.4 Model M3 — Charge model

Add:

$\vec{q}_{struct,k},\vec{q}_{eff,k},\vec{r}_{q,k}.$ (F.182)

Test whether relational typing improves prediction or reconciliation.

F.21.5 Model M4 — Action–ledger model

Add:

$\Psi_k,\Delta_{spin,k},T_{close,k}.$ (F.183)

Test unresolved closure and branch outcomes.

F.21.6 Model M5 — Frame-transport model

Add:

r_AB,k,ABFix_AB,k,E_recon,k. (F.184)

Test cross-frame failure.

F.21.7 Model M6 — Loop model

Add:

κ_loop,k,r_path,k,r_order,k. (F.185)

Test whether loop diagnostics outperform ordinary exception counts.

F.21.8 Model M7 — Full hybrid Gauge–Dirac model

Estimate:

(i$\not{D}$_fin − M_k)Ψ_k = $\mathscr{R}$_k (F.186)

with gate jumps and recursive parameters.

Compare complexity-adjusted performance against M0–M6.

F.22 Estimation Methods

F.22.1 Deterministic reconstruction

When all relevant variables are observed, estimate:

- Q;
- θ;
- B;
- charge balance;
- transport residual;
- spinor split

directly from declared equations.

This should precede latent-variable modelling.

F.22.2 State-space estimation

Use:

$s_{k+1} = F_k s_k + G_k u_k + w_k.$ (F.187)

$y_k = H_k s_k + v_k.$ (F.188)

Possible latent states include:

- θ;
- collateral stress;
- mass;
- closure capacity;
- hidden residual.

F.22.3 Switching state-space model

Let regime be:

$z_k \in \{Normal, Called, Liquidating, Defaulted\}.$ (F.189)

Then:

$s_{k+1} = F_{z_k} s_k + w_k.$ (F.190)

Gate observations inform transitions between regimes.

F.22.4 Survival model

For closure time:

$h_{close}(t \mid X) = h_0(t)\exp(\beta^T X).$ (F.191)

Candidate predictors include:

- Δ_{spin};
- κ_{loop};
- m_{eff};
- call size;
- market depth;
- branch;
- residual.

F.22.5 Queueing model

Estimate:

- action arrival rate λ_A;
- ledger service rate μ_L;
- backlog;
- overload threshold.

This is a necessary benchmark for c_P.

If queueing variables explain residual growth completely, the stronger semantic-cone terminology may be unnecessary.

F.22.6 Graph model

Use frame nodes and transport edges.

Estimate edge residual:

r_e,k = x_target,k − U_ex_source,k. (F.192)

A constrained graph model should preserve known identity and charge rules.

F.22.7 Constrained maximum likelihood

Estimate parameters:

Θ = (μ_θ,σ_θ,κ_I,M,c_P,U_AB,DelayParameters). (F.193)

subject to:

- h ∈ [0,1];
- charge signs fixed by contract;
- valid connection structure;
- nonnegative delays;
- declared mass stability conditions.

F.23 Parameter-Recovery Tests

F.23.1 Bias

For parameter θ_j:

Bias(θ^_j) := E[θ^_j − θ_j*]. (F.194)

F.23.2 Root mean square error

$$\mathrm{RMSE}(\hat{\theta}_j) := \sqrt{E[(\hat{\theta}_j - \theta_j^*)^2]}. \qquad (F.195)$$

F.23.3 Coverage

For confidence interval CI_j:

$$\mathrm{Coverage}_j := \Pr(\theta_j^* \in CI_j). \qquad (F.196)$$

F.23.4 Sign recovery

For parameters whose sign is structurally important:

$$\mathrm{SignRecovery}_j := \Pr[\mathrm{sign}(\hat{\theta}_j) = \mathrm{sign}(\theta_j^*)]. \qquad (F.197)$$

Examples include:

- phase-to-buffer coefficient;
- haircut sensitivity;
- action-to-ledger coupling;
- residual backreaction.

F.23.5 Regime recovery

For true regime z_k^* and estimated regime $\hat{z}_k$:

$$\mathrm{RegimeAccuracy} := \Pr(\hat{z}_k = z_k^*). \qquad (F.198)$$

Adjusted measures should account for label switching.

F.23.6 Gate recovery

Define:

$$\mathrm{GatePrecision} := \mathrm{TrueAdmittedGates}/\mathrm{EstimatedAdmittedGates}. \qquad (F.199)$$

$$\mathrm{GateRecall} := \mathrm{TrueAdmittedGates}/\mathrm{AllTrueAdmittedGates}. \qquad (F.200)$$

The estimator should distinguish threshold crossings from authorized calls.

F.24 Identification Experiments

F.24.1 Q identification

Vary:

- θ volatility;
- A volatility;
- observation noise;
- sample size.

Test whether Q is recoverable when:

R and A are independently observed. (F.201)

Q is not independently identifiable when A is unobserved and freely chosen.

That limitation must remain explicit.

F.24.2 Charge identification

Charge is identifiable only when the data preserve:

- carrier;
- sign;
- field;
- transfer;
- conversion;
- boundary.

Remove one of these elements in controlled experiments.

The model should then downgrade charge to exposure or obligation.

F.24.3 Spinor identification

Vary ledger delay:

$d_L = 0,1,2,\ldots$ (F.202)

When:

$d_L = 0$ (F.203)

and action and ledger are deterministically identical, the spinor should collapse toward a scalar model.

When:

$d_L > 0$ (F.204)

with independent residual, the doublet should become identifiable.

F.24.4 Connection identification

A transport operator U_AB is identifiable only when:

- source and target states vary;
- identity matching is known;
- timing is observed;
- enough independent dimensions exist.

If all observations lie on one line, many connection matrices may fit equally well.

F.24.5 Mass identification

Mass may be confounded with:

- damping;
- latency;
- low liquidity;
- processing capacity;
- gate rigidity.

Create simulations in which only one factor varies at a time.

The estimator should not label every slow process as mass.

F.24.6 c_P identification

Generate several action-throughput levels.

If all observations satisfy:

$$v_A \ll c_P, \qquad (F.205)$$

the capacity boundary is not identifiable.

The simulation must include observations near and above the threshold.

F.25 Ablation Experiments

F.25.1 Remove Q

Compare:

Model with R and Q (F.206)

against:

Model with R only. (F.207)

F.25.2 Remove phase

Compare:

(R,Q,θ) (F.208)

against:

Flexible nonlinear f(R,Q). (F.209)

F.25.3 Remove charge typing

Replace:

$\vec{q}$_struct (F.210)

with:

unsigned notional. (F.211)

F.25.4 Remove ledger independence

Replace:

$\Psi = [\psi_A,\psi_L]^T$ (F.212)

with:

one completion status. (F.213)

F.25.5 Remove frame transport

Replace:

x_B − U_ABx_A (F.214)

with:

x_B − x_A. (F.215)

This tests whether lawful connection modelling matters.

F.25.6 Remove loop structure

Replace:

κ_loop (F.216)

with:

sum of pairwise residuals. (F.217)

F.25.7 Remove mass matrix

Replace M with:

scalar damping parameter. (F.218)

F.25.8 Remove recursion

Set:

$\Theta_k_{+1} = \Theta_k$. (F.219)

Test whether ledger history still adds information.

F.26 Negative Controls

F.26.1 Random Q

Generate:

Q_random,k ⊥ R_k,Gate_k,Outcome_k. (F.220)

The model should not find stable incremental value.

F.26.2 Shuffled charge signs

Randomly permute:

sign(q_i). (F.221)

Charge-based performance should deteriorate.

F.26.3 False frame matching

Pair source state from account i with target state from account j:

$i \neq j$. (F.222)

A-B Fixedness should fall sharply.

F.26.4 Placebo loop

Construct a sequence of frames with no institutional connection.

Loop curvature should not be interpreted substantively.

F.26.5 Placebo gate

Choose a numerical threshold lacking authority.

Compare it with the valid margin gate.

The gate hypothesis expects:

Pr(PersistentConsequence | ValidGate) > Pr(PersistentConsequence | PlaceboCrossing). (F.223)

F.26.6 Future leakage control

Deliberately include future settlement information in one contaminated model.

Its apparent performance should rise.

The research pipeline must detect and reject this leakage.

F.27 Sample-Size Experiments

F.27.1 Number of accounts

Vary:

$N_{account} \in \{50,100,500,1,000,10,000\}$. (F.224)

F.27.2 Number of events

Vary:

$N_{event} \in \{100,500,1,000,5,000\}$. (F.225)

Rare default branches may require much larger samples.

F.27.3 Observation frequency

Vary:

$\Delta t \in \{intraday,daily,weekly\}$. (F.226)

High frequency improves timing resolution but may increase noise and asynchronous-frame mismatch.

F.27.4 Closure-depth observations

Vary the number of observed ledger stages.

The spinor model may be poorly identified when only:

- call time;
- final resolution

are observed.

Richer identification requires intermediate traces.

F.28 Missing-Data Experiments

F.28.1 Random missingness

Generate missingness independent of state:

$$\Pr(\text{Missing}_{j,k} = 1) = p_j. \qquad \text{(F.227)}$$

F.28.2 State-dependent missingness

Generate:

$$\Pr(\text{Missing}_{j,k} = 1) = \text{Logistic}(\alpha_j + \beta_j \text{Stress}_k). \qquad \text{(F.228)}$$

This is more realistic because stressed systems may produce worse records.

F.28.3 Residual-dependent concealment

Generate:

$$\Pr(\text{ResidualRecorded}_k = 0) = \text{Logistic}(\alpha_H + \beta_H \|\mathscr{R}_{\text{true},k}\|). \qquad \text{(F.229)}$$

This creates residual denial correlated with severity.

F.28.4 Evaluation

Compare:

- complete-case analysis;
- multiple imputation;
- state-space missing-data treatment;
- explicit missingness-as-residual model.

The original missingness indicator should remain available even after imputation.

F.29 Noise and Misspecification Experiments

F.29.1 Market noise

Add:

$$R_{obs,k} = R_{true,k} + \varepsilon_{R,k}. \qquad (F.230)$$

F.29.2 Haircut noise

Add:

$$h_{obs,k} = h_{true,k} + \varepsilon_{h,k}. \qquad (F.231)$$

with clipping to $[0,1]$.

F.29.3 Timestamp noise

Add:

$$t_{obs} = t_{true} + \varepsilon_{t}. \qquad (F.232)$$

This tests sensitivity of ledger-delay and loop reconstruction.

F.29.4 Identity error

With probability p_{ID}:

$$K_{obs,k} \neq K_{true,k}. \qquad (F.233)$$

The model should identify identity mismatch before interpreting value residual.

F.29.5 Structural misspecification

Estimate a linear-impact model on data generated by square-root impact.

Estimate fixed h on data generated by state-dependent h.

Estimate scalar mass on data generated by matrix mass.

The advanced model should report model residual rather than forcing exact closure.

F.30 Validation Design

F.30.1 Chronological split

Use:

Training < Validation < Test. (F.234)

Even in simulation, chronological testing is useful when:

- parameters drift;
- regimes change;
- ledger recursion creates dependence.

The related empirical programme recommends chronological splits and walk-forward validation rather than random shuffling when temporal dependence and regime evolution matter.

F.30.2 Walk-forward testing

For endpoint m:

Train on [1,m]. (F.235)

Test on [m + 1,m + h]. (F.236)

Then roll forward.

F.30.3 Monte Carlo replication

For each parameter setting:

Run r = 1,...,R independent replications. (F.237)

Report:

- mean performance;
- dispersion;
- failure rate;
- parameter recovery;
- model-selection frequency.

F.30.4 Stress holdout

Reserve one extreme stress regime not seen during training.

This tests whether:

- structural equations transport;

- model behaviour collapses;
- residual honestly signals extrapolation failure.

F.31 Evaluation Metrics

F.31.1 Call prediction

Use:

- log loss;
- Brier score;
- area under precision–recall curve;
- calibration slope;
- calibration intercept.

F.31.2 Buffer prediction

Use:

$$MAE_B := \mathrm{Mean}|B_{obs} - B_{pred}|. \qquad (F.238)$$

$$RMSE_B := \sqrt{\mathrm{Mean}(B_{obs} - B_{pred})^2}. \qquad (F.239)$$

F.31.3 Closure prediction

Use:

- concordance index;
- integrated Brier score;
- closure-time MAE;
- unresolved-event recall.

F.31.4 Spinor diagnostics

Define:

$$\mathrm{SpinorGain} := \mathrm{Performance}(\Delta_{spin},\mathrm{Controls}) - \mathrm{Performance}(\mathrm{Controls}). \qquad (F.240)$$

F.31.5 Transport diagnostics

Define:

TransportGain
:= Error_naive-difference − Error_connection-adjusted. (F.241)

F.31.6 Loop diagnostics

Define:

LoopGain
:= Performance(κ_loop,Controls) − Performance(PairwiseResiduals,Controls). (F.242)

F.31.7 Residual recall

ResidualRecall
:= FutureFailureTypes anticipated by $\mathscr{R}$_k
÷ Total FutureFailureTypes. (F.243)

This metric is also part of the empirical contract proposed in the main Financial Standard Model.

F.31.8 Complexity-adjusted utility

Define:

U_j = Performance_j + GovernanceGain_j − λComplexity_j. (F.244)

The advanced model survives only when:

U_advanced > U_benchmark. (F.245)

F.32 Recovery Criteria by Concept

F.32.1 Complex-state recovery

The complex layer passes simulation when:

1. A, R, Q, and θ are recovered within tolerance;
2. the complex constraint improves estimation when true;
3. the model reduces to a real pair when the constraint is false;
4. the estimator does not invent phase advantage under the null.

F.32.2 Charge recovery

The charge layer passes when:

1. carrier and sign are recovered;
2. ordinary transport preserves charge;
3. declared vertices explain conversion;
4. injected charge errors produce r_q;
5. unsigned notional performs worse when orientation matters.

F.32.3 Spin recovery

The spin layer passes when:

1. action and ledger components are independently recovered;
2. Δ_spin rises after outward action;
3. Δ_spin declines after verified ledger return;
4. false closure remains distinguishable;
5. the scalar model wins when ledger independence is removed.

F.32.4 Gauge recovery

The gauge layer passes when:

1. U_AB is recovered;
2. lawful local reparameterization leaves substantive diagnostics unchanged;
3. connection-adjusted residual outperforms raw difference;
4. invalid identity matching lowers A-B Fixedness;
5. loop residual detects injected path inconsistency.

F.32.5 Mass recovery

The mass layer passes when:

1. M parameters are identifiable;
2. estimated mass predicts transformation resistance;
3. mass remains distinct from latency and liquidity;
4. eigenmodes correspond to generated closure modes;
5. a generic friction index wins when matrix structure is absent.

F.32.6 Closure-capacity recovery

The c_P layer passes when:

1. residual growth changes near the true capacity threshold;
2. c_P is recovered with acceptable uncertainty;
3. ordinary queueing benchmarks are defeated or matched with clearer governance value;
4. no threshold is reported under a smooth null process.

F.32.7 Recursive recovery

The recursive layer passes when:

1. ledger history improves future-state prediction;
2. subjects with identical X but different L have distinguishable outcomes;
3. the estimator rejects recursion under the fixed-parameter null;
4. recursive parameters remain stable out of sample.

F.33 Model-Selection Outcomes

F.33.1 Full recovery

If M_7 is generated and M_7 is selected reliably:

Simulation supports identifiability of the full architecture. (F.246)

This still does not establish real-world validity.

F.33.2 Partial recovery

If simulation reliably recovers:

- Q;
- action–ledger split;
- frame residual;

but not:

- Γ algebra;
- matrix mass;
- gauge covariance,

retain:

Complex Hybrid Closure Model. (F.247)

F.33.3 Scalar dominance

If M_0 repeatedly matches or outperforms all advanced models:

Reduce to Conventional Margin Dynamics. (F.248)

F.33.4 Identification failure

If the true advanced parameters exist but cannot be recovered:

Do not proceed directly to real-data claims. (F.249)

Instead revise:

- data frequency;
- observation design;
- state definitions;
- parameterization;
- model complexity.

F.33.5 False-positive failure

If advanced structure is frequently detected under S_0 or S_1:

The inference procedure is invalid. (F.250)

This is more serious than low power.

A model that always discovers its preferred ontology is unfalsifiable.

F.34 Minimal Simulation Pilot

F.34.1 Narrow objective

The first simulation should not attempt to validate the whole Gauge–Dirac architecture.

It should test three distinctions:

Breach ≠ Gate. (F.251)

Action ≠ Ledger Return. (F.252)

Lawful Frame Difference ≠ Residual. (F.253)

F.34.2 Pilot system

Use:

- 1,000 simulated accounts;
- 500 time steps;
- one risky asset;
- one funding liability;
- one collateral factor;
- four gate branches;
- market, collateral, settlement, and accounting frames.

F.34.3 Pilot models

Compare:

M_0 — scalar margin model. (F.254)

M_1 — R–Q real-pair model. (F.255)

M_4 — action–ledger model. (F.256)

M_5 — frame-transport model. (F.257)

Do not include the Γ algebra in the first confirmatory pilot.

F.34.4 Pilot hypotheses

Pilot H_1

Authorized calls are distinguished from numerical breaches.

Pilot H_2

Action–ledger split predicts unresolved closure beyond call age and shortfall.

Pilot H_3

Connection-adjusted residual distinguishes lawful collateral haircut from true reconciliation error.

Pilot H_4

The models reduce correctly under null-generating processes.

F.34.5 Pilot success criteria

The pilot succeeds when:

- gate precision and recall exceed predeclared thresholds;
- Δ_{spin} adds out-of-sample closure information;
- transport adjustment reduces false reconciliation alarms;
- advanced layers are rejected under their corresponding nulls;
- all simulation parameters and outcomes remain reproducible.

F.35 Advanced Simulation Studies

F.35.1 Study 2 — Complex priority

Test:

Complex Model
versus
Flexible Real Pair. (F.258)

The complex model survives only if its geometric constraint produces reproducible gain.

F.35.2 Study 3 — Charge vertices

Inject:

- issuance;
- transfer;
- margin activation;
- liquidation;
- default.

Test charge conservation and conversion residual.

F.35.3 Study 4 — Closure modes

Generate synchronized and antisymmetric action–ledger modes.

Test whether estimated eigenmodes recover them.

F.35.4 Study 5 — Gauge covariance

Apply random lawful local transformations.

Test whether substantive event and residual classifications remain invariant.

F.35.5 Study 6 — Loop curvature

Generate:

- flat loops;
- fee-bearing lawful loops;
- genuinely inconsistent loops.

Test whether the method distinguishes all three.

F.35.6 Study 7 — Identity mass

Generate state-dependent matrix mass.

Test:

- recovery;
- closure-time effect;
- mode stability;
- distinction from ordinary friction.

F.35.7 Study 8 — Capacity crisis

Increase action arrival rate through c_P.

Test:

κ_cone
→ backlog
→ residual growth
→ A-B Fixedness decline. (F.259)

F.35.8 Study 9 — Network contagion

Simulate many accounts sharing:

- one asset;
- one clearing system;
- one collateral rule.

Test:

Individual Gate
→ Aggregate Liquidation
→ Price Impact
→ New Gates. (F.260)

F.35.9 Study 10 — Recursive world change

Allow completed events to change:

- future h;
- funding spread;
- mass;
- capacity;
- permissions.

Test whether the ledgered model predicts better than a fixed-parameter model.

F.36 Reproducibility Package

F.36.1 Required components

A complete simulation package should contain:

- parameter files;
- random seeds;
- data-generating code;
- simulated raw events;
- frame-local ledgers;
- ground-truth latent states;
- benchmark code;
- evaluation scripts;
- model-version ledger;
- reduction decisions.

F.36.2 Data layers

Export at least five tables.

Subject table

Identity and contractual properties.

State table

Market, collateral, funding, and buffer variables.

Event table

Gate, branch, authority, and timing.

Frame table

Frame-local representations and transport edges.

Ledger table

Action, settlement, recognition, residual, and closure.

F.36.3 Ground-truth table

Preserve a separate table containing:

- true Q and θ;
- true structural charge;
- true mass;
- true c_P;
- true connection matrices;
- true hidden residual;
- true identity conversions.

The estimation process should not access this table until evaluation.

F.36.4 Simulation-version ledger

Define:

L_sim,v := (CodeVersion,Parameters,Seed,Scenario,Outputs,Tests,Failures,Revision). (F.261)

When the simulation design changes, preserve prior versions.

F.37 Prospective Research Contract

The simulation programme commits to:

1. declare the generating level before fitting;

2. include null systems without the advanced structure;
3. use equally capable simpler benchmarks;
4. separate tuning from final evaluation;
5. preserve failed parameter-recovery attempts;
6. report false positives as well as power;
7. prevent future leakage;
8. preserve hidden residual in ground truth;
9. distinguish lawful economic cost from transport failure;
10. distinguish threshold crossing from authorized gate;
11. reduce unsupported terminology;
12. avoid investment-performance claims.

In compact form:

No Structure by Construction Alone.
No Advanced Layer without a Null.
No Recovery Claim without Parameter Recovery.
No Gauge Claim without Representation Tests.
No Spin Claim without Independent Ledger Data.
No Dirac Claim before Simpler Hybrid Benchmarks Fail. (F.262)

F.38 Transition from Simulation to Real Data

F.38.1 Readiness criteria

Proceed to real data only when:

- key parameters are recoverable;
- false-positive rates are controlled;
- model selection chooses the correct generating level;
- connection residual is separable from timing noise;
- charge and identity errors are detectable;
- hidden residual produces measurable false-closure signatures;
- benchmark implementation is stable.

F.38.2 First real-data target

The first real-data target should remain narrower than the full architecture.

A suitable pilot is:

> Does an independently measured action–ledger mismatch predict delayed or failed margin-call closure beyond shortfall, leverage, branch, and elapsed time?

This requires:

- call records;
- collateral instructions;
- settlement records;
- debt updates;
- accounting or risk recognition;
- verified closure status.

It does not initially require a full gauge algebra or Γ structure.

F.38.3 Second real-data target

A second pilot may test:

> Does lawful market-to-collateral transport reduce false reconciliation alarms and improve diagnosis of margin events?

This requires:

- market marks;
- haircut rules;
- eligibility;
- timestamps;
- observed collateral records;
- dispute outcomes.

F.38.4 Later targets

Only after the first pilots should research advance to:

- complex priority;
- charge-vertex accounting;
- loop curvature;
- mass estimation;
- closure capacity;

- recursive operator change;
- full Gauge–Dirac estimation.

F.39 Final Simulation Architecture

The complete synthetic programme may be summarized as:

Latent Financial Identity
→ Complex Market State
→ Collateral Filter
→ Funding and Buffer
→ Candidate Breach
→ Authorized Gate
→ Branch Action
→ Action–Ledger Split
→ Frame Transport
→ Settlement and Recognition
→ Residual
→ Closure Certificate
→ Recursive Parameter Update. (F.263)

The corresponding estimation programme is:

Scalar Baseline
→ Real Pair
→ Complex Phase
→ Charge Typing
→ Action–Ledger Doublet
→ Frame Transport
→ Loop Model
→ Gauge–Dirac System. (F.264)

The governing model-selection rule is:

Retain the Highest Layer that Is Recoverable when True and Rejected when False. (F.265)

F.40 Appendix Conclusion

The Financial Gauge–Dirac architecture becomes scientifically meaningful only when its parts can fail independently.

A valid simulation programme must therefore create worlds in which:

- Q exists and worlds in which it does not;
- charge is structurally stable and worlds in which it is merely sensitivity;
- action and ledger are independent and worlds in which one scalar suffices;
- frame transport is covariant and worlds in which it is ordinary reconciliation;
- loop residual is genuine and worlds in which differences are fully explained by lawful cost;
- mass affects identity-preserving change and worlds in which it is only friction;

- closure capacity has a threshold and worlds in which residual grows smoothly;
- ledger history changes future dynamics and worlds in which the process is stationary.

Only then can the model demonstrate that it knows when **not** to discover itself.

The simulation programme's deepest criterion is therefore:

A Scientific Financial Standard Model Must Recover Its Structures when Present, Reject Them when Absent, and Reduce Itself when Simpler Explanations Suffice. (F.266)

The next appendix will formulate a **minimal real-data pilot protocol**, including event inclusion rules, information-time controls, frame reconciliation tables, target variables, benchmark specifications, preregistered hypotheses, and publication-ready reporting templates.

Appendix G — Minimal Real-Data Pilot Protocol

G.1 Purpose

The preceding appendices constructed:

- a synthetic margin-account world;
- charge vertices;
- action–ledger matrices;
- frame connections;
- loop-closure diagnostics;
- simulation-based falsification tests.

A real-data study should not begin by estimating the complete Financial Gauge–Dirac equation.

The first empirical programme should ask a smaller and more executable question:

> **Does an independently measured action–ledger mismatch predict delayed or failed margin-call closure beyond conventional account variables, event size, branch type, and elapsed time?**

A second, linked question is:

> **Does governed market-to-collateral transport distinguish lawful valuation differences from genuine reconciliation errors better than direct numerical comparison?**

These questions test the minimum distinctive core:

Breach ≠ Gate. (G.1)

Action ≠ Ledger Return. (G.2)

Frame Difference ≠ Residual. (G.3)

The first study should not attempt to prove:

- physical financial spin;

- a complete gauge algebra;
- a universal mass operator;
- a Clifford law for markets;
- superior trading profitability.

The source research programme explicitly recommends a modular first programme containing one event gate, a residual audit, and one transport test rather than attempting to validate the entire architecture at once.

G.2 Pilot Claim Ceiling

G.2.1 Permitted claims

The pilot may responsibly test whether:

1. margin breaches and admitted margin calls are empirically distinguishable;
2. action and ledger states can be measured separately;
3. an action–ledger mismatch predicts unresolved closure;
4. market-to-collateral transport reduces false reconciliation alarms;
5. residual categories anticipate later failure modes;
6. ledger history adds information beyond the current balance sheet.

G.2.2 Claims not permitted from the pilot alone

The pilot cannot establish that:

- financial spin is physically identical to spin-½;
- margin accounts obey a universal Dirac equation;
- financial charge is equivalent to physical gauge charge;
- Q is a universal risk measure;
- the framework guarantees profit;
- the tested institution represents all financial systems.

The claim ceiling is:

Current Pilot Result
= Operational Closure and Transport Test, not Universal Financial Gauge–Dirac Validation. (G.4)

The main source likewise sets the current finance result at the level of a testable architecture rather than a validated universal model.

G.3 Primary Research Questions

G.3.1 Gate question

Does a validly admitted margin call predict persistent financial consequence more reliably than a raw numerical buffer breach?

Formally:

Pr(PersistentConsequence | Breach,ValidGate)

Pr(PersistentConsequence | Breach,NoValidGate). (G.5)

G.3.2 Spinor question

Does the transport-adjusted action–ledger split predict unresolved closure beyond conventional controls?

Formally:

I(Y_unresolved,H;Δ_spin | B,C_call,Leverage,Branch,ElapsedTime,Controls) > 0. (G.6)

G.3.3 Transport question

Does governed market-to-collateral transport reduce apparent mismatch relative to raw comparison?

Let:

r_MC^raw := x_C − x_M. (G.7)

Let:

r_MC^G := x_C − T_MCx_M. (G.8)

The transport hypothesis is:

ErrorClassification(r_MC^G)

ErrorClassification(r_MC^raw). (G.9)

Here “greater” means better discrimination between:

- lawful haircut difference;
- timing difference;
- eligibility adjustment;
- genuine reconciliation failure.

G.3.4 Residual question

Do predeclared residual categories anticipate later failure modes?

Residual recall is:

ResidualRecall
:= FutureFailureModes anticipated in $\mathscr{R}_t$
÷ Total FutureFailureModes. (G.10)

G.3.5 Ledger-memory question

For subjects with similar current financial states, does prior ledger history predict different future outcomes?

$I(Y_{t+H};L_t \mid X_t) > 0.$ (G.11)

G.4 Study Profiles

The pilot contains three nested profiles.

G.4.1 Profile P0 — Gate-only study

Required data:

- margin buffer;
- threshold;
- call status;
- authority;
- call time;
- outcome.

Permitted claim:

Valid margin gates can be distinguished from raw threshold crossings.

G.4.2 Profile P1 — Action–ledger closure study

Required data:

- P0 data;
- outward action records;
- independent ledger-return records;
- closure outcome;
- residual.

Permitted claim:

Two-stage closure provides measurable diagnostic value.

G.4.3 Profile P2 — Transported closure study

Required data:

- P1 data;
- market frame;
- collateral frame;
- declared transport rule;
- timestamps;
- transport residual.

Permitted claim:

Governed cross-frame transport improves reconciliation and closure diagnosis.

G.4.4 Excluded first-pilot profiles

The first confirmatory pilot should not require:

- full local gauge-group estimation;
- Γ-matrix estimation;
- matrix mass estimation;
- network contagion;
- multiple currencies;
- options;
- cross-margining;
- collateral rehypothecation chains.

These should remain later extensions.

G.5 Calibration Boundary

G.5.1 Minimum subject

The unit financial subject is:

S_i := one margin account carrying one risky position, one funding balance, and one governing

margin agreement. (G.12)

The preferred first sample should limit each observation to:

- one account;
- one base currency;
- one margining authority;
- one collateral rule;
- one primary risky asset or one predeclared position aggregate.

G.5.2 System boundary

The study boundary should include:

- account records;
- margin engine;
- collateral ledger;
- execution and settlement records;
- debt or funding ledger;
- closure-status record.

The boundary may exclude:

- unrelated portfolio positions;
- parent-company funding;
- downstream tax effects;
- external regulatory consequences;

unless those elements directly determine the margin event.

G.5.3 Boundary identifier

Every event record should contain:

BoundaryID := (Institution,LegalEntity,Account,Agreement,Currency,CollateralSet). (G.13)

No neutrality, charge balance, or closure claim should be made without this boundary.

G.6 Unit of Observation

G.6.1 Account-time record

For account i and timestamp t:

O_i,t^account := (X_i,t,L_i,t,Gate_i,t,Residual_i,t). (G.14)

This table supports:

- buffer modelling;
- call hazard;
- pre-event state reconstruction;
- ledger-memory tests.

G.6.2 Margin-event record

For event k:

O_k^event := (AccountID,CallID,t_candidate,t_gate,t_due,Branch,t_close,Outcome).
(G.15)

This is the primary unit for closure analysis.

G.6.3 Action record

For action a:

O_a^action := (CallID,ActionType,InstructionTime,Amount,Quantity,Authority,Status).
(G.16)

G.6.4 Ledger-return record

For ledger return l:

O_l^ledger := (CallID,LedgerType,RecognitionTime,Amount,Quantity,Status,Residual).
(G.17)

G.6.5 Transport-edge record

For edge A → B:

O_e^transport :=
(SubjectID,SourceFrame,TargetFrame,SourceTime,TargetTime,Protocol,ExpectedTarget,Observe
(G.18)

G.7 Event Inclusion Rules

G.7.1 Included events

A margin event is included when all of the following are available:

1. stable account identifier;
2. applicable margin agreement;
3. pre-event market or protocol mark;
4. collateral factor or eligibility rule;
5. debt or funding balance;
6. posted collateral balance;
7. candidate breach time;
8. admitted call status or explicit non-admission;
9. at least one post-event action record;
10. verified closure or censoring status.

G.7.2 Included non-call controls

The dataset should include control observations where:

$B < B_{target}$ (G.19)

but:

$ValidCall = 0.$ (G.20)

Possible reasons include:

- waiver;
- disputed mark;
- cure period;
- wrong agreement;
- operational non-admission.

These cases are necessary to test the distinction between breach and gate.

G.7.3 Included safe controls

The dataset should also include account-time observations satisfying:

$B \geq B_{target}.$ (G.21)

These observations support:

- call-hazard modelling;

- false-positive testing;
- phase and Q comparison.

G.7.4 Excluded events

Exclude or separately classify events where:

- identity cannot be matched;
- governing agreement is unknown;
- mark source is unavailable;
- timestamps are irrecoverably inconsistent;
- closure outcome is unobservable;
- manual account changes cannot be distinguished from margin actions;
- records were overwritten without version history.

Exclusion reasons must be reported.

They should not be hidden inside generic “data cleaning.”

G.7.5 Competing-event cases

If several calls overlap for one account, use one of three declared methods:

Method A — First-call cohort

Include only the first call until closure.

Method B — Episode aggregation

Combine overlapping calls into one episode.

Method C — Recurrent-event model

Model each call while preserving dependence.

The method must be selected before outcome analysis.

G.8 Event-Time Definitions

G.8.1 Candidate breach time

t_candidate := first evidence time at which B < B_target under the applicable calculation. (G.22)

G.8.2 Gate-admission time

t_gate := time at which authorized call status is committed. (G.23)

Therefore:

t_gate ≥ t_candidate in the ordinary case. (G.24)

G.8.3 Action time

t_action := time of collateral instruction, sale instruction, liquidation, or default action. (G.25)

G.8.4 Economic-effect time

t_effect := time when the action economically changes the account. (G.26)

Examples include:

- execution;
- receipt of funds;
- debt reduction;
- collateral control.

G.8.5 Ledger-return time

t_ledger := time when the consequence is recognized in the required ledger. (G.27)

G.8.6 Verified closure time

t_close := first time all mandatory closure conditions are satisfied. (G.28)

The closure duration is:

T_close := t_close – t_gate. (G.29)

G.8.7 Censoring

If closure has not occurred by observation end t_end:

T_close is right-censored at t_end – t_gate. (G.30)

Censored events should not be coded as successful closure or failure merely because observation ended.

G.9 Information-Time Consistency

G.9.1 Evidence cutoff

For each prediction time t:

$\mathcal{J}_t$:= all evidence available to the relevant observer by t. (G.31)

A valid predictor satisfies:

$X_{j,t} \in \mathcal{J}_t$. (G.32)

G.9.2 Versioned data

If a value is later corrected, preserve:

$X_t^{original}$. (G.33)

$X_t^{revised}$. (G.34)

The original gate study must use:

$X_t^{original}$. (G.35)

The revised value may be used to analyze:

- correction;
- residual;
- false gate;
- reopening.

G.9.3 Prohibited future leakage

The following must not enter a pre-call model:

- final settlement quantity;
- eventual collateral acceptance;
- later legal determination;
- final liquidation price;
- final accounting adjustment;
- eventual default classification.

G.9.4 Frozen claim record

At prediction time t, freeze:

Claim_t := (Protocol_t,Evidence_t,Gate_t,Residual_t,Transport_t,Invalidation_t). (G.36)

Outcome is observed later:

Outcome_t+H. (G.37)

The draft source treats such frozen pre-outcome records as essential protection against retrospective relabelling.

G.10 Protocol Preregistration

G.10.1 Protocol object

Before accessing outcomes, preregister:

P := (B,Δ,h,u,Φ,G,T,R,V). (G.38)

where:

- B = study boundary;
- Δ = aggregation and measurement rules;
- h = prediction and closure horizon;
- u = admissible interventions;
- Φ = feature maps;
- G = gate definitions;
- T = trace rules;
- R = residual rules;
- V = transport rules.

The main source defines the same protocol as the minimum condition of reproducibility.

G.10.2 Predeclared decisions

The preregistration should specify:

- account inclusion;
- event inclusion;
- call episode construction;
- time alignment;
- missing-data rules;

- closure definition;
- residual categories;
- transport maps;
- benchmark models;
- primary outcomes;
- statistical tests;
- complexity penalty;
- reduction conditions.

G.10.3 Exploratory changes

A post-registration change is permitted only when recorded as:

Revision_v2 := (OriginalRule,NewEvidence,Reason,NewRule,Impact). (G.39)

Confirmatory and exploratory results must remain separated.

G.11 Data Lineage

G.11.1 Raw-source registry

Define raw source set:

$\mathcal{D}$_raw
:=
{Market,MarginEngine,Collateral,Trading,Settlement,Treasury,Accounting,Legal,ResidualLog}.
(G.40)

G.11.2 Derived-variable lineage

Every derived variable X_j should record:

Lineage(X_j) := (RawSources,Transformation,Version,Timestamp). (G.41)

For example:

Lineage(B_t)
:= (C_t,n_t,h_t,R_mark,t,D_t,FormulaVersion,CalculationTime). (G.42)

G.11.3 Shared-lineage warning

Two features may appear independent while using the same raw inputs.

Define lineage overlap:

Ω_jk
:= SharedInputs(X_j,X_k)/TotalInputs(X_j ∪ X_k). (G.43)

High Ω_jk suggests confirmation redundancy.

The source empirical contract emphasizes that different indicator names do not create independent evidence when they share the same data lineage.

G.12 Minimum Data Tables

G.12.1 Table 1 — Account master

Field	Description
AccountID	Stable account identifier
AgreementID	Governing margin agreement
LegalEntity	Account legal entity
BaseCurrency	Declared currency
InstrumentID	Risky asset or aggregate
PositionSign	Long or short orientation
OpenDate	Account or position inception
CloseDate	Account termination, if any

G.12.2 Table 2 — Account state

Field	Description
AccountID	Subject identifier
Timestamp	Evidence time
Quantity	Signed position quantity
MarketMark	Applicable market or protocol mark
CollateralFactor	h
PostedCollateral	C
DebtBalance	D
TargetBuffer	B_target
CalculatedBuffer	B
CallAmount	C_call
Leverage	Declared leverage measure

G.12.3 Table 3 — Gate record

Field	Description
CallID	Margin event identifier
CandidateTime	First threshold crossing
GateTime	Authorized call time
Authority	Calling authority
AgreementRule	Rule applied

Field	Description
MarkVersion	Mark used
GateDecision	Call, waiver, reject, monitor
DueTime	Cure deadline
OriginalTrace	Immutable decision record

G.12.4 Table 4 — Action record

Field	Description
CallID	Margin event identifier
ActionID	Action identifier
ActionType	Collateral, sale, liquidation, default
InstructionTime	Action instruction
InstructionAmount	Requested amount
InstructionQuantity	Requested quantity
ExecutionTime	Economic execution
ExecutionAmount	Executed amount
Authority	Acting party

G.12.5 Table 5 — Ledger-return record

Field	Description
CallID	Margin event identifier
LedgerType	Collateral, settlement, debt, accounting, legal
RecognitionTime	Ledger time
RecordedAmount	Recognized amount
RecordedQuantity	Recognized quantity
Status	Open, partial, closed, disputed
ResidualCode	Predeclared residual category
RevisionID	Later correction, if any

G.12.6 Table 6 — Frame transport

Field	Description
SubjectID	Account or transaction
SourceFrame	Market, legal, settlement, etc.
TargetFrame	Collateral, accounting, etc.
SourceState	State used for transport
ConnectionVersion	Applicable map
ExpectedTarget	Transported value
ObservedTarget	Actual target state
EdgeResidual	Observed minus expected
SourceTime	Source timestamp
TargetTime	Target timestamp

G.12.7 Table 7 — Closure certificate

Field	Description
CallID	Event identifier
ClosureTime	Verified closure
Branch	Cure branch
AmountClosed	Amount criterion
SettlementClosed	Settlement criterion
DebtClosed	Debt criterion
AccountingClosed	Recognition criterion
LegalClosed	Legal criterion
ChargeClosed	Charge criterion
ResidualDisclosed	Residual criterion
ClosureStatus	Closed, partial, open, contested, default-converted

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G.13 Conventional Financial Variables

G.13.1 Buffer

$$B_t = C_t + n_{th_tR_mark,t} - D_t. \qquad (G.44)$$

G.13.2 Shortfall

$$S_{B,t} = [B_{target,t} - B_t]_+. \qquad (G.45)$$

G.13.3 Call amount

$$C_{call,t} = S_{B,t} \quad \text{when a valid call is admitted.} \qquad (G.46)$$

G.13.4 Market value

$$V_{M,t} = n_{tR_mark,t}. \qquad (G.47)$$

G.13.5 Collateral value

$$V_{C,t} = n_{th_tR_mark,t}. \qquad (G.48)$$

G.13.6 Equity or net account value

$$E_t = n_{tR_mark,t} + C_t - D_t. \qquad (G.49)$$

G.13.7 Leverage

One declared leverage measure is:

$\lambda_t = |n_t R_{mark,t}| / [|E_t| + \varepsilon]$. (G.50)

Alternative leverage definitions must be preregistered.

G.14 Optional Complex-CAPM Variables

G.14.1 Use as a secondary module

The first action–ledger pilot does not require Complex CAPM.

If sufficient inputs are available, add it as a preregistered secondary module.

G.14.2 Baseline amplitude

$A_t = CF_t / (1 + r_{base,t})^T$. (G.51)

G.14.3 CAPM-admitted value

$R_{CAPM,t} = CF_t / (1 + r_{base,t} + \beta_t ERP_t)^T$. (G.52)

G.14.4 Pressure coordinate

$Q_t = \sqrt{(A_t^2 - R_{CAPM,t}^2)}$. (G.53)

G.14.5 Phase

$\theta_t = \arccos(R_{CAPM,t} / A_t)$. (G.54)

G.14.6 Market-mark residual

The actual contractual margin mark may differ from CAPM value.

Define:

$r_{mark,t} := R_{mark,t} - R_{CAPM,t}$. (G.55)

The contractual call must use the applicable margin mark, not the research CAPM value.

The source Finance Geometry framework similarly distinguishes protocol-admitted value from alternative market-residual calibration and requires the declared finance protocol to determine R.

G.15 Structural Charge Coding

G.15.1 Asset charge

For the minimum account:

$q_{A,t} := n_{tq_0}(\text{InstrumentID})$. (G.56)

G.15.2 Funding charge

Define:

$q_{F,t} := -\text{ChargeMap}_F(D_t)$. (G.57)

G.15.3 Collateral charge

The structural collateral obligation is:

$q_C^{\text{struct}} :=$ contractual margin-performance orientation. (G.58)

The operational charge is:

$q_{C,t}^{\text{op}} := 1[\text{ValidCallOpen}_t] q_C^{\text{struct}}$. (G.59)

G.15.4 Charge residual

At event k:

$r_{q,k} := \Sigma q_{\text{in},k} + q_{\text{gate},k} - \Sigma q_{\text{out},k}$. (G.60)

The first pilot may report charge reconciliation descriptively without claiming a full conservation law.

G.16 Action–Ledger Feature Construction

G.16.1 Action vector

Define the minimum action vector:

$$\psi_{A,k} := \left[C_{\text{call},k},\ C_{\text{instruction},k},\ \text{SaleInstruction}_k,\ \text{SaleExecuted}_k,\ \text{ExpectedDebtReduction}_k,\ \text{EconomicLoss}_k,\ \text{GateStatus}_k \right]$$ (G.61)

G.16.2 Ledger vector

Define:

ψ_L,k :=
[
CollateralAdmitted_k
CollateralRecorded_k
SaleSettled_k
RecordedDebtReduction_k
RecognizedLoss_k
LegalStatus_k
LedgerStatus_k
]. (G.62)

G.16.3 Unit normalization

Raw coordinates use different units.

Define diagonal scale matrix:

S_scale := diag(s1,s2,...,s_m). (G.63)

Normalize:

$\tilde{\psi}_A := S_{scale}^{-1}\psi_A$. (G.64)

$\tilde{\psi}_L := S_{scale}^{-1}\psi_L$. (G.65)

Scale choices may use:

- call amount;
- account equity;
- position quantity;
- contractual tolerance;
- materiality threshold.

Scaling rules must be determined without using closure outcomes.

G.16.4 Expected action-to-ledger map

Define branch-specific transporter:

U_AL^(b). (G.66)

Examples include:

Collateral branch

Call amount → admitted collateral.

Deleveraging branch

Executed quantity → settled quantity.

Expected net proceeds → recorded debt reduction.

Default branch

Outstanding claim → recovery ledger identity.

G.16.5 Action–ledger residual

The residual vector is:

$$\delta_{spin,k} := \tilde{\psi}_{L,k} - U_{AL,k}^{(b)}\tilde{\psi}_{A,k}. \qquad (G.67)$$

G.16.6 Spinor-split score

Define:

$$\Delta_{spin,k}^2 := \delta_{spin,k}^{T} W_{spin} \delta_{spin,k}. \qquad (G.68)$$

The primary study should use the safer terminology:

Action–Ledger Closure Defect. (G.69)

The term spinor split may be used as a formal candidate, but exact spin-½ status is not tested in this pilot.

G.16.7 Age-adjusted defect

Some mismatch is expected immediately after action.

Define elapsed return time:

$$a_k(t) := t - t_{action,k}. \qquad (G.70)$$

Estimate expected ordinary delay:

$$m_b(a) := E[\Delta_{spin} \mid Branch = b, Elapsed = a, EventuallyCleanClosure]. \qquad (G.71)$$

Define excess defect:

$$\Delta_{spin,k}^{excess}(t) := \Delta_{spin,k}(t) - m_b[a_k(t)]. \qquad (G.72)$$

This prevents normal settlement latency from being labelled failure.

G.17 Market-to-Collateral Transport Test

G.17.1 Source state

Use the market or margin mark:

x_M,t := (InstrumentID,n_t,R_mark,t,t_M). (G.73)

G.17.2 Connection

The minimum connection is:

$\mathcal{A}$_MC,t :=
(h_t,Eligibility_t,ConcentrationAdjustment_t,CurrencyAdjustment_t,TimingRule_t).
(G.74)

G.17.3 Expected target

The expected collateral value is:

V^_C,t = n_tR_mark,th_te_t – Adj_concentration,t – Adj_currency,t. (G.75)

where:

e_t ∈ {0,1} (G.76)

is legal or protocol eligibility.

G.17.4 Observed target

Let:

V_C,t^obs = collateral value recorded by the collateral system. (G.77)

G.17.5 Edge residual

r_MC,t = V_C,t^obs – V^_C,t. (G.78)

G.17.6 Raw mismatch

The naive mismatch is:

r_MC,t^raw = V_C,t^obs – n_tR_mark,t. (G.79)

The raw difference will normally be large because it ignores the haircut.

It should not be classified automatically as reconciliation failure.

G.17.7 Timing-adjusted residual

If collateral recognition occurs at t_C > t_M:

V^_C,t_C
= T_MC[MarketPath(t_M,t_C),h_t_C,Eligibility_t_C]. (G.80)

Then:

r_MC,t_C^time
= V_C,t_C^obs – V^_C,t_C. (G.81)

G.17.8 Transport-test outcome

Human reviewers classify each apparent mismatch as:

- lawful haircut;
- timing difference;
- concentration adjustment;
- eligibility change;
- true data error;
- unresolved.

The test compares the classification accuracy of:

1. raw numerical difference;
2. governed transport residual.

G.18 Closure Outcome Definitions

G.18.1 Amount cure

Y_amount = 1[B ≥ B_target]. (G.82)

G.18.2 Operational cure

Y_operational = 1[RequiredActionCompleted]. (G.83)

G.18.3 Ledger cure

Y_ledger = 1[RequiredLedgersReturned]. (G.84)

G.18.4 Verified closure

Define:

Y_close = 1 (G.85)

only when:

AmountCured
∧ RequiredSettlementCompleted
∧ DebtUpdated
∧ CollateralRecognized
∧ ChargeReconciled
∧ RequiredTraceAccessible
∧ ResidualDisclosed. (G.86)

G.18.5 Valid default conversion

A default branch may close validly when:

Y_default-close = 1 (G.87)

and:

- margin identity has converted into recovery identity;
- claims and obligations are reconciled;
- legal and accounting status are recorded;
- residual is disclosed.

Default is not coded automatically as unresolved closure.

G.18.6 Partial closure

Define:

Y_partial = 1 (G.88)

when some but not all mandatory closure conditions have passed.

G.18.7 False closure

Define:

Y_false-close
:= 1[ReportedClosed = 1 ∧ VerifiedClosed = 0]. (G.89)

This is a principal governance outcome.

G.19 Residual Taxonomy

G.19.1 Minimum residual vector

Define:

$\mathscr{R}$_k :=
[
r_amount
r_quantity
r_cash
r_debt
r_collateral
r_timing
r_identity
r_authority
r_legal
r_accounting
r_model
]. (G.90)

G.19.2 Residual status

Each residual has:

Status(r_j)
∈ {Open,Investigating,Explained,Corrected,Accepted,Escalated}. (G.91)

G.19.3 Explained economic difference

A lawful fee may first appear as a debt mismatch.

Once identified and included in the expected transport:

r_debt → 0. (G.92)

The fee remains an economic cost.

It ceases to be unexplained residual.

G.19.4 No-material-residual state

A clean event should contain the explicit record:

NoMaterialResidualDetected = 1. (G.93)

Absence of a residual entry should not be assumed to mean zero residual.

G.20 Annotation and Reliability

G.20.1 Why manual annotation may be required

Some variables may require expert interpretation:

- valid authority;
- legal eligibility;
- branch type;
- identity conversion;
- closure status;
- residual cause.

These cannot always be inferred from numerical records alone.

G.20.2 Annotation manual

The manual should define:

- each gate state;
- each branch;
- each residual category;
- each closure condition;
- hierarchy for conflicting evidence;
- uncertainty labels;
- escalation rules.

G.20.3 Independent coders

At least two coders should independently annotate a validation subset.

G.20.4 Reliability statistics

Use:

- Cohen’s κ for two coders;
- Fleiss’ κ for multiple coders;
- Krippendorff’s α for missing or ordinal labels;
- confusion matrices for category-specific disagreement.

G.20.5 Reliability gate

A minimum criterion is:

Reliability_new-system

Reliability_conventional-status-labels. (G.94)

If not, the new taxonomy may be too ambiguous.

The source empirical contract proposes the same comparison between the governed grammar and conventional labels.

G.20.6 Adjudication

Disagreements should be preserved as:

OriginalCoderA. (G.95)

OriginalCoderB. (G.96)

AdjudicatedLabel. (G.97)

Reason. (G.98)

The original annotations should not be overwritten.

G.21 Primary Hypotheses

G.21.1 H1 — Gate separation

Valid calls have greater persistent consequence than ungated breaches:

Pr(Y_persistent = 1 | Breach,ValidCall)

Pr(Y_persistent = 1 | Breach,NoCall). (G.99)

G.21.2 H2 — Closure-defect value

Higher action–ledger defect predicts slower closure:

∂E[T_close]/∂Δ_spin^excess > 0. (G.100)

G.21.3 H3 — Unresolved-event prediction

Pr(Y_unresolved,H = 1 | Δ_spin^excess,Controls)
increases with Δ_spin^excess. (G.101)

G.21.4 H4 — False-closure detection

Δ_spin^excess predicts false closure beyond ordinary status fields:

I(Y_false-close;Δ_spin^excess | ReportedStatus,CallAge,Shortfall) > 0. (G.102)

G.21.5 H_5 — Transport improvement

Governed transport reduces false reconciliation classification:

$$\text{FalseAlarmRate}_G < \text{FalseAlarmRate}_{raw}. \quad (G.103)$$

G.21.6 H_6 — Residual recall

Pre-event residual categories anticipate later failure modes better than generic exception counts:

$$\text{ResidualRecall}_{typed} > \text{ResidualRecall}_{untyped}. \quad (G.104)$$

G.21.7 H_7 — Ledger memory

Ledger history adds information beyond the current state:

$$I(Y_{t+H}; L_t \mid X_t) > 0. \quad (G.105)$$

G.21.8 Secondary H_8 — Q value

Where Complex CAPM inputs exist:

$$I(\Delta B_{t+H}; Q_t, \theta_t \mid R_t, B_t, h_t, D_t, \text{Controls}) > 0. \quad (G.106)$$

This hypothesis is secondary and should compete with a flexible real-vector model.

G.22 Benchmark Models

G.22.1 Benchmark B_0 — Contractual rule

Use:

- current buffer;
- target buffer;
- call amount;
- contractual deadline.

No statistical estimation is required.

G.22.2 Benchmark B_1 — Conventional logistic model

For unresolved status:

$$\text{logit } \Pr(Y_{\text{unresolved},H} = 1) = \alpha + \beta_1 S_B + \beta_2 \lambda + \beta_3 \text{Volatility} + \beta_4 \text{Branch} + \beta_5 \text{ElapsedTime}. \quad (G.107)$$

G.22.3 Benchmark B_2 — Survival model

For closure hazard:

$$h_{\text{close}}(t) = h_0(t)\exp(\beta^T X). \quad (G.108)$$

G.22.4 Benchmark B_3 — Workflow-age model

Use:

- branch;
- call age;
- action age;
- number of open tasks;
- latest status.

This is a strong benchmark for the action–ledger defect.

G.22.5 Benchmark B_4 — Queueing model

Use:

- open-event arrivals;
- ledger-processing completions;
- backlog;
- service time.

This competes with closure-capacity interpretations.

G.22.6 Benchmark B_5 — Flexible machine-learning model

Use conventional raw variables with:

- gradient boosting;
- random forest;

- regularized nonlinear model;

subject to chronological validation.

This prevents the proposed structure from winning merely because the conventional benchmark is underfitted.

G.22.7 Benchmark B6 — Raw reconciliation model

Use:

|ObservedTarget – SourceValue|. (G.109)

Compare with governed transport residual:

|ObservedTarget – ExpectedTransportedTarget|. (G.110)

G.23 Proposed Models

G.23.1 Model M1 — Gate model

Add:

- candidate breach;
- authority;
- rule validity;
- gate decision;
- waiver.

G.23.2 Model M2 — Action–ledger model

Add:

- ψ_A;
- ψ_L;
- Δ_spin;
- residual types.

G.23.3 Model M3 — Transported closure model

Add:

- T_MC;

- r_MC;
- A-B Fixedness;
- frame-recognition status.

G.23.4 Model M_4 — Ledger-memory model

Add:

- prior calls;
- prior residual;
- prior false closures;
- prior settlement failures;
- current rule changes.

G.23.5 Model M_5 — Optional complex module

Add:

- A;
- R_CAPM;
- Q;
- θ;
- collateral phase.

The complex model must be compared with an equally flexible real-pair model.

G.24 Statistical Specifications

G.24.1 Call persistence model

logit Pr(Y_persistent,k = 1)
= α

- β_BBreach_k
- β_GValidGate_k
- β_BG(Breach_k × ValidGate_k)
- $\gamma^{T}X_k.$ (G.111)

The gate-separation hypothesis expects:

β_G > 0 or β_BG > 0, (G.112)

depending on coding.

G.24.2 Closure survival model

h_close,k(t)
= h0(t)exp[
β1Δ_spin,k^excess(t)

- β2S_B,k
- β3Branch_k
- β4Leverage_k
- β5Volatility_k
- β6Backlog_t
  ]. (G.113)

If higher hazard means faster closure, the expected sign is:

β1 < 0. (G.114)

G.24.3 False-closure model

logit Pr(Y_false-close,k = 1)
= α

- β1Δ_spin,k^excess
- β2r_MC,k
- β3ResidualCount_k
- β4ReportedStatus_k
- γ^TX_k. (G.115)

G.24.4 Ledger-memory model

Outcome_{i,t+H}
= f(X_i,t,L_i,t) + ε_i,t+H. (G.116)

Compare with:

Outcome_{i,t+H}
= f(X_i,t) + ε_i,t+H. (G.117)

Ledger-memory gain is:

ΔU_L := U(X,L) − U(X). (G.118)

G.24.5 Transport-classification model

For mismatch j:

$$\Pr(\text{TrueError}_j = 1) = \text{Logistic}[\alpha + \beta_1|r_{raw,j}| + \beta_2|r_{G,j}| + \beta_3 \text{Delay}_j + \beta_4 \text{Eligibility}_j]. \quad (G.119)$$

The governed residual should dominate raw mismatch after lawful adjustments.

G.25 Validation Design

G.25.1 Chronological split

Use:

$$\text{Training period} < \text{Validation period} < \text{Test period}. \quad (G.120)$$

Random row splitting is inappropriate when:

- calls recur;
- rules change;
- ledger history matters;
- events cluster in crises.

G.25.2 Account separation

Where possible, prevent the same account episode from appearing in both training and test sets.

G.25.3 Walk-forward validation

For cut point m:

$$\text{Fit on } [1,m]. \quad (G.121)$$

$$\text{Validate on } [m+1, m+h]. \quad (G.122)$$

Then move forward.

G.25.4 Regime holdout

Reserve at least one stress interval or rule regime for final testing.

This assesses transport beyond the calibration environment.

G.25.5 Nested tuning

Hyperparameter tuning occurs only inside the training and validation periods.

The final test period remains untouched until all model and variable decisions are frozen.

G.26 Evaluation Metrics

G.26.1 Binary outcomes

Report:

- precision;
- recall;
- area under precision–recall curve;
- Brier score;
- log loss;
- calibration slope;
- calibration intercept.

G.26.2 Closure time

Report:

- concordance index;
- integrated Brier score;
- time-dependent calibration;
- median closure-time error.

G.26.3 False closure

Report:

FalseClosureRate
:= FalseClosedEvents/ReportedClosedEvents. (G.123)

G.26.4 Gate separation

Report:

GatePersistenceGap

:= Pr(Persistent | Breach,Gate) − Pr(Persistent | Breach,NoGate). (G.124)

G.26.5 Transport value

Report:

TransportFalseAlarmReduction
:= FalseAlarm_raw − FalseAlarm_G. (G.125)

G.26.6 Residual value

Report:

ResidualRecall
and
ResidualPrecision. (G.126)

G.26.7 Complexity-adjusted utility

Define:

U_j
:= w_PP_j

- w_DD_j
- w_GG_j
  − λ_CC_j. (G.127)

where:

- P_j = predictive gain;
- D_j = diagnostic gain;
- G_j = governance gain;
- C_j = complexity cost.

The source benchmark rule similarly requires advanced performance net of complexity cost to exceed the simpler alternative.

G.27 Missing Data

G.27.1 Missingness flag

For every variable X_j:

M_j := 1[X_j missing]. (G.128)

G.27.2 Missingness as residual

When a variable required for closure cannot be observed:

r_missing,j := MissingRequiredState_j. (G.129)

The missingness remains visible even when values are imputed.

G.27.3 No automatic zero

A missing ledger entry is not:

LedgerValue = 0. (G.130)

It is:

LedgerValue = Unknown. (G.131)

G.27.4 Sensitivity analyses

Compare:

- complete cases;
- multiple imputation;
- inverse-probability weighting;
- explicit missing-state model;
- worst-case closure classification.

G.28 Confounding and Alternative Explanations

G.28.1 Event size

A large action–ledger defect may simply reflect a larger call.

Normalize by:

Δ_spin,normalized
:= Δ_spin/[CallSize + ε]. (G.132)

Both raw and normalized results should be reported.

G.28.2 Ordinary delay

Mismatch may reflect expected settlement latency.

Use branch- and institution-specific expected-delay controls.

G.28.3 Operational backlog

High Δ_{spin} may be caused by system-wide backlog rather than subject-specific instability.

Include:

$Backlog_t$. (G.133)

G.28.4 Market stress

Closure failure and mismatch may both rise during market stress.

Control for:

- volatility;
- liquidity;
- market depth;
- aggregate calls;
- stress-regime indicators.

G.28.5 Institution and rule effects

Include:

- institution fixed effects;
- agreement type;
- collateral rule version;
- settlement regime;
- calendar period.

G.29 Robustness Tests

G.29.1 Alternative closure tolerances

Vary:

ε_{spin}. (G.134)

ε_{amount}. (G.135)

$\varepsilon_{transport}$. (G.136)

G.29.2 Alternative normalization

Test scaling by:

- call amount;
- account equity;
- gross market value;
- contractual materiality threshold.

G.29.3 Alternative branch maps

Estimate separate U_AL for:

- collateral;
- deleveraging;
- forced liquidation;
- default.

Compare with one pooled transporter.

G.29.4 Alternative delay models

Use:

- fixed delay;
- branch-specific delay;
- distributed delay;
- survival-based expected return.

G.29.5 Alternative account aggregation

Compare:

- instrument-level;
- account-level;
- collateral-set-level.

G.29.6 Placebo thresholds

Construct numerical thresholds lacking contractual authority.

The valid gate should produce stronger consequence separation.

G.29.7 Shuffled action–ledger pairing

Randomly match actions and ledger returns from different events.

The closure-defect signal should disappear or materially weaken.

G.29.8 Raw-pair benchmark

Compare the full closure defect with a flexible model using all raw action and ledger variables separately.

If the raw model performs equally well, the compact spinor-defect score may offer interpretation but not predictive compression.

G.30 Preregistered Falsification Gates

G.30.1 Gate-model failure

Reject the gate layer if:

$$\text{GatePersistenceGap} \le 0 \qquad (G.137)$$

and valid gate status adds no diagnostic or predictive value beyond the raw breach.

Reduction:

$$\text{Gate} \rightarrow \text{Threshold Status.} \qquad (G.138)$$

G.30.2 Action–ledger failure

Reject the strong two-component layer if:

$$\Delta U_{spin} \le 0, \qquad (G.139)$$

where:

$$\Delta U_{spin} := U(\text{ActionLedgerModel}) - U(\text{WorkflowBenchmark}). \qquad (G.140)$$

Reduction:

$$\text{Spinor Candidate} \rightarrow \text{Workflow Model.} \qquad (G.141)$$

G.30.3 Transport failure

Reject the governed-transport claim if:

$$\text{FalseAlarm}_G \ge \text{FalseAlarm}_{raw} \qquad (G.142)$$

or if the transport map is unstable or retrospectively selected.

Reduction:

Governed Transport → Pairwise Reconciliation. (G.143)

G.30.4 Ledger-memory failure

Reject recursion if:

$\Delta U_L \le 0$. (G.144)

Reduction:

Ledgered Recursion → Current-State Model. (G.145)

G.30.5 Complex-module failure

Reject complex priority if:

$U_{complex} \le U_{real\text{-}pair}$. (G.146)

Reduction:

$Z = R + iQ \rightarrow (R,Q)$. (G.147)

If Q itself adds no value:

$(R,Q) \rightarrow R$. (G.148)

G.30.6 Residual-taxonomy failure

Reduce residual typing if:

$Reliability_{typed} \le Reliability_{generic}$ (G.149)

and:

$ResidualRecall_{typed} \le ResidualRecall_{generic}$. (G.150)

G.31 Claim-Status Object

For every main result, report:

ClaimStatus
:=
(Protocol,ClosureLevel,Admission,Residual,Transport,Authority,ModelLevel,ValidationStatus).
(G.151)

Example:

Protocol: One-currency margin account

Closure level: Event
Admission: Valid margin call
Residual: Material settlement delay
Transport: Market-to-collateral passed
Authority: Broker margin engine
Model level: Action–ledger doublet
Validation status: Prospective test supported

This format is adapted from the source's standard claim-status architecture, which requires closure, admission, residual, transport, authority, and model level to remain explicit.

G.32 Publication-Ready Main Results Table

Model	Predictive metric	Calibration	False-closure recall	Transport false alarms	Complexity	Retained status
Contractual baseline					Low	
Conventional logistic					Low	
Workflow-age model					Medium	
Action–ledger model					Medium	
Transported closure model					Medium–High	
Optional complex module					High	

The retained status should be one of:

- supported;
- partially supported;
- governance-only;
- exploratory;
- reduced;
- rejected.

G.33 Event Case-Study Template

Each published event case should use the following order.

1. Protocol

What boundary, agreement, mark, and closure rule applied?

2. Evidence cutoff

What was known at the candidate and gate times?

3. Candidate breach

What numerical threshold was crossed?

4. Gate

Was a call validly admitted, waived, rejected, or disputed?

5. Action

What outward instruction or intervention occurred?

6. Ledger return

Which consequences were recognized, settled, or legally completed?

7. Transport

How did market, collateral, settlement, and accounting frames map the subject?

8. Residual

What remained unresolved?

9. Closure status

Was the event closed, partially closed, contested, or validly converted into default?

10. Invalidation

What later evidence could reopen the result?

The source standard uses essentially the same report sequence: protocol, evidence cutoff, projection, typing, candidate, gate, residual, transport, claim ceiling, and invalidation.

G.34 Example of a Pilot Event Record

G.34.1 Protocol

Account A-104 is governed by Agreement M-7 in U.S. dollars.

The applicable rule is:

$$B = C + nhR_{mark} - D. \qquad (G.152)$$

A call is admitted when:

$B < 0$ (G.153)

and the broker's margin engine validates the mark and agreement.

G.34.2 Candidate

At 10:02:

$B = -\$425.$ (G.154)

A threshold breach exists.

G.34.3 Gate

At 10:05, the broker issues a valid call:

$C_{call} = \$425.$ (G.155)

Call status becomes Open.

G.34.4 Action

At 10:18, the client instructs a sale expected to reduce debt by:

$600. (G.156)

G.34.5 Ledger state

At 10:18:

$ExpectedDebtReduction = \$600.$ (G.157)

$RecordedDebtReduction = \$0.$ (G.158)

The action–ledger mismatch is nonzero.

G.34.6 Settlement

At 14:10, only $550 is recorded because of fees and partial execution.

After lawful adjustment for a $20 fee and $30 unexecuted amount:

$ExpectedRecordedReduction = \$550.$ (G.159)

The residual falls to zero.

G.34.7 Closure

At 14:15:

B = +$35. (G.160)

settlement is recorded, debt is updated, and no material residual remains.

The call is verified closed.

The economic fee remains a loss.

It is not a transport residual.

G.35 Data Governance

G.35.1 Confidentiality

Margin, collateral, and funding records are highly sensitive.

The pilot should use:

- de-identified account IDs;
- access-controlled datasets;
- minimum necessary fields;
- separated identity keys;
- audit logs;
- approved retention periods.

G.35.2 Human decision records

Where possible, preserve:

- automated decision;
- human override;
- authority;
- reason;
- timestamp.

A human override is not automatically error.

It is an institutional gate that must be typed and audited.

G.35.3 Model decision boundary

The pilot should not autonomously:

- issue margin calls;
- liquidate positions;
- deny collateral;
- declare default.

Its first role is:

- retrospective reconstruction;
- prospective warning;
- reconciliation support;
- closure auditing.

Any later operational use requires separate governance and validation.

G.36 Interpretation of Possible Outcomes

G.36.1 Outcome A — Gate and closure support

Valid gate status and action–ledger defect both add value.

Retain:

Governed Margin Event

- Action–Ledger Closure Model. (G.161)

G.36.2 Outcome B — Gate support, no doublet support

Valid calls differ from raw breaches, but the action–ledger defect adds no value beyond workflow age.

Retain:

Gate-Governed Workflow Model. (G.162)

Remove the stronger spinor claim.

G.36.3 Outcome C — Transport support only

Governed transport reduces false reconciliation alarms, but closure modelling adds little.

Retain:

Financial Frame-Transport Audit. (G.163)

G.36.4 Outcome D — Governance-only value

Prediction does not improve, but the framework identifies:

- false closure;
- authority errors;
- missing traces;
- unexplained residual.

Retain it as:

Closure-Governance Architecture. (G.164)

Do not report it as forecasting superiority.

G.36.5 Outcome E — Conventional dominance

Simple variables and workflow status perform equally well.

Reduce to:

Conventional Margin and Settlement Model. (G.165)

G.36.6 Outcome F — Measurement failure

Action and ledger cannot be reconstructed reliably.

The correct conclusion is:

The Current Dataset Cannot Test the Spin Hypothesis. (G.166)

It is not:

The Spin Hypothesis Is Confirmed or Refuted. (G.167)

G.37 Minimal Replication Package

A publishable pilot should release, where confidentiality permits:

- protocol specification;
- variable dictionary;
- synthetic or de-identified example data;
- annotation manual;

- model code;
- benchmark code;
- chronological split definition;
- preregistration;
- falsification thresholds;
- result tables;
- reduction certificate;
- model-version ledger.

G.37.1 Replication object

Define:

ReplicationPackage
:= (Protocol,DataSchema,Code,Seeds,Models,Tests,Claims,Residuals,Revisions). (G.168)

G.37.2 Immutable original result

When the model is later revised:

Result_v1 remains accessible. (G.169)

The new result is:

Result_v2 = Revision(Result_v1,NewEvidence). (G.170)

The source research constitution requires revision to link to prior versions rather than erase them.

G.38 Pilot Research Contract

The pilot commits to:

1. no hidden account boundary;
2. no future leakage;
3. no call event without an authority record;
4. no closure status based only on outward action;
5. no raw frame comparison without declared transport;
6. no missing ledger state silently treated as zero;
7. no residual erasure;

8. no retrospective change to charge signs;
9. no complex model without a real-pair benchmark;
10. no action–ledger model without a workflow benchmark;
11. no broad claim from one institution without transport evidence;
12. no stronger terminology after failed admission tests.

In compact form:

No Hidden Boundary.
No Future Leakage.
No Ungated Call.
No Action-Only Closure.
No Untransported Comparison.
No Missingness as Zero.
No Residual Erasure.
No Silent Revision.
No Advanced Model without a Simpler Benchmark. (G.171)

This follows the canonical research contract of the source framework.

G.39 Pilot Decision Algorithm

After the final test period:

Step 1 — Test gate separation

If unsupported:

Reduce to threshold modelling. (G.172)

Step 2 — Test independent action–ledger measurement

If unavailable:

Stop the spin test. (G.173)

Step 3 — Test incremental closure value

If unsupported:

Reduce to workflow status. (G.174)

Step 4 — Test transport adjustment

If unsupported:

Reduce to pairwise reconciliation. (G.175)

Step 5 — Test ledger memory

If unsupported:

Use current-state dynamics. (G.176)

Step 6 — Test optional complex module

If unsupported:

Reduce $Z \to (R,Q) \to R$. (G.177)

Step 7 — Issue claim certificate

Record:

SupportedClaims. (G.178)

ReducedClaims. (G.179)

RejectedClaims. (G.180)

ResidualQuestions. (G.181)

G.40 Pilot Conclusion

The first real-data study should not ask whether finance obeys a hidden physical Standard Model.

It should ask whether the proposed architecture improves three concrete distinctions:

A numerical breach is not yet an admitted event. (G.182)

An outward action is not yet accountable closure. (G.183)

A difference between financial frames is not yet an unexplained residual. (G.184)

These distinctions are measurable.

They require no assumption that:

- markets are quantum;
- margin accounts are physical spinors;
- all financial frames possess one universal gauge group.

The minimum empirical architecture is:

Account State
→ Candidate Breach
→ Authorized Gate
→ Outward Action
→ Expected Ledger Return
→ Observed Ledger Return
→ Transport-Adjusted Residual
→ Verified Closure or Reopening. (G.185)

The minimum study models are:

Conventional Margin Baseline
→ Gate Model
→ Action–Ledger Model
→ Governed Transport Model. (G.186)

The minimum success condition is:

The stronger model must improve prediction, diagnosis, or governance after complexity is penalized. (G.187)

The minimum failure condition is:

When the stronger layer adds no reproducible value, reduce it while preserving the lower structure that survives. (G.188)

The pilot's final governing principle is therefore:

> **Begin with one bounded account, one gate, one action–ledger return, and one frame-transport test. Permit the Financial Gauge–Dirac architecture to grow only after these minimum structures have been measured, benchmarked, and allowed to fail.**

Appendix H — Reference Implementation and Machine-Readable Runtime

H.1 Purpose

The preceding appendices defined:

- financial identities;
- structural charges;
- action–ledger closure states;
- financial frame graphs;
- transport residuals;
- simulation protocols;
- a minimal real-data pilot.

This appendix converts those theoretical objects into a reference software architecture.

The purpose is not to prescribe one programming language, database, vendor, or deployment model.

The purpose is to specify the smallest machine-readable runtime capable of enforcing the article's governing distinctions:

Breach ≠ Gate. (H.1)

Action ≠ Ledger Return. (H.2)

Frame Difference ≠ Residual. (H.3)

Reported Closure ≠ Verified Closure. (H.4)

Model Revision ≠ Historical Erasure. (H.5)

The implementation should therefore be judged not by the number of indicators, dashboards, or machine-learning models it contains, but by whether it can preserve:

- protocol;
- identity;
- evidence lineage;
- gate authority;
- action–ledger separation;
- residual;
- frame transport;
- version history;
- reduction to simpler models.

The source Periodic Grammar proposes an integrated runtime:

Declare
→ Observe
→ Classify
→ Gate
→ Trace
→ Preserve Residual
→ Transport
→ Compare
→ Revise. (H.6)

It also states that no theoretical object should be promoted merely because it is mathematically elegant or physically suggestive.

H.2 Implementation Claim Ceiling

H.2.1 What the reference implementation does

The runtime may:

- register financial protocols;
- freeze evidence snapshots;
- calculate derived states;
- type candidate claims;

- evaluate analytical gates;
- record authoritative decisions;
- separate outward actions from ledger return;
- calculate residuals;
- transport identities across frames;
- compare advanced models with simpler benchmarks;
- preserve revisions and failed claims.

H.2.2 What the reference implementation does not do automatically

The minimum runtime should not autonomously:

- issue legally binding margin calls;
- liquidate customer positions;
- declare contractual default;
- alter accounting records;
- overwrite legal ownership;
- approve collateral;
- claim universal financial charge;
- claim physical financial spin;
- infer a gauge group from correlation alone.

These actions require external authority and institution-specific governance.

The analytical runtime may recommend, classify, reconstruct, or warn.

It should not confuse analytical output with institutional permission.

H.2.3 Runtime status

The software is:

A Research and Governance Runtime for Constraint-Bearing Financial Claims. (H.7)

It is not:

An Autonomous Financial Authority. (H.8)

H.3 Reference Architecture

H.3.1 Core runtime

Define the Financial Closure Runtime:

FCR
:= ProtocolRegistry
× EvidenceVault
× IdentityRegistry
× ProjectionEngine
× ClaimCompiler
× GateEngine
× ActionLedgerEngine
× ResidualLedger
× TransportEngine
× BenchmarkEngine
× RevisionManager. (H.9)

The multiplication sign denotes architectural conjunction rather than numerical multiplication.

Each module performs a distinct function.

No single model or database table should silently absorb all functions.

H.3.2 Four-layer implementation stack

The implementation contains four principal layers.

Layer 1 — Ontology

Defines which objects may exist:

- protocol;
- financial identity;
- charge;
- frame;
- gate;
- trace;
- residual;
- ledger state;
- transport edge;
- closure certificate.

Layer 2 — Runtime

Executes:

- projections;
- gate evaluations;
- state transitions;
- transport;
- reconciliation;
- closure tests;
- revisions.

Layer 3 — Benchmark

Evaluates:

- predictive performance;
- diagnostic value;
- calibration;
- model complexity;
- false closure;
- transport accuracy;
- reduction conditions.

Layer 4 — Research ledger

Preserves:

- protocol versions;
- model versions;
- failed claims;
- rejected gates;
- residual history;
- revised conclusions;
- reduction certificates.

The source implementation architecture similarly separates ontology, runtime, benchmark, and research ledger rather than treating a model output as the complete research world.

H.3.3 Governance rails

Three governance rails operate across all layers.

Residual rail

No closure process may discard unresolved consequence.

Transport rail

No broad claim may be promoted without testing its survival across declared frames.

Revision rail

No later correction may erase the original evidence, decision, or model version.

H.4 Core Machine-Readable Objects

H.4.1 Protocol record

Every calculation and claim must reference a protocol record.

Define:

ProtocolRecord
:= (
ProtocolID,
Version,
Boundary,
ObservationRule,
Horizon,
AdmissibleActions,
FeatureMap,
GateRules,
TraceRules,
ResidualRules,
TransportRules,
EffectiveFrom,
EffectiveTo,
Authority
). (H.10)

The ProtocolID identifies the protocol family.

Version identifies the exact rule set used.

A protocol should be immutable after activation.

A change creates a new version.

H.4.2 Evidence snapshot

Define:

EvidenceSnapshot

:= (
SnapshotID,
ProtocolID,
SubjectID,
EvidenceTime,
SourceRecords,
SourceVersions,
DerivedInputs,
MissingFields,
Hash,
CreatedAt
). (H.11)

The snapshot freezes what was available at a particular evidence time.

It protects the distinction:

Evidence Available Then
≠ Information Learned Later. (H.12)

A later correction should not mutate the original snapshot.

It should create:

EvidenceSnapshot_v2. (H.13)

linked to:

EvidenceSnapshot_v1. (H.14)

H.4.3 Identity record

Define:

IdentityRecord
:= (
IdentityID,
IdentityClass,
InstrumentID,
ContractID,
Owner,
Obligor,
SignedQuantity,
Currency,
Maturity,
AccountID,
AgreementID,
TransactionLineage,
LifecycleStatus,
ParentIdentityID,
SuccessorIdentityID
). (H.15)

The identity record should distinguish:

- economic instrument;

- legal claim;
- account position;
- settlement obligation;
- collateral relation;
- recovery identity.

A valid identity conversion creates a successor link rather than overwriting the original identity.

H.4.4 Frame record

Define:

FrameRecord
:= (
FrameID,
FrameType,
Boundary,
Observer,
Authority,
CoordinateSchema,
TimeRule,
TraceSource,
InvariantSchema,
ResidualSchema
). (H.16)

Possible FrameType values include:

- Market;
- Collateral;
- Treasury;
- Risk;
- Settlement;
- Accounting;
- Legal;
- Regulatory.

H.4.5 Frame-state record

Define:

FrameState
:= (
FrameStateID,

FrameID,
IdentityID,
ProtocolID,
EconomicTime,
ObservationTime,
RecognitionTime,
Coordinates,
ChargeVector,
GateStatus,
TraceReferences,
ResidualReferences
). (H.17)

Coordinates should use a typed schema.

For example, a collateral frame may contain:

Coordinates_C
:= (
MarketMark,
CollateralFactor,
Eligibility,
ConcentrationAdjustment,
AdmittedValue,
Encumbrance,
MarginBuffer
). (H.18)

H.4.6 Projection record

A projection converts frozen evidence into a derived state.

Define:

ProjectionRecord
:= (
ProjectionID,
SnapshotID,
OperatorID,
OperatorVersion,
Inputs,
Outputs,
Units,
Assumptions,
Uncertainty,
Status,
CreatedAt
). (H.19)

Possible projections include:

- CAPM value;
- Q coordinate;
- phase;

- collateral value;
- margin buffer;
- expected ledger return;
- transport residual.

Every projection must identify:

- its source evidence;
- its operator version;
- its assumptions;
- its units.

H.4.7 Charge record

Define:

ChargeRecord
:= (
ChargeID,
IdentityID,
ProtocolID,
Field,
ChargeClass,
StructuralSign,
StructuralMagnitude,
OperationalStatus,
EffectiveMagnitude,
ScreeningRule,
Units,
Boundary,
ValidFrom,
ValidTo
). (H.20)

The record should distinguish:

StructuralCharge. (H.21)

OperationalCharge. (H.22)

EffectiveCharge. (H.23)

For example:

q_C^struct ≠ 0. (H.24)

q_C^op = 0 before call. (H.25)

q_C^op = q_C^struct after valid call. (H.26)

H.4.8 Candidate-claim record

Define:

ClaimRecord
:= (
ClaimID,
SubjectID,
ProtocolID,
SnapshotID,
ClaimType,
ClaimLevel,
ClaimContent,
SupportingEvidence,
ContradictingEvidence,
MaximumPermittedLevel,
Status,
InvalidationRule,
CreatedAt
). (H.27)

Possible ClaimType values include:

- buffer breach;
- margin-call candidate;
- admitted margin event;
- closure candidate;
- frame mismatch;
- charge mismatch;
- residual warning;
- recursive-risk warning.

Possible ClaimLevel values include:

- Measurement;
- Exposure;
- Candidate;
- Event;
- Episode;
- World.

The system should enforce:

ClaimLevel ≤ MaximumPermittedLevel. (H.28)

H.4.9 Gate-decision record

Define:

GateDecision
:= (
GateDecisionID,
ClaimID,
GateType,
GateRuleVersion,
AuthorityType,
AuthorityID,
EvidenceSnapshotID,
Decision,
DecisionTime,
Strength,
ConditionsPassed,
ConditionsFailed,
TraceID,
ResidualIDs,
ReopenRule
). (H.29)

Possible Decision values include:

- Rejected;
- Observed;
- Admitted;
- Deferred;
- Waived;
- Contested;
- Reopened.

A gate decision must contain:

- rule;
- authority;
- evidence;
- result;
- residual.

H.4.10 Action record

Define:

ActionRecord

:= (
ActionID,
GateDecisionID,
IdentityID,
ActionType,
InstructionTime,
ExecutionTime,
InstructionAmount,
ExecutionAmount,
InstructionQuantity,
ExecutionQuantity,
Authority,
Status,
ExpectedLedgerEffects
). (H.30)

Examples include:

- collateral request;
- collateral transfer;
- voluntary sale;
- forced liquidation;
- debt repayment;
- default declaration.

H.4.11 Ledger-return record

Define:

LedgerReturn
:= (
LedgerReturnID,
ActionID,
LedgerFrame,
RecognitionTime,
RecordedAmount,
RecordedQuantity,
RecordedStatus,
TransportRule,
ExpectedValue,
ObservedValue,
ResidualID,
CompletionStatus
). (H.31)

The action and ledger-return records must remain separate.

Joining them is permitted.

Collapsing them into one mutable status record is not.

H.4.12 Residual record

Define:

ResidualRecord
:= (
ResidualID,
SubjectID,
ProtocolID,
ResidualType,
SourceObjectType,
SourceObjectID,
ExpectedState,
ObservedState,
Difference,
Units,
Severity,
Materiality,
Status,
DetectedAt,
ResolvedAt,
ResolutionTraceID,
CarryForwardRule
). (H.32)

Possible ResidualType values include:

- Amount;
- Quantity;
- Timing;
- Identity;
- Charge;
- Authority;
- Transport;
- Legal;
- Accounting;
- Model;
- MissingEvidence;
- Loop;
- Gate.

H.4.13 Transport record

Define:

TransportRecord
:= (
TransportID,
IdentityID,
ProtocolID,
SourceFrameID,
TargetFrameID,
ConnectionID,
ConnectionVersion,
SourceStateID,
ExpectedTargetState,
ObservedTargetState,
TransportResidualID,
InvariantResult,
ChargeResult,
TimingResult,
Status
). (H.33)

H.4.14 Loop record

Define:

LoopRecord
:= (
LoopID,
IdentityID,
ProtocolID,
FramePath,
EdgeTransportIDs,
StartState,
ExpectedReturnState,
ObservedReturnState,
LoopResidualID,
InvariantResult,
AuthorityResult,
ClosureStatus
). (H.34)

H.4.15 Closure certificate

Define:

ClosureCertificate
:= (
CertificateID,
SubjectID,
EventID,
ProtocolID,
IdentityResult,
AmountResult,
SettlementResult,
DebtResult,

CollateralResult,
AccountingResult,
LegalResult,
ChargeResult,
TransportResult,
ResidualDisclosureResult,
FinalStatus,
CertifiedBy,
CertifiedAt,
ReopenConditions
). (H.35)

Possible FinalStatus values include:

- Closed;
- ClosedWithResidual;
- Partial;
- Open;
- Contested;
- DefaultConverted;
- Invalid.

H.4.16 Revision record

Define:

RevisionRecord
:= (
RevisionID,
ObjectType,
OriginalObjectID,
OriginalVersion,
RevisedObjectID,
RevisedVersion,
NewEvidence,
Reason,
Authority,
ImpactAssessment,
CreatedAt
). (H.36)

A revision must point backward.

It may not replace history silently.

H.4.17 Model-comparison record

Define:

ModelComparison
:= (
ComparisonID,
ProtocolID,
DatasetVersion,
TrainingPeriod,
TestPeriod,
AdvancedModelID,
BenchmarkModelIDs,
Metrics,
ComplexityPenalty,
UtilityScores,
Decision,
ReductionCertificateID
). (H.37)

H.4.18 Reduction certificate

Define:

ReductionCertificate
:= (
ReductionID,
ModelBefore,
FailedAdmissionCondition,
Evidence,
Residual,
ModelAfter,
ClaimsRetained,
ClaimsRemoved,
DecisionAuthority,
DecisionTime
). (H.38)

H.5 Identifier and Version Rules

H.5.1 Stable identity identifiers

An IdentityID should remain stable while the same bounded identity survives.

A new IdentityID is required when:

- ownership changes materially;
- obligor changes through novation;
- default converts a performing claim into a recovery claim;
- a contract is extinguished and replaced;
- aggregation changes the modeled subject.

H.5.2 Version identifiers

Objects that may change through corrected evidence or new rules should carry:

ObjectID. (H.39)

VersionNumber. (H.40)

ParentVersionID. (H.41)

The object family remains linked.

The prior version remains queryable.

H.5.3 Effective-time and system-time

Every material record should carry two time dimensions.

Effective time

When the financial state or rule applied.

System time

When the system learned or recorded it.

Define:

$t_{effective}$. (H.42)

$t_{recorded}$. (H.43)

In general:

$t_{effective} \neq t_{recorded}$. (H.44)

This distinction is essential for:

- delayed recognition;
- backdated corrections;
- settlement;
- restatement;
- legal reclassification;
- evidence-time consistency.

H.5.4 Bitemporal storage

A bitemporal record contains:

ValidFrom. (H.45)

ValidTo. (H.46)

RecordedFrom. (H.47)

RecordedTo. (H.48)

This permits the runtime to answer two different questions:

What is now believed to have been true at time t? (H.49)

What did the system believe at time t? (H.50)

The second question is required for prospective gate testing.

H.6 Append-Only Event Ledger

H.6.1 Why event sourcing is preferred

A mutable status table may show only:

CurrentStatus = Closed. (H.51)

It may conceal:

- initial breach;
- rejected call;
- revised mark;
- partial cure;
- settlement failure;
- reopening;
- final closure.

An append-only ledger preserves the full sequence.

The source implementation blueprint recommends event sourcing or versioned append-only tables specifically to prevent silent historical rewriting.

H.6.2 Ledger event

Define:

LedgerEvent
:= (
EventID,
EventType,
SubjectID,
ProtocolID,

EffectiveTime,
RecordedTime,
Actor,
Authority,
Payload,
ParentEventID,
SupersedesEventID,
EvidenceSnapshotID,
ResidualIDs
). (H.52)

H.6.3 Example event sequence

A margin episode may produce:

1. `BUFFER_CALCULATED`
2. `BREACH_CANDIDATE_CREATED`
3. `CALL_ADMITTED`
4. `COLLATERAL_INSTRUCTED`
5. `COLLATERAL_PARTIALLY_RECEIVED`
6. `COLLATERAL_REJECTED`
7. `SALE_INSTRUCTED`
8. `SALE_EXECUTED`
9. `SALE_SETTLED`
10. `DEBT_UPDATED`
11. `ACCOUNTING_RECOGNIZED`
12. `CLOSURE_CERTIFIED`

No event should be deleted merely because a later event supersedes it.

H.6.4 State reconstruction

The state at ledger tick k is:

$\Sigma_k = \text{Fold}(\text{Event}_1,\text{Event}_2,\ldots,\text{Event_k}).$ (H.53)

where Fold is the deterministic state-reconstruction operator.

The next state is:

$\Sigma_k_{+1} = \text{Apply}(\Sigma_k,\text{Event_k}_{+1}).$ (H.54)

This implements the recursive law:

$\text{Trace}_k + \text{Residual}_k \rightarrow \text{Ledger}_{k+1}.$ (H.55)

H.6.5 Event immutability

Once committed:

EventPayload_k is immutable. (H.56)

Correction occurs through:

CorrectionEvent_$k+1$. (H.57)

not through mutation of Event_k.

H.7 State Machines

H.7.1 Claim-state machine

Define claim states:

ClaimState
$\in$ {
Draft,
Measured,
Candidate,
GatePending,
Admitted,
Rejected,
Deferred,
TransportTesting,
Supported,
Reduced,
Reopened,
Invalidated
}. (H.58)

Valid transitions include:

Draft → Measured. (H.59)

Measured → Candidate. (H.60)

Candidate → GatePending. (H.61)

GatePending → Admitted. (H.62)

GatePending → Rejected. (H.63)

Admitted → TransportTesting. (H.64)

TransportTesting → Supported. (H.65)

TransportTesting → Reduced. (H.66)

Supported → Reopened. (H.67)

Reopened → Supported. (H.68)

Reopened → Invalidated. (H.69)

Invalid transitions should be rejected by the runtime.

For example:

Measured → Supported (H.70)

is invalid because no gate or transport test has occurred.

H.7.2 Margin-event state machine

Define:

MarginState
∈ {
Safe,
BreachCandidate,
CallPending,
CallOpen,
CureInProgress,
LiquidationInProgress,
DefaultPending,
PartiallyClosed,
Closed,
ClosedWithResidual,
DefaultConverted,
Contested,
Reopened
}. (H.71)

H.7.3 Residual-state machine

Define:

ResidualStatus
∈ {
Open,
Investigating,
Explained,
Corrected,
Accepted,
Escalated,
CarriedForward,
Closed
}. (H.72)

A residual may move:

Open → Explained. (H.73)

Explained → Closed. (H.74)

Open → Accepted. (H.75)

Accepted → CarriedForward. (H.76)

Closed → Reopened. (H.77)

A residual should never move directly from:

Open → Deleted. (H.78)

H.7.4 Identity lifecycle

Define:

IdentityStatus
∈ {
Proposed,
Active,
Restricted,
Defaulted,
Recovery,
Extinguished,
Disputed
}. (H.79)

Valid default conversion is:

Active → Defaulted → Recovery. (H.80)

Invalid silent deletion is:

Active → Missing. (H.81)

H.8 Runtime Pipeline

H.8.1 Stage 1 — Protocol resolution

Given a request and subject, resolve:

P* = ApplicableProtocol(Subject,EventTime,Authority). (H.82)

If no applicable protocol exists:

RuntimeStatus = ProtocolMissing. (H.83)

No advanced claim should proceed.

H.8.2 Stage 2 — Evidence freeze

Create:

Snapshot_t = FreezeEvidence($\mathcal{J}$_t). (H.84)

The snapshot includes missing fields and source versions.

H.8.3 Stage 3 — Identity resolution

Resolve:

$$K_t = \mathrm{ResolveIdentity}(\mathrm{Snapshot}_t, P^*). \quad \text{(H.85)}$$

If identity confidence falls below tolerance:

$$\mathrm{IdentityStatus} = \mathrm{Contested}. \quad \text{(H.86)}$$

The runtime should stop charge conservation or loop-closure claims until identity is resolved.

H.8.4 Stage 4 — Projection

Apply registered operators:

$$Y_j = \mathrm{Operator}_j(\mathrm{Snapshot}_t, P^*). \quad \text{(H.87)}$$

Possible outputs include:

- R;
- Q;
- θ;
- h;
- B;
- C_{call};
- $\vec{q}$;
- expected ledger state.

H.8.5 Stage 5 — Claim compilation

Translate outputs into a typed candidate:

$$\mathrm{Claim} = \mathrm{Compile}(Y, P^*, \mathrm{ClaimSchema}). \quad \text{(H.88)}$$

The compiler assigns:

- object type;
- claim level;
- supporting evidence;
- contradictions;
- maximum permitted level;

- invalidation rule.

H.8.6 Stage 6 — Gate evaluation

Evaluate:

GateResult = EvaluateGate(Claim,P*,Authority). (H.89)

The result includes:

- decision;
- passed conditions;
- failed conditions;
- trace;
- residual.

H.8.7 Stage 7 — Action monitoring

If the gate is admitted, monitor authorized actions:

ActionSet = ObserveActions(GateDecisionID). (H.90)

H.8.8 Stage 8 — Ledger-return monitoring

For each action, calculate:

ExpectedLedger = U_AL(Action,P*). (H.91)

Observe:

ActualLedger. (H.92)

Then:

δ_spin = ActualLedger − ExpectedLedger. (H.93)

H.8.9 Stage 9 — Frame transport

For each required edge A → B:

ExpectedState_B = U_AB(State_A). (H.94)

Residual_AB = State_B^obs − ExpectedState_B. (H.95)

H.8.10 Stage 10 — Closure certification

Compute:

ClosureCertificate = Certify(
Identity,
Gate,
Action,
LedgerReturn,
Charge,
Transport,
Residual
). (H.96)

H.8.11 Stage 11 — Benchmark comparison

Compare:

AdvancedModel. (H.97)

SimplerBenchmarks. (H.98)

If the advanced model fails its admission or utility threshold:

Issue ReductionCertificate. (H.99)

H.8.12 Stage 12 — Recursive update

Write:

Trace + Residual + ClosureCertificate (H.100)

to the event ledger.

Update:

$\Theta_k_{+1} = U_\Theta(\Theta_k, L_k_{+1}, \mathscr{R}_k)$. (H.101)

H.9 Machine-Readable Runtime Pseudocode

```
INPUT:
    subject_id
    evaluation_time
    requested_claim_type

1. protocol = resolve_protocol(subject_id, evaluation_time)

2. if protocol is missing:
       return status = "PROTOCOL_MISSING"

3. snapshot = freeze_evidence(
       subject_id,
       evaluation_time,
       protocol
   )

4. identity = resolve_identity(snapshot, protocol)
```

```
5. if identity.status == "CONTESTED":
       create_residual(type="IDENTITY")
       return status = "IDENTITY_UNRESOLVED"

6. projections = run_registered_operators(
       snapshot,
       protocol,
       requested_claim_type
   )

7. claim = compile_typed_claim(
       identity,
       projections,
       protocol
   )

8. gate_result = evaluate_gate(
       claim,
       snapshot,
       protocol
   )

9. record_gate_trace_and_residual(gate_result)

10. if gate_result.decision != "ADMITTED":
        compare_with_benchmarks(claim)
        return governed_claim_output(claim, gate_result)

11. actions = collect_authorized_actions(gate_result)

12. for each action:
        expected_ledger = transport_action_to_ledger(
            action,
            protocol
        )

        observed_ledger = read_ledger_returns(
            action,
            protocol
        )

        closure_defect = compare(
            expected_ledger,
            observed_ledger
        )

        record_residuals(closure_defect)

13. for each required frame edge:
        expected_target = transport_frame_state(
            source_state,
            edge_connection
        )

        observed_target = read_target_state()

        record_transport_residual(
            observed_target - expected_target
        )

14. certificate = evaluate_closure(
        identity,
        gate_result,
        actions,
        ledger_returns,
```

```
        charge_balance,
        transport_results,
        residuals
    )

15. benchmark_result = compare_advanced_and_simple_models()

16. if advanced_model_not_supported:
        issue_reduction_certificate()

17. append_all_events_to_immutable_ledger()

18. update_recursive_parameters()

OUTPUT:
    typed_claim
    gate_status
    action_status
    ledger_return_status
    transport_results
    residual_register
    closure_certificate
    benchmark_result
    reduction_or_revision_instruction
```

This runtime is a specialization of the source article's machine-readable process: validate the boundary and lineage, classify the object, evaluate the gate, preserve alternatives and residual, transport the claim, compare simpler models, and retain complete revision history.

H.10 Relational Database Schema

H.10.1 Minimum tables

A minimum relational implementation may contain:

1. `protocol`
2. `protocol_version`
3. `evidence_snapshot`
4. `evidence_source`
5. `identity`
6. `identity_relation`
7. `frame`
8. `frame_state`
9. `projection`
10. `charge`
11. `claim`
12. `gate_decision`
13. `action`
14. `ledger_return`

15. `residual`
16. `transport`
17. `loop_result`
18. `closure_certificate`
19. `model_run`
20. `model_comparison`
21. `revision`
22. `reduction_certificate`
23. `ledger_event`

H.10.2 Protocol table

Column	Type	Meaning
protocol_id	Identifier	Protocol family
version	Integer	Exact rule version
boundary_json	Structured	System boundary
observation_rule_json	Structured	Data construction
gate_rule_json	Structured	Admission rules
residual_rule_json	Structured	Residual typing
transport_rule_json	Structured	Frame connections
authority_json	Structured	Valid authorities
effective_from	Timestamp	Rule activation
effective_to	Timestamp	Rule expiry

H.10.3 Evidence snapshot table

Column	Type	Meaning
snapshot_id	Identifier	Immutable snapshot
subject_id	Identifier	Financial subject
protocol_id	Identifier	Applied protocol
protocol_version	Integer	Applied version
evidence_time	Timestamp	Information cutoff
recorded_time	Timestamp	Storage time
source_manifest	Structured	Source lineage
missing_fields	Structured	Required missing evidence
content_hash	Text	Integrity check

H.10.4 Gate-decision table

Column	Type	Meaning
gate_decision_id	Identifier	Decision record

Column	Type	Meaning
claim_id	Identifier	Candidate evaluated
gate_type	Text	Margin, settlement, accounting, etc.
decision	Text	Admitted, rejected, deferred, waived
authority_id	Identifier	Decision authority
decision_time	Timestamp	Commitment time
conditions_passed	Structured	Passed tests
conditions_failed	Structured	Failed tests
trace_event_id	Identifier	Persistent trace
reopen_rule	Structured	Invalidation condition

H.10.5 Residual table

Column	Type	Meaning
residual_id	Identifier	Residual record
subject_id	Identifier	Related subject
residual_type	Text	Amount, timing, identity, etc.
expected_state	Structured	Protocol expectation
observed_state	Structured	Observed state
difference	Structured	Typed mismatch
severity	Numeric or ordinal	Consequence
status	Text	Open, explained, accepted, etc.
detected_at	Timestamp	Detection time
resolved_at	Timestamp	Resolution time
carry_forward	Boolean	Enters next ledger

H.10.6 Event-ledger table

Column	Type	Meaning
event_id	Identifier	Immutable event
event_type	Text	State transition
subject_id	Identifier	Related identity
effective_time	Timestamp	Financial effective time
recorded_time	Timestamp	System record time
actor_id	Identifier	Actor
authority_id	Identifier	Authority
payload	Structured	Event data
parent_event_id	Identifier	Event lineage
supersedes_event_id	Identifier	Correction relation
evidence_snapshot_id	Identifier	Evidence basis

H.11 Document and Object Storage

H.11.1 Evidence documents

Some evidence does not fit naturally into relational columns:

- contracts;
- collateral agreements;
- statements;
- valuation reports;
- legal opinions;
- accounting support;
- screenshots;
- model files.

Store these in an evidence vault or object store.

The structured database should reference:

DocumentID. (H.102)

DocumentVersion. (H.103)

ContentHash. (H.104)

AccessPolicy. (H.105)

H.11.2 Evidence immutability

The original document should remain unchanged.

Annotations, extracted variables, or corrected versions should be stored as linked objects.

Define:

Document_v2 = Revision(Document_v1,NewEvidence). (H.106)

not:

Document_v1 := Overwrite(Document_v2). (H.107)

H.12 Schema Validation

H.12.1 Type validation

Every record should be validated against a machine-readable schema.

Possible tools include:

- JSON Schema;
- Pydantic;

- protocol buffers;
- relational constraints.

The technology is secondary.

The requirement is that invalid objects cannot enter the authoritative ledger silently.

H.12.2 Unit validation

Every numeric coordinate should carry a unit.

Examples include:

- USD;
- shares;
- percentage;
- radians;
- days;
- dimensionless score.

A comparison is invalid when:

Unit_A ≠ Unit_B (H.108)

unless a declared conversion exists.

H.12.3 Currency validation

When financial amounts use different currencies, transport requires:

FXConnection. (H.109)

Without an FX rule:

Amount_USD – Amount_GBP (H.110)

is not a meaningful residual.

The minimum calibration case avoids this complication by using one currency.

H.12.4 Boundary validation

A charge or neutrality calculation must reference:

BoundaryID. (H.111)

The runtime should reject:

ChargeBalance without BoundaryID. (H.112)

H.12.5 Time validation

The runtime should enforce:

EvidenceTime ≤ DecisionTime. (H.113)

ActionInstructionTime ≤ ActionExecutionTime. (H.114)

GateTime ≤ ClosureTime. (H.115)

Exceptions must create timing residuals rather than being silently accepted.

H.13 Invariants and Runtime Constraints

H.13.1 Identity invariant

For ordinary transport:

Identity_B = TransportIdentity_AB(Identity_A). (H.116)

A mismatch creates:

ResidualType = Identity. (H.117)

H.13.2 Charge invariant

For a non-converting edge:

$\vec{q}_B = \vec{q}_A$. (H.118)

A difference requires either:

- a declared charge-conversion vertex;
- a charge residual.

H.13.3 Gate invariant

An admitted event must contain:

ValidRule
∧ ValidAuthority
∧ EvidenceSnapshot
∧ DecisionTime
∧ Trace. (H.119)

H.13.4 Closure invariant

A closure certificate cannot be `Closed` when:

RequiredResidualDisclosure = False. (H.120)

or:

RequiredLedgerReturn = Open. (H.121)

or:

IdentityStatus = Contested. (H.122)

H.13.5 Revision invariant

A revised object must contain:

ParentVersionID ≠ Null. (H.123)

unless it is the first version.

H.13.6 Model-promotion invariant

An advanced model cannot be marked `Supported` without:

BenchmarkComparisonID. (H.124)

and:

ComplexityPenaltyDeclared = True. (H.125)

H.14 Authority Control

H.14.1 Analytical authority

Analytical services may:

- calculate;
- classify;
- estimate;
- warn;
- recommend review.

H.14.2 Institutional authority

Institutional actors may:

- issue calls;

- approve collateral;
- liquidate;
- recognize accounting entries;
- declare legal default;
- certify closure.

The software must distinguish:

ModelRecommendation. (H.126)

InstitutionalDecision. (H.127)

H.14.3 Authority matrix

Define:

$A_{role,action} \in \{0,1\}$. (H.128)

For example:

Role	Calculate buffer	Admit call	Execute sale	Certify accounting	Declare default
Research model	1	0	0	0	0
Margin engine	1	1	0	0	0
Broker operations	0	0	1	0	0
Accounting authority	0	0	0	1	0
Legal authority	0	0	0	0	1

The actual matrix is institution-specific.

H.14.4 Separation of duties

No single runtime component should be able to:

1. calculate the event;
2. approve the event;
3. modify the evidence;
4. certify closure;

without independent trace.

This reduces false self-confirmation.

H.15 Action–Ledger Engine

H.15.1 Branch-specific transporter registry

Define:

TransporterRegistry_AL
:= {
CollateralBranch ↦ U_AL^coll,
DeleveragingBranch ↦ U_AL^delev,
LiquidationBranch ↦ U_AL^liq,
DefaultBranch ↦ U_AL^def
}. (H.129)

The correct transporter is selected by:

- protocol;
- branch;
- identity class;
- effective date.

H.15.2 Expected ledger state

For action a:

L^_a = U_AL^(b)(A_a). (H.130)

H.15.3 Observed ledger state

Let:

L_a^obs = AssembleLedgerReturns(ActionID). (H.131)

H.15.4 Closure defect

Define:

δ_a = L_a^obs − L^_a. (H.132)

The weighted defect is:

Δ_a² = δ_a^⊤W_aδ_a. (H.133)

H.15.5 Age-aware expected state

At elapsed time s:

L^_a(s) = U_AL^(b)(s)A_a. (H.134)

Immediately after execution, incomplete settlement may be expected.

The runtime should therefore compare observed return with the expected return at the same elapsed closure time.

H.15.6 Nested closures

An outer margin event may depend on inner events:

MarginClosure
requires
TradeSettlement
∧ CollateralAdmission
∧ DebtUpdate
∧ AccountingReturn. (H.135)

The runtime should represent this as a dependency graph.

A parent event cannot close while a mandatory child event remains open.

H.16 Transport Engine

H.16.1 Connection registry

Define:

ConnectionRegistry
:= {
(SourceFrame,TargetFrame,ProtocolVersion)
↦ ConnectionDefinition
}. (H.136)

Each connection definition contains:

- input schema;
- output schema;
- units;
- timing rule;
- transformation;
- invariant;
- charge rule;
- expected residual tolerance.

H.16.2 Transport calculation

For source state x_A:

$\hat{x}_B = T_{AB}(x_A; \mathcal{A}_{AB})$. (H.137)

The result should store:

- source state;
- connection version;
- expected target;
- observed target;
- residual;
- invariant result.

H.16.3 Noninvertible transport

Collateral filtering and accounting aggregation may lose information.

Therefore:

T_{AB}^{-1} may not exist. (H.138)

The runtime should not manufacture an inverse.

Instead, it should define a reconstruction map:

$R_{BA} : \mathrm{Range}(T_{AB}) \to \mathrm{ComparableSubspace}_A$. (H.139)

Loop closure should then be evaluated only on the comparable invariant subspace.

H.16.4 Path transport

For path p:

$T_p = T_n \circ T_{n-1} \circ \dots \circ T_1$. (H.140)

The system should store every edge rather than only the final composite result.

This permits residual localization.

H.16.5 Loop transport

For loop ℓ:

$\hat{x}_{return} = H_\ell^P x_{start}$. (H.141)

The loop residual is:

$\mathscr{R}_\ell = x_{return}^{obs} - \hat{x}_{return}$. (H.142)

The runtime should separate:

- expected fee;
- expected timing effect;
- expected valuation adjustment;
- unexplained residual.

H.17 Residual Ledger

H.17.1 Residual is a first-class object

Residual must not be stored only as:

ErrorMessage. (H.143)

It is a typed object with:

- source;
- expected state;
- observed state;
- severity;
- status;
- future carry-forward rule.

H.17.2 Residual aggregation

Residuals may be aggregated only after transport into compatible units and frames.

For residuals r_j:

$$R_{total}^2 = \Sigma_j w_j \| T_{j\to*}(r_j) \|^2. \quad (H.144)$$

where $T_{j\to*}$ transports residual j into the declared comparison frame.

H.17.3 No cancellation across incompatible types

A positive monetary residual should not cancel a negative identity residual.

Therefore:

r_amount + r_identity (H.145)

is not a valid scalar sum by default.

Use a residual vector:

$\mathscr{R}$ = (r_amount,r_identity,r_timing,r_authority,...). (H.146)

H.17.4 Residual carry-forward

For residual r_k:

r_k_{+1}^inherited = CarryForward(r_k,L_k_{+1}). (H.147)

Possible CarryForward modes include:

- None;
- UntilResolved;
- PermanentTrace;
- ParameterPenalty;
- GateRestriction;
- IdentityDispute.

H.17.5 Residual reopening

A closed residual may be reopened when new evidence invalidates the prior explanation.

The runtime should create:

ResidualReopenedEvent. (H.148)

The previous resolution remains visible.

H.18 Benchmark and Reduction Engine

H.18.1 Model registry

Define:

ModelRegistry
:= (
ModelID,
Version,
InputSchema,
OutputSchema,
ClaimCeiling,
Parameters,
TrainingData,
Falsifiers,
BenchmarkRequirements
). (H.149)

H.18.2 Required benchmark mappings

The runtime should encode mandatory comparisons.

Complex model

Must compare with:

Flexible Real Pair. (H.150)

Spinor model

Must compare with:

Workflow Model. (H.151)

Gauge model

Must compare with:

Pairwise Reconciliation. (H.152)

Dirac model

Must compare with:

Hybrid State-Space Model. (H.153)

Recursive model

Must compare with:

Current-State or Fixed-Parameter Model. (H.154)

H.18.3 Utility calculation

For model j:

$U_j = w_PP_j + w_DD_j + w_GG_j - \lambda_CC_j.$ (H.155)

where:

- P_j = predictive utility;
- D_j = diagnostic utility;
- G_j = governance utility;
- C_j = complexity.

H.18.4 Promotion rule

Promote model M_a over benchmark M_b only when:

U_a − U_b > δ_U. (H.156)

where δ_U is a predeclared minimum gain.

H.18.5 Reduction rule

If:

U_a − U_b ≤ δ_U, (H.157)

issue:

ReductionCertificate(M_a → M_b). (H.158)

The software should not merely lower a confidence score while continuing to display the stronger terminology.

It should change the retained model label.

H.19 Research and Production Separation

H.19.1 Research environment

The research environment may:

- explore alternative definitions;
- estimate models;
- simulate systems;
- test new residual types;
- compare protocols.

H.19.2 Controlled runtime

The controlled runtime should execute only:

- approved protocol versions;
- registered operators;
- validated schemas;
- authorized connections;
- approved gate rules.

H.19.3 Promotion process

A research object enters the controlled runtime only after:

Definition
→ Test
→ Review
→ Versioned Approval
→ Deployment. (H.159)

H.19.4 No notebook authority

A notebook result should not directly become:

- a margin event;
- a legal conclusion;
- an accounting entry;
- a closure certificate.

It must pass through the appropriate gate and authority.

H.20 API Boundaries

H.20.1 Read-only evidence interface

The evidence service should expose:

- frozen snapshots;
- source lineage;
- document versions;
- timestamps;
- missing-field status.

It should not permit analytical models to alter source evidence.

H.20.2 Projection interface

Example operation:

```
POST /projections
```

Input:

- SnapshotID;

- OperatorID;
- ProtocolVersion.

Output:

- ProjectionRecord;
- uncertainty;
- assumptions;
- status.

H.20.3 Claim interface

Example operation:

```
POST /claims
```

Input:

- ProjectionIDs;
- ClaimType;
- ProtocolID.

Output:

- typed claim;
- maximum closure level;
- invalidation rule.

H.20.4 Gate interface

Example operation:

```
POST /gate-decisions
```

Input:

- ClaimID;
- AuthorityID;
- EvidenceSnapshotID.

Output:

- gate decision;
- trace;

- residuals;
- reopen rule.

H.20.5 Transport interface

Example operation:

```
POST /transports
```

Input:

- SourceStateID;
- TargetFrameID;
- ConnectionVersion.

Output:

- expected target;
- observed target;
- residual;
- invariant result.

H.20.6 Closure interface

Example operation:

```
POST /closure-certificates
```

Input:

- EventID;
- required action IDs;
- required ledger-return IDs;
- transport IDs;
- residual IDs.

Output:

- closure status;
- failed conditions;
- residual disclosure;
- reopen conditions.

H.21 Human-Readable Claim Output

Every system-generated claim should expose a standard card.

Claim

What is being asserted?

Protocol

Under which boundary, rule, horizon, and authority?

Evidence cutoff

What was known and when?

Identity

Which bounded financial subject is involved?

Projection

Which variables were calculated?

Gate

Was the candidate admitted?

Action

What outward consequence occurred?

Ledger return

What has been recognized or settled?

Transport

Which frames were tested?

Residual

What remains open?

Claim ceiling

What stronger interpretation is prohibited?

Invalidation

What evidence would reopen or reject the claim?

This mirrors the source architecture's insistence that protocol, gate, residual, transport, authority, and revision remain attached to every mature claim.

H.22 Audit Queries

H.22.1 Calls without authority

```
Find all admitted margin calls
where authority_id is missing
or authority was not valid at gate_time.
```

Expected result:

Zero valid events. (H.160)

H.22.2 Closures with open mandatory ledgers

```
Find all closure certificates marked Closed
where any required ledger-return status is Open,
Partial, Missing, or Contested.
```

These are potential false closures.

H.22.3 Residuals missing from closure certificates

```
Find all material residuals linked to an event
that are absent from the event's closure certificate.
```

H.22.4 Mutated evidence

```
Find evidence records with the same snapshot_id
but different content hashes.
```

Expected result:

Zero.

H.22.5 Unsupported advanced claims

```
Find all claims labelled Gauge, Spinor, Mass, or Dirac
without the required admission-test records.
```

H.22.6 Revisions without parent trace

```
Find revised protocols, models, or claims
whose parent_version_id is missing.
```

H.22.7 Charge changes without vertices

```
Find charge records whose structural charge changed
without a linked transfer, conversion, activation,
extinguishment, or default vertex.
```

H.22.8 Cross-frame claims without transport

```
Find claims involving more than one frame
without a TransportRecord or declared connection.
```

H.23 Test Suite

H.23.1 Unit tests

Test individual functions:

- buffer calculation;
- call amount;
- Q calculation;
- collateral transport;
- charge activation;
- residual construction;
- closure-condition logic.

H.23.2 Schema tests

Reject:

- missing ProtocolID;
- missing units;
- invalid timestamps;
- unknown authority;
- unsupported frame type;
- unversioned connection.

H.23.3 State-transition tests

Verify that invalid transitions fail.

Examples:

Safe → Closed without event. (H.161)

CallOpen → Closed without ledger return. (H.162)

OpenResidual → Deleted. (H.163)

H.23.4 Property tests

Useful properties include:

Buffer identity

$B = C + nhR - D.$ (H.164)

Charge preservation

Ordinary transport should not change structural charge.

Evidence immutability

A frozen snapshot's hash remains constant.

Revision lineage

Every noninitial version has a parent.

Closure monotonicity is not assumed

A closed event may reopen when new evidence appears.

Therefore:

Closed → Reopened is valid. (H.165)

H.23.5 Counterfactual tests

Create paired cases differing only in:

- authority;
- haircut;
- settlement;
- identity;
- residual disclosure.

The runtime should change only the corresponding result.

H.23.6 Null-model tests

The system must be able to conclude:

- no charge;
- no independent spinor;
- no gauge structure;
- no useful complex phase;
- no recursive effect.

A runtime that always returns the strongest term has failed.

H.24 Observability and Monitoring

H.24.1 Runtime health metrics

Track:

- unprocessed evidence;
- gate queue length;
- open residual count;
- unresolved identity count;
- transport failure rate;
- closure latency;
- false-closure rate;
- model-reduction frequency.

H.24.2 Action–ledger backlog

Define:

Backlog_t
:= OpenExpectedLedgerReturns_t − CompletedLedgerReturns_t. (H.166)

H.24.3 Residual velocity

Define:

v_$\mathscr{R}$,t := NewMaterialResiduals_t/Δt. (H.167)

H.24.4 Residual service rate

Define:

$\mu_{\mathscr{R},t} := \text{ResolvedMaterialResiduals}_t/\Delta t.$ (H.168)

Residual backlog grows when:

$v_{\mathscr{R},t} > \mu_{\mathscr{R},t}.$ (H.169)

H.24.5 Closure-capacity warning

A practical warning is:

$\text{ActionArrivalRate} > \text{VerifiedLedgerReturnRate}.$ (H.170)

This is the operational version of action outrunning coherent closure capacity.

H.25 Security and Access Control

H.25.1 Data sensitivity

The runtime may contain:

- positions;
- leverage;
- collateral;
- debt;
- customer identities;
- legal disputes;
- default status;
- accounting corrections.

Access should follow least privilege.

H.25.2 Role-based access

Suggested roles include:

- Researcher;
- RiskAnalyst;
- MarginOperator;

- TreasuryOperator;
- Accountant;
- LegalReviewer;
- Auditor;
- SystemAdministrator.

H.25.3 Field-level control

A user may be permitted to view:

- aggregated model metrics;

but not:

- customer identity;
- legal documents;
- raw transaction details.

H.25.4 Immutable audit trail

Every access or modification should record:

- actor;
- action;
- object;
- time;
- authority;
- prior and new versions.

H.25.5 Model-access control

Only approved model versions should be callable in the controlled runtime.

Exploratory models remain isolated.

H.26 Privacy-Preserving Research

H.26.1 De-identification

Replace direct identifiers with:

ResearchSubjectID. (H.171)

The mapping to real identity should be stored separately.

H.26.2 Data minimization

Collect only fields required by the declared protocol.

A broader dataset is not automatically a better dataset.

H.26.3 Synthetic development data

Use synthetic data for:

- interface development;
- unit testing;
- training;
- demonstration.

Real confidential data should enter only approved analytical environments.

H.27 Minimal Technology Choices

H.27.1 Modest implementation

A modest research implementation may use:

- Python for operators and model evaluation;
- PostgreSQL for structured records;
- object storage for evidence snapshots;
- JSON Schema or Pydantic for validation;
- a lightweight API framework;
- Git for protocol and code versioning;
- notebooks for reproducible analysis.

The source implementation appendix proposes essentially this technology-neutral minimum and explicitly states that the theory does not require blockchain, a graph database, or real-time microservices.

H.27.2 Optional graph layer

A graph database may help when:

- identity lineage is complex;
- many frame edges exist;
- charge vertices form large networks;
- nested closure dependencies are common.

It is optional.

A relational database can represent the first pilot adequately.

H.27.3 Optional event stream

Event streaming may help when:

- calls arrive continuously;
- settlement returns are asynchronous;
- real-time backlog monitoring is required.

It is unnecessary for a retrospective pilot.

H.27.4 Blockchain is not required

An append-only audit ledger does not require a public blockchain.

Possible implementations include:

- immutable database tables;
- signed event logs;
- write-once storage;
- database temporal tables;
- cryptographic hashes.

The requirement is trace integrity, not a particular technology brand.

H.28 Deployment Modes

H.28.1 Research notebook mode

Suitable for:

- simulations;
- equation checks;
- feature exploration;

- retrospective reconstruction.

Claim ceiling:

Exploratory. (H.172)

H.28.2 Batch audit mode

Suitable for:

- historical margin episodes;
- residual audits;
- false-closure detection;
- frame reconciliation.

Claim ceiling:

Retrospective diagnostic. (H.173)

H.28.3 Prospective monitoring mode

Suitable for:

- warning;
- gate preparation;
- closure monitoring;
- backlog detection.

Claim ceiling:

Prospective analytical support. (H.174)

H.28.4 Controlled operational mode

Suitable only after separate validation and governance.

It may integrate with:

- margin systems;
- collateral systems;
- settlement systems;
- accounting workflows.

Institutional decisions remain outside the research model unless explicitly authorized.

H.29 Implementation Roadmap

H.29.1 Phase 1 — Protocol and evidence

Build:

- ProtocolRegistry;
- EvidenceSnapshot;
- IdentityRegistry;
- versioning;
- append-only event ledger.

Do not begin with advanced equations.

H.29.2 Phase 2 — One margin gate

Implement:

$$B = C + nhR - D. \qquad (H.175)$$

$$C_{call} = [B_{target} - B]_+. \qquad (H.176)$$

Separate:

- candidate breach;
- admitted call;
- waived call;
- disputed call.

H.29.3 Phase 3 — Action–ledger closure

Implement one branch first, preferably collateral posting.

Track:

Call
→ Collateral Instruction
→ Collateral Receipt
→ Eligibility
→ Ledger Posting
→ Closure Certificate. (H.177)

H.29.4 Phase 4 — Market-to-collateral transport

Implement:

V^_C = nR_markhe – Adjustments. (H.178)

Compare observed collateral value.

Record typed residual.

H.29.5 Phase 5 — Benchmark layer

Compare:

- raw workflow status;
- action–ledger defect;
- raw numerical difference;
- governed transport residual.

H.29.6 Phase 6 — Additional branches

Add:

- deleveraging;
- forced liquidation;
- default conversion.

Each branch requires its own action-to-ledger transporter.

H.29.7 Phase 7 — Optional Complex CAPM

Add:

A. (H.179)

R_CAPM. (H.180)

Q. (H.181)

θ. (H.182)

Only after the real-pair benchmark is implemented.

H.29.8 Phase 8 — Multi-frame loops

Add:

- settlement;
- accounting;

- legal;
- regulatory frames.

Calculate loop residuals.

H.29.9 Phase 9 — Recursive updates

Allow closure history to alter:

- haircut;
- funding spread;
- limits;
- mass proxy;
- closure capacity.

H.29.10 Phase 10 — Candidate Gauge–Dirac model

Estimate the full first-order model only after the lower-level objects have proved measurable.

The correct implementation order is:

Protocol
→ Evidence
→ Gate
→ Action–Ledger
→ Transport
→ Residual
→ Benchmark
→ Recursion
→ Optional Dirac Kernel. (H.183)

Not:

Dirac Equation
→ Search for Data that Resembles It. (H.184)

H.30 Common Implementation Failures

H.30.1 Mutable current-status design

The application stores only the latest status.

Failure

Earlier gates, residuals, and reopenings disappear.

Repair

Use event sourcing or versioned append-only records.

H.30.2 Indicator-first design

The system begins with many features but lacks:

- protocol;
- identity;
- gate;
- closure;
- residual.

Failure

The implementation becomes a dashboard rather than a governed research runtime.

Repair

Implement one complete event lifecycle before expanding feature count.

H.30.3 Scalar-dashboard compression

The system reports one score:

$$\text{ClosureScore} = 82\%. \qquad (H.185)$$

Failure

A critical legal or settlement defect may be hidden inside an average.

Repair

Retain the closure vector and mandatory conditions.

H.30.4 Complex-module contamination

Every subject is forced to possess:

$$R, Q, \theta. \qquad (H.186)$$

Failure

The system invents complex structure where no defensible conjugate coordinate exists.

Repair

Make the complex module optional and admission-gated.

H.30.5 Automated-authority escalation

A probability estimate is treated as permission to commit an event.

Failure

Projection becomes authority.

Repair

Separate analytical outputs from institutional gate decisions.

H.30.6 Missingness as zero

A missing ledger record is stored as:

Amount = 0. (H.187)

Failure

Unknown and zero become indistinguishable.

Repair

Use explicit missing-state and missing-evidence residuals.

H.30.7 Residual deletion

An explained mismatch is deleted from the database.

Failure

The history of explanation and correction is lost.

Repair

Change residual status and link a resolution trace.

H.30.8 Silent protocol drift

A haircut or closure rule changes without a new protocol version.

Failure

Results before and after the change become incomparable.

Repair

Create a new immutable protocol version with an effective date.

H.30.9 Benchmark omission

The advanced model is evaluated without a strong simpler comparison.

Failure

Complexity appears as scientific gain.

Repair

Encode mandatory benchmark mappings in the model registry.

H.30.10 Frame collapse

Market, collateral, accounting, and legal values are stored in one column called `value`.

Failure

Distinct frame-local objects become indistinguishable.

Repair

Store the frame and connection explicitly.

H.30.11 Identity collapse

A position is identified only by ticker.

Failure

Owner, account, contract, quantity, and lifecycle are lost.

Repair

Use a bounded identity record with transaction lineage.

H.30.12 Closure by timeout

An event is marked closed automatically after a fixed period.

Failure

Elapsed time substitutes for ledger return.

Repair

Use timeout as an escalation gate, not as proof of closure.

H.31 Compliance Tests

A reference implementation is compliant only if it passes the following tests.

Test 1 — Protocol presence

Every claim references one exact protocol version.

Test 2 — Evidence lineage

Every projection references a frozen evidence snapshot.

Test 3 — Gate trace

Every admitted event has a gate decision and authority.

Test 4 — Residual output

Every gate may emit zero or more explicit residual records.

“No material residual” must be affirmative rather than assumed.

Test 5 — Action–ledger separation

Every closure event distinguishes outward action from ledger return.

Test 6 — Frame declaration

Every cross-frame comparison references a connection.

Test 7 — Identity preservation

Every transport records an invariant result.

Test 8 — Revision preservation

Every revision links to its original object.

Test 9 — Benchmark requirement

Every promoted advanced model has a simpler benchmark.

Test 10 — Reduction capability

Every advanced model can be demoted without destroying the lower-level records.

Test 11 — Authority control

Analytical services cannot perform unauthorized institutional actions.

Test 12 — Queryable failure

Rejected claims and failed models remain accessible.

H.32 Implementation Validity

Define implementation validity:

$$
\begin{aligned}
&V_{impl} \\
&:= V_{protocol} \\
&\times V_{lineage} \\
&\times V_{gate} \\
&\times V_{residual} \\
&\times V_{trace} \\
&\times V_{transport} \\
&\times V_{authority} \\
&\times V_{revision} \\
&\times V_{reduction}.
\end{aligned}
\tag{H.188}
$$

Each component lies in:

$$0 \le V_j \le 1. \tag{H.189}$$

The multiplicative form means that a complete failure in one mandatory component collapses overall validity.

A system with strong modelling but no evidence lineage is invalid.

A system with complete trace but no authority separation is invalid.

A system with accurate predictions but hidden residual is invalid under this architecture.

H.33 Minimal Reference Object Set

The smallest useful implementation contains only nine disciplined objects:

1. Protocol
2. Evidence Snapshot
3. Financial Identity
4. Typed Claim
5. Gate Decision
6. Trace Event
7. Residual Record

8. Transport Record
9. Revision Record

The first margin pilot adds:

10. Action Record
11. Ledger-Return Record
12. Closure Certificate

The theory does not require a large software platform before these objects become testable.

H.34 Compact Runtime Formula

The complete runtime may be represented:

Input_t
:= (P_v,K,Snapshot_t). (H.190)

Projection_t
:= Φ_v(Input_t). (H.191)

Claim_t
:= Compile(Projection_t,P_v). (H.192)

Gate_t
:= Admit(Claim_t,Authority_t). (H.193)

Action_t
:= ExecuteAuthorizedBranch(Gate_t). (H.194)

LedgerReturn_t+s
:= U_AL(Action_t). (H.195)

Residual_t+s
:= ObservedLedger_t+s – LedgerReturn_t+s. (H.196)

Transport_t
:= {U_AB(State_A)}. (H.197)

Closure_t+s
:= Certify(Gate,Action,Ledger,Charge,Transport,Residual). (H.198)

Ledger_k+1
:= Ledger_k ⊕ Trace_k ⊕ Residual_k ⊕ Closure_k. (H.199)

ModelStatus_k+1
:= CompareAndReduce(Model_k,Benchmarks_k). (H.200)

This is the machine-operational form of recursive financial closure.

H.35 Implementation Contract

A compliant implementation should satisfy:

1. Every calculation has a declared protocol.
2. Every claim has frozen evidence.
3. Every financial object has a bounded identity.
4. Every admitted event has an authority.
5. Every outward action has an expected ledger return.
6. Every mismatch is typed before aggregation.
7. Every cross-frame comparison has a connection.
8. Every closure has a certificate.
9. Every residual remains queryable.
10. Every revision preserves prior trace.
11. Every advanced model has a simpler benchmark.
12. Every failed advanced layer produces explicit reduction.
13. Every model output remains separate from institutional authority.
14. Every missing required state is recorded as missing rather than zero.
15. Every broad claim states its transport boundary.

In compact form:

ImplementationValidity
= Reproducibility
× Typing
× GateDiscipline
× ActionLedgerSeparation
× ResidualHonesty
× TraceIntegrity
× Transport
× AuthorityControl
× Revision
× Reduction. (H.201)

H.36 Appendix Conclusion

The Financial Gauge–Dirac–Gate–Ledger architecture does not first require a giant mathematical engine.

It first requires a disciplined record system.

The runtime must know:

- which financial identity is being discussed;

- under which protocol;
- from which evidence;
- through which projection;
- at which gate;
- under whose authority;
- producing which outward action;
- requiring which ledger return;
- transported through which frames;
- leaving which residual;
- closed under which certificate;
- revised under which later evidence.

The implementation sequence is:

Declare Protocol
→ Freeze Evidence
→ Resolve Identity
→ Calculate Projection
→ Compile Claim
→ Evaluate Gate
→ Record Action
→ Observe Ledger Return
→ Transport Across Frames
→ Preserve Residual
→ Certify Closure
→ Compare Benchmarks
→ Revise or Reduce. (H.202)

The architecture's deepest software rule is:

> **Never allow a mutable status field to erase the distinction between what was measured, what was admitted, what was done, what returned through the ledger, and what remained unresolved.**

The corresponding research rule is:

> **A machine-readable Financial Standard Model should not be judged by whether it can encode impressive terminology. It should be judged by whether it prevents unsupported promotion, preserves failed traces, distinguishes authority from prediction, and reliably reduces itself when a simpler model is sufficient.**

The next appendix will provide a **complete machine-readable worked example**, following one margin event from frozen evidence through breach calculation, gate admission, collateral and liquidation actions, ledger return, transport residual, closure certification, and recursive protocol update.

Appendix I — Complete Machine-Readable Worked

Example

I.1 Purpose

This appendix follows one leveraged margin account through a complete governed lifecycle:

Frozen Evidence
→ Identity Resolution
→ Complex Valuation
→ Collateral Transport
→ Buffer Breach
→ Margin-Gate Admission
→ Collateral Attempt
→ Partial Failure
→ Forced Deleveraging
→ Trade Settlement
→ Debt Recognition
→ Residual Reconciliation
→ Closure Certification
→ Recursive Protocol Update. (I.1)

The example demonstrates how the conceptual system may be represented simultaneously as:

- financial equations;
- machine-readable objects;
- append-only events;
- action–ledger states;
- frame transports;
- charge vertices;
- residual records;
- a final closure certificate.

All amounts are illustrative.

They do not represent investment advice, a production margin rule, or a validated universal Financial Gauge–Dirac law.

I.2 Scenario Overview

I.2.1 Financial subject

The subject is one leveraged account:

S_MA-104
:= one account holding 100 units of Asset X, financed partly by debt, under Margin Agreement M-7. (I.2)

The account operates in:

- one legal entity;
- one currency;
- one margin agreement;
- one collateral set;
- one settlement system.

I.2.2 Initial financial state

The pre-shock account contains:

$n_0 = 100$ units. (I.3)

$C_0 = \$300.00$. (I.4)

$D_0 = \$8{,}200.00$. (I.5)

$h_0 = 0.75$. (I.6)

$R_{M,0} = \$107.91$ per unit. (I.7)

The initial margin buffer is:

$$B_0 = C_0 + n_0 h_0 R_{M,0} - D_0. \quad \text{(I.8)}$$

Therefore:

$$B_0 = \$300.00 + 100 \times 0.75 \times \$107.91 - \$8{,}200.00. \quad \text{(I.9)}$$

Hence:

$B_0 = \$193.25$. (I.10)

The small difference from Appendix B reflects rounding to cents at the state-record level.

The account begins in the safe regime:

$B_0 > 0$. (I.11)

I.2.3 Shock state

At 10:02, the financial environment changes.

The new market value is:

$R_{M,1} = \$105.63$. (I.12)

The new collateral factor is:

$h_1 = 0.70$. (I.13)

The debt and posted collateral remain:

$D_1 = \$8{,}200.00.$ (I.14)

$C_1 = \$300.00.$ (I.15)

The resulting buffer is:

$B_1 = \$300.00 + 100 \times 0.70 \times \$105.63 - \$8{,}200.00.$ (I.16)

Therefore:

$B_1 = -\$505.90.$ (I.17)

The candidate shortfall is:

$S_{B,1} = [0 - (-\$505.90)]_+.$ (I.18)

Hence:

$S_{B,1} = \$505.90.$ (I.19)

I.3 Protocol Record

I.3.1 Human-readable protocol

The applicable protocol is:

Protocol ID: MARGIN-US-01
Version: 3
Boundary: Account MA-104 under Agreement M-7
Currency: USD
Margin rule: $B = C + nhR_{mark} - D$
Call condition: $B < 0$
Call amount: $[-B]_+$
Authority: Broker Margin Engine BME-2
Permitted cure branches: cash collateral, eligible securities, voluntary deleveraging, broker liquidation, default conversion
Closure rule: amount cured plus mandatory ledger return and residual disclosure

---

I.3.2 Machine-readable protocol object

```
{
  "protocol_id": "MARGIN-US-01",
  "version": 3,
  "boundary": {
    "legal_entity": "LE-01",
    "account_id": "MA-104",
    "agreement_id": "M-7",
    "collateral_set_id": "CS-1",
    "currency": "USD"
  },
  "observation_rule": {
    "position_aggregation": "single_asset",
    "market_mark_source": "MKT-X-CLOSE",
    "debt_source": "TREASURY-LEDGER",
    "collateral_source": "COLLATERAL-LEDGER"
  },
```

```
  "margin_rule": {
    "formula": "B = C + n*h*R_mark - D",
    "target_buffer": 0.0,
    "call_amount_formula": "max(target_buffer - B, 0)"
  },
  "gate_rule": {
    "required_authority": "BME-2",
    "requires_valid_mark": true,
    "requires_applicable_agreement": true,
    "requires_persistent_trace": true
  },
  "permitted_branches": [
    "CASH_COLLATERAL",
    "SECURITY_COLLATERAL",
    "VOLUNTARY_DELEVERAGING",
    "FORCED_LIQUIDATION",
    "DEFAULT_CONVERSION"
  ],
  "closure_rule": {
    "requires_amount_cure": true,
    "requires_settlement_return": true,
    "requires_debt_update": true,
    "requires_charge_reconciliation": true,
    "requires_residual_disclosure": true
  },
  "effective_from": "2026-01-01T00:00:00Z",
  "effective_to": null
}
```

I.4 Financial Identity Record

I.4.1 Identity kernel

The account identity is:

K_MA-104
:= (Asset X,100 units,Owner LE-01,Account MA-104,USD,Agreement M-7,Active). (I.20)

The identity is more than the ticker.

It includes:

- account;
- owner;
- signed quantity;
- governing contract;
- currency;
- transaction lineage;
- lifecycle status.

I.4.2 Machine-readable identity object

```
{
  "identity_id": "ID-MA-104-X-001",
  "identity_class": "LEVERAGED_MARGIN_POSITION",
  "instrument_id": "ASSET-X",
  "contract_id": "M-7",
  "owner": "LE-01",
  "obligor": "MA-104",
  "signed_quantity": 100.0,
  "currency": "USD",
  "account_id": "MA-104",
  "agreement_id": "M-7",
  "transaction_lineage": [
    "TRADE-X-8871",
    "SETTLEMENT-X-8871"
  ],
  "lifecycle_status": "ACTIVE",
  "parent_identity_id": null,
  "successor_identity_id": null
}
```

I.5 Evidence Snapshot Before the Breach

I.5.1 Frozen evidence time

The first frozen evidence time is:

to = 2026-07-15 09:55:00 UTC. (I.21)

The snapshot contains only information available at that time.

It does not include:

- the later margin call;
- later collateral rejection;
- later sale;
- later settlement;
- later debt update.

I.5.2 Evidence object


```
{
  "snapshot_id": "SNAP-MA104-095500",
  "protocol_id": "MARGIN-US-01",
  "protocol_version": 3,
  "subject_id": "ID-MA-104-X-001",
  "evidence_time": "2026-07-15T09:55:00Z",
  "source_records": {
    "market_mark": {
      "source": "MKT-X-CLOSE",
      "value": 107.91,
      "currency": "USD",
      "version": 1
    },
    "quantity": {
      "source": "POSITION-LEDGER",
```

```
      "value": 100.0,
      "units": "shares",
      "version": 17
    },
    "collateral_factor": {
      "source": "COLLATERAL-RULE",
      "value": 0.75,
      "version": 3
    },
    "posted_collateral": {
      "source": "COLLATERAL-LEDGER",
      "value": 300.0,
      "currency": "USD",
      "version": 8
    },
    "debt_balance": {
      "source": "TREASURY-LEDGER",
      "value": 8200.0,
      "currency": "USD",
      "version": 12
    }
  },
  "missing_fields": [],
  "content_hash": "sha256:example-initial-hash"
}
```

I.6 Initial Projection Records

I.6.1 Initial buffer calculation

The projection operator is:

Φ_buffer(n,h,R,C,D) := C + nhR − D. (I.22)

Applying the operator:

B0 = Φ_buffer(100,0.75,$107.91,$300.00,$8,200.00). (I.23)

Therefore:

B0 = $193.25. (I.24)

I.6.2 Projection object


```
{
  "projection_id": "PROJ-BUFFER-MA104-0001",
  "snapshot_id": "SNAP-MA104-095500",
  "operator_id": "BUFFER-OP",
  "operator_version": 3,
  "inputs": {
    "quantity": 100.0,
    "collateral_factor": 0.75,
    "market_mark": 107.91,
    "posted_collateral": 300.0,
    "debt_balance": 8200.0
  },
  "outputs": {
    "margin_buffer": 193.25,
    "candidate_shortfall": 0.0
```

```
  },
  "units": "USD",
  "status": "VALID"
}
```

I.6.3 Initial gate result

Because:

$B_0 > 0,$ (I.25)

the candidate condition fails.

The gate decision is:

$\text{Decision} = \text{NotTriggered}.$ (I.26)

No collateral obligation is operationally active:

$q_{C,0}^{op} = 0.$ (I.27)

The account remains:

$\text{MarginState} = \text{Safe}.$ (I.28)

I.7 Complex Valuation Snapshot

I.7.1 Baseline amplitude

Assume the research Complex CAPM module has:

$A_0 = \$115.38.$ (I.29)

The CAPM-admitted value is:

$R_{CAPM,0} = \$107.91.$ (I.30)

The conjugate coordinate is:

$Q_0 = \sqrt{(\$115.38^2 - \$107.91^2)}.$ (I.31)

Therefore:

$Q_0 \approx \$40.84.$ (I.32)

The phase is:

$\theta_0 = \arccos(\$107.91/\$115.38).$ (I.33)

Hence:

$\theta_0 \approx 0.3618 \text{ radians}.$ (I.34)

I.7.2 Complex projection record

```
{
  "projection_id": "PROJ-COMPLEX-MA104-0001",
  "snapshot_id": "SNAP-MA104-095500",
  "operator_id": "COMPLEX-CAPM-OP",
  "operator_version": 1,
  "inputs": {
    "baseline_amplitude": 115.38,
    "capm_value": 107.91
  },
  "outputs": {
    "R": 107.91,
    "Q": 40.84,
    "theta_radians": 0.3618
  },
  "units": {
    "R": "USD",
    "Q": "USD",
    "theta": "radian"
  },
  "status": "RESEARCH_ONLY"
}
```


The contractual margin engine does not use this research Q value to issue the call.

The operational gate continues to use the contractual market mark and margin rule.

I.8 Shock Evidence Snapshot

I.8.1 New evidence time

At:

t_1 = 2026-07-15 10:02:00 UTC, (I.35)

the market mark and collateral factor change.

I.8.2 Shock snapshot


```
{
  "snapshot_id": "SNAP-MA104-100200",
  "protocol_id": "MARGIN-US-01",
  "protocol_version": 3,
  "subject_id": "ID-MA-104-X-001",
  "evidence_time": "2026-07-15T10:02:00Z",
  "source_records": {
    "market_mark": {
      "source": "MKT-X-LIVE",
      "value": 105.63,
      "currency": "USD",
      "version": 24
    },
    "quantity": {
      "source": "POSITION-LEDGER",
      "value": 100.0,
      "units": "shares",
      "version": 17
    },
    "collateral_factor": {
      "source": "COLLATERAL-RULE",
```

```
      "value": 0.70,
      "version": 4
    },
    "posted_collateral": {
      "source": "COLLATERAL-LEDGER",
      "value": 300.0,
      "currency": "USD",
      "version": 8
    },
    "debt_balance": {
      "source": "TREASURY-LEDGER",
      "value": 8200.0,
      "currency": "USD",
      "version": 12
    }
  },
  "missing_fields": [],
  "content_hash": "sha256:example-shock-hash"
}
```

I.9 Breach Projection

I.9.1 Buffer

The new buffer is:

$$B_1 = \$300.00 + 100 \times 0.70 \times \$105.63 - \$8{,}200.00. \qquad (I.36)$$

Therefore:

$$B_1 = -\$505.90. \qquad (I.37)$$

I.9.2 Candidate shortfall

The candidate call amount is:

$$C_{call,candidate} = [-B_1]_+. \qquad (I.38)$$

Therefore:

$$C_{call,candidate} = \$505.90. \qquad (I.39)$$

I.9.3 Candidate-claim object


```
{
  "claim_id": "CLAIM-MA104-BREACH-0001",
  "subject_id": "ID-MA-104-X-001",
  "protocol_id": "MARGIN-US-01",
  "snapshot_id": "SNAP-MA104-100200",
  "claim_type": "MARGIN_BREACH_CANDIDATE",
  "claim_level": "CANDIDATE",
  "claim_content": {
    "margin_buffer": -505.90,
    "target_buffer": 0.0,
    "candidate_shortfall": 505.90
  },
```

```
  "supporting_evidence": [
    "MKT-X-LIVE:24",
    "POSITION-LEDGER:17",
    "COLLATERAL-RULE:4",
    "COLLATERAL-LEDGER:8",
    "TREASURY-LEDGER:12"
  ],
  "contradicting_evidence": [],
  "maximum_permitted_level": "EVENT",
  "status": "GATE_PENDING",
  "invalidation_rule": {
    "type": "MARK_OR_RULE_REVISION",
    "deadline": "2026-07-15T10:05:00Z"
  }
}
```

At this stage:

BreachCandidate = 1. (I.40)

but:

ValidMarginCall is not yet established. (I.41)

I.10 Gate Admission

I.10.1 Gate conditions

The margin gate checks:

1. Is the buffer below target?
2. Is the market mark valid?
3. Is Agreement M-7 applicable?
4. Does BME-2 possess authority?
5. Has the decision been committed to trace?

All five conditions pass.

I.10.2 Gate time

The call is admitted at:

t_gate = 2026-07-15 10:05:00 UTC. (I.42)

I.10.3 Gate-decision object

```
{
  "gate_decision_id": "GATE-MA104-CALL-0001",
  "claim_id": "CLAIM-MA104-BREACH-0001",
  "gate_type": "MARGIN_CALL",
  "gate_rule_version": 3,
```

```
  "authority_type": "AUTOMATED_MARGIN_AUTHORITY",
  "authority_id": "BME-2",
  "evidence_snapshot_id": "SNAP-MA104-100200",
  "decision": "ADMITTED",
  "decision_time": "2026-07-15T10:05:00Z",
  "strength": 1.0,
  "conditions_passed": [
    "BUFFER_BELOW_TARGET",
    "MARK_VALID",
    "AGREEMENT_APPLICABLE",
    "AUTHORITY_VALID",
    "TRACE_WRITTEN"
  ],
  "conditions_failed": [],
  "trace_id": "TRACE-CALL-MA104-0001",
  "residual_ids": [],
  "reopen_rule": {
    "type": "MATERIAL_MARK_OR_AGREEMENT_REVISION"
  }
}
```

I.10.4 State transition

The account moves:

Safe
→ BreachCandidate
→ CallOpen. (I.43)

The closure sign becomes:

$s_{close} : +1 \to -1.$ (I.44)

The operational collateral charge activates:

$q_C^{op} : 0 \to q_C^{struct}.$ (I.45)

I.11 Charge Records Before and After the Gate

I.11.1 Pre-gate charge state

Before the call:

$\vec{q}_{struct,-} = (q_A,q_F,q_C).$ (I.46)

The operational vector is:

$\vec{q}_{op,-} = (q_A,q_F,0).$ (I.47)

I.11.2 Post-gate charge state

After call admission:

$\vec{q}_{op,+} = (q_A,q_F,q_C).$ (I.48)

The gate does not create the collateral clause.

It activates the previously dormant contractual obligation.

I.11.3 Machine-readable activation record


```
{
  "charge_id": "CHARGE-MA104-COLLATERAL-0001",
  "identity_id": "ID-MA-104-X-001",
  "protocol_id": "MARGIN-US-01",
  "field": "COLLATERAL",
  "charge_class": "MARGIN_PERFORMANCE_OBLIGATION",
  "structural_sign": -1,
  "structural_magnitude": 1.0,
  "operational_status": "ACTIVE",
  "effective_magnitude": 505.90,
  "screening_rule": "NONE",
  "units": "USD_OBLIGATION_CLASS",
  "boundary": "MA-104/M-7",
  "valid_from": "2026-07-15T10:05:00Z",
  "valid_to": null
}
```


The value $505.90 is the present required amount.

The structural sign and obligation class are separate from the monetary amount.

I.12 First Cure Attempt — Cash Collateral Instruction

I.12.1 Available cash

The account has only:

CashAvailable = $300.00. (I.49)

The client instructs transfer of:

C_instr,1 = $300.00. (I.50)

The remaining expected shortfall is:

$505.90 − $300.00 = $205.90. (I.51)

I.12.2 Action object


```
{
  "action_id": "ACTION-MA104-CASH-0001",
  "gate_decision_id": "GATE-MA104-CALL-0001",
  "identity_id": "ID-MA-104-X-001",
  "action_type": "CASH_COLLATERAL_INSTRUCTION",
  "instruction_time": "2026-07-15T10:12:00Z",
  "execution_time": null,
  "instruction_amount": 300.0,
  "execution_amount": 0.0,
  "instruction_quantity": null,
  "execution_quantity": null,
```

```
  "authority": "CLIENT-MA104",
  "status": "INSTRUCTED",
  "expected_ledger_effects": {
    "collateral_admitted": 300.0,
    "margin_buffer_change": 300.0
  }
}
```

I.12.3 Action-state component

Define the first action vector:

ψ_A,1 =
[
$505.90
$300.00
0
$300.00
0
0
CallOpen
]. (I.52)

The components represent:

- call amount;
- collateral instruction;
- sale instruction;
- expected collateral admission;
- expected debt reduction;
- economic loss;
- gate state.

I.12.4 Initial ledger component

Immediately after instruction:

ψ_L,1 =
[
$0.00
$0.00
0
$0.00
0
0
LedgerOpen
]. (I.53)

The instruction has occurred.

The collateral has not yet been admitted.

Therefore:

ActionCompletedPartially = 1. (I.54)

LedgerReturned = 0. (I.55)

I.13 Collateral Transport and Partial Rejection

I.13.1 Expected cash-collateral transport

For cash collateral, the expected map is:

C_expected,ledger = C_instruction. (I.56)

Thus:

C_expected,ledger = $300.00. (I.57)

I.13.2 Observed ledger result

At 10:20, only:

C_admitted = $250.00. (I.58)

is admitted.

The remaining:

$50.00 (I.59)

is rejected because the transfer contains an incorrect reference.

I.13.3 Ledger-return object


```
{
  "ledger_return_id": "LEDGER-MA104-CASH-0001",
  "action_id": "ACTION-MA104-CASH-0001",
  "ledger_frame": "COLLATERAL",
  "recognition_time": "2026-07-15T10:20:00Z",
  "recorded_amount": 250.0,
  "recorded_quantity": null,
  "recorded_status": "PARTIAL",
  "transport_rule": "CASH-INSTRUCTION-TO-COLLATERAL-ADMISSION-v2",
  "expected_value": 300.0,
  "observed_value": 250.0,
  "residual_id": "RES-MA104-CASH-0001",
  "completion_status": "PARTIAL"
}
```

I.13.4 Collateral residual

The residual is:

$r_{collateral,1} = \$250.00 - \$300.00.$ (I.60)

Therefore:

$r_{collateral,1} = -\$50.00.$ (I.61)

This is not market loss.

It is a failed collateral-admission residual.

I.13.5 Residual object


```
{
  "residual_id": "RES-MA104-CASH-0001",
  "subject_id": "ID-MA-104-X-001",
  "protocol_id": "MARGIN-US-01",
  "residual_type": "COLLATERAL_ADMISSION",
  "source_object_type": "LEDGER_RETURN",
  "source_object_id": "LEDGER-MA104-CASH-0001",
  "expected_state": {
    "admitted_cash": 300.0
  },
  "observed_state": {
    "admitted_cash": 250.0
  },
  "difference": {
    "amount": -50.0
  },
  "units": "USD",
  "severity": "MATERIAL",
  "materiality": 0.0988,
  "status": "OPEN",
  "detected_at": "2026-07-15T10:20:00Z",
  "resolved_at": null,
  "resolution_trace_id": null,
  "carry_forward_rule": "UNTIL_RESOLVED"
}
```

I.14 Buffer After Partial Collateral Admission

I.14.1 New collateral balance

The original posted collateral was:

$C_1 = \$300.00.$ (I.62)

The admitted additional collateral is:

$\$250.00.$ (I.63)

Therefore:

$C_2 = \$550.00.$ (I.64)

I.14.2 New margin buffer

$B_2 = \$550.00 + 100 \times 0.70 \times \$105.63 - \$8{,}200.00.$ (I.65)

Therefore:

$B_2 = -\$255.90.$ (I.66)

The account remains below target.

The open shortfall is:

$C_{call,remaining} = \$255.90.$ (I.67)

The margin cycle remains open.

I.15 Branch Escalation to Deleveraging

I.15.1 Required sale quantity

Assume the expected net proceeds per sold unit are:

$p_{net,expected} = \$103.50.$ (I.68)

The collateral value removed per unit is:

$hR_M = 0.70 \times \$105.63.$ (I.69)

Therefore:

$hR_M = \$73.94.$ (I.70)

The approximate buffer improvement per sold unit is:

$\Delta B_{unit} = \$103.50 - \$73.94.$ (I.71)

Hence:

$\Delta B_{unit} = \$29.56.$ (I.72)

The approximate quantity required is:

$\ell^* = \$255.90/\$29.56.$ (I.73)

Therefore:

$\ell^* \approx 8.66 \text{ units}.$ (I.74)

The instruction is rounded to:

$\ell_{instr} = 9 \text{ units}.$ (I.75)

I.15.2 Sale instruction object

```
{
  "action_id": "ACTION-MA104-SALE-0001",
  "gate_decision_id": "GATE-MA104-CALL-0001",
  "identity_id": "ID-MA-104-X-001",
  "action_type": "VOLUNTARY_DELEVERAGING",
  "instruction_time": "2026-07-15T10:28:00Z",
  "execution_time": null,
  "instruction_amount": null,
  "execution_amount": null,
  "instruction_quantity": 9.0,
  "execution_quantity": 0.0,
  "authority": "CLIENT-MA104",
  "status": "INSTRUCTED",
  "expected_ledger_effects": {
    "position_reduction": 9.0,
    "expected_net_proceeds": 931.5,
    "expected_debt_reduction": 931.5
  }
}
```

I.16 Sale Execution

I.16.1 Execution result

At 10:31, all 9 units are executed.

The gross execution price is:

$p_{exec} = \$104.00.$ (I.76)

The per-unit fee is:

$Fee_{unit} = \$0.50.$ (I.77)

The net price is:

$p_{net} = \$103.50.$ (I.78)

Total net proceeds are:

$P_{net} = 9 \times \$103.50.$ (I.79)

Therefore:

$P_{net} = \$931.50.$ (I.80)

I.16.2 Execution event


```
{
  "event_id": "EVENT-MA104-SALE-EXEC-0001",
  "event_type": "SALE_EXECUTED",
  "subject_id": "ID-MA-104-X-001",
  "protocol_id": "MARGIN-US-01",
  "effective_time": "2026-07-15T10:31:00Z",
  "recorded_time": "2026-07-15T10:31:02Z",
  "actor": "EXECUTION-VENUE-1",
  "authority": "CLIENT-MA104",
  "payload": {
```

```
    "quantity": 9.0,
    "gross_price_per_unit": 104.0,
    "fee_per_unit": 0.5,
    "net_proceeds": 931.5
  },
  "parent_event_id": "ACTION-MA104-SALE-0001",
  "supersedes_event_id": null,
  "evidence_snapshot_id": "SNAP-MA104-100200",
  "residual_ids": []
}
```

I.16.3 Position charge change

Before execution:

$$q_{A,-} = 100q_0. \qquad \text{(I.81)}$$

After execution:

$$q_{A,+} = 91q_0. \qquad \text{(I.82)}$$

The structural asset charge changes through a valid sale vertex:

$$\Delta q_A = -9q_0. \qquad \text{(I.83)}$$

However, legal and settlement completion are still pending.

The market-action frame has moved ahead of the settlement ledger.

I.17 Action–Ledger Spinor After Execution

I.17.1 Updated action vector

Define:

$$\psi_{A,2} = [\$505.90\ \$300.00\ 9\ \$250.00\ \$931.50\ \$4.50\ \text{SaleExecuted}]. \qquad \text{(I.84)}$$

The economic execution fee is:

$$9 \times \$0.50 = \$4.50. \qquad \text{(I.85)}$$

I.17.2 Ledger vector immediately after execution

Before settlement:

ψ_L,2 =
[
$250.00
$250.00
0
$0.00
$0.00
PositionStillUnsettled
LedgerOpen
]. (I.86)

The action has produced an expected position and debt update.

The settlement and treasury ledgers have not yet returned.

I.17.3 Expected action-to-ledger image

The expected ledger state after full settlement is:

ψ^_L,2 =
[
$250.00
$250.00
9
$931.50
$4.50
Settled
LedgerReturned
]. (I.87)

I.17.4 Immediate closure defect

The raw action–ledger defect is:

δ_spin,2 = ψ_L,2 − ψ^_L,2. (I.88)

Its material components include:

SettlementQuantityResidual = 0 − 9 = −9 units. (I.89)

DebtReductionResidual = $0.00 − $931.50 = −$931.50. (I.90)

LossRecognitionResidual = $0.00 − $4.50 = −$4.50. (I.91)

This mismatch is expected immediately after execution.

It is an open return state, not necessarily a failure.

The correct comparison must be age-adjusted.

I.18 Settlement Return

I.18.1 Settlement event

At 13:55, all 9 units settle.

```
{
  "ledger_return_id": "LEDGER-MA104-SETTLEMENT-0001",
  "action_id": "ACTION-MA104-SALE-0001",
  "ledger_frame": "SETTLEMENT",
  "recognition_time": "2026-07-15T13:55:00Z",
  "recorded_amount": 931.5,
  "recorded_quantity": 9.0,
  "recorded_status": "SETTLED",
  "transport_rule": "TRADE-TO-SETTLEMENT-v4",
  "expected_value": {
    "quantity": 9.0,
    "cash": 931.5
  },
  "observed_value": {
    "quantity": 9.0,
    "cash": 931.5
  },
  "residual_id": null,
  "completion_status": "COMPLETE"
}
```

The settlement residual is:

r_settlement = 0. (I.92)

I.19 Treasury Debt Update

I.19.1 Expected debt after settlement

The expected debt is:

D_expected = $8,200.00 − $931.50. (I.93)

Therefore:

D_expected = $7,268.50. (I.94)

I.19.2 Observed debt

At 14:03, treasury records:

D_observed = $7,270.00. (I.95)

The initial debt residual is:

r_debt = $7,270.00 − $7,268.50. (I.96)

Therefore:

r_debt = $1.50. (I.97)

I.19.3 Debt residual object


```
{
  "residual_id": "RES-MA104-DEBT-0001",
  "subject_id": "ID-MA-104-X-001",
  "protocol_id": "MARGIN-US-01",
  "residual_type": "DEBT_RECOGNITION",
  "source_object_type": "LEDGER_RETURN",
  "source_object_id": "LEDGER-MA104-TREASURY-0001",
  "expected_state": {
    "debt_balance": 7268.5
  },
  "observed_state": {
    "debt_balance": 7270.0
  },
  "difference": {
    "amount": 1.5
  },
  "units": "USD",
  "severity": "LOW",
  "materiality": 0.0002,
  "status": "INVESTIGATING",
  "detected_at": "2026-07-15T14:03:00Z",
  "resolved_at": null,
  "resolution_trace_id": null,
  "carry_forward_rule": "UNTIL_EXPLAINED"
}
```

I.20 Residual Explanation

I.20.1 Additional settlement charge

Investigation identifies a separate:

SettlementCharge = $1.50. (I.98)

This charge was not included in the original expected transport map.

The corrected expected debt is:

$$D_{expected,revised} = \$8{,}200.00 - \$931.50 + \$1.50. \quad \text{(I.99)}$$

Therefore:

$$D_{expected,revised} = \$7{,}270.00. \quad \text{(I.100)}$$

The governed residual becomes:

$$r_{debt,revised} = \$7{,}270.00 - \$7{,}270.00. \quad \text{(I.101)}$$

Hence:

$$r_{debt,revised} = \$0.00. \quad \text{(I.102)}$$

The $1.50 remains an economic cost.

It is no longer an unexplained transport residual.

I.20.2 Residual-resolution event


```
{
  "event_id": "EVENT-MA104-RESOLUTION-0001",
  "event_type": "RESIDUAL_EXPLAINED",
  "subject_id": "ID-MA-104-X-001",
  "protocol_id": "MARGIN-US-01",
  "effective_time": "2026-07-15T14:05:00Z",
  "recorded_time": "2026-07-15T14:05:10Z",
  "actor": "TREASURY-OPS-4",
  "authority": "TREASURY-OPS-4",
  "payload": {
    "residual_id": "RES-MA104-DEBT-0001",
    "explanation": "SETTLEMENT_CHARGE",
    "amount": 1.5,
    "revised_expected_debt": 7270.0
  },
  "parent_event_id": "RES-MA104-DEBT-0001",
  "supersedes_event_id": null,
  "evidence_snapshot_id": "SNAP-MA104-100200",
  "residual_ids": []
}
```


The residual status becomes:

Investigating → Explained → Closed. (I.103)

I.21 Position and Debt After Settlement

I.21.1 Position quantity

The settled quantity is:

$n_3 = 100 - 9.$ (I.104)

Therefore:

$n_3 = 91$ units. (I.105)

I.21.2 Debt

The recognized debt is:

$D_3 = \$7{,}270.00.$ (I.106)

I.21.3 Posted collateral

The admitted collateral is:

$C_3 = \$550.00.$ (I.107)

I.21.4 Updated buffer

Assuming the mark and collateral factor remain:

$R_{M,3} = \$105.63.$ (I.108)

$h_3 = 0.70.$ (I.109)

then:

$B_3 = \$550.00 + 91 \times 0.70 \times \$105.63 - \$7{,}270.00.$ (I.110)

The admitted collateral value is:

$91 \times 0.70 \times \$105.63 = \$6{,}729.65.$ (I.111)

Therefore:

$B_3 = \$550.00 + \$6{,}729.65 - \$7{,}270.00.$ (I.112)

Hence:

$B_3 = \$9.65.$ (I.113)

The amount condition is now satisfied:

$B_3 \geq 0.$ (I.114)

I.22 Market-to-Collateral Transport

I.22.1 Expected collateral value

The expected market-to-collateral transport is:

$\hat{V}_C = nhR_M.$ (I.115)

Therefore:

$\hat{V}_C = 91 \times 0.70 \times \$105.63.$ (I.116)

Hence:

$\hat{V}_C = \$6{,}729.65.$ (I.117)

I.22.2 Observed collateral value

The collateral system records:

$V_C^{obs} = \$6{,}728.00.$ (I.118)

The initial transport residual is:

$r_{MC} = \$6{,}728.00 - \$6{,}729.65.$ (I.119)

Therefore:

$r_{MC} = -\$1.65.$ (I.120)

I.22.3 Residual investigation

The collateral engine rounds each unit’s admitted value before aggregation:

R_C,rounded = Round($105.63 × 0.70,2). (I.121)

Therefore:

R_C,rounded = $73.94. (I.122)

For 91 units:

V_C,rounded = 91 × $73.94. (I.123)

Hence:

V_C,rounded = $6,728.54. (I.124)

A further:

$0.54 (I.125)

is removed by an account-level concentration adjustment.

Thus:

V_C,expected-final = $6,728.00. (I.126)

The governed transport residual becomes:

r_MC^G = $0.00. (I.127)

The original $1.65 difference was not a reconciliation failure.

It was a combination of:

- unit-level rounding;
- concentration adjustment.

I.22.4 Transport object


```
{
  "transport_id": "TRANS-MA104-MC-0001",
  "identity_id": "ID-MA-104-X-001",
  "protocol_id": "MARGIN-US-01",
  "source_frame_id": "FRAME-MARKET",
  "target_frame_id": "FRAME-COLLATERAL",
  "connection_id": "CONN-MARKET-COLLATERAL",
  "connection_version": 4,
  "source_state_id": "FRAMESTATE-MA104-MKT-140500",
  "expected_target_state": {
    "quantity": 91.0,
    "unit_market_mark": 105.63,
    "collateral_factor": 0.70,
    "rounded_unit_collateral_value": 73.94,
    "gross_collateral_value": 6728.54,
```

```
    "concentration_adjustment": -0.54,
    "net_collateral_value": 6728.00
  },
  "observed_target_state": {
    "net_collateral_value": 6728.00
  },
  "transport_residual_id": null,
  "invariant_result": "PASSED",
  "charge_result": "PASSED",
  "timing_result": "PASSED",
  "status": "CLOSED"
}
```

I.23 Action–Ledger State at Final Return

I.23.1 Final action state

The complete outward-action state is:

ψ_A,final =
[
$505.90
$300.00
9
$250.00
$931.50
$6.00
CallAndSaleCompleted
]. (I.128)

The total economic cost includes:

- execution fee of $4.50;
- settlement charge of $1.50.

Therefore:

EconomicCost = $6.00. (I.129)

I.23.2 Final ledger state

The final ledger state is:

ψ_L,final =
[
$250.00
$250.00
9
$930.00
$6.00
SettledAndRecognized
LedgerReturned
]. (I.130)

The debt reduction recorded is:

$8,200.00 − $7,270.00 = $930.00. (I.131)

This differs from net sale proceeds of $931.50 because of the additional $1.50 settlement charge.

The lawful action-to-ledger transporter includes that cost.

I.23.3 Final expected ledger image

The expected ledger image is:

U_ALψ_A,final =
[
$250.00
$250.00
9
$930.00
$6.00
SettledAndRecognized
LedgerReturned
]. (I.132)

Therefore:

δ_spin,final = ψ_L,final − U_ALψ_A,final. (I.133)

Hence:

δ_spin,final = 0. (I.134)

within the declared tolerance.

I.24 Charge Reconciliation

I.24.1 Asset charge

Before sale:

q_A,− = 100q0. (I.135)

After sale:

q_A,+ = 91q0. (I.136)

Transferred asset charge is:

q_A,transferred = 9q0. (I.137)

The asset-charge vertex is:

100q0 = 91q0 + 9q0 + r_q,A. (I.138)

Therefore:

r_q,A = 0. (I.139)

I.24.2 Funding charge

Before debt reduction:

D_{-} = $8,200.00. (I.140)

After debt reduction and charges:

D_{+} = $7,270.00. (I.141)

The funding obligation decreases by:

$930.00. (I.142)

The funding charge changes according to the declared ChargeMap_F.

No unexplained funding-charge residual remains.

I.24.3 Collateral charge

The active collateral obligation required:

$505.90. (I.143)

Resolution came through:

- $250.00 admitted collateral;
- sufficient deleveraging and debt reduction;
- final positive buffer of $9.65.

Therefore the operational collateral obligation closes:

q_C^op → 0. (I.144)

The structural contractual obligation remains dormant for future calls:

q_C^struct ≠ 0. (I.145)

I.25 Closure Certificate

I.25.1 Closure conditions

The event passes:

Identity condition

The post-event account remains the same account under Agreement M-7, with quantity reduced from 100 to 91.

Amount condition

$B_3 = \$9.65 \geq 0.$ (I.146)

Settlement condition

All 9 sold units settled.

Debt condition

Debt updated to $7,270.00.

Collateral condition

$250.00 additional cash collateral admitted.

Charge condition

Asset, funding, and collateral charge transitions reconcile.

Transport condition

Market-to-collateral transport closes after lawful rounding and concentration adjustment.

Residual condition

The $50.00 rejected collateral residual remains explicitly recorded as a failed first branch, but it no longer prevents closure because the later deleveraging branch resolved the amount shortfall.

The $1.50 debt residual was explained and closed.

I.25.2 Important residual distinction

The rejected $50.00 instruction does not disappear.

Its status becomes:

Open
→ Superseded by Alternative Cure
→ Accepted Historical Residual. (I.147)

It remains part of the event trace.

The final event can therefore be:

ClosedWithResidual. (I.148)

rather than falsely represented as a perfectly frictionless closure.

I.25.3 Closure-certificate object

{

```
  "certificate_id": "CERT-MA104-CALL-0001",
  "subject_id": "ID-MA-104-X-001",
  "event_id": "GATE-MA104-CALL-0001",
  "protocol_id": "MARGIN-US-01",
  "identity_result": "PRESERVED_WITH_MODIFICATION",
  "amount_result": {
    "status": "PASSED",
    "final_buffer": 9.65
  },
  "settlement_result": {
    "status": "PASSED",
    "settled_quantity": 9.0
  },
  "debt_result": {
    "status": "PASSED",
    "final_debt": 7270.0
  },
  "collateral_result": {
    "status": "PASSED_WITH_PRIOR_PARTIAL_FAILURE",
    "additional_collateral_admitted": 250.0
  },
  "accounting_result": {
    "status": "PASSED",
    "recognized_cost": 6.0
  },
  "legal_result": {
    "status": "PASSED"
  },
  "charge_result": {
    "status": "PASSED",
    "asset_charge_units_remaining": 91.0,
    "operational_collateral_charge": 0.0
  },
  "transport_result": {
    "status": "PASSED",
    "market_to_collateral_residual": 0.0
  },
  "residual_disclosure_result": {
    "status": "PASSED",
    "accepted_historical_residuals": [
      "RES-MA104-CASH-0001"
    ],
    "closed_residuals": [
      "RES-MA104-DEBT-0001"
    ]
  },
  "final_status": "CLOSED_WITH_RESIDUAL",
  "certified_by": "MARGIN-CLOSURE-CONTROLLER-1",
  "certified_at": "2026-07-15T14:15:00Z",
  "reopen_conditions": [
    "SETTLEMENT_REVERSAL",
    "MATERIAL_MARK_REVISION",
    "LEGAL_OWNERSHIP_DISPUTE",
    "DEBT_RESTATEMENT"
  ]
}
```

I.26 Append-Only Event Sequence

The complete event history is:

```
09:55  BUFFER_CALCULATED_SAFE
10:02  BUFFER_BREACH_CANDIDATE
10:05  MARGIN_CALL_ADMITTED
```

```
10:12  CASH_COLLATERAL_INSTRUCTED
10:20  CASH_COLLATERAL_PARTIALLY_ADMITTED
10:20  COLLATERAL_RESIDUAL_OPENED
10:28  VOLUNTARY_SALE_INSTRUCTED
10:31  SALE_EXECUTED
13:55  SALE_SETTLED
14:03  DEBT_UPDATED_WITH_DIFFERENCE
14:03  DEBT_RESIDUAL_OPENED
14:05  SETTLEMENT_CHARGE_IDENTIFIED
14:05  DEBT_RESIDUAL_EXPLAINED
14:08  MARKET_COLLATERAL_TRANSPORT_RECONCILED
14:12  CHARGE_RECONCILIATION_PASSED
14:15  CLOSURE_CERTIFIED_WITH_RESIDUAL
```

No event is removed.

The final status is reconstructed from the whole sequence.

I.27 State Reconstruction

I.27.1 Initial state

$\Sigma_0 =$
(
n = 100,
C = $300.00,
D = $8,200.00,
B = $193.25,
CallStatus = None,
ClosureStatus = Closed
). (I.149)

I.27.2 Post-shock state

$\Sigma_1 =$
(
n = 100,
C = $300.00,
D = $8,200.00,
B = −$505.90,
CallStatus = Open,
ClosureStatus = Open
). (I.150)

I.27.3 Partial-collateral state

$\Sigma_2 =$
(
n = 100,
C = $550.00,
D = $8,200.00,
B = −$255.90,
CallStatus = Open,

ClosureStatus = Partial
). (I.151)

I.27.4 Post-execution pre-settlement state

Σ_3 =
(
n_economic = 91,
n_ledger = 100,
C = $550.00,
D_ledger = $8,200.00,
ExpectedDebt = $7,268.50,
CallStatus = Open,
ClosureStatus = OpenReturn
). (I.152)

This state visibly separates:

- economic action;
- ledger recognition.

I.27.5 Final state

Σ_4 =
(
n = 91,
C = $550.00,
D = $7,270.00,
B = $9.65,
CallStatus = Closed,
ClosureStatus = ClosedWithResidual
). (I.153)

I.28 Recursive Protocol Update

I.28.1 Why closure changes the future world

Although the call has closed, the account now has:

- fewer asset units;
- less debt;
- $250.00 more encumbered cash;
- a recorded collateral-processing failure;
- a recorded margin event;
- lower free liquidity.

The next period is not governed by exactly the same effective parameters.

I.28.2 Proposed next-period haircut

Suppose the broker applies a tighter collateral factor:

$h_4 = 0.68.$ (I.154)

The prior factor was:

$h_3 = 0.70.$ (I.155)

Thus:

$\Delta h = -0.02.$ (I.156)

I.28.3 Proposed funding spread update

Suppose the funding spread rises by:

$\Delta s_F = 25$ basis points. (I.157)

This is a ledger-conditioned change.

I.28.4 Mass-proxy update

Suppose identity inertia increases because future actions require enhanced review.

Define the prior mass proxy:

$m_{eff,3} = 1.00.$ (I.158)

The residual and margin event add:

$\Delta m = 0.15.$ (I.159)

Therefore:

$m_{eff,4} = 1.15.$ (I.160)

I.28.5 Closure-capacity update

Suppose the collateral-processing incident reduces effective closure capacity temporarily:

$c_{P,3} = 1.00.$ (I.161)

$c_{P,4} = 0.90.$ (I.162)

The account's next permitted action rate may therefore be reduced.

I.28.6 Recursive update object


```
{
  "event_id": "EVENT-MA104-RECURSIVE-UPDATE-0001",
  "event_type": "POST_CLOSURE_PARAMETER_UPDATE",
  "subject_id": "ID-MA-104-X-001",
  "protocol_id": "MARGIN-US-01",
  "effective_time": "2026-07-15T14:15:00Z",
  "recorded_time": "2026-07-15T14:16:00Z",
  "actor": "RISK-POLICY-ENGINE-1",
  "authority": "RISK-COMMITTEE-DELEGATE",
  "payload": {
    "collateral_factor_before": 0.70,
    "collateral_factor_after": 0.68,
    "funding_spread_change_bps": 25,
    "mass_proxy_before": 1.00,
    "mass_proxy_after": 1.15,
    "closure_capacity_before": 1.00,
    "closure_capacity_after": 0.90,
    "reason_codes": [
      "RECENT_MARGIN_EVENT",
      "COLLATERAL_PROCESSING_RESIDUAL",
      "REDUCED_FREE_LIQUIDITY"
    ]
  },
  "parent_event_id": "CERT-MA104-CALL-0001",
  "supersedes_event_id": null,
  "evidence_snapshot_id": "SNAP-MA104-100200",
  "residual_ids": [
    "RES-MA104-CASH-0001"
  ]
}
```

I.29 Recursive State Law

The completed event generates:

$L_4 = L_3 \oplus \mathrm{Trace_call} \oplus \mathrm{Trace_collateral} \oplus \mathrm{Trace_sale} \oplus \mathrm{Residuals} \oplus \mathrm{ClosureCertificate}.$ (I.163)

The next parameter set is:

$\Theta_4 = U_{\Theta}(\Theta_3, L_4, \mathscr{R}_3).$ (I.164)

Therefore:

$\Theta_4 \neq \Theta_3.$ (I.165)

The next-period account begins with:

$K_4 = \mathrm{Modified}(K_3).$ (I.166)

$\vec{q}_{\mathrm{struct},4} = \mathrm{UpdatedStructuralCharge}.$ (I.167)

$\vec{q}_{\mathrm{eff},4} = \Lambda(X_4, L_4, \mathscr{R}_4)\vec{q}_{\mathrm{struct},4}.$ (I.168)

$M_4 = \mathrm{UpdatedIdentityInertia}.$ (I.169)

$c_{P,4} = \mathrm{UpdatedClosureCapacity}.$ (I.170)

I.30 Claim-Status Object

The final governed claim is:


```
{
  "claim_status": {
    "protocol": {
      "id": "MARGIN-US-01",
      "version": 3
    },
    "closure_level": "EVENT",
    "admission": {
      "candidate_breach": true,
      "valid_margin_call": true
    },
    "residual": {
      "open_material_residual": false,
      "accepted_historical_residual": true,
      "residual_ids": [
        "RES-MA104-CASH-0001"
      ]
    },
    "transport": {
      "market_to_collateral": "PASSED",
      "trade_to_settlement": "PASSED",
      "settlement_to_treasury": "PASSED"
    },
    "authority": {
      "gate_authority": "BME-2",
      "closure_authority": "MARGIN-CLOSURE-CONTROLLER-1"
    },
    "model_level": "ACTION_LEDGER_WITH_GOVERNED_TRANSPORT",
    "validation_status": "ILLUSTRATIVE_WORKED_EXAMPLE",
    "final_claim": "MARGIN_EVENT_CLOSED_WITH_RESIDUAL",
    "prohibited_claims": [
      "PHYSICAL_SPIN_CONFIRMED",
      "UNIVERSAL_GAUGE_CHARGE_CONFIRMED",
      "DIRAC_MODEL_EMPIRICALLY_VALIDATED"
    ]
  }
}
```

I.31 Comparison with a Mutable-Status Implementation

I.31.1 Weak implementation

A conventional mutable table might contain only:


```
{
  "account_id": "MA-104",
  "margin_status": "CLOSED",
  "current_buffer": 9.65
}
```


This record loses:

- the original breach;
- the admitted call;

- the rejected $50.00 collateral;
- the sale;
- the settlement delay;
- the $1.50 debt difference;
- the explanation;
- the charge transition;
- the frame transport;
- the recursive update.

I.31.2 Governed implementation

The governed runtime preserves:

Measured State
→ Candidate
→ Gate
→ Action
→ Partial Failure
→ Alternative Branch
→ Ledger Return
→ Residual Explanation
→ Closure
→ Recursive Consequence. (I.171)

The difference is not merely greater data volume.

It is a different ontology of financial completion.

I.32 Audit Queries Applied to the Example

I.32.1 Was the call admitted by valid authority?

Result:

Yes. (I.172)

Authority:

BME-2. (I.173)

I.32.2 Was the account marked closed before settlement?

Result:

No. (I.174)

The closure certificate was issued only after:

- sale settlement;
- debt recognition;
- transport reconciliation;
- residual disclosure.

I.32.3 Did the $50.00 rejected collateral disappear?

Result:

No. (I.175)

It remains as an accepted historical residual.

I.32.4 Was the $1.50 debt mismatch deleted?

Result:

No. (I.176)

It was:

- detected;
- investigated;
- explained;
- closed through a resolution trace.

I.32.5 Did structural charge change without a vertex?

Result:

No. (I.177)

The asset-charge reduction is linked to the sale vertex.

The collateral-charge activation is linked to the call gate.

The collateral-charge deactivation is linked to verified cure.

I.32.6 Was market-to-collateral mismatch measured by raw subtraction alone?

Result:

No. (I.178)

The transport included:

- haircut;
- unit-level rounding;
- concentration adjustment.

I.33 Closure-Defect Timeline

Define normalized closure defect:

$\Delta_{close}(t)$. (I.179)

A stylized event sequence is:

Before call:

$\Delta_{close} \approx 0$. (I.180)

After call:

$\Delta_{close} \uparrow$. (I.181)

After partial collateral:

Δ_{close} decreases but remains material. (I.182)

After sale execution:

Δ_{close} may increase temporarily because action outruns settlement. (I.183)

After settlement:

Δ_{close} decreases sharply. (I.184)

After debt reconciliation:

$\Delta_{close} \to 0$ within tolerance. (I.185)

The trajectory illustrates why one-cycle action is not sufficient.

I.34 Financial Gauge–Dirac Interpretation

I.34.1 Continuous pre-gate propagation

Before the call, the account may be represented by:

$(i\mathcal{D}_{fin} - M)\Psi = \mathscr{R}_{cont}$. (I.186)

The market phase and collateral factor move continuously or quasi-continuously.

I.34.2 Gate jump

At 10:05:

$\Psi^{+} = G_call\Psi^{-} + \eta_call.$ (I.187)

The gate:

- activates q_C^op;
- changes account status;
- opens the return cycle;
- creates a persistent trace.

I.34.3 Branch operators

The first branch is:

G_collateral. (I.188)

It produces partial cure and a $50.00 residual.

The second branch is:

G_deleveraging. (I.189)

It changes:

- asset charge;
- quantity;
- expected debt;
- settlement obligations.

I.34.4 Gauge transport

The financial identity moves through:

Market
→ Collateral
→ Settlement
→ Treasury
→ Closure Controller. (I.190)

Each edge has its own connection and residual test.

I.34.5 Ledger return

The ledger return transforms:

Action-Complete but Ledger-Open (I.191)

into:

Accountable Successor Identity. (I.192)

I.34.6 Recursive backreaction

The closure certificate alters future:

- haircut;
- funding spread;
- mass proxy;
- closure capacity.

Thus the event changes the operator governing the next period.

I.35 What the Example Demonstrates

The example demonstrates eight distinct principles.

1. A breach is not yet a call

At 10:02, a candidate existed.

The event was admitted only at 10:05 by valid authority.

---

2. A call is not closure

The call opened an obligation.

It did not resolve it.

---

3. An instruction is not ledger return

The $300.00 cash instruction produced only $250.00 admitted collateral.

---

4. An execution is not settlement

The 9-unit sale changed the economic state before the settlement and treasury ledgers returned.

---

5. Economic cost is not residual

The $1.50 settlement charge was an economic cost.

Once included in the expected transport, it ceased to be unexplained residual.

---

6. Frame difference is not necessarily inconsistency

The collateral value differed from raw market value because of:

- haircut;
- rounding;
- concentration adjustment.

---

7. Closure can preserve residual

The event closed with an accepted historical record of the failed $50.00 collateral portion.

---

8. Closure changes the future system

The account emerged with:

- fewer units;
- lower debt;
- more encumbered cash;
- tighter haircut;
- higher funding spread;
- lower closure capacity.

I.36 Master Machine-Readable Recurrence

The event may be summarized as:

```
STATE_0:
    safe account

MARKET_AND_COLLATERAL_SHOCK:
    R decreases
    h decreases
    B becomes negative

CANDIDATE:
    breach calculated

GATE:
    margin call admitted
    collateral charge activated
    closure state opened
```

```
BRANCH_1:
    cash collateral instructed
    partial admission
    residual recorded

BRANCH_2:
    sale instructed
    sale executed
    asset charge reduced
    settlement obligation created

LEDGER_RETURN:
    sale settled
    debt updated
    cost recognized
    transport reconciled

CLOSURE:
    final buffer positive
    charge reconciled
    residual disclosed
    certificate issued

RECURSION:
    haircut tightened
    funding spread increased
    mass proxy increased
    closure capacity reduced
```

In mathematical form:

$\mathcal{S}_0$
$\to \Phi_{\text{market,collateral}}$
$\to \text{Candidate}_1$
$\to G_{\text{call}}$
$\to G_{\text{collateral}}$
$\to \mathscr{R}_{\text{collateral}}$
$\to G_{\text{deleveraging}}$
$\to U_{\text{settlement}}$
$\to U_{\text{treasury}}$
$\to U_{\text{reconciliation}}$
$\to \text{Cert}_{\text{close}}$
$\to U_{\Theta}$
$\to \mathcal{S}_4.$ (I.193)

I.37 Compact Final State Object

```
{
  "subject_id": "ID-MA-104-X-001",
  "state_time": "2026-07-15T14:16:00Z",
  "financial_state": {
    "quantity": 91.0,
    "market_mark": 105.63,
    "collateral_factor": 0.68,
    "posted_collateral": 550.0,
    "debt_balance": 7270.0,
    "margin_buffer_at_closure_rule": 9.65,
    "currency": "USD"
  },
  "charge_state": {
```

```
    "asset_charge_units": 91.0,
    "funding_charge_amount": -7270.0,
    "collateral_charge_structural": -1.0,
    "collateral_charge_operational": 0.0
  },
  "closure_state": {
    "status": "CLOSED_WITH_RESIDUAL",
    "closure_sign": 1,
    "action_ledger_defect": 0.0
  },
  "transport_state": {
    "market_to_collateral": "PASSED",
    "trade_to_settlement": "PASSED",
    "settlement_to_treasury": "PASSED"
  },
  "residual_state": {
    "open_material_residuals": [],
    "accepted_historical_residuals": [
      "RES-MA104-CASH-0001"
    ],
    "closed_residuals": [
      "RES-MA104-DEBT-0001"
    ]
  },
  "recursive_parameters": {
    "next_collateral_factor": 0.68,
    "funding_spread_change_bps": 25,
    "mass_proxy": 1.15,
    "closure_capacity": 0.90
  }
}
```

I.38 Appendix Conclusion

The worked example shows what a machine-readable Financial Gauge–Dirac–Gate–Ledger system would actually need to preserve.

It must preserve:

- the safe pre-event state;
- the evidence available at breach time;
- the distinction between breach and admitted call;
- the authority that committed the call;
- the collateral instruction;
- the partial collateral failure;
- the sale instruction;
- execution;
- settlement;
- debt recognition;
- explained economic cost;

- cross-frame transport;
- structural charge changes;
- accepted residual;
- closure certificate;
- future parameter update.

The complete lifecycle is:

Safe Identity
→ Pressure and Constraint
→ Candidate Breach
→ Authoritative Gate
→ Operational Charge Activation
→ Outward Action
→ Partial Failure
→ Alternative Branch
→ Settlement
→ Ledger Return
→ Residual Explanation
→ Accountable Closure
→ Revised Financial World. (I.194)

The core machine-readable principle is:

> **The final status must be reconstructible from immutable evidence, gates, actions, ledger returns, transports, and residuals—not merely asserted by the latest mutable field.**

The core financial principle is:

> **A margin account is not closed when the call is sent, collateral is attempted, or a trade is executed. It is closed when the transformed identity has returned through the required ledgers, its charges reconcile, its frame transports are accountable, and its unresolved consequences remain visible.**

The next appendix will consolidate the article into a **formal definitions, propositions, conjectures, and falsifiers register**, distinguishing exact identities from proposed constructions, empirical hypotheses, and metaphorical correspondences.

Appendix J — Formal Definitions, Propositions, Conjectures, and Falsifiers Register

J.1 Purpose

The preceding article developed a layered architecture containing:

- Complex CAPM valuation;
- structural financial charge;
- authoritative margin gates;

- action–ledger double closure;
- governed financial frame transport;
- loop residual and curvature candidates;
- identity mass;
- a first-order Gauge–Dirac kernel;
- discrete gate jumps;
- recursive ledger backreaction.

These objects do not all possess the same epistemic status.

Some relations are exact consequences of definitions.

Some are modelling choices.

Some are proposed structural interpretations.

Some are empirical conjectures.

Some are cross-domain analogies.

This appendix separates them formally.

The governing editorial rule is:

Claim Strength ≤ Demonstrated Mathematical and Empirical Support. (J.1)

The source Financial Standard Model makes the same distinction: its present result is a structural research programme and testable architecture, not a completed physical derivation or validated universal financial law.

J.2 Claim Classes

J.2.1 Class I — Mathematical identity

A Class I statement follows exactly from declared definitions and ordinary algebra.

Notation:

[I]. (J.2)

Example:

$A^2 = R^2 + Q^2$ (J.3)

after defining:

$Q := \sqrt{(A^2 - R^2)}$. (J.4)

A Class I statement may still have limited practical relevance if its definitions are poorly chosen.

Exactness does not establish usefulness.

J.2.2 Class D — Definition

A Class D statement assigns meaning to a symbol or object.

Notation:

[D]. (J.5)

Example:

B := C + nhR − D. (J.6)

A definition cannot be falsified in the same manner as an empirical hypothesis.

It can instead be judged:

- coherent;
- incoherent;
- operational;
- unobservable;
- useful;
- redundant;
- misleading.

J.2.3 Class C — Proposed construction

A Class C statement introduces a mathematical architecture whose internal consequences may be exact once the construction is accepted.

Notation:

[C]. (J.7)

Example:

$\Psi := [\psi_A, \psi_L]^T$. (J.8)

The matrix algebra following equation (J.8) may be exact.

But the decision to represent the financial subject through this doublet remains a proposed construction.

J.2.4 Class T — Structural interpretation

A Class T statement assigns a domain interpretation to a mathematical or operational object.

Notation:

[T]. (J.9)

Example:

Q = first-order valuation-phase exposure. (J.10)

This interpretation is supported by:

$\partial R/\partial \theta = -Q$. (J.11)

But the interpretation remains specific to the declared valuation geometry.

J.2.5 Class H — Empirical hypothesis

A Class H statement may be supported, weakened, or rejected by observations.

Notation:

[H]. (J.12)

A complete hypothesis must declare:

- protocol;
- variables;
- outcome;
- horizon;
- benchmark;
- tolerance;
- falsification rule.

J.2.6 Class A — Analogy

A Class A statement maps a structural relation from another domain.

Notation:

[A]. (J.13)

Example:

Action–ledger double closure is spin-like. (J.14)

An analogy must state:

- what is preserved;
- what changes;
- what is excluded;

- what evidence would justify a stronger correspondence.

Therefore:

Analogy ≠ Mechanism. (J.15)

Analogy ≠ Ontological Identity. (J.16)

J.2.7 Class G — Governance rule

A Class G statement defines a discipline for constructing, promoting, reducing, or revising claims.

Notation:

[G]. (J.17)

Example:

No Independent Ledger Return ⇒ No Financial Spin Claim. (J.18)

The source framework imposes this discipline explicitly for charge, spinor, gauge, complex, and other advanced terminology.

J.3 Formal Definition Register

Definition J-D1 — Protocol

A financial protocol is:

P := (B,Δ,h,u,Φ,G,T,R,V). (J.19)

where:

- B = system boundary;
- Δ = observation and aggregation rule;
- h = declared horizon;
- u = admissible intervention set;
- Φ = feature and projection map;
- G = gate rules;
- T = trace rules;
- R = residual rules;
- V = transport rules.

Status: [D]

A financial claim without a declared protocol is incomplete.

Definition J-D2 — Bounded financial identity

A bounded financial identity is:

K := (Instrument,Quantity,Owner,Obligor,Account,Contract,Currency,Maturity,Lineage,Status). (J.20)

The identity is bounded when the protocol determines:

- what is inside;
- what is outside;
- which transformations preserve it;
- which transformations convert it;
- which events extinguish it.

Status: [D/C]

Definition J-D3 — Identity equivalence

Two states K_1 and K_2 are identity-equivalent under protocol P when:

$K_1 \simeq_P K_2$. (J.21)

This means that all protocol-mandatory invariants agree within tolerance.

Numerical equality is not required.

Status: [D]

Definition J-D4 — Baseline valuation amplitude

For a future cash flow CF_T:

A_T := CF_T/(1 + r_base)^T. (J.22)

Status: [D]

Definition J-D5 — CAPM-admitted value

Let:

r_CAPM := r_base + βERP. (J.23)

Then:

R_T := CF_T/(1 + r_CAPM)^T. (J.24)

Status: [D]

Definition J-D6 — Valuation phase

When:

$0 \le R \le A,$ (J.25)

define:

$\theta := \arccos(R/A).$ (J.26)

Status: [D]

Definition J-D7 — Conjugate pressure coordinate

Define:

$Q := \sqrt{(A^2 - R^2)}.$ (J.27)

Then the complex valuation state is:

$Z := R + iQ.$ (J.28)

Status: [D/C]

This Q is specific to the declared amplitude and admission construction.

It is not the generic name for every unobserved financial risk.

Definition J-D8 — Collateral-admitted value

For collateral factor h:

$R_C := hR_M.$ (J.29)

where:

$0 \le h \le 1.$ (J.30)

Status: [D]

Definition J-D9 — Collateral pressure coordinate

Under the shared-amplitude convention:

$Q_C := \sqrt{(A^2 - R_C^2)}.$ (J.31)

Status: [C]

The shared-amplitude convention is a proposed sequential-filter construction, not a universal collateral law.

Definition J-D10 — Margin buffer

For position quantity n, collateral factor h, market mark R_M, posted collateral C, and debt D:

$B := C + nhR_M - D.$ (J.32)

Status: [D]

Definition J-D11 — Margin shortfall

Given target buffer B_{target}:

$S_B := [B_{target} - B]_+.$ (J.33)

where:

$[x]_+ := \max(x,0).$ (J.34)

Status: [D]

Definition J-D12 — Candidate breach

A candidate breach exists when:

$B < B_{target}.$ (J.35)

Status: [D]

A candidate breach is not yet an admitted margin event.

Definition J-D13 — Authoritative gate

A gate is:

$G := (Rule, Authority, Decision, Trace, Residual).$ (J.36)

A gate is valid when:

$$\begin{aligned} &RuleValid \\ &\wedge AuthorityValid \\ &\wedge EvidenceValid \\ &\wedge DecisionCommitted \\ &\wedge TraceWritten. \end{aligned}$$ (J.37)

Status: [D/G]

A threshold crossing without authority or trace is not a complete gate.

Definition J-D14 — Trace

A trace is a persistent, versioned record of:

- evidence;
- decision;
- authority;
- time;
- action;
- consequence;
- residual.

Denote:

$T_k := \mathrm{Trace}(\mathrm{Event}_k).$ (J.38)

Status: [D]

Definition J-D15 — Residual

Residual is the unresolved difference remaining after a declared projection, gate, transport, action, or closure:

$\mathscr{R}_P := \mathrm{ObservedState} - \mathrm{ProtocolExpectedState}.$ (J.39)

Where direct subtraction is inappropriate:

$\mathscr{R}_P := \mathrm{Diff}_P(\mathrm{ObservedState},\mathrm{ExpectedState}).$ (J.40)

Status: [D]

Residual is not automatically error.

It may represent:

- missing evidence;
- timing difference;
- disputed authority;
- lawful but unintegrated consequence;
- model inadequacy;
- unresolved identity mismatch.

Definition J-D16 — Ledger

The ledger is the ordered, versioned inheritance of admitted trace and residual:

$L_{k+1} := U_L(L_k,T_k,\mathscr{R}_k).$ (J.41)

Status: [C/T]

A database is not automatically a ledger in this stronger sense.

The ledger must affect future admissibility or interpretation.

Definition J-D17 — Structural financial charge

A candidate structural financial charge is a field-indexed relational orientation carried by a bounded financial identity:

$q_r := \mathrm{Orientation}(K \mid \mathrm{Field}_r, P).$ (J.42)

Status: [C/T]

A candidate q_r is admitted as charge only if it possesses:

- stable carrier;
- field;
- sign;
- coupling;
- transport;
- vertex;
- residual register.

The source charge audit demands precisely these additional obligations before a coordinate should be promoted from sensitivity or exposure to charge.

Definition J-D18 — Structural charge vector

Define:

$\vec{q}_{\mathrm{struct}} := (q_A, q_F, q_C, \ldots).$ (J.43)

where coordinates refer to distinct fields such as:

- asset claim;
- funding;
- collateral;
- liquidity;
- optionality;
- reporting.

Status: [C]

Coordinates from different fields do not cancel automatically.

Definition J-D19 — Operational charge

For activation function a_r:

$q_r^{op} := a_r(X,L,G)q_r^{struct}.$ (J.44)

where:

$0 \le a_r \le 1.$ (J.45)

Status: [C]

For a binary margin-call activation:

$$a_C = \begin{cases} 0, & \text{call inactive;} \\ 1, & \text{call active.} \end{cases}$$ (J.46)

Definition J-D20 — Effective charge

Define:

$\vec{q}_{eff} := \Lambda(X,L,Regime)\vec{q}_{struct}.$ (J.47)

where Λ is a screening and amplification operator.

Status: [C/H]

Possible elements of Λ include:

- leverage;
- beta;
- collateral;
- hedging;
- liquidity;
- concentration;
- netting;
- ledger history.

Definition J-D21 — Charge vertex

A charge vertex is an admitted event transforming financial identities and charge states:

$\mathcal{V}_k := (K_{in}, \vec{q}_{in}, G_k, K_{out}, \vec{q}_{out}, T_k, \mathscr{R}_k).$ (J.48)

Status: [C]

Definition J-D22 — Action state

The action state records outward intervention:

$\psi_A := \Phi_A(\text{ActionEvidence}).$ (J.49)

Examples include:

- margin call;
- collateral instruction;
- sale;
- exercise;
- liquidation;
- default declaration.

Status: [D/C]

Definition J-D23 — Ledger-return state

The ledger-return state records institutional integration of the action:

$\psi_L := \Phi_L(\text{LedgerEvidence}).$ (J.50)

Examples include:

- settlement;
- collateral admission;
- debt update;
- accounting recognition;
- legal perfection;
- residual registration.

Status: [D/C]

Definition J-D24 — Financial action–ledger doublet

Define:

$\Psi := [\psi_A, \psi_L]^{\mathsf{T}}.$ (J.51)

Status: [C]

This object should not be called a financial spinor merely because it has two components.

Definition J-D25 — Expected ledger return

Let:

$U_{AL} : \psi_A \to \psi\hat{}_L.$ (J.52)

Then:

$\psi\hat{}_L := U_{AL}\psi_A.$ (J.53)

Status: [C]

Definition J-D26 — Action–ledger closure defect

Define:

$\delta_{spin} := \psi_L - U_{AL}\psi_A.$ (J.54)

The weighted closure defect is:

$\Delta_{spin}^2 := \delta_{spin}^{\dagger} W \delta_{spin}.$ (J.55)

Status: [C]

The safer operational name is:

Action–Ledger Closure Defect. (J.56)

The stronger term spinor split is conditional upon admission tests.

Definition J-D27 — Financial spin

Financial spin is the closure class of a bounded identity whose outward action creates obligations requiring an independent ledger return before accountable successor identity is restored.

In compressed form:

Spin_fin
:= Identity

- Action
- Consequence
- IndependentReturn
- Residual
- AccountableSuccessor. (J.57)

Status: [T/C]

The source spinor admission criteria require bounded identity, consequential outward transition, independent ledger return, double closure, residual, and cross-frame recognition.

Definition J-D28 — Financial frame

A financial frame is:

$F_f := (Boundary_f, Observation_f, Horizon_f, Authority_f, Trace_f).$ (J.58)

Status: [D]

A frame is not merely a different table or software system.

It must apply a distinct observation, admission, authority, or recognition rule.

Definition J-D29 — Frame transport

For frames A and B:

$T_{AB} : S_A \to \hat{S}_B.$ (J.59)

The expected target is:

$\hat{S}_B := T_{AB}(S_A; \mathcal{A}_{AB}).$ (J.60)

where $\mathcal{A}_{AB}$ is the connection.

Status: [D/C]

Definition J-D30 — Edge transport residual

Define:

$r_{AB} := S_B^{obs} - T_{AB}(S_A).$ (J.61)

Status: [D]

Definition J-D31 — Financial gauge candidate

A financial transport system is gauge-like when it contains:

- multiple legitimate local frames;
- a stable identity invariant;
- a declared connection;
- lawful local transformations;
- covariance;

- loop tests;
- residual honesty.

Status: [C/T]

Without these conditions, use:

Governed Frame Transport. (J.62)

Definition J-D32 — A-B Fixedness

A-B Fixedness holds when two frames identify the same financial event or identity after lawful translation:

ABFix_P(S;A,B)
⇔ IdentityMatch
∧ InvariantPreservation
∧ RecordAccessibility
∧ ResidualDisclosure. (J.63)

Status: [C/T]

The generalized Dirac source defines A-B Fixedness through frame translation, compatible identification, and accessible record rather than simple numerical equality.

Definition J-D33 — Loop holonomy

For loop ℓ with directed transports U_e:

H_ℓ := ∏_{e∈ℓ}U_e. (J.64)

Status: [C]

Definition J-D34 — Governed loop residual

Let H_ℓ^P include authorized time, fee, and conversion effects.

Then:

$\mathscr{R}$_ℓ := S_return^obs − H_ℓ^PS_start. (J.65)

Status: [C]

Definition J-D35 — Financial curvature candidate

A loop residual may be called curvature-like only when:

- the connection is defined;
- the loop is meaningful;

- expected path effects are removed;
- transport order matters;
- the residual is stable and informative.

Status: [T/H]

Otherwise use:

Loop Transport Residual. (J.66)

Definition J-D36 — Identity mass

Financial identity mass is the cost or inertia associated with changing a bounded financial identity while preserving recognizability:

m_fin ~ Cost of Identity-Preserving Change/Identity Displacement. (J.67)

Status: [T/C]

The source generalized Dirac framework treats mass as a compiled constraint bundle and, more specifically, as the coupling that prevents action and ledger from separating freely.

Definition J-D37 — Financial mass operator

Define:

M := m_II₂ + m_Cσ_x + m_Δσ_z. (J.68)

Explicitly:

M = [[m_I + m_Δ,m_C],[m_C,m_I − m_Δ]]. (J.69)

Status: [C]

Definition J-D38 — Coherent closure capacity

Define:

c_P := sup{v | Residual remains bounded and ABFix remains stable under protocol P}. (J.70)

Status: [C/H]

A weaker operational term is:

Verified Processing Capacity. (J.71)

Definition J-D39 — Financial Γ algebra

A candidate 1+1 closure representation is:

$\Gamma^0 := \sigma_z.$ (J.72)

$\Gamma^1 := i\sigma_y.$ (J.73)

with:

$\{\Gamma^a,\Gamma^b\} = 2\eta^{ab}I_2.$ (J.74)

Status: [C]

The algebraic identity is exact for the chosen matrices.

Its financial necessity is an empirical question.

Definition J-D40 — Financial Gauge–Dirac kernel

Define:

$\mathcal{D}_fin := \Gamma^0\nabla_\tau\hat{}P + c_P\Gamma^1\mathfrak{D}_G.$ (J.75)

The proposed continuous kernel is:

$(i\mathcal{D}_fin - M)\Psi = \mathscr{R}_cont.$ (J.76)

Status: [C/A]

This is a structural generalization, not a derivation from particle physics.

Definition J-D41 — Hybrid gate jump

At event time τ_k:

$\mathcal{S}(\tau_k^+) = \mathcal{J}_k[\mathcal{S}(\tau_k^-),L_k] + \mathscr{R}_jump,_k.$ (J.77)

Status: [C]

Definition J-D42 — Recursive financial closure

Define the complete recurrence:

$(K_k,\vec{q}_k,\Psi_k,\Theta_k,L_k)$
$\rightarrow \text{Flow}_k$
$\rightarrow \text{Gate}_k$
$\rightarrow \text{Action}_k$
$\rightarrow \text{LedgerReturn}_k$
$\rightarrow \text{Residual}_k$
$\rightarrow (K_{k+1},\vec{q}_{k+1},\Psi_{k+1},\Theta_{k+1},L_{k+1}).$ (J.78)

Status: [C/T]

J.4 Exact Propositions

The propositions in this section are exact only under their declared definitions and regularity conditions.

They do not by themselves establish empirical usefulness.

Proposition J-P1 — Complex valuation identity

Given definitions:

$\theta := \arccos(R/A)$, (J.79)

$Q := \sqrt{(A^2 - R^2)}$, (J.80)

and:

$0 \leq R \leq A$, (J.81)

then:

$R = A \cos \theta$. (J.82)

$Q = A \sin \theta$. (J.83)

$A^2 = R^2 + Q^2$. (J.84)

$Z = Ae^{xp(i\theta)}$. (J.85)

Status: [I]

Proof: Direct substitution.

Proposition J-P2 — Phase derivative

Holding A constant:

$\partial R/\partial \theta = -A \sin \theta$. (J.86)

Using $Q = A \sin \theta$:

$\partial R/\partial \theta = -Q$. (J.87)

Status: [I]

Interpretive consequence: Q is the magnitude of first-order real-value exposure to phase movement under the declared geometry.

Proposition J-P3 — Differential decomposition of admitted value

For:

$R = A \cos \theta$, (J.88)

the total differential is:

$$dR = \cos\theta dA - A\sin\theta d\theta. \qquad (J.89)$$

Therefore:

$$dR = (R/A)dA - Qd\theta. \qquad (J.90)$$

Status: [I]

Proposition J-P4 — Scalar haircut relation

Let:

$$H := A - R. \qquad (J.91)$$

Then:

$$Q^2 = A^2 - R^2. \qquad (J.92)$$

Therefore:

$$Q^2 = (A - R)(A + R). \qquad (J.93)$$

Hence:

$$Q^2 = H(A + R). \qquad (J.94)$$

Status: [I]

Consequence: Q is generally not equal to the scalar haircut H.

Proposition J-P5 — Collateral-pressure decomposition

Given:

$$R_C = hR_M, \qquad (J.95)$$

$$Q_M^2 = A^2 - R_M^2, \qquad (J.96)$$

$$Q_C^2 = A^2 - h^2R_M^2, \qquad (J.97)$$

then:

$$Q_C^2 = Q_M^2 + (1 - h^2)R_M^2. \qquad (J.98)$$

Status: [I under the shared-amplitude construction]

Proposition J-P6 — Collateral-phase relation

Given:

$$R_M = A\cos\theta_M, \qquad (J.99)$$

and:

$$R_C = hR_M = A \cos \theta_C, \qquad (J.100)$$

then:

$$\cos \theta_C = h \cos \theta_M. \qquad (J.101)$$

Therefore:

$$\theta_C = \arccos(h \cos \theta_M). \qquad (J.102)$$

Status: [I under the shared-amplitude construction]

Proposition J-P7 — Margin-buffer differential

Given:

$$B = C + nhR_M - D, \qquad (J.103)$$

the differential is:

$$dB = dC + nhdR_M + nR_Mdh + hR_Mdn - dD. \qquad (J.104)$$

Substituting equation (J.90):

$$dB = dC + nh(R_M/A)dA - nhQ_Md\theta_M + nR_Mdh + hR_Mdn - dD. \qquad (J.105)$$

Status: [I]

Proposition J-P8 — Phase-to-buffer sensitivity

Holding A, C, h, n, and D constant:

$$\partial B/\partial \theta_M = nh\partial R_M/\partial \theta_M. \qquad (J.106)$$

Using equation (J.87):

$$\partial B/\partial \theta_M = -nhQ_M. \qquad (J.107)$$

Status: [I]

This is a local identity, not a claim that phase is the dominant empirical driver of buffer movement.

Proposition J-P9 — Local phase distance to the margin boundary

For small adverse phase displacement and fixed A, C, h, n, and D:

$$\Delta B \approx -nhQ_M\Delta\theta_M. \qquad (J.108)$$

If the current surplus above target is:

$B - B_{target} > 0,$ (J.109)

then the approximate adverse phase displacement reaching the gate is:

$\Delta\theta_M^* \approx (B - B_{target})/(nhQ_M).$ (J.110)

Status: [I as a first-order approximation]

Proposition J-P10 — Charge balance as a declared accounting identity

Define vertex residual:

$\vec{r}_q := \Sigma\vec{q}_{in} + \vec{q}_{gate} - \Sigma\vec{q}_{out}.$ (J.111)

Then:

$\Sigma\vec{q}_{in} + \vec{q}_{gate} = \Sigma\vec{q}_{out} + \vec{r}_q.$ (J.112)

Status: [I by definition]

Limitation: Equation (J.112) does not prove physical conservation.

It defines the unresolved imbalance under the declared charge map.

Proposition J-P11 — Zero closure defect under exact return

If:

$\psi_L = U_{AL}\psi_A,$ (J.113)

then:

$\delta_{spin} = \psi_L - U_{AL}\psi_A = 0.$ (J.114)

Therefore:

$\Delta_{spin} = 0.$ (J.115)

for any positive semidefinite W.

Status: [I]

The empirical question is whether U_{AL} is meaningful, prospective, and independently measurable.

Proposition J-P12 — Covariance of the transport defect

Let local frame transformations be:

$S_{A'} = G_A S_A.$ (J.116)

$S_{B'} = G_{BS_B}$. (J.117)

Let:

$U_{AB'} = G_{BU_{ABG_A^{-1}}}$. (J.118)

Then:

$r_{AB'} = S_{B'} - U_{AB'S_{A'}}$. (J.119)

Substitution gives:

$r_{AB'} = G_B(S_B - U_{ABS_A})$. (J.120)

Therefore:

$r_{AB'} = G_{Br_{AB}}$. (J.121)

Status: [I]

A compatible norm may render the residual magnitude representation-invariant.

Proposition J-P13 — Flat governed loop

If:

$S_{return}^{obs} = H_{\ell}^{PS_{start}}$, (J.122)

then:

$\mathscr{R}_{\ell} = 0$. (J.123)

Status: [I]

This does not require:

$S_{return}^{obs} = S_{start}$. (J.124)

Authorized costs, time effects, and identity conversion may legitimately change the returning state.

Proposition J-P14 — Mass-operator eigenvalues

For:

$M = [[m_I + m_\Delta, m_C], [m_C, m_I - m_\Delta]]$, (J.125)

the characteristic equation gives:

$m_{\pm} = m_I \pm \sqrt{(m_C^2 + m_\Delta^2)}$. (J.126)

Status: [I]

The financial interpretation of these eigenmodes remains proposed.

Proposition J-P15 — Synchronized and opposed closure modes

For symmetric coupling matrix:

$K = [[k_A,k_C],[k_C,k_L]],$ (J.127)

the eigenvalues are:

$\lambda_{\pm} = (k_A + k_L)/2 \pm \sqrt{[((k_A - k_L)/2)^2 + k_C^2]}.$ (J.128)

When:

$k_A = k_L,$ (J.129)

the associated eigenvectors are proportional to:

$e_+ = [1,1]^T.$ (J.130)

$e_- = [1,-1]^T.$ (J.131)

Status: [I]

Their interpretation as synchronized and mismatch modes is [T/H].

Proposition J-P16 — Recursive state dependence implies non-Markovianity in reduced state

Suppose the full state is:

$\mathcal{S}_k^{full} := (X_k,L_k).$ (J.132)

and:

$X_{k+1} = F(X_k,L_k,\varepsilon_{k+1}).$ (J.133)

If L_k cannot be written as a function solely of X_k, then the reduced process X_k is generally not first-order Markov.

Status: [I under the stated conditions]

The full state may remain Markov when L_k is included.

J.5 Structural Propositions

These propositions are not algebraic theorems.

They are methodological consequences of the declared ontology.

Proposition J-SP1 — Nonlinearity is insufficient for charge

A nonlinear mapping:

$y = f(x)$ (J.134)

does not by itself define:

- a stable carrier;
- a field;
- a relational sign;
- a transport law;
- a vertex.

Therefore:

Nonlinearity Alone $\nRightarrow$ Financial Charge. (J.135)

Status: [T/G]

Proposition J-SP2 — Constraint is insufficient for spin

A scalar constraint:

$g(X) \geq 0$ (J.136)

does not by itself create:

- outward action;
- consequential obligation;
- independent ledger return;
- double closure.

Therefore:

Constraint Alone $\nRightarrow$ Financial Spin. (J.137)

Status: [T/G]

Proposition J-SP3 — Charge and spin are logically independent

A system may possess a stable claim–obligation orientation without an action–ledger double closure.

A system may possess two-stage action–ledger closure without a useful charge algebra.

Therefore:

Charge $\nRightarrow$ Spin. (J.138)

Spin ⇏ Charge. (J.139)

Status: [T]

Proposition J-SP4 — Threshold and gate are distinct

A threshold is a numerical condition.

A gate requires:

- applicability;
- authority;
- decision;
- trace;
- residual.

Therefore:

Threshold Crossing ≠ Admitted Event. (J.140)

Status: [T/G]

Proposition J-SP5 — Action and closure are distinct

An outward act may occur while:

- settlement remains open;
- debt remains unchanged;
- accounting remains unposted;
- legal status remains incomplete.

Therefore:

Action Completion ≠ Accountable Closure. (J.141)

Status: [T/G]

Proposition J-SP6 — Frame difference and residual are distinct

For frames A and B:

$S_B - S_A$ (J.142)

is not necessarily a meaningful residual.

The proper residual is:

S_B – T_ABS_A. (J.143)

Therefore:

Frame Difference ≠ Transport Failure. (J.144)

Status: [T/G]

Proposition J-SP7 — Economic loss and residual are distinct

A known fee or realized loss may be included in the expected transport.

Then the event may have:

EconomicLoss > 0, (J.145)

while:

Residual = 0. (J.146)

Therefore:

Economic Loss ≠ Unexplained Residual. (J.147)

Status: [T]

Proposition J-SP8 — Closure and restoration are distinct

A closed financial identity may differ numerically from its pre-event state:

Ψ_after ≠ Ψ_before. (J.148)

Closure requires:

Ψ_after ≃_P Ψ_expected-successor. (J.149)

Therefore:

Closure ≠ Return to Original Numbers. (J.150)

Status: [T]

Proposition J-SP9 — Residual and failure are distinct

A disclosed, classified, and inherited residual may coexist with accountable closure.

Therefore:

Residual ≠ Automatic Failure. (J.151)

But:

Hidden Material Residual ⇒ Governance Failure. (J.152)

Status: [T/G]

Proposition J-SP10 — Gauge language creates additional obligations

Calling a mapping gauge-like requires more than multiple frames.

It requires:

- invariant;
- connection;
- covariance;
- loop test;
- residual.

Therefore:

Multiple Representations Alone ⇏ Gauge Structure. (J.153)

Status: [G]

The source correspondence ladder similarly requires a progression from metaphor to role analogy, structural homology, operator correspondence, invariant correspondence, and only then restricted isomorphism.

Proposition J-SP11 — Dirac language creates additional obligations

Calling a model Dirac-like requires:

- irreducible multicomponent identity;
- first-order propagation;
- meaningful Γ structure;
- mass-like constraint;
- frame covariance;
- incremental value.

Therefore:

Two Variables + First-Order Equation ⇏ Financial Dirac System. (J.154)

Status: [G]

J.6 Empirical Conjectures

Conjecture J-C1 — Q incremental-information conjecture

Under a declared valuation protocol, Q contains information about future buffer deterioration beyond R and conventional controls:

$$I(\Delta B_{t+H}; Q_t \mid R_t, B_t, h_t, D_t, \text{Leverage}_t, \text{Volatility}_t) > 0. \qquad (J.155)$$

Status: [H]

Falsifier: Out-of-sample incremental information is nonpositive across predeclared samples.

Reduction: Q becomes descriptive notation or is removed.

Conjecture J-C2 — Phase-to-buffer transmission conjecture

The phase term:

$$-nhQ_M \Delta\theta_M \qquad (J.156)$$

provides a useful decomposition of observed margin-buffer movement beyond amplitude, haircut, debt, quantity, and collateral channels.

Status: [H]

Falsifier: The estimated phase channel is unstable, wrongly signed, or adds no out-of-sample explanatory value.

Conjecture J-C3 — Collateral-phase conjecture

Collateral phase θ_C and phase separation:

$$\Delta\theta_{MC} := \theta_C - \theta_M \qquad (J.157)$$

improve detection of margin vulnerability beyond market phase alone.

Status: [H]

Falsifier: θ_C and $\Delta\theta_{MC}$ add no information beyond h, R_M, and B.

Conjecture J-C4 — Gate-separation conjecture

An authorized gate separates persistent consequential events from ungated numerical candidates:

Pr(Persistence | Candidate,Gate)

Pr(Persistence | Candidate,¬Gate). (J.158)

Status: [H]

This conjecture is part of the source empirical contract.

Falsifier: Valid gate status adds no persistent-consequence or calibration gain beyond the raw threshold.

Conjecture J-C5 — Charge-typing conjecture

Typed structural charges improve prediction or reconciliation beyond unsigned notional and ordinary exposure labels:

$U(\vec{q}_{struct},\vec{q}_{eff}) > U(Notional,ConventionalExposure)$. (J.159)

Status: [H]

Falsifier: Charge coding adds no incremental value or cannot be reproduced reliably.

Reduction: Charge → Signed Exposure or Contractual Obligation.

Conjecture J-C6 — Charge-vertex residual conjecture

Unexplained charge residual:

$|\vec{r}_q| > 0$ (J.160)

predicts later:

- settlement failure;
- ownership dispute;
- collateral mismatch;
- accounting correction;
- legal reopening.

Status: [H]

Falsifier: Charge residual is unrelated to later failure after controlling for ordinary exception measures.

Conjecture J-C7 — Action–ledger closure conjecture

Transport-adjusted action–ledger defect predicts delayed or failed closure:

$\partial E[T_{close}]/\partial \Delta_{spin}^{excess} > 0$. (J.161)

Status: [H]

Falsifier: Δ_spin adds no information beyond call age, branch, shortfall, event size, and workflow status.

Reduction: Spinor → Two-Stage Workflow.

Conjecture J-C8 — False-closure conjecture

A large action–ledger defect identifies reported closure that remains protocol-open:

Pr(FalseClosure | Δ_spin high)

> Pr(FalseClosure | Δ_spin low). (J.162)

Status: [H]

Falsifier: The defect cannot distinguish verified from false closure.

Conjecture J-C9 — Governed-transport conjecture

Connection-adjusted residual produces fewer false reconciliation alarms than raw numerical difference:

FalseAlarm_GovernedTransport
< FalseAlarm_RawDifference. (J.163)

Status: [H]

Falsifier: Governed transport is no more accurate, stable, or interpretable than ordinary matching.

Conjecture J-C10 — A-B Fixedness conjecture

Higher A-B Fixedness predicts lower future cross-frame failure:

∂Pr(FutureFrameFailure)/∂ABFix < 0. (J.164)

Status: [H]

Falsifier: ABFix adds no information beyond ordinary data-quality or reconciliation metrics.

Conjecture J-C11 — Loop-residual conjecture

Normalized governed loop residual:

κ_loop := ‖$\mathscr{R}$_loop‖/[‖ExpectedReturn‖ + ε] (J.165)

predicts:

- settlement break;

- accounting restatement;
- margin escalation;
- legal dispute;
- false closure.

Status: [H]

Falsifier: κ_loop adds no value beyond pairwise residuals and ordinary exception counts.

Reduction: Curvature → Loop Reconciliation Error.

Conjecture J-C12 — Order-effect conjecture

Noncommuting financial gates create economically relevant path dependence:

$G_X G_Y \mathcal{S} \neq G_Y G_X \mathcal{S}$. (J.166)

Status: [H]

Falsifier: After controlling for known timing and state differences, gate order has no reproducible effect.

Conjecture J-C13 — Identity-mass conjecture

Higher identity mass predicts greater cost or time required for identity-preserving transformation:

$\partial E[T_{close}]/\partial m_{eff} > 0$. (J.167)

Status: [H]

Falsifier: The proposed mass proxy adds no information beyond liquidity, event size, complexity, and operational delay.

Reduction: Mass → Friction Index.

Conjecture J-C14 — Coupling-mass conjecture

The off-diagonal mass parameter m_C captures action–ledger binding distinct from ordinary delay and damping.

Status: [H]

Falsifier: m_C is unidentifiable or fully absorbed by simpler coupling and latency parameters.

Conjecture J-C15 — Finite closure-capacity conjecture

There exists a protocol-dependent threshold c_P such that:

v_action ≤ c_P ⇒ Residual remains bounded. (J.168)

v_action > c_P ⇒ Residual growth or A-B Fixedness deterioration accelerates. (J.169)

Status: [H]

The source generalized Dirac programme treats c_P as a preliminary empirical quantity defined by bounded residual and stable A-B Fixedness, not as a universal physical constant.

Falsifier: No stable threshold exists or ordinary queueing capacity explains all effects.

Conjecture J-C16 — Ledger-memory conjecture

Past trace and residual affect future financial behaviour after the current observable state is controlled:

$I(Y_t_{+}H;L_t \mid X_t) > 0.$ (J.170)

Status: [H]

Falsifier: Ledger variables provide no incremental information beyond current and finite-lag state variables.

Reduction: Recursive Ledger Model → Current-State or Finite-Lag Model.

Conjecture J-C17 — Residual-backreaction conjecture

Unresolved residual changes future:

- haircut;
- funding spread;
- effective charge;
- mass;
- permissions;
- closure capacity.

Formally:

$\Theta_{k+1} = U_\Theta(\Theta_k, L_{k+1}, \mathscr{R}_k)$ (J.171)

with:

$\partial\Theta_{k+1}/\partial\mathscr{R}_k \neq 0.$ (J.172)

Status: [H]

Falsifier: Residual has no reproducible effect after current financial state and known regime are controlled.

Conjecture J-C18 — Margin-cascade conjecture

Forced liquidation generates destabilizing feedback when:

$$\partial B_{next}/\partial \ell < 0 \qquad (J.173)$$

over a relevant liquidation interval.

The causal loop is:

$$B\downarrow \rightarrow Liquidation\uparrow \rightarrow MarketImpact\uparrow \rightarrow R\downarrow \rightarrow B\downarrow. \qquad (J.174)$$

Status: [H]

Falsifier: Liquidation consistently improves buffer after fees and impact, with no destabilizing region.

Conjecture J-C19 — Complex-priority conjecture

The constrained complex representation:

$$Z = R + iQ \qquad (J.175)$$

provides greater utility than a flexible unconstrained two-real-variable model.

Status: [H]

Falsifier:

$$U_{complex} \leq U_{real\text{-}pair}. \qquad (J.176)$$

Reduction:

$$Z \rightarrow (R,Q). \qquad (J.177)$$

Conjecture J-C20 — Γ-increment conjecture

The Γ algebra imposes useful and testable restrictions unavailable from generic two-component coupling matrices.

Status: [H]

Falsifier: Arbitrary matrix bases perform equally well and the Γ-derived squared relation yields no distinct observable consequence.

Reduction: Γ Algebra → Generic Coupling Matrix.

Conjecture J-C21 — Financial Dirac increment conjecture

The full hybrid Financial Gauge–Dirac model provides greater complexity-adjusted utility than:

- scalar margin model;
- workflow model;
- hybrid state-space model;
- multi-frame reconciliation model.

Formally:

U_GD > max(U_scalar,U_workflow,U_hybrid,U_transport). (J.178)

Status: [H]

Falsifier: Equation (J.178) fails prospectively.

Reduction: Gauge–Dirac → Lowest surviving simpler model.

Conjecture J-C22 — Constraint-generated transformation-memory conjecture

Financial charge and spin tend to arise when constraints become:

- identity-bearing;
- relational;
- authoritative;
- history-writing.

In compressed form:

Identity-Bearing Relational Constraint

- Authority
- Trace
  → Candidate Charge and Spin. (J.179)

Status: [H/T]

Falsifier: Systems satisfying these conditions show no greater charge stability or double-closure structure than ordinary unconstrained nonlinear systems.

J.7 Strong Correspondence Conjectures

These conjectures require substantially more evidence than the operational hypotheses above.

Conjecture J-SC1 — Restricted gauge correspondence

A subset of financial frame transformations forms a stable representation with:

- identifiable local transformations G_f;
- connections U_AB;
- covariance;
- nontrivial loop holonomy;
- frame-independent observables.

Status: [H/A]

Promotion requirement: Reproducible covariance across more than one protocol family.

Conjecture J-SC2 — Restricted spin-½ correspondence

A financial closure class exhibits an operationally meaningful distinction between:

2π-like incomplete return (J.180)

and:

4π-like accountable return. (J.181)

Status: [H/A]

Falsifier: An ordinary two-stage workflow captures the entire observable structure.

Conjecture J-SC3 — Restricted Dirac correspondence

A financial subject possesses an empirically useful first-order multicomponent propagation law whose Γ structure, mass modes, and covariant derivatives survive benchmark comparison.

Status: [H/A]

Falsifier: The model reduces without loss to a generic hybrid state-space system.

Conjecture J-SC4 — Financial spectrum conjecture

Stable classes of bounded financial identities and collective modes can eventually be classified by:

- transformation law;
- charge;
- closure topology;
- mass;
- binding;
- gate behaviour;

- residual signature.

Status: [H]

This is the broadest Financial Standard Model research target.

It is not established by the present margin calibration.

J.8 Falsifier Register

J.8.1 Complex-state falsifiers

Reject complex priority when:

1. Q is only a residual error bucket;
2. A is chosen retrospectively;
3. phase is unstable under reasonable normalization;
4. the real pair performs equally well;
5. episode alignment does not improve;
6. gate prediction does not improve.

Reduction:

Complex State
→ Real Pair
→ Scalar State. (J.182)

J.8.2 Charge falsifiers

Reject charge status when:

1. no stable carrier exists;
2. no field is declared;
3. sign changes to fit outcomes;
4. transport cannot be specified;
5. no conversion vertex exists;
6. no residual can be measured;
7. charge coding is indistinguishable from ordinary exposure.

Reduction:

Charge
→ Coupling Orientation
→ Signed Exposure

→ Descriptive Coordinate. (J.183)

J.8.3 Spin falsifiers

Reject financial spin when:

1. action and ledger are not independently measurable;
2. no first-cycle incompleteness exists;
3. the ledger return has no independent consequence;
4. no meaningful residual arises from failed return;
5. scalar status performs equally well;
6. cross-frame identity is irrelevant.

Reduction:

Spinor
→ Action–Ledger Doublet
→ Workflow
→ Scalar Status. (J.184)

J.8.4 Gauge falsifiers

Reject gauge terminology when:

1. frames are merely different column names;
2. no invariant kernel exists;
3. connections are retrospectively chosen;
4. local reparameterization changes substantive conclusions;
5. no loop can be constructed;
6. pairwise reconciliation performs equally well.

Reduction:

Gauge
→ Governed Transport
→ Multi-Ledger Reconciliation
→ Data Matching. (J.185)

J.8.5 Curvature falsifiers

Reject curvature terminology when:

1. the loop is not closed;

2. the starting and returning objects are incomparable;
3. lawful costs explain the full difference;
4. path order does not matter;
5. loop residual is unstable;
6. pairwise exception counts perform equally well.

Reduction:

Curvature
→ Loop Transport Residual. (J.186)

J.8.6 Mass falsifiers

Reject financial mass when:

1. transformation inertia cannot be measured;
2. the quantity is only position size or volatility;
3. it has no effect on propagation or closure;
4. it is confounded completely with latency or liquidity;
5. the matrix modes are not identifiable.

Reduction:

Mass
→ Identity Inertia
→ Friction Index. (J.187)

J.8.7 Closure-capacity falsifiers

Reject c_P as a distinct construct when:

1. no threshold behaviour exists;
2. residual grows smoothly without capacity transition;
3. A-B Fixedness is unrelated to throughput;
4. c_P is not reproducible;
5. a standard queueing model is sufficient.

Reduction:

Semantic or Coherent Closure Capacity
→ Operational Processing Capacity. (J.188)

J.8.8 Γ-algebra falsifiers

Reject Γ structure when:

1. basis choice controls the result;
2. anticommutation adds no restriction;
3. the component interpretation is arbitrary;
4. the squared equation adds no testable modes;
5. generic coupling matrices perform equally well.

Reduction:

Γ Algebra
→ Generic Two-Component Operator. (J.189)

J.8.9 Recursive-closure falsifiers

Reject ledger recursion when:

1. history adds no information beyond current state;
2. residual has no future effect;
3. subjects with identical current states but different histories behave identically;
4. fixed-parameter models perform equally well.

Reduction:

Recursive Ledger Model
→ Finite-Lag Model
→ Stationary Current-State Model. (J.190)

J.8.10 Full-system falsifier

Reject the complete Financial Gauge–Dirac–Gate–Ledger system as the preferred model when:

$U_{full} \leq \max_j U_{simpler,j}$. (J.191)

This does not automatically reject:

- the margin-buffer identity;
- gate discipline;
- residual preservation;
- action–ledger separation;
- governed transport.

The framework is modular.

J.9 Dependency Register

Advanced claims depend upon earlier claims.

The dependency chain is:

Protocol
→ Identity
→ Projection
→ Gate
→ Trace
→ Residual
→ Transport
→ Charge
→ Spin
→ Mass
→ Gauge
→ Dirac
→ Recursive Spectrum. (J.192)

The ordering is not strictly linear in every implementation.

But certain dependencies are mandatory.

J.9.1 Identity dependency

No stable identity implies:

No Reliable Charge Carrier. (J.193)

No Reliable Cross-Frame Invariant. (J.194)

No Accountable Successor Identity. (J.195)

J.9.2 Gate dependency

No authoritative gate implies:

Candidate Only, not Event. (J.196)

No Event Trace. (J.197)

No Event-Level Ledger Backreaction. (J.198)

J.9.3 Ledger dependency

No independent ledger return implies:

No Strong Spin Claim. (J.199)

No Recursive Closure Claim. (J.200)

J.9.4 Frame dependency

No frame map implies:

No Gauge Claim. (J.201)

No Loop Curvature Claim. (J.202)

J.9.5 Benchmark dependency

No simpler benchmark implies:

No Advanced-Model Promotion. (J.203)

J.10 Claim-Promotion Ladder

A claim may be promoted only after passing the next gate.

Observation
→ Measured Relation
→ Candidate
→ Admitted Event
→ Closed Event
→ Episode Pattern
→ Cross-Frame Regularity
→ Recursive World Law. (J.204)

The promotion conditions are:

Relation → Candidate

Requires declared protocol and threshold.

Candidate → Event

Requires authoritative gate and trace.

Event → Closed Event

Requires ledger return and residual disclosure.

Closed Event → Episode

Requires persistent ordering across multiple events.

Episode → Cross-Frame Regularity

Requires transport.

Cross-Frame Regularity → World Law

Requires authority, ledger backreaction, and changed future admissibility.

J.11 Claim-Downgrading Ladder

A claim should be downgraded when mandatory evidence fails.

Universal Law
→ Cross-Protocol Regularity. (J.205)

Cross-Protocol Regularity
→ Local Protocol Model. (J.206)

Event
→ Candidate Warning. (J.207)

Phase Time
→ Phase Coordinate. (J.208)

Complex State
→ Real Pair. (J.209)

Charge
→ Exposure. (J.210)

Spinor
→ Workflow. (J.211)

Gauge
→ Transport Map. (J.212)

Curvature
→ Loop Residual. (J.213)

Mass
→ Friction. (J.214)

Dirac
→ Hybrid State Space. (J.215)

Downgrading is not failure of scientific integrity.

It is the mechanism preserving it.

J.12 Major Claim Matrix

Claim	Class	Current status	Required evidence	Reduction if unsupported
$A^2 = R^2 + Q^2$	I	Exact under definitions	Correct declaration of A and R	None; reject definitions if inappropriate
$\partial R/\partial \theta = -Q$	I/T	Exact and interpretable	Practical incremental value	Keep as local identity only

Claim	Class	Current status	Required evidence	Reduction if unsupported
Q predicts buffer stress	H	Proposed	Out-of-sample benchmark gain	Remove Q or retain as descriptive
$Q_C^2 = Q_M^2 + (1 - h^2)R_M^2$	I/C	Exact under shared amplitude	Justification of shared amplitude	Use separate real variables
$B = C + nhR - D$	D/I	Exact margin construction	Applicable contractual protocol	Replace with actual margin rule
Threshold crossing is not gate	T/G	Strong methodological claim	Gate-comparison study	Treat threshold as event if institution truly does
Structural charge	C/T	Proposed	Carrier, field, vertex, transport	Exposure or obligation
Charge-vertex balance	I/C	Accounting identity by definition	Empirical residual usefulness	Ordinary transaction reconciliation
Action–ledger doublet	C	Operationally plausible	Independent measurement	Workflow state
Financial spin	T/H	Strong candidate	Double closure and incremental gain	Action–ledger terminology
A-B Fixedness	C/H	Proposed diagnostic	Reliability and predictive value	Identity matching
Gauge covariance	C/H	Unvalidated	Representation tests	Governed transport
Loop curvature	T/H	Unvalidated	Stable loop and benchmark gain	Loop residual
Identity mass	T/H	Proposed	Measurable inertia distinct from friction	Friction index
c_P	C/H	Proposed	Reproducible overload threshold	Processing capacity
Γ algebra	C/H	Algebra exact, finance unvalidated	Distinct observable consequences	Generic matrices
Financial Dirac kernel	C/A/H	Coherent construction	Full benchmark superiority	Hybrid state space
Gate jumps	C	Strong modelling fit	Event-level calibration	Switching model
Ledger recursion	C/H	Proposed	History-dependent future dynamics	Finite-lag model
Financial spectrum	H	Long-term research target	Stable classifications across domains	Local case taxonomy

J.13 Deterministic versus Probabilistic Falsification

J.13.1 Deterministic claims

A deterministic claim may be falsified by one valid counterexample.

Example:

“All ordinary transport preserves owner identity.” (J.216)

A lawful ordinary edge that changes owner without a conversion vertex would contradict the claim.

J.13.2 Probabilistic claims

A probabilistic claim concerns distributions.

Example:

“Higher Δ_spin predicts slower closure.” (J.217)

One rapidly closed high-defect event does not falsify the claim.

The relevant test concerns:

- estimated coefficient;
- uncertainty;
- calibration;
- out-of-sample performance;
- predeclared tolerance.

J.13.3 Model-family claims

A model-family claim is falsified when:

- it cannot represent its intended cases;
- parameters are not identifiable;
- benchmark utility is lower;
- terminology exceeds demonstrated structure.

J.14 Locality Register

Every supported claim should state its scope.

Define:

Scope(Claim)
:= (Institution,AssetClass,Protocol,FrameSet,Horizon,Regime,Outcome). (J.218)

A claim surviving only one protocol should be stated as:

Local Regularity_P. (J.219)

not:

Universal Financial Law. (J.220)

Transport failure produces:

Claim_global → Claim_local,P. (J.221)

J.15 Invalidation Register

Every strong claim should specify what later evidence can reopen it.

Define:

Invalidate(Claim_k)
:= NewEvidence
∧ MaterialConflict
∧ ValidAuthority
∧ PreservedPriorTrace. (J.222)

Possible invalidators include:

- revised market mark;
- corrected quantity;
- legal ownership dispute;
- settlement reversal;
- accounting restatement;
- newly discovered charge vertex;
- protocol misclassification;
- hidden residual;
- benchmark failure.

Reopening is:

Claim_supported,v1
→ Claim_reopened,v2. (J.223)

The original claim remains in the ledger.

J.16 Provenance Register

Every major statement should identify its provenance.

J.16.1 Source-derived

Directly stated or mathematically constructed in the source materials.

Example:

Identity remembers what remains recognizable.

Charge remembers how identity rotates.

Spin remembers how identity returns.

Mass measures the cost of identity-preserving change.

J.16.2 Adapted source architecture

A source object translated into the present margin-account system.

Example:

The generalized action–ledger spinor adapted to margin closure.

J.16.3 Present-article synthesis

A new combination of source concepts.

Example:

The Financial Gauge–Dirac–Gate–Ledger architecture.

J.16.4 Present empirical proposal

A test introduced by the present article.

Example:

Δ_spin predicting margin-call closure duration.

J.16.5 Cross-domain analogy

A comparison with physical gauge or Dirac structures.

The source main article expressly requires these categories to remain distinct rather than being merged into one undifferentiated claim.

J.17 Formal Claim Object

A publication-ready claim should be represented as:

Claim_j
:= (
Statement_j,
Class_j,
Protocol_j,
Scope_j,
Definitions_j,
Evidence_j,
Benchmark_j,
Tolerance_j,
Falsifier_j,
Reduction_j,

Invalidation_j,
Provenance_j
). (J.224)

A claim lacking one of the mandatory fields should not be promoted beyond exploratory status.

J.18 Example Claim Objects

J.18.1 Q phase-exposure claim

Statement

Q is the first-order phase exposure of R under fixed A.

Class

[I/T]

Protocol

Declared Complex CAPM geometry.

Equation

$\partial R/\partial \theta = -Q.$ (J.225)

Scope

Local differential interpretation.

Falsifier

The mathematical identity is not empirically falsifiable once definitions are accepted.

The practical-value claim is falsified by no benchmark gain.

Reduction

Retain equation as geometry only.

J.18.2 Financial-charge claim

Statement

The margin account carries structural asset, funding, and collateral charges.

Class

[C/T/H]

Protocol

One account, one asset, one debt, one margin agreement.

Required evidence

- stable carrier;
- contractual orientation;
- transport;
- vertices;
- residual.

Falsifier

Coding cannot survive ordinary transaction and frame transport.

Reduction

Signed claim and obligation classes.

J.18.3 Financial-spin claim

Statement

Margin closure is spin-like because the admitted call opens an action–ledger cycle that closes only after independent return through collateral, settlement, debt, accounting, and legal ledgers.

Class

[T/H/A]

Required evidence

- independent components;
- measurable first-cycle incompleteness;
- return map;
- residual;
- benchmark gain.

Falsifier

Workflow age and simple status explain all outcomes.

Reduction

Two-stage closure workflow.

J.18.4 Gauge claim

Statement

Market and collateral representations form a restricted gauge-like transport system.

Class

[C/H/A]

Required evidence

- frames;
- invariant;
- connection;
- covariance;
- loop test.

Falsifier

Substantive conclusions change under presentation-only transformation.

Reduction

Market-to-collateral reconciliation map.

J.18.5 Dirac claim

Statement

The account admits a Financial Gauge–Dirac representation between gates.

Class

[C/H/A]

Required evidence

- irreducible doublet;
- first-order propagation;
- Γ usefulness;
- mass identification;
- frame covariance;
- benchmark gain.

Falsifier

Hybrid state-space model performs equally well.

Reduction

Hybrid action–ledger state-space model.

J.19 Minimal Admission Tests

J.19.1 Complex admission

AdmitComplex(Z)
:= IndependentR
∧ IndependentQ
∧ StableGeometry
∧ PhaseRelevance
∧ BenchmarkGain. (J.226)

J.19.2 Charge admission

AdmitCharge(q)
:= Carrier
∧ Field
∧ Sign
∧ Coupling
∧ Transport
∧ Vertex
∧ Residual. (J.227)

J.19.3 Spin admission

AdmitSpin(Ψ)
:= Identity
∧ IndependentAction
∧ IndependentLedger
∧ FirstCycleIncomplete
∧ ReturnMap
∧ Residual
∧ BenchmarkGain. (J.228)

J.19.4 Gauge admission

AdmitGauge
:= Frames
∧ Invariant
∧ Connection
∧ Covariance
∧ Loop
∧ ResidualHonesty. (J.229)

J.19.5 Mass admission

AdmitMass
:= IdentitySpecificInertia
∧ Measurability
∧ DynamicRole
∧ DistinctionFromFriction
∧ BenchmarkGain. (J.230)

J.19.6 Dirac admission

AdmitDirac
:= Identity
∧ IrreducibleDoublet
∧ FirstOrderPropagation
∧ ΓConstraint
∧ Mass
∧ Covariance
∧ Residual
∧ IncrementalGain. (J.231)

J.20 Canonical Claim Ceiling

For any claim:

L_claim
≤ min(
L_evidence,
L_gate,
L_trace,
L_transport,
L_authority,
L_model,
L_benchmark
). (J.232)

A strong result in one dimension cannot compensate for a missing mandatory dimension.

Examples:

Strong Q Geometry

- No Gate Utility
  → Phase Diagnostic Only. (J.233)

Valid Margin Event

- No Independent Ledger Return
  → Event Model, not Spin Model. (J.234)

Stable Frame Map

- No Covariance
  → Governed Transport, not Gauge. (J.235)

Useful Doublet

- No Γ Increment
  → Coupled State Space, not Dirac. (J.236)

J.21 Formal Research Sequence

The cheapest and least speculative claims should be tested first.

The correct sequence is:

1. Validate protocol and identity.
2. Validate gate separation.
3. Validate residual typing.
4. Validate action–ledger independence.
5. Validate frame transport.
6. Test Q and phase increment.
7. Test charge stability.
8. Test loop residual.
9. Estimate mass and capacity.
10. Test Γ and Dirac increment.
11. Test recursive spectrum.

In compressed form:

Operational Definition
→ Minimal Test
→ Null Comparison
→ Residual Audit
→ Conditional Promotion. (J.237)

This ordering follows the source research roadmap's rule to test the cheapest falsifier before constructing the most elaborate model.

J.22 Final Formal Thesis

The article's final thesis can now be stated with explicit epistemic structure.

J.22.1 Exact mathematical core

Under declared definitions:

$A^2 = R^2 + Q^2.$ (J.238)

$\partial R/\partial \theta = -Q.$ (J.239)

$dB = dC + nh(R/A)dA - nhQd\theta + nRdh + hRdn - dD.$ (J.240)

These are exact local identities.

J.22.2 Operational financial core

A mature margin event contains:

Bounded Identity
→ Candidate Breach
→ Authoritative Gate
→ Outward Action
→ Independent Ledger Return
→ Residual
→ Accountable Successor. (J.241)

This is a proposed but directly operational architecture.

J.22.3 Structural interpretation

When rights and obligations create stable field-indexed transformation orientation, the system admits a charge-like description.

When outward action requires independent return through ledgers, the system admits a spin-like closure description.

When one identity is transported lawfully across several legitimate frames, the system admits a gauge-like description.

When identity-preserving change has measurable inertia, the system admits a mass-like description.

These are conditional structural interpretations.

J.22.4 Strong mathematical proposal

When:

- the doublet is irreducible;
- propagation is first-order;
- the Γ algebra is useful;
- mass is measurable;
- transport is covariant;
- residual is explicit;

the financial subject admits the candidate continuous kernel:

$(i\not{D}_{fin} - M)\Psi = \mathscr{R}$. (J.242)

At authoritative gates:

$\mathcal{S}^{+} = \mathcal{J}(\mathcal{S}^{-}, L) + \mathscr{R}_{jump}$. (J.243)

After return to ledger:

$L_{k+1} = U_L(L_k, T_k, \mathscr{R}_k)$. (J.244)

This is a proposed hybrid model family.

J.22.5 Empirical ceiling

The full model is supported only when:

$$U_{full} > U_{simpler} \qquad (J.245)$$

under prospective, complexity-penalized testing.

Until that condition is met:

Financial Gauge–Dirac
= Research Construction, not Validated Universal Law. (J.246)

J.23 Compact Definitions and Status Table

Object	Compact definition	Status
Protocol P	Boundary, observation, horizon, intervention, gate, trace, residual, transport	Definition
Identity K	Recognizable financial carrier across permitted transformation	Definition/construction
Q	$\sqrt{(A^2 - R^2)}$	Exact derived coordinate
θ	arccos(R/A)	Exact derived coordinate
Charge q	Field-indexed relational orientation	Proposed interpretation
Gate G	Rule + authority + decision + trace + residual	Operational definition
Trace T	Persistent record of admitted event	Operational definition
Residual $\mathscr{R}$	Observed minus protocol-expected state	Operational definition
Ledger L	Ordered inherited trace and residual	Proposed operational construction
Spin Ψ	Action–ledger double closure	Proposed structural interpretation
Frame F	Local financial observation and authority protocol	Definition
Transport U_{AB}	Lawful map between frame representations	Construction
ABFix	Cross-frame identity recognition after transport	Proposed diagnostic
Holonomy H_ℓ	Product of loop transports	Mathematical construction
Curvature κ	Governed loop residual with path significance	Empirical candidate
Mass M	Identity-preserving transformation inertia	Proposed interpretation
c_P	Maximum coherent closure rate	Empirical candidate
Γ algebra	Closure-space anticommutation structure	Mathematical construction
Financial Dirac kernel	First-order charged doublet propagation	Proposed model
Gate jump $\mathcal{J}$	Discrete authoritative state transition	Hybrid construction
Recursive closure	Ledger-conditioned future operator update	Empirical model

J.24 Final Register Principle

The completed article should never present all of its equations as though they possessed one evidential status.

Its disciplined structure is:

Exact Identity
≠ Operational Definition
≠ Proposed Construction
≠ Structural Interpretation
≠ Empirical Conjecture
≠ Cross-Domain Analogy. (J.247)

The final rule is:

> **Use mathematics to make the obligations of the theory explicit, not to make every obligation appear already satisfied.**

The compact scientific constitution is:

Define before interpreting. (J.248)

Declare before measuring. (J.249)

Gate before promoting. (J.250)

Transport before generalizing. (J.251)

Preserve residual before closing. (J.252)

Benchmark before celebrating complexity. (J.253)

Reduce when evidence fails. (J.254)

Revise without erasing history. (J.255)

Appendix K — Research Constitution, Publication Blueprint, and Closing Declaration

K.1 Purpose

The article has now developed a complete conceptual and operational architecture containing:

- Complex CAPM valuation geometry;
- a bounded leveraged financial subject;
- structural and effective financial charge;
- margin constraints and authoritative gates;
- action–ledger double closure;
- financial frame transport;

- A-B Fixedness;
- loop residual and curvature candidates;
- identity mass;
- coherent closure capacity;
- a candidate Financial Gauge–Dirac kernel;
- discontinuous gate jumps;
- recursive ledger backreaction;
- synthetic and real-data testing protocols;
- a machine-readable implementation;
- a formal definitions and falsifiers register.

The remaining task is not to add another mathematical layer.

It is to govern the architecture as a research programme.

A theory of recursive financial closure must itself be capable of:

- remembering what it originally claimed;
- preserving contrary evidence;
- distinguishing definition from discovery;
- reducing unsupported terminology;
- reopening conclusions without rewriting history;
- closing unproductive research branches.

The source Financial Standard Model imposes the same ceiling: the present result is a **testable architecture**, not a validated universal financial model, and imported physical vocabulary is justified only when it creates measurable structural obligations.

The constitutional rule is:

A Theory of Accountable Closure Must Be Accountable for Its Own Closure. (K.1)

K.2 The Article in One Sentence

The article's complete proposal is:

> **A class of financial systems may be modelled as bounded identity-propagation systems in which valuation pressure encounters contractual constraints, authoritative gates activate relational obligations, outward actions require independent ledger return, cross-frame transport preserves accountable identity, residual records nonclosure, and completed events recursively alter the conditions governing future financial behaviour.**

In compressed form:

Value
→ Identity-Bearing Constraint
→ Charge
→ Gate
→ Action
→ Ledger Return
→ Transport
→ Residual
→ Recursive Financial World. (K.2)

This is a present-article synthesis.

It should not be attributed to any one source as a previously established theorem.

K.3 What Has Been Established Inside the Article

K.3.1 Exact mathematical identities

Under the declared Complex CAPM construction:

$A^2 = R^2 + Q^2$. (K.3)

$R = A \cos \theta$. (K.4)

$Q = A \sin \theta$. (K.5)

$Z = R + iQ = Ae^{xp(i\theta)}$. (K.6)

$\partial R/\partial \theta = -Q$. (K.7)

For the margin subject:

$B = C + nhR - D$. (K.8)

and:

$dB = dC + nh(R/A)dA - nhQd\theta + nRdh + hRdn - dD$. (K.9)

Therefore:

$\partial B/\partial \theta = -nhQ$. (K.10)

These results are exact under their definitions and differentiability assumptions.

They do not independently establish empirical superiority.

K.3.2 Operational distinctions

The article has defined and maintained the following distinctions:

Candidate Breach ≠ Admitted Margin Call. (K.11)

Admitted Call ≠ Cure Action. (K.12)

Cure Action ≠ Ledger Return. (K.13)

Execution ≠ Settlement. (K.14)

Economic Cost ≠ Transport Residual. (K.15)

Frame Difference ≠ Financial Inconsistency. (K.16)

Reported Closure ≠ Verified Closure. (K.17)

Residual ≠ Automatic Failure. (K.18)

Revision ≠ Historical Erasure. (K.19)

These distinctions are not dependent upon a physical analogy.

They constitute the minimum operational core of the model.

K.3.3 Coherent proposed constructions

The article has constructed:

Structural charge vector:

$\vec{q}_{struct} = (q_A, q_F, q_C, \ldots)$. (K.20)

Operational charge:

$q_r^{op} = a_r q_r^{struct}$. (K.21)

Effective charge:

$\vec{q}_{eff} = \Lambda \vec{q}_{struct}$. (K.22)

Action–ledger doublet:

$\Psi = [\psi_A, \psi_L]^T$. (K.23)

Closure defect:

$\delta_{spin} = \psi_L - U_{AL}\psi_A$. (K.24)

Frame residual:

$r_{AB} = S_B - U_{ABS_A}$. (K.25)

Loop residual:

$\mathscr{R}_\ell = S_{return}^{obs} - H_\ell^{PS_{start}}$. (K.26)

Mass operator:

$M = m_{II_2} + m_C \sigma_x + m_\Delta \sigma_z$. (K.27)

Financial Gauge–Dirac kernel:

$(i\not{D}_{fin} - M)\Psi = \mathscr{R}_{cont}$. (K.28)

Gate jump:

$\mathcal{S}^{+}_{k} = \mathcal{J}_{k}(\mathcal{S}^{-}_{k}, L_{k}) + \mathscr{R}_\text{jump},_{k}.$ (K.29)

Ledger recursion:

$L_{k+1} = U_L(L_{k}, T_{k}, \mathscr{R}_{k}).$ (K.30)

These constructions are mathematically executable.

Their necessity and empirical utility remain testable.

K.4 What Remains Unestablished

The article does not yet establish that:

1. Q universally measures financial pressure;
2. the shared-amplitude collateral construction is uniquely correct;
3. every financial claim possesses charge;
4. every institutional workflow possesses spin;
5. the proposed charge vector is complete;
6. the financial frame graph possesses a universal gauge group;
7. loop residual is best interpreted as curvature;
8. identity mass is separable from liquidity, latency, and friction;
9. c_P is a stable cross-institutional quantity;
10. the selected Γ matrices are uniquely appropriate;
11. the Gauge–Dirac model outperforms mature hybrid state-space models;
12. the architecture improves investment returns;
13. the architecture is a physical theory of markets.

Therefore:

Current Result
= Structural and Empirical Research Architecture. (K.31)

Not:

Current Result
= Validated Universal Financial Law. (K.32)

The source framework explicitly requires this ceiling and warns that typed financial claims do not by themselves constitute profitable trading rules.

K.5 The Minimal Surviving Core

The full architecture can fail at several advanced levels while leaving a useful lower structure.

The minimum core is:

1. declare the protocol;
2. identify the financial subject;
3. freeze the available evidence;
4. distinguish candidate from admitted event;
5. record authority and trace;
6. separate action from ledger return;
7. transport before comparing frames;
8. preserve residual;
9. retain version history;
10. reduce unsupported complexity.

In compact form:

Protocol

- Identity
- Gate
- Trace
- Residual
- Transport
- Revision. (K.33)

This core remains useful even if:

- Q fails;
- charge language fails;
- spin language fails;
- gauge language fails;
- Γ algebra fails;
- the Dirac kernel fails.

K.6 The Model Hierarchy

K.6.1 Level M0 — Conventional scalar finance

State:

x_t. (K.34)

Use when one scalar or ordinary vector sufficiently describes the subject.

K.6.2 Level M1 — Real-pair finance

State:

$x_t = (R_t,Q_t)$. (K.35)

Use when two independently useful real coordinates exist but complex priority is unsupported.

K.6.3 Level M2 — Complex valuation finance

State:

$Z_t = R_t + iQ_t$. (K.36)

Use when:

- the conjugate relation is stable;
- phase is meaningful;
- the complex constraint adds value.

K.6.4 Level M3 — Constraint-bearing state model

State:

$X_t = (Z_t,n_t,h_t,C_t,D_t,B_t)$. (K.37)

Use when valuation and balance-sheet constraints jointly determine admissibility.

K.6.5 Level M4 — Hybrid gate model

Between events:

$\dot{X} = F(X,u)$. (K.38)

At gates:

$X^{+} = J(X^{-})$. (K.39)

Use when authoritative thresholds produce discontinuous status transitions.

K.6.6 Level M5 — Action–ledger model

State:

$\Psi = [\psi_A,\psi_L]^{T}.$ (K.40)

Use when outward action and ledger return are independently measurable and consequential.

K.6.7 Level M6 — Governed frame-transport model

State:

$\{S_f,U_{AB},r_{AB}\}.$ (K.41)

Use when the same identity exists across several legitimate institutional frames.

K.6.8 Level M7 — Gauge-like transport model

Add:

- local frame transformations;
- covariance;
- connection;
- loop holonomy;
- invariant observables.

Use only when these structures are demonstrated.

K.6.9 Level M8 — Gauge–Dirac model

Use:

$(i\not{D}_{fin} - M)\Psi = \mathscr{R}.$ (K.42)

only when:

- the doublet is irreducible;
- first-order propagation is appropriate;
- Γ structure constrains the model;
- mass is measurable;
- covariance survives;
- benchmark gain is positive.

K.6.10 Level M9 — Recursive financial-world model

Use when closure history changes:

- future rules;
- future permissions;
- future effective charge;
- future mass;
- future capacity;
- future available branches.

Then:

$\mathcal{L}_{k+1} \neq \mathcal{L}_k$. (K.43)

K.6.11 Model-selection law

The governing law is:

Retain the Highest Model Level Whose Mandatory Conditions Survive. (K.44)

A failed higher model does not invalidate every lower level.

The source Periodic Grammar expresses the same principle through linked closure and state-model reduction ladders.

K.7 The Reduction Ladder

K.7.1 Complex reduction

Time-Bearing Financial World
→ Phase-Sensitive Recursive Model
→ Complex State
→ Real Pair
→ Scalar State. (K.45)

K.7.2 Institutional reduction

Financial Gauge–Dirac System
→ Gauge-Covariant Closure Model
→ Governed Frame Transport
→ Action–Ledger Workflow
→ Hybrid Gate Model
→ Conventional Margin Model. (K.46)

K.7.3 Vocabulary reduction

Charge
→ Coupling Orientation
→ Signed Exposure
→ Contractual Obligation. (K.47)

Spin
→ Action–Ledger Double Closure
→ Two-Stage Workflow. (K.48)

Gauge
→ Governed Frame Transport
→ Cross-Ledger Reconciliation. (K.49)

Curvature
→ Loop Transport Residual. (K.50)

Mass
→ Identity Inertia
→ Friction Index. (K.51)

Dirac
→ First-Order Coupled Identity Model
→ Hybrid State-Space Model. (K.52)

The source minimal-terminology rule likewise requires choosing the weaker term until the stronger structure is demonstrated.

K.8 The Research Constitution

Law 1 — Protocol Law

Every strong financial claim is protocol-bound.

Claim = Claim_P. (K.53)

No boundary, no reproducible object.

Law 2 — Filtration Law

Later evidence cannot be used as though it had been available at the original decision time.

Evidence_t ⊆ $\mathscr{J}$_t. (K.54)

Future information may revise a claim.

It cannot retroactively justify it.

Law 3 — Identity Law

No charge, transport, or closure claim is valid without a bounded financial identity.

No Stable Carrier ⇒ No Stable Transformation Memory. (K.55)

Law 4 — Projection Law

Every valuation, indicator, or model output is a partial projection.

Projection_P(S) ≠ S. (K.56)

A model output should not be mistaken for the whole financial subject.

Law 5 — Gate Law

A candidate becomes an admitted event only through a declared gate.

Candidate + Authority + Decision + Trace → Event. (K.57)

Law 6 — Action–Ledger Law

Outward action does not by itself close identity.

Financial Closure
= Outward Action

- Independent Ledger Return. (K.58)

This is the central closure law of the source action–ledger architecture.

Law 7 — Transport Law

Two frame-local values should not be compared as residual until lawful transport has been applied.

Residual_AB = Observed_B − T_AB(State_A). (K.59)

Law 8 — Residual Law

Every mature closure process must explicitly state what remains unresolved.

Closure = Admitted Trace + Preserved Nonclosure. (K.60)

Residual may be resolved, accepted, carried, or invalidating.

It may not disappear without explanation.

Law 9 — Trace Law

Commitment must enter persistent history.

Decision without Trace ⇒ No Accountable Commitment. (K.61)

Law 10 — Benchmark Law

Every advanced representation must compete against a capable simpler model.

$U_{advanced} > U_{simpler} + \delta_U.$ (K.62)

Otherwise reduce.

Law 11 — Revision Law

When new evidence invalidates closure:

$Closed_{v1} \rightarrow Reopened_{v2}$ with PriorTrace Preserved. (K.63)

The source framework expressly requires preserving the original decision, evidence, new evidence, residual, and revision rather than rewriting history.

Law 12 — Reduction Law

No model is entitled to retain terminology that its evidence cannot support.

Unsupported Higher Layer
→ Strongest Surviving Lower Layer. (K.64)

K.9 Research-Branch Governance

K.9.1 Branch object

Each research branch should be represented as:

Branch_j
:= (
Question,
Protocol,
Definitions,
Evidence,
Benchmark,
Falsifier,
Residual,
Status,
ReductionPath
). (K.65)

K.9.2 Branch statuses

A branch may be:

- Proposed;

- Operationalized;
- Simulated;
- Pilot-Tested;
- Replicated;
- Supported;
- Locally Supported;
- Reduced;
- Suspended;
- Closed.

K.9.3 Promotion

A branch may be promoted only when:

AdmissionPassed
∧ BenchmarkPassed
∧ ResidualAcceptable
∧ ReplicationAdequate. (K.66)

K.9.4 Reduction

A branch is reduced when:

CoreOperationalValue survives
but
StrongInterpretation fails. (K.67)

Example:

Financial Spin
→ Action–Ledger Closure Model. (K.68)

K.9.5 Suspension

A branch is suspended when:

- evidence is insufficient;
- identification is impossible;
- required data do not exist;
- a dependency remains unresolved.

Suspension is not confirmation or rejection.

K.9.6 Closure of a branch

A branch should be closed when:

1. repeated adequately powered tests fail;
2. the construct cannot be operationalized;
3. a simpler model dominates consistently;
4. the branch survives only through post-hoc redefinition;
5. the cost of testing exceeds plausible scientific value.

A self-revising theory must be capable of closing its own branches.

K.10 Staged Research Roadmap

The source research roadmap requires progression from operational definition to minimal test, null comparison, residual audit, and only then conditional promotion.

The present article's roadmap is therefore divided into eight stages.

Stage 1 — Measurement and gate discipline

Objective

Validate:

- buffer construction;
- candidate breach;
- admitted call;
- authority;
- trace;
- closure outcome.

Principal hypothesis

Valid gates separate consequential events from numerical crossings.

Required data

- account state;
- agreement;
- mark;

- gate decision;
- outcome.

Strongest permitted term

Governed Margin Event.

Stage 2 — Action–ledger separation

Objective

Measure:

- outward action;
- expected ledger return;
- observed ledger return;
- closure defect;
- false closure.

Principal hypothesis

Δ_spin predicts delayed or failed closure beyond ordinary workflow variables.

Strongest permitted term

Action–Ledger Closure Model.

Use financial spin only provisionally.

Stage 3 — Governed frame transport

Objective

Test:

- market-to-collateral transport;
- trade-to-settlement transport;
- settlement-to-accounting transport;
- identity preservation;
- edge residual.

Principal hypothesis

Transport-adjusted residual reduces false alarms.

Strongest permitted term

Governed Financial Transport.

Stage 4 — Complex valuation priority

Objective

Compare:

Complex CAPM
versus
Flexible Real Pair.

Principal hypotheses

- Q adds information;
- θ improves episode alignment;
- complex constraints improve calibration.

Strongest permitted term

Complex Financial State.

Do not promote phase to a clock without independent phase-time tests.

The secondary-time source explicitly requires rejection or reduction when Q is merely an error bucket, phase lacks simplification, or flexible real models perform as well.

Stage 5 — Charge and vertex system

Objective

Test whether:

- structural signs are stable;
- ordinary transport preserves charge;
- declared vertices explain conversion;
- charge residual predicts later failure.

Strongest permitted term

Financial Charge Candidate.

Stage 6 — Loop and curvature studies

Objective

Construct:

- meaningful closed frame loops;
- protocol-expected return operators;
- lawful cost controls;
- pairwise residual benchmarks.

Principal hypothesis

Loop residual adds information beyond ordinary reconciliation.

Strongest permitted term

Financial Loop Residual.

Use curvature only after order and holonomy tests survive.

Stage 7 — Mass, capacity, and Γ structure

Objective

Estimate:

- identity inertia;
- action–ledger coupling modes;
- closure-capacity threshold;
- Γ-specific restrictions.

Principal comparisons

- mass versus friction;
- c_P versus queueing capacity;
- Γ algebra versus generic matrices.

Strongest permitted term

First-Order Coupled Identity Model.

Stage 8 — Recursive Financial Gauge–Dirac model

Objective

Test the full architecture:

Continuous propagation

- Gate jumps
- Frame transport
- Residual
- Ledger-conditioned parameter update.

Required evidence

- lower layers validated;
- independent replication;
- prospective benchmark gain;
- cross-protocol survival.

Strongest permitted term

Restricted Financial Gauge–Dirac System.

Not universal Financial Standard Model.

K.11 Publication Sequence

The full research programme should not first be published as one empirical claim.

A defensible sequence is:

Paper 1

Margin Calls as Governed Events

Candidate breach versus admitted gate.

Paper 2

Action Is Not Closure

Action–ledger mismatch and closure duration.

Paper 3

From Market Mark to Collateral Value

Governed frame transport and reconciliation residual.

Paper 4

Complex CAPM Under Empirical Pressure

Q and phase versus real-pair alternatives.

Paper 5

Financial Charge as Relational Contract Memory

Carrier, field, vertex, transport, and residual tests.

Paper 6

Financial Holonomy and Cross-Ledger Failure

Loop residual and path dependence.

Paper 7

Identity Inertia and Closure Capacity

Mass and c_P estimation.

Paper 8

A Restricted Financial Gauge–Dirac Model

Integrated prospective test.

The sequence protects the architecture from being judged entirely on its most speculative layer before its operational core has been tested.

K.12 Publication Structure for the Present Article

The article's publication form may be divided into five major components.

Part I — Problem and valuation kernel

Why smooth valuation is insufficient.

Complex CAPM and the Q coordinate.

Part II — Constraint, charge, and gate

The leveraged subject.

Margin geometry.

Charge activation and transaction vertices.

Part III — Spin, transport, and Dirac structure

Action–ledger closure.

Frame graphs.

Mass, Γ algebra, and first-order propagation.

Part IV — Nonlinearity and recursion

Gate jumps.

Forced liquidation.

Ledger backreaction.

General constraint-bearing finance.

Part V — Scientific contract

Benchmarks.

Falsifiers.

Simulation.

Real-data pilot.

Implementation.

Formal claim register.

The appendices may remain as a technical companion if the main article becomes too long for a single publication.

K.13 Recommended Separation into Main Article and Technical Supplement

K.13.1 Main article

Retain:

- Abstract;
- Reader's Guide;
- Parts I–V;
- central equations;
- one worked margin example;
- empirical roadmap;
- final thesis.

Target emphasis:

Conceptual clarity and falsifiable architecture.

K.13.2 Technical supplement

Move:

- full formula compendium;
- charge-vertex catalogue;
- matrix operators;
- frame graph catalogue;
- simulation blueprint;
- real-data protocol;
- machine-readable runtime;
- worked JSON example;
- definitions and conjectures register;
- research constitution.

The supplement should be cited as an integral methodological component rather than treated as optional decoration.

K.14 Publication-Ready Abstract Compression

A compressed abstract may read:

> Classical finance models value, sensitivity, and risk effectively, but many institutional financial processes also depend on identity, authority, conditional obligation, cross-frame recognition, and ledgered closure. This article develops a protocol-bound architecture beginning from Complex CAPM, where admitted value R and conjugate pressure Q form $Z = R + iQ$, and extends it to a leveraged margin account carrying asset, funding, and collateral relations. A numerical margin breach becomes an event only through an authoritative gate; the resulting action is not closed until its consequences return through collateral, settlement, treasury, accounting, legal, and risk ledgers. The account is therefore represented provisionally by an action–ledger doublet transported across governed financial frames. Under stronger admission conditions, the model admits candidate charge, gauge, mass, and first-order Dirac-like structures. The complete system combines continuous propagation, discrete gate jumps, residual registration, and recursive ledger update. These constructions are presented as a falsifiable research architecture, not a validated universal law. Simulation, real-data pilots, benchmark ladders, reduction rules, and a machine-readable runtime are supplied to ensure that each advanced term survives only when it adds measurable value beyond simpler models.

K.15 Publication-Ready Standard Disclaimer

Scientific and Financial Disclaimer
The physical terminology used in this article denotes proposed structural correspondences, formal candidates, and research hypotheses. It does not imply that financial markets literally instantiate particle-physics charge, quantum spin, gauge fields, relativistic spacetime, fermionic statistics, or the Standard Model. Each correspondence must be evaluated independently through declared protocols, transformations, invariants, measurements, benchmarks, and falsifiers. The Complex CAPM, charge, spin, gauge, mass, and Financial Gauge–Dirac constructions are conceptual and methodological unless separately supported by empirical evidence. This article does not provide investment advice, trading recommendations, portfolio instructions, or guarantees of financial performance.

This disclaimer is consistent with the source publication discipline and claim ceilings.

K.16 Reviewer Audit

A reviewer should be able to answer the following questions.

K.16.1 Valuation geometry

1. Is A independently defined?
2. Is R protocol-admitted rather than selected retrospectively?
3. Is Q independently meaningful?
4. Does the real-pair benchmark receive equal flexibility?

K.16.2 Charge

1. What carries charge?
2. In which field?
3. What does the sign mean?
4. How does it transport?
5. Which vertices change it?
6. What residual reveals failure?

K.16.3 Spin

1. Are action and ledger independently measurable?
2. Does the first cycle leave a meaningful open state?
3. What completes the return?
4. Does a workflow model perform equally well?

K.16.4 Gauge

1. What are the frames?
2. What invariant survives?
3. What is the connection?
4. Which changes are representational?
5. What loop is tested?
6. What simpler reconciliation model is the benchmark?

K.16.5 Mass and Dirac structure

1. What observable quantity represents identity inertia?
2. How is it distinguished from delay and liquidity?
3. Why are Γ matrices required?
4. What does the squared equation predict?
5. Does a generic state-space model perform equally well?

K.16.6 Recursion

1. Does ledger history change future outcomes after controlling for current state?
2. Which parameters change?
3. Are revisions preserved?
4. Can the recursive claim reduce to a finite-lag model?

K.17 Practitioner Compression

A practitioner does not need the complete operator algebra.

The minimum diagnostic card is:

Protocol

Which agreement, boundary, mark, and rule apply?

Current object

Is this:

- safe state;

- candidate breach;
- admitted call;
- action in progress;
- ledger-open event;
- closed event;
- contested event?

Gate

Which authority committed the status?

Action

What has actually been done?

Ledger return

What has settled, posted, or been recognized?

Residual

What remains unresolved?

Claim ceiling

What stronger conclusion is still prohibited?

Invalidation

What evidence would reopen the result?

This compression preserves the distinctions that control the claim.

K.18 Researcher Compression

A researcher should maintain the following record.

Claim

What exact relation is being tested?

Protocol

Which boundary and rule define the objects?

Evidence cutoff

What information was available?

Baseline

Which simpler models compete?

Gate

What admission condition applies?

Falsifier

What result rejects the claim?

Reduction

What weaker model survives?

Result

What happened out of sample?

Residual

What remained unexplained?

Replication

Did the result survive another period, institution, or protocol?

Strongest retained term

What vocabulary does the evidence currently permit?

K.19 LLM Compression

An LLM using this framework to analyze a financial event should follow:

Step 1 — Identify the subject

Which bounded account, instrument, obligation, or episode is being discussed?

Step 2 — Recover the protocol

What boundary, timeframe, valuation rule, agreement, and evidence cutoff apply?

Step 3 — Separate observation from interpretation

State the measured condition before naming the event.

Step 4 — Identify the gate

Do not call a threshold crossing a confirmed institutional event without authority and trace.

Step 5 — Separate action from ledger return

Determine whether the consequence has settled, posted, or been legally recognized.

Step 6 — Transport frame-local values

Do not treat market, collateral, accounting, and legal numbers as directly comparable.

Step 7 — Surface residual

Name missing, conflicting, delayed, or disputed evidence.

Step 8 — State the claim ceiling

Use the strongest term whose conditions survive.

Step 9 — State invalidation

Explain what evidence would alter the conclusion.

Step 10 — Reduce when necessary

Prefer a simpler model when it preserves all supported distinctions.

K.20 The Final Twelve Questions

Before any advanced financial claim is accepted, ask:

1. **What is the protocol?**
2. **What is the bounded identity?**
3. **What is directly observed?**
4. **What is mathematically constructed?**
5. **What is merely interpreted?**
6. **Which gate admitted the event?**
7. **Who had authority?**

8. **What outward action occurred?**
9. **Which ledger return remains required?**
10. **How was the identity transported across frames?**
11. **What residual remains?**
12. **Which simpler model could explain the same result?**

If these questions cannot be answered, the advanced terminology should be withheld.

K.21 The Final Financial Standard Model Candidate

The strongest compressed form of the model is:

Valuation state

$Z_k = R_k + iQ_k.$ (K.69)

Financial identity

K_k = bounded carrier of rights, obligations, and lineage. (K.70)

Charge state

$\vec{q}_k$ = field-indexed relational orientation. (K.71)

Closure state

$\Psi_k = [\psi_A,_k,\psi_L,_k]^T.$ (K.72)

Frame transport

S_B = U_ABS_A + r_AB. (K.73)

Continuous propagation

$(i\mathcal{D}_fin[\Theta_k] - M[\Theta_k])\Psi_k = \mathscr{R}_cont,_k.$ (K.74)

Gate condition

$g_j(\mathcal{S}_k) \geq 0.$ (K.75)

Gate jump

$\mathcal{S}^+_k = \mathcal{J}_j(\mathcal{S}^-_k,L_k) + \mathscr{R}_jump,_k.$ (K.76)

Ledger update

$L_{k+1} = U_L(L_k, T_k, \mathscr{R}_k)$. (K.77)

Recursive operator update

$\Theta_{k+1} = U_\Theta(\Theta_k, L_{k+1}, \mathscr{R}_k)$. (K.78)

Together:

FSM_candidate
$:= (Z, K, \vec{q}, \Psi, U, G, M, \mathscr{R}, L, U_\Theta)$. (K.79)

This is a candidate **Financial Standard Model architecture**.

It is not yet a standardized empirical model.

K.22 The Final Causal Thesis

The article's central causal thesis is:

> Financial charge and spin do not arise from nonlinearity itself. They arise when constraints become identity-bearing, relational, authoritative, and history-writing.

The generative sequence is:

Identity
→ Right or Obligation
→ Conditional Constraint
→ Gate Authority
→ Action
→ Independent Ledger Return
→ Residual
→ Recursive Future Constraint. (K.80)

Nonlinearity then appears through:

- threshold activation;
- complementarity;
- branch selection;
- leverage;
- market impact;
- network contagion;
- path dependence;
- recursive parameter change.

Therefore:

Identity-Bearing Constraint Architecture
→ Candidate Charge and Spin
→ Nonlinear Recursive Finance. (K.81)

Not:

Nonlinearity Alone
→ Charge and Spin. (K.82)

K.23 The Final Memory Thesis

The architecture can be understood as a theory of financial memory.

Identity remembers what remains recognizable. (K.83)

Charge remembers how identity is oriented under a field. (K.84)

Spin remembers how identity returns through action and ledger. (K.85)

Mass remembers the cost of remaining recognizable while changing. (K.86)

Gauge remembers how identity survives another frame. (K.87)

The gate remembers which possibility became institutionally consequential. (K.88)

The trace remembers what was committed. (K.89)

Residual remembers what the commitment failed to contain. (K.90)

The ledger remembers what the future must inherit. (K.91)

Recursion remembers that the next financial world is built from the closure of the previous one. (K.92)

K.24 Closing Declaration

This article began with a conventional valuation problem.

A future financial claim was discounted under a baseline rate and a risk-adjusted rate.

The difference was completed geometrically:

$A^2 = R^2 + Q^2$. (K.93)

That completion revealed a phase:

$Z = Ae^{xp(i\theta)}$. (K.94)

But valuation geometry alone was not enough.

The financial subject also carried:

- ownership;
- debt;
- collateral obligations;
- authority relations;

- settlement consequences;
- accounting recognition;
- legal status;
- residual history.

The addition of margin constraints did not mechanically create charge or spin.

What created the possibility of those structures was more specific:

- a bounded identity;
- relational claims and obligations;
- conditional contractual activation;
- authoritative gates;
- independent action and ledger surfaces;
- cross-frame transport;
- trace-preserving recursion.

The leveraged margin account then became more than a discounted asset.

It became an accountable financial subject.

Its complete event was not:

Price Falls
→ Margin Call. (K.95)

It was:

Market and Collateral Movement
→ Buffer Breach
→ Authorized Call
→ Activated Obligation
→ Collateral or Sale Action
→ Settlement
→ Debt and Accounting Return
→ Residual Reconciliation
→ Closure Certificate
→ Changed Future Rules. (K.96)

The proposed Financial Gauge–Dirac system is an attempt to represent that fuller process.

Its value will not be determined by the elegance of its vocabulary.

It will be determined by whether it can:

- identify events more accurately;
- detect false closure;
- reconcile frames more honestly;

- preserve residual;
- predict closure failure;
- survive benchmark comparison;
- reduce itself when unsupported.

The article's final scientific law is therefore not an equation of market destiny.

It is a discipline of model responsibility:

> **Name only what can be bounded, transported, measured, falsified, and risked.**

The final methodological law is:

> **When evidence cannot support the full architecture, preserve the smallest useful structure and close the unsupported remainder.**

And the final financial law proposed for future testing is:

Value Becomes Financial History
Only When Identity Passes Through a Gate, Returns Through Its Ledgers, and Leaves Its Residual Visible. (K.97)

Manuscript Completion Status

The conceptual article and technical appendices are now complete:

- **Main text:** Parts I–XI
- **Appendix A:** Formula Compendium and Symbol Ledger
- **Appendix B:** Numerical Margin Calibration
- **Appendix C:** Charge Vertices and Transaction Grammar
- **Appendix D:** Action–Ledger Matrices
- **Appendix E:** Frame Graphs and Loop Closure
- **Appendix F:** Simulation and Estimation Blueprint
- **Appendix G:** Minimal Real-Data Pilot
- **Appendix H:** Machine-Readable Runtime
- **Appendix I:** Complete Worked Runtime Example
- **Appendix J:** Definitions, Propositions, Conjectures, and Falsifiers
- **Appendix K:** Research Constitution and Closing Declaration

All planned sections are finished.

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.